SOTRASER - Control Combustible

Codigo Fuente Completo

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57 archivos

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```

shared/schema.ts

401 lineas

```
1 | import { sql } from "drizzle-orm";
2 | import { pgTable, text, varchar, integer, real, boolean, date, serial, unique, numeric, timestamp, jsonb } from "drizzle
3 | import { createInsertSchema } from "drizzle-zod";
4 | import { z } from "zod";
5 |
6 | export const faenas = pgTable("faenas", {
7 |   id: serial("id").primaryKey(),
8 |   nombre: text("nombre").notNull(),
9 |   color: text("color").notNull().default("#1A8FFF"),
10 | });
11 |
12 | export const insertFaenaSchema = createInsertSchema(faenas).omit({ id: true });
13 | export type InsertFaena = z.infer<typeof insertFaenaSchema>;
14 | export type Faena = typeof faenas.$inferSelect;
15 |
16 | export const camiones = pgTable("camiones", {
17 |   id: serial("id").primaryKey(),
18 |   patente: text("patente").notNull(),
19 |   modelo: text("modelo").notNull(),
20 |   faenaId: integer("faena_id").notNull(),
21 |   metaKmL: real("meta_km_l").notNull().default(2.1),
22 |   vin: text("vin"),
23 |   odometro: integer("odometro"),
24 |   horasMotor: integer("horas_motor"),
25 |   horasRalenti: integer("horas_ralenti"),
26 |   velPromedio: integer("vel_promedio"),
27 |   conductor: text("conductor"),
28 |   syncOk: boolean("sync_ok").notNull().default(false),
29 |   syncAt: text("sync_at"),
30 |   taraKg: numeric("tara_kg", { precision: 8, scale: 1 }),
31 |   pesoMaximoKg: numeric("peso_maximo_kg", { precision: 8, scale: 1 }),
32 |   capacidadCargaKg: numeric("capacidad_carga_kg", { precision: 8, scale: 1 }),
33 |   tipoVehiculo: varchar("tipo_vehiculo", { length: 50 }),
34 |   anioFabricacion: integer("anio_fabricacion"),
35 |   configuracionEjes: varchar("configuracion_ejes", { length: 20 }),
36 | });
37 |
38 | export const insertCamionSchema = createInsertSchema(camiones).omit({ id: true });
39 | export type InsertCamion = z.infer<typeof insertCamionSchema>;
40 | export type Camion = typeof camiones.$inferSelect;
41 |
42 | export const cargas = pgTable("cargas", {
43 |   id: serial("id").primaryKey(),
44 |   camionId: integer("camion_id").notNull(),
45 |   fecha: text("fecha").notNull(),
46 |   litrosSurtidor: real("litros_surtidor").notNull(),
47 |   litrosEcu: real("litros_ecu").notNull(),
48 |   kmAnterior: integer("km_anterior").notNull(),
49 |   kmActual: integer("km_actual").notNull(),
50 |   proveedor: text("proveedor").notNull().default("COPEC"),
51 | });
52 |
53 | export const insertCargaSchema = createInsertSchema(cargas).omit({ id: true });
54 | export type InsertCarga = z.infer<typeof insertCargaSchema>;
55 | export type Carga = typeof cargas.$inferSelect;
56 |
57 | export const desviacionChecks = pgTable("desviacion_checks", {
58 |   id: serial("id").primaryKey(),
59 |   fleetNum: text("fleet_num").notNull(),
60 |   tipo: text("tipo").notNull(),
61 |   gestionado: boolean("gestionado").notNull().default(false),
62 |   gestionadoAt: text("gestionado_at"),
63 |   nota: text("nota"),
64 | }, (table) => [
65 |   unique("desviacion_checks_fleet_tipo").on(table.fleetNum, table.tipo),
66 | ]);
67 |
68 | export const insertDesviacionCheckSchema = createInsertSchema(desviacionChecks).omit({ id: true });
69 | export type InsertDesviacionCheck = z.infer<typeof insertDesviacionCheckSchema>;
70 | export type DesviacionCheck = typeof desviacionChecks.$inferSelect;
71 |
72 | export const volvoFuelSnapshots = pgTable("volvo_fuel_snapshots", {
73 |   id: serial("id").primaryKey(),
74 |   vin: text("vin").notNull(),
75 |   totalFuelUsed: real("total_fuel_used").notNull(),
76 |   totalDistance: real("total_distance"),
77 |   capturedAt: text("captured_at").notNull(),
78 | }, (table) => [
79 |   unique("volvo_fuel_snap_vin_hour").on(table.vin, table.capturedAt),
80 | ]);
81 |
82 | export const insertVolvoFuelSnapshotSchema = createInsertSchema(volvoFuelSnapshots).omit({ id: true });
83 | export type InsertVolvoFuelSnapshot = z.infer<typeof insertVolvoFuelSnapshotSchema>;
84 | export type VolvoFuelSnapshot = typeof volvoFuelSnapshots.$inferSelect;
85 |
86 | export const parametros = pgTable("parametros", {
87 |   clave: varchar("clave", { length: 50 }).primaryKey(),
88 |   valor: varchar("valor", { length: 200 }).notNull(),
89 |   updatedAt: timestamp("updated_at").defaultNow(),
90 | });
91 |
92 | export const insertParametroSchema = createInsertSchema(parametros);
93 | export type InsertParametro = z.infer<typeof insertParametroSchema>;
94 | export type Parametro = typeof parametros.$inferSelect;
95 |
96 | export const puntosRuta = pgTable("puntos_ruta", {
97 |   id: serial("id").primaryKey(),
98 |   camionId: integer("camion_id"),
99 |   lat: numeric("lat", { precision: 10, scale: 7 }).notNull(),
100 |   lng: numeric("lng", { precision: 10, scale: 7 }).notNull(),
101 |   tipo: varchar("tipo", { length: 20 }),
102 |   nombreInferido: varchar("nombre_inferido", { length: 200 }),
103 |   duracionMin: integer("duracion_min"),
104 |   fecha: timestamp("fecha"),
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105 |   confirmado: boolean("confirmado").default(false),
106 |   confirmadoPor: varchar("confirmado_por", { length: 80 }),
107 |   createdAt: timestamp("created_at").defaultNow(),
108 | });
109 |
110 | export const insertPuntoRutaSchema = createInsertSchema(puntosRuta).omit({ id: true, createdAt: true });
111 | export type InsertPuntoRuta = z.infer<typeof insertPuntoRutaSchema>;
112 | export type PuntoRuta = typeof puntosRuta.$inferSelect;
113 |
114 | export const tarifasRuta = pgTable("tarifas_ruta", {
115 |   id: serial("id").primaryKey(),
116 |   nombreRuta: varchar("nombre_ruta", { length: 200 }).notNull(),
117 |   puntoOrigenId: integer("punto_origen_id"),
118 |   puntoDestinoId: integer("punto_destino_id"),
119 |   origenNombre: varchar("origen_nombre", { length: 200 }),
120 |   destinoNombre: varchar("destino_nombre", { length: 200 }),
121 |   distanciaKm: numeric("distancia_km", { precision: 8, scale: 1 }),
122 |   litrosPromedio: numeric("litros_promedio", { precision: 8, scale: 2 }),
123 |   tiempoHoras: numeric("tiempo_horas", { precision: 5, scale: 2 }),
124 |   tarifaClp: numeric("tarifa_clp", { precision: 12, scale: 0 }),
125 |   tarifaUsd: numeric("tarifa_usd", { precision: 10, scale: 2 }),
126 |   notas: text("notas"),
127 |   activa: boolean("activa").default(true),
128 |   createdAt: timestamp("created_at").defaultNow(),
129 | });
130 |
131 | export const insertTarifaRutaSchema = createInsertSchema(tarifasRuta).omit({ id: true, createdAt: true });
132 | export type InsertTarifaRuta = z.infer<typeof insertTarifaRutaSchema>;
133 | export type TarifaRuta = typeof tarifasRuta.$inferSelect;
134 |
135 | export const tmsContratos = pgTable("tms_contratos", {
136 |   id: serial("id").primaryKey(),
137 |   faenaId: integer("faena_id").references(() => faenas.id),
138 |   nombre: varchar("nombre", { length: 200 }).notNull(),
139 |   cliente: varchar("cliente", { length: 200 }),
140 |   descripcion: text("descripcion"),
141 |   fechaInicio: date("fecha_inicio").default("2026-03-01"),
142 |   activo: boolean("activo").default(true),
143 |   totalViajes: integer("total_viajes").default(0),
144 |   kmTotal: numeric("km_total", { precision: 12, scale: 1 }).default("0"),
145 |   litrosTotal: numeric("litros_total", { precision: 12, scale: 2 }).default("0"),
146 |   createdAt: timestamp("created_at").defaultNow(),
147 | });
148 |
149 | export const insertTmsContratoSchema = createInsertSchema(tmsContratos).omit({ id: true, createdAt: true });
150 | export type InsertTmsContrato = z.infer<typeof insertTmsContratoSchema>;
151 | export type TmsContrato = typeof tmsContratos.$inferSelect;
152 |
153 | export const tmsViajes = pgTable("tms_viajes", {
154 |   id: serial("id").primaryKey(),
155 |   contratoId: integer("contrato_id").references(() => tmsContratos.id),
156 |   camionId: integer("camion_id").references(() => camiones.id),
157 |   codigo: varchar("codigo", { length: 30 }).notNull().unique(),
158 |   conductor: varchar("conductor", { length: 100 }),
159 |   origenNombre: varchar("origen_nombre", { length: 200 }),
160 |   origenLat: numeric("origen_lat", { precision: 10, scale: 7 }),
161 |   origenLng: numeric("origen_lng", { precision: 10, scale: 7 }),
162 |   destinoNombre: varchar("destino_nombre", { length: 200 }),
163 |   destinoLat: numeric("destino_lat", { precision: 10, scale: 7 }),
164 |   destinoLng: numeric("destino_lng", { precision: 10, scale: 7 }),
165 |   fechaSalida: timestamp("fecha_salida"),
166 |   fechaLlegada: timestamp("fecha_llegada"),
167 |   fechaCierre: timestamp("fecha_cierre"),
168 |   estado: varchar("estado", { length: 20 }).notNull().default("DETECTADO"),
169 |   kmInicio: numeric("km_inicio", { precision: 10, scale: 1 }),
170 |   kmCierre: numeric("km_cierre", { precision: 10, scale: 1 }),
171 |   kmRecorridos: numeric("km_recorridos", { precision: 10, scale: 1 }),
172 |   litrosSigetra: numeric("litros_sigetra", { precision: 8, scale: 2 }),
173 |   litrosEcu: numeric("litros_ecu", { precision: 8, scale: 2 }),
174 |   diferenciaLitros: numeric("diferencia_litros", { precision: 8, scale: 2 }),
175 |   rendimientoReal: numeric("rendimiento_real", { precision: 6, scale: 2 }),
176 |   detectadoPorIa: boolean("detectado_por_ia").default(false),
177 |   confirmado: boolean("confirmado").default(false),
178 |   notas: text("notas"),
179 |   createdAt: timestamp("created_at").defaultNow(),
180 | });
181 |
182 | export const insertTmsViajeSchema = createInsertSchema(tmsViajes).omit({ id: true, createdAt: true });
183 | export type InsertTmsViaje = z.infer<typeof insertTmsViajeSchema>;
184 | export type TmsViaje = typeof tmsViajes.$inferSelect;
185 |
186 | export const tmsParadas = pgTable("tms_paradas", {
187 |   id: serial("id").primaryKey(),
188 |   viajeId: integer("viaje_id").references(() => tmsViajes.id),
189 |   orden: integer("orden").notNull(),
190 |   nombre: varchar("nombre", { length: 200 }),
191 |   direccion: varchar("direccion", { length: 300 }),
192 |   lat: numeric("lat", { precision: 10, scale: 7 }),
193 |   lng: numeric("lng", { precision: 10, scale: 7 }),
194 |   tipo: varchar("tipo", { length: 20 }).notNull().default("ENTREGA"),
195 |   estado: varchar("estado", { length: 20 }).notNull().default("PENDIENTE"),
196 |   horaEstimada: timestamp("hora_estimada"),
197 |   horaReal: timestamp("hora_real"),
198 |   notas: text("notas"),
199 | });
200 |
201 | export const insertTmsParadaSchema = createInsertSchema(tmsParadas).omit({ id: true });
202 | export type InsertTmsParada = z.infer<typeof insertTmsParadaSchema>;
203 | export type TmsParada = typeof tmsParadas.$inferSelect;
204 |
205 | export const tmsPuntos = pgTable("tms_puntos", {
206 |   id: serial("id").primaryKey(),
207 |   contratoId: integer("contrato_id").references(() => tmsContratos.id),
208 |   camionId: integer("camion_id").references(() => camiones.id),
209 |   lat: numeric("lat", { precision: 10, scale: 7 }).notNull(),
210 |   lng: numeric("lng", { precision: 10, scale: 7 }).notNull(),
211 |   nombreInferido: varchar("nombre_inferido", { length: 200 }),
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212 |   tipo: varchar("tipo", { length: 20 }),
213 |   vecesVisitado: integer("veces_visitado").default(1),
214 |   duracionPromedioMin: integer("duracion_promedio_min"),
215 |   primeraVisita: timestamp("primera_visita"),
216 |   ultimaVisita: timestamp("ultima_visita"),
217 |   confirmado: boolean("confirmado").default(false),
218 | })
219 |
220 | export const insertTmsPuntoSchema = createInsertSchema(tmsPuntos).omit({ id: true });
221 | export type InsertTmsPunto = z.infer<typeof insertTmsPuntoSchema>;
222 | export type TmsPunto = typeof tmsPuntos.$inferSelect;
223 |
224 | // %P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P
225 | // GEOVALIDATOR TABLES
226 | // %P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P
227 |
228 | export const geoTrayectorias = pgTable("geo_trayectorias", {
229 |   id: serial("id").primaryKey(),
230 |   camionId: integer("camion_id").references(() => camiones.id),
231 |   patente: varchar("patente", { length: 20 }).notNull(),
232 |   fechaInicio: timestamp("fecha_inicio").notNull(),
233 |   fechaFin: timestamp("fecha_fin"),
234 |   estado: varchar("estado", { length: 20 }).default("ACTIVA"),
235 |   totalPuntos: integer("total_puntos").default(0),
236 |   kmCalculados: numeric("km_calculados", { precision: 10, scale: 2 }).default("0"),
237 |   creadaAt: timestamp("creada_at").defaultNow(),
238 | });
239 |
240 | export const insertGeoTrayectoriaSchema = createInsertSchema(geoTrayectorias).omit({ id: true });
241 | export type InsertGeoTrayectoria = z.infer<typeof insertGeoTrayectoriaSchema>;
242 | export type GeoTrayectoria = typeof geoTrayectorias.$inferSelect;
243 |
244 | export const geoPuntos = pgTable("geo_puntos", {
245 |   id: serial("id").primaryKey(),
246 |   trayectoriaId: integer("trayectoria_id").references(() => geoTrayectorias.id),
247 |   camionId: integer("camion_id").references(() => camiones.id),
248 |   patente: varchar("patente", { length: 20 }),
249 |   lat: numeric("lat", { precision: 10, scale: 7 }).notNull(),
250 |   lng: numeric("lng", { precision: 10, scale: 7 }).notNull(),
251 |   timestampPunto: timestamp("timestamp_punto").notNull(),
252 |   velocidadKmh: numeric("velocidad_kmh", { precision: 6, scale: 1 }).default("0"),
253 |   rumboGrados: numeric("rumbo_grados", { precision: 5, scale: 1 }),
254 |   kmOdometro: numeric("km_odometro", { precision: 10, scale: 1 }),
255 |   fuente: varchar("fuente", { length: 20 }).default("VOLVO"),
256 |   creadoAt: timestamp("creado_at").defaultNow(),
257 | });
258 |
259 | export const insertGeoPuntoSchema = createInsertSchema(geoPuntos).omit({ id: true });
260 | export type InsertGeoPunto = z.infer<typeof insertGeoPuntoSchema>;
261 | export type GeoPunto = typeof geoPuntos.$inferSelect;
262 |
263 | export const geoViajes = pgTable("geo_viajes", {
264 |   id: serial("id").primaryKey(),
265 |   camionId: integer("camion_id").references(() => camiones.id),
266 |   patente: varchar("patente", { length: 20 }).notNull(),
267 |   contrato: varchar("contrato", { length: 100 }),
268 |   origenLat: numeric("origen_lat", { precision: 10, scale: 7 }),
269 |   origenLng: numeric("origen_lng", { precision: 10, scale: 7 }),
270 |   origenNombre: varchar("origen_nombre", { length: 200 }),
271 |   origenTimestamp: timestamp("origen_timestamp"),
272 |   destinoLat: numeric("destino_lat", { precision: 10, scale: 7 }),
273 |   destinoLng: numeric("destino_lng", { precision: 10, scale: 7 }),
274 |   destinoNombre: varchar("destino_nombre", { length: 200 }),
275 |   destinoTimestamp: timestamp("destino_timestamp"),
276 |   paradas: jsonb("paradas").default([]),
277 |   kmGps: numeric("km_gps", { precision: 10, scale: 2 }),
278 |   kmOdometroInicio: numeric("km_odometro_inicio", { precision: 10, scale: 1 }),
279 |   kmOdometroFin: numeric("km_odometro_fin", { precision: 10, scale: 1 }),
280 |   kmOdometroDelta: numeric("km_odometro_delta", { precision: 10, scale: 1 }),
281 |   duracionMinutos: integer("duracion_minutos"),
282 |   velocidadPromedio: numeric("velocidad_promedio", { precision: 6, scale: 1 }),
283 |   velocidadMaxima: numeric("velocidad_maxima", { precision: 6, scale: 1 }),
284 |   tiempoDetenidoMin: integer("tiempo_detenido_min").default(0),
285 |   tiempoMovimientoMin: integer("tiempo_movimiento_min").default(0),
286 |   validacionEstado: varchar("validacion_estado", { length: 20 }).default("PENDIENTE"),
287 |   validacionDetalle: jsonb("validacion_detalle"),
288 |   sigetraCargaId: integer("sigetra_carga_id"),
289 |   sigetraKmDeltaPct: numeric("sigetra_km_delta_pct", { precision: 5, scale: 2 }),
290 |   sigetraSurtidorEnRuta: boolean("sigetra_surtidor_en_ruta"),
291 |   sigetraLitros: numeric("sigetra_litros", { precision: 8, scale: 2 }),
292 |   validadoManualmente: boolean("validado_manualmente").default(false),
293 |   validadoPor: varchar("validado_por", { length: 100 }),
294 |   notas: text("notas"),
295 |   trayectoriaId: integer("trayectoria_id").references(() => geoTrayectorias.id),
296 |   destinoLugarId: integer("destino_lugar_id"),
297 |   origenLugarId: integer("origen_lugar_id"),
298 |   paradasLugares: jsonb("paradas_lugares").default([]),
299 |   creadoAt: timestamp("creado_at").defaultNow(),
300 |   actualizadoAt: timestamp("actualizado_at").defaultNow(),
301 | });
302 |
303 | export const insertGeoViajeSchema = createInsertSchema(geoViajes).omit({ id: true });
304 | export type InsertGeoViaje = z.infer<typeof insertGeoViajeSchema>;
305 | export type GeoViaje = typeof geoViajes.$inferSelect;
306 |
307 | export const geoBases = pgTable("geo_bases", {
308 |   id: serial("id").primaryKey(),
309 |   nombre: varchar("nombre", { length: 200 }).notNull(),
310 |   lat: numeric("lat", { precision: 10, scale: 7 }).notNull(),
311 |   lng: numeric("lng", { precision: 10, scale: 7 }).notNull(),
312 |   radioMetros: integer("radio_metros").default(3000),
313 |   contrato: varchar("contrato", { length: 100 }),
314 |   activa: boolean("activa").default(true),
315 | });
316 |
317 | export const insertGeoBaseSchema = createInsertSchema(geoBases).omit({ id: true });
318 | export type InsertGeoBase = z.infer<typeof insertGeoBaseSchema>;
```

```
319 | export type GeoBase = typeof geoBases.$inferSelect;
320 |
321 | export const geoGeocache = pgTable("geo_geocache", {
322 |   id: serial("id").primaryKey(),
323 |   lat: numeric("lat", { precision: 10, scale: 5 }).notNull(),
324 |   lng: numeric("lng", { precision: 10, scale: 5 }).notNull(),
325 |   nombre: varchar("nombre", { length: 300 }),
326 |   ciudad: varchar("ciudad", { length: 100 }),
327 |   region: varchar("region", { length: 100 }),
328 |   creadoAt: timestamp("creado_at").defaultNow(),
329 | });
330 |
331 | export const geoLugares = pgTable("geo_lugares", {
332 |   id: serial("id").primaryKey(),
333 |   nombre: varchar("nombre", { length: 200 }),
334 |   nombreConfirmado: varchar("nombre_confirmado", { length: 200 }),
335 |   tipo: varchar("tipo", { length: 50 }),
336 |   lat: numeric("lat", { precision: 10, scale: 7 }).notNull(),
337 |   lng: numeric("lng", { precision: 10, scale: 7 }).notNull(),
338 |   radioMetros: integer("radio_metros").default(500),
339 |   direccion: varchar("direccion", { length: 300 }),
340 |   comuna: varchar("comuna", { length: 100 }),
341 |   region: varchar("region", { length: 100 }),
342 |   detectadoVia: varchar("detectado_via", { length: 50 }),
343 |   confianzaPct: integer("confianza_pct").default(0),
344 |   vecesVisitado: integer("veces_visitado").default(0),
345 |   primeraVisita: date("primera_visita"),
346 |   ultimaVisita: date("ultima_visita"),
347 |   confirmado: boolean("confirmado").default(false),
348 |   activo: boolean("activo").default(true),
349 |   creadoAt: timestamp("creado_at").defaultNow(),
350 | });
351 |
352 | export const insertGeoLugarSchema = createInsertSchema(geoLugares).omit({ id: true });
353 | export type InsertGeoLugar = z.infer<typeof insertGeoLugarSchema>;
354 | export type GeoLugar = typeof geoLugares.$inferSelect;
355 |
356 | export const geoVisitas = pgTable("geo_visitas", {
357 |   id: serial("id").primaryKey(),
358 |   lugarId: integer("lugar_id").references(() => geoLugares.id),
359 |   camionId: integer("camion_id").references(() => camiones.id),
360 |   patente: varchar("patente", { length: 20 }),
361 |   viajeId: integer("viaje_id").references(() => geoViajes.id),
362 |   llegada: timestamp("llegada"),
363 |   salida: timestamp("salida"),
364 |   minutosDetenido: integer("minutos_detenido"),
365 |   latExacta: numeric("lat_exacta", { precision: 10, scale: 7 }),
366 |   lngExacta: numeric("lng_exacta", { precision: 10, scale: 7 }),
367 |   creadoAt: timestamp("creado_at").defaultNow(),
368 | });
369 |
370 | export const insertGeoVisitaSchema = createInsertSchema(geoVisitas).omit({ id: true });
371 | export type InsertGeoVisita = z.infer<typeof insertGeoVisitaSchema>;
372 | export type GeoVisita = typeof geoVisitas.$inferSelect;
373 |
374 | export const geoAnalisisIa = pgTable("geo_analisis_ia", {
375 |   id: serial("id").primaryKey(),
376 |   tipo: varchar("tipo", { length: 50 }),
377 |   periodoDesde: date("periodo_desde"),
378 |   periodoHasta: date("periodo_hasta"),
379 |   resultadoJson: jsonb("resultado_json"),
380 |   resumenTexto: text("resumen_texto"),
381 |   generadoAt: timestamp("generado_at").defaultNow(),
382 | });
383 |
384 | export const insertGeoAnalisisIaSchema = createInsertSchema(geoAnalisisIa).omit({ id: true });
385 | export type InsertGeoAnalisisIa = z.infer<typeof insertGeoAnalisisIaSchema>;
386 | export type GeoAnalisisIa = typeof geoAnalisisIa.$inferSelect;
387 |
388 | export const users = pgTable("users", {
389 |   id: varchar("id").primaryKey().default(sql`gen_random_uuid()`),
390 |   username: text("username").notNull().unique(),
391 |   password: text("password").notNull(),
392 | });
393 |
394 | export const insertUserSchema = createInsertSchema(users).pick({
395 |   username: true,
396 |   password: true,
397 | });
398 |
399 | export type InsertUser = z.infer<typeof insertUserSchema>;
400 | export type User = typeof users.$inferSelect;
401 |
```

server/index.ts

104 lineas

```
1 | import express, { type Request, Response, NextFunction } from "express";
2 | import { registerRoutes } from "./routes";
3 | import { serveStatic } from "./static";
4 | import { createServer } from "http";
5 |
6 | const app = express();
7 | const httpServer = createServer(app);
8 |
9 | declare module "http" {
10 |   interface IncomingMessage {
11 |     rawBody: unknown;
12 |   }
13 | }
14 |
15 | app.use(
16 |   express.json({
17 |     verify: (req, _res, buf) => {
18 |       req.rawBody = buf;
19 |     },
20 |   }),
21 | );
22 |
23 | app.use(express.urlencoded({ extended: false }));
24 |
25 | export function log(message: string, source = "express") {
26 |   const formattedTime = new Date().toLocaleTimeString("en-US", {
27 |     hour: "numeric",
28 |     minute: "2-digit",
29 |     second: "2-digit",
30 |     hour12: true,
31 |   });
32 |
33 |   console.log(`${formattedTime} [${source}] ${message}`);
34 | }
35 |
36 | app.use((req, res, next) => {
37 |   const start = Date.now();
38 |   const path = req.path;
39 |   let capturedJsonResponse: Record<string, any> | undefined = undefined;
40 |
41 |   const originalResJson = res.json;
42 |   res.json = function (bodyJson, ...args) {
43 |     capturedJsonResponse = bodyJson;
44 |     return originalResJson.apply(res, [bodyJson, ...args]);
45 |   };
46 |
47 |   res.on("finish", () => {
48 |     const duration = Date.now() - start;
49 |     if (path.startsWith("/api")) {
50 |       let logLine = `${req.method} ${path} ${res.statusCode} in ${duration}ms`;
51 |       if (capturedJsonResponse) {
52 |         logLine += ` :: ${JSON.stringify(capturedJsonResponse)}`;
53 |       }
54 |
55 |       log(logLine);
56 |     }
57 |   });
58 |
59 |   next();
60 | });
61 |
62 | (async () => {
63 |   await registerRoutes(httpServer, app);
64 |
65 |   app.use((err: any, _req: Request, res: Response, next: NextFunction) => {
66 |     const status = err.status || err.statusCode || 500;
67 |     const message = err.message || "Internal Server Error";
68 |
69 |     console.error("Internal Server Error:", err);
70 |
71 |     if (res.headersSent) {
72 |       return next(err);
73 |     }
74 |
75 |     return res.status(status).json({ message });
76 |   });
77 |
78 |   // importantly only setup vite in development and after
79 |   // setting up all the other routes so the catch-all route
80 |   // doesn't interfere with the other routes
81 |   if (process.env.NODE_ENV === "production") {
82 |     serveStatic(app);
83 |   } else {
84 |     const { setupVite } = await import("./vite");
85 |     await setupVite(httpServer, app);
86 |   }
87 |
88 |   // ALWAYS serve the app on the port specified in the environment variable PORT
89 |   // Other ports are firewalled. Default to 5000 if not specified.
90 |   // this serves both the API and the client.
91 |   // It is the only port that is not firewalled.
92 |   const port = parseInt(process.env.PORT || "5000", 10);
93 |   httpServer.listen(
94 |     {
95 |       port,
96 |       host: "0.0.0.0",
97 |       reusePort: true,
98 |     },
99 |     () => {
100 |       log(`serving on port ${port}`);
101 |     },
102 |   );
103 | })();
104 |
```

server/db.ts

14 lineas

```
1 | import { drizzle } from "drizzle-orm/node-postgres";
2 | import pg from "pg";
3 | import * as schema from "@shared/schema";
4 |
5 | if (!process.env.DATABASE_URL) {
6 |   throw new Error("DATABASE_URL must be set. Did you forget to provision a database?");
7 | }
8 |
9 | export const pool = new pg.Pool({
10 |   connectionString: process.env.DATABASE_URL,
11 | });
12 |
13 | export const db = drizzle(pool, { schema });
14 |
```

server/storage.ts

358 lineas

```
1 | import {
2 |   type User, type InsertUser,
3 |   type Faena, type InsertFaena,
4 |   type Camion, type InsertCamion,
5 |   type Carga, type InsertCarga,
6 |   type DesviacionCheck, type InsertDesviacionCheck,
7 |   type VolvoFuelSnapshot, type InsertVolvoFuelSnapshot,
8 |   type Parametro,
9 |   type PuntoRuta, type InsertPuntoRuta,
10 |   type TarifaRuta, type InsertTarifaRuta,
11 |   type TmsContrato, type InsertTmsContrato,
12 |   type TmsViaje, type InsertTmsViaje,
13 |   type TmsParada, type InsertTmsParada,
14 |   type TmsPunto, type InsertTmsPunto,
15 |   users, faenas, camiones, cargas, desviacionChecks, volvoFuelSnapshots,
16 |   parametros, puntosRuta, tarifasRuta, tmsContratos, tmsViajes, tmsParadas, tmsPuntos,
17 | } from "@shared/schema";
18 | import { db } from "./db";
19 | import { eq, and, gte, lte, inArray, desc, asc } from "drizzle-orm";
20 |
21 | export interface IStorage {
22 |   getUser(id: string): Promise<User | undefined>;
23 |   getUserByUsername(username: string): Promise<User | undefined>;
24 |   createUser(user: InsertUser): Promise<User>;
25 |
26 |   getFaenas(): Promise<Faena[]>;
27 |   getFaena(id: number): Promise<Faena | undefined>;
28 |   createFaena(faena: InsertFaena): Promise<Faena>;
29 |   updateFaena(id: number, faena: Partial<InsertFaena>): Promise<Faena | undefined>;
30 |   deleteFaena(id: number): Promise<void>;
31 |
32 |   getCamiones(): Promise<Camion[]>;
33 |   getCamion(id: number): Promise<Camion | undefined>;
34 |   getCamionByVin(vin: string): Promise<Camion | undefined>;
35 |   createCamion(camion: InsertCamion): Promise<Camion>;
36 |   updateCamion(id: number, camion: Partial<InsertCamion>): Promise<Camion | undefined>;
37 |   deleteCamion(id: number): Promise<void>;
38 |
39 |   getCargas(camionId?: number): Promise<Carga[]>;
40 |   getCarga(id: number): Promise<Carga | undefined>;
41 |   createCarga(carga: InsertCarga): Promise<Carga>;
42 |   deleteCarga(id: number): Promise<void>;
43 |
44 |   getDesviacionChecks(): Promise<DesviacionCheck[]>;
45 |   upsertDesviacionCheck(fleetNum: string, tipo: string, gestionado: boolean, nota?: string): Promise<DesviacionCheck>;
46 |
47 |   saveVolvoFuelSnapshots(snapshots: InsertVolvoFuelSnapshot[]): Promise<void>;
48 |   getVolvoFuelSnapshotsInRange(vins: string[], from: Date, to: Date): Promise<VolvoFuelSnapshot[]>;
49 |
50 |   getParametros(): Promise<Parametro[]>;
51 |   getParametro(clave: string): Promise<Parametro | undefined>;
52 |   upsertParametro(clave: string, valor: string): Promise<Parametro>;
53 |
54 |   getPuntosRuta(): Promise<PuntoRuta[]>;
55 |   getPuntosRutaByCamion(camionId: number): Promise<PuntoRuta[]>;
56 |   createPuntoRuta(punto: InsertPuntoRuta): Promise<PuntoRuta>;
57 |   updatePuntoRuta(id: number, data: Partial<InsertPuntoRuta>): Promise<PuntoRuta | undefined>;
58 |   getConfirmedPuntos(): Promise<PuntoRuta[]>;
59 |
60 |   getTarifas(): Promise<TarifaRuta[]>;
61 |   createTarifa(tarifa: InsertTarifaRuta): Promise<TarifaRuta>;
62 |   updateTarifa(id: number, data: Partial<InsertTarifaRuta>): Promise<TarifaRuta | undefined>;
63 |   deactivateTarifa(id: number): Promise<void>;
64 |
65 |   getTmsContratos(): Promise<TmsContrato[]>;
66 |   getTmsContrato(id: number): Promise<TmsContrato | undefined>;
67 |   createTmsContrato(data: InsertTmsContrato): Promise<TmsContrato>;
68 |   updateTmsContrato(id: number, data: Partial<InsertTmsContrato>): Promise<TmsContrato | undefined>;
69 |
70 |   getTmsViajes(contratoId: number, filters?: { estado?: string; camionId?: number; fechaDesde?: Date; fechaHasta?: Date
71 |   getTmsViaje(id: number): Promise<TmsViaje | undefined>;
72 |   createTmsViaje(data: InsertTmsViaje): Promise<TmsViaje>;
73 |   updateTmsViaje(id: number, data: Partial<InsertTmsViaje>): Promise<TmsViaje | undefined>;
74 |
75 |   getTmsParadas(viajeId: number): Promise<TmsParada[]>;
76 |   createTmsParada(data: InsertTmsParada): Promise<TmsParada>;
77 |   updateTmsParada(id: number, data: Partial<InsertTmsParada>): Promise<TmsParada | undefined>;
78 |
79 |   getTmsPuntos(contratoId: number): Promise<TmsPunto[]>;
80 |   createTmsPunto(data: InsertTmsPunto): Promise<TmsPunto>;
81 |   updateTmsPunto(id: number, data: Partial<InsertTmsPunto>): Promise<TmsPunto | undefined>;
82 |
83 |   deleteTmsViajesByContrato(contratoId: number): Promise<void>;
84 |   deleteTmsPuntosByContrato(contratoId: number): Promise<void>;
85 | }
86 |
87 | export class DatabaseStorage implements IStorage {
88 |   async getUser(id: string): Promise<User | undefined> {
89 |     const [user] = await db.select().from(users).where(eq(users.id, id));
90 |     return user;
91 |   }
92 |
93 |   async getUserByUsername(username: string): Promise<User | undefined> {
94 |     const [user] = await db.select().from(users).where(eq(users.username, username));
95 |     return user;
96 |   }
97 |
98 |   async createUser(insertUser: InsertUser): Promise<User> {
99 |     const [user] = await db.insert(users).values(insertUser).returning();
100 |     return user;
101 |   }
102 |
103 |   async getFaenas(): Promise<Faena[]> {
104 |     return db.select().from(faenas);
```

```
105 |   }
106 |
107 |   async getFaena(id: number): Promise<Faena | undefined> {
108 |     const [faena] = await db.select().from(faenas).where(eq(faenas.id, id));
109 |     return faena;
110 |   }
111 |
112 |   async createFaena(faena: InsertFaena): Promise<Faena> {
113 |     const [created] = await db.insert(faenas).values(faena).returning();
114 |     return created;
115 |   }
116 |
117 |   async updateFaena(id: number, data: Partial<InsertFaena>): Promise<Faena | undefined> {
118 |     const [updated] = await db.update(faenas).set(data).where(eq(faenas.id, id)).returning();
119 |     return updated;
120 |   }
121 |
122 |   async deleteFaena(id: number): Promise<void> {
123 |     await db.delete(faenas).where(eq(faenas.id, id));
124 |   }
125 |
126 |   async getCamiones(): Promise<Camion[]> {
127 |     return db.select().from(camiones);
128 |   }
129 |
130 |   async getCamion(id: number): Promise<Camion | undefined> {
131 |     const [camion] = await db.select().from(camiones).where(eq(camiones.id, id));
132 |     return camion;
133 |   }
134 |
135 |   async getCamionByVin(vin: string): Promise<Camion | undefined> {
136 |     const [camion] = await db.select().from(camiones).where(eq(camiones.vin, vin));
137 |     return camion;
138 |   }
139 |
140 |   async createCamion(camion: InsertCamion): Promise<Camion> {
141 |     const [created] = await db.insert(camiones).values(camion).returning();
142 |     return created;
143 |   }
144 |
145 |   async updateCamion(id: number, data: Partial<InsertCamion>): Promise<Camion | undefined> {
146 |     const [updated] = await db.update(camiones).set(data).where(eq(camiones.id, id)).returning();
147 |     return updated;
148 |   }
149 |
150 |   async deleteCamion(id: number): Promise<void> {
151 |     await db.delete(cargas).where(eq(cargas.camionId, id));
152 |     await db.delete(camiones).where(eq(camiones.id, id));
153 |   }
154 |
155 |   async getCargas(camionId?: number): Promise<Carga[]> {
156 |     if (camionId) {
157 |       return db.select().from(cargas).where(eq(cargas.camionId, camionId));
158 |     }
159 |     return db.select().from(cargas);
160 |   }
161 |
162 |   async getCarga(id: number): Promise<Carga | undefined> {
163 |     const [carga] = await db.select().from(cargas).where(eq(cargas.id, id));
164 |     return carga;
165 |   }
166 |
167 |   async createCarga(carga: InsertCarga): Promise<Carga> {
168 |     const [created] = await db.insert(cargas).values(carga).returning();
169 |     return created;
170 |   }
171 |
172 |   async deleteCarga(id: number): Promise<void> {
173 |     await db.delete(cargas).where(eq(cargas.id, id));
174 |   }
175 |
176 |   async getDesviacionChecks(): Promise<DesviacionCheck[]> {
177 |     return db.select().from(desviacionChecks);
178 |   }
179 |
180 |   async upsertDesviacionCheck(fleetNum: string, tipo: string, gestionado: boolean, nota?: string): Promise<DesviacionChe
181 |     const now = new Date().toLocaleString("es-CL", { timeZone: "America/Santiago" });
182 |     const [existing] = await db.select().from(desviacionChecks)
183 |       .where(and(eq(desviacionChecks.fleetNum, fleetNum), eq(desviacionChecks.tipo, tipo)));
184 |
185 |     if (existing) {
186 |       const [updated] = await db.update(desviacionChecks)
187 |         .set({ gestionado, gestionadoAt: gestionado ? now : null, nota: nota ?? existing.nota })
188 |         .where(eq(desviacionChecks.id, existing.id))
189 |         .returning();
190 |       return updated;
191 |     }
192 |
193 |     const [created] = await db.insert(desviacionChecks)
194 |       .values({ fleetNum, tipo, gestionado, gestionadoAt: gestionado ? now : null, nota: nota ?? null })
195 |       .returning();
196 |     return created;
197 |   }
198 |
199 |   async saveVolvoFuelSnapshots(snapshots: InsertVolvoFuelSnapshot[]): Promise<void> {
200 |     if (snapshots.length === 0) return;
201 |     await db.insert(volvoFuelSnapshots).values(snapshots).onConflictDoNothing();
202 |   }
203 |
204 |   async getVolvoFuelSnapshotsInRange(vins: string[], from: Date, to: Date): Promise<VolvoFuelSnapshot[]> {
205 |     if (vins.length === 0) return [];
206 |     return db.select().from(volvoFuelSnapshots)
207 |       .where(and(
208 |         inArray(volvoFuelSnapshots.vin, vins),
209 |         gte(volvoFuelSnapshots.capturedAt, from.toISOString()),
210 |         lte(volvoFuelSnapshots.capturedAt, to.toISOString()),
211 |       ));
```

```
212 |   }
213 |
214 |   async getParametros(): Promise<Parametro[]> {
215 |     return db.select().from(parametros);
216 |   }
217 |
218 |   async getParametro(clave: string): Promise<Parametro | undefined> {
219 |     const [row] = await db.select().from(parametros).where(eq(parametros.clave, clave));
220 |     return row;
221 |   }
222 |
223 |   async upsertParametro(clave: string, valor: string): Promise<Parametro> {
224 |     const existing = await this.getParametro(clave);
225 |     if (existing) {
226 |       const [updated] = await db.update(parametros).set({ valor, updatedAt: new Date() }).where(eq(parametros.clave, cla
227 |       return updated;
228 |     }
229 |     const [created] = await db.insert(parametros).values({ clave, valor }).returning();
230 |     return created;
231 |   }
232 |
233 |   async getPuntosRuta(): Promise<PuntoRuta[]> {
234 |     return db.select().from(puntosRuta);
235 |   }
236 |
237 |   async getPuntosRutaByCamion(camionId: number): Promise<PuntoRuta[]> {
238 |     return db.select().from(puntosRuta).where(eq(puntosRuta.camionId, camionId));
239 |   }
240 |
241 |   async createPuntoRuta(punto: InsertPuntoRuta): Promise<PuntoRuta> {
242 |     const [created] = await db.insert(puntosRuta).values(punto).returning();
243 |     return created;
244 |   }
245 |
246 |   async updatePuntoRuta(id: number, data: Partial<InsertPuntoRuta>): Promise<PuntoRuta | undefined> {
247 |     const [updated] = await db.update(puntosRuta).set(data).where(eq(puntosRuta.id, id)).returning();
248 |     return updated;
249 |   }
250 |
251 |   async getConfirmedPuntos(): Promise<PuntoRuta[]> {
252 |     return db.select().from(puntosRuta).where(eq(puntosRuta.confirmado, true));
253 |   }
254 |
255 |   async getTarifas(): Promise<TarifaRuta[]> {
256 |     return db.select().from(tarifasRuta);
257 |   }
258 |
259 |   async createTarifa(tarifa: InsertTarifaRuta): Promise<TarifaRuta> {
260 |     const [created] = await db.insert(tarifasRuta).values(tarifa).returning();
261 |     return created;
262 |   }
263 |
264 |   async updateTarifa(id: number, data: Partial<InsertTarifaRuta>): Promise<TarifaRuta | undefined> {
265 |     const [updated] = await db.update(tarifasRuta).set(data).where(eq(tarifasRuta.id, id)).returning();
266 |     return updated;
267 |   }
268 |
269 |   async deactivateTarifa(id: number): Promise<void> {
270 |     await db.update(tarifasRuta).set({ activa: false }).where(eq(tarifasRuta.id, id));
271 |   }
272 |
273 |   async getTmsContratos(): Promise<TmsContrato[]> {
274 |     return db.select().from(tmsContratos).where(eq(tmsContratos.activo, true)).orderBy(desc(tmsContratos.createdAt));
275 |   }
276 |
277 |   async getTmsContrato(id: number): Promise<TmsContrato | undefined> {
278 |     const [c] = await db.select().from(tmsContratos).where(eq(tmsContratos.id, id));
279 |     return c;
280 |   }
281 |
282 |   async createTmsContrato(data: InsertTmsContrato): Promise<TmsContrato> {
283 |     const [created] = await db.insert(tmsContratos).values(data).returning();
284 |     return created;
285 |   }
286 |
287 |   async updateTmsContrato(id: number, data: Partial<InsertTmsContrato>): Promise<TmsContrato | undefined> {
288 |     const [updated] = await db.update(tmsContratos).set(data).where(eq(tmsContratos.id, id)).returning();
289 |     return updated;
290 |   }
291 |
292 |   async getTmsViajes(contratoId: number, filters?: { estado?: string; camionId?: number; fechaDesde?: Date; fechaHasta?:
293 |     const conditions = [eq(tmsViajes.contratoId, contratoId)];
294 |     if (filters?.estado) conditions.push(eq(tmsViajes.estado, filters.estado));
295 |     if (filters?.camionId) conditions.push(eq(tmsViajes.camionId, filters.camionId));
296 |     if (filters?.fechaDesde) conditions.push(gte(tmsViajes.fechaSalida, filters.fechaDesde));
297 |     if (filters?.fechaHasta) conditions.push(lte(tmsViajes.fechaSalida, filters.fechaHasta));
298 |     return db.select().from(tmsViajes).where(and(...conditions)).orderBy(desc(tmsViajes.createdAt));
299 |   }
300 |
301 |   async getTmsViaje(id: number): Promise<TmsViaje | undefined> {
302 |     const [v] = await db.select().from(tmsViajes).where(eq(tmsViajes.id, id));
303 |     return v;
304 |   }
305 |
306 |   async createTmsViaje(data: InsertTmsViaje): Promise<TmsViaje> {
307 |     const [created] = await db.insert(tmsViajes).values(data).returning();
308 |     return created;
309 |   }
310 |
311 |   async updateTmsViaje(id: number, data: Partial<InsertTmsViaje>): Promise<TmsViaje | undefined> {
312 |     const [updated] = await db.update(tmsViajes).set(data).where(eq(tmsViajes.id, id)).returning();
313 |     return updated;
314 |   }
315 |
316 |   async getTmsParadas(viajeId: number): Promise<TmsParada[]> {
317 |     return db.select().from(tmsParadas).where(eq(tmsParadas.viajeId, viajeId)).orderBy(asc(tmsParadas.orden));
318 |   }
```

```
319 |
320 |   async createTmsParada(data: InsertTmsParada): Promise<TmsParada> {
321 |     const [created] = await db.insert(tmsParadas).values(data).returning();
322 |     return created;
323 |   }
324 |
325 |   async updateTmsParada(id: number, data: Partial<InsertTmsParada>): Promise<TmsParada | undefined> {
326 |     const [updated] = await db.update(tmsParadas).set(data).where(eq(tmsParadas.id, id)).returning();
327 |     return updated;
328 |   }
329 |
330 |   async getTmsPuntos(contratoId: number): Promise<TmsPunto[]> {
331 |     return db.select().from(tmsPuntos).where(eq(tmsPuntos.contratoId, contratoId)).orderBy(desc(tmsPuntos.vecesVisitado)
332 |   }
333 |
334 |   async createTmsPunto(data: InsertTmsPunto): Promise<TmsPunto> {
335 |     const [created] = await db.insert(tmsPuntos).values(data).returning();
336 |     return created;
337 |   }
338 |
339 |   async updateTmsPunto(id: number, data: Partial<InsertTmsPunto>): Promise<TmsPunto | undefined> {
340 |     const [updated] = await db.update(tmsPuntos).set(data).where(eq(tmsPuntos.id, id)).returning();
341 |     return updated;
342 |   }
343 |
344 |   async deleteTmsViajesByContrato(contratoId: number): Promise<void> {
345 |     const viajes = await this.getTmsViajes(contratoId);
346 |     for (const v of viajes) {
347 |       await db.delete(tmsParadas).where(eq(tmsParadas.viajeId, v.id));
348 |     }
349 |     await db.delete(tmsViajes).where(eq(tmsViajes.contratoId, contratoId));
350 |   }
351 |
352 |   async deleteTmsPuntosByContrato(contratoId: number): Promise<void> {
353 |     await db.delete(tmsPuntos).where(eq(tmsPuntos.contratoId, contratoId));
354 |   }
355 | }
356 |
357 | export const storage = new DatabaseStorage();
358 |
```

server/routes.ts

1909 lineas

```
1 | import type { Express } from "express";
2 | import { createServer, type Server } from "http";
3 | import { storage } from "./storage";
4 | import { insertFaenaSchema, insertCamionSchema, insertCargaSchema, insertTarifaRutaSchema, camiones, volvoFuelSnapshots
5 | import { z } from "zod";
6 | import { getVehicles, getVehicleStatuses, getVehiclePositions, getFleetStatus, getSingleVehicleStatus } from "./volvo-ap
7 | import { checkSigetraConnection, getCachedFuelData, getSigetraFuelSummary } from "./sigetra-api";
8 | import { registerIARoutes } from "./ia-routes";
9 | import { registerTMSRoutes } from "./tms-routes";
10 | import { registerCEORoutes } from "./ceo-routes";
11 | import { registerGeoRoutes } from "./geo-routes";
12 | import { getCencosudFaenaId, getCencosudPatentes, getCencosudCamionIds } from "./cencosud-filter";
13 | import { getAllContracts, getContractConfig, getContractPatentes, getContractCamiones, invalidateCache as invalidateFaen
14 | import { db } from "./db";
15 | import { getSigetraFuelData } from "./sigetra-api";
16 | import { eq } from "drizzle-orm";
17 |
18 | export async function registerRoutes(
19 |   httpServer: Server,
20 |   app: Express
21 | ): Promise<Server> {
22 |
23 |   app.get("/api/faenas", async (_req, res) => {
24 |     const faenas = await storage.getFaenas();
25 |     res.json(faenas);
26 |   });
27 |
28 |   app.get("/api/contratos", async (_req, res) => {
29 |     try {
30 |       const contracts = await getAllContracts();
31 |       res.json(contracts);
32 |     } catch (error: any) {
33 |       res.status(500).json({ message: error.message });
34 |     }
35 |   });
36 |
37 |   app.post("/api/contratos/importar-sigetra", async (req, res) => {
38 |     try {
39 |       const { contrato } = req.body;
40 |       if (!contrato) return res.status(400).json({ message: "contrato required" });
41 |
42 |       const from = new Date("2026-03-01");
43 |       const to = new Date();
44 |       const allFuel = await getSigetraFuelData(from, to);
45 |       const faenas = await storage.getFaenas();
46 |       const existingCamiones = await storage.getCamiones();
47 |       const existingPatentes = new Set(existingCamiones.map(c => c.patente));
48 |
49 |       let importedCount = 0;
50 |       const matchFaenas = faenas.filter(f => f.nombre.toUpperCase().includes(contrato.toUpperCase()));
51 |
52 |       for (const faena of matchFaenas) {
53 |         const faenaFuel = allFuel.filter(r => r.faena === faena.nombre);
54 |         const truckMap = new Map<string, { modelo: string; marca: string }>();
55 |
56 |         for (const r of faenaFuel) {
57 |           const numVeh = String(r.numVeh || "").trim();
58 |           if (!numVeh || numVeh === "null" || numVeh === "0") continue;
59 |           if (!truckMap.has(numVeh)) {
60 |             truckMap.set(numVeh, {
61 |               modelo: r.modelo || r.marca || "Camion",
62 |               marca: r.marca || "Volvo",
63 |             });
64 |           }
65 |         }
66 |
67 |         for (const [patente, info] of truckMap) {
68 |           if (existingPatentes.has(patente)) continue;
69 |           await storage.createCamion({
70 |             patente,
71 |             modelo: info.modelo ? `${info.marca || ""} ${info.modelo}`.trim() : "Camion",
72 |             faenaId: faena.id,
73 |             metaKmL: 2.1,
74 |             vin: null,
75 |             odometro: null,
76 |             horasMotor: null,
77 |             horasRalenti: null,
78 |             velPromedio: null,
79 |             conductor: null,
80 |             syncOk: false,
81 |             syncAt: null,
82 |           });
83 |           existingPatentes.add(patente);
84 |           importedCount++;
85 |         }
86 |       }
87 |
88 |       invalidateFaenaCache();
89 |       const updatedContracts = await getAllContracts();
90 |       res.json({ success: true, imported: importedCount, faenas: matchFaenas.map(f => f.nombre), contracts: updatedContr
91 |     } catch (error: any) {
92 |       console.error("[importar] Error:", error.message);
93 |       res.status(500).json({ message: error.message });
94 |     }
95 |   });
96 |
97 |   app.post("/api/faenas", async (req, res) => {
98 |     const parsed = insertFaenaSchema.safeParse(req.body);
99 |     if (!parsed.success) return res.status(400).json({ message: parsed.error.message });
100 |     const faena = await storage.createFaena(parsed.data);
101 |     res.json(faena);
102 |   });
103 |
104 |   app.patch("/api/faenas/:id", async (req, res) => {
```

```
105 |     const id = parseInt(req.params.id);
106 |     if (isNaN(id)) return res.status(400).json({ message: "Invalid ID" });
107 |     const parsed = insertFaenaSchema.partial().safeParse(req.body);
108 |     if (!parsed.success) return res.status(400).json({ message: parsed.error.message });
109 |     const updated = await storage.updateFaena(id, parsed.data);
110 |     if (!updated) return res.status(404).json({ message: "Faena not found" });
111 |     res.json(updated);
112 |   });
113 |
114 |   app.delete("/api/faenas/:id", async (req, res) => {
115 |     const id = parseInt(req.params.id);
116 |     if (isNaN(id)) return res.status(400).json({ message: "Invalid ID" });
117 |     await storage.deleteFaena(id);
118 |     res.json({ success: true });
119 |   });
120 |
121 |   app.get("/api/camiones", async (_req, res) => {
122 |     const cams = await storage.getCamiones();
123 |     const faenaId = await getCencosudFaenaId();
124 |     res.json(cams.filter(c => c.faenaId === faenaId));
125 |   });
126 |
127 |   app.get("/api/camiones/preview-asignacion", async (_req, res) => {
128 |     try {
129 |       const camiones = await storage.getCamiones();
130 |       const faenas = await storage.getFaenas();
131 |       const now = new Date();
132 |       const from = new Date("2026-03-01");
133 |       const summaries = await getSigetraFuelSummary(from, now);
134 |
135 |       const faenaNameMap = new Map(faenas.map(f => [f.nombre.toLowerCase().trim(), f]));
136 |
137 |       const detalle = camiones.map(cam => {
138 |         const pNorm = cam.patente.trim();
139 |         const summary = summaries.find(s => String(s.numVeh).trim() === pNorm || s.patente?.trim() === pNorm);
140 |         const clienteSigetra = summary?.faenaPri?.trim() || null;
141 |         const litros = summary ? Math.round(summary.cantidadLt) : 0;
142 |         const faenaActual = faenas.find(f => f.id === cam.faenaId);
143 |
144 |         let faenaPropuesta: string | null = null;
145 |         let confianza: "alta" | "media" | "sin_datos" = "sin_datos";
146 |
147 |         if (clienteSigetra) {
148 |           faenaPropuesta = clienteSigetra;
149 |           confianza = litros >= 500 ? "alta" : "media";
150 |         }
151 |
152 |         const yaAsignado = faenaActual && faenaActual.id !== 0 &&
153 |           faenaPropuesta && faenaActual.nombre.toLowerCase().trim() === faenaPropuesta.toLowerCase().trim();
154 |
155 |         return {
156 |           camionId: cam.id,
157 |           patente: cam.patente,
158 |           clienteSigetra,
159 |           litrosSigetra: litros,
160 |           faenaActualId: cam.faenaId,
161 |           faenaActualNombre: faenaActual?.nombre || "Sin asignar",
162 |           faenaPropuesta,
163 |           confianza,
164 |           accion: !clienteSigetra ? "SIN_DATOS" : yaAsignado ? "YA_ASIGNADO" : "ASIGNAR",
165 |         };
166 |       });
167 |
168 |       detalle.sort((a, b) => {
169 |         const order = { ASIGNAR: 0, YA_ASIGNADO: 1, SIN_DATOS: 2 };
170 |         return (order[a.accion as keyof typeof order] ?? 3) - (order[b.accion as keyof typeof order] ?? 3);
171 |       });
172 |
173 |       res.json({
174 |         total: detalle.length,
175 |         porAsignar: detalle.filter(d => d.accion === "ASIGNAR").length,
176 |         yaAsignados: detalle.filter(d => d.accion === "YA_ASIGNADO").length,
177 |         sinDatos: detalle.filter(d => d.accion === "SIN_DATOS").length,
178 |         detalle,
179 |       });
180 |     } catch (err: any) {
181 |       console.error("[asignacion] Error preview:", err.message);
182 |       res.status(500).json({ message: err.message });
183 |     }
184 |   });
185 |
186 |   app.get("/api/camiones/:id", async (req, res) => {
187 |     const id = parseInt(req.params.id);
188 |     const cam = await storage.getCamion(id);
189 |     if (!cam) return res.status(404).json({ message: "Camion not found" });
190 |     res.json(cam);
191 |   });
192 |
193 |   app.post("/api/camiones", async (req, res) => {
194 |     const parsed = insertCamionSchema.safeParse(req.body);
195 |     if (!parsed.success) return res.status(400).json({ message: parsed.error.message });
196 |     const cam = await storage.createCamion(parsed.data);
197 |     res.json(cam);
198 |   });
199 |
200 |   app.patch("/api/camiones/:id", async (req, res) => {
201 |     const id = parseInt(req.params.id);
202 |     if (isNaN(id)) return res.status(400).json({ message: "Invalid ID" });
203 |     const parsed = insertCamionSchema.partial().safeParse(req.body);
204 |     if (!parsed.success) return res.status(400).json({ message: parsed.error.message });
205 |     const updated = await storage.updateCamion(id, parsed.data);
206 |     if (!updated) return res.status(404).json({ message: "Camion not found" });
207 |     res.json(updated);
208 |   });
209 |
210 |   app.delete("/api/camiones/:id", async (req, res) => {
211 |     const id = parseInt(req.params.id);
```

```
212 |     await storage.deleteCamion(id);
213 |     res.json({ success: true });
214 |   });
215 |
216 |   app.get("/api/cargas", async (req, res) => {
217 |     const camionId = req.query.camionId ? parseInt(req.query.camionId as string) : undefined;
218 |     const cargasList = await storage.getCargas(camionId);
219 |     const ids = await getCencosudCamionIds();
220 |     res.json(cargasList.filter(c => ids.has(c.camionId)));
221 |   });
222 |
223 |   app.post("/api/cargas", async (req, res) => {
224 |     const parsed = insertCargaSchema.safeParse(req.body);
225 |     if (!parsed.success) return res.status(400).json({ message: parsed.error.message });
226 |     const carga = await storage.createCarga(parsed.data);
227 |     res.json(carga);
228 |   });
229 |
230 |   app.delete("/api/cargas/:id", async (req, res) => {
231 |     const id = parseInt(req.params.id);
232 |     await storage.deleteCarga(id);
233 |     res.json({ success: true });
234 |   });
235 |
236 |   app.post("/api/camiones/asignar-faenas-sigetra", async (_req, res) => {
237 |     try {
238 |       const camiones = await storage.getCamiones();
239 |       const faenas = await storage.getFaenas();
240 |       const now = new Date();
241 |       const from = new Date("2026-03-01");
242 |       const summaries = await getSigetraFuelSummary(from, now);
243 |
244 |       const faenaNameMap = new Map<string, number>();
245 |       faenas.forEach(f => faenaNameMap.set(f.nombre.toLowerCase().trim(), f.id));
246 |
247 |       let asignados = 0;
248 |       let sinDatosSigetra = 0;
249 |       let faenasCreadas = 0;
250 |       const detalle: any[] = [];
251 |       const colorsPool = ["#1A8FFF", "#FF6B35", "#00C49A", "#FFD93D", "#6C5CE7", "#E17055", "#00B894", "#FD79A8", "#74B9
252 |
253 |       for (const cam of camiones) {
254 |         const pNorm = cam.patente.trim();
255 |         const summary = summaries.find(s => String(s.numVeh).trim() === pNorm || s.patente?.trim() === pNorm);
256 |         const clienteSigetra = summary?.faenaPri?.trim() || null;
257 |
258 |         if (!clienteSigetra) {
259 |           sinDatosSigetra++;
260 |           detalle.push({ patente: cam.patente, cliente_detectado: null, faena_asignada: null, accion: "SIN_DATOS" });
261 |           continue;
262 |         }
263 |
264 |         const key = clienteSigetra.toLowerCase().trim();
265 |         let faenaId = faenaNameMap.get(key);
266 |
267 |         if (faenaId == null) {
268 |           const newFaena = await storage.createFaena({
269 |             nombre: clienteSigetra,
270 |             color: colorsPool[faenasCreadas % colorsPool.length],
271 |           });
272 |           faenaId = newFaena.id;
273 |           faenaNameMap.set(key, faenaId);
274 |           faenasCreadas++;
275 |         }
276 |
277 |         if (cam.faenaId !== faenaId) {
278 |           await storage.updateCamion(cam.id, { faenaId });
279 |           asignados++;
280 |           detalle.push({ patente: cam.patente, cliente_detectado: clienteSigetra, faena_asignada: clienteSigetra, accion
281 |         } else {
282 |           detalle.push({ patente: cam.patente, cliente_detectado: clienteSigetra, faena_asignada: clienteSigetra, accion
283 |         }
284 |       }
285 |
286 |       res.json({
287 |         procesados: camiones.length,
288 |         asignados,
289 |         sin_datos_sigetra: sinDatosSigetra,
290 |         faenas_creadas: faenasCreadas,
291 |         detalle,
292 |       });
293 |     } catch (err: any) {
294 |       console.error("[asignacion] Error asignar:", err.message);
295 |       res.status(500).json({ message: err.message });
296 |     }
297 |   });
298 |
299 |   app.get("/api/volvo/status", async (_req, res) => {
300 |     const user = process.env.VOLVO_CONNECT_USER || "";
301 |     const hasPassword = !!process.env.VOLVO_CONNECT_PASSWORD;
302 |     const configured = !!user && hasPassword;
303 |
304 |     if (!configured) {
305 |       return res.json({ configured, status: "not_configured", user: "", message: "Credenciales no configuradas" });
306 |     }
307 |
308 |     try {
309 |       const authHeader = "Basic " + Buffer.from(`${user}:${process.env.VOLVO_CONNECT_PASSWORD}`).toString("base64");
310 |       const controller = new AbortController();
311 |       const timeout = setTimeout(() => controller.abort(), 10000);
312 |       const testRes = await fetch("https://api.volvotrucks.com/rfms/vehicles", {
313 |         headers: {
314 |           Authorization: authHeader,
315 |           Accept: "application/vnd.fmsstandard.com.Vehicles.v2.1+json",
316 |         },
317 |         signal: controller.signal,
318 |       });
```

```
319 |       clearTimeout(timeout);
320 |
321 |       if (testRes.status === 200) {
322 |         return res.json({ configured, status: "connected", user, message: "Conectado a Volvo rFMS API" });
323 |       } else if (testRes.status === 403) {
324 |         return res.json({ configured, status: "forbidden", user, message: "Credenciales v\u00e1lidas pero acceso API no
325 |       } else if (testRes.status === 401) {
326 |         return res.json({ configured, status: "unauthorized", user, message: "Credenciales inv\u00e1lidas (401 Unauthori
327 |       } else {
328 |         return res.json({ configured, status: "error", user, message: `Respuesta inesperada: ${testRes.status}` });
329 |       }
330 |     } catch (error: any) {
331 |       return res.json({ configured, status: "error", user, message: `Error de conexi\u00f3n: ${error.message}` });
332 |     }
333 |   });
334 |
335 |   app.get("/api/volvo/vehicles", async (_req, res) => {
336 |     try {
337 |       const vehicles = await getVehicles();
338 |       const allCams = await storage.getCamiones();
339 |       const faenaId = await getCencosudFaenaId();
340 |       const cencosudVins = new Set(allCams.filter(c => c.faenaId === faenaId && c.vin).map(c => c.vin));
341 |       res.json(vehicles.filter((v: any) => cencosudVins.has(v.vin)));
342 |     } catch (error: any) {
343 |       console.error("[volvo] Error fetching vehicles:", error.message);
344 |       res.status(502).json({ message: error.message });
345 |     }
346 |   });
347 |
348 |   app.get("/api/volvo/vehicle-status", async (req, res) => {
349 |     try {
350 |       const vin = req.query.vin as string | undefined;
351 |       const statuses = await getVehicleStatuses(vin, true);
352 |       res.json(statuses);
353 |     } catch (error: any) {
354 |       console.error("[volvo] Error fetching statuses:", error.message);
355 |       res.status(502).json({ message: error.message });
356 |     }
357 |   });
358 |
359 |   app.get("/api/volvo/vehicle-positions", async (req, res) => {
360 |     try {
361 |       const vin = req.query.vin as string | undefined;
362 |       const positions = await getVehiclePositions(vin, true);
363 |       res.json(positions);
364 |     } catch (error: any) {
365 |       console.error("[volvo] Error fetching positions:", error.message);
366 |       res.status(502).json({ message: error.message });
367 |     }
368 |   });
369 |
370 |   app.get("/api/volvo/fleet-status", async (_req, res) => {
371 |     try {
372 |       const fleet = await getFleetStatus();
373 |       const nowDate = new Date();
374 |       const hourKey = nowDate.toISOString().slice(0, 13) + ":00:00.000Z";
375 |       const snapsToSave: { vin: string; totalFuelUsed: number; totalDistance: number | null; capturedAt: string }[] = []
376 |       for (const vs of fleet) {
377 |         if (vs.totalFuelUsed != null) {
378 |           snapsToSave.push({ vin: vs.vin, totalFuelUsed: vs.totalFuelUsed, totalDistance: vs.totalDistance ?? null, capt
379 |         }
380 |       }
381 |       if (snapsToSave.length > 0) {
382 |         storage.saveVolvoFuelSnapshots(snapsToSave).catch((err: any) => console.error("[fleet-status] snapshot save erro
383 |       }
384 |       const allCams = await storage.getCamiones();
385 |       const faenaId = await getCencosudFaenaId();
386 |       const cencosudVins = new Set(allCams.filter(c => c.faenaId === faenaId && c.vin).map(c => c.vin));
387 |       res.json(fleet.filter((v: any) => cencosudVins.has(v.vin)));
388 |     } catch (error: any) {
389 |       console.error("[volvo] Error fetching fleet status:", error.message);
390 |       res.status(502).json({ message: error.message });
391 |     }
392 |   });
393 |
394 |   app.get("/api/volvo/vehicle-status/:vin", async (req, res) => {
395 |     try {
396 |       const { vin } = req.params;
397 |       if (!vin) return res.status(400).json({ message: "VIN is required" });
398 |       const status = await getSingleVehicleStatus(vin);
399 |       res.json(status);
400 |     } catch (error: any) {
401 |       console.error("[volvo] Error fetching vehicle status:", error.message);
402 |       res.status(502).json({ message: error.message });
403 |     }
404 |   });
405 |
406 |   app.get("/api/volvo/truck-locations/:patente", async (req, res) => {
407 |     try {
408 |       const { patente } = req.params;
409 |       const fromStr = req.query.from as string | undefined;
410 |       const toStr = req.query.to as string | undefined;
411 |       const from = fromStr ? new Date(fromStr) : new Date(Date.now() - 30 * 24 * 60 * 60 * 1000);
412 |       const to = toStr ? new Date(toStr) : new Date();
413 |       if (isNaN(from.getTime()) || isNaN(to.getTime())) {
414 |         return res.status(400).json({ message: "Parametros from/to invalidos" });
415 |       }
416 |
417 |       const camion = (await storage.getCamiones()).find(c => c.patente === patente);
418 |       if (!camion) return res.status(404).json({ message: "Camion no encontrado" });
419 |
420 |       let currentGps: { latitude: number | null; longitude: number | null; speed: number | null; positionDateTime: strin
421 |       if (camion.vin) {
422 |         try {
423 |           const status = await getSingleVehicleStatus(camion.vin);
424 |           if (status?.gps) {
425 |             currentGps = {
```

```
426 |               latitude: status.gps.latitude,
427 |               longitude: status.gps.longitude,
428 |               speed: status.gps.speed,
429 |               positionDateTime: status.gps.positionDateTime,
430 |             };
431 |           }
432 |         } catch { }
433 |       }
434 |
435 |       const fuelData = await getCachedFuelData(from, to);
436 |       const numVeh = parseInt(patente, 10);
437 |       const truckCargas = fuelData.filter(c =>
438 |         (!isNaN(numVeh) && c.numVeh === numVeh) || c.patente === patente
439 |       );
440 |
441 |       const locations = truckCargas.map(c => ({
442 |         fecha: c.fechaConsumo,
443 |         lugar: c.lugarConsumo,
444 |         litros: c.cantidadLt,
445 |         odometro: c.odometroActual,
446 |         kmRecorrido: c.kmRecorrido,
447 |         conductor: c.nombreConductor,
448 |         faena: c.faena,
449 |         numGuia: c.numGuia,
450 |       }));
451 |
452 |       locations.sort((a, b) => new Date(b.fecha).getTime() - new Date(a.fecha).getTime());
453 |
454 |       const lugarSummary = new Map<string, { count: number; totalLitros: number; lastDate: string }>();
455 |       for (const loc of locations) {
456 |         const key = loc.lugar || "Desconocido";
457 |         const existing = lugarSummary.get(key) || { count: 0, totalLitros: 0, lastDate: loc.fecha };
458 |         existing.count++;
459 |         existing.totalLitros += loc.litros;
460 |         if (new Date(loc.fecha) > new Date(existing.lastDate)) existing.lastDate = loc.fecha;
461 |         lugarSummary.set(key, existing);
462 |       }
463 |
464 |       res.json({
465 |         patente,
466 |         vin: camion.vin,
467 |         modelo: camion.modelo,
468 |         currentGps,
469 |         locations,
470 |         lugarSummary: Array.from(lugarSummary.entries()).map(([lugar, data]) => ({
471 |           lugar,
472 |           ...data,
473 |           totalLitros: Math.round(data.totalLitros * 100) / 100,
474 |         })),
475 |         totalCargas: locations.length,
476 |         periodo: { from: from.toISOString().slice(0, 10), to: to.toISOString().slice(0, 10) },
477 |       });
478 |     } catch (error: any) {
479 |       console.error("[volvo] Error fetching truck locations:", error.message);
480 |       res.status(500).json({ message: error.message });
481 |     }
482 |   });
483 |
484 |   app.post("/api/volvo/sync", async (_req, res) => {
485 |     try {
486 |       const [vehicles, statuses] = await Promise.all([
487 |         getVehicles(),
488 |         getVehicleStatuses(undefined, true),
489 |       ]);
490 |
491 |       const statusMap = new Map(statuses.map(s => [s.Vin, s]));
492 |       const now = new Date().toLocaleString("es-CL", { timeZone: "America/Santiago" });
493 |       let synced = 0;
494 |
495 |       for (const v of vehicles) {
496 |         const status = statusMap.get(v.VIN);
497 |         const existing = await storage.getCamionByVin(v.VIN);
498 |
499 |         const camionData = {
500 |           patente: v.CustomerVehicleName || v.VIN.substring(v.VIN.length - 6),
501 |           modelo: v.Model ? `${v.Brand || "Volvo"} ${v.Model}` : (v.Brand || "Volvo"),
502 |           faenaId: existing?.faenaId || 0,
503 |           metaKmL: existing?.metaKmL || 2.0,
504 |           vin: v.VIN,
505 |           odometro: status?.HRTotalVehicleDistance ? Math.round(status.HRTotalVehicleDistance / 1000) : (existing?.odome
506 |           horasMotor: status?.TotalEngineHours ? Math.round(status.TotalEngineHours) : (existing?.horasMotor || null),
507 |           horasRalenti: existing?.horasRalenti || null,
508 |           velPromedio: existing?.velPromedio || null,
509 |           conductor: existing?.conductor || null,
510 |           syncOk: true,
511 |           syncAt: now,
512 |         };
513 |
514 |         if (existing) {
515 |           await storage.updateCamion(existing.id, camionData);
516 |         } else {
517 |           await storage.createCamion(camionData);
518 |         }
519 |         synced++;
520 |       }
521 |
522 |       res.json({ success: true, synced, total: vehicles.length });
523 |     } catch (error: any) {
524 |       console.error("[volvo] Sync error:", error.message);
525 |       res.status(502).json({ success: false, message: error.message });
526 |     }
527 |   });
528 |
529 |   app.get("/api/desviaciones/checks", async (_req, res) => {
530 |     try {
531 |       const checks = await storage.getDesviacionChecks();
532 |       res.json(checks);
```

```
533 |     } catch (error: any) {
534 |       res.status(500).json({ message: error.message });
535 |     }
536 |   });
537 |
538 |   app.post("/api/desviaciones/check", async (req, res) => {
539 |     try {
540 |       const { fleetNum, tipo, gestionado, nota } = req.body;
541 |       if (!fleetNum || typeof fleetNum !== "string") return res.status(400).json({ message: "fleetNum (string) required"
542 |       if (!tipo || typeof tipo !== "string") return res.status(400).json({ message: "tipo (string) required" });
543 |       if (typeof gestionado !== "boolean") return res.status(400).json({ message: "gestionado (boolean) required" });
544 |       const check = await storage.upsertDesviacionCheck(fleetNum, tipo, gestionado, typeof nota === "string" ? nota : un
545 |       res.json(check);
546 |     } catch (error: any) {
547 |       res.status(500).json({ message: error.message });
548 |     }
549 |   });
550 |
551 |   app.get("/api/sigetra/status", async (_req, res) => {
552 |     try {
553 |       const status = await checkSigetraConnection();
554 |       res.json(status);
555 |     } catch (error: any) {
556 |       res.json({ connected: false, message: error.message, user: "" });
557 |     }
558 |   });
559 |
560 |   app.get("/api/sigetra/fuel-data", async (req, res) => {
561 |     try {
562 |       const fromStr = req.query.from as string | undefined;
563 |       const toStr = req.query.to as string | undefined;
564 |       const from = fromStr ? new Date(fromStr) : new Date(Date.now() - 7 * 24 * 60 * 60 * 1000);
565 |       const to = toStr ? new Date(toStr) : new Date();
566 |       if (isNaN(from.getTime()) || isNaN(to.getTime())) {
567 |         return res.status(400).json({ message: "Parametros from/to invalidos" });
568 |       }
569 |       const data = await getCachedFuelData(from, to);
570 |       const patentes = await getCencosudPatentes();
571 |       const filtered = data.filter((r: any) => {
572 |         return patentes.has(String(r.numVeh || ""));
573 |       });
574 |       res.json(filtered);
575 |     } catch (error: any) {
576 |       console.error("[sigetra] Error fetching fuel data:", error.message);
577 |       res.status(502).json({ message: error.message });
578 |     }
579 |   });
580 |
581 |   app.get("/api/sigetra/fuel-summary", async (req, res) => {
582 |     try {
583 |       const from = req.query.from ? new Date(req.query.from as string) : new Date(Date.now() - 7 * 24 * 60 * 60 * 1000);
584 |       const to = req.query.to ? new Date(req.query.to as string) : new Date();
585 |       const data = await getSigetraFuelSummary(from, to);
586 |       const patentes = await getCencosudPatentes();
587 |       const filtered = (data as any[]).filter((r: any) => {
588 |         return patentes.has(String(r.numVeh || ""));
589 |       });
590 |       res.json(filtered);
591 |     } catch (error: any) {
592 |       console.error("[sigetra] Error fetching fuel summary:", error.message);
593 |       res.status(502).json({ message: error.message });
594 |     }
595 |   });
596 |
597 |   app.get("/api/parametros", async (_req, res) => {
598 |     try {
599 |       const params = await storage.getParametros();
600 |       res.json(params);
601 |     } catch (error: any) {
602 |       res.status(500).json({ message: error.message });
603 |     }
604 |   });
605 |
606 |   app.get("/api/flota/percentiles", async (_req, res) => {
607 |     try {
608 |       const cached = await storage.getParametro("percentil_90");
609 |       const now = Date.now();
610 |       const sixHours = 6 * 60 * 60 * 1000;
611 |       if (cached && cached.updatedAt && (now - new Date(cached.updatedAt).getTime()) < sixHours) {
612 |         const p90 = parseFloat((await storage.getParametro("percentil_90"))?.valor || "0");
613 |         const p75 = parseFloat((await storage.getParametro("percentil_75"))?.valor || "0");
614 |         const p50 = parseFloat((await storage.getParametro("percentil_50"))?.valor || "0");
615 |         const totalCamiones = parseInt((await storage.getParametro("percentil_total_camiones"))?.valor || "0");
616 |         const fechaCalculo = cached.updatedAt;
617 |         return res.json({ p50, p75, p90, totalCamiones, fechaCalculo, calibrando: totalCamiones < 10 });
618 |       }
619 |
620 |       const { pool } = await import("./db");
621 |       const cencFIdP = await getCencosudFaenaId();
622 |       const result = await pool.query(`
623 |         WITH truck_rend AS (
624 |           SELECT c.patente,
625 |                  AVG(
626 |                    CASE WHEN cg.km_actual - cg.km_anterior > 0 AND cg.litros_ecu > 0
627 |                         THEN (cg.km_actual - cg.km_anterior)::float / cg.litros_ecu
628 |                         ELSE NULL END
629 |                  ) AS avg_rend,
630 |                  COUNT(*) AS num_cargas
631 |           FROM camiones c
632 |           JOIN cargas cg ON cg.camion_id = c.id
633 |           WHERE cg.fecha >= '2026-03-01'
634 |             AND c.faena_id = ${cencFIdP}
635 |           GROUP BY c.patente
636 |           HAVING COUNT(*) >= 3
637 |         )
638 |         SELECT
639 |           PERCENTILE_CONT(0.90) WITHIN GROUP (ORDER BY avg_rend) AS p90,
```

```
640 |           PERCENTILE_CONT(0.75) WITHIN GROUP (ORDER BY avg_rend) AS p75,
641 |           PERCENTILE_CONT(0.50) WITHIN GROUP (ORDER BY avg_rend) AS p50,
642 |           COUNT(*) AS total_camiones
643 |         FROM truck_rend
644 |         WHERE avg_rend > 0 AND avg_rend < 15
645 |       `);
646 |
647 |       const row = result.rows[0];
648 |       const p90 = row?.p90 ? Math.round(parseFloat(row.p90) * 100) / 100 : 0;
649 |       const p75 = row?.p75 ? Math.round(parseFloat(row.p75) * 100) / 100 : 0;
650 |       const p50 = row?.p50 ? Math.round(parseFloat(row.p50) * 100) / 100 : 0;
651 |       const totalCamiones = parseInt(row?.total_camiones || "0");
652 |
653 |       await storage.upsertParametro("percentil_90", String(p90));
654 |       await storage.upsertParametro("percentil_75", String(p75));
655 |       await storage.upsertParametro("percentil_50", String(p50));
656 |       await storage.upsertParametro("percentil_total_camiones", String(totalCamiones));
657 |
658 |       const fechaCalculo = new Date();
659 |       res.json({ p50, p75, p90, totalCamiones, fechaCalculo, calibrando: totalCamiones < 10 });
660 |     } catch (error: any) {
661 |       console.error("[percentiles] Error:", error.message);
662 |       res.status(500).json({ message: error.message });
663 |     }
664 |   });
665 |
666 |   app.get("/api/puntos-ruta", async (req, res) => {
667 |     try {
668 |       const camionId = req.query.camionId ? parseInt(req.query.camionId as string) : undefined;
669 |       const puntos = camionId ? await storage.getPuntosRutaByCamion(camionId) : await storage.getPuntosRuta();
670 |       res.json(puntos);
671 |     } catch (error: any) {
672 |       res.status(500).json({ message: error.message });
673 |     }
674 |   });
675 |
676 |   app.post("/api/puntos-ruta/:id/confirmar", async (req, res) => {
677 |     try {
678 |       const id = parseInt(req.params.id);
679 |       if (isNaN(id)) return res.status(400).json({ message: "Invalid ID" });
680 |       const updated = await storage.updatePuntoRuta(id, { confirmado: true, confirmadoPor: req.body.confirmadoPor || "ad
681 |       if (!updated) return res.status(404).json({ message: "Punto not found" });
682 |       res.json(updated);
683 |     } catch (error: any) {
684 |       res.status(500).json({ message: error.message });
685 |     }
686 |   });
687 |
688 |   app.put("/api/puntos-ruta/:id", async (req, res) => {
689 |     try {
690 |       const id = parseInt(req.params.id);
691 |       if (isNaN(id)) return res.status(400).json({ message: "Invalid ID" });
692 |       const { nombreInferido, tipo } = req.body;
693 |       const data: any = {};
694 |       if (nombreInferido !== undefined) data.nombreInferido = nombreInferido;
695 |       if (tipo !== undefined) data.tipo = tipo;
696 |       const updated = await storage.updatePuntoRuta(id, data);
697 |       if (!updated) return res.status(404).json({ message: "Punto not found" });
698 |       res.json(updated);
699 |     } catch (error: any) {
700 |       res.status(500).json({ message: error.message });
701 |     }
702 |   });
703 |
704 |   app.get("/api/tarifas/puntos-confirmados", async (_req, res) => {
705 |     try {
706 |       const puntos = await storage.getConfirmedPuntos();
707 |       res.json(puntos);
708 |     } catch (error: any) {
709 |       res.status(500).json({ message: error.message });
710 |     }
711 |   });
712 |
713 |   app.get("/api/tarifas", async (_req, res) => {
714 |     try {
715 |       const tarifas = await storage.getTarifas();
716 |       res.json(tarifas);
717 |     } catch (error: any) {
718 |       res.status(500).json({ message: error.message });
719 |     }
720 |   });
721 |
722 |   app.post("/api/tarifas", async (req, res) => {
723 |     try {
724 |       const parsed = insertTarifaRutaSchema.safeParse(req.body);
725 |       if (!parsed.success) return res.status(400).json({ message: parsed.error.message });
726 |       const tarifa = await storage.createTarifa(parsed.data);
727 |       res.json(tarifa);
728 |     } catch (error: any) {
729 |       res.status(500).json({ message: error.message });
730 |     }
731 |   });
732 |
733 |   app.put("/api/tarifas/:id", async (req, res) => {
734 |     try {
735 |       const id = parseInt(req.params.id);
736 |       if (isNaN(id)) return res.status(400).json({ message: "Invalid ID" });
737 |       const allowed = ["nombreRuta", "origenNombre", "destinoNombre", "distanciaKm", "litrosPromedio", "tiempoHoras", "t
738 |       const data: any = {};
739 |       for (const key of allowed) {
740 |         if (req.body[key] !== undefined) data[key] = req.body[key];
741 |       }
742 |       if (Object.keys(data).length === 0) return res.status(400).json({ message: "No valid fields provided" });
743 |       const updated = await storage.updateTarifa(id, data);
744 |       if (!updated) return res.status(404).json({ message: "Tarifa not found" });
745 |       res.json(updated);
746 |     } catch (error: any) {
```

```
747 |       res.status(500).json({ message: error.message });
748 |     }
749 |   });
750 |
751 |   app.delete("/api/tarifas/:id", async (req, res) => {
752 |     try {
753 |       const id = parseInt(req.params.id);
754 |       if (isNaN(id)) return res.status(400).json({ message: "Invalid ID" });
755 |       await storage.deactivateTarifa(id);
756 |       res.json({ success: true });
757 |     } catch (error: any) {
758 |       res.status(500).json({ message: error.message });
759 |     }
760 |   });
761 |
762 |   app.get("/api/dashboard/hero", async (_req, res) => {
763 |     try {
764 |       const [allCamiones, allFaenas, faenaId, patentes] = await Promise.all([
765 |         storage.getCamiones(),
766 |         storage.getFaenas(),
767 |         getCencosudFaenaId(),
768 |         getCencosudPatentes(),
769 |       ]);
770 |       const camionesCenc = allCamiones.filter(c => c.faenaId === faenaId);
771 |
772 |       const to = new Date();
773 |       const from30 = new Date(Date.now() - 30 * 24 * 60 * 60 * 1000);
774 |
775 |       let fuelData: any[] = [];
776 |       try {
777 |         const allFuel = await getCachedFuelData(from30, to);
778 |         fuelData = allFuel.filter((r: any) => patentes.has(String(r.numVeh || "")));
779 |       } catch {}
780 |
781 |       let fleetStatus: any[] = [];
782 |       try {
783 |         const allFleet = await getFleetStatus();
784 |         const cencVins = new Set(camionesCenc.map(c => c.vin).filter(Boolean));
785 |         fleetStatus = allFleet.filter((v: any) => cencVins.has(v.vin));
786 |       } catch {}
787 |
788 |       const totalCamiones = camionesCenc.length;
789 |       const totalFaenas = 1;
790 |       const litros30d = Math.round(fuelData.reduce((s: number, c: any) => s + c.cantidadLt, 0));
791 |       const cargas30d = fuelData.length;
792 |
793 |       let km30d = 0;
794 |       for (const rec of fuelData) {
795 |         if (rec.kmRecorrido != null && rec.kmRecorrido > 0 && rec.kmRecorrido < 10000) {
796 |           km30d += rec.kmRecorrido;
797 |         }
798 |       }
799 |       km30d = Math.round(km30d);
800 |
801 |       const validRend = fuelData.filter((c: any) => c.rendReal != null && c.rendReal > 0 && c.rendReal < 100);
802 |       const rendPromedio = validRend.length > 0
803 |         ? Math.round((validRend.reduce((s: number, c: any) => s + c.rendReal, 0) / validRend.length) * 100) / 100
804 |         : 0;
805 |
806 |       const camionesConVin = camionesCenc.filter(c => c.vin).length;
807 |       const camionesOnline = fleetStatus.length;
808 |
809 |       const conductoresSet = new Set<string>();
810 |       for (const rec of fuelData) {
811 |         if (rec.nombreConductor) conductoresSet.add(rec.nombreConductor);
812 |       }
813 |
814 |       res.json({
815 |         totalCamiones,
816 |         totalFaenas,
817 |         litros30d,
818 |         km30d,
819 |         cargas30d,
820 |         rendPromedio,
821 |         camionesConVin,
822 |         camionesOnline,
823 |         conductores: conductoresSet.size,
824 |       });
825 |     } catch (error: any) {
826 |       console.error("[dashboard/hero] Error:", error.message);
827 |       res.status(500).json({ message: error.message });
828 |     }
829 |   });
830 |
831 |   app.get("/api/dashboard/faenas", async (_req, res) => {
832 |     try {
833 |       const [allFaenas, allCamiones] = await Promise.all([
834 |         storage.getFaenas(),
835 |         storage.getCamiones(),
836 |       ]);
837 |
838 |       const to = new Date();
839 |       const from30 = new Date(Date.now() - 30 * 24 * 60 * 60 * 1000);
840 |       let fuelData: any[] = [];
841 |       try {
842 |         fuelData = await getCachedFuelData(from30, to);
843 |       } catch {}
844 |
845 |       const camionByPatente = new Map(allCamiones.map(c => [c.patente, c]));
846 |
847 |       const fuelByFaena = new Map<string, any[]>();
848 |       for (const rec of fuelData) {
849 |         const faenaName = rec.faena || "Sin Faena";
850 |         const arr = fuelByFaena.get(faenaName) || [];
851 |         arr.push(rec);
852 |         fuelByFaena.set(faenaName, arr);
853 |       }
```

```
854 |
855 |       const cencId = await getCencosudFaenaId();
856 |       const faenaResults = allFaenas.filter(f => f.id === cencId).map(f => {
857 |         const camionesEnFaena = allCamiones.filter(c => c.faenaId === f.id);
858 |         const patentesEnFaena = new Set(camionesEnFaena.map(c => c.patente));
859 |
860 |         const fuelRecs = fuelByFaena.get(f.nombre) || [];
861 |         const totalLitros = Math.round(fuelRecs.reduce((s: number, c: any) => s + c.cantidadLt, 0));
862 |         const validRend = fuelRecs.filter((c: any) => c.rendReal != null && c.rendReal > 0 && c.rendReal < 100);
863 |         const rendPromedio = validRend.length > 0
864 |           ? Math.round((validRend.reduce((s: number, c: any) => s + c.rendReal, 0) / validRend.length) * 100) / 100
865 |           : 0;
866 |
867 |         const criticos = validRend.filter((c: any) => c.rendReal < 1.5).length;
868 |         const alertas = validRend.filter((c: any) => c.rendReal >= 1.5 && c.rendReal < 2.0).length;
869 |
870 |         const patentesFuel = new Set(fuelRecs.map((c: any) => String(c.numVeh || c.patente)));
871 |         const totalTrucks = Math.max(camionesEnFaena.length, patentesFuel.size);
872 |
873 |         let estado: "operativa" | "alerta" | "critica" = "operativa";
874 |         if (criticos > totalTrucks * 0.3) estado = "critica";
875 |         else if (alertas > totalTrucks * 0.2 || criticos > 0) estado = "alerta";
876 |
877 |         return {
878 |           id: f.id,
879 |           nombre: f.nombre,
880 |           color: f.color,
881 |           camiones: totalTrucks,
882 |           litros30d: totalLitros,
883 |           rendPromedio,
884 |           criticos,
885 |           alertas,
886 |           estado,
887 |         };
888 |       });
889 |
890 |       res.json(faenaResults);
891 |     } catch (error: any) {
892 |       console.error("[dashboard/faenas] Error:", error.message);
893 |       res.status(500).json({ message: error.message });
894 |     }
895 |   });
896 |
897 |   app.get("/api/dashboard/faena/:id", async (req, res) => {
898 |     try {
899 |       const faenaId = parseInt(req.params.id);
900 |       if (isNaN(faenaId)) return res.status(400).json({ message: "Invalid ID" });
901 |
902 |       const faena = await storage.getFaena(faenaId);
903 |       if (!faena) return res.status(404).json({ message: "Faena no encontrada" });
904 |
905 |       const allCamiones = await storage.getCamiones();
906 |       const camionesEnFaena = allCamiones.filter(c => c.faenaId === faenaId);
907 |
908 |       const to = new Date();
909 |       const from30 = new Date(Date.now() - 30 * 24 * 60 * 60 * 1000);
910 |       let fuelData: any[] = [];
911 |       try {
912 |         fuelData = await getCachedFuelData(from30, to);
913 |       } catch {}
914 |
915 |       const fuelRecs = fuelData.filter((r: any) => r.faena === faena.nombre);
916 |       const totalLitros = Math.round(fuelRecs.reduce((s: number, c: any) => s + (c.cantidadLt || 0), 0));
917 |       const validRend = fuelRecs.filter((c: any) => c.rendReal != null && c.rendReal > 0 && c.rendReal < 100);
918 |       const rendPromedio = validRend.length > 0
919 |         ? Math.round((validRend.reduce((s: number, c: any) => s + c.rendReal, 0) / validRend.length) * 100) / 100
920 |         : 0;
921 |       const criticos = validRend.filter((c: any) => c.rendReal < 1.5).length;
922 |       const alertas = validRend.filter((c: any) => c.rendReal >= 1.5 && c.rendReal < 2.0).length;
923 |
924 |       const contratos = await storage.getTmsContratos();
925 |       const contrato = contratos.find(c => c.faenaId === faenaId);
926 |
927 |       let viajesData: any[] = [];
928 |       let totalViajes = 0;
929 |       let totalKmViajes = 0;
930 |       if (contrato) {
931 |         const viajes = await storage.getTmsViajes(contrato.id);
932 |         totalViajes = viajes.length;
933 |         totalKmViajes = Math.round(viajes.reduce((s, v) => s + (parseFloat(v.kmRecorridos || "0") || 0), 0) * 10) / 10;
934 |
935 |         const camionMap = new Map(camionesEnFaena.map(c => [c.id, c]));
936 |
937 |         viajesData = viajes
938 |           .sort((a, b) => new Date(b.fechaSalida || 0).getTime() - new Date(a.fechaSalida || 0).getTime())
939 |           .slice(0, 20)
940 |           .map(v => ({
941 |             id: v.id,
942 |             codigo: v.codigo,
943 |             camionPatente: camionMap.get(v.camionId)?.patente || "?",
944 |             fechaSalida: v.fechaSalida,
945 |             estado: v.estado,
946 |             km: parseFloat(v.kmRecorridos || "0") || 0,
947 |             rendimiento: parseFloat(v.rendimientoReal || "0") || 0,
948 |             litros: parseFloat(v.litrosSigetra || "0") || 0,
949 |           }));
950 |       }
951 |
952 |       const fuelByPatente = new Map<string, any[]>();
953 |       for (const r of fuelRecs) {
954 |         const key = String(r.numVeh || r.patente);
955 |         const arr = fuelByPatente.get(key) || [];
956 |         arr.push(r);
957 |         fuelByPatente.set(key, arr);
958 |       }
959 |
960 |       const viajeCountByPatente = new Map<string, number>();
```

```
961 |       for (const v of viajesData) {
962 |         viajeCountByPatente.set(v.camionPatente, (viajeCountByPatente.get(v.camionPatente) || 0) + 1);
963 |       }
964 |
965 |       const camionesConStats = camionesEnFaena.map(cam => {
966 |         const camFuel = fuelByPatente.get(cam.patente) || [];
967 |         const camLitros = Math.round(camFuel.reduce((s: number, c: any) => s + (c.cantidadLt || 0), 0));
968 |         const camRendVals = camFuel.filter((c: any) => c.rendReal > 0 && c.rendReal < 100).map((c: any) => c.rendReal);
969 |         const camRend = camRendVals.length > 0 ? Math.round(camRendVals.reduce((s: number, v: number) => s + v, 0) / cam
970 |
971 |         return {
972 |           ...cam,
973 |           litros30d: camLitros,
974 |           rendimiento: camRend,
975 |           cargas: camFuel.length,
976 |           viajes: viajeCountByPatente.get(cam.patente) || 0,
977 |         };
978 |       }).sort((a, b) => b.litros30d - a.litros30d);
979 |
980 |       const patentesFuel = new Set(fuelRecs.map((c: any) => String(c.numVeh || c.patente)));
981 |       const totalTrucks = Math.max(camionesEnFaena.length, patentesFuel.size);
982 |
983 |       res.json({
984 |         faena,
985 |         camiones: camionesConStats,
986 |         contrato: contrato ? { id: contrato.id, nombre: contrato.nombre, totalViajes: contrato.totalViajes } : null,
987 |         viajes: viajesData,
988 |         resumen: {
989 |           totalCamiones: totalTrucks,
990 |           litros30d: totalLitros,
991 |           rendPromedio,
992 |           criticos,
993 |           alertas,
994 |           totalViajes,
995 |           totalKmViajes,
996 |           cargas: fuelRecs.length,
997 |         },
998 |       });
999 |     } catch (error: any) {
1000 |       console.error("[dashboard/faena] Error:", error.message);
1001 |       res.status(500).json({ message: error.message });
1002 |     }
1003 |   });
1004 |
1005 |   app.get("/api/faenas/resumen", async (_req, res) => {
1006 |     try {
1007 |       const allFaenas = await storage.getFaenas();
1008 |       const allCamiones = await storage.getCamiones();
1009 |       const contratos = await storage.getTmsContratos();
1010 |
1011 |       const to = new Date();
1012 |       const from30 = new Date(Date.now() - 30 * 24 * 60 * 60 * 1000);
1013 |       let fuelData: any[] = [];
1014 |       try { fuelData = await getCachedFuelData(from30, to); } catch {}
1015 |
1016 |       const cencFIdR = await getCencosudFaenaId();
1017 |       const results = await Promise.all(allFaenas.filter(f => f.id === cencFIdR).map(async f => {
1018 |         const camionesEnFaena = allCamiones.filter(c => c.faenaId === f.id);
1019 |         if (camionesEnFaena.length === 0) return null;
1020 |
1021 |         const fuelRecs = fuelData.filter((r: any) => r.faena === f.nombre);
1022 |         const totalLitros = Math.round(fuelRecs.reduce((s: number, c: any) => s + (c.cantidadLt || 0), 0));
1023 |         const validRend = fuelRecs.filter((c: any) => c.rendReal != null && c.rendReal > 0 && c.rendReal < 100);
1024 |         const rendPromedio = validRend.length > 0
1025 |           ? Math.round((validRend.reduce((s: number, c: any) => s + c.rendReal, 0) / validRend.length) * 100) / 100 : 0;
1026 |
1027 |         const contrato = contratos.find(c => c.faenaId === f.id);
1028 |         let totalViajes = 0;
1029 |         let totalKm = 0;
1030 |         let viajesPorDia: Record<string, { total: number; km: number; camiones: string[] }> = {};
1031 |         let camionResumen: { patente: string; modelo: string; viajes: number; km: number; litros: number; rendimiento: n
1032 |
1033 |         if (contrato) {
1034 |           const viajes = await storage.getTmsViajes(contrato.id);
1035 |           totalViajes = viajes.length;
1036 |           totalKm = Math.round(viajes.reduce((s, v) => s + (parseFloat(v.kmRecorridos || "0") || 0), 0) * 10) / 10;
1037 |
1038 |           const camionMap = new Map(camionesEnFaena.map(c => [c.id, c]));
1039 |
1040 |           for (const v of viajes) {
1041 |             const fecha = v.fechaSalida ? new Date(v.fechaSalida).toISOString().split("T")[0] : "sin-fecha";
1042 |             if (!viajesPorDia[fecha]) viajesPorDia[fecha] = { total: 0, km: 0, camiones: [] };
1043 |             viajesPorDia[fecha].total++;
1044 |             viajesPorDia[fecha].km += parseFloat(v.kmRecorridos || "0") || 0;
1045 |             const pat = camionMap.get(v.camionId)?.patente || "?";
1046 |             if (!viajesPorDia[fecha].camiones.includes(pat)) viajesPorDia[fecha].camiones.push(pat);
1047 |           }
1048 |
1049 |           for (const k of Object.keys(viajesPorDia)) {
1050 |             viajesPorDia[k].km = Math.round(viajesPorDia[k].km * 10) / 10;
1051 |           }
1052 |
1053 |           const viajesByCamion = new Map<number, typeof viajes>();
1054 |           for (const v of viajes) {
1055 |             const arr = viajesByCamion.get(v.camionId) || [];
1056 |             arr.push(v);
1057 |             viajesByCamion.set(v.camionId, arr);
1058 |           }
1059 |
1060 |           const fuelByPatente = new Map<string, any[]>();
1061 |           for (const r of fuelRecs) {
1062 |             const key = String(r.numVeh || r.patente);
1063 |             const arr = fuelByPatente.get(key) || [];
1064 |             arr.push(r);
1065 |             fuelByPatente.set(key, arr);
1066 |           }
1067 |
```

```
1068 |           camionResumen = camionesEnFaena.map(cam => {
1069 |             const cv = viajesByCamion.get(cam.id) || [];
1070 |             const cf = fuelByPatente.get(cam.patente) || [];
1071 |             const camLitros = Math.round(cf.reduce((s: number, c: any) => s + (c.cantidadLt || 0), 0));
1072 |             const rendVals = cf.filter((c: any) => c.rendReal > 0 && c.rendReal < 100).map((c: any) => c.rendReal);
1073 |             const rend = rendVals.length > 0 ? Math.round(rendVals.reduce((a: number, b: number) => a + b, 0) / rendVals
1074 |             return {
1075 |               patente: cam.patente,
1076 |               modelo: cam.modelo,
1077 |               viajes: cv.length,
1078 |               km: Math.round(cv.reduce((s, v) => s + (parseFloat(v.kmRecorridos || "0") || 0), 0) * 10) / 10,
1079 |               litros: camLitros,
1080 |               rendimiento: rend,
1081 |             };
1082 |           }).filter(c => c.viajes > 0 || c.litros > 0).sort((a, b) => b.viajes - a.viajes || b.km - a.km);
1083 |         }
1084 |
1085 |         return {
1086 |           id: f.id,
1087 |           nombre: f.nombre,
1088 |           color: f.color,
1089 |           totalCamiones: camionesEnFaena.length,
1090 |           litros30d: totalLitros,
1091 |           rendPromedio,
1092 |           contrato: contrato ? { id: contrato.id, nombre: contrato.nombre } : null,
1093 |           totalViajes,
1094 |           totalKm,
1095 |           viajesPorDia,
1096 |           camionResumen,
1097 |         };
1098 |       }));
1099 |
1100 |       res.json(results.filter(Boolean).sort((a: any, b: any) => b.totalViajes - a.totalViajes || b.litros30d - a.litros3
1101 |     } catch (error: any) {
1102 |       console.error("[faenas/resumen] Error:", error.message);
1103 |       res.status(500).json({ message: error.message });
1104 |     }
1105 |   });
1106 |
1107 |   app.get("/api/faenas/en-movimiento", async (_req, res) => {
1108 |     try {
1109 |       const allCamiones = await storage.getCamiones();
1110 |       const allFaenas = await storage.getFaenas();
1111 |
1112 |       let fleet: any[] = [];
1113 |       try {
1114 |         const { getFleetStatus } = await import("./volvo-api");
1115 |         fleet = await getFleetStatus();
1116 |       } catch {}
1117 |
1118 |       const vinToFleet = new Map(fleet.map(v => [v.vin, v]));
1119 |       const now = Date.now();
1120 |
1121 |       const faenaMap = new Map(allFaenas.map(f => [f.id, f]));
1122 |       const byFaena = new Map<number, any[]>();
1123 |
1124 |       const cencFIdM = await getCencosudFaenaId();
1125 |       const cencCamiones = allCamiones.filter(c => c.faenaId === cencFIdM);
1126 |       for (const cam of cencCamiones) {
1127 |         if (!cam.faenaId || !cam.vin) continue;
1128 |         const volvo = vinToFleet.get(cam.vin);
1129 |         if (!volvo?.gps) continue;
1130 |
1131 |         const gpsAge = volvo.gps.positionDateTime
1132 |           ? (now - new Date(volvo.gps.positionDateTime).getTime()) / (1000 * 60)
1133 |           : 999;
1134 |
1135 |         const isMoving = (volvo.gps.speed || 0) > 2;
1136 |         const isRecent = gpsAge < 60;
1137 |         const isOnline = gpsAge < 360;
1138 |
1139 |         if (!isOnline) continue;
1140 |
1141 |         const truck = {
1142 |           id: cam.id,
1143 |           patente: cam.patente,
1144 |           modelo: cam.modelo,
1145 |           vin: cam.vin,
1146 |           lat: volvo.gps.latitude,
1147 |           lng: volvo.gps.longitude,
1148 |           speed: volvo.gps.speed || 0,
1149 |           heading: volvo.gps.heading,
1150 |           gpsTime: volvo.gps.positionDateTime,
1151 |           gpsAgeMin: Math.round(gpsAge),
1152 |           fuelLevel: volvo.fuelLevel,
1153 |           engineHours: volvo.engineHours,
1154 |           totalDistance: volvo.totalDistance,
1155 |           driverWorkingState: volvo.driverWorkingState,
1156 |           isMoving,
1157 |           isRecent,
1158 |           estado: isMoving ? "EN_RUTA" : isRecent ? "DETENIDO" : "INACTIVO",
1159 |         };
1160 |
1161 |         const arr = byFaena.get(cam.faenaId) || [];
1162 |         arr.push(truck);
1163 |         byFaena.set(cam.faenaId, arr);
1164 |       }
1165 |
1166 |       const result = Array.from(byFaena.entries()).map(([faenaId, trucks]) => {
1167 |         const faena = faenaMap.get(faenaId);
1168 |         if (!faena) return null;
1169 |         const enRuta = trucks.filter(t => t.estado === "EN_RUTA").length;
1170 |         const detenidos = trucks.filter(t => t.estado === "DETENIDO").length;
1171 |         const inactivos = trucks.filter(t => t.estado === "INACTIVO").length;
1172 |         return {
1173 |           faenaId,
1174 |           nombre: faena.nombre,
```

```
1175 |           color: faena.color,
1176 |           trucks: trucks.sort((a, b) => {
1177 |             const order: Record<string, number> = { EN_RUTA: 0, DETENIDO: 1, INACTIVO: 2 };
1178 |             return (order[a.estado] ?? 3) - (order[b.estado] ?? 3);
1179 |           }),
1180 |           enRuta,
1181 |           detenidos,
1182 |           inactivos,
1183 |           totalOnline: trucks.length,
1184 |         };
1185 |       }).filter(Boolean).sort((a: any, b: any) => b.enRuta - a.enRuta || b.totalOnline - a.totalOnline);
1186 |
1187 |       const totals = {
1188 |         totalOnline: result.reduce((s: number, f: any) => s + f.totalOnline, 0),
1189 |         enRuta: result.reduce((s: number, f: any) => s + f.enRuta, 0),
1190 |         detenidos: result.reduce((s: number, f: any) => s + f.detenidos, 0),
1191 |         inactivos: result.reduce((s: number, f: any) => s + f.inactivos, 0),
1192 |         faenasActivas: result.length,
1193 |       };
1194 |
1195 |       res.json({ faenas: result, totals });
1196 |     } catch (error: any) {
1197 |       console.error("[faenas/en-movimiento] Error:", error.message);
1198 |       res.status(500).json({ message: error.message });
1199 |     }
1200 |   });
1201 |
1202 |   app.get("/api/dashboard/sistema", async (_req, res) => {
1203 |     try {
1204 |       let volvoStatus: any = { status: "unknown" };
1205 |       try {
1206 |         const user = process.env.VOLVO_CONNECT_USER || "";
1207 |         const hasPassword = !!process.env.VOLVO_CONNECT_PASSWORD;
1208 |         const configured = !!user && hasPassword;
1209 |         if (configured) {
1210 |           const fleet = await getFleetStatus();
1211 |           volvoStatus = { status: "connected", vehiculos: fleet.length, configured: true };
1212 |         } else {
1213 |           volvoStatus = { status: "not_configured", vehiculos: 0, configured: false };
1214 |         }
1215 |       } catch (e: any) {
1216 |         volvoStatus = { status: "error", vehiculos: 0, configured: true, error: e.message };
1217 |       }
1218 |
1219 |       let sigetraStatus: any = { status: "unknown" };
1220 |       try {
1221 |         const sigetra = await checkSigetraConnection();
1222 |         sigetraStatus = {
1223 |           status: sigetra.connected ? "connected" : "disconnected",
1224 |           connected: sigetra.connected,
1225 |           user: sigetra.user,
1226 |         };
1227 |       } catch (e: any) {
1228 |         sigetraStatus = { status: "error", connected: false, error: e.message };
1229 |       }
1230 |
1231 |       const iaConfigured = !!process.env.ANTHROPIC_API_KEY;
1232 |       const iaStatus = {
1233 |         status: iaConfigured ? "available" : "not_configured",
1234 |         configured: iaConfigured,
1235 |         model: iaConfigured ? "Claude Sonnet" : null,
1236 |       };
1237 |
1238 |       const now = new Date();
1239 |       const ultimoSync = now.toLocaleString("es-CL", { timeZone: "America/Santiago" });
1240 |
1241 |       const healthVolvo = volvoStatus.status === "connected" ? 100 : volvoStatus.status === "not_configured" ? 0 : 30;
1242 |       const healthSigetra = sigetraStatus.status === "connected" ? 100 : sigetraStatus.status === "disconnected" ? 0 : 3
1243 |       const healthIA = iaConfigured ? 100 : 0;
1244 |
1245 |       res.json({
1246 |         volvo: { ...volvoStatus, health: healthVolvo },
1247 |         sigetra: { ...sigetraStatus, health: healthSigetra },
1248 |         ia: { ...iaStatus, health: healthIA },
1249 |         ultimoSync,
1250 |         healthGeneral: Math.round((healthVolvo + healthSigetra + healthIA) / 3),
1251 |       });
1252 |     } catch (error: any) {
1253 |       console.error("[dashboard/sistema] Error:", error.message);
1254 |       res.status(500).json({ message: error.message });
1255 |     }
1256 |   });
1257 |
1258 |   registerIARoutes(app);
1259 |   registerTMSRoutes(app);
1260 |   registerCEORoutes(app);
1261 |   registerGeoRoutes(app);
1262 |
1263 |   app.get("/api/sigetra/fusion", async (req, res) => {
1264 |     try {
1265 |       const fromStr = req.query.from as string | undefined;
1266 |       const toStr = req.query.to as string | undefined;
1267 |       const from = fromStr ? new Date(fromStr) : new Date(Date.now() - 7 * 24 * 60 * 60 * 1000);
1268 |       const to = toStr ? new Date(toStr) : new Date();
1269 |       if (isNaN(from.getTime()) || isNaN(to.getTime())) {
1270 |         return res.status(400).json({ message: "Parametros from/to invalidos" });
1271 |       }
1272 |
1273 |       const [allFuelData, fleetStatus, allCamionesRaw, cencFaenaIdF, cencPatentesF] = await Promise.all([
1274 |         getCachedFuelData(from, to),
1275 |         getFleetStatus().catch(() => []),
1276 |         storage.getCamiones(),
1277 |         getCencosudFaenaId(),
1278 |         getCencosudPatentes(),
1279 |       ]);
1280 |
1281 |       const allCamiones = allCamionesRaw.filter(c => c.faenaId === cencFaenaIdF);
```

```
1282 |        const fuelData = allFuelData.filter((r: any) => cencPatentesF.has(String(r.numVeh || "")));
1283 |
1284 |        const volvoByVin = new Map(fleetStatus.map(s => [s.vin, s]));
1285 |        const camionByPatente = new Map(allCamiones.map(c => [c.patente, c]));
1286 |
1287 |        const nowDate = new Date();
1288 |        const hourKey = nowDate.toISOString().slice(0, 13) + ":00:00.000Z";
1289 |        const snapshotsToSave: { vin: string; totalFuelUsed: number; totalDistance: number | null; capturedAt: string }[]
1290 |        for (const vs of fleetStatus) {
1291 |          if (vs.totalFuelUsed != null) {
1292 |            snapshotsToSave.push({
1293 |              vin: vs.vin,
1294 |              totalFuelUsed: vs.totalFuelUsed,
1295 |              totalDistance: vs.totalDistance ?? null,
1296 |              capturedAt: hourKey,
1297 |            });
1298 |          }
1299 |        }
1300 |        if (snapshotsToSave.length > 0) {
1301 |          try {
1302 |            await storage.saveVolvoFuelSnapshots(snapshotsToSave);
1303 |          } catch (err: any) {
1304 |            console.error("[fusion] Error saving volvo snapshots:", err.message);
1305 |          }
1306 |        }
1307 |
1308 |        const allVins = allCamiones.map(c => c.vin).filter((v): v is string => v != null);
1309 |        const snapshotEnd = to < nowDate ? to : nowDate;
1310 |        const TWELVE_H = 12 * 60 * 60 * 1000;
1311 |        const expandedFrom = new Date(from.getTime() - TWELVE_H);
1312 |        const expandedTo = new Date(snapshotEnd.getTime() + TWELVE_H);
1313 |        const historicalSnapshots = allVins.length > 0
1314 |          ? await storage.getVolvoFuelSnapshotsInRange(allVins, expandedFrom, expandedTo)
1315 |          : [];
1316 |        const snapshotsByVin = new Map<string, typeof historicalSnapshots>();
1317 |        for (const snap of historicalSnapshots) {
1318 |          const arr = snapshotsByVin.get(snap.vin) || [];
1319 |          arr.push(snap);
1320 |          snapshotsByVin.set(snap.vin, arr);
1321 |        }
1322 |        for (const [vin, snaps] of snapshotsByVin) {
1323 |          snapshotsByVin.set(vin, snaps.sort((a: any, b: any) => new Date(a.capturedAt).getTime() - new Date(b.capturedAt)
1324 |        }
1325 |
1326 |        const computeVolvoFuelForWindow = (vin: string | null, windowStart: Date, windowEnd: Date, expand: boolean): numbe
1327 |          if (!vin) return null;
1328 |          const snaps = snapshotsByVin.get(vin);
1329 |          if (!snaps || snaps.length < 2) return null;
1330 |
1331 |          let filtered = snaps.filter((s: any) => {
1332 |            const t = new Date(s.capturedAt).getTime();
1333 |            return t >= windowStart.getTime() && t <= windowEnd.getTime();
1334 |          });
1335 |
1336 |          if (filtered.length < 2 && expand) {
1337 |            const expandedStart = new Date(windowStart.getTime() - TWELVE_H);
1338 |            const expandedEnd = new Date(windowEnd.getTime() + TWELVE_H);
1339 |            filtered = snaps.filter((s: any) => {
1340 |              const t = new Date(s.capturedAt).getTime();
1341 |              return t >= expandedStart.getTime() && t <= expandedEnd.getTime();
1342 |            });
1343 |          }
1344 |
1345 |          if (filtered.length < 2) return null;
1346 |          const earliest = filtered[0];
1347 |          const latest = filtered[filtered.length - 1];
1348 |          const delta = latest.totalFuelUsed - earliest.totalFuelUsed;
1349 |          return delta >= 0 ? Math.round(delta / 1000) : null;
1350 |        };
1351 |
1352 |        const fuelByNumVeh = new Map<number, typeof fuelData>();
1353 |        for (const rec of fuelData) {
1354 |          if (rec.numVeh == null) continue;
1355 |          const existing = fuelByNumVeh.get(rec.numVeh) || [];
1356 |          existing.push(rec);
1357 |          fuelByNumVeh.set(rec.numVeh, existing);
1358 |        }
1359 |
1360 |        const fusion: any[] = [];
1361 |        let totalViajesMatched = 0;
1362 |        let totalViajesSinEcu = 0;
1363 |        let totalCamionesOffline = 0;
1364 |
1365 |        for (const [numVeh, cargas] of fuelByNumVeh) {
1366 |          const camion = camionByPatente.get(String(numVeh));
1367 |          if (!camion) continue;
1368 |
1369 |          const volvo = camion.vin ? volvoByVin.get(camion.vin) : null;
1370 |
1371 |          const sortedCargas = [...cargas].sort((a: any, b: any) => new Date(a.fechaConsumo).getTime() - new Date(b.fechaC
1372 |          const totalLitrosSigetra = sortedCargas.reduce((s: number, c: any) => s + c.cantidadLt, 0);
1373 |          const totalCargasCount = sortedCargas.length;
1374 |          const lastCarga = sortedCargas[sortedCargas.length - 1];
1375 |          const rendPromedio = sortedCargas.filter((c: any) => c.rendReal != null && c.rendReal > 0 && c.rendReal < 100).r
1376 |
1377 |          const odometroSigetra = lastCarga?.odometroActual;
1378 |          const odometroVolvo = volvo?.totalDistance != null ? Math.round(volvo.totalDistance / 1000) : camion.odometro;
1379 |          const deltaOdometro = (odometroSigetra != null && odometroVolvo != null) ? odometroSigetra - odometroVolvo : nul
1380 |
1381 |          const fuelLevelVolvo = volvo?.fuelLevel ?? null;
1382 |          const totalFuelUsedVolvo = volvo?.totalFuelUsed != null ? Math.round(volvo.totalFuelUsed / 1000) : null;
1383 |
1384 |          let litrosVolvoPeriodo: number | null = null;
1385 |          let litrosSigetraViajes = 0;
1386 |          let viajesMatchedTruck = 0;
1387 |          let viajesSinEcuTruck = 0;
1388 |          let matchMethod: "viaje" | "periodo" | null = null;
```

```
1389 |
1390 |         if (camion.vin && sortedCargas.length >= 2) {
1391 |           let totalVolvoViajes = 0;
1392 |           for (let i = 0; i < sortedCargas.length - 1; i++) {
1393 |             const cargaI = sortedCargas[i];
1394 |             const cargaNext = sortedCargas[i + 1];
1395 |             const viajeInicio = new Date(cargaI.fechaConsumo);
1396 |             const viajeFin = new Date(cargaNext.fechaConsumo);
1397 |             const litrosSigetraViaje = cargaNext.cantidadLt;
1398 |
1399 |             const volvoLitrosViaje = computeVolvoFuelForWindow(camion.vin, viajeInicio, viajeFin, true);
1400 |
1401 |             if (volvoLitrosViaje !== null) {
1402 |               totalVolvoViajes += volvoLitrosViaje;
1403 |               litrosSigetraViajes += litrosSigetraViaje;
1404 |               viajesMatchedTruck++;
1405 |             } else {
1406 |               viajesSinEcuTruck++;
1407 |             }
1408 |           }
1409 |           if (viajesMatchedTruck > 0) {
1410 |             litrosVolvoPeriodo = totalVolvoViajes;
1411 |             matchMethod = "viaje";
1412 |           }
1413 |         }
1414 |
1415 |         if (litrosVolvoPeriodo === null && camion.vin) {
1416 |           const snaps = snapshotsByVin.get(camion.vin);
1417 |           if (snaps && snaps.length >= 2) {
1418 |             const earliest = snaps[0];
1419 |             const latest = snaps[snaps.length - 1];
1420 |             const snapStart = new Date(earliest.capturedAt);
1421 |             const snapEnd = new Date(latest.capturedAt);
1422 |             const delta = latest.totalFuelUsed - earliest.totalFuelUsed;
1423 |             if (delta > 0) {
1424 |               litrosVolvoPeriodo = Math.round(delta / 1000);
1425 |               const cargasInWindow = sortedCargas.filter((c: any) => {
1426 |                 const cf = new Date(c.fechaConsumo);
1427 |                 return cf >= snapStart && cf <= snapEnd;
1428 |               });
1429 |               if (cargasInWindow.length > 0) {
1430 |                 litrosSigetraViajes = cargasInWindow.reduce((s: number, c: any) => s + c.cantidadLt, 0);
1431 |               } else {
1432 |                 const expandedStart = new Date(snapStart.getTime() - TWELVE_H);
1433 |                 const expandedEnd = new Date(snapEnd.getTime() + TWELVE_H);
1434 |                 const cargasExpanded = sortedCargas.filter((c: any) => {
1435 |                   const cf = new Date(c.fechaConsumo);
1436 |                   return cf >= expandedStart && cf <= expandedEnd;
1437 |                 });
1438 |                 litrosSigetraViajes = cargasExpanded.length > 0
1439 |                   ? cargasExpanded.reduce((s: number, c: any) => s + c.cantidadLt, 0)
1440 |                   : totalLitrosSigetra;
1441 |               }
1442 |               matchMethod = "periodo";
1443 |             }
1444 |           }
1445 |         }
1446 |
1447 |         if (!camion.vin) {
1448 |           totalCamionesOffline++;
1449 |         }
1450 |
1451 |         totalViajesMatched += viajesMatchedTruck;
1452 |         totalViajesSinEcu += viajesSinEcuTruck;
1453 |
1454 |         let alertLevel: "ok" | "alerta" | "critico" = "ok";
1455 |         if (deltaOdometro != null) {
1456 |           const absDelta = Math.abs(deltaOdometro);
1457 |           if (absDelta > 5000) alertLevel = "critico";
1458 |           else if (absDelta > 1000) alertLevel = "alerta";
1459 |         }
1460 |
1461 |         fusion.push({
1462 |           fleetNum: String(numVeh),
1463 |           patenteReal: lastCarga?.patente || null,
1464 |           vin: camion.vin,
1465 |           modeloVolvo: camion.modelo,
1466 |           faenaSigetra: lastCarga?.faena || null,
1467 |           conductorSigetra: lastCarga?.nombreConductor || null,
1468 |           totalLitrosSigetra: Math.round(totalLitrosSigetra * 100) / 100,
1469 |           totalCargas: totalCargasCount,
1470 |           rendPromedio: Math.round(rendPromedio * 100) / 100,
1471 |           odometroSigetra,
1472 |           odometroVolvo,
1473 |           deltaOdometro,
1474 |           fuelLevelVolvo,
1475 |           totalFuelUsedVolvo,
1476 |           litrosVolvoPeriodo,
1477 |           litrosSigetraViajes: Math.round(litrosSigetraViajes * 100) / 100,
1478 |           viajesMatched: viajesMatchedTruck,
1479 |           viajesSinEcu: viajesSinEcuTruck,
1480 |           matchMethod,
1481 |           engineHoursVolvo: volvo?.engineHours != null ? Math.round(volvo.engineHours) : camion.horasMotor,
1482 |           gpsVolvo: volvo?.gps || null,
1483 |           alertLevel,
1484 |           cargas: cargas.map((c: any) => ({
1485 |             numGuia: c.numGuia,
1486 |             fecha: c.fechaConsumo,
1487 |             litros: c.cantidadLt,
1488 |             odometro: c.odometroActual,
1489 |             kmRecorrido: c.kmRecorrido,
1490 |             rendimiento: c.rendReal,
1491 |             lugar: c.lugarConsumo,
1492 |             tarjeta: c.tipoTarjeta,
1493 |             conductor: c.nombreConductor,
1494 |           })),
1495 |         });
```

```
1496 |       }
1497 |
1498 |       fusion.sort((a: any, b: any) => {
1499 |         const order: Record<string, number> = { critico: 0, alerta: 1, ok: 2 };
1500 |         return (order[a.alertLevel] ?? 2) - (order[b.alertLevel] ?? 2);
1501 |       });
1502 |
1503 |       const trucksWithVin = fusion.filter((t: any) => t.vin);
1504 |       const trucksWithoutVin = fusion.filter((t: any) => !t.vin);
1505 |
1506 |       res.json({
1507 |         from: from.toISOString(),
1508 |         to: to.toISOString(),
1509 |         totalTrucksMatched: fusion.length,
1510 |         totalLitros: Math.round(fusion.reduce((s: number, f: any) => s + f.totalLitrosSigetra, 0)),
1511 |         totalCargas: fusion.reduce((s: number, f: any) => s + f.totalCargas, 0),
1512 |         alertCount: fusion.filter((f: any) => f.alertLevel !== "ok").length,
1513 |         matchBreakdown: {
1514 |           viajesMatched: totalViajesMatched,
1515 |           viajesSinEcu: totalViajesSinEcu,
1516 |           camionesOffline: totalCamionesOffline,
1517 |         },
1518 |         trucks: fusion,
1519 |       });
1520 |     } catch (error: any) {
1521 |       console.error("[sigetra] Fusion error:", error.message);
1522 |       res.status(502).json({ message: error.message });
1523 |     }
1524 |   });
1525 |
1526 |   // %P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P
1527 |   // DATOS — MICRO-CARGAS (suspicious fuel loads)
1528 |   // %P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P%P
1529 |   app.get("/api/datos/micro-cargas", async (_req, res) => {
1530 |     try {
1531 |       const from = new Date("2026-03-01");
1532 |       const to = new Date();
1533 |       const allFuelData = await getCachedFuelData(from, to);
1534 |       const allCamiones = await storage.getCamiones();
1535 |       const faenas = await storage.getFaenas();
1536 |       const cencFaenaId = await getCencosudFaenaId();
1537 |       const cencPatentes = await getCencosudPatentes();
1538 |       const camiones = allCamiones.filter(c => c.faenaId === cencFaenaId);
1539 |       const fuelData = allFuelData.filter((r: any) => cencPatentes.has(String(r.numVeh || "")));
1540 |
1541 |       const params = await storage.getParametros();
1542 |       const litrosMicro = parseFloat(params.find(p => p.clave === "litros_micro_carga")?.valor || "100");
1543 |       const nivelTanqueSospechoso = parseFloat(params.find(p => p.clave === "nivel_tanque_sospechoso_pct")?.valor || "80
1544 |
1545 |       let fleet: any[] = [];
1546 |       try { fleet = await getFleetStatus(); } catch {}
1547 |
1548 |       const camionMap = new Map(camiones.map(c => [c.patente, c]));
1549 |       const faenaMap = new Map(faenas.map(f => [f.id, f]));
1550 |       const vinFuelLevel = new Map(fleet.map((v: any) => [v.vin, v.fuelLevel]));
1551 |
1552 |       const smallLoads = fuelData.filter(r => r.cantidadLt < litrosMicro && r.cantidadLt > 0);
1553 |
1554 |       // Signal A: Multiple small loads same day at different stations
1555 |       const dayGroups = new Map<string, typeof smallLoads>();
1556 |       for (const r of smallLoads) {
1557 |         const dateStr = r.fechaConsumo.split(" ")[0];
1558 |         const key = `${r.patente}|${dateStr}`;
1559 |         if (!dayGroups.has(key)) dayGroups.set(key, []);
1560 |         dayGroups.get(key)!.push(r);
1561 |       }
1562 |       const signalA = new Set<number>();
1563 |       for (const [, group] of dayGroups) {
1564 |         if (group.length >= 2) {
1565 |           const distinctPlaces = new Set(group.map(r => r.lugarConsumo || "").filter(Boolean));
1566 |           if (distinctPlaces.size >= 2) {
1567 |             for (const r of group) signalA.add(r.numGuia);
1568 |           }
1569 |         }
1570 |       }
1571 |
1572 |       // Signal B: Fuel load without movement — check volvo_fuel_snapshots for distance changes
1573 |       const signalB = new Set<number>();
1574 |       const horasVentana = parseFloat(params.find(p => p.clave === "horas_ventana_gps")?.valor || "4");
1575 |       {
1576 |         const allSnapshots = await db.select().from(
1577 |           (await import("@shared/schema")).volvoFuelSnapshots
1578 |         );
1579 |         const snapByVin = new Map<string, { capturedAt: string; totalDistance: number | null }[]>();
1580 |         for (const s of allSnapshots) {
1581 |           if (!snapByVin.has(s.vin)) snapByVin.set(s.vin, []);
1582 |           snapByVin.get(s.vin)!.push({ capturedAt: s.capturedAt, totalDistance: s.totalDistance });
1583 |         }
1584 |
1585 |         for (const r of smallLoads) {
1586 |           const cam = camionMap.get(r.patente);
1587 |           if (!cam?.vin) continue;
1588 |           const snaps = snapByVin.get(cam.vin);
1589 |           if (!snaps || snaps.length < 2) continue;
1590 |
1591 |           const loadTime = new Date(r.fechaConsumo).getTime();
1592 |           const windowMs = horasVentana * 60 * 60 * 1000;
1593 |           const nearbySnaps = snaps.filter(s => {
1594 |             const snapTime = new Date(s.capturedAt).getTime();
1595 |             return Math.abs(snapTime - loadTime) <= windowMs && s.totalDistance != null;
1596 |           });
1597 |
1598 |           if (nearbySnaps.length >= 2) {
1599 |             const distances = nearbySnaps.map(s => s.totalDistance!);
1600 |             const distChange = Math.max(...distances) - Math.min(...distances);
1601 |             if (distChange < 1000) {
1602 |               signalB.add(r.numGuia);
```

```
1603 |             }
1604 |           } else if (nearbySnaps.length === 0) {
1605 |             const volvo = fleet.find((v: any) => v.vin === cam.vin);
1606 |             if (volvo) {
1607 |               const gpsTime = volvo.gps?.positionDateTime;
1608 |               if (!gpsTime) { signalB.add(r.numGuia); continue; }
1609 |               const gpsAgeH = (Date.now() - new Date(gpsTime).getTime()) / (1000 * 60 * 60);
1610 |               if (gpsAgeH > 12 && (volvo.gps?.speed || 0) === 0) {
1611 |                 signalB.add(r.numGuia);
1612 |               }
1613 |             }
1614 |           }
1615 |         }
1616 |       }
1617 |
1618 |       // Signal C: Fuel load when tank is nearly full
1619 |       const signalC = new Set<number>();
1620 |       for (const r of smallLoads) {
1621 |         const cam = camionMap.get(r.patente);
1622 |         if (!cam?.vin) continue;
1623 |         const currentLevel = vinFuelLevel.get(cam.vin);
1624 |         if (currentLevel != null && currentLevel > nivelTanqueSospechoso) {
1625 |           signalC.add(r.numGuia);
1626 |         }
1627 |       }
1628 |
1629 |       const suspicious: any[] = [];
1630 |       for (const r of smallLoads) {
1631 |         const signals: string[] = [];
1632 |         if (signalA.has(r.numGuia)) signals.push("MULTIPLE_DIA");
1633 |         if (signalB.has(r.numGuia)) signals.push("SIN_MOVIMIENTO");
1634 |         if (signalC.has(r.numGuia)) signals.push("TANQUE_LLENO");
1635 |         if (signals.length === 0) continue;
1636 |
1637 |         const cam = camionMap.get(r.patente);
1638 |         const faena = cam ? faenaMap.get(cam.faenaId) : null;
1639 |
1640 |         suspicious.push({
1641 |           numGuia: r.numGuia,
1642 |           patente: r.patente,
1643 |           conductor: r.nombreConductor || null,
1644 |           fecha: r.fechaConsumo,
1645 |           litros: r.cantidadLt,
1646 |           lugar: r.lugarConsumo,
1647 |           faena: faena?.nombre || r.faena || null,
1648 |           signals,
1649 |           riesgo: signals.length >= 2 ? "CRITICO" : "SOSPECHOSO",
1650 |           fuelLevelActual: cam?.vin ? vinFuelLevel.get(cam.vin) ?? null : null,
1651 |         });
1652 |       }
1653 |
1654 |       suspicious.sort((a, b) => b.signals.length - a.signals.length || a.fecha.localeCompare(b.fecha));
1655 |
1656 |       const criticos = suspicious.filter(s => s.riesgo === "CRITICO").length;
1657 |       const sospechosos = suspicious.filter(s => s.riesgo === "SOSPECHOSO").length;
1658 |       const normales = smallLoads.length - suspicious.length;
1659 |
1660 |       res.json({
1661 |         registros: suspicious,
1662 |         totals: {
1663 |           criticos,
1664 |           sospechosos,
1665 |           normales,
1666 |           totalCargas: fuelData.length,
1667 |           cargasPequenas: smallLoads.length,
1668 |         },
1669 |         desde: "2026-03-01",
1670 |       });
1671 |     } catch (error: any) {
1672 |       console.error("[datos] micro-cargas error:", error.message);
1673 |       res.status(500).json({ message: error.message });
1674 |     }
1675 |   });
1676 |
1677 |   // %P%P%P%
```

```
1710 |           fecha: v.gps?.positionDateTime || v.createdDateTime || new Date().toISOString(),
1711 |           lat: v.gps?.latitude || null,
1712 |           lng: v.gps?.longitude || null,
1713 |           riesgo: speed > 120 ? "ALTO" : speed > 100 ? "MEDIO" : "BAJO",
1714 |         });
1715 |       }
1716 |
1717 |       eventos.sort((a, b) => new Date(b.fecha).getTime() - new Date(a.fecha).getTime());
1718 |
1719 |       const porCamion = new Map<string, any>();
1720 |       for (const e of eventos) {
1721 |         if (!porCamion.has(e.patente)) {
1722 |           porCamion.set(e.patente, {
1723 |             patente: e.patente,
1724 |             modelo: e.modelo,
1725 |             conductor: e.conductor,
1726 |             faena: e.faena,
1727 |             velocidadMaxima: e.velocidadMaxima,
1728 |             totalExcesos: 0,
1729 |             ultimoExceso: e.fecha,
1730 |             eventos: [],
1731 |             riesgo: "BAJO",
1732 |           });
1733 |         }
1734 |         const entry = porCamion.get(e.patente)!;
1735 |         entry.totalExcesos++;
1736 |         entry.eventos.push(e);
1737 |         if (e.velocidadMaxima > entry.velocidadMaxima) entry.velocidadMaxima = e.velocidadMaxima;
1738 |         if (entry.velocidadMaxima > 120 || entry.totalExcesos > 10) entry.riesgo = "ALTO";
1739 |         else if (entry.velocidadMaxima > 100 || entry.totalExcesos > 2) entry.riesgo = "MEDIO";
1740 |       }
1741 |
1742 |       const camionesConExceso = Array.from(porCamion.values()).sort((a, b) => b.totalExcesos - a.totalExcesos);
1743 |
1744 |       const totalCamiones = camiones.filter(c => c.vin).length;
1745 |       const camionesNunca = totalCamiones - camionesConExceso.length;
1746 |       const camionesOcasionales = camionesConExceso.filter(c => c.riesgo === "BAJO" || c.riesgo === "MEDIO").length;
1747 |       const camionesFrecuentes = camionesConExceso.filter(c => c.riesgo === "ALTO").length;
1748 |
1749 |       const masGrave = camionesConExceso.length > 0 ? camionesConExceso[0] : null;
1750 |
1751 |       res.json({
1752 |         eventos,
1753 |         porCamion: camionesConExceso,
1754 |         totals: {
1755 |           camionesConExceso: camionesConExceso.length,
1756 |           eventosTotal: eventos.length,
1757 |           masGrave: masGrave ? { patente: masGrave.patente, velocidad: masGrave.velocidadMaxima } : null,
1758 |           hoy: eventos.filter(e => new Date(e.fecha).toDateString() === new Date().toDateString()).length,
1759 |           limiteKmh,
1760 |         },
1761 |         resumen: {
1762 |           nuncaExcedieron: camionesNunca,
1763 |           ocasionales: camionesOcasionales,
1764 |           frecuentes: camionesFrecuentes,
1765 |         },
1766 |         desde: "2026-03-01",
1767 |       });
1768 |     } catch (error: any) {
1769 |       console.error("[datos] excesos-velocidad error:", error.message);
1770 |       res.status(500).json({ message: error.message });
1771 |     }
1772 |   });
1773 |
1774 |   // %P%P%P%
```

```
1817 |           cargaEstimada = Math.max(0, pesoVolvoKg - taraNum);
1818 |           pesoTotal = pesoVolvoKg;
1819 |         }
1820 |
1821 |         if (pesoTotal != null && pesoMaxNum != null) {
1822 |           const pct = (pesoTotal / pesoMaxNum) * 100;
1823 |           if (pct > 100) cumple = "EXCEDE";
1824 |           else if (pct >= 95) cumple = "AL_LIMITE";
1825 |           else cumple = "CUMPLE";
1826 |         }
1827 |
1828 |         return {
1829 |           id: c.id,
1830 |           patente: c.patente,
1831 |           modelo: c.modelo,
1832 |           faena: faena?.nombre || null,
1833 |           tipoVehiculo: c.tipoVehiculo || null,
1834 |           taraKg: taraNum,
1835 |           pesoMaximoKg: pesoMaxNum,
1836 |           capacidadCargaKg: c.capacidadCargaKg ? parseFloat(c.capacidadCargaKg as string) : null,
1837 |           cargaEstimada,
1838 |           pesoTotal,
1839 |           pesoVolvoKg,
1840 |           cumple,
1841 |           ultimaCarga: ultimaCarga ? {
1842 |             fecha: ultimaCarga.fechaConsumo,
1843 |             litros: ultimaCarga.cantidadLt,
1844 |             lugar: ultimaCarga.lugarConsumo,
1845 |           } : null,
1846 |           anioFabricacion: c.anioFabricacion || null,
1847 |           configuracionEjes: c.configuracionEjes || null,
1848 |         };
1849 |       });
1850 |
1851 |       registros.sort((a, b) => {
1852 |         const order: Record<string, number> = { EXCEDE: 0, AL_LIMITE: 1, SIN_DATOS: 2, CUMPLE: 3 };
1853 |         return (order[a.cumple] ?? 2) - (order[b.cumple] ?? 2);
1854 |       });
1855 |
1856 |       const totals = {
1857 |         conTara: conTara.length,
1858 |         cumplen: registros.filter(r => r.cumple === "CUMPLE").length,
1859 |         exceden: registros.filter(r => r.cumple === "EXCEDE").length,
1860 |         alLimite: registros.filter(r => r.cumple === "AL_LIMITE").length,
1861 |         sinDatos: registros.filter(r => r.cumple === "SIN_DATOS").length,
1862 |         totalCamiones: camiones.length,
1863 |       };
1864 |
1865 |       res.json({ registros, totals });
1866 |     } catch (error: any) {
1867 |       console.error("[datos] peso-tara error:", error.message);
1868 |       res.status(500).json({ message: error.message });
1869 |     }
1870 |   });
1871 |
1872 |   const taraUpdateSchema = z.object({
1873 |     taraKg: z.number().min(1000).max(30000).nullable().optional(),
1874 |     pesoMaximoKg: z.number().min(5000).max(100000).nullable().optional(),
1875 |     capacidadCargaKg: z.number().min(0).max(80000).nullable().optional(),
1876 |     tipoVehiculo: z.enum(["CAMION RIGIDO", "ARTICULADO", "TRACTO-CAMION", "OTRO"]).nullable().optional(),
1877 |     anioFabricacion: z.number().int().min(1990).max(2030).nullable().optional(),
1878 |     configuracionEjes: z.string().max(20).nullable().optional(),
1879 |   });
1880 |
1881 |   app.patch("/api/camiones/:id/tara", async (req, res) => {
1882 |     try {
1883 |       const id = parseInt(req.params.id);
1884 |       if (isNaN(id)) return res.status(400).json({ message: "Invalid ID" });
1885 |
1886 |       const parsed = taraUpdateSchema.safeParse(req.body);
1887 |       if (!parsed.success) return res.status(400).json({ message: parsed.error.message });
1888 |       const body = parsed.data;
1889 |
1890 |       const updateData: Record<string, any> = {};
1891 |       if (body.taraKg != null) updateData.taraKg = String(body.taraKg);
1892 |       if (body.pesoMaximoKg != null) updateData.pesoMaximoKg = String(body.pesoMaximoKg);
1893 |       if (body.capacidadCargaKg != null) updateData.capacidadCargaKg = String(body.capacidadCargaKg);
1894 |       if (body.tipoVehiculo != null) updateData.tipoVehiculo = body.tipoVehiculo;
1895 |       if (body.anioFabricacion != null) updateData.anioFabricacion = body.anioFabricacion;
1896 |       if (body.configuracionEjes != null) updateData.configuracionEjes = body.configuracionEjes;
1897 |
1898 |       const [updated] = await db.update(camiones).set(updateData).where(eq(camiones.id, id)).returning();
1899 |       if (!updated) return res.status(404).json({ message: "Camion not found" });
1900 |       res.json(updated);
1901 |     } catch (error: any) {
1902 |       console.error("[datos] update tara error:", error.message);
1903 |       res.status(500).json({ message: error.message });
1904 |     }
1905 |   });
1906 |
1907 |   return httpServer;
1908 | }
1909 |
```

server/geo-routes.ts

1231 lineas

```
1 | import type { Express, Request, Response } from "express";
2 | import { storage } from "./storage";
3 | import { db, pool } from "./db";
4 | import { geoPuntos, geoViajes, geoBases, geoTrayectorias, geoGeocache, camiones, geoLugares, geoVisitas, geoAnalisisIa }
5 | import { eq, desc, and, gte, lte, sql, inArray, asc } from "drizzle-orm";
6 | import { getFleetStatus } from "./volvo-api";
7 | import { getCachedFuelData } from "./sigetra-api";
8 | import { getCencosudFaenaId } from "./cencosud-filter";
9 | import { getContractConfig, getContractPatentes, getContractCamiones } from "./faena-filter";
10 | import { detectarLugar, registrarVisita, analizarHistoricoCompleto, generarAnalisisIA } from "./geo-lugares-service";
11 | import { obtenerHistorialCamion, procesarFlotaHistorico, obtenerResumenFlota } from "./geo-reconstruccion-service";
12 | import { detectarVisitasFlota, obtenerResumenVisitas, obtenerVisitasCamion, inicializarPerfilGPS } from "./geo-visitas-s
13 |
14 | function haversineKm(lat1: number, lon1: number, lat2: number, lon2: number): number {
15 |   const R = 6371;
16 |   const dLat = (lat2 - lat1) * Math.PI / 180;
17 |   const dLon = (lon2 - lon1) * Math.PI / 180;
18 |   const a = Math.sin(dLat / 2) ** 2 +
19 |     Math.cos(lat1 * Math.PI / 180) * Math.cos(lat2 * Math.PI / 180) *
20 |     Math.sin(dLon / 2) ** 2;
21 |   return R * 2 * Math.asin(Math.sqrt(a));
22 | }
23 |
24 | async function geocodificar(lat: number, lng: number): Promise<{ nombre: string; ciudad: string; region: string }> {
25 |   const roundLat = Math.round(lat * 100) / 100;
26 |   const roundLng = Math.round(lng * 100) / 100;
27 |
28 |   const cached = await db.select().from(geoGeocache)
29 |     .where(and(
30 |       eq(geoGeocache.lat, String(roundLat)),
31 |       eq(geoGeocache.lng, String(roundLng))
32 |     )).limit(1);
33 |
34 |   if (cached.length > 0) {
35 |     return { nombre: cached[0].nombre || "", ciudad: cached[0].ciudad || "", region: cached[0].region || "" };
36 |   }
37 |
38 |   try {
39 |     const res = await fetch(`https://nominatim.openstreetmap.org/reverse?lat=${lat}&lon=${lng}&format=json&zoom=14`, {
40 |       headers: { "User-Agent": "GEOVALIDATOR-Sotraser/1.0" }
41 |     });
42 |     const data = await res.json();
43 |     const nombre = data.display_name?.split(",").slice(0, 2).join(",").trim() || `${lat.toFixed(4)}, ${lng.toFixed(4)}`;
44 |     const ciudad = data.address?.city || data.address?.town || data.address?.village || "";
45 |     const region = data.address?.state || "";
46 |
47 |     await db.insert(geoGeocache).values({
48 |       lat: String(roundLat),
49 |       lng: String(roundLng),
50 |       nombre, ciudad, region,
51 |     }).onConflictDoNothing();
52 |
53 |     return { nombre, ciudad, region };
54 |   } catch {
55 |     return { nombre: `${lat.toFixed(4)}, ${lng.toFixed(4)}`, ciudad: "", region: "" };
56 |   }
57 | }
58 |
59 | async function getCencosudCamiones() {
60 |   const faenaId = await getCencosudFaenaId();
61 |   const allCamiones = await storage.getCamiones();
62 |   return allCamiones.filter(c => c.faenaId === faenaId);
63 | }
64 |
65 | export function registerGeoRoutes(app: Express) {
66 |
67 |   app.get("/api/geo/camiones-live", async (_req: Request, res: Response) => {
68 |     try {
69 |       const allCamiones = await storage.getCamiones();
70 |       const withVin = allCamiones.filter(c => c.vin);
71 |       let fleet: any[] = [];
72 |       try { fleet = await getFleetStatus(); } catch {}
73 |
74 |       const vinToFleet = new Map(fleet.map(v => [v.vin, v]));
75 |
76 |       const result = withVin.map(c => {
77 |         const volvo = c.vin ? vinToFleet.get(c.vin) : null;
78 |         const lat = volvo?.gps?.latitude || null;
79 |         const lng = volvo?.gps?.longitude || null;
80 |         const speed = volvo?.wheelBasedSpeed ?? volvo?.gps?.speed ?? 0;
81 |         const heading = volvo?.gps?.heading ?? 0;
82 |         const gpsTime = volvo?.gps?.positionDateTime || volvo?.createdDateTime || null;
83 |         const ageMinutes = gpsTime ? (Date.now() - new Date(gpsTime).getTime()) / 60000 : 999;
84 |
85 |         let estado = "SIN_SEÑAL";
86 |         if (ageMinutes < 60) {
87 |           estado = speed > 5 ? "EN_MOVIMIENTO" : (ageMinutes < 30 ? "DETENIDO_RECIENTE" : "DETENIDO");
88 |         }
89 |
90 |         return {
91 |           camionId: c.id,
92 |           patente: c.patente,
93 |           modelo: c.modelo,
94 |           conductor: c.conductor,
95 |           vin: c.vin,
96 |           lat, lng,
97 |           velocidad: Math.round(speed * 10) / 10,
98 |           rumbo: heading,
99 |           timestamp: gpsTime,
100 |           estado,
101 |           ageMinutes: Math.round(ageMinutes),
102 |           fuelLevel: volvo?.fuelLevel ?? null,
103 |         };
104 |       }).sort((a, b) => {
```

```
105 |         const order: Record<string, number> = { EN_MOVIMIENTO: 0, DETENIDO_RECIENTE: 1, DETENIDO: 2, SIN_SEÑAL: 3 };
106 |         return (order[a.estado] ?? 9) - (order[b.estado] ?? 9);
107 |       });
108 |
109 |       res.json(result);
110 |     } catch (error: any) {
111 |       console.error("[geo] camiones-live error:", error.message);
112 |       res.status(500).json({ message: error.message });
113 |     }
114 |   });
115 |
116 |   app.get("/api/geo/stats", async (_req: Request, res: Response) => {
117 |     try {
118 |       const totalPuntos = await pool.query(`SELECT COUNT(*) as cnt FROM geo_puntos`);
119 |       const totalCamiones = await pool.query(`SELECT COUNT(DISTINCT camion_id) as cnt FROM geo_puntos`);
120 |       const today = await pool.query(`SELECT COUNT(*) as cnt FROM geo_puntos WHERE timestamp_punto >= CURRENT_DATE`);
121 |       const last7d = await pool.query(`SELECT COUNT(*) as cnt FROM geo_puntos WHERE timestamp_punto >= NOW() - INTERVAL
122 |       const byDay = await pool.query(`
123 |         SELECT DATE(timestamp_punto) as dia, COUNT(*) as puntos, COUNT(DISTINCT camion_id) as camiones
124 |         FROM geo_puntos WHERE timestamp_punto >= NOW() - INTERVAL '14 days'
125 |         GROUP BY DATE(timestamp_punto) ORDER BY dia DESC
126 |       `);
127 |       const oldest = await pool.query(`SELECT MIN(timestamp_punto) as fecha FROM geo_puntos`);
128 |       const newest = await pool.query(`SELECT MAX(timestamp_punto) as fecha FROM geo_puntos`);
129 |
130 |       res.json({
131 |         totalPuntos: parseInt(totalPuntos.rows[0]?.cnt) || 0,
132 |         totalCamiones: parseInt(totalCamiones.rows[0]?.cnt) || 0,
133 |         puntosHoy: parseInt(today.rows[0]?.cnt) || 0,
134 |         puntos7d: parseInt(last7d.rows[0]?.cnt) || 0,
135 |         primerPunto: oldest.rows[0]?.fecha,
136 |         ultimoPunto: newest.rows[0]?.fecha,
137 |         porDia: byDay.rows.map((r: any) => ({ dia: r.dia, puntos: parseInt(r.puntos), camiones: parseInt(r.camiones) }))
138 |       });
139 |     } catch (error: any) {
140 |       res.status(500).json({ message: error.message });
141 |     }
142 |   });
143 |
144 |   app.get("/api/geo/trayectoria/:camionId", async (req: Request, res: Response) => {
145 |     try {
146 |       const camionId = parseInt(req.params.camionId);
147 |       const fecha = req.query.fecha as string || new Date().toISOString().split("T")[0];
148 |
149 |       const dayStart = new Date(fecha + "T00:00:00");
150 |       const dayEnd = new Date(fecha + "T23:59:59");
151 |
152 |       const puntos = await db.select().from(geoPuntos)
153 |         .where(and(
154 |           eq(geoPuntos.camionId, camionId),
155 |           gte(geoPuntos.timestampPunto, dayStart),
156 |           lte(geoPuntos.timestampPunto, dayEnd)
157 |         ))
158 |         .orderBy(geoPuntos.timestampPunto);
159 |
160 |       res.json(puntos);
161 |     } catch (error: any) {
162 |       console.error("[geo] trayectoria error:", error.message);
163 |       res.status(500).json({ message: error.message });
164 |     }
165 |   });
166 |
167 |   app.get("/api/geo/bases", async (_req: Request, res: Response) => {
168 |     try {
169 |       const bases = await db.select().from(geoBases).where(eq(geoBases.activa, true));
170 |       res.json(bases);
171 |     } catch (error: any) {
172 |       res.status(500).json({ message: error.message });
173 |     }
174 |   });
175 |
176 |   app.post("/api/geo/bases", async (req: Request, res: Response) => {
177 |     try {
178 |       const { nombre, lat, lng, radioMetros, contrato } = req.body;
179 |       if (!nombre || !lat || !lng) return res.status(400).json({ message: "nombre, lat, lng requeridos" });
180 |       const [base] = await db.insert(geoBases).values({
181 |         nombre, lat: String(lat), lng: String(lng),
182 |         radioMetros: radioMetros || 3000,
183 |         contrato: contrato || "CENCOSUD",
184 |       }).returning();
185 |       res.json(base);
186 |     } catch (error: any) {
187 |       res.status(500).json({ message: error.message });
188 |     }
189 |   });
190 |
191 |   app.get("/api/geo/viajes", async (req: Request, res: Response) => {
192 |     try {
193 |       const desde = req.query.desde as string;
194 |       const hasta = req.query.hasta as string;
195 |       const patente = req.query.patente as string;
196 |       const estado = req.query.estado as string;
197 |
198 |       let query = db.select().from(geoViajes).orderBy(desc(geoViajes.creadoAt));
199 |
200 |       const conditions: any[] = [];
201 |       if (desde) conditions.push(gte(geoViajes.origenTimestamp, new Date(desde)));
202 |       if (hasta) conditions.push(lte(geoViajes.origenTimestamp, new Date(hasta)));
203 |       if (patente) conditions.push(eq(geoViajes.patente, patente));
204 |       if (estado) conditions.push(eq(geoViajes.validacionEstado, estado));
205 |
206 |       const viajes = conditions.length > 0
207 |         ? await db.select().from(geoViajes).where(and(...conditions)).orderBy(desc(geoViajes.creadoAt)).limit(200)
208 |         : await db.select().from(geoViajes).orderBy(desc(geoViajes.creadoAt)).limit(200);
209 |
210 |       res.json(viajes);
211 |     } catch (error: any) {
```

```
212 |       console.error("[geo] viajes error:", error.message);
213 |       res.status(500).json({ message: error.message });
214 |     }
215 |   });
216 |
217 |   app.get("/api/geo/viajes/:id", async (req: Request, res: Response) => {
218 |     try {
219 |       const id = parseInt(req.params.id);
220 |       const [viaje] = await db.select().from(geoViajes).where(eq(geoViajes.id, id));
221 |       if (!viaje) return res.status(404).json({ message: "Viaje no encontrado" });
222 |
223 |       let puntos: any[] = [];
224 |       if (viaje.trayectoriaId) {
225 |         puntos = await db.select().from(geoPuntos)
226 |           .where(eq(geoPuntos.trayectoriaId, viaje.trayectoriaId))
227 |           .orderBy(geoPuntos.timestampPunto);
228 |       }
229 |
230 |       res.json({ viaje, puntos });
231 |     } catch (error: any) {
232 |       res.status(500).json({ message: error.message });
233 |     }
234 |   });
235 |
236 |   app.post("/api/geo/viajes/:id/validar", async (req: Request, res: Response) => {
237 |     try {
238 |       const id = parseInt(req.params.id);
239 |       const { estado, notas, validadoPor } = req.body;
240 |       if (!["VALIDADO", "REVISAR", "ANOMALIA", "PENDIENTE"].includes(estado)) {
241 |         return res.status(400).json({ message: "Estado invalido" });
242 |       }
243 |       await db.update(geoViajes).set({
244 |         validacionEstado: estado,
245 |         notas: notas || null,
246 |         validadoManualmente: true,
247 |         validadoPor: validadoPor || "manual",
248 |         actualizadoAt: new Date(),
249 |       }).where(eq(geoViajes.id, id));
250 |       res.json({ success: true });
251 |     } catch (error: any) {
252 |       res.status(500).json({ message: error.message });
253 |     }
254 |   });
255 |
256 |   app.post("/api/geo/viajes/aprobar-masivo", async (req: Request, res: Response) => {
257 |     try {
258 |       const result = await db.update(geoViajes).set({
259 |         validadoManualmente: true,
260 |         validadoPor: "auto-aprobado",
261 |         actualizadoAt: new Date(),
262 |       }).where(and(
263 |         eq(geoViajes.validacionEstado, "VALIDADO"),
264 |         eq(geoViajes.validadoManualmente, false)
265 |       ));
266 |       res.json({ success: true });
267 |     } catch (error: any) {
268 |       res.status(500).json({ message: error.message });
269 |     }
270 |   });
271 |
272 |   app.get("/api/geo/validacion/stats", async (req: Request, res: Response) => {
273 |     try {
274 |       const dias = parseInt(req.query.dias as string) || 7;
275 |       const desde = new Date(Date.now() - dias * 24 * 60 * 60 * 1000);
276 |
277 |       const stats = await pool.query(`
278 |         SELECT
279 |           validacion_estado,
280 |           COUNT(*) as count,
281 |           ROUND(AVG(km_gps::numeric), 1) as avg_km,
282 |           ROUND(AVG(duracion_minutos::numeric), 0) as avg_duracion,
283 |           ROUND(AVG(velocidad_promedio::numeric), 1) as avg_velocidad
284 |         FROM geo_viajes
285 |         WHERE creado_at >= $1
286 |         GROUP BY validacion_estado
287 |       `, [desde]);
288 |
289 |       const totalViajes = stats.rows.reduce((s: number, r: any) => s + parseInt(r.count), 0);
290 |       const kmTotal = await pool.query(`
291 |         SELECT COALESCE(SUM(km_gps::numeric), 0) as total_km
292 |         FROM geo_viajes WHERE creado_at >= $1
293 |       `, [desde]);
294 |
295 |       const hoyCount = await pool.query(`
296 |         SELECT COUNT(*) as count FROM geo_viajes
297 |         WHERE creado_at >= CURRENT_DATE
298 |       `);
299 |
300 |       res.json({
301 |         porEstado: stats.rows.map((r: any) => ({
302 |           estado: r.validacion_estado || "PENDIENTE",
303 |           count: parseInt(r.count),
304 |           avgKm: parseFloat(r.avg_km) || 0,
305 |           avgDuracion: parseInt(r.avg_duracion) || 0,
306 |           pct: totalViajes > 0 ? Math.round(parseInt(r.count) / totalViajes * 1000) / 10 : 0,
307 |         })),
308 |         totalViajes,
309 |         kmTotal: Math.round(parseFloat(kmTotal.rows[0]?.total_km) || 0),
310 |         hoyActivos: parseInt(hoyCount.rows[0]?.count) || 0,
311 |       });
312 |     } catch (error: any) {
313 |       console.error("[geo] validacion stats error:", error.message);
314 |       res.status(500).json({ message: error.message });
315 |     }
316 |   });
317 |
318 |   app.post("/api/geo/puntos/ingest", async (req: Request, res: Response) => {
```

```
319 |     try {
320 |       const puntos = req.body;
321 |       if (!Array.isArray(puntos) || puntos.length === 0) {
322 |         return res.status(400).json({ message: "Array de puntos requerido" });
323 |       }
324 |
325 |       const inserted = [];
326 |       for (const p of puntos) {
327 |         const [punto] = await db.insert(geoPuntos).values({
328 |           camionId: p.camionId || null,
329 |           patente: p.patente,
330 |           lat: String(p.lat),
331 |           lng: String(p.lng),
332 |           timestampPunto: new Date(p.timestamp),
333 |           velocidadKmh: String(p.velocidad_kmh || 0),
334 |           kmOdometro: p.km_odometro ? String(p.km_odometro) : null,
335 |           fuente: p.fuente || "MANUAL",
336 |         }).returning();
337 |         inserted.push(punto);
338 |       }
339 |
340 |       res.json({ inserted: inserted.length });
341 |     } catch (error: any) {
342 |       console.error("[geo] ingest error:", error.message);
343 |       res.status(500).json({ message: error.message });
344 |     }
345 |   });
346 |
347 |   async function ingestAllVolvo(): Promise<{ ingested: number; camiones: number }> {
348 |     const allCamiones = await storage.getCamiones();
349 |     const withVin = allCamiones.filter(c => c.vin);
350 |     let fleet: any[] = [];
351 |     try { fleet = await getFleetStatus(); } catch {}
352 |
353 |     const vinToFleet = new Map(fleet.map(v => [v.vin, v]));
354 |     let ingested = 0;
355 |
356 |     for (const cam of withVin) {
357 |       const volvo = vinToFleet.get(cam.vin!);
358 |       if (!volvo?.gps?.latitude || !volvo?.gps?.longitude) continue;
359 |
360 |       const gpsTime = volvo.gps.positionDateTime || volvo.createdDateTime;
361 |       if (!gpsTime) continue;
362 |
363 |       const existing = await db.select({ id: geoPuntos.id }).from(geoPuntos)
364 |         .where(and(
365 |           eq(geoPuntos.camionId, cam.id),
366 |           eq(geoPuntos.timestampPunto, new Date(gpsTime))
367 |         )).limit(1);
368 |
369 |       if (existing.length > 0) continue;
370 |
371 |       await db.insert(geoPuntos).values({
372 |         camionId: cam.id,
373 |         patente: cam.patente,
374 |         lat: String(volvo.gps.latitude),
375 |         lng: String(volvo.gps.longitude),
376 |         timestampPunto: new Date(gpsTime),
377 |         velocidadKmh: String(volvo.wheelBasedSpeed ?? volvo.gps.speed ?? 0),
378 |         rumboGrados: volvo.gps.heading != null ? String(volvo.gps.heading) : null,
379 |         kmOdometro: volvo.totalDistance != null ? String(Math.round(volvo.totalDistance / 1000)) : null,
380 |         fuente: "VOLVO",
381 |       });
382 |       ingested++;
383 |     }
384 |
385 |     return { ingested, camiones: withVin.length };
386 |   }
387 |
388 |   setInterval(async () => {
389 |     try {
390 |       const result = await ingestAllVolvo();
391 |       if (result.ingested > 0) {
392 |         console.log(`[geo-auto] Ingested ${result.ingested} GPS points from ${result.camiones} trucks`);
393 |       }
394 |     } catch (err: any) {
395 |       console.error("[geo-auto] Ingestion error:", err.message);
396 |     }
397 |   }, 5 * 60 * 1000);
398 |
399 |   setTimeout(async () => {
400 |     try {
401 |       const result = await ingestAllVolvo();
402 |       console.log(`[geo-auto] Initial ingestion: ${result.ingested} points from ${result.camiones} trucks`);
403 |     } catch (err: any) {
404 |       console.error("[geo-auto] Initial ingestion error:", err.message);
405 |     }
406 |   }, 10_000);
407 |
408 |   app.post("/api/geo/ingest-volvo", async (_req: Request, res: Response) => {
409 |     try {
410 |       const result = await ingestAllVolvo();
411 |       res.json(result);
412 |     } catch (error: any) {
413 |       console.error("[geo] ingest-volvo error:", error.message);
414 |       res.status(500).json({ message: error.message });
415 |     }
416 |   });
417 |
418 |   app.post("/api/geo/detectar-viajes", async (_req: Request, res: Response) => {
419 |     try {
420 |       const allCamiones = await storage.getCamiones();
421 |       const withVin = allCamiones.filter(c => c.vin);
422 |       const bases = await db.select().from(geoBases).where(eq(geoBases.activa, true));
423 |       const faenasResult = await pool.query(`SELECT id, nombre FROM faenas`);
424 |       const faenaMap = new Map<number, string>(faenasResult.rows.map((r: any) => [r.id, r.nombre]));
425 |       let detected = 0;
```

```
426 |
427 |       for (const cam of withVin) {
428 |         const puntos = await db.select().from(geoPuntos)
429 |           .where(and(
430 |             eq(geoPuntos.camionId, cam.id),
431 |             gte(geoPuntos.timestampPunto, new Date(Date.now() - 7 * 24 * 60 * 60 * 1000))
432 |           ))
433 |           .orderBy(geoPuntos.timestampPunto);
434 |
435 |         if (puntos.length < 3) continue;
436 |
437 |         let viajeInicio: any = null;
438 |         let maxSpeed = 0;
439 |         let totalKm = 0;
440 |         let prevPoint: any = null;
441 |
442 |         for (let i = 0; i < puntos.length; i++) {
443 |           const p = puntos[i];
444 |           const lat = parseFloat(p.lat as string);
445 |           const lng = parseFloat(p.lng as string);
446 |           const speed = parseFloat(p.velocidadKmh as string) || 0;
447 |
448 |           const inBase = bases.some(b => {
449 |             const dist = haversineKm(lat, lng, parseFloat(b.lat as string), parseFloat(b.lng as string)) * 1000;
450 |             return dist <= (b.radioMetros || 3000);
451 |           });
452 |
453 |           if (!viajeInicio && !inBase && speed > 5) {
454 |             viajeInicio = p;
455 |             maxSpeed = speed;
456 |             totalKm = 0;
457 |             prevPoint = p;
458 |             continue;
459 |           }
460 |
461 |           if (viajeInicio && prevPoint) {
462 |             const d = haversineKm(
463 |               parseFloat(prevPoint.lat as string), parseFloat(prevPoint.lng as string),
464 |               lat, lng
465 |             );
466 |             if (d < 500) totalKm += d;
467 |             if (speed > maxSpeed) maxSpeed = speed;
468 |           }
469 |
470 |           if (viajeInicio && inBase && speed < 5 && totalKm > 10) {
471 |             const origenLat = parseFloat(viajeInicio.lat as string);
472 |             const origenLng = parseFloat(viajeInicio.lng as string);
473 |             const [origenGeo, destinoGeo] = await Promise.all([
474 |               geocodificar(origenLat, origenLng),
475 |               geocodificar(lat, lng),
476 |             ]);
477 |
478 |             const duracion = Math.round(
479 |               (new Date(p.timestampPunto).getTime() - new Date(viajeInicio.timestampPunto).getTime()) / 60000
480 |             );
481 |
482 |             const existingViaje = await db.select({ id: geoViajes.id }).from(geoViajes)
483 |               .where(and(
484 |                 eq(geoViajes.camionId, cam.id),
485 |                 eq(geoViajes.origenTimestamp, new Date(viajeInicio.timestampPunto))
486 |               )).limit(1);
487 |
488 |             if (existingViaje.length === 0) {
489 |               await db.insert(geoViajes).values({
490 |                 camionId: cam.id,
491 |                 patente: cam.patente,
492 |                 contrato: faenaMap.get(cam.faenaId) || "DESCONOCIDO",
493 |                 origenLat: viajeInicio.lat,
494 |                 origenLng: viajeInicio.lng,
495 |                 origenNombre: origenGeo.nombre,
496 |                 origenTimestamp: new Date(viajeInicio.timestampPunto),
497 |                 destinoLat: String(lat),
498 |                 destinoLng: String(lng),
499 |                 destinoNombre: destinoGeo.nombre,
500 |                 destinoTimestamp: new Date(p.timestampPunto),
501 |                 kmGps: String(Math.round(totalKm * 100) / 100),
502 |                 duracionMinutos: duracion,
503 |                 velocidadPromedio: String(Math.round(totalKm / (duracion / 60) * 10) / 10),
504 |                 velocidadMaxima: String(Math.round(maxSpeed * 10) / 10),
505 |                 validacionEstado: "PENDIENTE",
506 |               });
507 |               detected++;
508 |             }
509 |
510 |             viajeInicio = null;
511 |             prevPoint = null;
512 |             totalKm = 0;
513 |             maxSpeed = 0;
514 |             continue;
515 |           }
516 |
517 |           prevPoint = p;
518 |         }
519 |       }
520 |
521 |       res.json({ detected });
522 |     } catch (error: any) {
523 |       console.error("[geo] detectar-viajes error:", error.message);
524 |       res.status(500).json({ message: error.message });
525 |     }
526 |   });
527 |
528 |   app.post("/api/geo/validar-sigetra", async (_req: Request, res: Response) => {
529 |     try {
530 |       const pendientes = await db.select().from(geoViajes)
531 |         .where(eq(geoViajes.validacionEstado, "PENDIENTE"))
532 |         .limit(50);
```

```
533 |
534 |       if (pendientes.length === 0) return res.json({ validated: 0 });
535 |
536 |       const from = new Date("2026-03-01");
537 |       const to = new Date();
538 |       let fuelData: any[] = [];
539 |       try { fuelData = await getCachedFuelData(from, to); } catch {}
540 |
541 |       let validated = 0;
542 |
543 |       for (const viaje of pendientes) {
544 |         const viajeStart = viaje.origenTimestamp ? new Date(viaje.origenTimestamp) : null;
545 |         const viajeEnd = viaje.destinoTimestamp ? new Date(viaje.destinoTimestamp) : null;
546 |         if (!viajeStart) continue;
547 |
548 |         const dayBefore = new Date(viajeStart.getTime() - 24 * 60 * 60 * 1000);
549 |         const dayAfter = viajeEnd ? new Date(viajeEnd.getTime() + 24 * 60 * 60 * 1000) : new Date(viajeStart.getTime() +
550 |
551 |         const matchingCargas = fuelData.filter((c: any) => {
552 |           if (c.patente !== viaje.patente && String(c.numVeh) !== viaje.patente) return false;
553 |           const cargaDate = new Date(c.fechaConsumo);
554 |           return cargaDate >= dayBefore && cargaDate <= dayAfter;
555 |         });
556 |
557 |         if (matchingCargas.length === 0) {
558 |           continue;
559 |         }
560 |
561 |         const carga = matchingCargas[0];
562 |         const checks: any = {};
563 |         let estado = "VALIDADO";
564 |
565 |         const kmGps = parseFloat(viaje.kmGps as string) || 0;
566 |         const kmSigetra = carga.kmRecorrido || carga.kmAjustado || 0;
567 |         if (kmGps > 0 && kmSigetra > 0) {
568 |           const deltaPct = Math.abs(kmGps - kmSigetra) / kmGps * 100;
569 |           checks.kmDeltaPct = Math.round(deltaPct * 10) / 10;
570 |           if (deltaPct > 25) { checks.kmStatus = "ANOMALIA"; estado = "ANOMALIA"; }
571 |           else if (deltaPct > 10) { checks.kmStatus = "REVISAR"; if (estado !== "ANOMALIA") estado = "REVISAR"; }
572 |           else checks.kmStatus = "OK";
573 |         }
574 |
575 |         const cargaDate = new Date(carga.fechaConsumo);
576 |         const timeDiffH = Math.abs(cargaDate.getTime() - viajeStart.getTime()) / (1000 * 60 * 60);
577 |         if (timeDiffH <= 4) checks.timingStatus = "OK";
578 |         else { checks.timingStatus = "REVISAR"; if (estado === "VALIDADO") estado = "REVISAR"; }
579 |
580 |         checks.surtidorStatus = "OK";
581 |
582 |         await db.update(geoViajes).set({
583 |           validacionEstado: estado,
584 |           validacionDetalle: checks,
585 |           sigetraKmDeltaPct: checks.kmDeltaPct != null ? String(checks.kmDeltaPct) : null,
586 |           sigetraLitros: carga.cantidadLt ? String(carga.cantidadLt) : null,
587 |           sigetraSurtidorEnRuta: checks.surtidorStatus === "OK",
588 |           actualizadoAt: new Date(),
589 |         }).where(eq(geoViajes.id, viaje.id));
590 |
591 |         validated++;
592 |       }
593 |
594 |       res.json({ validated, pending: pendientes.length });
595 |     } catch (error: any) {
596 |       console.error("[geo] validar-sigetra error:", error.message);
597 |       res.status(500).json({ message: error.message });
598 |     }
599 |   });
600 |
601 |   app.get("/api/geo/lugares", async (req, res) => {
602 |     try {
603 |       const todos = await db.select().from(geoLugares).where(eq(geoLugares.activo, true)).orderBy(desc(geoLugares.vecesV
604 |       const visitas = await db.select().from(geoVisitas);
605 |       const result = todos.map(l => {
606 |         const visitasLugar = visitas.filter(v => v.lugarId === l.id);
607 |         const camionesUnicos = new Set(visitasLugar.map(v => v.camionId)).size;
608 |         const tiempoPromedio = visitasLugar.length > 0
609 |           ? Math.round(visitasLugar.reduce((s, v) => s + (v.minutosDetenido || 0), 0) / visitasLugar.length)
610 |           : 0;
611 |         const ultimoCamion = visitasLugar.length > 0
612 |           ? visitasLugar.sort((a, b) => new Date(b.llegada || 0).getTime() - new Date(a.llegada || 0).getTime())[0]?.pat
613 |           : null;
614 |         return { ...l, camionesUnicos, tiempoPromedio, ultimoCamion, totalVisitas: visitasLugar.length };
615 |       });
616 |       res.json(result);
617 |     } catch (error: any) {
618 |       res.status(500).json({ message: error.message });
619 |     }
620 |   });
621 |
622 |   app.post("/api/geo/lugares", async (req, res) => {
623 |     try {
624 |       const { nombre, tipo, lat, lng, radioMetros, direccion, comuna, region } = req.body;
625 |       const [inserted] = await db.insert(geoLugares).values({
626 |         nombre,
627 |         tipo: tipo || "OTRO",
628 |         lat: String(lat),
629 |         lng: String(lng),
630 |         radioMetros: radioMetros || 500,
631 |         direccion,
632 |         comuna,
633 |         region,
634 |         detectadoVia: "MANUAL",
635 |         confianzaPct: 100,
636 |         confirmado: true,
637 |         activo: true,
638 |       }).returning();
639 |       res.json(inserted);
```

```
640 |     } catch (error: any) {
641 |       res.status(500).json({ message: error.message });
642 |     }
643 |   });
644 |
645 |   app.put("/api/geo/lugares/:id", async (req, res) => {
646 |     try {
647 |       const id = parseInt(req.params.id);
648 |       const updates: any = {};
649 |       if (req.body.nombreConfirmado !== undefined) updates.nombreConfirmado = req.body.nombreConfirmado;
650 |       if (req.body.nombre !== undefined) updates.nombre = req.body.nombre;
651 |       if (req.body.tipo !== undefined) updates.tipo = req.body.tipo;
652 |       if (req.body.confirmado !== undefined) updates.confirmado = req.body.confirmado;
653 |       if (req.body.radioMetros !== undefined) updates.radioMetros = req.body.radioMetros;
654 |
655 |       const [updated] = await db.update(geoLugares).set(updates).where(eq(geoLugares.id, id)).returning();
656 |       res.json(updated);
657 |     } catch (error: any) {
658 |       res.status(500).json({ message: error.message });
659 |     }
660 |   });
661 |
662 |   app.delete("/api/geo/lugares/:id", async (req, res) => {
663 |     try {
664 |       const id = parseInt(req.params.id);
665 |       await db.update(geoLugares).set({ activo: false }).where(eq(geoLugares.id, id));
666 |       res.json({ ok: true });
667 |     } catch (error: any) {
668 |       res.status(500).json({ message: error.message });
669 |     }
670 |   });
671 |
672 |   app.get("/api/geo/lugares/:id/visitas", async (req, res) => {
673 |     try {
674 |       const id = parseInt(req.params.id);
675 |       const result = await db.select().from(geoVisitas)
676 |         .where(eq(geoVisitas.lugarId, id))
677 |         .orderBy(desc(geoVisitas.llegada));
678 |       res.json(result);
679 |     } catch (error: any) {
680 |       res.status(500).json({ message: error.message });
681 |     }
682 |   });
683 |
684 |   app.get("/api/geo/analisis-ia", async (req, res) => {
685 |     try {
686 |       const [latest] = await db.select().from(geoAnalisisIa)
687 |         .orderBy(desc(geoAnalisisIa.generadoAt))
688 |         .limit(1);
689 |       res.json(latest || null);
690 |     } catch (error: any) {
691 |       res.status(500).json({ message: error.message });
692 |     }
693 |   });
694 |
695 |   app.post("/api/geo/analisis-ia/generar", async (req, res) => {
696 |     try {
697 |       const result = await generarAnalisisIA();
698 |       res.json(result);
699 |     } catch (error: any) {
700 |       console.error("[geo] analisis-ia error:", error.message);
701 |       res.status(500).json({ message: error.message });
702 |     }
703 |   });
704 |
705 |   app.post("/api/geo/analizar-historico", async (req, res) => {
706 |     try {
707 |       res.setHeader("Content-Type", "text/event-stream");
708 |       res.setHeader("Cache-Control", "no-cache");
709 |       res.setHeader("Connection", "keep-alive");
710 |
711 |       const result = await analizarHistoricoCompleto((msg) => {
712 |         res.write(`data: ${JSON.stringify(msg)}\n\n`);
713 |       });
714 |
715 |       res.write(`data: ${JSON.stringify({ paso: "DONE", ...result })}\n\n`);
716 |       res.end();
717 |     } catch (error: any) {
718 |       console.error("[geo] analizar-historico error:", error.message);
719 |       if (!res.headersSent) {
720 |         res.status(500).json({ message: error.message });
721 |       } else {
722 |         res.write(`data: ${JSON.stringify({ paso: "ERROR", error: error.message })}\n\n`);
723 |         res.end();
724 |       }
725 |     }
726 |   });
727 |
728 |   app.get("/api/geo/lugares/stats", async (req, res) => {
729 |     try {
730 |       const todos = await db.select().from(geoLugares).where(eq(geoLugares.activo, true));
731 |       const stats = {
732 |         total: todos.length,
733 |         jumbos: todos.filter(l => l.tipo === "LOCAL_JUMBO").length,
734 |         santaIsabel: todos.filter(l => l.tipo === "LOCAL_SANTA_ISABEL").length,
735 |         cds: todos.filter(l => l.tipo === "CD_CENCOSUD" || l.tipo === "BASE_ORIGEN").length,
736 |         easy: todos.filter(l => l.tipo === "LOCAL_EASY").length,
737 |         paris: todos.filter(l => l.tipo === "LOCAL_PARIS").length,
738 |         localesCencosud: todos.filter(l => l.tipo === "LOCAL_CENCOSUD").length,
739 |         sinIdentificar: todos.filter(l => l.tipo === "PUNTO_FRECUENTE").length,
740 |         pendienteConfirmar: todos.filter(l => !l.confirmado).length,
741 |         confirmados: todos.filter(l => l.confirmado === true).length,
742 |       };
743 |       res.json(stats);
744 |     } catch (error: any) {
745 |       res.status(500).json({ message: error.message });
746 |     }
```

```
747 |   });
748 |
749 |   app.get("/api/geo/camion/:patente/historial", async (req: Request, res: Response) => {
750 |     try {
751 |       const { patente } = req.params;
752 |       const desde = new Date(req.query.desde as string || "2026-03-01");
753 |       const hasta = req.query.hasta ? new Date(req.query.hasta as string) : new Date();
754 |
755 |       const resultado = await obtenerHistorialCamion(patente, desde, hasta);
756 |       if (!resultado) {
757 |         return res.status(404).json({ message: `Camion ${patente} no encontrado` });
758 |       }
759 |       res.json(resultado);
760 |     } catch (error: any) {
761 |       console.error("[geo] historial error:", error.message);
762 |       res.status(500).json({ message: error.message });
763 |     }
764 |   });
765 |
766 |   app.post("/api/geo/procesar-historico-flota", async (req: Request, res: Response) => {
767 |     try {
768 |       const { desde, patentes } = req.body;
769 |       const fechaDesde = new Date(desde || "2026-03-01");
770 |       const patentesList = patentes || [];
771 |
772 |       res.writeHead(200, {
773 |         "Content-Type": "text/event-stream",
774 |         "Cache-Control": "no-cache",
775 |         "Connection": "keep-alive",
776 |       });
777 |
778 |       await procesarFlotaHistorico(fechaDesde, patentesList, (data) => {
779 |         res.write(`data: ${JSON.stringify(data)}\n\n`);
780 |       });
781 |
782 |       res.end();
783 |     } catch (error: any) {
784 |       console.error("[geo] procesar-historico-flota error:", error.message);
785 |       if (!res.headersSent) {
786 |         res.status(500).json({ message: error.message });
787 |       } else {
788 |         res.write(`data: ${JSON.stringify({ estado: "error", mensaje: error.message })}\n\n`);
789 |         res.end();
790 |       }
791 |     }
792 |   });
793 |
794 |   app.get("/api/geo/resumen-flota", async (req: Request, res: Response) => {
795 |     try {
796 |       const desde = new Date(req.query.desde as string || "2026-03-01");
797 |       const resumen = await obtenerResumenFlota(desde);
798 |       res.json(resumen);
799 |     } catch (error: any) {
800 |       console.error("[geo] resumen-flota error:", error.message);
801 |       res.status(500).json({ message: error.message });
802 |     }
803 |   });
804 |
805 |   app.get("/api/geo/camion/:camionId/puntos", async (req: Request, res: Response) => {
806 |     try {
807 |       const camionId = parseInt(req.params.camionId);
808 |       const desde = new Date(req.query.desde as string || "2026-03-01");
809 |       const hasta = req.query.hasta ? new Date(req.query.hasta as string) : new Date();
810 |
811 |       const puntos = await db.select({
812 |         lat: geoPuntos.lat,
813 |         lng: geoPuntos.lng,
814 |         ts: geoPuntos.timestampPunto,
815 |         velocidad: geoPuntos.velocidadKmh,
816 |       }).from(geoPuntos)
817 |         .where(and(
818 |           eq(geoPuntos.camionId, camionId),
819 |           gte(geoPuntos.timestampPunto, desde),
820 |           lte(geoPuntos.timestampPunto, hasta)
821 |         ))
822 |         .orderBy(asc(geoPuntos.timestampPunto))
823 |         .limit(5000);
824 |
825 |       res.json(puntos);
826 |     } catch (error: any) {
827 |       res.status(500).json({ message: error.message });
828 |     }
829 |   });
830 |
831 |   inicializarPerfilGPS().catch(err => console.error("[geo-routes] Error init perfil GPS:", err));
832 |
833 |   app.post("/api/geo/detectar-visitas", async (req: Request, res: Response) => {
834 |     try {
835 |       const { desde = "2026-03-01", hasta, patentes = [] } = req.body;
836 |       const hastaFinal = hasta || new Date().toISOString().split("T")[0];
837 |       const resultado = await detectarVisitasFlota(desde, hastaFinal, patentes);
838 |       res.json(resultado);
839 |     } catch (error: any) {
840 |       console.error("[visitas] Error detectando visitas:", error);
841 |       res.status(500).json({ message: error.message });
842 |     }
843 |   });
844 |
845 |   app.get("/api/geo/resumen-visitas", async (_req: Request, res: Response) => {
846 |     try {
847 |       const desde = _req.query.desde as string || "2026-03-01";
848 |       const hasta = _req.query.hasta as string || new Date().toISOString().split("T")[0];
849 |       const resumen = await obtenerResumenVisitas(desde, hasta);
850 |       res.json(resumen);
851 |     } catch (error: any) {
852 |       console.error("[visitas] Error obteniendo resumen:", error);
853 |       res.status(500).json({ message: error.message });
```

```
854 |     }
855 |   });
856 |
857 |   app.get("/api/geo/camion/:patente/visitas", async (req: Request, res: Response) => {
858 |     try {
859 |       const { patente } = req.params;
860 |       const visitas = await obtenerVisitasCamion(patente);
861 |       res.json(visitas);
862 |     } catch (error: any) {
863 |       console.error("[visitas] Error obteniendo visitas camion:", error);
864 |       res.status(500).json({ message: error.message });
865 |     }
866 |   });
867 |
868 |   const ESTACIONES_COMBUSTIBLE: Record<string, { lat: number; lng: number; ciudad: string }> = {
869 |     "LOS ANGELES 3": { lat: -37.4695, lng: -72.3538, ciudad: "Los Angeles" },
870 |     "QUILICURA": { lat: -33.3583, lng: -70.7250, ciudad: "Santiago" },
871 |     "PANAMERICANA NORTE, CRUCE CAMINO HU": { lat: -33.32, lng: -70.73, ciudad: "Huechuraba" },
872 |     "RUTA 5 SUR KM 906": { lat: -40.57, lng: -73.13, ciudad: "Osorno" },
873 |     "CARRETERA PANAMERICANA NORTE KM 455": { lat: -30.40, lng: -70.98, ciudad: "La Serena" },
874 |     "CARR. PANAMERICANA NORTE KM.811": { lat: -40.57, lng: -73.13, ciudad: "Puerto Montt" },
875 |     "CARRETERA LONGITUDINAL SUR KM 275,3": { lat: -35.84, lng: -71.69, ciudad: "Linares" },
876 |   };
877 |
878 |   app.get("/api/geo/cargas-combustible", async (req: Request, res: Response) => {
879 |     try {
880 |       const desde = req.query.desde as string || "2026-03-01";
881 |       const hasta = req.query.hasta as string || new Date().toISOString().split("T")[0];
882 |       const patenteFilter = req.query.patente as string || "";
883 |       const { getCachedFuelData } = await import("./sigetra-api");
884 |       const { getCencosudPatentes } = await import("./cencosud-filter");
885 |       const patentes = await getCencosudPatentes();
886 |       const allFuel = await getCachedFuelData(new Date(desde), new Date(hasta + "T23:59:59"));
887 |       const cencFuel = allFuel.filter((r: any) => patentes.has(String(r.numVeh || "")));
888 |       const filtered = patenteFilter
889 |         ? cencFuel.filter((r: any) => String(r.numVeh) === patenteFilter)
890 |         : cencFuel;
891 |
892 |       const porEstacion: Record<string, { cargas: number; litros: number; patentes: Set<string>; ultimaCarga: string; la
893 |       for (const r of filtered) {
894 |         const lugar = r.lugarConsumo || "DESCONOCIDO";
895 |         if (!porEstacion[lugar]) {
896 |           const est = ESTACIONES_COMBUSTIBLE[lugar];
897 |           porEstacion[lugar] = { cargas: 0, litros: 0, patentes: new Set(), ultimaCarga: "", lat: est?.lat || null, lng:
898 |         }
899 |         porEstacion[lugar].cargas++;
900 |         porEstacion[lugar].litros += r.cantidadLt || 0;
901 |         porEstacion[lugar].patentes.add(String(r.numVeh));
902 |         const fecha = r.fechaConsumo || "";
903 |         if (fecha > porEstacion[lugar].ultimaCarga) porEstacion[lugar].ultimaCarga = fecha;
904 |       }
905 |
906 |       const estaciones = Object.entries(porEstacion).map(([nombre, d]) => ({
907 |         nombre,
908 |         cargas: d.cargas,
909 |         litros: Math.round(d.litros),
910 |         camiones: d.patentes.size,
911 |         ultimaCarga: d.ultimaCarga,
912 |         lat: d.lat,
913 |         lng: d.lng,
914 |         ciudad: d.ciudad,
915 |       })).sort((a, b) => b.cargas - a.cargas);
916 |
917 |       const cargas = filtered.map((r: any) => ({
918 |         numVeh: r.numVeh,
919 |         fecha: r.fechaConsumo,
920 |         litros: r.cantidadLt,
921 |         lugar: r.lugarConsumo,
922 |         odometro: r.odometroActual,
923 |         kmRecorrido: r.kmRecorrido,
924 |         rendimiento: r.rendReal,
925 |         conductor: r.nombreConductor,
926 |         lat: ESTACIONES_COMBUSTIBLE[r.lugarConsumo]?.lat || null,
927 |         lng: ESTACIONES_COMBUSTIBLE[r.lugarConsumo]?.lng || null,
928 |       }));
929 |
930 |       res.json({
931 |         totalCargas: filtered.length,
932 |         totalLitros: Math.round(filtered.reduce((s: number, r: any) => s + (r.cantidadLt || 0), 0)),
933 |         estaciones,
934 |         cargas,
935 |       });
936 |     } catch (error: any) {
937 |       console.error("[geo] Error cargas combustible:", error);
938 |       res.status(500).json({ message: error.message });
939 |     }
940 |   });
941 |
942 |   app.get("/api/geo/cruce-mensual", async (req: Request, res: Response) => {
943 |     try {
944 |       const desde = req.query.desde as string || "2026-03-01";
945 |       const hasta = req.query.hasta as string || new Date().toISOString().split("T")[0];
946 |       const contrato = (req.query.contrato as string || "CENCOSUD").toUpperCase();
947 |       const subfaena = req.query.subfaena as string || "";
948 |       const { getCachedFuelData } = await import("./sigetra-api");
949 |
950 |       let patentes: Set<string>;
951 |       let faenaIds: number[];
952 |       if (contrato === "CENCOSUD") {
953 |         const { getCencosudPatentes, getCencosudFaenaId } = await import("./cencosud-filter");
954 |         patentes = await getCencosudPatentes();
955 |         faenaIds = [await getCencosudFaenaId()];
956 |       } else {
957 |         const config = await getContractConfig(contrato);
958 |         if (subfaena) {
959 |           const subId = config.faenaIds.find((_, i) => config.faenaNames[i] === subfaena);
960 |           if (subId) {
```

```
961 |             const { getFaenaPatentes } = await import("./faena-filter");
962 |             patentes = await getFaenaPatentes(subId);
963 |             faenaIds = [subId];
964 |           } else {
965 |             patentes = await getContractPatentes(contrato);
966 |             faenaIds = config.faenaIds;
967 |           }
968 |         } else {
969 |           patentes = await getContractPatentes(contrato);
970 |           faenaIds = config.faenaIds;
971 |         }
972 |       }
973 |
974 |       const allFuel = await getCachedFuelData(new Date(desde), new Date(hasta + "T23:59:59"));
975 |       const filteredFuel = allFuel.filter((r: any) => patentes.has(String(r.numVeh || "")));
976 |
977 |       const sigetraPorCamion: Record<string, { litros: number; cargas: number; conductores: Set<string>; odoMin: number;
978 |       for (const r of filteredFuel) {
979 |         const p = String(r.numVeh);
980 |         if (!sigetraPorCamion[p]) sigetraPorCamion[p] = { litros: 0, cargas: 0, conductores: new Set(), odoMin: Infinity
981 |         sigetraPorCamion[p].litros += r.cantidadLt || 0;
982 |         sigetraPorCamion[p].cargas++;
983 |         if (r.nombreConductor) sigetraPorCamion[p].conductores.add(r.nombreConductor);
984 |         const odo = r.odometroActual || 0;
985 |         if (odo > 0) {
986 |           if (odo < sigetraPorCamion[p].odoMin) sigetraPorCamion[p].odoMin = odo;
987 |           if (odo > sigetraPorCamion[p].odoMax) sigetraPorCamion[p].odoMax = odo;
988 |         }
989 |       }
990 |
991 |       const faenaPlaceholders = faenaIds.map((_, i) => `$${i + 1}`).join(",");
992 |       const ecuResult = await pool.query(`
993 |         SELECT c.patente,
994 |           SUM(COALESCE(v.litros_ecu, 0)) as total_ecu,
995 |           SUM(COALESCE(v.km_recorridos, 0)) as total_km,
996 |           COUNT(*) as total_viajes
997 |         FROM tms_viajes v
998 |         JOIN camiones c ON v.camion_id = c.id
999 |         WHERE c.faena_id IN (${faenaPlaceholders}) AND v.fecha_salida >= $${faenaIds.length + 1} AND v.fecha_salida <= (
1000 |         GROUP BY c.patente
1001 |       `, [...faenaIds, desde, hasta]);
1002 |
1003 |       const ecuPorCamion: Record<string, { litrosEcu: number; km: number; viajes: number }> = {};
1004 |       for (const row of ecuResult.rows) {
1005 |         ecuPorCamion[row.patente] = {
1006 |           litrosEcu: parseFloat(row.total_ecu) || 0,
1007 |           km: parseFloat(row.total_km) || 0,
1008 |           viajes: parseInt(row.total_viajes) || 0,
1009 |         };
1010 |       }
1011 |
1012 |       let allPatentes = new Set([...Object.keys(sigetraPorCamion), ...Object.keys(ecuPorCamion)]);
1013 |       const volvoPatentes = new Set(Object.keys(ecuPorCamion));
1014 |       allPatentes = new Set([...allPatentes].filter(p => volvoPatentes.has(p)));
1015 |       const cruce = [...allPatentes].sort((a, b) => parseInt(a) - parseInt(b)).map(p => {
1016 |         const sig = sigetraPorCamion[p] || { litros: 0, cargas: 0, conductores: new Set(), odoMin: 0, odoMax: 0 };
1017 |         const ecu = ecuPorCamion[p] || { litrosEcu: 0, km: 0, viajes: 0 };
1018 |         const diff = sig.litros - ecu.litrosEcu;
1019 |         const diffPct = ecu.litrosEcu > 0 ? (diff / ecu.litrosEcu) * 100 : 0;
1020 |         const kmOdometro = sig.odoMax > sig.odoMin && sig.odoMin !== Infinity ? sig.odoMax - sig.odoMin : 0;
1021 |         const rendSigetra = kmOdometro > 0 && sig.litros > 0 ? kmOdometro / sig.litros : 0;
1022 |         const rendEcu = ecu.km > 0 && ecu.litrosEcu > 0 ? ecu.km / ecu.litrosEcu : 0;
1023 |         return {
1024 |           patente: p,
1025 |           conductores: [...sig.conductores],
1026 |           litrosSigetra: Math.round(sig.litros * 100) / 100,
1027 |           cargasSigetra: sig.cargas,
1028 |           litrosEcu: Math.round(ecu.litrosEcu * 100) / 100,
1029 |           kmEcu: Math.round(ecu.km * 100) / 100,
1030 |           kmOdometro,
1031 |           odoInicio: sig.odoMin !== Infinity ? sig.odoMin : 0,
1032 |           odoFin: sig.odoMax,
1033 |           viajesEcu: ecu.viajes,
1034 |           diferencia: Math.round(diff * 100) / 100,
1035 |           diferenciaPct: Math.round(diffPct * 10) / 10,
1036 |           rendimientoSigetra: Math.round(rendSigetra * 100) / 100,
1037 |           rendimientoEcu: Math.round(rendEcu * 100) / 100,
1038 |         };
1039 |       });
1040 |
1041 |       const totalSigetra = cruce.reduce((s, c) => s + c.litrosSigetra, 0);
1042 |       const totalEcu = cruce.reduce((s, c) => s + c.litrosEcu, 0);
1043 |
1044 |       res.json({
1045 |         desde,
1046 |         hasta,
1047 |         totalCamiones: cruce.length,
1048 |         totalSigetra: Math.round(totalSigetra),
1049 |         totalEcu: Math.round(totalEcu),
1050 |         diferencia: Math.round(totalSigetra - totalEcu),
1051 |         camiones: cruce,
1052 |       });
1053 |     } catch (error: any) {
1054 |       console.error("[geo] Error cruce mensual:", error);
1055 |       res.status(500).json({ message: error.message });
1056 |     }
1057 |   });
1058 |
1059 |   app.get("/api/geo/conductores", async (req: Request, res: Response) => {
1060 |     try {
1061 |       const desde = req.query.desde as string || "2026-03-01";
1062 |       const hasta = req.query.hasta as string || new Date().toISOString().split("T")[0];
1063 |       const contrato = (req.query.contrato as string || "CENCOSUD").toUpperCase();
1064 |       const subfaena = req.query.subfaena as string || "";
1065 |       const { getCachedFuelData } = await import("./sigetra-api");
1066 |
1067 |       let patentes: Set<string>;
```

```
      if (contrato === "CENCOSUD") {
        const { getCencosudPatentes } = await import("./cencosud-filter");
        patentes = await getCencosudPatentes();
      } else if (subfaena) {
        const config = await getContractConfig(contrato);
        const subId = config.faenaIds.find((_, i) => config.faenaNames[i] === subfaena);
        if (subId) {
          const { getFaenaPatentes } = await import("./faena-filter");
          patentes = await getFaenaPatentes(subId);
        } else {
          patentes = await getContractPatentes(contrato);
        }
      } else {
        patentes = await getContractPatentes(contrato);
      }

      const allFuel = await getCachedFuelData(new Date(desde), new Date(hasta + "T23:59:59"));
      let filteredFuel = allFuel.filter((r: any) => patentes.has(String(r.numVeh || "")));

      {
        let volvoFaenaIds: number[];
        if (contrato === "CENCOSUD") {
          const { getCencosudFaenaId } = await import("./cencosud-filter");
          volvoFaenaIds = [await getCencosudFaenaId()];
        } else if (subfaena) {
          const config = await getContractConfig(contrato);
          const subId = config.faenaIds.find((_, i) => config.faenaNames[i] === subfaena);
          volvoFaenaIds = subId ? [subId] : config.faenaIds;
        } else {
          const config = await getContractConfig(contrato);
          volvoFaenaIds = config.faenaIds;
        }
        const fp = volvoFaenaIds.map((_, i) => `$${i + 1}`).join(",");
        const ecuCheck = await pool.query(`
          SELECT DISTINCT c.patente FROM tms_viajes v
          JOIN camiones c ON v.camion_id = c.id
          WHERE c.faena_id IN (${fp}) AND v.fecha_salida >= $${volvoFaenaIds.length + 1}
        `, [...volvoFaenaIds, desde]);
        const volvoPatentes = new Set(ecuCheck.rows.map((r: any) => r.patente));
        filteredFuel = filteredFuel.filter((r: any) => volvoPatentes.has(String(r.numVeh || "")));
      }

      const byConductor: Record<string, {
        litros: number; cargas: number; kmTotal: number; litrosValidos: number;
        patentes: Set<string>; primerCarga: string; ultimaCarga: string;
        cargasDetalle: { fecha: string; litros: number; km: number; rend: number; patente: string; lugar: string }[];
      }> = {};

      for (const r of filteredFuel) {
        const cond = r.nombreConductor || "SIN CONDUCTOR";
        if (!byConductor[cond]) byConductor[cond] = {
          litros: 0, cargas: 0, kmTotal: 0, litrosValidos: 0,
          patentes: new Set(), primerCarga: "", ultimaCarga: "",
          cargasDetalle: [],
        };
        const d = byConductor[cond];
        d.litros += r.cantidadLt || 0;
        d.cargas++;
        d.patentes.add(String(r.numVeh));
        const fecha = r.fechaConsumo || "";
        if (!d.primerCarga || fecha < d.primerCarga) d.primerCarga = fecha;
        if (fecha > d.ultimaCarga) d.ultimaCarga = fecha;
        const km = r.kmRecorrido || 0;
        const rend = r.rendReal || 0;
        if (km > 0 && rend > 0 && rend < 100) {
          d.kmTotal += km;
          d.litrosValidos += r.cantidadLt || 0;
        }
        d.cargasDetalle.push({
          fecha,
          litros: r.cantidadLt || 0,
          km,
          rend,
          patente: String(r.numVeh),
          lugar: r.lugarConsumo || "",
        });
      }

      const conductores = Object.entries(byConductor)
        .map(([nombre, d]) => ({
          nombre,
          litrosTotales: Math.round(d.litros * 100) / 100,
          cargas: d.cargas,
          kmTotales: d.kmTotal,
          rendimiento: d.kmTotal > 0 && d.litrosValidos > 0 ? Math.round((d.kmTotal / d.litrosValidos) * 100) / 100 : 0,
          camiones: [...d.patentes].sort((a, b) => parseInt(a) - parseInt(b)),
          primerCarga: d.primerCarga,
          ultimaCarga: d.ultimaCarga,
          cargasDetalle: d.cargasDetalle.sort((a, b) => b.fecha.localeCompare(a.fecha)),
        }))
        .sort((a, b) => b.litrosTotales - a.litrosTotales);

      res.json({
        desde,
        hasta,
        totalConductores: conductores.length,
        conductores,
      });
    } catch (error: any) {
      console.error("[geo] Error conductores:", error);
      res.status(500).json({ message: error.message });
    }
  });

  app.get("/api/geo/viajes-tms", async (req: Request, res: Response) => {
    try {
      const patenteFilter = req.query.patente as string || "";
```

```
1175 |       const desde = req.query.desde as string || "2026-03-01";
1176 |       const { getCencosudFaenaId } = await import("./cencosud-filter");
1177 |       const faenaId = await getCencosudFaenaId();
1178 |
1179 |       let query = `
1180 |         SELECT v.id, v.codigo, c.patente, v.conductor, v.estado,
1181 |           v.origen_nombre, v.destino_nombre,
1182 |           v.fecha_salida, v.fecha_llegada, v.fecha_cierre,
1183 |           v.km_inicio, v.km_cierre, v.km_recorridos,
1184 |           v.litros_sigetra, v.litros_ecu, v.diferencia_litros,
1185 |           v.rendimiento_real, v.detectado_por_ia, v.notas
1186 |         FROM tms_viajes v
1187 |         JOIN camiones c ON v.camion_id = c.id
1188 |         WHERE c.faena_id = $1 AND v.fecha_salida >= $2
1189 |       `;
1190 |       const params: any[] = [faenaId, desde];
1191 |
1192 |       if (patenteFilter) {
1193 |         query += ` AND c.patente = $3`;
1194 |         params.push(patenteFilter);
1195 |       }
1196 |       query += ` ORDER BY v.fecha_salida DESC`;
1197 |
1198 |       const result = await pool.query(query, params);
1199 |       const viajes = result.rows.map(v => ({
1200 |         id: v.id,
1201 |         codigo: v.codigo,
1202 |         patente: v.patente,
1203 |         conductor: v.conductor || null,
1204 |         estado: v.estado,
1205 |         origenNombre: v.origen_nombre || null,
1206 |         destinoNombre: v.destino_nombre || null,
1207 |         fechaSalida: v.fecha_salida,
1208 |         fechaLlegada: v.fecha_llegada || null,
1209 |         fechaCierre: v.fecha_cierre || null,
1210 |         kmInicio: parseFloat(v.km_inicio) || 0,
1211 |         kmCierre: parseFloat(v.km_cierre) || 0,
1212 |         kmRecorridos: parseFloat(v.km_recorridos) || 0,
1213 |         litrosSigetra: parseFloat(v.litros_sigetra) || null,
1214 |         litrosEcu: parseFloat(v.litros_ecu) || null,
1215 |         diferenciaLitros: parseFloat(v.diferencia_litros) || null,
1216 |         rendimiento: parseFloat(v.rendimiento_real) || null,
1217 |         detectadoPorIa: v.detectado_por_ia,
1218 |         notas: v.notas,
1219 |       }));
1220 |
1221 |       const patentes = [...new Set(viajes.map(v => v.patente))].sort((a, b) => parseInt(a) - parseInt(b));
1222 |
1223 |       res.json({ total: viajes.length, patentes, viajes });
1224 |     } catch (error: any) {
1225 |       console.error("[geo] Error viajes TMS:", error);
1226 |       res.status(500).json({ message: error.message });
1227 |     }
1228 |   });
1229 |
1230 | }
1231 |
```

server/ceo-routes.ts

529 lineas

```
1 | import type { Express, Request, Response } from "express";
2 | import Anthropic from "@anthropic-ai/sdk";
3 | import { storage } from "./storage";
4 | import { getCachedFuelData } from "./sigetra-api";
5 | import { getFleetStatus } from "./volvo-api";
6 | import { getCencosudFaenaId, getCencosudPatentes, filterCamiones, filterFuelByPatentes } from "./cencosud-filter";
7 |
8 | const MODEL = "claude-sonnet-4-20250514";
9 | const DATA_START = new Date("2026-03-01T00:00:00");
10 | const CEO_CACHE_TTL = 30 * 60 * 1000;
11 |
12 | let ceoDiagCache: { texto: string; timestamp: number } | null = null;
13 |
14 | function getClient(): Anthropic {
15 |   return new Anthropic();
16 | }
17 |
18 | function parsePeriodo(p: string): number {
19 |   if (p === "1d" || p === "hoy") return 1;
20 |   if (p === "7d") return 7;
21 |   if (p === "30d") return 30;
22 |   return 7;
23 | }
24 |
25 | async function getCoreData(dias: number) {
26 |   const now = new Date();
27 |   let from = new Date(Date.now() - dias * 24 * 60 * 60 * 1000);
28 |   if (from < DATA_START) from = new Date(DATA_START);
29 |   const prevFrom = new Date(from.getTime() - dias * 24 * 60 * 60 * 1000);
30 |   const prevTo = from;
31 |   let prevFuelFrom = prevFrom < DATA_START ? new Date(DATA_START) : prevFrom;
32 |
33 |   const [allCamiones, faenas, allFuelData, allFuelDataPrev, fleet, params, cencosudFaenaId, cencosudPatentes] = await Pr
34 |     storage.getCamiones(),
35 |     storage.getFaenas(),
36 |     getCachedFuelData(from, now).catch(() => []),
37 |     getCachedFuelData(prevFuelFrom, prevTo).catch(() => []),
38 |     getFleetStatus().catch(() => []),
39 |     storage.getParametros(),
40 |     getCencosudFaenaId(),
41 |     getCencosudPatentes(),
42 |   ]);
43 |
44 |   const camiones = filterCamiones(allCamiones, cencosudFaenaId);
45 |   const fuelData = filterFuelByPatentes(allFuelData, cencosudPatentes);
46 |   const fuelDataPrev = filterFuelByPatentes(allFuelDataPrev, cencosudPatentes);
47 |
48 |   const cencosudVins = new Set(camiones.map(c => c.vin).filter((v): v is string => v != null));
49 |   const vinsOnline = new Set(fleet.filter((v: any) => cencosudVins.has(v.vin)).map((v: any) => v.vin));
50 |   const camionesActivos = camiones.filter(c => c.vin && vinsOnline.has(c.vin)).length;
51 |
52 |   const litrosPeriodo = fuelData.reduce((s: number, r: any) => s + (r.cantidadLt || 0), 0);
53 |   const litrosPrev = fuelDataPrev.reduce((s: number, r: any) => s + (r.cantidadLt || 0), 0);
54 |   const litrosCambioPct = litrosPrev > 0 ? +((litrosPeriodo - litrosPrev) / litrosPrev * 100).toFixed(1) : 0;
55 |
56 |   function matchPatente(record: any): string | null {
57 |     const numVeh = record.numVeh != null ? String(record.numVeh) : null;
58 |     const pat = record.patente || "";
59 |     for (const cam of camiones) {
60 |       if (numVeh && cam.patente === numVeh) return cam.patente;
61 |       if (pat && cam.patente === pat) return cam.patente;
62 |     }
63 |     return numVeh || pat || null;
64 |   }
65 |
66 |   const byCamion = new Map<string, { litros: number; rendVals: number[]; desvMax: number }>();
67 |   for (const r of fuelData as any[]) {
68 |     const p = matchPatente(r);
69 |     if (!p) continue;
70 |     const t = byCamion.get(p) || { litros: 0, rendVals: [], desvMax: 0 };
71 |     t.litros += r.cantidadLt || 0;
72 |     if (r.rendReal && r.rendReal > 0 && r.rendReal < 20) t.rendVals.push(r.rendReal);
73 |     const dev = Math.abs(r.desviacion || 0);
74 |     if (dev > t.desvMax) t.desvMax = dev;
75 |     byCamion.set(p, t);
76 |   }
77 |
78 |   const stats = camiones.map(cam => {
79 |     const d = byCamion.get(cam.patente) || { litros: 0, rendVals: [], desvMax: 0 };
80 |     const rend = d.rendVals.length > 0 ? +(d.rendVals.reduce((a, b) => a + b, 0) / d.rendVals.length).toFixed(2) : 0;
81 |     const estado = d.desvMax > 40 ? "CRITICO" : d.desvMax > 15 ? "ALERTA" : "NORMAL";
82 |     return { camionId: cam.id, faenaId: cam.faenaId, patente: cam.patente, modelo: cam.modelo, conductor: cam.conductor,
83 |   });
84 |
85 |   const criticos = stats.filter(s => s.estado === "CRITICO").length;
86 |   const rendValues = stats.map(s => s.rend).filter(r => r > 0);
87 |   const rendPromedio = rendValues.length > 0 ? +(rendValues.reduce((a, b) => a + b, 0) / rendValues.length).toFixed(2) :
88 |
89 |   const rendPrevByPatente = new Map<string, number[]>();
90 |   for (const r of fuelDataPrev as any[]) {
91 |     const p = String(r.patente || "");
92 |     if (r.rendReal && r.rendReal > 0 && r.rendReal < 20) {
93 |       const arr = rendPrevByPatente.get(p) || [];
94 |       arr.push(r.rendReal);
95 |       rendPrevByPatente.set(p, arr);
96 |     }
97 |   }
98 |   const rendPrevAvgs = [...rendPrevByPatente.values()].map(v => v.reduce((a, b) => a + b, 0) / v.length);
99 |   const rendPrev = rendPrevAvgs.length > 0 ? rendPrevAvgs.reduce((a, b) => a + b, 0) / rendPrevAvgs.length : 0;
100 |   const rendCambioPct = rendPrev > 0 ? +((rendPromedio - rendPrev) / rendPrev * 100).toFixed(1) : 0;
101 |
102 |   const limiteKmh = parseFloat(params.find(p => p.clave === "vel_max_kmh")?.valor || "90");
103 |   let excesosVel = 0;
104 |   const cencosudVinsForSpeed = new Set(camiones.map(c => c.vin).filter(Boolean));
```

```
105 |   for (const v of fleet) {
106 |     if (!cencosudVinsForSpeed.has((v as any).vin)) continue;
107 |     const speed = (v as any).wheelBasedSpeed ?? (v as any).gps?.speed ?? 0;
108 |     if (speed > limiteKmh) excesosVel++;
109 |   }
110 |
111 |   const litrosMicro = parseFloat(params.find(p => p.clave === "litros_micro_carga")?.valor || "100");
112 |   const cargasPequenas = fuelData.filter((r: any) => (r.cantidadLt || 0) <= litrosMicro && (r.cantidadLt || 0) > 0);
113 |   const microSospechas = cargasPequenas.length;
114 |
115 |   return {
116 |     camiones, faenas, fleet, stats, fuelData, params,
117 |     totals: {
118 |       camionesTotal: camiones.length,
119 |       camionesActivos,
120 |       rendPromedio,
121 |       rendCambioPct,
122 |       litrosPeriodo: Math.round(litrosPeriodo),
123 |       litrosCambioPct,
124 |       alertasCriticas: criticos,
125 |       excesosVel,
126 |       microSospechas,
127 |     }
128 |   };
129 | }
130 |
131 | export function registerCEORoutes(app: Express) {
132 |
133 |   app.get("/api/ceo/kpis", async (req: Request, res: Response) => {
134 |     try {
135 |       const dias = parsePeriodo((req.query.periodo as string) || "7d");
136 |       const { totals } = await getCoreData(dias);
137 |       res.json(totals);
138 |     } catch (error: any) {
139 |       console.error("[ceo/kpis] Error:", error.message);
140 |       res.status(500).json({ message: error.message });
141 |     }
142 |   });
143 |
144 |   app.get("/api/ceo/contratos-estado", async (_req: Request, res: Response) => {
145 |     try {
146 |       const { faenas, stats, camiones, fleet } = await getCoreData(7);
147 |       const vinsOnline = new Set(fleet.map((v: any) => v.vin));
148 |
149 |       const allRend = stats.map(s => s.rend).filter(r => r > 0).sort((a, b) => a - b);
150 |       const p50 = allRend.length > 3 ? allRend[Math.floor(allRend.length * 0.5)] : 0;
151 |
152 |       const result = faenas
153 |         .map(f => {
154 |           const fStats = stats.filter(s => s.faenaId === f.id);
155 |           if (fStats.length === 0) return null;
156 |           const alertas = fStats.filter(s => s.estado === "CRITICO").length;
157 |           const rendVals = fStats.map(s => s.rend).filter(r => r > 0);
158 |           const rendProm = rendVals.length > 0 ? +(rendVals.reduce((a, b) => a + b, 0) / rendVals.length).toFixed(2) : 0
159 |
160 |           const fCamiones = camiones.filter(c => c.faenaId === f.id);
161 |           const camActivos = fCamiones.filter(c => c.vin && vinsOnline.has(c.vin)).length;
162 |
163 |           const estado = alertas >= 3 ? "CRITICO"
164 |             : (alertas >= 1 || (p50 > 0 && rendProm > 0 && rendProm < p50)) ? "ATENCION"
165 |             : "NORMAL";
166 |
167 |           return {
168 |             nombre: f.nombre,
169 |             color: f.color,
170 |             camionesTotal: fStats.length,
171 |             camionesActivos: camActivos,
172 |             rendPromedio: rendProm,
173 |             alertasCriticas: alertas,
174 |             estado,
175 |           };
176 |         })
177 |         .filter(Boolean)
178 |         .sort((a: any, b: any) => {
179 |           const order: Record<string, number> = { CRITICO: 0, ATENCION: 1, NORMAL: 2 };
180 |           return (order[a.estado] ?? 2) - (order[b.estado] ?? 2);
181 |         });
182 |       res.json(result);
183 |     } catch (error: any) {
184 |       console.error("[ceo/contratos-estado] Error:", error.message);
185 |       res.status(500).json({ message: error.message });
186 |     }
187 |   });
188 |
189 |   app.get("/api/ceo/contrato-detalle", async (req: Request, res: Response) => {
190 |     try {
191 |       const nombre = (req.query.nombre as string) || "";
192 |       if (!nombre) return res.status(400).json({ message: "Se requiere parametro 'nombre'" });
193 |
194 |       const { camiones, faenas, fuelData, fleet, stats } = await getCoreData(1);
195 |       const faena = faenas.find(f => f.nombre === nombre);
196 |       if (!faena) return res.status(404).json({ message: "Contrato no encontrado" });
197 |
198 |       const vinsOnline = new Set(fleet.map((v: any) => v.vin));
199 |       const fCamiones = camiones.filter(c => c.faenaId === faena.id);
200 |       const fStats = stats.filter(s => s.faenaId === faena.id);
201 |
202 |       const now = new Date();
203 |       const hace24h = new Date(Date.now() - 24 * 60 * 60 * 1000);
204 |
205 |       const fuelHoy = (fuelData as any[]).filter(r => {
206 |         const fecha = new Date(r.fechaConsumo || r.fecha);
207 |         if (fecha < hace24h || fecha > now) return false;
208 |         return fCamiones.some(c => c.patente === String(r.numVeh || ""));
209 |       });
210 |
211 |       const litrosHoy = fuelHoy.reduce((s: number, r: any) => s + (r.cantidadLt || 0), 0);
```

```
212 |       const rendVals = fuelHoy.map((r: any) => r.rendReal).filter((v: any) => v > 0 && v < 20);
213 |       const rendHoy = rendVals.length > 0 ? +(rendVals.reduce((a: number, b: number) => a + b, 0) / rendVals.length).toF
214 |       const cargasHoy = fuelHoy.length;
215 |
216 |       const camionesDetalle = fCamiones.map(c => {
217 |         const st = fStats.find(s => s.patente === c.patente);
218 |         const online = c.vin ? vinsOnline.has(c.vin) : false;
219 |         const volvo = fleet.find((v: any) => v.vin === c.vin);
220 |         const speed = volvo ? ((volvo as any).wheelBasedSpeed ?? (volvo as any).gps?.speed ?? 0) : 0;
221 |
222 |         const fuelCam = fuelHoy.filter((r: any) => {
223 |           return String(r.numVeh || "") === c.patente;
224 |         });
225 |         const litrosCam = fuelCam.reduce((s: number, r: any) => s + (r.cantidadLt || 0), 0);
226 |
227 |         return {
228 |           patente: c.patente,
229 |           modelo: c.modelo,
230 |           conductor: c.conductor || "Sin asignar",
231 |           online,
232 |           velocidad: Math.round(speed),
233 |           rendimiento: st?.rend ?? 0,
234 |           litrosHoy: Math.round(litrosCam),
235 |           estado: st?.estado ?? "NORMAL",
236 |           cargasHoy: fuelCam.length,
237 |         };
238 |       }).sort((a, b) => {
239 |         const order: Record<string, number> = { CRITICO: 0, ALERTA: 1, NORMAL: 2 };
240 |         return (order[a.estado] ?? 2) - (order[b.estado] ?? 2);
241 |       });
242 |
243 |       const alertasCriticas = fStats.filter(s => s.estado === "CRITICO").length;
244 |       const alertasAlerta = fStats.filter(s => s.estado === "ALERTA").length;
245 |
246 |       res.json({
247 |         nombre: faena.nombre,
248 |         color: faena.color,
249 |         camionesTotal: fCamiones.length,
250 |         camionesOnline: camionesDetalle.filter(c => c.online).length,
251 |         litrosHoy: Math.round(litrosHoy),
252 |         rendHoy,
253 |         cargasHoy,
254 |         alertasCriticas,
255 |         alertasAlerta,
256 |         camiones: camionesDetalle,
257 |       });
258 |     } catch (error: any) {
259 |       console.error("[ceo/contrato-detalle] Error:", error.message);
260 |       res.status(500).json({ message: error.message });
261 |     }
262 |   });
263 |
264 |   app.post("/api/ceo/contrato-ia-resumen", async (req: Request, res: Response) => {
265 |     try {
266 |       const { nombre } = req.body;
267 |       if (!nombre) return res.status(400).json({ message: "Se requiere 'nombre'" });
268 |
269 |       const detailRes = await new Promise<any>((resolve, reject) => {
270 |         const mockReq = { query: { nombre } } as any;
271 |         const mockRes = {
272 |           json: (data: any) => resolve(data),
273 |           status: (code: number) => ({ json: (data: any) => reject(new Error(data.message || `Error ${code}`)) }),
274 |         } as any;
275 |         (async () => {
276 |           const { camiones, faenas, fuelData, fleet, stats } = await getCoreData(1);
277 |           const faena = faenas.find((f: any) => f.nombre === nombre);
278 |           if (!faena) return reject(new Error("Contrato no encontrado"));
279 |
280 |           const vinsOnline = new Set(fleet.map((v: any) => v.vin));
281 |           const fCamiones = camiones.filter((c: any) => c.faenaId === faena.id);
282 |           const fStats = stats.filter((s: any) => s.faenaId === faena.id);
283 |           const hace24h = new Date(Date.now() - 24 * 60 * 60 * 1000);
284 |
285 |           const fuelHoy = (fuelData as any[]).filter(r => {
286 |             const fecha = new Date(r.fechaConsumo || r.fecha);
287 |             if (fecha < hace24h) return false;
288 |             return fCamiones.some((c: any) => c.patente === String(r.numVeh || ""));
289 |           });
290 |
291 |           const litrosHoy = fuelHoy.reduce((s: number, r: any) => s + (r.cantidadLt || 0), 0);
292 |           const criticos = fStats.filter((s: any) => s.estado === "CRITICO");
293 |
294 |           resolve({
295 |             nombre: faena.nombre,
296 |             camiones: fCamiones.length,
297 |             online: fCamiones.filter((c: any) => c.vin && vinsOnline.has(c.vin)).length,
298 |             litrosHoy: Math.round(litrosHoy),
299 |             cargas: fuelHoy.length,
300 |             criticos: criticos.map((c: any) => c.patente),
301 |           });
302 |         })();
303 |       });
304 |
305 |       const client = getClient();
306 |       const response = await client.messages.create({
307 |         model: MODEL,
308 |         max_tokens: 600,
309 |         messages: [{
310 |           role: "user",
311 |           content: `Eres un analista de flotas de camiones. Genera un resumen ejecutivo de 3-4 lineas sobre el contrato
312 |
313 | Datos del contrato "${nombre}" (ultimas 24h):
314 | - Camiones asignados: ${detailRes.camiones} (${detailRes.online} en linea)
315 | - Litros cargados hoy: ${detailRes.litrosHoy}
316 | - Cargas realizadas: ${detailRes.cargas}
317 | - Camiones criticos: ${detailRes.criticos.length > 0 ? detailRes.criticos.join(", ") : "Ninguno"}
318 |
```

```
319 | Indica si hay algo preocupante o si la operacion esta normal.`
320 |         }],
321 |       });
322 |
323 |       const text = response.content[0].type === "text" ? response.content[0].text : "";
324 |       res.json({ resumen: text });
325 |     } catch (error: any) {
326 |       console.error("[ceo/contrato-ia-resumen] Error:", error.message);
327 |       res.status(500).json({ message: error.message });
328 |     }
329 |   });
330 |
331 |   app.get("/api/ceo/tendencia", async (req: Request, res: Response) => {
332 |     try {
333 |       const dias = parseInt((req.query.dias as string) || "7");
334 |       const { fuelData } = await getCoreData(dias);
335 |
336 |       const byDay = new Map<string, { litros: number; km: number; count: number }>();
337 |       for (let i = 0; i < dias; i++) {
338 |         const d = new Date(Date.now() - (dias - 1 - i) * 24 * 60 * 60 * 1000);
339 |         const key = d.toISOString().split("T")[0];
340 |         byDay.set(key, { litros: 0, km: 0, count: 0 });
341 |       }
342 |
343 |       for (const r of fuelData as any[]) {
344 |         const dateStr = r.fechaConsumo || "";
345 |         if (!dateStr) continue;
346 |         const key = new Date(dateStr).toISOString().split("T")[0];
347 |         const day = byDay.get(key);
348 |         if (day) {
349 |           day.litros += r.cantidadLt || 0;
350 |           if (r.rendReal && r.rendReal > 0 && r.rendReal < 20) {
351 |             day.km += r.rendReal;
352 |             day.count++;
353 |           }
354 |         }
355 |       }
356 |
357 |       const tendencia = [...byDay.entries()].map(([fecha, d]) => ({
358 |         fecha,
359 |         litrosDia: Math.round(d.litros),
360 |         rendimientoDia: d.count > 0 ? +(d.km / d.count).toFixed(2) : 0,
361 |       }));
362 |
363 |       res.json(tendencia);
364 |     } catch (error: any) {
365 |       console.error("[ceo/tendencia] Error:", error.message);
366 |       res.status(500).json({ message: error.message });
367 |     }
368 |   });
369 |
370 |   app.get("/api/ceo/alertas", async (req: Request, res: Response) => {
371 |     try {
372 |       const limite = parseInt((req.query.limite as string) || "5");
373 |       const { stats, camiones, faenas, fleet, params } = await getCoreData(7);
374 |
375 |       const faenaMap = new Map(faenas.map(f => [f.id, f.nombre]));
376 |       const vinsOnline = new Set(fleet.map((v: any) => v.vin));
377 |
378 |       const alertas: { tipo: string; nivel: string; titulo: string; detalle: string; hace: string; gravedad: number }[]
379 |
380 |       for (const s of stats) {
381 |         if (s.estado === "CRITICO") {
382 |           const faena = faenaMap.get(s.faenaId ?? 0) || "Sin faena";
383 |           alertas.push({
384 |             tipo: "desviacion",
385 |             nivel: "CRITICO",
386 |             titulo: `Cam ${s.patente} — ${faena}`,
387 |             detalle: `Desviacion grave en consumo de combustible. Rendimiento: ${s.rend > 0 ? s.rend.toFixed(2) : "N/D"}
388 |             hace: "periodo actual",
389 |             gravedad: 10,
390 |           });
391 |         }
392 |       }
393 |
394 |       const limiteKmh = parseFloat(params.find(p => p.clave === "vel_max_kmh")?.valor || "90");
395 |       for (const v of fleet) {
396 |         const speed = (v as any).wheelBasedSpeed ?? (v as any).gps?.speed ?? 0;
397 |         if (speed > 120) {
398 |           const cam = camiones.find(c => c.vin === (v as any).vin);
399 |           if (cam) {
400 |             const faena = faenaMap.get(cam.faenaId ?? 0) || "Sin faena";
401 |             alertas.push({
402 |               tipo: "exceso_velocidad",
403 |               nivel: "CRITICO",
404 |               titulo: `Cam ${cam.patente} — ${faena}`,
405 |               detalle: `Exceso de velocidad grave: ${speed.toFixed(0)} km/h (limite ${limiteKmh} km/h)`,
406 |               hace: "en tiempo real",
407 |               gravedad: 9,
408 |             });
409 |           }
410 |         }
411 |       }
412 |
413 |       try {
414 |         const microRes = await fetch("http://localhost:5000/api/datos/micro-cargas");
415 |         if (microRes.ok) {
416 |           const microData = await microRes.json();
417 |           const sosp = (microData.registros || []).filter((r: any) => r.nivel === "CRITICO" || r.nivel === "SOSPECHOSO")
418 |           for (const r of sosp.slice(0, 3)) {
419 |             const faena = r.faena || "Sin faena";
420 |             alertas.push({
421 |               tipo: "micro_carga",
422 |               nivel: r.nivel === "CRITICO" ? "CRITICO" : "ALERTA",
423 |               titulo: `Cam ${r.patente} — ${faena}`,
424 |               detalle: `Micro-carga sospechosa: ${r.litros}L con ${r.senales?.length || 0} senales de alerta`,
425 |               hace: r.fecha || "periodo actual",
```

```
426 |               gravedad: r.nivel === "CRITICO" ? 8 : 6,
427 |             });
428 |           }
429 |         }
430 |       } catch {}
431 |
432 |       for (const cam of camiones) {
433 |         if (!cam.vin) continue;
434 |         if (!vinsOnline.has(cam.vin)) {
435 |           const faena = faenaMap.get(cam.faenaId ?? 0) || "Sin faena";
436 |           if (cam.syncAt) {
437 |             const horasOffline = (Date.now() - new Date(cam.syncAt).getTime()) / (1000 * 60 * 60);
438 |             if (horasOffline > 12) {
439 |               alertas.push({
440 |                 tipo: "offline",
441 |                 nivel: "ALERTA",
442 |                 titulo: `Cam ${cam.patente} — ${faena}`,
443 |                 detalle: `Camion sin conexion hace ${Math.round(horasOffline)} horas`,
444 |                 hace: `${Math.round(horasOffline)}h offline`,
445 |                 gravedad: 5,
446 |               });
447 |             }
448 |           }
449 |         }
450 |       }
451 |
452 |       const rendSorted = stats.map(s => s.rend).filter(r => r > 0).sort((a, b) => a - b);
453 |       const p25 = rendSorted.length > 3 ? rendSorted[Math.floor(rendSorted.length * 0.25)] : 0;
454 |       if (p25 > 0) {
455 |         for (const s of stats) {
456 |           if (s.rend > 0 && s.rend < p25) {
457 |             const faena = faenaMap.get(s.faenaId ?? 0) || "Sin faena";
458 |             alertas.push({
459 |               tipo: "rendimiento_bajo",
460 |               nivel: "ALERTA",
461 |               titulo: `Cam ${s.patente} — ${faena}`,
462 |               detalle: `Rendimiento bajo: ${s.rend.toFixed(2)} km/L (P25 = ${p25.toFixed(2)} km/L)`,
463 |               hace: "periodo actual",
464 |               gravedad: 3,
465 |             });
466 |           }
467 |         }
468 |       }
469 |
470 |       alertas.sort((a, b) => b.gravedad - a.gravedad);
471 |       res.json(alertas.slice(0, limite));
472 |     } catch (error: any) {
473 |       console.error("[ceo/alertas] Error:", error.message);
474 |       res.status(500).json({ message: error.message });
475 |     }
476 |   });
477 |
478 |   app.get("/api/ceo/diagnostico-ia", async (req: Request, res: Response) => {
479 |     try {
480 |       const force = req.query.force === "true";
481 |
482 |       if (!force && ceoDiagCache && Date.now() - ceoDiagCache.timestamp < CEO_CACHE_TTL) {
483 |         const minAgo = Math.round((Date.now() - ceoDiagCache.timestamp) / 60000);
484 |         return res.json({ texto: ceoDiagCache.texto, generadoHaceMin: minAgo, cached: true });
485 |       }
486 |
487 |       const { totals } = await getCoreData(7);
488 |
489 |       const prompt = `Eres el analista ejecutivo de SOTRASER, empresa de transporte de carga en Chile.
490 | Genera diagnostico ejecutivo del contrato CENCOSUD para el CEO.
491 |
492 | Datos actuales del contrato CENCOSUD:
493 | - Camiones activos: ${totals.camionesActivos} de ${totals.camionesTotal}
494 | - Rendimiento flota: ${totals.rendPromedio} km/L
495 | - Cambio vs semana anterior: ${totals.rendCambioPct}%
496 | - Litros periodo: ${totals.litrosPeriodo}L
497 | - Alertas criticas: ${totals.alertasCriticas}
498 | - Excesos velocidad detectados: ${totals.excesosVel}
499 | - Micro-cargas sospechosas: ${totals.microSospechas}
500 |
501 | Responde exactamente en 4 lineas en espanol:
502 | ESTADO: [estado general de la operacion hoy]
503 | POSITIVO: [algo concreto que esta funcionando bien]
504 | ATENCION: [el problema mas urgente con datos]
505 | ACCION: [una accion especifica recomendada para hoy]
506 |
507 | Sin formato adicional. Solo las 4 lineas.`;
508 |
509 |       const client = getClient();
510 |       const response = await client.messages.create({
511 |         model: MODEL,
512 |         max_tokens: 300,
513 |         messages: [{ role: "user", content: prompt }],
514 |       });
515 |
516 |       const texto = response.content[0].type === "text" ? response.content[0].text : "Sin datos disponibles.";
517 |       ceoDiagCache = { texto, timestamp: Date.now() };
518 |       res.json({ texto, generadoHaceMin: 0, cached: false });
519 |     } catch (error: any) {
520 |       console.error("[ceo/diagnostico-ia] Error:", error.message);
521 |       if (ceoDiagCache) {
522 |         const minAgo = Math.round((Date.now() - ceoDiagCache.timestamp) / 60000);
523 |         return res.json({ texto: ceoDiagCache.texto, generadoHaceMin: minAgo, cached: true, error: error.message });
524 |       }
525 |       res.status(500).json({ message: error.message });
526 |     }
527 |   });
528 | }
529 |
```

server/ia-routes.ts

1078 lineas

```
1 | import type { Express, Request, Response } from "express";
2 | import Anthropic from "@anthropic-ai/sdk";
3 | import { getCachedFuelData } from "./sigetra-api";
4 | import { getFleetStatus } from "./volvo-api";
5 | import { storage } from "./storage";
6 | import { getCencosudFaenaId, getCencosudPatentes } from "./cencosud-filter";
7 |
8 | const MODEL = "claude-sonnet-4-20250514";
9 |
10 | function getClient(): Anthropic {
11 |   return new Anthropic();
12 | }
13 |
14 | const DATA_START_DATE = new Date("2026-03-01T00:00:00");
15 |
16 | async function getFusionContext(days: number = 7): Promise<string> {
17 |   const to = new Date();
18 |   let from = new Date(Date.now() - days * 24 * 60 * 60 * 1000);
19 |   if (from < DATA_START_DATE) from = new Date(DATA_START_DATE);
20 |
21 |   const [allFuelData, fleetStatus, allCamionesRaw, faenas, cencFaenaId, cencPatentes] = await Promise.all([
22 |     getCachedFuelData(from, to).catch(() => []),
23 |     getFleetStatus().catch(() => []),
24 |     storage.getCamiones(),
25 |     storage.getFaenas(),
26 |     getCencosudFaenaId(),
27 |     getCencosudPatentes(),
28 |   ]);
29 |
30 |   const allCamiones = allCamionesRaw.filter((c: any) => c.faenaId === cencFaenaId);
31 |   const fuelData = allFuelData.filter((r: any) => cencPatentes.has(String(r.numVeh || "")));
32 |
33 |   const volvoByVin = new Map(fleetStatus.map((s: any) => [s.vin, s]));
34 |   const camionByPatente = new Map(allCamiones.map((c: any) => [c.patente, c]));
35 |
36 |   const allVins = allCamiones.map((c: any) => c.vin).filter((v: any): v is string => v != null);
37 |   const snapshotEnd = to;
38 |   const TWELVE_H = 12 * 60 * 60 * 1000;
39 |   const expandedFrom = new Date(from.getTime() - TWELVE_H);
40 |   const expandedTo = new Date(snapshotEnd.getTime() + TWELVE_H);
41 |   const historicalSnapshots = allVins.length > 0
42 |     ? await storage.getVolvoFuelSnapshotsInRange(allVins, expandedFrom, expandedTo)
43 |     : [];
44 |   const snapshotsByVin = new Map<string, typeof historicalSnapshots>();
45 |   for (const snap of historicalSnapshots) {
46 |     const arr = snapshotsByVin.get(snap.vin) || [];
47 |     arr.push(snap);
48 |     snapshotsByVin.set(snap.vin, arr);
49 |   }
50 |   for (const [vin, snaps] of snapshotsByVin) {
51 |     snapshotsByVin.set(vin, snaps.sort((a: any, b: any) => new Date(a.capturedAt).getTime() - new Date(b.capturedAt).get
52 |   }
53 |
54 |   const computeVolvoFuelForWindow = (vin: string | null, windowStart: Date, windowEnd: Date): number | null => {
55 |     if (!vin) return null;
56 |     const snaps = snapshotsByVin.get(vin);
57 |     if (!snaps || snaps.length < 2) return null;
58 |     let filtered = snaps.filter((s: any) => {
59 |       const t = new Date(s.capturedAt).getTime();
60 |       return t >= windowStart.getTime() && t <= windowEnd.getTime();
61 |     });
62 |     if (filtered.length < 2) {
63 |       const expStart = new Date(windowStart.getTime() - TWELVE_H);
64 |       const expEnd = new Date(windowEnd.getTime() + TWELVE_H);
65 |       filtered = snaps.filter((s: any) => {
66 |         const t = new Date(s.capturedAt).getTime();
67 |         return t >= expStart.getTime() && t <= expEnd.getTime();
68 |       });
69 |     }
70 |     if (filtered.length < 2) return null;
71 |     const delta = filtered[filtered.length - 1].totalFuelUsed - filtered[0].totalFuelUsed;
72 |     return delta >= 0 ? Math.round(delta / 1000) : null;
73 |   };
74 |
75 |   const fuelByNumVeh = new Map<number, typeof fuelData>();
76 |   for (const rec of fuelData) {
77 |     if (rec.numVeh == null) continue;
78 |     const existing = fuelByNumVeh.get(rec.numVeh) || [];
79 |     existing.push(rec);
80 |     fuelByNumVeh.set(rec.numVeh, existing);
81 |   }
82 |
83 |   const trucks: any[] = [];
84 |   for (const [numVeh, cargas] of fuelByNumVeh) {
85 |     const camion = camionByPatente.get(String(numVeh));
86 |     if (!camion) continue;
87 |     const volvo = camion.vin ? volvoByVin.get(camion.vin) : null;
88 |     const totalLitrosSigetra = cargas.reduce((s: number, c: any) => s + c.cantidadLt, 0);
89 |     const rendPromedio = cargas.filter((c: any) => c.rendReal != null && c.rendReal > 0 && c.rendReal < 100).reduce((s:
90 |
91 |     const sortedCargas = [...cargas].sort((a: any, b: any) => new Date(a.fechaConsumo).getTime() - new Date(b.fechaConsu
92 |     let litrosVolvoPeriodo: number | null = null;
93 |     let litrosSigetraViajes = 0;
94 |     if (camion.vin && sortedCargas.length >= 2) {
95 |       let totalVolvoViajes = 0;
96 |       let matchedCount = 0;
97 |       for (let i = 0; i < sortedCargas.length - 1; i++) {
98 |         const viajeInicio = new Date(sortedCargas[i].fechaConsumo);
99 |         const viajeFin = new Date(sortedCargas[i + 1].fechaConsumo);
100 |         const volvoLitros = computeVolvoFuelForWindow(camion.vin, viajeInicio, viajeFin);
101 |         if (volvoLitros !== null) {
102 |           totalVolvoViajes += volvoLitros;
103 |           litrosSigetraViajes += sortedCargas[i + 1].cantidadLt;
104 |           matchedCount++;
```

```
105 |         }
106 |       }
107 |       if (matchedCount > 0) litrosVolvoPeriodo = totalVolvoViajes;
108 |     }
109 |
110 |     if (litrosVolvoPeriodo === null && camion.vin) {
111 |       const snaps = snapshotsByVin.get(camion.vin);
112 |       if (snaps && snaps.length >= 2) {
113 |         const earliest = snaps[0];
114 |         const latest = snaps[snaps.length - 1];
115 |         const snapStart = new Date(earliest.capturedAt);
116 |         const snapEnd = new Date(latest.capturedAt);
117 |         const delta = latest.totalFuelUsed - earliest.totalFuelUsed;
118 |         if (delta > 0) {
119 |           litrosVolvoPeriodo = Math.round(delta / 1000);
120 |           const cargasInWindow = sortedCargas.filter((c: any) => {
121 |             const cf = new Date(c.fechaConsumo);
122 |             return cf >= snapStart && cf <= snapEnd;
123 |           });
124 |           if (cargasInWindow.length > 0) {
125 |             litrosSigetraViajes = cargasInWindow.reduce((s: number, c: any) => s + c.cantidadLt, 0);
126 |           } else {
127 |             const expStart = new Date(snapStart.getTime() - TWELVE_H);
128 |             const expEnd = new Date(snapEnd.getTime() + TWELVE_H);
129 |             const cargasExp = sortedCargas.filter((c: any) => {
130 |               const cf = new Date(c.fechaConsumo);
131 |               return cf >= expStart && cf <= expEnd;
132 |             });
133 |             litrosSigetraViajes = cargasExp.length > 0
134 |               ? cargasExp.reduce((s: number, c: any) => s + c.cantidadLt, 0)
135 |               : totalLitrosSigetra;
136 |           }
137 |         }
138 |       }
139 |     }
140 |
141 |     const lastCarga = sortedCargas[sortedCargas.length - 1];
142 |
143 |     let cuadratura: number | null = null;
144 |     if (litrosVolvoPeriodo != null && litrosVolvoPeriodo > 0 && litrosSigetraViajes > 0) {
145 |       cuadratura = Math.round((Math.min(litrosVolvoPeriodo, litrosSigetraViajes) / Math.max(litrosVolvoPeriodo, litrosSi
146 |     }
147 |
148 |     const odometroSigetra = lastCarga?.odometroActual;
149 |     const odometroVolvo = volvo?.totalDistance != null ? Math.round(volvo.totalDistance / 1000) : camion.odometro;
150 |     const deltaOdometro = (odometroSigetra != null && odometroVolvo != null) ? odometroSigetra - odometroVolvo : null;
151 |
152 |     trucks.push({
153 |       interno: String(numVeh),
154 |       patente: lastCarga?.patente || camion.patente,
155 |       faena: lastCarga?.faena || null,
156 |       conductor: lastCarga?.nombreConductor || null,
157 |       litrosSigetra: Math.round(totalLitrosSigetra),
158 |       litrosVolvo: litrosVolvoPeriodo,
159 |       cuadratura,
160 |       rendimiento: Math.round(rendPromedio * 100) / 100,
161 |       totalCargas: cargas.length,
162 |       deltaOdometro,
163 |       lugares: [...new Set(cargas.map((c: any) => c.lugarConsumo).filter(Boolean))].slice(0, 5),
164 |     });
165 |   }
166 |
167 |   const totalLitros = trucks.reduce((s: number, t: any) => s + t.litrosSigetra, 0);
168 |   const totalCargas = fuelData.length;
169 |   const trucksConCuadratura = trucks.filter((t: any) => t.cuadratura != null);
170 |   const cuadraturaBaja = trucksConCuadratura.filter((t: any) => t.cuadratura < 95);
171 |   const alertasOdo = trucks.filter((t: any) => t.deltaOdometro != null && Math.abs(t.deltaOdometro) > 1000);
172 |
173 |   return `DATOS DE LA FLOTA SOTRASER (ultimos ${days} dias):
174 | - Total camiones con datos: ${trucks.length}
175 | - Total litros Sigetra: ${totalLitros.toLocaleString("es-CL")} L
176 | - Total cargas: ${totalCargas}
177 | - Faenas activas: ${faenas.length} (${faenas.map((f: any) => f.nombre).join(", ")})
178 | - Camiones con cuadratura <95%: ${cuadraturaBaja.length} de ${trucksConCuadratura.length} evaluables
179 | - Camiones con delta odometro >1000km: ${alertasOdo.length}
180 |
181 | DETALLE POR CAMION (top 50 por litros):
182 | ${trucks.sort((a: any, b: any) => b.litrosSigetra - a.litrosSigetra).slice(0, 50).map((t: any) =>
183 |   `N.${t.interno} Pat:${t.patente} Faena:${t.faena || "N/A"} Cond:${t.conductor || "N/A"} L.Sig:${t.litrosSigetra} L.Vol
184 | ).join("\n")}
185 |
186 | CAMIONES CON CUADRATURA BAJA (<95%):
187 | ${cuadraturaBaja.map((t: any) => `N.${t.interno} Sig:${t.litrosSigetra}L Vol:${t.litrosVolvo}L Cuad:${t.cuadratura}% Dif
188 |
189 | ALERTAS ODOMETRO (delta >1000km):
190 | ${alertasOdo.map((t: any) => `N.${t.interno} Delta:${t.deltaOdometro}km`).join("\n") || "Ninguna"}`;
191 | }
192 |
193 | let dashboardInsightCache: { text: string; timestamp: number } | null = null;
194 | const DASHBOARD_INSIGHT_TTL = 30 * 60 * 1000;
195 |
196 | export function registerIARoutes(app: Express) {
197 |   app.get("/api/ia/resumen-dashboard", async (_req: Request, res: Response) => {
198 |     try {
199 |       if (dashboardInsightCache && Date.now() - dashboardInsightCache.timestamp < DASHBOARD_INSIGHT_TTL) {
200 |         return res.json({ resumen: dashboardInsightCache.text, cached: true });
201 |       }
202 |
203 |       const context = await getFusionContext(7);
204 |       const client = getClient();
205 |       const response = await client.messages.create({
206 |         model: MODEL,
207 |         max_tokens: 300,
208 |         messages: [{
209 |           role: "user",
210 |           content: `Eres un analista de flotas de camiones de Sotraser en Chile. Genera exactamente 2 oraciones cortas r
211 |
```

```
212 | Datos:
213 | ${context}`
214 |         }],
215 |       });
216 |
217 |       const text = response.content[0].type === "text" ? response.content[0].text : "Sin datos disponibles para generar
218 |       dashboardInsightCache = { text, timestamp: Date.now() };
219 |       res.json({ resumen: text, cached: false });
220 |     } catch (error: any) {
221 |       console.error("[ia/resumen-dashboard] Error:", error.message);
222 |       if (dashboardInsightCache) {
223 |         return res.json({ resumen: dashboardInsightCache.text, cached: true, error: error.message });
224 |       }
225 |       res.status(500).json({ message: error.message, resumen: "No se pudo generar el resumen de IA." });
226 |     }
227 |   });
228 |
229 |   app.post("/api/ia/analizar", async (_req: Request, res: Response) => {
230 |     try {
231 |       const context = await getFusionContext(7);
232 |       const client = getClient();
233 |       const response = await client.messages.create({
234 |         model: MODEL,
235 |         max_tokens: 2000,
236 |         messages: [{
237 |           role: "user",
238 |           content: `Eres un analista experto en flotas de camiones y combustible en Chile. Analiza estos datos de la flo
239 | {
240 |   "resumen": "Resumen ejecutivo de 2-3 oraciones sobre el estado actual de la flota",
241 |   "litros_en_riesgo": "Litros estimados en riesgo por desvios y cuadraturas bajas. Formato: X.XXX L",
242 |   "alertas_criticas": [
243 |     {"patente": "XXXX", "problema": "descripcion breve", "accion": "accion recomendada", "urgencia": "INMEDIATA|24H|SEMA
244 |   ],
245 |   "recomendacion_principal": "La accion mas importante a tomar esta semana"
246 | }
247 |
248 | Datos:
249 | ${context}`
250 |         }],
251 |       });
252 |
253 |       const text = response.content[0].type === "text" ? response.content[0].text : "";
254 |       const jsonMatch = text.match(/\{[\s\S]*\}/);
255 |       if (jsonMatch) {
256 |         const parsed = JSON.parse(jsonMatch[0]);
257 |         res.json(parsed);
258 |       } else {
259 |         res.json({ resumen: text, litros_en_riesgo: "No calculado", alertas_criticas: [], recomendacion_principal: text
260 |       }
261 |     } catch (error: any) {
262 |       console.error("[ia/analizar] Error:", error.message);
263 |       res.status(500).json({ message: error.message });
264 |     }
265 |   });
266 |
267 |   app.post("/api/ia/chat", async (req: Request, res: Response) => {
268 |     try {
269 |       const { messages } = req.body;
270 |       if (!messages || !Array.isArray(messages) || messages.length === 0) {
271 |         return res.status(400).json({ message: "Se requiere campo 'messages' (array)" });
272 |       }
273 |
274 |       const context = await getFusionContext(14);
275 |       const client = getClient();
276 |
277 |       const systemPrompt = `Eres SOTRA IA, el asistente inteligente de SOTRASER, empresa de transporte de carga en Chile
278 |
279 | Cuando el usuario pregunte sobre la flota, responde con datos concretos si los tienes. Cuando no tengas datos exactos, d
280 |
281 | Datos actuales de la flota:
282 | ${context}`;
283 |
284 |       const response = await client.messages.create({
285 |         model: MODEL,
286 |         max_tokens: 1500,
287 |         system: systemPrompt,
288 |         messages: messages.map((m: any) => ({
289 |           role: m.role === "user" ? "user" as const : "assistant" as const,
290 |           content: m.content,
291 |         })),
292 |       });
293 |
294 |       const text = response.content[0].type === "text" ? response.content[0].text : "";
295 |       res.json({ respuesta: text });
296 |     } catch (error: any) {
297 |       console.error("[ia/chat] Error:", error.message);
298 |       res.status(500).json({ message: error.message });
299 |     }
300 |   });
301 |
302 |   app.post("/api/ia/consulta", async (req: Request, res: Response) => {
303 |     try {
304 |       const { pregunta } = req.body;
305 |       if (!pregunta || typeof pregunta !== "string") {
306 |         return res.status(400).json({ message: "Se requiere campo 'pregunta'" });
307 |       }
308 |
309 |       const context = await getFusionContext(14);
310 |       const client = getClient();
311 |       const response = await client.messages.create({
312 |         model: MODEL,
313 |         max_tokens: 1500,
314 |         messages: [{
315 |           role: "user",
316 |           content: `Eres un analista experto en flotas de camiones de la empresa Sotraser en Chile. Responde la siguient
317 |
318 | Pregunta del usuario: ${pregunta}
```

```
319 |
320 | Datos actuales de la flota:
321 | ${context}`
322 |         }],
323 |       });
324 |
325 |       const text = response.content[0].type === "text" ? response.content[0].text : "";
326 |       res.json({ respuesta: text });
327 |     } catch (error: any) {
328 |       console.error("[ia/consulta] Error:", error.message);
329 |       res.status(500).json({ message: error.message });
330 |     }
331 |   });
332 |
333 |   app.post("/api/ia/patrones", async (_req: Request, res: Response) => {
334 |     try {
335 |       const context = await getFusionContext(30);
336 |       const client = getClient();
337 |       const response = await client.messages.create({
338 |         model: MODEL,
339 |         max_tokens: 2000,
340 |         messages: [{
341 |           role: "user",
342 |           content: `Eres un detective de fraudes de combustible en flotas de camiones en Chile. Sistema de medicion acti
343 | - Mismo conductor con desvios repetidos en misma ubicacion
344 | - Cuadraturas consistentemente bajas en ciertos camiones
345 | - Patrones de consumo anomalos (litros excesivos para la distancia)
346 | - Conductores que siempre cargan en el mismo lugar con rendimiento bajo
347 |
348 | Genera un JSON con exactamente esta estructura (sin texto adicional, solo JSON):
349 | {
350 |   "patrones": [
351 |     {
352 |       "descripcion": "Descripcion del patron detectado",
353 |       "nivel_sospecha": "ALTO|MEDIO|BAJO",
354 |       "evidencia": "Datos concretos que soportan la sospecha (fechas, litros, camiones involucrados)",
355 |       "camiones_involucrados": ["N.XXXX"],
356 |       "conductor": "Nombre si aplica",
357 |       "litros_perdidos": "Litros estimados perdidos"
358 |     }
359 |   ],
360 |   "resumen": "Resumen de hallazgos en 2-3 oraciones"
361 | }
362 |
363 | Datos de la flota (ultimos 30 dias):
364 | ${context}`
365 |         }],
366 |       });
367 |
368 |       const text = response.content[0].type === "text" ? response.content[0].text : "";
369 |       const jsonMatch = text.match(/\{[\s\S]*\}/);
370 |       if (jsonMatch) {
371 |         res.json(JSON.parse(jsonMatch[0]));
372 |       } else {
373 |         res.json({ patrones: [], resumen: text });
374 |       }
375 |     } catch (error: any) {
376 |       console.error("[ia/patrones] Error:", error.message);
377 |       res.status(500).json({ message: error.message });
378 |     }
379 |   });
380 |
381 |   app.post("/api/ia/resumen-semanal", async (_req: Request, res: Response) => {
382 |     try {
383 |       const context7 = await getFusionContext(7);
384 |       const context14 = await getFusionContext(14);
385 |       const client = getClient();
386 |       const response = await client.messages.create({
387 |         model: MODEL,
388 |         max_tokens: 2000,
389 |         messages: [{
390 |           role: "user",
391 |           content: `Eres un analista de flotas que genera reportes ejecutivos semanales. Genera un resumen de los ultimo
392 |
393 | Genera un JSON con exactamente esta estructura (sin texto adicional, solo JSON):
394 | {
395 |   "periodo": "DD/MM/YYYY - DD/MM/YYYY",
396 |   "total_litros": 0,
397 |   "litros_desviados": 0,
398 |   "top3_camiones_problema": [
399 |     {"interno": "XXXX", "problema": "descripcion", "litros_desviados": 0}
400 |   ],
401 |   "top3_conductores_problema": [
402 |     {"nombre": "XXXX", "problema": "descripcion"}
403 |   ],
404 |   "comparativa_semana_anterior": "Texto comparando esta semana vs la anterior",
405 |   "acciones_prioritarias": [
406 |     "Accion 1",
407 |     "Accion 2",
408 |     "Accion 3"
409 |   ],
410 |   "resumen_ejecutivo": "Resumen de 3-4 oraciones para presentar a gerencia"
411 | }
412 |
413 | Datos ultima semana:
414 | ${context7}
415 |
416 | Datos ultimas 2 semanas (para comparar):
417 | ${context14}`
418 |         }],
419 |       });
420 |
421 |       const text = response.content[0].type === "text" ? response.content[0].text : "";
422 |       const jsonMatch = text.match(/\{[\s\S]*\}/);
423 |       if (jsonMatch) {
424 |         res.json(JSON.parse(jsonMatch[0]));
425 |       } else {
```

```
426 |         res.json({ resumen_ejecutivo: text, acciones_prioritarias: [] });
427 |       }
428 |     } catch (error: any) {
429 |       console.error("[ia/resumen-semanal] Error:", error.message);
430 |       res.status(500).json({ message: error.message });
431 |     }
432 |   });
433 |
434 |   app.post("/api/ia/camion", async (req: Request, res: Response) => {
435 |     try {
436 |       const { camionId, dias } = req.body;
437 |       if (!camionId) return res.status(400).json({ message: "camionId requerido" });
438 |       const period = dias || 56;
439 |
440 |       const camion = await storage.getCamion(camionId);
441 |       if (!camion) return res.status(404).json({ message: "Camion no encontrado" });
442 |
443 |       const to = new Date();
444 |       const from = new Date(Date.now() - period * 24 * 60 * 60 * 1000);
445 |       const faenas = await storage.getFaenas();
446 |       const faena = faenas.find(f => f.id === camion.faenaId);
447 |
448 |       const fuelData = await getCachedFuelData(from, to).catch(() => []);
449 |       const numVeh = parseInt(camion.patente, 10);
450 |       const truckCargas = fuelData.filter((c: any) =>
451 |         (!isNaN(numVeh) && c.numVeh === numVeh) || c.patente === camion.patente
452 |       );
453 |
454 |       let volvoSnaps: any[] = [];
455 |       if (camion.vin) {
456 |         volvoSnaps = await storage.getVolvoFuelSnapshotsInRange([camion.vin], from, to);
457 |       }
458 |
459 |       const weeklyData: any[] = [];
460 |       for (let w = 0; w < 8; w++) {
461 |         const weekEnd = new Date(to.getTime() - w * 7 * 24 * 60 * 60 * 1000);
462 |         const weekStart = new Date(weekEnd.getTime() - 7 * 24 * 60 * 60 * 1000);
463 |         const weekCargas = truckCargas.filter((c: any) => {
464 |           const d = new Date(c.fechaConsumo);
465 |           return d >= weekStart && d < weekEnd;
466 |         });
467 |         const litrosSurtidor = weekCargas.reduce((s: number, c: any) => s + c.cantidadLt, 0);
468 |         const validRend = weekCargas.filter((c: any) => c.rendReal != null && c.rendReal > 0 && c.rendReal < 100);
469 |         const rendPromedio = validRend.length > 0 ? validRend.reduce((s: number, c: any) => s + c.rendReal, 0) / validRe
470 |
471 |         const weekSnaps = volvoSnaps.filter(s => {
472 |           const d = new Date(s.capturedAt);
473 |           return d >= weekStart && d < weekEnd;
474 |         }).sort((a: any, b: any) => new Date(a.capturedAt).getTime() - new Date(b.capturedAt).getTime());
475 |         let litrosEcu = null;
476 |         if (weekSnaps.length >= 2) {
477 |           const delta = weekSnaps[weekSnaps.length - 1].totalFuelUsed - weekSnaps[0].totalFuelUsed;
478 |           if (delta > 0) litrosEcu = Math.round(delta / 1000);
479 |         }
480 |
481 |         weeklyData.unshift({
482 |           semana: `S${8 - w}`,
483 |           desde: weekStart.toISOString().slice(0, 10),
484 |           hasta: weekEnd.toISOString().slice(0, 10),
485 |           litrosSurtidor: Math.round(litrosSurtidor),
486 |           litrosEcu,
487 |           rendimiento: rendPromedio ? Math.round(rendPromedio * 100) / 100 : null,
488 |           cargas: weekCargas.length,
489 |         });
490 |       }
491 |
492 |       const totalLitros = truckCargas.reduce((s: number, c: any) => s + c.cantidadLt, 0);
493 |       const allRend = truckCargas.filter((c: any) => c.rendReal != null && c.rendReal > 0 && c.rendReal < 100);
494 |       const rendGeneral = allRend.length > 0 ? allRend.reduce((s: number, c: any) => s + c.rendReal, 0) / allRend.length
495 |       const rendValues = allRend.map((c: any) => c.rendReal as number);
496 |       const rendStdDev = rendValues.length > 1 ? Math.sqrt(rendValues.reduce((s, v) => s + Math.pow(v - (rendGeneral ||
497 |
498 |       const truckContext = `CAMION N.${camion.patente} (${camion.modelo})
499 | Faena: ${faena?.nombre || "N/A"}
500 | Conductor: ${camion.conductor || "N/A"}
501 | Meta: ${camion.metaKmL} km/L
502 | Odometro: ${camion.odometro || "N/A"} km
503 | Horas motor: ${camion.horasMotor || "N/A"}
504 | Horas ralenti: ${camion.horasRalenti || "N/A"}
505 |
506 | DATOS ULTIMAS 8 SEMANAS:
507 | ${weeklyData.map(w => `${w.semana} (${w.desde} a ${w.hasta}): Surt=${w.litrosSurtidor}L ECU=${w.litrosEcu ?? "N/A"}L Ren
508 |
509 | RESUMEN PERIODO:
510 | - Total litros: ${Math.round(totalLitros)}L
511 | - Rendimiento promedio: ${rendGeneral ? rendGeneral.toFixed(2) : "N/A"} km/L
512 | - Variacion rendimiento (std): ${rendStdDev.toFixed(2)}
513 | - Total cargas: ${truckCargas.length}
514 | - Lugares frecuentes: ${[...new Set(truckCargas.map((c: any) => c.lugarConsumo).filter(Boolean))].slice(0, 5).join(", ")
515 |
516 |       const client = getClient();
517 |       const response = await client.messages.create({
518 |         model: MODEL,
519 |         max_tokens: 2000,
520 |         messages: [{
521 |           role: "user",
522 |           content: `Eres un analista experto en flotas de camiones mineros en Chile. Analiza los datos de este camion in
523 | {
524 |   "tendencia": "MEJORANDO|ESTABLE|EMPEORANDO|INSUFICIENTE",
525 |   "resumen": "Resumen de 2-3 oraciones sobre el estado del camion",
526 |   "anomalias": ["Lista de anomalias detectadas"],
527 |   "causa_probable": "Causa mas probable de problemas (si hay)",
528 |   "recomendacion": "Accion recomendada principal",
529 |   "score_salud": 85,
530 |   "detalle_semanas": "Analisis breve de la tendencia semanal"
531 | }
532 |
```

```
533 | score_salud: 0-100 donde 100=perfecto. Considera rendimiento vs meta, variacion, diferencia surtidor/ECU.
534 |
535 | Datos del camion:
536 | ${truckContext}`
537 |         }],
538 |       });
539 |
540 |       const text = response.content[0].type === "text" ? response.content[0].text : "";
541 |       const jsonMatch = text.match(/\{[\s\S]*\}/);
542 |       let iaAnalisis = { tendencia: "INSUFICIENTE", resumen: text, anomalias: [], causa_probable: "", recomendacion: "",
543 |       if (jsonMatch) {
544 |         try { iaAnalisis = JSON.parse(jsonMatch[0]); } catch {}
545 |       }
546 |
547 |       res.json({
548 |         camion: {
549 |           id: camion.id,
550 |           patente: camion.patente,
551 |           modelo: camion.modelo,
552 |           faena: faena?.nombre || null,
553 |           conductor: camion.conductor,
554 |           metaKmL: camion.metaKmL,
555 |           odometro: camion.odometro,
556 |           horasMotor: camion.horasMotor,
557 |           horasRalenti: camion.horasRalenti,
558 |         },
559 |         weeklyData,
560 |         metricas: {
561 |           rendPromedio: rendGeneral ? Math.round(rendGeneral * 100) / 100 : null,
562 |           variacion: Math.round(rendStdDev * 100) / 100,
563 |           totalLitros: Math.round(totalLitros),
564 |           totalCargas: truckCargas.length,
565 |           eventosCriticos: weeklyData.filter(w => w.rendimiento !== null && w.rendimiento < (camion.metaKmL * 0.7)).leng
566 |           horasRalenti: camion.horasRalenti,
567 |         },
568 |         ia: iaAnalisis,
569 |       });
570 |     } catch (error: any) {
571 |       console.error("[ia/camion] Error:", error.message);
572 |       res.status(500).json({ message: error.message });
573 |     }
574 |   });
575 |
576 |   app.post("/api/ia/contrato", async (req: Request, res: Response) => {
577 |     try {
578 |       const { faenaId, dias } = req.body;
579 |       if (!faenaId) return res.status(400).json({ message: "faenaId requerido" });
580 |       const period = dias || 56;
581 |
582 |       const faenas = await storage.getFaenas();
583 |       const faena = faenas.find(f => f.id === faenaId);
584 |       if (!faena) return res.status(404).json({ message: "Contrato no encontrado" });
585 |
586 |       const allCamiones = await storage.getCamiones();
587 |       const camiones = allCamiones.filter(c => c.faenaId === faenaId);
588 |       if (camiones.length === 0) return res.json({ faena: faena.nombre, camiones: [], ia: null });
589 |
590 |       const to = new Date();
591 |       let from = new Date(Date.now() - period * 24 * 60 * 60 * 1000);
592 |       if (from < new Date("2026-03-01")) from = new Date("2026-03-01");
593 |
594 |       const fuelData = await getCachedFuelData(from, to).catch(() => []);
595 |       const fleet = await getFleetStatus().catch(() => []);
596 |       const vinsOnline = new Set(fleet.map((v: any) => v.vin));
597 |
598 |       const camionResults = camiones.map(cam => {
599 |         const numVeh = parseInt(cam.patente, 10);
600 |         const truckCargas = (fuelData as any[]).filter((c: any) =>
601 |           (!isNaN(numVeh) && c.numVeh === numVeh) || c.patente === cam.patente
602 |         );
603 |
604 |         const totalLitros = truckCargas.reduce((s: number, c: any) => s + (c.cantidadLt || 0), 0);
605 |         const allRend = truckCargas.filter((c: any) => c.rendReal != null && c.rendReal > 0 && c.rendReal < 100);
606 |         const rendProm = allRend.length > 0 ? +(allRend.reduce((s: number, c: any) => s + c.rendReal, 0) / allRend.lengt
607 |         const online = cam.vin ? vinsOnline.has(cam.vin) : false;
608 |         const cargas = truckCargas.length;
609 |
610 |         const weeklyData: any[] = [];
611 |         for (let w = 0; w < Math.min(8, Math.ceil(period / 7)); w++) {
612 |           const weekEnd = new Date(to.getTime() - w * 7 * 24 * 60 * 60 * 1000);
613 |           const weekStart = new Date(weekEnd.getTime() - 7 * 24 * 60 * 60 * 1000);
614 |           const weekCargas = truckCargas.filter((c: any) => {
615 |             const d = new Date(c.fechaConsumo);
616 |             return d >= weekStart && d < weekEnd;
617 |           });
618 |           const litros = weekCargas.reduce((s: number, c: any) => s + (c.cantidadLt || 0), 0);
619 |           const validRend = weekCargas.filter((c: any) => c.rendReal > 0 && c.rendReal < 100);
620 |           const rend = validRend.length > 0 ? +(validRend.reduce((s: number, c: any) => s + c.rendReal, 0) / validRend.l
621 |           weeklyData.unshift({
622 |             semana: `S${Math.min(8, Math.ceil(period / 7)) - w}`,
623 |             litros: Math.round(litros),
624 |             rendimiento: rend,
625 |             cargas: weekCargas.length,
626 |           });
627 |         }
628 |
629 |         return {
630 |           patente: cam.patente,
631 |           modelo: cam.modelo,
632 |           conductor: cam.conductor || "Sin asignar",
633 |           metaKmL: cam.metaKmL,
634 |           online,
635 |           rendPromedio: rendProm,
636 |           totalLitros: Math.round(totalLitros),
637 |           cargas,
638 |           estado: rendProm !== null && rendProm < cam.metaKmL * 0.7 ? "CRITICO" : rendProm !== null && rendProm < cam.me
639 |           weeklyData,
```

```
640 |         };
641 |       }).sort((a, b) => {
642 |         const order: Record<string, number> = { CRITICO: 0, ALERTA: 1, NORMAL: 2 };
643 |         return (order[a.estado] ?? 2) - (order[b.estado] ?? 2);
644 |       });
645 |
646 |       const totalLitrosContrato = camionResults.reduce((s, c) => s + c.totalLitros, 0);
647 |       const rendVals = camionResults.filter(c => c.rendPromedio !== null).map(c => c.rendPromedio as number);
648 |       const rendContratoAvg = rendVals.length > 0 ? +(rendVals.reduce((a, b) => a + b, 0) / rendVals.length).toFixed(2)
649 |       const criticos = camionResults.filter(c => c.estado === "CRITICO").length;
650 |
651 |       const contratoContext = `CONTRATO: ${faena.nombre}
652 | Camiones: ${camiones.length} (${camionResults.filter(c => c.online).length} en linea)
653 | Periodo: ${period} dias
654 | Rendimiento promedio contrato: ${rendContratoAvg ?? "N/A"} km/L
655 | Total litros: ${totalLitrosContrato}
656 | Camiones criticos: ${criticos}
657 |
658 | DETALLE POR CAMION:
659 | ${camionResults.map(c => `- ${c.patente} (${c.modelo}) Conductor:${c.conductor} Rend:${c.rendPromedio ?? "N/A"}km/L Litr
660 |
661 |       const client = getClient();
662 |       const response = await client.messages.create({
663 |         model: MODEL,
664 |         max_tokens: 2000,
665 |         messages: [{
666 |           role: "user",
667 |           content: `Eres un analista experto en flotas de camiones mineros en Chile. Analiza los datos de este CONTRATO
668 | {
669 |   "estado_general": "NORMAL|ATENCION|CRITICO",
670 |   "resumen": "Resumen ejecutivo de 3-4 oraciones sobre el contrato",
671 |   "rendimiento_contrato": "Evaluacion del rendimiento general",
672 |   "camiones_problema": ["Lista de patentes con problemas y por que"],
673 |   "camion_destacado": "Patente del mejor camion y por que",
674 |   "riesgos": ["Riesgos detectados en el contrato"],
675 |   "recomendaciones": ["3-4 acciones concretas priorizadas"],
676 |   "score_contrato": 75
677 | }
678 |
679 | score_contrato: 0-100 donde 100=operacion perfecta. Considera rendimiento promedio, camiones criticos, variacion entre c
680 |
681 | ${contratoContext}`
682 |         }],
683 |       });
684 |
685 |       const text = response.content[0].type === "text" ? response.content[0].text : "";
686 |       const jsonMatch = text.match(/\{[\s\S]*\}/);
687 |       let iaResult = {
688 |         estado_general: "NORMAL", resumen: text, rendimiento_contrato: "",
689 |         camiones_problema: [] as string[], camion_destacado: "", riesgos: [] as string[],
690 |         recomendaciones: [] as string[], score_contrato: 0
691 |       };
692 |       if (jsonMatch) {
693 |         try { iaResult = JSON.parse(jsonMatch[0]); } catch {}
694 |       }
695 |
696 |       res.json({
697 |         faena: faena.nombre,
698 |         color: faena.color,
699 |         camionesTotal: camiones.length,
700 |         camionesOnline: camionResults.filter(c => c.online).length,
701 |         totalLitros: totalLitrosContrato,
702 |         rendPromedio: rendContratoAvg,
703 |         criticos,
704 |         camiones: camionResults,
705 |         ia: iaResult,
706 |       });
707 |     } catch (error: any) {
708 |       console.error("[ia/contrato] Error:", error.message);
709 |       res.status(500).json({ message: error.message });
710 |     }
711 |   });
712 |
713 |   app.post("/api/ia/detectar-puntos", async (req: Request, res: Response) => {
714 |     try {
715 |       const { camionId } = req.body;
716 |       if (!camionId) return res.status(400).json({ message: "camionId requerido" });
717 |
718 |       const camion = await storage.getCamion(camionId);
719 |       if (!camion) return res.status(404).json({ message: "Camion no encontrado" });
720 |
721 |       const to = new Date();
722 |       const from = new Date(Date.now() - 30 * 24 * 60 * 60 * 1000);
723 |       const fuelData = await getCachedFuelData(from, to).catch(() => []);
724 |       const numVeh = parseInt(camion.patente, 10);
725 |       const truckCargas = fuelData.filter((c: any) =>
726 |         (!isNaN(numVeh) && c.numVeh === numVeh) || c.patente === camion.patente
727 |       );
728 |
729 |       const locations = truckCargas.map((c: any) => ({
730 |         fecha: c.fechaConsumo,
731 |         lugar: c.lugarConsumo,
732 |         litros: c.cantidadLt,
733 |         odometro: c.odometroActual,
734 |       }));
735 |
736 |       const client = getClient();
737 |       const response = await client.messages.create({
738 |         model: MODEL,
739 |         max_tokens: 2000,
740 |         messages: [{
741 |           role: "user",
742 |           content: `Eres un analista de rutas de camiones mineros en Chile. Analiza los puntos de carga de combustible d
743 |
744 | Genera un JSON con exactamente esta estructura (sin texto adicional, solo JSON):
745 | {
746 |   "puntos": [
```

```
747 |     {
748 |       "tipo": "COMBUSTIBLE|CARGA|ENTREGA|DESCANSO",
749 |       "nombreInferido": "Nombre del punto inferido (ej: Estacion Copec Antofagasta, Faena Spence, etc)",
750 |       "frecuencia": 5,
751 |       "litrosPromedio": 200
752 |     }
753 |   ],
754 |   "resumen_ruta": "Descripcion de la ruta habitual del camion"
755 | }
756 |
757 | Datos de cargas del camion N.${camion.patente}:
758 | ${locations.map((l: any) => `Fecha:${l.fecha} Lugar:${l.lugar} Litros:${l.litros} Odo:${l.odometro}`).join("\n")}`
759 |         }],
760 |       });
761 |
762 |       const text = response.content[0].type === "text" ? response.content[0].text : "";
763 |       const jsonMatch = text.match(/\{[\s\S]*\}/);
764 |       if (jsonMatch) {
765 |         const parsed = JSON.parse(jsonMatch[0]);
766 |         const savedPuntos = [];
767 |         for (const p of (parsed.puntos || [])) {
768 |           const saved = await storage.createPuntoRuta({
769 |             camionId,
770 |             lat: "0",
771 |             lng: "0",
772 |             tipo: p.tipo,
773 |             nombreInferido: p.nombreInferido,
774 |             duracionMin: null,
775 |             fecha: null,
776 |             confirmado: false,
777 |             confirmadoPor: null,
778 |           });
779 |           savedPuntos.push({ ...saved, frecuencia: p.frecuencia, litrosPromedio: p.litrosPromedio });
780 |         }
781 |         res.json({ puntos: savedPuntos, resumen: parsed.resumen_ruta });
782 |       } else {
783 |         res.json({ puntos: [], resumen: text });
784 |       }
785 |     } catch (error: any) {
786 |       console.error("[ia/detectar-puntos] Error:", error.message);
787 |       res.status(500).json({ message: error.message });
788 |     }
789 |   });
790 |
791 |   app.get("/api/ia/proyeccion-mes", async (_req: Request, res: Response) => {
792 |     try {
793 |       const context = await getFusionContext(14);
794 |       const client = getClient();
795 |       const response = await client.messages.create({
796 |         model: MODEL,
797 |         max_tokens: 1500,
798 |         messages: [{
799 |           role: "user",
800 |           content: `Eres un analista de flotas que hace proyecciones de consumo. Con los datos de los ultimos 14 dias, p
801 |
802 | Genera un JSON con exactamente esta estructura (sin texto adicional, solo JSON):
803 | {
804 |   "litros_14d": 0,
805 |   "litros_proyectados_mes": 0,
806 |   "camiones_empeorando": [
807 |     {"interno": "XXXX", "tendencia": "descripcion de la tendencia negativa", "rendimiento_actual": 0}
808 |   ],
809 |   "camiones_mejorando": [
810 |     {"interno": "XXXX", "tendencia": "descripcion positiva", "rendimiento_actual": 0}
811 |   ],
812 |   "proyeccion_texto": "Texto de 2-3 oraciones explicando la proyeccion"
813 | }
814 |
815 | Datos ultimos 14 dias:
816 | ${context}`
817 |         }],
818 |       });
819 |
820 |       const text = response.content[0].type === "text" ? response.content[0].text : "";
821 |       const jsonMatch = text.match(/\{[\s\S]*\}/);
822 |       if (jsonMatch) {
823 |         res.json(JSON.parse(jsonMatch[0]));
824 |       } else {
825 |         res.json({ proyeccion_texto: text });
826 |       }
827 |     } catch (error: any) {
828 |       console.error("[ia/proyeccion-mes] Error:", error.message);
829 |       res.status(500).json({ message: error.message });
830 |     }
831 |   });
832 |
833 |   function haversineKm(lat1: number, lng1: number, lat2: number, lng2: number): number {
834 |     const R = 6371;
835 |     const dLat = (lat2 - lat1) * Math.PI / 180;
836 |     const dLng = (lng2 - lng1) * Math.PI / 180;
837 |     const a = Math.sin(dLat / 2) ** 2 + Math.cos(lat1 * Math.PI / 180) * Math.cos(lat2 * Math.PI / 180) * Math.sin(dLng
838 |     return R * 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));
839 |   }
840 |
841 |   interface RouteCluster {
842 |     routeId: string;
843 |     camionId: number;
844 |     patente: string;
845 |     faena: string;
846 |     origenNombre: string;
847 |     destinoNombre: string;
848 |     origenLat: number;
849 |     origenLng: number;
850 |     destinoLat: number;
851 |     destinoLng: number;
852 |     viajes: {
853 |       id: number;
```

```
      codigo: string;
      fecha: string;
      kmRecorridos: number;
      litrosSigetra: number;
      litrosEcu: number;
      rendimiento: number;
      desviacionPct: number;
      esAnomalia: boolean;
    }[];
    totalViajes: number;
    promedioRendimiento: number;
    promedioKm: number;
    promedioLitros: number;
    desviacionMax: number;
    anomalias: number;
  }

  app.get("/api/ia/combustibles-reales", async (_req: Request, res: Response) => {
    try {
      const contratos = await storage.getTmsContratos();
      const camiones = await storage.getCamiones();
      const faenas = await storage.getFaenas();

      const camionMap = new Map(camiones.map(c => [c.id, c]));
      const faenaMap = new Map(faenas.map(f => [f.id, f]));

      let allTrips: any[] = [];
      for (const contrato of contratos) {
        const viajes = await storage.getTmsViajes(contrato.id);
        allTrips.push(...viajes);
      }

      const validTrips = allTrips.filter(v =>
        v.origenLat && v.origenLng && v.destinoLat && v.destinoLng &&
        Number(v.kmRecorridos) > 10 &&
        (Number(v.litrosEcu) > 0 || Number(v.litrosSigetra) > 0) &&
        Number(v.rendimientoReal) > 0 && Number(v.rendimientoReal) < 15
      );

      const tripsByCamion = new Map<number, typeof validTrips>();
      for (const trip of validTrips) {
        const arr = tripsByCamion.get(trip.camionId) || [];
        arr.push(trip);
        tripsByCamion.set(trip.camionId, arr);
      }

      const ROUTE_TOLERANCE_KM = 5;
      const ANOMALY_THRESHOLD_PCT = 20;
      const MIN_TRIPS_FOR_BASELINE = 3;

      const clusters: RouteCluster[] = [];
      let clusterIdx = 0;

      for (const [camionId, trips] of tripsByCamion) {
        const cam = camionMap.get(camionId);
        if (!cam) continue;
        const faena = faenaMap.get(cam.faenaId);

        const assigned: boolean[] = new Array(trips.length).fill(false);

        for (let i = 0; i < trips.length; i++) {
          if (assigned[i]) continue;

          const group = [trips[i]];
          assigned[i] = true;

          const oLat = Number(trips[i].origenLat);
          const oLng = Number(trips[i].origenLng);
          const dLat = Number(trips[i].destinoLat);
          const dLng = Number(trips[i].destinoLng);

          for (let j = i + 1; j < trips.length; j++) {
            if (assigned[j]) continue;
            const distOrig = haversineKm(oLat, oLng, Number(trips[j].origenLat), Number(trips[j].origenLng));
            const distDest = haversineKm(dLat, dLng, Number(trips[j].destinoLat), Number(trips[j].destinoLng));
            if (distOrig < ROUTE_TOLERANCE_KM && distDest < ROUTE_TOLERANCE_KM) {
              group.push(trips[j]);
              assigned[j] = true;
            }
          }

          if (group.length < MIN_TRIPS_FOR_BASELINE) continue;

          const rendimientos = group.map(v => Number(v.rendimientoReal));
          const avgRend = rendimientos.reduce((a, b) => a + b, 0) / rendimientos.length;
          const avgKm = group.reduce((a, v) => a + Number(v.kmRecorridos), 0) / group.length;
          const avgLitros = group.reduce((a, v) => a + Number(v.litrosEcu || v.litrosSigetra), 0) / group.length;

          const viajesDetail = group.map(v => {
            const rend = Number(v.rendimientoReal);
            const desvPct = avgRend > 0 ? ((rend - avgRend) / avgRend) * 100 : 0;
            return {
              id: v.id,
              codigo: v.codigo,
              fecha: v.fechaSalida ? new Date(v.fechaSalida).toISOString().split("T")[0] : "",
              kmRecorridos: Math.round(Number(v.kmRecorridos)),
              litrosSigetra: Math.round(Number(v.litrosSigetra || 0)),
              litrosEcu: Math.round(Number(v.litrosEcu || 0)),
              rendimiento: Math.round(rend * 100) / 100,
              desviacionPct: Math.round(desvPct * 10) / 10,
              esAnomalia: Math.abs(desvPct) > ANOMALY_THRESHOLD_PCT,
            };
          });

          viajesDetail.sort((a, b) => a.fecha.localeCompare(b.fecha));

          const anomalias = viajesDetail.filter(v => v.esAnomalia).length;
```

```
961 |           const maxDesv = Math.max(...viajesDetail.map(v => Math.abs(v.desviacionPct)));
962 |
963 |           clusterIdx++;
964 |           clusters.push({
965 |             routeId: `R-${cam.patente}-${String(clusterIdx).padStart(3, "0")}`,
966 |             camionId,
967 |             patente: cam.patente,
968 |             faena: faena?.nombre || "Sin asignar",
969 |             origenNombre: group[0].origenNombre || "Origen desconocido",
970 |             destinoNombre: group[0].destinoNombre || "Destino desconocido",
971 |             origenLat: oLat,
972 |             origenLng: oLng,
973 |             destinoLat: dLat,
974 |             destinoLng: dLng,
975 |             viajes: viajesDetail,
976 |             totalViajes: group.length,
977 |             promedioRendimiento: Math.round(avgRend * 100) / 100,
978 |             promedioKm: Math.round(avgKm),
979 |             promedioLitros: Math.round(avgLitros),
980 |             desviacionMax: Math.round(maxDesv * 10) / 10,
981 |             anomalias,
982 |           });
983 |         }
984 |       }
985 |
986 |       clusters.sort((a, b) => b.anomalias - a.anomalias || b.desviacionMax - a.desviacionMax);
987 |
988 |       const totalAnomalias = clusters.reduce((s, c) => s + c.anomalias, 0);
989 |       const totalViajes = clusters.reduce((s, c) => s + c.totalViajes, 0);
990 |       const rutasConAnomalias = clusters.filter(c => c.anomalias > 0).length;
991 |
992 |       res.json({
993 |         rutas: clusters,
994 |         resumen: {
995 |           totalRutas: clusters.length,
996 |           totalViajes,
997 |           totalAnomalias,
998 |           rutasConAnomalias,
999 |           camionesAnalizados: tripsByCamion.size,
1000 |           toleranciaKm: ROUTE_TOLERANCE_KM,
1001 |           umbralAnomaliaPct: ANOMALY_THRESHOLD_PCT,
1002 |           minimoViajes: MIN_TRIPS_FOR_BASELINE,
1003 |         },
1004 |       });
1005 |     } catch (error: any) {
1006 |       console.error("[ia/combustibles-reales] Error:", error.message);
1007 |       res.status(500).json({ message: error.message });
1008 |     }
1009 |   });
1010 |
1011 |   app.post("/api/ia/combustibles-reales/consulta", async (req: Request, res: Response) => {
1012 |     try {
1013 |       const { pregunta } = req.body;
1014 |       if (!pregunta || typeof pregunta !== "string") {
1015 |         return res.status(400).json({ message: "Se requiere campo 'pregunta'" });
1016 |       }
1017 |
1018 |       const contratos = await storage.getTmsContratos();
1019 |       const camiones = await storage.getCamiones();
1020 |       const faenas = await storage.getFaenas();
1021 |       const camionMap = new Map(camiones.map(c => [c.id, c]));
1022 |       const faenaMap = new Map(faenas.map(f => [f.id, f]));
1023 |
1024 |       let allTrips: any[] = [];
1025 |       for (const contrato of contratos) {
1026 |         const viajes = await storage.getTmsViajes(contrato.id);
1027 |         allTrips.push(...viajes);
1028 |       }
1029 |
1030 |       const validTrips = allTrips.filter(v =>
1031 |         Number(v.kmRecorridos) > 10 &&
1032 |         (Number(v.litrosEcu) > 0 || Number(v.litrosSigetra) > 0) &&
1033 |         Number(v.rendimientoReal) > 0
1034 |       );
1035 |
1036 |       const tripSummaries = validTrips.slice(0, 200).map(v => {
1037 |         const cam = camionMap.get(v.camionId);
1038 |         const faena = cam ? faenaMap.get(cam.faenaId) : null;
1039 |         return `Patente:${cam?.patente||"?"} Faena:${faena?.nombre||"?"} Ruta:${v.origenNombre||"?"}>${v.destinoNombre||
1040 |       });
1041 |
1042 |       const fusionCtx = await getFusionContext(14);
1043 |
1044 |       const contextStr = `VIAJES TMS DETECTADOS (${validTrips.length} total, mostrando ${tripSummaries.length}):\n${trip
1045 |
1046 |       const client = getClient();
1047 |       const response = await client.messages.create({
1048 |         model: MODEL,
1049 |         max_tokens: 2000,
1050 |         messages: [{
1051 |           role: "user",
1052 |           content: `Eres un analista experto en combustibles de flotas de camiones de la empresa Sotraser en Chile. Tu e
1053 |
1054 | REGLAS:
1055 | - Compara siempre el consumo de un camion con SU PROPIO historial en la MISMA ruta (origen-destino similar)
1056 | - Minimo 3 viajes en la misma ruta para establecer un baseline confiable
1057 | - Una desviacion mayor al 20% del baseline propio es sospechosa
1058 | - Considera diferencias entre litros Surtidor (manual) y litros ECU (telematica) como indicador de posible fraude
1059 | - Responde en espanol, con datos concretos (patentes, rutas, litros, porcentajes)
1060 | - No menciones costos ni valores en CLP
1061 | - Se directo y conciso
1062 |
1063 | Pregunta del usuario: ${pregunta}
1064 |
1065 | Datos disponibles:
1066 | ${contextStr}`
1067 |         }],
```

```
      });

      const text = response.content[0].type === "text" ? response.content[0].text : "";
      res.json({ respuesta: text });
    } catch (error: any) {
      console.error("[ia/combustibles-reales/consulta] Error:", error.message);
      res.status(500).json({ message: error.message });
    }
  });
}

```

server/volvo-api.ts

283 lineas

```
1 | const RFMS_BASE = "https://api.volvotrucks.com/rfms";
2 |
3 | function getAuthHeader(): string {
4 |   const user = process.env.VOLVO_CONNECT_USER;
5 |   const pass = process.env.VOLVO_CONNECT_PASSWORD;
6 |   if (!user || !pass) throw new Error("Volvo Connect credentials not configured");
7 |   return "Basic " + Buffer.from(`${user}:${pass}`).toString("base64");
8 | }
9 |
10 | async function rfmsGet<T>(path: string, accept: string, params?: Record<string, string>): Promise<T> {
11 |   const url = new URL(`${RFMS_BASE}${path}`);
12 |   if (params) {
13 |     for (const [k, v] of Object.entries(params)) {
14 |       if (v) url.searchParams.set(k, v);
15 |     }
16 |   }
17 |
18 |   console.log(`[volvo-api] GET ${url.toString()}`);
19 |   const controller = new AbortController();
20 |   const timeout = setTimeout(() => controller.abort(), 15000);
21 |
22 |   try {
23 |     const res = await fetch(url.toString(), {
24 |       headers: {
25 |         Authorization: getAuthHeader(),
26 |         Accept: accept,
27 |       },
28 |       signal: controller.signal,
29 |     });
30 |
31 |     if (!res.ok) {
32 |       const body = await res.text();
33 |       console.error(`[volvo-api] ${res.status} ${res.statusText}: ${body}`);
34 |       throw new Error(`Volvo API ${res.status}: ${res.statusText}`);
35 |     }
36 |
37 |     return res.json() as Promise<T>;
38 |   } catch (err: any) {
39 |     if (err.name === "AbortError") {
40 |       throw new Error("Volvo API timeout (15s)");
41 |     }
42 |     throw err;
43 |   } finally {
44 |     clearTimeout(timeout);
45 |   }
46 | }
47 |
48 | export interface RfmsVehicle {
49 |   VIN: string;
50 |   CustomerVehicleName?: string;
51 |   Brand?: string;
52 |   Model?: string;
53 |   Type?: string;
54 |   ChassisType?: string;
55 |   BodyType?: string;
56 |   EmissionLevel?: string;
57 |   NoOfAxles?: number;
58 |   TotalFuelTankVolume?: number;
59 |   GearboxType?: string;
60 |   PossibleFuelType?: string[];
61 |   ProductionDate?: { Year?: number; Month?: number; Day?: number };
62 | }
63 |
64 | export interface RfmsVehiclesResponse {
65 |   Vehicle?: RfmsVehicle[];
66 |   MoreDataAvailable?: boolean;
67 | }
68 |
69 | export interface RfmsGNSSPosition {
70 |   Latitude?: number;
71 |   Longitude?: number;
72 |   Altitude?: number;
73 |   Heading?: number;
74 |   Speed?: number;
75 |   PositionDateTime?: string;
76 | }
77 |
78 | export interface RfmsSnapshot {
79 |   FuelLevel1?: number;
80 |   EngineSpeed?: number;
81 |   WheelBasedSpeed?: number;
82 |   TachographSpeed?: number;
83 |   AmbientAirTemperature?: number;
84 |   CatalystFuelLevel?: number;
85 |   GNSSPosition?: RfmsGNSSPosition;
86 |   Driver1WorkingState?: string;
87 | }
88 |
89 | export interface RfmsVehicleStatus {
90 |   Vin: string;
91 |   CreatedDateTime: string;
92 |   ReceivedDateTime: string;
93 |   HRTotalVehicleDistance?: number;
94 |   TotalEngineHours?: number;
95 |   EngineTotalFuelUsed?: number;
96 |   GrossCombinationVehicleWeight?: number;
97 |   Driver1Id?: { OemDriverIdentification?: { IdType?: string; DriverIdentification?: string } };
98 |   SnapshotData?: RfmsSnapshot;
99 |   TriggerType?: { TriggerType: string; Context: string };
100 | }
101 |
102 | export interface RfmsVehicleStatusesResponse {
103 |   VehicleStatus?: RfmsVehicleStatus[];
104 |   MoreDataAvailable?: boolean;
```

```
105 |   RequestServerDateTime?: string;
106 | }
107 |
108 | export interface RfmsVehiclePosition {
109 |   VIN: string;
110 |   CreatedDateTime: string;
111 |   ReceivedDateTime: string;
112 |   GNSSPosition?: RfmsGNSSPosition;
113 |   WheelBasedSpeed?: number;
114 |   TachographSpeed?: number;
115 |   TriggerType?: { TriggerType: string; Context: string };
116 | }
117 |
118 | export interface RfmsVehiclePositionsResponse {
119 |   VehiclePosition?: RfmsVehiclePosition[];
120 |   MoreDataAvailable?: boolean;
121 |   RequestServerDateTime?: string;
122 | }
123 |
124 | export async function getVehicles(): Promise<RfmsVehicle[]> {
125 |   const allVehicles: RfmsVehicle[] = [];
126 |   let lastVin: string | undefined;
127 |   let hasMore = true;
128 |   const MAX_PAGES = 50;
129 |   let page = 0;
130 |
131 |   while (hasMore && page < MAX_PAGES) {
132 |     const params: Record<string, string> = {};
133 |     if (lastVin) params.lastVin = lastVin;
134 |
135 |     const data = await rfmsGet<RfmsVehiclesResponse>(
136 |       "/vehicles",
137 |       "application/vnd.fmsstandard.com.Vehicles.v2.1+json",
138 |       params
139 |     );
140 |
141 |     if (!data.Vehicle || data.Vehicle.length === 0) break;
142 |
143 |     const newLastVin = data.Vehicle[data.Vehicle.length - 1].VIN;
144 |     if (newLastVin === lastVin) break;
145 |
146 |     allVehicles.push(...data.Vehicle);
147 |     lastVin = newLastVin;
148 |     hasMore = data.MoreDataAvailable === true;
149 |     page++;
150 |   }
151 |
152 |   console.log(`[volvo-api] Fetched ${allVehicles.length} vehicles in ${page} pages`);
153 |   return allVehicles;
154 | }
155 |
156 | export async function getVehicleStatuses(vin?: string, latestOnly = true): Promise<RfmsVehicleStatus[]> {
157 |   const params: Record<string, string> = {};
158 |   if (vin) params.vin = vin;
159 |   if (latestOnly) params.latestOnly = "true";
160 |
161 |   const data = await rfmsGet<RfmsVehicleStatusesResponse>(
162 |     "/vehiclestatuses",
163 |     "application/vnd.fmsstandard.com.Vehiclestatuses.v2.1+json",
164 |     params
165 |   );
166 |
167 |   return data.VehicleStatus || [];
168 | }
169 |
170 | export async function getVehiclePositions(vin?: string, latestOnly = true): Promise<RfmsVehiclePosition[]> {
171 |   const params: Record<string, string> = {};
172 |   if (vin) params.vin = vin;
173 |   if (latestOnly) params.latestOnly = "true";
174 |
175 |   const data = await rfmsGet<RfmsVehiclePositionsResponse>(
176 |     "/vehiclepositions",
177 |     "application/vnd.fmsstandard.com.Vehiclepositions.v2.1+json",
178 |     params
179 |   );
180 |
181 |   return data.VehiclePosition || [];
182 | }
183 |
184 | export interface UnifiedVehicleStatus {
185 |   vin: string;
186 |   createdDateTime: string | null;
187 |   gps: {
188 |     latitude: number | null;
189 |     longitude: number | null;
190 |     altitude: number | null;
191 |     heading: number | null;
192 |     speed: number | null;
193 |     positionDateTime: string | null;
194 |   };
195 |   fuelLevel: number | null;
196 |   engineHours: number | null;
197 |   totalDistance: number | null;
198 |   totalFuelUsed: number | null;
199 |   engineSpeed: number | null;
200 |   wheelBasedSpeed: number | null;
201 |   ambientTemperature: number | null;
202 |   driverId: string | null;
203 |   driverWorkingState: string | null;
204 |   catalystFuelLevel: number | null;
205 |   grossWeight: number | null;
206 | }
207 |
208 | function buildUnifiedStatus(
209 |   status: RfmsVehicleStatus | undefined,
210 |   position: RfmsVehiclePosition | undefined,
211 |   vin: string
```

```
212 | ): UnifiedVehicleStatus {
213 |   const snap = status?.SnapshotData;
214 |   const gpsFromStatus = snap?.GNSSPosition;
215 |   const gpsFromPosition = position?.GNSSPosition;
216 |   const gps = gpsFromStatus || gpsFromPosition;
217 |
218 |   return {
219 |     vin,
220 |     createdDateTime: status?.CreatedDateTime || position?.CreatedDateTime || null,
221 |     gps: {
222 |       latitude: gps?.Latitude ?? null,
223 |       longitude: gps?.Longitude ?? null,
224 |       altitude: gps?.Altitude ?? null,
225 |       heading: gps?.Heading ?? null,
226 |       speed: gps?.Speed ?? null,
227 |       positionDateTime: gps?.PositionDateTime ?? null,
228 |     },
229 |     fuelLevel: snap?.FuelLevel1 ?? null,
230 |     engineHours: status?.TotalEngineHours ?? null,
231 |     totalDistance: status?.HRTotalVehicleDistance ?? null,
232 |     totalFuelUsed: status?.EngineTotalFuelUsed ?? null,
233 |     engineSpeed: snap?.EngineSpeed ?? null,
234 |     wheelBasedSpeed: snap?.WheelBasedSpeed ?? position?.WheelBasedSpeed ?? null,
235 |     ambientTemperature: snap?.AmbientAirTemperature ?? null,
236 |     driverId: status?.Driver1Id?.OemDriverIdentification?.DriverIdentification ?? null,
237 |     driverWorkingState: snap?.Driver1WorkingState ?? null,
238 |     catalystFuelLevel: snap?.CatalystFuelLevel ?? null,
239 |     grossWeight: status?.GrossCombinationVehicleWeight ?? null,
240 |   };
241 | }
242 |
243 | let fleetCache: { data: UnifiedVehicleStatus[]; timestamp: number } | null = null;
244 | const FLEET_CACHE_TTL = 5 * 60 * 1000;
245 |
246 | export async function getFleetStatus(): Promise<UnifiedVehicleStatus[]> {
247 |   if (fleetCache && Date.now() - fleetCache.timestamp < FLEET_CACHE_TTL) {
248 |     console.log("[volvo-api] Returning cached fleet status");
249 |     return fleetCache.data;
250 |   }
251 |
252 |   const [statuses, positions] = await Promise.all([
253 |     getVehicleStatuses(undefined, true),
254 |     getVehiclePositions(undefined, true),
255 |   ]);
256 |
257 |   const statusMap = new Map(statuses.map(s => [s.Vin, s]));
258 |   const positionMap = new Map(positions.map(p => [p.VIN, p]));
259 |
260 |   const allVins = new Set(Array.from(statusMap.keys()).concat(Array.from(positionMap.keys())));
261 |   const result: UnifiedVehicleStatus[] = [];
262 |
263 |   for (const vin of Array.from(allVins)) {
264 |     result.push(buildUnifiedStatus(statusMap.get(vin), positionMap.get(vin), vin));
265 |   }
266 |
267 |   fleetCache = { data: result, timestamp: Date.now() };
268 |   console.log(`[volvo-api] Fleet status cached: ${result.length} vehicles`);
269 |   return result;
270 | }
271 |
272 | export async function getSingleVehicleStatus(vin: string): Promise<UnifiedVehicleStatus> {
273 |   const [statuses, positions] = await Promise.all([
274 |     getVehicleStatuses(vin, true),
275 |     getVehiclePositions(vin, true),
276 |   ]);
277 |
278 |   const status = statuses.find(s => s.Vin === vin);
279 |   const position = positions.find(p => p.VIN === vin);
280 |
281 |   return buildUnifiedStatus(status, position, vin);
282 | }
283 |
```

server/sigetra-api.ts

363 lineas

```
1 | const SIGETRA_BASE = process.env.SIGETRA_URL || "http://sigetraweb.sotraser.cl";
2 |
3 | export interface SigetraFuelRecord {
4 |   numGuia: number;
5 |   fechaConsumo: string;
6 |   patente: string;
7 |   numVeh: number | null;
8 |   grupoVeh: string | null;
9 |   tipoVeh: string | null;
10 |   marca: string | null;
11 |   modelo: string | null;
12 |   yearFab: number | null;
13 |   faena: string | null;
14 |   subfaena: string | null;
15 |   runConductor: string | null;
16 |   nombreConductor: string | null;
17 |   tipoTarjeta: string | null;
18 |   numTarjeta: number | null;
19 |   lugarConsumo: string | null;
20 |   cantidadLt: number;
21 |   odometroActual: number | null;
22 |   odometroPrevio: number | null;
23 |   kmRecorrido: number | null;
24 |   factorAjuste: number | null;
25 |   kmAjustado: number | null;
26 |   rendReal: number | null;
27 |   rendEsperado: number | null;
28 |   desviacion: number | null;
29 |   tipoRend: string | null;
30 |   producto: string | null;
31 |   observacion: string | null;
32 |   difdia: number | null;
33 | }
34 |
35 | export interface SigetraFuelSummary {
36 |   patente: string;
37 |   numVeh: number | null;
38 |   grupoVeh: string | null;
39 |   tipoVeh: string | null;
40 |   marca: string | null;
41 |   modelo: string | null;
42 |   yearFab: number | null;
43 |   faenaPri: string | null;
44 |   cantidadLt: number;
45 |   odometroMin: number | null;
46 |   odometroMax: number | null;
47 |   kmRecorrido: number | null;
48 |   kmAjustado: number | null;
49 |   rendReal: number | null;
50 |   rendEsperado: number | null;
51 |   desviacion: number | null;
52 |   tipoRend: string | null;
53 |   producto: string | null;
54 |   cantidadLtAdb: number | null;
55 | }
56 |
57 | let sessionCookie: string | null = null;
58 | let sessionExpiry = 0;
59 | const SESSION_TTL = 30 * 60 * 1000;
60 |
61 | async function sigetraLogin(): Promise<string> {
62 |   const user = process.env.SIGETRA_USER;
63 |   const pass = process.env.SIGETRA_PASSWORD;
64 |   if (!user || !pass) throw new Error("Sigetra credentials not configured");
65 |
66 |   console.log("[sigetra] Authenticating...");
67 |
68 |   const validateRes = await fetch(
69 |     `${SIGETRA_BASE}/modulos/seguridad/ui_seg_login_backend.aspx/ValidateUser`,
70 |     {
71 |       method: "POST",
72 |       headers: { "Content-Type": "application/json; charset=utf-8" },
73 |       body: JSON.stringify({ username: user, password: pass }),
74 |     }
75 |   );
76 |
77 |   if (!validateRes.ok) throw new Error(`Sigetra login failed: ${validateRes.status}`);
78 |
79 |   const validateData: any = await validateRes.json();
80 |   const parsed = JSON.parse(validateData.d);
81 |   if (parsed.State !== "OK") throw new Error(`Sigetra auth rejected: ${parsed.Message}`);
82 |
83 |   const redirectRes = await fetch(
84 |     `${SIGETRA_BASE}/modulos/seguridad/ui_seg_login_redirect.aspx`,
85 |     {
86 |       method: "POST",
87 |       headers: { "Content-Type": "application/x-www-form-urlencoded" },
88 |       body: `username=${encodeURIComponent(user)}&returnurl=../../ui_menu.aspx`,
89 |       redirect: "manual",
90 |     }
91 |   );
92 |
93 |   let authCookie = "";
94 |   const redirectCookies = redirectRes.headers.getSetCookie?.() || [];
95 |   for (const c of redirectCookies) {
96 |     if (c.includes(".ASPXAUTH")) {
97 |       authCookie = c.split(";")[0];
98 |       break;
99 |     }
100 |   }
101 |
102 |   if (!authCookie) {
103 |     const rawHeader = redirectRes.headers.get("set-cookie") || "";
104 |     const match = rawHeader.match(/\.ASPXAUTH=([^;]+)/);
```

```
105 |     if (match && match[1].length > 10) authCookie = `.ASPXAUTH=${match[1]}`;
106 |   }
107 |
108 |   if (!authCookie || authCookie.length < 20) throw new Error("Sigetra: no auth cookie received");
109 |
110 |   console.log("[sigetra] Authenticated successfully");
111 |   return authCookie;
112 | }
113 |
114 | async function getSession(): Promise<string> {
115 |   if (sessionCookie && Date.now() < sessionExpiry) return sessionCookie;
116 |   sessionCookie = await sigetraLogin();
117 |   sessionExpiry = Date.now() + SESSION_TTL;
118 |   return sessionCookie;
119 | }
120 |
121 | async function sigetraPost<T>(url: string, data: Record<string, string>): Promise<T> {
122 |   const cookie = await getSession();
123 |
124 |   const res = await fetch(`${SIGETRA_BASE}${url}`, {
125 |     method: "POST",
126 |     headers: {
127 |       "Content-Type": "application/json; charset=utf-8",
128 |       Cookie: cookie,
129 |     },
130 |     body: JSON.stringify(data),
131 |   });
132 |
133 |   if (res.status === 302 || res.status === 401) {
134 |     console.log("[sigetra] Session expired, re-authenticating...");
135 |     sessionCookie = null;
136 |     sessionExpiry = 0;
137 |     const newCookie = await getSession();
138 |     const retryRes = await fetch(`${SIGETRA_BASE}${url}`, {
139 |       method: "POST",
140 |       headers: {
141 |         "Content-Type": "application/json; charset=utf-8",
142 |         Cookie: newCookie,
143 |       },
144 |       body: JSON.stringify(data),
145 |     });
146 |     if (!retryRes.ok) throw new Error(`Sigetra retry failed: ${retryRes.status}`);
147 |     const retryData: any = await retryRes.json();
148 |     return JSON.parse(retryData.d) as T;
149 |   }
150 |
151 |   if (!res.ok) {
152 |     const text = await res.text();
153 |     throw new Error(`Sigetra request failed: ${res.status} - ${text.slice(0, 200)}`);
154 |   }
155 |
156 |   const responseData: any = await res.json();
157 |   return JSON.parse(responseData.d) as T;
158 | }
159 |
160 | function formatSigetraDate(date: Date): string {
161 |   const d = date.getDate().toString().padStart(2, "0");
162 |   const m = (date.getMonth() + 1).toString().padStart(2, "0");
163 |   const y = date.getFullYear();
164 |   return `${d}-${m}-${y}`;
165 | }
166 |
167 | interface RawFuelRecord {
168 |   num_guia: number;
169 |   fechahr_consumo: string;
170 |   patente: string;
171 |   num_veh: number | null;
172 |   grupo_veh: string | null;
173 |   tipo_veh: string | null;
174 |   marca: string | null;
175 |   modelo: string | null;
176 |   year_fab: number | null;
177 |   faena: string | null;
178 |   subfaena: string | null;
179 |   run_conductor: string | null;
180 |   nombre_conductor: string | null;
181 |   tipo_tarjeta: string | null;
182 |   num_tarjeta: number | null;
183 |   lugar_consumo: string | null;
184 |   cantidad_lt: number;
185 |   odometro_actual: number | null;
186 |   odometro_previo: number | null;
187 |   km_recorrido: number | null;
188 |   factor_ajuste: number | null;
189 |   km_ajustado: number | null;
190 |   rend_real: number | null;
191 |   rend_esperado: number | null;
192 |   desviacion: number | null;
193 |   tipo_rend: string | null;
194 |   producto: string | null;
195 |   observacion: string | null;
196 |   difdia: number | null;
197 | }
198 |
199 | interface RawFuelSummary {
200 |   patente: string;
201 |   num_veh: number | null;
202 |   grupo_veh: string | null;
203 |   tipo_veh: string | null;
204 |   marca: string | null;
205 |   modelo: string | null;
206 |   year_fab: number | null;
207 |   faena_pri: string | null;
208 |   cantidad_lt: number;
209 |   odometro_min: number | null;
210 |   odometro_max: number | null;
211 |   km_recorrido: number | null;
```

```
212 |   km_ajustado: number | null;
213 |   rend_real: number | null;
214 |   rend_esperado: number | null;
215 |   desviacion: number | null;
216 |   tipo_rend: string | null;
217 |   producto: string | null;
218 |   cantidad_lt_adb: number | null;
219 | }
220 |
221 | function mapFuelRecord(r: RawFuelRecord): SigetraFuelRecord {
222 |   return {
223 |     numGuia: r.num_guia,
224 |     fechaConsumo: r.fechahr_consumo,
225 |     patente: r.patente,
226 |     numVeh: r.num_veh,
227 |     grupoVeh: r.grupo_veh,
228 |     tipoVeh: r.tipo_veh,
229 |     marca: r.marca,
230 |     modelo: r.modelo,
231 |     yearFab: r.year_fab,
232 |     faena: r.faena,
233 |     subfaena: r.subfaena,
234 |     runConductor: r.run_conductor,
235 |     nombreConductor: r.nombre_conductor,
236 |     tipoTarjeta: r.tipo_tarjeta,
237 |     numTarjeta: r.num_tarjeta,
238 |     lugarConsumo: r.lugar_consumo,
239 |     cantidadLt: r.cantidad_lt,
240 |     odometroActual: r.odometro_actual,
241 |     odometroPrevio: r.odometro_previo,
242 |     kmRecorrido: r.km_recorrido,
243 |     factorAjuste: r.factor_ajuste,
244 |     kmAjustado: r.km_ajustado,
245 |     rendReal: r.rend_real,
246 |     rendEsperado: r.rend_esperado,
247 |     desviacion: r.desviacion,
248 |     tipoRend: r.tipo_rend,
249 |     producto: r.producto,
250 |     observacion: r.observacion,
251 |     difdia: r.difdia,
252 |   };
253 | }
254 |
255 | function mapFuelSummary(r: RawFuelSummary): SigetraFuelSummary {
256 |   return {
257 |     patente: r.patente,
258 |     numVeh: r.num_veh,
259 |     grupoVeh: r.grupo_veh,
260 |     tipoVeh: r.tipo_veh,
261 |     marca: r.marca,
262 |     modelo: r.modelo,
263 |     yearFab: r.year_fab,
264 |     faenaPri: r.faena_pri,
265 |     cantidadLt: r.cantidad_lt,
266 |     odometroMin: r.odometro_min,
267 |     odometroMax: r.odometro_max,
268 |     kmRecorrido: r.km_recorrido,
269 |     kmAjustado: r.km_ajustado,
270 |     rendReal: r.rend_real,
271 |     rendEsperado: r.rend_esperado,
272 |     desviacion: r.desviacion,
273 |     tipoRend: r.tipo_rend,
274 |     producto: r.producto,
275 |     cantidadLtAdb: r.cantidad_lt_adb,
276 |   };
277 | }
278 |
279 | export async function getSigetraFuelData(
280 |   from: Date,
281 |   to: Date,
282 |   filters?: { codVeh?: string; codFaena?: string; producto?: string }
283 | ): Promise<SigetraFuelRecord[]> {
284 |   const fechaIni = formatSigetraDate(from) + " 00:00";
285 |   const fechaFin = formatSigetraDate(to) + " 23:59";
286 |
287 |   console.log(`[sigetra] Fetching fuel data ${fechaIni} - ${fechaFin}`);
288 |
289 |   const raw = await sigetraPost<RawFuelRecord[]>(
290 |     "/modulos/combustible/ui_cmb_consumo_combustible_backend.aspx/RetrieveConsultaConsumo",
291 |     {
292 |       fechaIni,
293 |       fechaFin,
294 |       codVeh: filters?.codVeh || "",
295 |       grupoVeh: "0",
296 |       tipoVeh: "0",
297 |       codFaena: filters?.codFaena || "0",
298 |       conductor: "",
299 |       producto: filters?.producto || "DIESEL",
300 |       tipoConsumo: "",
301 |     }
302 |   );
303 |
304 |   console.log(`[sigetra] Got ${raw.length} fuel records`);
305 |   return raw.map(mapFuelRecord);
306 | }
307 |
308 | export async function getSigetraFuelSummary(
309 |   from: Date,
310 |   to: Date,
311 |   filters?: { codVeh?: string; codFaena?: string; producto?: string }
312 | ): Promise<SigetraFuelSummary[]> {
313 |   const fechaIni = formatSigetraDate(from) + " 00:00";
314 |   const fechaFin = formatSigetraDate(to) + " 23:59";
315 |
316 |   console.log(`[sigetra] Fetching fuel summary ${fechaIni} - ${fechaFin}`);
317 |
318 |   const raw = await sigetraPost<RawFuelSummary[]>(
```

```
319 |     "/modulos/combustible/ui_cmb_consumo_combustible_resumen_backend.aspx/RetrieveConsultaConsumoResumen",
320 |     {
321 |       fechaIni,
322 |       fechaFin,
323 |       codVeh: filters?.codVeh || "",
324 |       grupoVeh: "0",
325 |       tipoVeh: "0",
326 |       codFaena: filters?.codFaena || "0",
327 |       producto: filters?.producto || "DIESEL",
328 |     }
329 |   );
330 |
331 |   console.log(`[sigetra] Got ${raw.length} summary records`);
332 |   return raw.map(mapFuelSummary);
333 | }
334 |
335 | export async function checkSigetraConnection(): Promise<{ connected: boolean; message: string; user: string }> {
336 |   try {
337 |     const user = process.env.SIGETRA_USER;
338 |     if (!user || !process.env.SIGETRA_PASSWORD) {
339 |       return { connected: false, message: "Credenciales Sigetra no configuradas", user: "" };
340 |     }
341 |     await getSession();
342 |     return { connected: true, message: "Conectado a Sigetra Web", user: user };
343 |   } catch (err: any) {
344 |     return { connected: false, message: err.message || "Error de conexion", user: process.env.SIGETRA_USER || "" };
345 |   }
346 | }
347 |
348 | let fuelDataCache: { data: SigetraFuelRecord[]; from: string; to: string; timestamp: number } | null = null;
349 | const FUEL_CACHE_TTL = 10 * 60 * 1000;
350 |
351 | export async function getCachedFuelData(from: Date, to: Date): Promise<SigetraFuelRecord[]> {
352 |   const fromStr = from.toISOString().slice(0, 10);
353 |   const toStr = to.toISOString().slice(0, 10);
354 |
355 |   if (fuelDataCache && fuelDataCache.from === fromStr && fuelDataCache.to === toStr && Date.now() - fuelDataCache.timest
356 |     return fuelDataCache.data;
357 |   }
358 |
359 |   const data = await getSigetraFuelData(from, to);
360 |   fuelDataCache = { data, from: fromStr, to: toStr, timestamp: Date.now() };
361 |   return data;
362 | }
363 |
```

server/faena-filter.ts

121 lineas

```
1 | import { storage } from "./storage";
2 | import { pool } from "./db";
3 |
4 | export interface ContractConfig {
5 |   name: string;
6 |   faenaIds: number[];
7 |   faenaNames: string[];
8 | }
9 |
10 | let cachedFaenas: Map<number, string> = new Map();
11 | let cachedCamiones: any[] = [];
12 | let cacheTime = 0;
13 | const CACHE_TTL = 60_000;
14 |
15 | async function ensureCache() {
16 |   if (Date.now() - cacheTime < CACHE_TTL && cachedCamiones.length > 0) return;
17 |   const faenas = await storage.getFaenas();
18 |   cachedFaenas = new Map(faenas.map(f => [f.id, f.nombre]));
19 |   cachedCamiones = await storage.getCamiones();
20 |   cacheTime = Date.now();
21 | }
22 |
23 | export async function getContractConfig(contractName: string): Promise<ContractConfig> {
24 |   await ensureCache();
25 |   const faenas = await storage.getFaenas();
26 |
27 |   if (contractName.toUpperCase() === "ANGLO") {
28 |     const angloFaenas = faenas.filter(f =>
29 |       f.nombre.toUpperCase().includes("ANGLO")
30 |     );
31 |     return {
32 |       name: "ANGLO",
33 |       faenaIds: angloFaenas.map(f => f.id),
34 |       faenaNames: angloFaenas.map(f => f.nombre),
35 |     };
36 |   }
37 |
38 |   const faena = faenas.find(f =>
39 |     f.nombre.toUpperCase().includes(contractName.toUpperCase())
40 |   );
41 |   if (!faena) throw new Error(`Contrato ${contractName} no encontrado`);
42 |   return {
43 |     name: contractName.toUpperCase(),
44 |     faenaIds: [faena.id],
45 |     faenaNames: [faena.nombre],
46 |   };
47 | }
48 |
49 | export async function getContractPatentes(contractName: string): Promise<Set<string>> {
50 |   await ensureCache();
51 |   const config = await getContractConfig(contractName);
52 |   const faenaIdSet = new Set(config.faenaIds);
53 |   return new Set(cachedCamiones.filter(c => faenaIdSet.has(c.faenaId)).map(c => c.patente));
54 | }
55 |
56 | export async function getContractCamiones(contractName: string): Promise<any[]> {
57 |   await ensureCache();
58 |   const config = await getContractConfig(contractName);
59 |   const faenaIdSet = new Set(config.faenaIds);
60 |   return cachedCamiones.filter(c => faenaIdSet.has(c.faenaId));
61 | }
62 |
63 | export async function getFaenaPatentes(faenaId: number): Promise<Set<string>> {
64 |   await ensureCache();
65 |   return new Set(cachedCamiones.filter(c => c.faenaId === faenaId).map(c => c.patente));
66 | }
67 |
68 | export async function getAllContracts(): Promise<{ name: string; faenas: { id: number; nombre: string }[]; camiones: num
69 |   await ensureCache();
70 |   const faenas = await storage.getFaenas();
71 |
72 |   const angloFaenas = faenas.filter(f => f.nombre.toUpperCase().includes("ANGLO"));
73 |   const angloIds = angloFaenas.map(f => f.id);
74 |
75 |   let angloCamionesVolvo = 0;
76 |   if (angloIds.length > 0) {
77 |     const fp = angloIds.map((_, i) => `$${i + 1}`).join(",");
78 |     const result = await pool.query(`
79 |       SELECT COUNT(DISTINCT c.patente) as cnt FROM tms_viajes v
80 |       JOIN camiones c ON v.camion_id = c.id
81 |       WHERE c.faena_id IN (${fp}) AND v.fecha_salida >= '2026-03-01'
82 |     `, angloIds);
83 |     angloCamionesVolvo = parseInt(result.rows[0]?.cnt) || 0;
84 |   }
85 |
86 |   const cencosudFaena = faenas.find(f => f.nombre.toUpperCase().includes("CENCOSUD"));
87 |   let cencosudCamiones = 0;
88 |   if (cencosudFaena) {
89 |     const cResult = await pool.query(`
90 |       SELECT COUNT(DISTINCT c.patente) as cnt FROM tms_viajes v
91 |       JOIN camiones c ON v.camion_id = c.id
92 |       WHERE c.faena_id = $1 AND v.fecha_salida >= '2026-03-01'
93 |     `, [cencosudFaena.id]);
94 |     cencosudCamiones = parseInt(cResult.rows[0]?.cnt) || 0;
95 |   }
96 |
97 |   const contracts = [];
98 |
99 |   if (cencosudFaena && cencosudCamiones > 0) {
100 |     contracts.push({
101 |       name: "CENCOSUD",
102 |       faenas: [{ id: cencosudFaena.id, nombre: cencosudFaena.nombre }],
103 |       camiones: cencosudCamiones,
104 |     });
```

```
105 |   }
106 |
107 |   if (angloCamionesVolvo > 0) {
108 |     contracts.push({
109 |       name: "ANGLO",
110 |       faenas: angloFaenas.map(f => ({ id: f.id, nombre: f.nombre })),
111 |       camiones: angloCamionesVolvo,
112 |     });
113 |   }
114 |
115 |   return contracts;
116 | }
117 |
118 | export function invalidateCache() {
119 |   cacheTime = 0;
120 | }
121 |
```

server/cencosud-filter.ts

52 lineas

```
1 | import { storage } from "./storage";
2 |
3 | const CENCOSUD_FAENA_NAME = "CENCOSUD";
4 |
5 | let cachedFaenaId: number | null = null;
6 | let cachedPatentes: Set<string> | null = null;
7 | let cachedCamionIds: Set<number> | null = null;
8 | let cacheTime = 0;
9 | const CACHE_TTL = 60_000;
10 |
11 | async function ensureCache() {
12 |   if (cachedFaenaId && Date.now() - cacheTime < CACHE_TTL) return;
13 |   const faenas = await storage.getFaenas();
14 |   const faena = faenas.find(f => f.nombre.toUpperCase().includes(CENCOSUD_FAENA_NAME));
15 |   if (!faena) throw new Error("Faena CENCOSUD no encontrada");
16 |   cachedFaenaId = faena.id;
17 |   const camiones = await storage.getCamiones();
18 |   const filtered = camiones.filter(c => c.faenaId === faena.id);
19 |   cachedPatentes = new Set(filtered.map(c => c.patente));
20 |   cachedCamionIds = new Set(filtered.map(c => c.id));
21 |   cacheTime = Date.now();
22 | }
23 |
24 | export async function getCencosudFaenaId(): Promise<number> {
25 |   await ensureCache();
26 |   return cachedFaenaId!;
27 | }
28 |
29 | export async function getCencosudPatentes(): Promise<Set<string>> {
30 |   await ensureCache();
31 |   return cachedPatentes!;
32 | }
33 |
34 | export async function getCencosudCamionIds(): Promise<Set<number>> {
35 |   await ensureCache();
36 |   return cachedCamionIds!;
37 | }
38 |
39 | export function filterCamiones<T extends { faenaId: number | null }>(camiones: T[], faenaId: number): T[] {
40 |   return camiones.filter(c => c.faenaId === faenaId);
41 | }
42 |
43 | export function filterFuelByPatentes(fuelData: any[], patentes: Set<string>): any[] {
44 |   return fuelData.filter((r: any) => {
45 |     return patentes.has(String(r.numVeh || ""));
46 |   });
47 | }
48 |
49 | export function filterFleetByVins(fleet: any[], vins: Set<string>): any[] {
50 |   return fleet.filter((v: any) => vins.has(v.vin));
51 | }
52 |
```

server/geo-reconstruccion-service.ts

691 lineas

```
1 | import { db } from "./db";
2 | import { geoPuntos, geoViajes, geoGeocache, geoLugares, camiones, cargas } from "@shared/schema";
3 | import { eq, and, gte, lte, sql, desc, asc } from "drizzle-orm";
4 | import { getCencosudFaenaId } from "./cencosud-filter";
5 |
6 | function haversineKm(lat1: number, lon1: number, lat2: number, lon2: number): number {
7 |   const R = 6371;
8 |   const dLat = (lat2 - lat1) * Math.PI / 180;
9 |   const dLon = (lon2 - lon1) * Math.PI / 180;
10 |   const a = Math.sin(dLat / 2) ** 2 +
11 |     Math.cos(lat1 * Math.PI / 180) * Math.cos(lat2 * Math.PI / 180) *
12 |     Math.sin(dLon / 2) ** 2;
13 |   return R * 2 * Math.asin(Math.sqrt(a));
14 | }
15 |
16 | interface PuntoGPS {
17 |   lat: number;
18 |   lng: number;
19 |   ts: Date;
20 |   velocidadKmh: number;
21 |   kmOdometro: number | null;
22 | }
23 |
24 | interface ParadaDetectada {
25 |   lat: number;
26 |   lng: number;
27 |   inicio: Date;
28 |   fin: Date;
29 |   minutos: number;
30 |   nombre: string;
31 |   tipo: string;
32 |   lugarId: number | null;
33 | }
34 |
35 | interface ViajeReconstruido {
36 |   origenLat: number;
37 |   origenLng: number;
38 |   origenNombre: string;
39 |   origenTimestamp: Date;
40 |   origenLugarId: number | null;
41 |   destinoLat: number;
42 |   destinoLng: number;
43 |   destinoNombre: string;
44 |   destinoTimestamp: Date;
45 |   destinoLugarId: number | null;
46 |   kmGps: number;
47 |   kmOdometroInicio: number | null;
48 |   kmOdometroFin: number | null;
49 |   kmOdometroDelta: number | null;
50 |   duracionMinutos: number;
51 |   velocidadPromedio: number;
52 |   velocidadMaxima: number;
53 |   tiempoDetenidoMin: number;
54 |   tiempoMovimientoMin: number;
55 |   paradas: ParadaDetectada[];
56 |   paradasLugares: any[];
57 |   totalPuntos: number;
58 | }
59 |
60 | async function geocodificarPunto(lat: number, lng: number): Promise<string> {
61 |   const roundLat = Math.round(lat * 1000) / 1000;
62 |   const roundLng = Math.round(lng * 1000) / 1000;
63 |
64 |   const cached = await db.select().from(geoGeocache)
65 |     .where(and(
66 |       eq(geoGeocache.lat, String(roundLat)),
67 |       eq(geoGeocache.lng, String(roundLng))
68 |     )).limit(1);
69 |
70 |   if (cached.length > 0) {
71 |     const c = cached[0];
72 |     if (c.ciudad && c.region) return `${c.ciudad}, ${c.region}`;
73 |     if (c.nombre) return c.nombre;
74 |     return `${lat.toFixed(4)}, ${lng.toFixed(4)}`;
75 |   }
76 |
77 |   try {
78 |     await new Promise(r => setTimeout(r, 1100));
79 |     const res = await fetch(
80 |       `https://nominatim.openstreetmap.org/reverse?lat=${lat}&lon=${lng}&format=json&zoom=10&accept-language=es`,
81 |       { headers: { "User-Agent": "GEOVALIDATOR-Sotraser/1.0" } }
82 |     );
83 |     const data: any = await res.json();
84 |     const ciudad = data.address?.city || data.address?.town || data.address?.village || data.address?.county || "";
85 |     const region = data.address?.state || "";
86 |     const nombre = ciudad && region ? `${ciudad}, ${region}` :
87 |                    data.display_name?.split(",").slice(0, 2).join(",").trim() ||
88 |                    `${lat.toFixed(4)}, ${lng.toFixed(4)}`;
89 |
90 |     await db.insert(geoGeocache).values({
91 |       lat: String(roundLat),
92 |       lng: String(roundLng),
93 |       nombre,
94 |       ciudad,
95 |       region,
96 |     }).onConflictDoNothing();
97 |
98 |     return ciudad && region ? `${ciudad}, ${region}` : nombre;
99 |   } catch {
100 |     return `${lat.toFixed(4)}, ${lng.toFixed(4)}`;
101 |   }
102 | }
103 |
104 | async function buscarLugarCercano(lat: number, lng: number, radioKm: number = 1): Promise<{ id: number; nombre: string;
```

```
105 |   const lugares = await db.select().from(geoLugares)
106 |     .where(eq(geoLugares.activo, true));
107 |
108 |   for (const lugar of lugares) {
109 |     const dist = haversineKm(lat, lng, parseFloat(lugar.lat as string), parseFloat(lugar.lng as string));
110 |     if (dist <= radioKm) {
111 |       return {
112 |         id: lugar.id,
113 |         nombre: lugar.nombreConfirmado || lugar.nombre || "Lugar",
114 |         tipo: lugar.tipo || "PUNTO_FRECUENTE"
115 |       };
116 |     }
117 |   }
118 |   return null;
119 | }
120 |
121 | export async function reconstruirViajesCamion(
122 |   camionId: number,
123 |   desde: Date,
124 |   hasta: Date
125 | ): Promise<{ tieneGps: boolean; totalPuntos: number; viajes: ViajeReconstruido[] }> {
126 |
127 |   const puntos = await db.select().from(geoPuntos)
128 |     .where(and(
129 |       eq(geoPuntos.camionId, camionId),
130 |       gte(geoPuntos.timestampPunto, desde),
131 |       lte(geoPuntos.timestampPunto, hasta),
132 |       sql`${geoPuntos.lat} IS NOT NULL`,
133 |       sql`${geoPuntos.lng} IS NOT NULL`
134 |     ))
135 |     .orderBy(asc(geoPuntos.timestampPunto));
136 |
137 |   if (puntos.length === 0) {
138 |     return { tieneGps: false, totalPuntos: 0, viajes: [] };
139 |   }
140 |
141 |   const pts: PuntoGPS[] = puntos.map(p => ({
142 |     lat: parseFloat(p.lat as string),
143 |     lng: parseFloat(p.lng as string),
144 |     ts: new Date(p.timestampPunto),
145 |     velocidadKmh: parseFloat(p.velocidadKmh as string) || 0,
146 |     kmOdometro: p.kmOdometro ? parseFloat(p.kmOdometro as string) : null,
147 |   }));
148 |
149 |   const viajes = detectarViajes(pts);
150 |
151 |   const viajesFiltrados = viajes.filter(v =>
152 |     v.kmGps > 30 &&
153 |     v.duracionMinutos > 60 &&
154 |     v.duracionMinutos < 7200 &&
155 |     v.totalPuntos >= 5
156 |   );
157 |
158 |   for (const viaje of viajesFiltrados) {
159 |     const lugarOrigen = await buscarLugarCercano(viaje.origenLat, viaje.origenLng);
160 |     if (lugarOrigen) {
161 |       viaje.origenNombre = lugarOrigen.nombre;
162 |       viaje.origenLugarId = lugarOrigen.id;
163 |     } else {
164 |       viaje.origenNombre = await geocodificarPunto(viaje.origenLat, viaje.origenLng);
165 |     }
166 |
167 |     const lugarDestino = await buscarLugarCercano(viaje.destinoLat, viaje.destinoLng);
168 |     if (lugarDestino) {
169 |       viaje.destinoNombre = lugarDestino.nombre;
170 |       viaje.destinoLugarId = lugarDestino.id;
171 |     } else {
172 |       viaje.destinoNombre = await geocodificarPunto(viaje.destinoLat, viaje.destinoLng);
173 |     }
174 |
175 |     for (const parada of viaje.paradas) {
176 |       if (parada.minutos >= 60) {
177 |         const lugarParada = await buscarLugarCercano(parada.lat, parada.lng);
178 |         if (lugarParada) {
179 |           parada.nombre = lugarParada.nombre;
180 |           parada.tipo = lugarParada.tipo;
181 |           parada.lugarId = lugarParada.id;
182 |         } else {
183 |           parada.nombre = await geocodificarPunto(parada.lat, parada.lng);
184 |         }
185 |       }
186 |     }
187 |
188 |     viaje.paradasLugares = viaje.paradas
189 |       .filter(p => p.lugarId)
190 |       .map(p => ({ lugarId: p.lugarId, nombre: p.nombre, minutos: p.minutos }));
191 |   }
192 |
193 |   return { tieneGps: true, totalPuntos: pts.length, viajes: viajesFiltrados };
194 | }
195 |
196 | function detectarViajes(pts: PuntoGPS[]): ViajeReconstruido[] {
197 |   if (pts.length < 5) return [];
198 |
199 |   const viajes: ViajeReconstruido[] = [];
200 |   let viajeInicio: number | null = null;
201 |   let detenidoDesde: number | null = null;
202 |
203 |   for (let i = 0; i < pts.length; i++) {
204 |     const p = pts[i];
205 |     const enMovimiento = p.velocidadKmh > 5;
206 |
207 |     if (viajeInicio === null) {
208 |       if (enMovimiento) {
209 |         const esPrimerPuntoDia = i === 0 ||
210 |           (pts[i].ts.getDate() !== pts[i - 1].ts.getDate());
211 |
```

```
212 |         let detenidoPrevio = false;
213 |         if (i > 0) {
214 |           let j = i - 1;
215 |           while (j >= 0 && pts[j].velocidadKmh <= 5) j--;
216 |           if (j < i - 1) {
217 |             const minutosParado = (pts[i].ts.getTime() - pts[j + 1].ts.getTime()) / 60000;
218 |             detenidoPrevio = minutosParado >= 20;
219 |           }
220 |         }
221 |
222 |         if (esPrimerPuntoDia || detenidoPrevio || i === 0) {
223 |           viajeInicio = i;
224 |           detenidoDesde = null;
225 |         }
226 |       }
227 |       continue;
228 |     }
229 |
230 |     if (!enMovimiento) {
231 |       if (detenidoDesde === null) detenidoDesde = i;
232 |       const minutosDetenido = (p.ts.getTime() - pts[detenidoDesde].ts.getTime()) / 60000;
233 |
234 |       const distOrigen = haversineKm(
235 |         pts[viajeInicio].lat, pts[viajeInicio].lng,
236 |         p.lat, p.lng
237 |       );
238 |
239 |       const sinPuntosFuturos = i === pts.length - 1 ||
240 |         (i < pts.length - 1 && (pts[i + 1].ts.getTime() - p.ts.getTime()) > 4 * 3600000);
241 |
242 |       if ((minutosDetenido >= 20 && distOrigen > 10) || sinPuntosFuturos) {
243 |         const viaje = construirViaje(pts, viajeInicio, i);
244 |         if (viaje) viajes.push(viaje);
245 |         viajeInicio = null;
246 |         detenidoDesde = null;
247 |       }
248 |     } else {
249 |       detenidoDesde = null;
250 |     }
251 |   }
252 |
253 |   if (viajeInicio !== null) {
254 |     const viaje = construirViaje(pts, viajeInicio, pts.length - 1);
255 |     if (viaje) viajes.push(viaje);
256 |   }
257 |
258 |   return viajes;
259 | }
260 |
261 | function construirViaje(pts: PuntoGPS[], inicio: number, fin: number): ViajeReconstruido | null {
262 |   if (fin - inicio < 4) return null;
263 |
264 |   const ptsViaje = pts.slice(inicio, fin + 1);
265 |   let kmGps = 0;
266 |   let velocidadMaxima = 0;
267 |   let tiempoDetenido = 0;
268 |   let tiempoMovimiento = 0;
269 |   const paradas: ParadaDetectada[] = [];
270 |
271 |   let detenidoDesdeIdx: number | null = null;
272 |
273 |   for (let i = 1; i < ptsViaje.length; i++) {
274 |     const prev = ptsViaje[i - 1];
275 |     const curr = ptsViaje[i];
276 |     const gapMin = (curr.ts.getTime() - prev.ts.getTime()) / 60000;
277 |
278 |     if (curr.velocidadKmh > velocidadMaxima) velocidadMaxima = curr.velocidadKmh;
279 |
280 |     if (curr.velocidadKmh > 5) {
281 |       if (detenidoDesdeIdx !== null) {
282 |         const minutosParada = (ptsViaje[i - 1].ts.getTime() - ptsViaje[detenidoDesdeIdx].ts.getTime()) / 60000;
283 |         if (minutosParada >= 30) {
284 |           const tipo = minutosParada >= 60 ? "PARADA_LARGA" : "PARADA";
285 |           paradas.push({
286 |             lat: ptsViaje[detenidoDesdeIdx].lat,
287 |             lng: ptsViaje[detenidoDesdeIdx].lng,
288 |             inicio: ptsViaje[detenidoDesdeIdx].ts,
289 |             fin: ptsViaje[i - 1].ts,
290 |             minutos: Math.round(minutosParada),
291 |             nombre: "",
292 |             tipo,
293 |             lugarId: null,
294 |           });
295 |         }
296 |         tiempoDetenido += minutosParada;
297 |         detenidoDesdeIdx = null;
298 |       }
299 |
300 |       if (gapMin < 15) {
301 |         kmGps += haversineKm(prev.lat, prev.lng, curr.lat, curr.lng);
302 |       }
303 |       tiempoMovimiento += gapMin;
304 |     } else {
305 |       if (detenidoDesdeIdx === null) detenidoDesdeIdx = i;
306 |     }
307 |   }
308 |
309 |   if (detenidoDesdeIdx !== null) {
310 |     const minutosParada = (ptsViaje[ptsViaje.length - 1].ts.getTime() - ptsViaje[detenidoDesdeIdx].ts.getTime()) / 60000
311 |     tiempoDetenido += minutosParada;
312 |   }
313 |
314 |   const duracionMinutos = Math.round((ptsViaje[ptsViaje.length - 1].ts.getTime() - ptsViaje[0].ts.getTime()) / 60000);
315 |   const velocidadPromedio = duracionMinutos > 0 ? (kmGps / (duracionMinutos / 60)) : 0;
316 |
317 |   const kmOdoInicio = ptsViaje[0].kmOdometro;
318 |   const kmOdoFin = ptsViaje[ptsViaje.length - 1].kmOdometro;
```

```
319 |   const kmOdoDelta = kmOdoInicio !== null && kmOdoFin !== null ? kmOdoFin - kmOdoInicio : null;
320 |
321 |   return {
322 |     origenLat: ptsViaje[0].lat,
323 |     origenLng: ptsViaje[0].lng,
324 |     origenNombre: "",
325 |     origenTimestamp: ptsViaje[0].ts,
326 |     origenLugarId: null,
327 |     destinoLat: ptsViaje[ptsViaje.length - 1].lat,
328 |     destinoLng: ptsViaje[ptsViaje.length - 1].lng,
329 |     destinoNombre: "",
330 |     destinoTimestamp: ptsViaje[ptsViaje.length - 1].ts,
331 |     destinoLugarId: null,
332 |     kmGps: Math.round(kmGps * 10) / 10,
333 |     kmOdometroInicio: kmOdoInicio,
334 |     kmOdometroFin: kmOdoFin,
335 |     kmOdometroDelta: kmOdoDelta ? Math.round(kmOdoDelta * 10) / 10 : null,
336 |     duracionMinutos,
337 |     velocidadPromedio: Math.round(velocidadPromedio * 10) / 10,
338 |     velocidadMaxima: Math.round(velocidadMaxima * 10) / 10,
339 |     tiempoDetenidoMin: Math.round(tiempoDetenido),
340 |     tiempoMovimientoMin: Math.round(tiempoMovimiento),
341 |     paradas,
342 |     paradasLugares: [],
343 |     totalPuntos: ptsViaje.length,
344 |   };
345 | }
346 |
347 | export async function guardarViajesReconstruidos(
348 |   camionId: number,
349 |   patente: string,
350 |   viajes: ViajeReconstruido[]
351 | ): Promise<number> {
352 |   let guardados = 0;
353 |
354 |   for (const v of viajes) {
355 |     const fechaViaje = v.origenTimestamp;
356 |     const existing = await db.select({ id: geoViajes.id, kmGps: geoViajes.kmGps })
357 |       .from(geoViajes)
358 |       .where(and(
359 |         eq(geoViajes.camionId, camionId),
360 |         gte(geoViajes.origenTimestamp, new Date(fechaViaje.getTime() - 3600000)),
361 |         lte(geoViajes.origenTimestamp, new Date(fechaViaje.getTime() + 3600000))
362 |       ));
363 |
364 |     const duplicado = existing.find(e => {
365 |       const kmExist = parseFloat(e.kmGps as string) || 0;
366 |       const delta = Math.abs(kmExist - v.kmGps) / Math.max(kmExist, v.kmGps, 1);
367 |       return delta < 0.2;
368 |     });
369 |
370 |     if (duplicado) {
371 |       await db.update(geoViajes)
372 |         .set({
373 |           origenLat: String(v.origenLat),
374 |           origenLng: String(v.origenLng),
375 |           origenNombre: v.origenNombre,
376 |           origenTimestamp: v.origenTimestamp,
377 |           origenLugarId: v.origenLugarId,
378 |           destinoLat: String(v.destinoLat),
379 |           destinoLng: String(v.destinoLng),
380 |           destinoNombre: v.destinoNombre,
381 |           destinoTimestamp: v.destinoTimestamp,
382 |           destinoLugarId: v.destinoLugarId,
383 |           kmGps: String(v.kmGps),
384 |           kmOdometroInicio: v.kmOdometroInicio !== null ? String(v.kmOdometroInicio) : null,
385 |           kmOdometroFin: v.kmOdometroFin !== null ? String(v.kmOdometroFin) : null,
386 |           kmOdometroDelta: v.kmOdometroDelta !== null ? String(v.kmOdometroDelta) : null,
387 |           duracionMinutos: v.duracionMinutos,
388 |           velocidadPromedio: String(v.velocidadPromedio),
389 |           velocidadMaxima: String(v.velocidadMaxima),
390 |           tiempoDetenidoMin: v.tiempoDetenidoMin,
391 |           tiempoMovimientoMin: v.tiempoMovimientoMin,
392 |           paradas: v.paradas,
393 |           paradasLugares: v.paradasLugares,
394 |           actualizadoAt: new Date(),
395 |         })
396 |         .where(eq(geoViajes.id, duplicado.id));
397 |     } else {
398 |       await db.insert(geoViajes).values({
399 |         camionId,
400 |         patente,
401 |         contrato: "CENCOSUD",
402 |         origenLat: String(v.origenLat),
403 |         origenLng: String(v.origenLng),
404 |         origenNombre: v.origenNombre,
405 |         origenTimestamp: v.origenTimestamp,
406 |         origenLugarId: v.origenLugarId,
407 |         destinoLat: String(v.destinoLat),
408 |         destinoLng: String(v.destinoLng),
409 |         destinoNombre: v.destinoNombre,
410 |         destinoTimestamp: v.destinoTimestamp,
411 |         destinoLugarId: v.destinoLugarId,
412 |         kmGps: String(v.kmGps),
413 |         kmOdometroInicio: v.kmOdometroInicio !== null ? String(v.kmOdometroInicio) : null,
414 |         kmOdometroFin: v.kmOdometroFin !== null ? String(v.kmOdometroFin) : null,
415 |         kmOdometroDelta: v.kmOdometroDelta !== null ? String(v.kmOdometroDelta) : null,
416 |         duracionMinutos: v.duracionMinutos,
417 |         velocidadPromedio: String(v.velocidadPromedio),
418 |         velocidadMaxima: String(v.velocidadMaxima),
419 |         tiempoDetenidoMin: v.tiempoDetenidoMin,
420 |         tiempoMovimientoMin: v.tiempoMovimientoMin,
421 |         paradas: v.paradas,
422 |         paradasLugares: v.paradasLugares,
423 |         validacionEstado: "PENDIENTE",
424 |       });
425 |     }
```

```
426 |     guardados++;
427 |   }
428 |
429 |   return guardados;
430 | }
431 |
432 | export async function obtenerHistorialCamion(patente: string, desde: Date, hasta: Date) {
433 |   const [camion] = await db.select().from(camiones).where(eq(camiones.patente, patente)).limit(1);
434 |   if (!camion) return null;
435 |
436 |   const totalPuntos = await db.select({ count: sql<number>`count(*)::int` })
437 |     .from(geoPuntos)
438 |     .where(and(
439 |       eq(geoPuntos.camionId, camion.id),
440 |       gte(geoPuntos.timestampPunto, desde),
441 |       lte(geoPuntos.timestampPunto, hasta)
442 |     ));
443 |
444 |   const puntosCount = totalPuntos[0]?.count || 0;
445 |
446 |   let primerPunto: Date | null = null;
447 |   let ultimoPunto: Date | null = null;
448 |   if (puntosCount > 0) {
449 |     const [first] = await db.select({ ts: geoPuntos.timestampPunto })
450 |       .from(geoPuntos)
451 |       .where(and(eq(geoPuntos.camionId, camion.id), gte(geoPuntos.timestampPunto, desde)))
452 |       .orderBy(asc(geoPuntos.timestampPunto))
453 |       .limit(1);
454 |     const [last] = await db.select({ ts: geoPuntos.timestampPunto })
455 |       .from(geoPuntos)
456 |       .where(and(eq(geoPuntos.camionId, camion.id), lte(geoPuntos.timestampPunto, hasta)))
457 |       .orderBy(desc(geoPuntos.timestampPunto))
458 |       .limit(1);
459 |     primerPunto = first?.ts || null;
460 |     ultimoPunto = last?.ts || null;
461 |   }
462 |
463 |   let viajesDB = await db.select().from(geoViajes)
464 |     .where(and(
465 |       eq(geoViajes.camionId, camion.id),
466 |       gte(geoViajes.origenTimestamp, desde),
467 |       lte(geoViajes.origenTimestamp, hasta)
468 |     ))
469 |     .orderBy(desc(geoViajes.origenTimestamp));
470 |
471 |   if (viajesDB.length === 0 && puntosCount > 0) {
472 |     const result = await reconstruirViajesCamion(camion.id, desde, hasta);
473 |     if (result.viajes.length > 0) {
474 |       await guardarViajesReconstruidos(camion.id, camion.patente, result.viajes);
475 |       viajesDB = await db.select().from(geoViajes)
476 |         .where(and(
477 |           eq(geoViajes.camionId, camion.id),
478 |           gte(geoViajes.origenTimestamp, desde),
479 |           lte(geoViajes.origenTimestamp, hasta)
480 |         ))
481 |         .orderBy(desc(geoViajes.origenTimestamp));
482 |     }
483 |   }
484 |
485 |   let cargasSigetra: any[] = [];
486 |   try {
487 |     cargasSigetra = await db.select().from(cargas)
488 |       .where(eq(cargas.camionId, camion.id));
489 |   } catch {}
490 |
491 |   const totalKmGps = viajesDB.reduce((sum, v) => sum + (parseFloat(v.kmGps as string) || 0), 0);
492 |   const destinos = viajesDB.map(v => v.destinoNombre).filter(Boolean);
493 |   const destinoFrecuencias = destinos.reduce((acc: Record<string, number>, d) => {
494 |     acc[d!] = (acc[d!] || 0) + 1;
495 |     return acc;
496 |   }, {});
497 |   const destinoMasFrecuente = Object.entries(destinoFrecuencias).sort((a, b) => b[1] - a[1])[0]?.[0] || null;
498 |
499 |   return {
500 |     camion: {
501 |       id: camion.id,
502 |       patente: camion.patente,
503 |       modelo: camion.modelo,
504 |       conductor: camion.conductor,
505 |       contrato: "CENCOSUD",
506 |       tiene_gps: puntosCount > 0,
507 |       total_puntos_gps: puntosCount,
508 |       primer_punto: primerPunto,
509 |       ultimo_punto: ultimoPunto,
510 |     },
511 |     viajes: viajesDB.map(v => ({
512 |       id: v.id,
513 |       fecha: v.origenTimestamp ? new Date(v.origenTimestamp).toISOString().split("T")[0] : null,
514 |       origen_nombre: v.origenNombre,
515 |       destino_nombre: v.destinoNombre,
516 |       km_gps: parseFloat(v.kmGps as string) || 0,
517 |       km_odometro: v.kmOdometroDelta ? parseFloat(v.kmOdometroDelta as string) : null,
518 |       duracion_horas: v.duracionMinutos ? Math.round((v.duracionMinutos / 60) * 10) / 10 : 0,
519 |       duracion_minutos: v.duracionMinutos || 0,
520 |       velocidad_maxima: parseFloat(v.velocidadMaxima as string) || 0,
521 |       velocidad_promedio: parseFloat(v.velocidadPromedio as string) || 0,
522 |       tiempo_detenido_min: v.tiempoDetenidoMin || 0,
523 |       paradas: v.paradas || [],
524 |       validacion_estado: v.validacionEstado || "PENDIENTE",
525 |       sigetra_match: v.sigetraCargaId ? {
526 |         encontrado: true,
527 |         litros: parseFloat(v.sigetraLitros as string) || 0,
528 |         km_delta_pct: parseFloat(v.sigetraKmDeltaPct as string) || 0,
529 |       } : { encontrado: false },
530 |       origen_lat: parseFloat(v.origenLat as string),
531 |       origen_lng: parseFloat(v.origenLng as string),
532 |       destino_lat: parseFloat(v.destinoLat as string),
```

```
533 |       destino_lng: parseFloat(v.destinoLng as string),
534 |     })),
535 |     sigetra: cargasSigetra.length > 0 ? {
536 |       total_cargas: cargasSigetra.length,
537 |       total_litros: cargasSigetra.reduce((s, c) => s + (c.litrosSurtidor || 0), 0),
538 |       proveedores: [...new Set(cargasSigetra.map(c => c.proveedor).filter(Boolean))],
539 |     } : null,
540 |     resumen: {
541 |       total_viajes: viajesDB.length,
542 |       total_km_gps: Math.round(totalKmGps),
543 |       km_promedio_por_viaje: viajesDB.length > 0 ? Math.round(totalKmGps / viajesDB.length) : 0,
544 |       destino_mas_frecuente: destinoMasFrecuente,
545 |       viajes_validados: viajesDB.filter(v => v.validacionEstado === "VALIDADO").length,
546 |       viajes_anomalia: viajesDB.filter(v => v.validacionEstado === "ANOMALIA").length,
547 |     },
548 |     mensaje: puntosCount === 0 ? `Este camion no tiene registros GPS desde ${desde.toISOString().split("T")[0]}` : null,
549 |   };
550 | }
551 |
552 | export async function procesarFlotaHistorico(
553 |   desde: Date,
554 |   patentes: string[],
555 |   onProgress: (data: any) => void
556 | ) {
557 |   const faenaId = await getCencosudFaenaId();
558 |   const allCamiones = await db.select().from(camiones).where(eq(camiones.faenaId, faenaId));
559 |   const hasta = new Date();
560 |
561 |   const flotaFiltrada = patentes.length > 0
562 |     ? allCamiones.filter(c => patentes.includes(c.patente))
563 |     : allCamiones;
564 |
565 |   let conGps = 0;
566 |   let sinGps = 0;
567 |   let totalViajes = 0;
568 |
569 |   for (const cam of flotaFiltrada) {
570 |     onProgress({
571 |       patente: cam.patente,
572 |       estado: "procesando",
573 |       puntos: 0,
574 |       viajes_detectados: 0,
575 |     });
576 |
577 |     const result = await reconstruirViajesCamion(cam.id, desde, hasta);
578 |
579 |     if (!result.tieneGps) {
580 |       sinGps++;
581 |       onProgress({
582 |         patente: cam.patente,
583 |         estado: "sin_gps",
584 |         puntos: 0,
585 |         viajes_detectados: 0,
586 |       });
587 |       continue;
588 |     }
589 |
590 |     conGps++;
591 |     let guardados = 0;
592 |     if (result.viajes.length > 0) {
593 |       guardados = await guardarViajesReconstruidos(cam.id, cam.patente, result.viajes);
594 |     }
595 |     totalViajes += guardados;
596 |
597 |     onProgress({
598 |       patente: cam.patente,
599 |       estado: "listo",
600 |       puntos: result.totalPuntos,
601 |       viajes_detectados: guardados,
602 |     });
603 |   }
604 |
605 |   onProgress({
606 |     estado: "completo",
607 |     total_camiones: flotaFiltrada.length,
608 |     con_gps: conGps,
609 |     sin_gps: sinGps,
610 |     total_viajes: totalViajes,
611 |   });
612 | }
613 |
614 | export async function obtenerResumenFlota(desde: Date) {
615 |   const faenaId = await getCencosudFaenaId();
616 |   const allCamiones = await db.select().from(camiones).where(eq(camiones.faenaId, faenaId));
617 |   const hasta = new Date();
618 |
619 |   const resumen = [];
620 |
621 |   for (const cam of allCamiones) {
622 |     const puntosResult = await db.select({ count: sql<number>`count(*)::int` })
623 |       .from(geoPuntos)
624 |       .where(and(
625 |         eq(geoPuntos.camionId, cam.id),
626 |         gte(geoPuntos.timestampPunto, desde)
627 |       ));
628 |     const puntos = puntosResult[0]?.count || 0;
629 |
630 |     const viajes = await db.select().from(geoViajes)
631 |       .where(and(
632 |         eq(geoViajes.camionId, cam.id),
633 |         gte(geoViajes.origenTimestamp, desde),
634 |         lte(geoViajes.origenTimestamp, hasta)
635 |       ));
636 |
637 |     const totalKm = viajes.reduce((s, v) => s + (parseFloat(v.kmGps as string) || 0), 0);
638 |     const destinos = viajes.map(v => v.destinoNombre).filter(Boolean);
639 |     const destinoFrec = destinos.reduce((acc: Record<string, number>, d) => {
```

```
640 |       acc[d!] = (acc[d!] || 0) + 1;
641 |       return acc;
642 |     }, {});
643 |     const topDestino = Object.entries(destinoFrec).sort((a, b) => b[1] - a[1])[0]?.[0] || null;
644 |
645 |     let cargasSigetra = 0;
646 |     let litrosSigetra = 0;
647 |     try {
648 |       const cResult = await db.select().from(cargas).where(eq(cargas.camionId, cam.id));
649 |       cargasSigetra = cResult.length;
650 |       litrosSigetra = cResult.reduce((s, c) => s + (c.litrosSurtidor || 0), 0);
651 |     } catch {}
652 |
653 |     resumen.push({
654 |       patente: cam.patente,
655 |       camionId: cam.id,
656 |       conductor: cam.conductor,
657 |       modelo: cam.modelo,
658 |       tiene_gps: puntos > 0,
659 |       puntos_gps: puntos,
660 |       total_viajes: viajes.length,
661 |       total_km: Math.round(totalKm),
662 |       destino_top: topDestino,
663 |       validados: viajes.filter(v => v.validacionEstado === "VALIDADO").length,
664 |       anomalias: viajes.filter(v => v.validacionEstado === "ANOMALIA").length,
665 |       sigetra_cargas: cargasSigetra,
666 |       sigetra_litros: Math.round(litrosSigetra),
667 |     });
668 |   }
669 |
670 |   const conGps = resumen.filter(r => r.tiene_gps).length;
671 |   const sinGps = resumen.filter(r => !r.tiene_gps);
672 |   const activos = resumen.filter(r => r.tiene_gps);
673 |   const totalViajes = activos.reduce((s, r) => s + r.total_viajes, 0);
674 |   const totalKm = activos.reduce((s, r) => s + r.total_km, 0);
675 |
676 |   activos.sort((a, b) => b.total_viajes - a.total_viajes || b.puntos_gps - a.puntos_gps);
677 |
678 |   return {
679 |     kpis: {
680 |       total_camiones: resumen.length,
681 |       con_gps: conGps,
682 |       sin_gps: sinGps.length,
683 |       total_viajes: totalViajes,
684 |       total_km: totalKm,
685 |       ocultos_sin_gps: sinGps.length,
686 |     },
687 |     camiones: activos,
688 |     camiones_sin_gps: sinGps.map(c => ({ patente: c.patente, camionId: c.camionId, modelo: c.modelo, conductor: c.conduc
689 |   };
690 | }
691 |
```

server/geo-lugares-service.ts

388 lineas

```
1 | import { db } from "./db";
2 | import { geoLugares, geoVisitas, geoPuntos, geoViajes, camiones } from "@shared/schema";
3 | import { eq, and, gte, lte, sql, desc, asc } from "drizzle-orm";
4 | import { getCencosudFaenaId } from "./cencosud-filter";
5 |
6 | function haversineKm(lat1: number, lon1: number, lat2: number, lon2: number): number {
7 |   const R = 6371;
8 |   const dLat = (lat2 - lat1) * Math.PI / 180;
9 |   const dLon = (lon2 - lon1) * Math.PI / 180;
10 |   const a = Math.sin(dLat / 2) ** 2 +
11 |     Math.cos(lat1 * Math.PI / 180) * Math.cos(lat2 * Math.PI / 180) *
12 |     Math.sin(dLon / 2) ** 2;
13 |   return R * 2 * Math.asin(Math.sqrt(a));
14 | }
15 |
16 | const CENCOSUD_BRANDS = [
17 |   { pattern: /jumbo/i, tipo: "LOCAL_JUMBO", confianza: 90 },
18 |   { pattern: /santa\s*isabel/i, tipo: "LOCAL_SANTA_ISABEL", confianza: 90 },
19 |   { pattern: /easy/i, tipo: "LOCAL_EASY", confianza: 85 },
20 |   { pattern: /paris/i, tipo: "LOCAL_PARIS", confianza: 85 },
21 |   { pattern: /johnson/i, tipo: "LOCAL_PARIS", confianza: 85 },
22 |   { pattern: /cencosud/i, tipo: "LOCAL_CENCOSUD", confianza: 85 },
23 | ];
24 |
25 | async function queryOverpass(lat: number, lng: number): Promise<{ name: string; tipo: string; confianza: number } | null
26 |   const query = `[out:json];(node["name"~"Jumbo|Santa Isabel|Cencosud|Easy|Paris|Johnson|Disco",i](around:500,${lat},${l
27 |   try {
28 |     const res = await fetch(`https://overpass-api.de/api/interpreter?data=${encodeURIComponent(query)}`, {
29 |       headers: { "User-Agent": "GEOVALIDATOR-Sotraser/1.0" },
30 |       signal: AbortSignal.timeout(15000),
31 |     });
32 |     const data = await res.json();
33 |     if (!data.elements || data.elements.length === 0) return null;
34 |
35 |     for (const el of data.elements) {
36 |       const name = el.tags?.name || "";
37 |       for (const brand of CENCOSUD_BRANDS) {
38 |         if (brand.pattern.test(name)) {
39 |           return { name, tipo: brand.tipo, confianza: brand.confianza };
40 |         }
41 |       }
42 |     }
43 |     return null;
44 |   } catch (e) {
45 |     console.log("[geo-lugares] Overpass error:", (e as Error).message);
46 |     return null;
47 |   }
48 | }
49 |
50 | async function reverseGeocode(lat: number, lng: number): Promise<{ display: string; road: string; city: string; region:
51 |   try {
52 |     const res = await fetch(`https://nominatim.openstreetmap.org/reverse?lat=${lat}&lon=${lng}&format=json&zoom=16`, {
53 |       headers: { "User-Agent": "GEOVALIDATOR-Sotraser/1.0" },
54 |       signal: AbortSignal.timeout(10000),
55 |     });
56 |     const data = await res.json();
57 |     return {
58 |       display: data.display_name?.split(",").slice(0, 3).join(",").trim() || "",
59 |       road: data.address?.road || "",
60 |       city: data.address?.city || data.address?.town || data.address?.village || "",
61 |       region: data.address?.state || "",
62 |       comuna: data.address?.suburb || data.address?.city_district || data.address?.town || "",
63 |     };
64 |   } catch {
65 |     return { display: `${lat.toFixed(4)}, ${lng.toFixed(4)}`, road: "", city: "", region: "", comuna: "" };
66 |   }
67 | }
68 |
69 | export async function detectarLugar(lat: number, lng: number, minutosDetenido: number): Promise<{ lugarId: number; nombr
70 |   const allLugares = await db.select().from(geoLugares).where(eq(geoLugares.activo, true));
71 |
72 |   for (const lugar of allLugares) {
73 |     const dist = haversineKm(lat, lng, parseFloat(lugar.lat), parseFloat(lugar.lng));
74 |     const radioKm = (lugar.radioMetros || 500) / 1000;
75 |     if (dist <= radioKm) {
76 |       await db.update(geoLugares).set({
77 |         vecesVisitado: (lugar.vecesVisitado || 0) + 1,
78 |         ultimaVisita: new Date().toISOString().split("T")[0],
79 |       }).where(eq(geoLugares.id, lugar.id));
80 |       return { lugarId: lugar.id, nombre: lugar.nombreConfirmado || lugar.nombre || "Sin nombre", tipo: lugar.tipo || "O
81 |     }
82 |   }
83 |
84 |   const geo = await reverseGeocode(lat, lng);
85 |
86 |   await new Promise(r => setTimeout(r, 1200));
87 |
88 |   const osm = await queryOverpass(lat, lng);
89 |
90 |   let tipo = "PUNTO_FRECUENTE";
91 |   let nombre = `Punto frecuente · ${geo.city || geo.comuna}${geo.road ? ", " + geo.road : ""}`;
92 |   let confianza = 50;
93 |   let detectadoVia = "COMPORTAMIENTO";
94 |
95 |   if (osm) {
96 |     tipo = osm.tipo;
97 |     nombre = `${osm.name} · ${geo.city || geo.comuna}`;
98 |     confianza = osm.confianza;
99 |     detectadoVia = "OSM";
100 |   } else if (minutosDetenido < 120) {
101 |     tipo = "PUNTO_FRECUENTE";
102 |     confianza = 40;
103 |   }
104 |
```

```
105 |   const [inserted] = await db.insert(geoLugares).values({
106 |     nombre,
107 |     tipo,
108 |     lat: String(lat),
109 |     lng: String(lng),
110 |     radioMetros: 500,
111 |     direccion: geo.display,
112 |     comuna: geo.comuna,
113 |     region: geo.region,
114 |     detectadoVia,
115 |     confianzaPct: confianza,
116 |     vecesVisitado: 1,
117 |     primeraVisita: new Date().toISOString().split("T")[0],
118 |     ultimaVisita: new Date().toISOString().split("T")[0],
119 |     confirmado: false,
120 |     activo: true,
121 |   }).returning();
122 |
123 |   return { lugarId: inserted.id, nombre, tipo, nuevo: true };
124 | }
125 |
126 | export async function registrarVisita(params: {
127 |   lugarId: number;
128 |   camionId: number;
129 |   patente: string;
130 |   viajeId?: number;
131 |   llegada: Date;
132 |   salida?: Date;
133 |   minutosDetenido: number;
134 |   lat: number;
135 |   lng: number;
136 | }) {
137 |   await db.insert(geoVisitas).values({
138 |     lugarId: params.lugarId,
139 |     camionId: params.camionId,
140 |     patente: params.patente,
141 |     viajeId: params.viajeId || null,
142 |     llegada: params.llegada,
143 |     salida: params.salida || null,
144 |     minutosDetenido: params.minutosDetenido,
145 |     latExacta: String(params.lat),
146 |     lngExacta: String(params.lng),
147 |   });
148 | }
149 |
150 | interface ProgressCallback {
151 |   (msg: { paso: string; progreso: number; total: number; detalles?: string }): void;
152 | }
153 |
154 | export async function analizarHistoricoCompleto(onProgress?: ProgressCallback) {
155 |   const faenaId = await getCencosudFaenaId();
156 |   const cencosudCamiones = await db.select().from(camiones).where(eq(camiones.faenaId, faenaId));
157 |   const total = cencosudCamiones.length;
158 |   let lugaresDetectados = 0;
159 |   let viajesReconstruidos = 0;
160 |   let puntosAnalizados = 0;
161 |
162 |   for (let i = 0; i < cencosudCamiones.length; i++) {
163 |     const cam = cencosudCamiones[i];
164 |     onProgress?.({
165 |       paso: `Procesando ${cam.patente}`,
166 |       progreso: i + 1,
167 |       total,
168 |       detalles: `Camion ${i + 1}/${total}`,
169 |     });
170 |
171 |     const puntos = await db.select().from(geoPuntos)
172 |       .where(and(
173 |         eq(geoPuntos.camionId, cam.id),
174 |         gte(geoPuntos.timestampPunto, new Date("2026-03-01")),
175 |       ))
176 |       .orderBy(asc(geoPuntos.timestampPunto));
177 |
178 |     puntosAnalizados += puntos.length;
179 |     if (puntos.length === 0) continue;
180 |
181 |     const paradas = detectarParadasConsecutivas(puntos, cam.id, cam.patente);
182 |
183 |     for (const parada of paradas) {
184 |       if (parada.minutosDetenido >= 60) {
185 |         try {
186 |           const result = await detectarLugar(parada.lat, parada.lng, parada.minutosDetenido);
187 |           if (result.nuevo) lugaresDetectados++;
188 |
189 |           await registrarVisita({
190 |             lugarId: result.lugarId,
191 |             camionId: cam.id,
192 |             patente: cam.patente,
193 |             llegada: parada.inicio,
194 |             salida: parada.fin,
195 |             minutosDetenido: parada.minutosDetenido,
196 |             lat: parada.lat,
197 |             lng: parada.lng,
198 |           });
199 |
200 |           await new Promise(r => setTimeout(r, 1500));
201 |         } catch (e) {
202 |           console.log(`[geo-lugares] Error detectando lugar para ${cam.patente}:`, (e as Error).message);
203 |         }
204 |       }
205 |     }
206 |   }
207 |
208 |   onProgress?.({
209 |     paso: "Completado",
210 |     progreso: total,
211 |     total,
```

```
212 |     detalles: `Lugares detectados: ${lugaresDetectados} · Puntos analizados: ${puntosAnalizados}`,
213 |   });
214 |
215 |   return { lugaresDetectados, viajesReconstruidos, puntosAnalizados, camionesAnalizados: total };
216 | }
217 |
218 | interface Parada {
219 |   lat: number;
220 |   lng: number;
221 |   inicio: Date;
222 |   fin: Date;
223 |   minutosDetenido: number;
224 |   puntoCount: number;
225 | }
226 |
227 | function detectarParadasConsecutivas(
228 |   puntos: any[],
229 |   camionId: number,
230 |   patente: string,
231 | ): Parada[] {
232 |   const paradas: Parada[] = [];
233 |   if (puntos.length === 0) return paradas;
234 |
235 |   let grupoInicio = 0;
236 |   let grupoLats: number[] = [];
237 |   let grupoLngs: number[] = [];
238 |
239 |   for (let i = 0; i < puntos.length; i++) {
240 |     const vel = parseFloat(puntos[i].velocidadKmh || "0");
241 |     const lat = parseFloat(puntos[i].lat);
242 |     const lng = parseFloat(puntos[i].lng);
243 |
244 |     if (vel < 5) {
245 |       if (grupoLats.length === 0) grupoInicio = i;
246 |       grupoLats.push(lat);
247 |       grupoLngs.push(lng);
248 |     } else {
249 |       if (grupoLats.length >= 2) {
250 |         const centroLat = grupoLats.reduce((a, b) => a + b, 0) / grupoLats.length;
251 |         const centroLng = grupoLngs.reduce((a, b) => a + b, 0) / grupoLngs.length;
252 |
253 |         const spread = Math.max(
254 |           ...grupoLats.map(l => haversineKm(l, centroLng, centroLat, centroLng))
255 |         );
256 |
257 |         if (spread < 0.3) {
258 |           const inicio = new Date(puntos[grupoInicio].timestampPunto);
259 |           const fin = new Date(puntos[i - 1].timestampPunto);
260 |           const minutosDetenido = Math.round((fin.getTime() - inicio.getTime()) / 60000);
261 |
262 |           if (minutosDetenido >= 30) {
263 |             paradas.push({
264 |               lat: centroLat,
265 |               lng: centroLng,
266 |               inicio,
267 |               fin,
268 |               minutosDetenido,
269 |               puntoCount: grupoLats.length,
270 |             });
271 |           }
272 |         }
273 |       }
274 |       grupoLats = [];
275 |       grupoLngs = [];
276 |     }
277 |   }
278 |
279 |   if (grupoLats.length >= 2) {
280 |     const centroLat = grupoLats.reduce((a, b) => a + b, 0) / grupoLats.length;
281 |     const centroLng = grupoLngs.reduce((a, b) => a + b, 0) / grupoLngs.length;
282 |     const inicio = new Date(puntos[grupoInicio].timestampPunto);
283 |     const fin = new Date(puntos[puntos.length - 1].timestampPunto);
284 |     const minutosDetenido = Math.round((fin.getTime() - inicio.getTime()) / 60000);
285 |     if (minutosDetenido >= 30) {
286 |       paradas.push({ lat: centroLat, lng: centroLng, inicio, fin, minutosDetenido, puntoCount: grupoLats.length });
287 |     }
288 |   }
289 |
290 |   return paradas;
291 | }
292 |
293 | export async function generarAnalisisIA(): Promise<{ resumen: string; datos: any }> {
294 |   const faenaId = await getCencosudFaenaId();
295 |   const cencosudCamiones = await db.select().from(camiones).where(eq(camiones.faenaId, faenaId));
296 |   const camionIds = cencosudCamiones.map(c => c.id);
297 |
298 |   const lugares = await db.select().from(geoLugares).where(eq(geoLugares.activo, true));
299 |   const viajes = await db.select().from(geoViajes).where(gte(geoViajes.creadoAt, new Date("2026-03-01")));
300 |   const cencosudViajes = viajes.filter(v => camionIds.includes(v.camionId || 0));
301 |
302 |   const visitas = await db.select().from(geoVisitas);
303 |   const cencosudVisitas = visitas.filter(v => camionIds.includes(v.camionId || 0));
304 |
305 |   const lugarStats = lugares.map(l => ({
306 |     nombre: l.nombreConfirmado || l.nombre,
307 |     tipo: l.tipo,
308 |     vecesVisitado: l.vecesVisitado,
309 |     caminonesDistintos: new Set(cencosudVisitas.filter(v => v.lugarId === l.id).map(v => v.camionId)).size,
310 |     tiempoPromedioMin: Math.round(
311 |       cencosudVisitas.filter(v => v.lugarId === l.id).reduce((s, v) => s + (v.minutosDetenido || 0), 0) /
312 |       Math.max(1, cencosudVisitas.filter(v => v.lugarId === l.id).length)
313 |     ),
314 |   })).filter(l => (l.vecesVisitado || 0) > 0).sort((a, b) => (b.vecesVisitado || 0) - (a.vecesVisitado || 0));
315 |
316 |   const anomaliasKm = cencosudViajes
317 |     .filter(v => v.sigetraKmDeltaPct && Math.abs(parseFloat(v.sigetraKmDeltaPct)) > 10)
318 |     .map(v => ({
```

```
319 |       patente: v.patente,
320 |       fecha: v.origenTimestamp,
321 |       kmGps: v.kmGps,
322 |       deltaPct: v.sigetraKmDeltaPct,
323 |     }));
324 |
325 |   const validacionResumen = {
326 |     total: cencosudViajes.length,
327 |     validados: cencosudViajes.filter(v => v.validacionEstado === "VALIDADO").length,
328 |     revisar: cencosudViajes.filter(v => v.validacionEstado === "REVISAR").length,
329 |     anomalia: cencosudViajes.filter(v => v.validacionEstado === "ANOMALIA").length,
330 |     pendiente: cencosudViajes.filter(v => v.validacionEstado === "PENDIENTE").length,
331 |   };
332 |
333 |   const datos = {
334 |     periodo: "01-03-2026 a hoy",
335 |     camionesAnalizados: cencosudCamiones.length,
336 |     totalViajes: cencosudViajes.length,
337 |     totalKmGps: Math.round(cencosudViajes.reduce((s, v) => s + (parseFloat(v.kmGps || "0") || 0), 0)),
338 |     lugaresDetectados: lugarStats.slice(0, 15),
339 |     anomaliasKm: anomaliasKm.slice(0, 10),
340 |     lugaresTotal: lugares.length,
341 |     lugaresSinIdentificar: lugares.filter(l => l.tipo === "PUNTO_FRECUENTE" && !l.confirmado).length,
342 |     validacionResumen,
343 |   };
344 |
345 |   const prompt = `Eres analista de operaciones de SOTRASER, empresa de transporte para CENCOSUD en Chile. Analiza estos
346 | ${JSON.stringify(datos, null, 2)}
347 |
348 | Genera analisis en espanol con estas secciones:
349 |
350 | PATRONES DE RUTA:
351 | Que rutas se repiten mas? Hay patrones claros de dias y horarios?
352 |
353 | LOCALES CENCOSUD DETECTADOS:
354 | Que locales/CDs visitan con mas frecuencia? Hay alguno que recibe mas visitas de lo esperado?
355 |
356 | TIEMPOS DE PERMANENCIA:
357 | Hay locales donde los camiones pasan demasiado tiempo? Cual es el tiempo normal?
358 |
359 | ANOMALIAS DETECTADAS:
360 | Hay viajes con km muy distintos entre GPS y Sigetra? Hay patrones sospechosos?
361 |
362 | RECOMENDACIONES:
363 | 3 acciones concretas para optimizar la operacion basadas en los datos.
364 |
365 | Maximo 5 parrafos totales. Sin markdown. Solo texto directo con los titulos indicados.`;
366 |
367 |   const Anthropic = (await import("@anthropic-ai/sdk")).default;
368 |   const anthropic = new Anthropic();
369 |
370 |   const response = await anthropic.messages.create({
371 |     model: "claude-sonnet-4-20250514",
372 |     max_tokens: 2000,
373 |     messages: [{ role: "user", content: prompt }],
374 |   });
375 |
376 |   const resumen = (response.content[0] as any).text || "";
377 |
378 |   await db.insert(geoAnalisisIa).values({
379 |     tipo: "LUGARES",
380 |     periodoDesde: "2026-03-01",
381 |     periodoHasta: new Date().toISOString().split("T")[0],
382 |     resultadoJson: datos,
383 |     resumenTexto: resumen,
384 |   });
385 |
386 |   return { resumen, datos };
387 | }
388 |
```

server/geo-visitas-service.ts

678 lineas

```
1 | import { db, pool } from "./db";
2 | import { geoLugares, geoPuntos, geoViajes, camiones } from "@shared/schema";
3 | import { eq, and, gte, lte, sql, desc, asc } from "drizzle-orm";
4 | import { getCencosudFaenaId } from "./cencosud-filter";
5 |
6 | function haversineKm(lat1: number, lon1: number, lat2: number, lon2: number): number {
7 |   if (!lat1 || !lon1 || !lat2 || !lon2) return 9999;
8 |   const R = 6371;
9 |   const dLat = (lat2 - lat1) * Math.PI / 180;
10 |   const dLon = (lon2 - lon1) * Math.PI / 180;
11 |   const a =
12 |     Math.sin(dLat / 2) ** 2 +
13 |     Math.cos(lat1 * Math.PI / 180) * Math.cos(lat2 * Math.PI / 180) *
14 |     Math.sin(dLon / 2) ** 2;
15 |   return R * 2 * Math.asin(Math.sqrt(a));
16 | }
17 |
18 | interface DestinoCencosud {
19 |   id: number;
20 |   nombre: string;
21 |   lat: number;
22 |   lng: number;
23 |   radioMetros: number;
24 | }
25 |
26 | interface PuntoGPS {
27 |   id: number;
28 |   camionId: number;
29 |   lat: number;
30 |   lng: number;
31 |   timestamp: Date;
32 |   velocidadKmh: number | null;
33 |   kmOdometro: number | null;
34 | }
35 |
36 | interface VisitaDetectada {
37 |   camionId: number;
38 |   patente: string;
39 |   destinoId: number;
40 |   destinoNombre: string;
41 |   destinoLat: number;
42 |   destinoLng: number;
43 |   fechaVisita: Date;
44 |   kmViaje: number;
45 |   velocidadEnDestino: number | null;
46 |   tiempoEnDestinoMin: number | null;
47 |   confianza: "ALTA" | "MEDIA" | "SIN_DATOS";
48 |   puntosEnRadio: number;
49 |   origenNombre: string | null;
50 |   origenLat: number | null;
51 |   origenLng: number | null;
52 |   metodo: "GPS" | "ODOMETRO" | "SIGETRA";
53 |   tarifaClp: number | null;
54 |   tmsViajeId: number | null;
55 | }
56 |
57 | const RADIO_DETECCION_KM = 2.0;
58 | const VELOCIDAD_PARADA_KMH = 15;
59 | const TIEMPO_MINIMO_MIN = 30;
60 | const DISTANCIA_VIAJE_MIN_KM = 50;
61 |
62 | let destinosCencosud: DestinoCencosud[] = [];
63 | let perfilGpsFlota: Map<number, { totalPuntos: number; desde: Date | null; hasta: Date | null; conCoords: number; kmDelt
64 | let fuenteGps: string = "geo_puntos";
65 |
66 | const TARIFAS_POR_DESTINO: Record<string, number> = {
67 |   "CD Cencosud La Serena": 810000,
68 |   "CD Cencosud Coquimbo": 810000,
69 |   "CD Cencosud Chillan": 610000,
70 |   "CD Cencosud Chillán": 610000,
71 |   "CD Cencosud Puerto Montt": 1400000,
72 |   "CD Vespucio": 120000,
73 |   "CD Lo Aguirre": 120000,
74 |   "Base Santiago CENCOSUD": 0,
75 | };
76 |
77 | const DISTANCIAS_REFERENCIA_KM: Record<string, number> = {
78 |   "CD Cencosud La Serena": 470,
79 |   "CD Cencosud Coquimbo": 460,
80 |   "CD Cencosud Chillan": 400,
81 |   "CD Cencosud Chillán": 400,
82 |   "CD Cencosud Puerto Montt": 1020,
83 |   "CD Vespucio": 15,
84 |   "CD Lo Aguirre": 25,
85 | };
86 |
87 | export async function inicializarPerfilGPS(): Promise<void> {
88 |   try {
89 |     const faenaId = await getCencosudFaenaId();
90 |
91 |     const lugares = await db.select().from(geoLugares).where(
92 |       sql`${geoLugares.activo} = true`
93 |     );
94 |     destinosCencosud = lugares
95 |       .filter(l => l.tipo === "CD_CENCOSUD" || l.tipo === "BASE_ORIGEN")
96 |       .map(l => ({
97 |         id: l.id,
98 |         nombre: l.nombre,
99 |         lat: parseFloat(l.lat),
100 |         lng: parseFloat(l.lng),
101 |         radioMetros: l.radioMetros || 2000,
102 |       }));
103 |
104 |     const perfilResult = await pool.query(`
```

```
105 |       SELECT c.id as camion_id, c.patente, COUNT(g.id) as total_puntos,
106 |         MIN(g.timestamp_punto) as desde, MAX(g.timestamp_punto) as hasta,
107 |         SUM(CASE WHEN g.lat IS NOT NULL AND g.lng IS NOT NULL THEN 1 ELSE 0 END) as puntos_con_coords,
108 |         COALESCE(MAX(g.km_odometro) - MIN(g.km_odometro), 0) as km_delta_periodo
109 |       FROM camiones c
110 |       LEFT JOIN geo_puntos g ON g.camion_id = c.id AND g.timestamp_punto >= '2026-03-01'
111 |       WHERE c.faena_id = $1
112 |       GROUP BY c.id, c.patente
113 |     `, [faenaId]);
114 |
115 |     perfilGpsFlota.clear();
116 |     for (const r of perfilResult.rows) {
117 |       perfilGpsFlota.set(r.camion_id, {
118 |         totalPuntos: parseInt(r.total_puntos) || 0,
119 |         desde: r.desde ? new Date(r.desde) : null,
120 |         hasta: r.hasta ? new Date(r.hasta) : null,
121 |         conCoords: parseInt(r.puntos_con_coords) || 0,
122 |         kmDelta: parseFloat(r.km_delta_periodo) || 0,
123 |       });
124 |     }
125 |
126 |     console.log(`[visitas] Perfil GPS cargado: ${perfilGpsFlota.size} camiones, ${destinosCencosud.length} destinos CENC
127 |     console.log(`[visitas] Fuente GPS: ${fuenteGps}`);
128 |   } catch (err) {
129 |     console.error("[visitas] Error inicializando perfil GPS:", err);
130 |   }
131 | }
132 |
133 | async function obtenerPuntosGPS(camionId: number, desde: string, hasta: string): Promise<PuntoGPS[]> {
134 |   const result = await pool.query(`
135 |     SELECT id, camion_id, lat, lng, timestamp_punto, velocidad_kmh, km_odometro
136 |     FROM geo_puntos
137 |     WHERE camion_id = $1
138 |       AND timestamp_punto >= $2
139 |       AND timestamp_punto <= $3
140 |       AND lat IS NOT NULL AND lng IS NOT NULL
141 |     ORDER BY timestamp_punto ASC
142 |   `, [camionId, desde, hasta]);
143 |
144 |   return result.rows.map(r => ({
145 |     id: r.id,
146 |     camionId: r.camion_id,
147 |     lat: parseFloat(r.lat),
148 |     lng: parseFloat(r.lng),
149 |     timestamp: new Date(r.timestamp_punto),
150 |     velocidadKmh: r.velocidad_kmh !== null ? parseFloat(r.velocidad_kmh) : null,
151 |     kmOdometro: r.km_odometro !== null ? parseFloat(r.km_odometro) : null,
152 |   }));
153 | }
154 |
155 | function detectarVisitasGPS(
156 |   puntos: PuntoGPS[],
157 |   patente: string,
158 |   camionId: number
159 | ): VisitaDetectada[] {
160 |   if (puntos.length === 0) return [];
161 |
162 |   const visitas: VisitaDetectada[] = [];
163 |
164 |   for (const destino of destinosCencosud) {
165 |     if (destino.nombre === "Base Santiago CENCOSUD") continue;
166 |
167 |     const puntosEnRadio: PuntoGPS[] = [];
168 |
169 |     for (const p of puntos) {
170 |       const dist = haversineKm(p.lat, p.lng, destino.lat, destino.lng);
171 |       if (dist <= RADIO_DETECCION_KM) {
172 |         puntosEnRadio.push(p);
173 |       }
174 |     }
175 |
176 |     if (puntosEnRadio.length === 0) continue;
177 |
178 |     const grupos = agruparConsecutivos(puntosEnRadio, puntos);
179 |
180 |     for (const grupo of grupos) {
181 |       const resultado = evaluarVisita(grupo, puntos, destino, patente, camionId);
182 |       if (resultado) {
183 |         visitas.push(resultado);
184 |       }
185 |     }
186 |   }
187 |
188 |   return visitas;
189 | }
190 |
191 | function agruparConsecutivos(
192 |   puntosEnRadio: PuntoGPS[],
193 |   todosPuntos: PuntoGPS[]
194 | ): PuntoGPS[][] {
195 |   if (puntosEnRadio.length === 0) return [];
196 |
197 |   const grupos: PuntoGPS[][] = [];
198 |   let grupoActual: PuntoGPS[] = [puntosEnRadio[0]];
199 |
200 |   for (let i = 1; i < puntosEnRadio.length; i++) {
201 |     const diffHoras = (puntosEnRadio[i].timestamp.getTime() - puntosEnRadio[i - 1].timestamp.getTime()) / (1000 * 60 * 6
202 |     if (diffHoras <= 6) {
203 |       grupoActual.push(puntosEnRadio[i]);
204 |     } else {
205 |       grupos.push(grupoActual);
206 |       grupoActual = [puntosEnRadio[i]];
207 |     }
208 |   }
209 |   grupos.push(grupoActual);
210 |
211 |   return grupos;
```

```
212 | }
213 |
214 | function evaluarVisita(
215 |   grupo: PuntoGPS[],
216 |   todosPuntos: PuntoGPS[],
217 |   destino: DestinoCencosud,
218 |   patente: string,
219 |   camionId: number
220 | ): VisitaDetectada | null {
221 |   const primerPunto = grupo[0];
222 |   const ultimoPunto = grupo[grupo.length - 1];
223 |
224 |   const velocidadEnDestino = grupo.reduce((min, p) =>
225 |     p.velocidadKmh !== null ? Math.min(min, p.velocidadKmh) : min, 999);
226 |
227 |   const criterio1 = true;
228 |
229 |   const criterio2 = velocidadEnDestino <= VELOCIDAD_PARADA_KMH || velocidadEnDestino === 999;
230 |
231 |   let criterio3 = false;
232 |   let tiempoEnDestino = 0;
233 |
234 |   if (grupo.length >= 2) {
235 |     tiempoEnDestino = (ultimoPunto.timestamp.getTime() - primerPunto.timestamp.getTime()) / (1000 * 60);
236 |     criterio3 = tiempoEnDestino >= TIEMPO_MINIMO_MIN;
237 |   } else {
238 |     const idx = todosPuntos.findIndex(p => p.id === primerPunto.id);
239 |     const anterior = idx > 0 ? todosPuntos[idx - 1] : null;
240 |     const siguiente = idx < todosPuntos.length - 1 ? todosPuntos[idx + 1] : null;
241 |
242 |     const distAnterior = anterior ? haversineKm(anterior.lat, anterior.lng, primerPunto.lat, primerPunto.lng) : 0;
243 |     const distSiguiente = siguiente ? haversineKm(primerPunto.lat, primerPunto.lng, siguiente.lat, siguiente.lng) : 0;
244 |
245 |     if (distAnterior > DISTANCIA_VIAJE_MIN_KM && distSiguiente > DISTANCIA_VIAJE_MIN_KM) {
246 |       criterio3 = true;
247 |       tiempoEnDestino = -1;
248 |     } else if (distAnterior > DISTANCIA_VIAJE_MIN_KM || distSiguiente > DISTANCIA_VIAJE_MIN_KM) {
249 |       criterio3 = true;
250 |       tiempoEnDestino = -1;
251 |     }
252 |   }
253 |
254 |   let criterio4 = false;
255 |   const idx = todosPuntos.findIndex(p => p.id === primerPunto.id);
256 |   if (idx > 0) {
257 |     for (let i = idx - 1; i >= 0; i--) {
258 |       const dist = haversineKm(todosPuntos[i].lat, todosPuntos[i].lng, destino.lat, destino.lng);
259 |       if (dist > RADIO_DETECCION_KM) {
260 |         criterio4 = dist > DISTANCIA_VIAJE_MIN_KM;
261 |         break;
262 |       }
263 |     }
264 |   } else {
265 |     criterio4 = true;
266 |   }
267 |
268 |   const criteriosCumplidos = [criterio1, criterio2, criterio3, criterio4].filter(Boolean).length;
269 |
270 |   if (criteriosCumplidos < 3) return null;
271 |
272 |   const confianza: "ALTA" | "MEDIA" = criteriosCumplidos === 4 && grupo.length >= 2 ? "ALTA" : "MEDIA";
273 |
274 |   let origenLat: number | null = null;
275 |   let origenLng: number | null = null;
276 |   let origenNombre: string | null = null;
277 |   let kmViaje = 0;
278 |
279 |   if (idx > 0) {
280 |     const origenPunto = todosPuntos[0];
281 |     origenLat = origenPunto.lat;
282 |     origenLng = origenPunto.lng;
283 |     kmViaje = haversineKm(origenLat, origenLng, destino.lat, destino.lng) * 1.15;
284 |   } else {
285 |     kmViaje = (DISTANCIAS_REFERENCIA_KM[destino.nombre] || 0) * 2;
286 |   }
287 |
288 |   if (primerPunto.kmOdometro && todosPuntos[0]?.kmOdometro) {
289 |     const delta = Math.abs(primerPunto.kmOdometro - todosPuntos[0].kmOdometro);
290 |     if (delta > 10) kmViaje = delta;
291 |   }
292 |
293 |   return {
294 |     camionId,
295 |     patente,
296 |     destinoId: destino.id,
297 |     destinoNombre: destino.nombre,
298 |     destinoLat: destino.lat,
299 |     destinoLng: destino.lng,
300 |     fechaVisita: primerPunto.timestamp,
301 |     kmViaje: Math.round(kmViaje * 10) / 10,
302 |     velocidadEnDestino: velocidadEnDestino === 999 ? null : velocidadEnDestino,
303 |     tiempoEnDestinoMin: tiempoEnDestino >= 0 ? Math.round(tiempoEnDestino) : null,
304 |     confianza,
305 |     puntosEnRadio: grupo.length,
306 |     origenNombre,
307 |     origenLat,
308 |     origenLng,
309 |     metodo: "GPS",
310 |     tarifaClp: TARIFAS_POR_DESTINO[destino.nombre] || null,
311 |     tmsViajeId: null,
312 |   };
313 | }
314 |
315 | async function detectarVisitasPorOdometro(
316 |   camionId: number,
317 |   patente: string,
318 |   desde: string,
```

```
  hasta: string
): Promise<VisitaDetectada[]> {
  const visitas: VisitaDetectada[] = [];

  const trips = await pool.query(`
    SELECT id, km_inicio, km_cierre, km_recorridos, fecha_salida, fecha_llegada,
      litros_sigetra, litros_ecu, rendimiento_real,
      destino_lat, destino_lng, destino_nombre
    FROM tms_viajes
    WHERE camion_id = $1 AND contrato_id = 2
      AND fecha_salida >= $2
    ORDER BY fecha_salida ASC
  `, [camionId, desde]);

  for (const trip of trips.rows) {
    const km = parseFloat(trip.km_recorridos) || 0;
    if (km < 30) continue;

    let mejorDestino: DestinoCencosud | null = null;
    let menorDiffKm = Infinity;

    for (const d of destinosCencosud) {
      if (d.nombre === "Base Santiago CENCOSUD") continue;
      const refKm = DISTANCIAS_REFERENCIA_KM[d.nombre] || 0;
      if (refKm === 0) continue;

      const refKmRoundTrip = refKm * 2;
      const tolerancia = refKmRoundTrip * 0.30;
      const diff = Math.abs(km - refKmRoundTrip);

      if (diff <= tolerancia && diff < menorDiffKm) {
        menorDiffKm = diff;
        mejorDestino = d;
      }
    }

    if (trip.destino_lat && trip.destino_lng) {
      const lat = parseFloat(trip.destino_lat);
      const lng = parseFloat(trip.destino_lng);
      for (const d of destinosCencosud) {
        const dist = haversineKm(lat, lng, d.lat, d.lng);
        if (dist <= 10) {
          mejorDestino = d;
          menorDiffKm = 0;
          break;
        }
      }
    }

    if (mejorDestino) {
      visitas.push({
        camionId,
        patente,
        destinoId: mejorDestino.id,
        destinoNombre: mejorDestino.nombre,
        destinoLat: mejorDestino.lat,
        destinoLng: mejorDestino.lng,
        fechaVisita: new Date(trip.fecha_salida),
        kmViaje: km,
        velocidadEnDestino: null,
        tiempoEnDestinoMin: null,
        confianza: menorDiffKm < (DISTANCIAS_REFERENCIA_KM[mejorDestino.nombre] || 999) * 0.15 ? "MEDIA" : "MEDIA",
        puntosEnRadio: 0,
        origenNombre: "Santiago (estimado)",
        origenLat: -33.4489,
        origenLng: -70.6693,
        metodo: "ODOMETRO",
        tarifaClp: TARIFAS_POR_DESTINO[mejorDestino.nombre] || null,
        tmsViajeId: trip.id,
      });
    }
  }

  return visitas;
}

export async function detectarVisitasFlota(
  desde: string,
  hasta: string,
  patentes: string[]
): Promise<{
  procesados: number;
  con_gps: number;
  sin_gps: number;
  visitas_detectadas: number;
  visitas_alta_confianza: number;
  visitas_media_confianza: number;
  por_destino: Record<string, number>;
  visitas: VisitaDetectada[];
}> {
  if (destinosCencosud.length === 0) {
    await inicializarPerfilGPS();
  }

  const faenaId = await getCencosudFaenaId();
  let camionesQuery: any[];

  if (patentes.length > 0) {
    const result = await pool.query(
      `SELECT id, patente FROM camiones WHERE faena_id = $1 AND patente = ANY($2)`,
      [faenaId, patentes]
    );
    camionesQuery = result.rows;
  } else {
    const result = await pool.query(
      `SELECT id, patente FROM camiones WHERE faena_id = $1`,
      [faenaId]
```

```
426 |     );
427 |     camionesQuery = result.rows;
428 |   }
429 |
430 |   let conGps = 0;
431 |   let sinGps = 0;
432 |   const todasVisitas: VisitaDetectada[] = [];
433 |
434 |   for (const cam of camionesQuery) {
435 |     const perfil = perfilGpsFlota.get(cam.id);
436 |     const tieneDatosGPS = perfil && perfil.conCoords > 2;
437 |
438 |     if (tieneDatosGPS) {
439 |       conGps++;
440 |       const puntos = await obtenerPuntosGPS(cam.id, desde, hasta);
441 |       const visitasGPS = detectarVisitasGPS(puntos, cam.patente, cam.id);
442 |       todasVisitas.push(...visitasGPS);
443 |
444 |       if (visitasGPS.length === 0) {
445 |         const visitasOdo = await detectarVisitasPorOdometro(cam.id, cam.patente, desde, hasta);
446 |         todasVisitas.push(...visitasOdo);
447 |       }
448 |     } else {
449 |       sinGps++;
450 |       const visitasOdo = await detectarVisitasPorOdometro(cam.id, cam.patente, desde, hasta);
451 |       todasVisitas.push(...visitasOdo);
452 |     }
453 |   }
454 |
455 |   const porDestino: Record<string, number> = {};
456 |   for (const v of todasVisitas) {
457 |     const nombre = v.destinoNombre.replace("CD Cencosud ", "").replace("CD ", "");
458 |     porDestino[nombre] = (porDestino[nombre] || 0) + 1;
459 |   }
460 |
461 |   await guardarVisitas(todasVisitas);
462 |
463 |   return {
464 |     procesados: camionesQuery.length,
465 |     con_gps: conGps,
466 |     sin_gps: sinGps,
467 |     visitas_detectadas: todasVisitas.length,
468 |     visitas_alta_confianza: todasVisitas.filter(v => v.confianza === "ALTA").length,
469 |     visitas_media_confianza: todasVisitas.filter(v => v.confianza === "MEDIA").length,
470 |     por_destino: porDestino,
471 |     visitas: todasVisitas,
472 |   };
473 | }
474 |
475 | async function guardarVisitas(visitas: VisitaDetectada[]): Promise<void> {
476 |   if (visitas.length === 0) return;
477 |
478 |   await pool.query(`DELETE FROM geo_viajes WHERE contrato = 'CENCOSUD'`);
479 |
480 |   for (const v of visitas) {
481 |     await pool.query(`
482 |       INSERT INTO geo_viajes (
483 |         camion_id, patente, contrato,
484 |         origen_lat, origen_lng, origen_nombre, origen_timestamp,
485 |         destino_lat, destino_lng, destino_nombre, destino_timestamp,
486 |         km_gps, validacion_estado, validacion_detalle,
487 |         destino_lugar_id, creado_at
488 |       ) VALUES (
489 |         $1, $2, 'CENCOSUD',
490 |         $3, $4, $5, $6,
491 |         $7, $8, $9, $10,
492 |         $11, $12, $13,
493 |         $14, NOW()
494 |       )
495 |     `, [
496 |       v.camionId, v.patente,
497 |       v.origenLat, v.origenLng, v.origenNombre || "Santiago", v.fechaVisita,
498 |       v.destinoLat, v.destinoLng, v.destinoNombre, v.fechaVisita,
499 |       v.kmViaje,
500 |       v.confianza === "ALTA" ? "VALIDADO" : "PENDIENTE",
501 |       JSON.stringify({
502 |         confianza: v.confianza,
503 |         metodo: v.metodo,
504 |         puntosEnRadio: v.puntosEnRadio,
505 |         velocidadEnDestino: v.velocidadEnDestino,
506 |         tiempoEnDestinoMin: v.tiempoEnDestinoMin,
507 |         tarifaClp: v.tarifaClp,
508 |         tmsViajeId: v.tmsViajeId,
509 |       }),
510 |       v.destinoId,
511 |     ]);
512 |   }
513 |
514 |   console.log(`[visitas] Guardadas ${visitas.length} visitas en geo_viajes`);
515 | }
516 |
517 | export async function obtenerResumenVisitas(desde: string, hasta: string): Promise<any> {
518 |   const visitas = await pool.query(`
519 |     SELECT gv.*, gl.nombre as lugar_nombre
520 |     FROM geo_viajes gv
521 |     LEFT JOIN geo_lugares gl ON gv.destino_lugar_id = gl.id
522 |     WHERE gv.contrato = 'CENCOSUD'
523 |       AND gv.origen_timestamp >= $1
524 |     ORDER BY gv.destino_nombre, gv.origen_timestamp
525 |   `, [desde]);
526 |
527 |   const destinosMap = new Map<string, {
528 |     nombre: string;
529 |     lugarId: number | null;
530 |     lat: number;
531 |     lng: number;
532 |     tarifaClp: number;
```

```
533 |     kmReferencia: number;
534 |     visitas: any[];
535 |     visitasAltaConfianza: number;
536 |     visitasMediaConfianza: number;
537 |     camionesDistintos: Set<string>;
538 |     ultimoViaje: string | null;
539 |     kmTotal: number;
540 |   }>();
541 |
542 |   let sinDestinoDetectado = 0;
543 |
544 |   for (const v of visitas.rows) {
545 |     const nombre = v.destino_nombre || "Sin detectar";
546 |     if (nombre === "Sin detectar") {
547 |       sinDestinoDetectado++;
548 |       continue;
549 |     }
550 |
551 |     if (!destinosMap.has(nombre)) {
552 |       const destInfo = destinosCencosud.find(d => d.nombre === nombre);
553 |       destinosMap.set(nombre, {
554 |         nombre,
555 |         lugarId: v.destino_lugar_id,
556 |         lat: parseFloat(v.destino_lat) || destInfo?.lat || 0,
557 |         lng: parseFloat(v.destino_lng) || destInfo?.lng || 0,
558 |         tarifaClp: TARIFAS_POR_DESTINO[nombre] || 0,
559 |         kmReferencia: (DISTANCIAS_REFERENCIA_KM[nombre] || 0) * 2,
560 |         visitas: [],
561 |         visitasAltaConfianza: 0,
562 |         visitasMediaConfianza: 0,
563 |         camionesDistintos: new Set(),
564 |         ultimoViaje: null,
565 |         kmTotal: 0,
566 |       });
567 |     }
568 |
569 |     const dest = destinosMap.get(nombre)!;
570 |     const detalle = typeof v.validacion_detalle === "string" ? JSON.parse(v.validacion_detalle) : v.validacion_detalle |
571 |
572 |     const confianza = detalle.confianza || (v.validacion_estado === "VALIDADO" ? "ALTA" : "MEDIA");
573 |     if (confianza === "ALTA") dest.visitasAltaConfianza++;
574 |     else dest.visitasMediaConfianza++;
575 |
576 |     dest.camionesDistintos.add(v.patente);
577 |     const fechaStr = v.origen_timestamp ? new Date(v.origen_timestamp).toISOString().split("T")[0] : null;
578 |     if (fechaStr && (!dest.ultimoViaje || fechaStr > dest.ultimoViaje)) {
579 |       dest.ultimoViaje = fechaStr;
580 |     }
581 |
582 |     const km = parseFloat(v.km_gps) || 0;
583 |     dest.kmTotal += km;
584 |
585 |     dest.visitas.push({
586 |       id: v.id,
587 |       patente: v.patente,
588 |       fecha: fechaStr,
589 |       km: km,
590 |       confianza,
591 |       metodo: detalle.metodo || "GPS",
592 |       tarifaClp: detalle.tarifaClp || TARIFAS_POR_DESTINO[nombre] || 0,
593 |       velocidadEnDestino: detalle.velocidadEnDestino,
594 |       tiempoEnDestinoMin: detalle.tiempoEnDestinoMin,
595 |     });
596 |   }
597 |
598 |   const destinos = Array.from(destinosMap.values()).map(d => ({
599 |     nombre: d.nombre.replace("CD Cencosud ", "").replace("CD ", ""),
600 |     nombreCompleto: d.nombre,
601 |     lugarId: d.lugarId,
602 |     lat: d.lat,
603 |     lng: d.lng,
604 |     tarifaClp: d.tarifaClp,
605 |     kmReferencia: d.kmReferencia,
606 |     totalVisitas: d.visitas.length,
607 |     visitasAltaConfianza: d.visitasAltaConfianza,
608 |     visitasMediaConfianza: d.visitasMediaConfianza,
609 |     camionesDistintos: d.camionesDistintos.size,
610 |     ultimoViaje: d.ultimoViaje,
611 |     kmPromedioReal: d.visitas.length > 0 ? Math.round(d.kmTotal / d.visitas.length) : 0,
612 |     visitas: d.visitas,
613 |   }));
614 |
615 |   destinos.sort((a, b) => b.totalVisitas - a.totalVisitas);
616 |
617 |   const totalVisitas = destinos.reduce((s, d) => s + d.totalVisitas, 0);
618 |
619 |   return {
620 |     desde,
621 |     hasta,
622 |     destinos,
623 |     totalVisitas,
624 |     sinDestinoDetectado,
625 |     destinosSinVisitas: destinosCencosud
626 |       .filter(d => d.nombre !== "Base Santiago CENCOSUD" && !destinosMap.has(d.nombre))
627 |       .map(d => ({
628 |         nombre: d.nombre.replace("CD Cencosud ", "").replace("CD ", ""),
629 |         nombreCompleto: d.nombre,
630 |         lat: d.lat,
631 |         lng: d.lng,
632 |         tarifaClp: TARIFAS_POR_DESTINO[d.nombre] || 0,
633 |         kmReferencia: (DISTANCIAS_REFERENCIA_KM[d.nombre] || 0) * 2,
634 |       })),
635 |   };
636 | }
637 |
638 | export async function obtenerVisitasCamion(patente: string): Promise<any[]> {
639 |   const result = await pool.query(`
```

```
640 |     SELECT gv.*, gl.nombre as lugar_nombre
641 |     FROM geo_viajes gv
642 |     LEFT JOIN geo_lugares gl ON gv.destino_lugar_id = gl.id
643 |     WHERE gv.patente = $1 AND gv.contrato = 'CENCOSUD'
644 |     ORDER BY gv.origen_timestamp DESC
645 |   `, [patente]);
646 |
647 |   return result.rows.map(v => {
648 |     const detalle = typeof v.validacion_detalle === "string" ? JSON.parse(v.validacion_detalle) : v.validacion_detalle |
649 |     return {
650 |       id: v.id,
651 |       patente: v.patente,
652 |       destino: v.destino_nombre,
653 |       destinoCorto: (v.destino_nombre || "").replace("CD Cencosud ", "").replace("CD ", ""),
654 |       destinoLat: parseFloat(v.destino_lat) || null,
655 |       destinoLng: parseFloat(v.destino_lng) || null,
656 |       origenNombre: v.origen_nombre,
657 |       origenLat: parseFloat(v.origen_lat) || null,
658 |       origenLng: parseFloat(v.origen_lng) || null,
659 |       fecha: v.origen_timestamp ? new Date(v.origen_timestamp).toISOString().split("T")[0] : null,
660 |       km: parseFloat(v.km_gps) || 0,
661 |       confianza: detalle.confianza || (v.validacion_estado === "VALIDADO" ? "ALTA" : "MEDIA"),
662 |       metodo: detalle.metodo || "GPS",
663 |       tarifaClp: detalle.tarifaClp || 0,
664 |       velocidadEnDestino: detalle.velocidadEnDestino,
665 |       tiempoEnDestinoMin: detalle.tiempoEnDestinoMin,
666 |       puntosEnRadio: detalle.puntosEnRadio || 0,
667 |     };
668 |   });
669 | }
670 |
671 | export function getPerfilGPS() {
672 |   return perfilGpsFlota;
673 | }
674 |
675 | export function getDestinosCencosud() {
676 |   return destinosCencosud;
677 | }
678 |
```

server/vite.ts

59 lineas

```
1 | import { type Express } from "express";
2 | import { createServer as createViteServer, createLogger } from "vite";
3 | import { type Server } from "http";
4 | import viteConfig from "../vite.config";
5 | import fs from "fs";
6 | import path from "path";
7 | import { nanoid } from "nanoid";
8 |
9 | const viteLogger = createLogger();
10 |
11 | export async function setupVite(server: Server, app: Express) {
12 |   const serverOptions = {
13 |     middlewareMode: true,
14 |     hmr: { server, path: "/vite-hmr" },
15 |     allowedHosts: true as const,
16 |   };
17 |
18 |   const vite = await createViteServer({
19 |     ...viteConfig,
20 |     configFile: false,
21 |     customLogger: {
22 |       ...viteLogger,
23 |       error: (msg, options) => {
24 |         viteLogger.error(msg, options);
25 |         process.exit(1);
26 |       },
27 |     },
28 |     server: serverOptions,
29 |     appType: "custom",
30 |   });
31 |
32 |   app.use(vite.middlewares);
33 |
34 |   app.use("/{*path}", async (req, res, next) => {
35 |     const url = req.originalUrl;
36 |
37 |     try {
38 |       const clientTemplate = path.resolve(
39 |         import.meta.dirname,
40 |         "..",
41 |         "client",
42 |         "index.html",
43 |       );
44 |
45 |       // always reload the index.html file from disk incase it changes
46 |       let template = await fs.promises.readFile(clientTemplate, "utf-8");
47 |       template = template.replace(
48 |         `src="/src/main.tsx"`,
49 |         `src="/src/main.tsx?v=${nanoid()}"`,
50 |       );
51 |       const page = await vite.transformIndexHtml(url, template);
52 |       res.status(200).set({ "Content-Type": "text/html" }).end(page);
53 |     } catch (e) {
54 |       vite.ssrFixStacktrace(e as Error);
55 |       next(e);
56 |     }
57 |   });
58 | }
59 |
```

server/static.ts

20 lineas

```
1 | import express, { type Express } from "express";
2 | import fs from "fs";
3 | import path from "path";
4 |
5 | export function serveStatic(app: Express) {
6 |   const distPath = path.resolve(__dirname, "public");
7 |   if (!fs.existsSync(distPath)) {
8 |     throw new Error(
9 |       `Could not find the build directory: ${distPath}, make sure to build the client first`,
10 |     );
11 |   }
12 |
13 |   app.use(express.static(distPath));
14 |
15 |   // fall through to index.html if the file doesn't exist
16 |   app.use("/{*path}", (_req, res) => {
17 |     res.sendFile(path.resolve(distPath, "index.html"));
18 |   });
19 | }
20 |
```

client/src/main.tsx

6 lineas

```
1 | import { createRoot } from "react-dom/client";
2 | import App from "./App";
3 | import "./index.css";
4 |
5 | createRoot(document.getElementById("root")!).render(<App />);
6 |
```

client/src/App.tsx

616 lineas

```
1 | import { useState, useMemo, useEffect, useRef } from "react";
2 | import { queryClient, apiRequest } from "./lib/queryClient";
3 | import { QueryClientProvider, useQuery } from "@tanstack/react-query";
4 | import { Toaster } from "@/components/ui/toaster";
5 | import { TooltipProvider } from "@/components/ui/tooltip";
6 | import Hoy from "@/pages/hoy";
7 | import Flota from "@/pages/flota";
8 | import Configuration from "@/pages/configuration";
9 | import IATab from "@/pages/ia-tab";
10 | import GeoValidator from "@/pages/geovalidator";
11 | import { X, Loader2, Send, Search, HelpCircle, Home, Truck, Settings, AlertTriangle, Brain, Globe } from "lucide-react";
12 |
13 | interface AnalisisResult {
14 |   resumen: string;
15 |   litros_en_riesgo: string;
16 |   alertas_criticas: { patente: string; problema: string; accion: string; urgencia: string }[];
17 |   recomendacion_principal: string;
18 | }
19 |
20 | interface ConsultaResult {
21 |   respuesta: string;
22 | }
23 |
24 | const PREGUNTAS_SUGERIDAS = [
25 |   "Peor rendimiento?",
26 |   "Patrones sospechosos?",
27 |   "Faena que mas consume?",
28 |   "Conductores con alertas?",
29 |   "Cuantos bajo meta?",
30 |   "Resumen ejecutivo",
31 | ];
32 |
33 | const LOADER_TEXTS = [
34 |   "Leyendo datos de flota...",
35 |   "Analizando camiones...",
36 |   "Detectando patrones...",
37 |   "Generando diagnostico...",
38 | ];
39 |
40 | function AnalisisModal({ onClose }: { onClose: () => void }) {
41 |   const [data, setData] = useState<AnalisisResult | null>(null);
42 |   const [loading, setLoading] = useState(true);
43 |   const [error, setError] = useState<string | null>(null);
44 |   const [loaderText, setLoaderText] = useState(0);
45 |
46 |   useEffect(() => {
47 |     apiRequest("POST", "/api/ia/analizar")
48 |       .then(res => res.json())
49 |       .then(setData)
50 |       .catch(e => setError(e.message))
51 |       .finally(() => setLoading(false));
52 |   }, []);
53 |
54 |   useEffect(() => {
55 |     if (!loading) return;
56 |     const id = setInterval(() => {
57 |       setLoaderText(p => (p + 1) % LOADER_TEXTS.length);
58 |     }, 800);
59 |     return () => clearInterval(id);
60 |   }, [loading]);
61 |
62 |   return (
63 |     <div className="fixed inset-0 z-50 flex items-center justify-center bg-black/80" onClick={onClose} data-testid="moda
64 |       <div className="w-full max-w-2xl max-h-[85vh] overflow-y-auto mx-4" style={{ background: '#020508', border: '1px s
65 |         <div className="flex items-center justify-between p-4 border-b" style={{ borderColor: '#0d2035' }}>
66 |           <div className="flex items-center gap-2">
67 |             <span className="text-sm">&#9889;</span>
68 |             <span className="font-space text-sm font-bold tracking-[0.15em]" style={{ color: '#00d4ff' }}>ANALISIS IA DE
69 |           </div>
70 |           <button onClick={onClose} className="p-1 cursor-pointer hover:opacity-70" data-testid="btn-close-analisis"><X
71 |         </div>
72 |
73 |         <div className="p-5 space-y-4">
74 |           {loading && (
75 |             <div className="flex flex-col items-center gap-4 py-12">
76 |               <div className="w-12 h-12 octagonal flex items-center justify-center" style={{ border: '1px solid rgba(0,2
77 |                 <Loader2 className="w-6 h-6 animate-spin" style={{ color: '#00d4ff' }} />
78 |               </div>
79 |               <span className="font-exo text-sm" style={{ color: '#3a6080' }}>{LOADER_TEXTS[loaderText]}</span>
80 |             </div>
81 |           )}
82 |
83 |           {error && (
84 |             <div className="p-4 border" style={{ borderColor: 'rgba(255,34,68,0.3)', background: 'rgba(255,34,68,0.05)'
85 |               <span className="font-rajdhani text-sm" style={{ color: '#ff2244' }}>{error}</span>
86 |             </div>
87 |           )}
88 |
89 |           {data && (
90 |             <>
91 |               <div className="p-4 border" style={{ borderColor: '#0d2035' }}>
92 |                 <div className="font-exo text-[8px] tracking-[0.2em] mb-2" style={{ color: '#3a6080' }}>RESUMEN</div>
93 |                 <p className="font-rajdhani text-sm leading-relaxed" style={{ color: '#c8e8ff' }}>{data.resumen}</p>
94 |               </div>
95 |
96 |               <div className="p-4 border" style={{ borderColor: 'rgba(255,34,68,0.3)', background: 'rgba(255,34,68,0.04)
97 |                 <div className="font-exo text-[8px] tracking-[0.2em] mb-2" style={{ color: '#ff2244' }}>LITROS EN RIESGO
98 |                 <p className="text-xl font-space font-bold" style={{ color: '#ff2244' }} data-testid="text-litros-en-rie
99 |               </div>
100 |
101 |               {data.alertas_criticas.length > 0 && (
102 |                 <div>
103 |                   <div className="font-exo text-[8px] tracking-[0.2em] mb-2" style={{ color: '#3a6080' }}>ALERTAS CRITIC
104 |                   <div className="space-y-2">
```

```
105 |                     {data.alertas_criticas.map((a, i) => (
106 |                       <div key={i} className="p-3 border flex items-start gap-3" style={{ borderColor: '#0d2035' }} data
107 |                         <span className={`px-2 py-0.5 text-[9px] font-space font-bold flex-shrink-0 border ${
108 |                           a.urgencia === "INMEDIATA" ? "border-[rgba(255,34,68,0.4)] text-[#ff2244] bg-[rgba(255,34,68,0
109 |                           a.urgencia === "24H" ? "border-[rgba(255,204,0,0.4)] text-[#ffcc00] bg-[rgba(255,204,0,0.1)]"
110 |                           "border-[rgba(0,212,255,0.4)] text-[#00d4ff] bg-[rgba(0,212,255,0.1)]"
111 |                         }`}>{a.urgencia}</span>
112 |                         <div className="min-w-0">
113 |                           <div className="font-space text-xs font-bold" style={{ color: '#c8e8ff' }}>{a.patente}: {a.pro
114 |                           <div className="font-rajdhani text-[11px] mt-1" style={{ color: '#3a6080' }}>{a.accion}</div>
115 |                         </div>
116 |                       </div>
117 |                     ))}
118 |                   </div>
119 |                 </div>
120 |               )}
121 |
122 |               <div className="p-4 border" style={{ borderColor: 'rgba(0,255,136,0.3)', background: 'rgba(0,255,136,0.04)
123 |                 <div className="font-exo text-[8px] tracking-[0.2em] mb-2" style={{ color: '#00ff88' }}>RECOMENDACION PR
124 |                 <p className="font-rajdhani text-sm leading-relaxed" style={{ color: '#00ff88' }} data-testid="text-reco
125 |               </div>
126 |             </>
127 |           )}
128 |         </div>
129 |       </div>
130 |     </div>
131 |   );
132 | }
133 |
134 | function IASidebar({ onClose }: { onClose: () => void }) {
135 |   const [pregunta, setPregunta] = useState("");
136 |   const [respuesta, setRespuesta] = useState<string | null>(null);
137 |   const [loading, setLoading] = useState(false);
138 |   const [error, setError] = useState<string | null>(null);
139 |   const [inputFocused, setInputFocused] = useState(false);
140 |
141 |   const enviar = async (q: string) => {
142 |     if (!q.trim()) return;
143 |     setPregunta(q);
144 |     setLoading(true);
145 |     setError(null);
146 |     setRespuesta(null);
147 |     try {
148 |       const res = await apiRequest("POST", "/api/ia/consulta", { pregunta: q });
149 |       const data: ConsultaResult = await res.json();
150 |       setRespuesta(data.respuesta);
151 |     } catch (e: any) {
152 |       setError(e.message);
153 |     } finally {
154 |       setLoading(false);
155 |     }
156 |   };
157 |
158 |   return (
159 |     <>
160 |       <div className="fixed inset-0 z-40 bg-black/50" onClick={onClose} />
161 |       <div className="fixed top-0 right-0 h-full w-full max-w-md z-50 overflow-y-auto"
162 |         style={{ background: '#020508', borderLeft: '1px solid #0d2035' }}
163 |         data-testid="sidebar-ia">
164 |
165 |         <div className="flex items-center justify-between p-4 border-b" style={{ borderColor: '#0d2035' }}>
166 |           <div className="flex items-center gap-2">
167 |             <Search className="w-4 h-4" style={{ color: '#00d4ff' }} />
168 |             <span className="font-space text-sm font-bold tracking-[0.15em]" style={{ color: '#c8e8ff' }}>SOTRA IA</span
169 |           </div>
170 |           <button onClick={onClose} className="p-1 cursor-pointer hover:opacity-70" data-testid="btn-close-ia-sidebar">
171 |             <X className="w-5 h-5" style={{ color: '#3a6080' }} />
172 |           </button>
173 |         </div>
174 |
175 |         <div className="p-4 space-y-4">
176 |           <div class="flex gap-2">
177 |             <input
178 |               value={pregunta}
179 |               onChange={e => setPregunta(e.target.value)}
180 |               onKeyDown={e => e.key === "Enter" && enviar(pregunta)}
181 |               onFocus={() => setInputFocused(true)}
182 |               onBlur={() => setInputFocused(false)}
183 |               placeholder="Que quieres saber de tu flota?"
184 |               className="flex-1 px-3 py-2.5 font-rajdhani text-sm bg-transparent border outline-none transition-all"
185 |               style={{
186 |                 borderColor: inputFocused ? '#00d4ff' : '#0d2035',
187 |                 color: '#c8e8ff',
188 |                 boxShadow: inputFocused ? '0 0 12px rgba(0,212,255,0.15)' : undefined
189 |               }}
190 |               data-testid="input-consulta-sidebar"
191 |               disabled={loading}
192 |             />
193 |             <button
194 |               onClick={() => enviar(pregunta)}
195 |               disabled={loading || !pregunta.trim()}
196 |               className="px-4 py-2.5 font-space text-sm font-bold cursor-pointer disabled:opacity-40 transition-all"
197 |               style={{ background: 'rgba(0,212,255,0.1)', border: '1px solid rgba(0,212,255,0.3)', color: '#00d4ff' }}
198 |               data-testid="btn-enviar-sidebar"
199 |             >
200 |               {loading ? <Loader2 className="w-4 h-4 animate-spin" /> : <Send class="w-4 h-4" />}
201 |             </button>
202 |           </div>
203 |
204 |           <div class="flex flex-wrap gap-2">
205 |             {PREGUNTAS_SUGERIDAS.map((q, i) => (
206 |               <button
207 |                 key={i}
208 |                 onClick={() => { setPregunta(q); enviar(q); }}
209 |                 class="px-3 py-1.5 font-exo text-[10px] border cursor-pointer transition-all hover:border-[#00d4ff]
210 |                 style={{ borderColor: '#0d2035', color: '#3a6080' }}
211 |                 data-testid={`chip-pregunta-${i}`}
```

```
212 |                 disabled={loading}
213 |               >
214 |                 {q}
215 |               </button>
216 |             ))}
217 |           </div>
218 |
219 |           {loading && (
220 |             <div className="flex flex-col items-center gap-3 py-8">
221 |               <Loader2 className="w-6 h-6 animate-spin" style={{ color: '#00d4ff' }} />
222 |               <span className="font-exo text-sm" style={{ color: '#3a6080' }}>Consultando IA...</span>
223 |             </div>
224 |           )}
225 |
226 |           {error && (
227 |             <div className="p-4 border" style={{ borderColor: 'rgba(255,34,68,0.3)', background: 'rgba(255,34,68,0.05)'
228 |               <span className="font-rajdhani text-sm" style={{ color: '#ff2244' }}>{error}</span>
229 |             </div>
230 |           )}
231 |
232 |           {respuesta && (
233 |             <div className="p-4 border" style={{ borderColor: '#0d2035' }}>
234 |               <div className="font-exo text-[8px] tracking-[0.2em] mb-3" style={{ color: '#3a6080' }}>RESPUESTA IA</div>
235 |               <div className="font-rajdhani text-sm leading-relaxed whitespace-pre-wrap" style={{ color: '#c8e8ff' }} da
236 |             </div>
237 |           )}
238 |         </div>
239 |       </div>
240 |     </>
241 |   );
242 | }
243 |
244 | const ONBOARDING_SLIDES = [
245 |   {
246 |     content: (
247 |       <div className="flex flex-col items-center text-center py-6">
248 |         <div className="octagonal w-[80px] h-[80px] flex items-center justify-center mb-4" style={{
249 |           background: 'radial-gradient(circle, rgba(0,212,255,0.08) 0%, transparent 70%)',
250 |           border: '1px solid rgba(0,212,255,0.3)',
251 |           animation: 'glowPulse 3s ease-in-out infinite'
252 |         }}>
253 |           <span className="text-3xl">&#9981;</span>
254 |         </div>
255 |         <div className="font-space text-2xl font-bold tracking-[0.3em] mb-1" style={{ color: '#00d4ff' }}>SOTRASER</div>
256 |         <div className="font-exo text-[11px] tracking-[0.3em] font-extralight mb-4" style={{ color: '#3a6080' }}>Tu torr
257 |       </div>
258 |     ),
259 |     button: "Siguiente"
260 |   },
261 |   {
262 |     content: (
263 |       <div className="space-y-3 py-4">
264 |         {[
265 |           { icon: "\ud83c\udfe0", label: "HOY", desc: "Ve el estado de tu flota ahora mismo" },
266 |           { icon: "\ud83d\ude9b", label: "FLOTA", desc: "Revisa camiones, viajes y conductores" },
267 |           { icon: "\u2699\ufe0f", label: "CONFIG", desc: "Ajusta parametros del sistema" },
268 |         ].map(item => (
269 |           <div key={item.label} className="flex items-center gap-3 p-3 border" style={{ borderColor: '#0d2035', backgrou
270 |             <span className="text-xl">{item.icon}</span>
271 |             <div>
272 |               <div className="font-space text-[11px] font-bold tracking-wider" style={{ color: '#00d4ff' }}>{item.label}
273 |               <div className="font-rajdhani text-[12px]" style={{ color: '#3a6080' }}>{item.desc}</div>
274 |             </div>
275 |           </div>
276 |         ))}
277 |       </div>
278 |     ),
279 |     button: "Siguiente"
280 |   },
281 |   {
282 |     content: (
283 |       <div className="flex flex-col items-center text-center py-6">
284 |         <div className="text-3xl mb-3">&#9989;</div>
285 |         <div className="font-space text-lg font-bold tracking-wider mb-2" style={{ color: '#00d4ff' }}>Todo listo</div>
286 |         <div className="font-rajdhani text-sm" style={{ color: '#3a6080' }}>
287 |           Tienes 23 camiones CENCOSUD conectados<br />
288 |           Empieza por la pestana HOY
289 |         </div>
290 |       </div>
291 |     ),
292 |     button: "Entrar"
293 |   }
294 | ];
295 |
296 | function OnboardingModal({ onClose }: { onClose: () => void }) {
297 |   const [page, setPage] = useState(0);
298 |
299 |   const next = () => {
300 |     if (page < ONBOARDING_SLIDES.length - 1) setPage(page + 1);
301 |     else {
302 |       localStorage.setItem("sotraser_onboarded", "true");
303 |       onClose();
304 |     }
305 |   };
306 |
307 |   return (
308 |     <div className="fixed inset-0 z-[60] flex items-center justify-center bg-black/85" data-testid="modal-onboarding">
309 |       <div className="w-full max-w-md mx-4 relative" style={{ background: '#020508', border: '1px solid rgba(0,212,255,0
310 |         <button onClick={() => { localStorage.setItem("sotraser_onboarded", "true"); onClose(); }}
311 |           className="absolute top-3 right-3 p-1 cursor-pointer hover:opacity-70 z-10" data-testid="btn-close-onboarding"
312 |           <X className="w-4 h-4" style={{ color: '#3a6080' }} />
313 |         </button>
314 |
315 |         <div className="p-6">
316 |           {ONBOARDING_SLIDES[page].content}
317 |
318 |           <button onClick={next} className="w-full mt-4 py-2.5 font-space text-xs font-bold tracking-[0.15em] cursor-poi
```

```
319 |             background: 'rgba(0,212,255,0.08)', border: '1px solid rgba(0,212,255,0.3)', color: '#00d4ff'
320 |           }} data-testid="btn-onboarding-next">
321 |             {ONBOARDING_SLIDES[page].button} &rarr;
322 |           </button>
323 |
324 |           <div className="flex justify-center gap-2 mt-4">
325 |             {ONBOARDING_SLIDES.map((_, i) => (
326 |               <button key={i} onClick={() => setPage(i)} className="w-2 h-2 rounded-full cursor-pointer transition-all"
327 |                 background: i === page ? '#00d4ff' : '#0d2035'
328 |               }} data-testid={`onboarding-dot-${i}`} />
329 |             ))}
330 |           </div>
331 |         </div>
332 |       </div>
333 |     </div>
334 |   );
335 | }
336 |
337 | function SplashScreen({ onDone }: { onDone: () => void }) {
338 |   const [progress, setProgress] = useState(0);
339 |   const [textIdx, setTextIdx] = useState(0);
340 |   const texts = ["Conectando Volvo rFMS...", "Sincronizando Sigetra...", "Procesando flota...", "Sistema listo"];
341 |
342 |   useEffect(() => {
343 |     const dur = 2500;
344 |     const interval = 30;
345 |     let elapsed = 0;
346 |     const timer = setInterval(() => {
347 |       elapsed += interval;
348 |       setProgress(Math.min((elapsed / dur) * 100, 100));
349 |       const idx = Math.min(Math.floor((elapsed / dur) * texts.length), texts.length - 1);
350 |       setTextIdx(idx);
351 |       if (elapsed >= dur) {
352 |         clearInterval(timer);
353 |         setTimeout(onDone, 500);
354 |       }
355 |     }, interval);
356 |     return () => clearInterval(timer);
357 |   }, []);
358 |
359 |   return (
360 |     <div className="fixed inset-0 z-[100] flex flex-col items-center justify-center" style={{ background: '#020508' }} d
361 |       <div className="octagonal w-[80px] h-[80px] flex items-center justify-center mb-6" style={{
362 |         background: 'radial-gradient(circle, rgba(0,212,255,0.08) 0%, transparent 70%)',
363 |         border: '1px solid rgba(0,212,255,0.3)',
364 |         animation: 'glowPulse 2s ease-in-out infinite'
365 |       }}>
366 |         <span className="text-3xl">&#9981;</span>
367 |       </div>
368 |       <div className="font-space text-[32px] font-bold tracking-[0.3em] mb-1" style={{ color: '#00d4ff', textShadow: '0
369 |         SOTRASER
370 |       </div>
371 |       <div className="font-exo text-[11px] tracking-[0.3em] font-extralight mb-8" style={{ color: '#3a6080' }}>
372 |         CONTROL OPERACIONAL DE FLOTA
373 |       </div>
374 |       <div className="w-64 h-1 overflow-hidden mb-4" style={{ background: '#0d2035' }}>
375 |         <div className="h-full transition-all duration-100" style={{ width: `${progress}%`, background: '#00d4ff' }} />
376 |       </div>
377 |       <div className="font-exo text-[11px] tracking-wider h-5" style={{ color: '#3a6080' }}>
378 |         {texts[textIdx]}
379 |       </div>
380 |     </div>
381 |   );
382 | }
383 |
384 | function LiveClock() {
385 |   const [time, setTime] = useState("");
386 |
387 |   useEffect(() => {
388 |     const update = () => {
389 |       const now = new Date();
390 |       const months = ["ENE", "FEB", "MAR", "ABR", "MAY", "JUN", "JUL", "AGO", "SEP", "OCT", "NOV", "DIC"];
391 |       const m = months[now.getMonth()];
392 |       const d = now.getDate();
393 |       const h = String(now.getHours()).padStart(2, "0");
394 |       const min = String(now.getMinutes()).padStart(2, "0");
395 |       setTime(`${m} ${d} \u00b7 ${h}:${min}`);
396 |     };
397 |     update();
398 |     const id = setInterval(update, 60000);
399 |     return () => clearInterval(id);
400 |   }, []);
401 |
402 |   return <span className="font-space text-[11px]" style={{ color: '#3a6080' }}>{time}</span>;
403 | }
404 |
405 | type Tab = "hoy" | "flota" | "ia" | "config" | "geo";
406 |
407 | function SemaforoBar({ criticos }: { criticos: number }) {
408 |   const badge = (() => {
409 |     if (criticos === 0) return { label: "OPERACION NORMAL", color: "#00ff88", blink: false };
410 |     if (criticos <= 3) return { label: "ATENCION", color: "#ffcc00", blink: true };
411 |     return { label: "ALERTA ACTIVA", color: "#ff2244", blink: true };
412 |   })();
413 |
414 |   const { data: kpis } = useQuery<any>({
415 |     queryKey: ["/api/ceo/kpis", "7d"],
416 |     queryFn: () => fetch("/api/ceo/kpis?periodo=7d").then(r => r.json()),
417 |     refetchInterval: 120000,
418 |   });
419 |
420 |   return (
421 |     <div className="flex items-center justify-between px-5 py-1.5" style={{ background: 'rgba(6,13,20,0.6)', borderBotto
422 |       data-testid="semaforo-bar">
423 |       <div className="flex items-center gap-2">
424 |         <span className="inline-block w-2.5 h-2.5 rounded-full" style={{
425 |           background: badge.color,
```

```
426 |           boxShadow: `0 0 8px ${badge.color}`,
427 |           animation: badge.blink ? "blinkFast 0.6s infinite" : undefined,
428 |         }} />
429 |         <span className="font-space text-[10px] font-bold tracking-[0.1em]" style={{ color: badge.color }}>
430 |           {badge.label}
431 |         </span>
432 |       </div>
433 |       <div className="flex items-center gap-3 font-exo text-[10px]" style={{ color: "#3a6080" }}>
434 |         {kpis && (
435 |           <>
436 |             <span>{kpis.camionesActivos} activos</span>
437 |             <span>&middot;</span>
438 |             <span>{kpis.rendPromedio} km/L</span>
439 |             <span>&middot;</span>
440 |             <span>{criticos} alertas criticas</span>
441 |           </>
442 |         )}
443 |       </div>
444 |     </div>
445 |   );
446 | }
447 |
448 | function AppShell() {
449 |   const [tab, setTab] = useState<Tab>("hoy");
450 |   const [showSplash, setShowSplash] = useState(true);
451 |   const [showOnboarding, setShowOnboarding] = useState(false);
452 |   const [showAnalisis, setShowAnalisis] = useState(false);
453 |   const [showIASidebar, setShowIASidebar] = useState(false);
454 |
455 |   const { data: alertasData } = useQuery<any[]>({
456 |     queryKey: ["/api/ceo/alertas"],
457 |     queryFn: () => fetch("/api/ceo/alertas?limite=20").then(r => r.json()),
458 |     refetchInterval: 120000,
459 |   });
460 |
461 |   const criticos = useMemo(() => {
462 |     if (!alertasData) return 0;
463 |     return alertasData.filter(a => a.nivel === "CRITICO").length;
464 |   }, [alertasData]);
465 |
466 |   useEffect(() => {
467 |     if (!showSplash && !localStorage.getItem("sotraser_onboarded")) {
468 |       setShowOnboarding(true);
469 |     }
470 |   }, [showSplash]);
471 |
472 |   const handleAlertBadgeClick = () => {
473 |     setTab("hoy");
474 |     setTimeout(() => {
475 |       const el = document.querySelector('[data-testid="hoy-alertas"]');
476 |       el?.scrollIntoView({ behavior: "smooth", block: "start" });
477 |     }, 100);
478 |   };
479 |
480 |   const [flotaInitialSub, setFlotaInitialSub] = useState<"envivo" | "conductores" | "combustible" | undefined>();
481 |
482 |   const handleNavigateFlota = (contrato?: string) => {
483 |     setTab("flota");
484 |     setFlotaInitialSub("envivo");
485 |   };
486 |
487 |   if (showSplash) {
488 |     return <SplashScreen onDone={() => setShowSplash(false)} />;
489 |   }
490 |
491 |   const tabs: { id: Tab; label: string; icon: typeof Home }[] = [
492 |     { id: "hoy", label: "HOY", icon: Home },
493 |     { id: "flota", label: "FLOTA", icon: Truck },
494 |     { id: "ia", label: "IA", icon: Brain },
495 |     { id: "geo", label: "GEO", icon: Globe },
496 |     { id: "config", label: "CONFIG", icon: Settings },
497 |   ];
498 |
499 |   return (
500 |     <div className="min-h-screen" style={{ background: '#020508', color: '#c8e8ff' }}>
501 |       <div className="sticky top-0 z-40" style={{ background: 'rgba(2,5,8,0.85)', backdropFilter: 'blur(12px)', borderBo
502 |         <div className="h-[50px] flex items-center justify-between px-5">
503 |           <div className="flex items-center gap-3">
504 |             <div className="octagonal w-8 h-8 flex items-center justify-center" style={{
505 |               border: '1px solid rgba(0,212,255,0.4)',
506 |               animation: 'glowPulse 3s ease-in-out infinite'
507 |             }}>
508 |               <span className="text-sm">&#9981;</span>
509 |             </div>
510 |             <div className="flex items-center gap-3">
511 |               <div className="font-space text-[15px] font-bold tracking-[0.3em]" style={{ color: '#00d4ff' }} data-testi
512 |                 SOTRASER
513 |               </div>
514 |               <div className="font-exo text-[11px] font-bold tracking-wider" style={{
515 |                 background: 'rgba(0,212,255,0.12)',
516 |                 border: '1px solid #00d4ff',
517 |                 color: '#00d4ff',
518 |                 borderRadius: '6px',
519 |                 padding: '3px 10px',
520 |               }} data-testid="badge-cencosud">
521 |                 CENCOSUD
522 |               </div>
523 |             </div>
524 |           </div>
525 |
526 |           <div className="flex items-center gap-1">
527 |             {tabs.map(t => {
528 |               const Icon = t.icon;
529 |               return (
530 |                 <button key={t.id} onClick={() => setTab(t.id)}
531 |                   data-testid={`tab-${t.id}`}
532 |                   className={`flex items-center gap-1.5 px-4 py-2 font-exo text-[11px] font-bold tracking-[0.15em] curso
```

```
533 |                     tab === t.id
534 |                       ? "border-[#00d4ff] text-[#00d4ff]"
535 |                       : "border-transparent text-[#3a6080] hover:text-[#c8e8ff]"
536 |                   }`}>
537 |                   <Icon className="w-3.5 h-3.5" />
538 |                   {t.label}
539 |                 </button>
540 |               );
541 |             })}
542 |
543 |             {criticos > 0 && (
544 |               <button onClick={handleAlertBadgeClick}
545 |                 className="ml-2 flex items-center gap-1.5 px-2.5 py-1 cursor-pointer transition-all"
546 |                 style={{ background: 'rgba(255,34,68,0.12)', border: '1px solid rgba(255,34,68,0.4)' }}
547 |                 data-testid="alert-badge">
548 |                 <span className="w-1.5 h-1.5 rounded-full" style={{ background: '#ff2244', animation: 'blinkFast 0.6s in
549 |                 <span className="font-space text-[11px] font-bold" style={{ color: '#ff2244' }}>
550 |                   {criticos}
551 |                 </span>
552 |               </button>
553 |             )}
554 |           </div>
555 |
556 |           <div className="flex items-center gap-3">
557 |             <div className="flex items-center gap-1.5">
558 |               <span className="w-2 h-2 rounded-full" style={{ background: '#00ff88', animation: 'blink 2s infinite', box
559 |               <span className="font-exo text-[10px] font-bold tracking-wider" style={{ color: '#00ff88' }}>EN VIVO</span
560 |             </div>
561 |             <LiveClock />
562 |
563 |             <button
564 |               onClick={() => setShowAnalisis(true)}
565 |               className="octagonal flex items-center gap-1.5 px-3 py-1.5 font-space text-[10px] font-bold cursor-pointer
566 |               style={{ background: 'rgba(0,212,255,0.06)', border: '1px solid rgba(0,212,255,0.3)', color: '#00d4ff' }}
567 |               data-testid="btn-header-analisis"
568 |             >
569 |               &#9889; IA
570 |             </button>
571 |             <button
572 |               onClick={() => setShowIASidebar(true)}
573 |               className="flex items-center gap-1.5 px-3 py-1.5 font-space text-[10px] font-bold cursor-pointer transitio
574 |               style={{ borderColor: '#0d2035', color: '#3a6080' }}
575 |               data-testid="btn-header-consultar"
576 |             >
577 |               <Search className="w-3 h-3" /> CONSULTAR
578 |             </button>
579 |             <button onClick={() => setShowOnboarding(true)} className="p-1.5 cursor-pointer hover:opacity-70" data-testi
580 |               <HelpCircle className="w-4 h-4" style={{ color: '#3a6080' }} />
581 |             </button>
582 |           </div>
583 |         </div>
584 |
585 |         {(tab === "hoy" || tab === "flota") && <SemaforoBar criticos={criticos} />}
586 |       </div>
587 |
588 |       <div className="p-5 max-w-[1400px] mx-auto">
589 |         {tab === "hoy" && <Hoy onNavigateFlota={handleNavigateFlota} onOpenIA={() => setShowIASidebar(true)} />}
590 |         {tab === "flota" && <Flota initialSub={flotaInitialSub} />}
591 |
592 |         {tab === "ia" && <IATab />}
593 |         {tab === "geo" && <GeoValidator />}
594 |         {tab === "config" && <Configuration />}
595 |       </div>
596 |
597 |       {showOnboarding && <OnboardingModal onClose={() => setShowOnboarding(false)} />}
598 |       {showAnalisis && <AnalisisModal onClose={() => setShowAnalisis(false)} />}
599 |       {showIASidebar && <IASidebar onClose={() => setShowIASidebar(false)} />}
600 |     </div>
601 |   );
602 | }
603 |
604 | function App() {
605 |   return (
606 |     <QueryClientProvider client={queryClient}>
607 |       <TooltipProvider>
608 |         <Toaster />
609 |         <AppShell />
610 |       </TooltipProvider>
611 |     </QueryClientProvider>
612 |   );
613 | }
614 |
615 | export default App;
616 |
```

client/src/index.css

454 lineas

```
1 | @tailwind base;
2 | @tailwind components;
3 | @tailwind utilities;
4 |
5 | /* LIGHT MODE */
6 | :root {
7 |   --button-outline: rgba(0,0,0, .10);
8 |   --badge-outline: rgba(0,0,0, .05);
9 |   --opaque-button-border-intensity: -8;
10 |   --elevate-1: rgba(0,0,0, .03);
11 |   --elevate-2: rgba(0,0,0, .08);
12 |   --background: 222 40% 4%;
13 |   --foreground: 210 40% 90%;
14 |   --border: 218 35% 14%;
15 |   --card: 220 38% 6%;
16 |   --card-foreground: 210 40% 90%;
17 |   --card-border: 218 35% 12%;
18 |   --sidebar: 220 35% 7%;
19 |   --sidebar-foreground: 210 40% 90%;
20 |   --sidebar-border: 218 35% 14%;
21 |   --sidebar-primary: 210 100% 55%;
22 |   --sidebar-primary-foreground: 0 0% 100%;
23 |   --sidebar-accent: 220 30% 12%;
24 |   --sidebar-accent-foreground: 210 40% 90%;
25 |   --sidebar-ring: 210 100% 55%;
26 |   --popover: 220 35% 8%;
27 |   --popover-foreground: 210 40% 90%;
28 |   --popover-border: 218 35% 14%;
29 |   --primary: 210 100% 55%;
30 |   --primary-foreground: 0 0% 100%;
31 |   --secondary: 220 30% 12%;
32 |   --secondary-foreground: 210 40% 90%;
33 |   --muted: 220 25% 13%;
34 |   --muted-foreground: 215 25% 45%;
35 |   --accent: 220 25% 11%;
36 |   --accent-foreground: 210 40% 90%;
37 |   --destructive: 350 90% 55%;
38 |   --destructive-foreground: 0 0% 100%;
39 |   --input: 218 35% 14%;
40 |   --ring: 210 100% 55%;
41 |   --chart-1: 210 100% 55%;
42 |   --chart-2: 155 100% 40%;
43 |   --chart-3: 38 100% 56%;
44 |   --chart-4: 280 65% 55%;
45 |   --chart-5: 350 90% 55%;
46 |   --font-sans: 'Barlow', system-ui, sans-serif;
47 |   --font-serif: Georgia, serif;
48 |   --font-mono: 'JetBrains Mono', 'Courier New', monospace;
49 |   --radius: .5rem;
50 |   --shadow-2xs: 0px 2px 0px 0px hsl(220 18% 90% / 0.00);
51 |   --shadow-xs: 0px 2px 0px 0px hsl(220 18% 90% / 0.00);
52 |   --shadow-sm: 0px 2px 0px 0px hsl(220 18% 90% / 0.00), 0px 1px 2px -1px hsl(220 18% 90% / 0.00);
53 |   --shadow: 0px 2px 0px 0px hsl(220 18% 90% / 0.00), 0px 1px 2px -1px hsl(220 18% 90% / 0.00);
54 |   --shadow-md: 0px 2px 0px 0px hsl(220 18% 90% / 0.00), 0px 2px 4px -1px hsl(220 18% 90% / 0.00);
55 |   --shadow-lg: 0px 2px 0px 0px hsl(220 18% 90% / 0.00), 0px 4px 6px -1px hsl(220 18% 90% / 0.00);
56 |   --shadow-xl: 0px 2px 0px 0px hsl(220 18% 90% / 0.00), 0px 8px 10px -1px hsl(220 18% 90% / 0.00);
57 |   --shadow-2xl: 0px 2px 0px 0px hsl(220 18% 90% / 0.00);
58 |   --tracking-normal: 0em;
59 |   --spacing: 0.25rem;
60 |
61 | /* Fallback for older browsers */
62 |   --sidebar-primary-border: hsl(var(--sidebar-primary));
63 |   --sidebar-primary-border: hsl(from hsl(var(--sidebar-primary)) h s calc(l + var(--opaque-button-border-intensity)) / a
64 |
65 |   /* Fallback for older browsers */
66 |   --sidebar-accent-border: hsl(var(--sidebar-accent));
67 |   --sidebar-accent-border: hsl(from hsl(var(--sidebar-accent)) h s calc(l + var(--opaque-button-border-intensity)) / alp
68 |
69 |   /* Fallback for older browsers */
70 |   --primary-border: hsl(var(--primary));
71 |   --primary-border: hsl(from hsl(var(--primary)) h s calc(l + var(--opaque-button-border-intensity)) / alpha);
72 |
73 |   /* Fallback for older browsers */
74 |   --secondary-border: hsl(var(--secondary));
75 |   --secondary-border: hsl(from hsl(var(--secondary)) h s calc(l + var(--opaque-button-border-intensity)) / alpha);
76 |
77 |   /* Fallback for older browsers */
78 |   --muted-border: hsl(var(--muted));
79 |   --muted-border: hsl(from hsl(var(--muted)) h s calc(l + var(--opaque-button-border-intensity)) / alpha);
80 |
81 |   /* Fallback for older browsers */
82 |   --accent-border: hsl(var(--accent));
83 |   --accent-border: hsl(from hsl(var(--accent)) h s calc(l + var(--opaque-button-border-intensity)) / alpha);
84 |
85 |   /* Fallback for older browsers */
86 |   --destructive-border: hsl(var(--destructive));
87 |   --destructive-border: hsl(from hsl(var(--destructive)) h s calc(l + var(--opaque-button-border-intensity)) / alpha);
88 | }
89 |
90 | .dark {
91 |   --button-outline: rgba(255,255,255, .10);
92 |   --badge-outline: rgba(255,255,255, .05);
93 |   --opaque-button-border-intensity: 9;
94 |   --elevate-1: rgba(255,255,255, .04);
95 |   --elevate-2: rgba(255,255,255, .09);
96 |   --background: 222 40% 4%;
97 |   --foreground: 210 40% 90%;
98 |   --border: 218 35% 14%;
99 |   --card: 220 38% 6%;
100 |   --card-foreground: 210 40% 90%;
101 |   --card-border: 218 35% 12%;
102 |   --sidebar: 220 35% 7%;
103 |   --sidebar-foreground: 210 40% 90%;
104 |   --sidebar-border: 218 35% 14%;
```

```
105 |   --sidebar-primary: 210 100% 55%;
106 |   --sidebar-primary-foreground: 0 0% 100%;
107 |   --sidebar-accent: 220 30% 12%;
108 |   --sidebar-accent-foreground: 210 40% 90%;
109 |   --sidebar-ring: 210 100% 55%;
110 |   --popover: 220 35% 8%;
111 |   --popover-foreground: 210 40% 90%;
112 |   --popover-border: 218 35% 14%;
113 |   --primary: 210 100% 55%;
114 |   --primary-foreground: 0 0% 100%;
115 |   --secondary: 220 30% 12%;
116 |   --secondary-foreground: 210 40% 90%;
117 |   --muted: 220 25% 13%;
118 |   --muted-foreground: 215 25% 45%;
119 |   --accent: 220 25% 11%;
120 |   --accent-foreground: 210 40% 90%;
121 |   --destructive: 350 90% 55%;
122 |   --destructive-foreground: 0 0% 100%;
123 |   --input: 218 35% 14%;
124 |   --ring: 210 100% 55%;
125 |   --chart-1: 210 100% 55%;
126 |   --chart-2: 155 100% 40%;
127 |   --chart-3: 38 100% 56%;
128 |   --chart-4: 280 65% 55%;
129 |   --chart-5: 350 90% 55%;
130 |   --shadow-2xs: 0px 2px 0px 0px hsl(220 30% 10% / 0.00);
131 |   --shadow-xs: 0px 2px 0px 0px hsl(220 30% 10% / 0.00);
132 |   --shadow-sm: 0px 2px 0px 0px hsl(220 30% 10% / 0.00), 0px 1px 2px -1px hsl(220 30% 10% / 0.00);
133 |   --shadow: 0px 2px 0px 0px hsl(220 30% 10% / 0.00), 0px 1px 2px -1px hsl(220 30% 10% / 0.00);
134 |   --shadow-md: 0px 2px 0px 0px hsl(220 30% 10% / 0.00), 0px 2px 4px -1px hsl(220 30% 10% / 0.00);
135 |   --shadow-lg: 0px 2px 0px 0px hsl(220 30% 10% / 0.00), 0px 4px 6px -1px hsl(220 30% 10% / 0.00);
136 |   --shadow-xl: 0px 2px 0px 0px hsl(220 30% 10% / 0.00), 0px 8px 10px -1px hsl(220 30% 10% / 0.00);
137 |   --shadow-2xl: 0px 2px 0px 0px hsl(220 30% 10% / 0.00);
138 |
139 | /* Fallback for older browsers */
140 |   --sidebar-primary-border: hsl(var(--sidebar-primary));
141 |   --sidebar-primary-border: hsl(from hsl(var(--sidebar-primary)) h s calc(l + var(--opaque-button-border-intensity)) / a
142 |
143 |   /* Fallback for older browsers */
144 |   --sidebar-accent-border: hsl(var(--sidebar-accent));
145 |   --sidebar-accent-border: hsl(from hsl(var(--sidebar-accent)) h s calc(l + var(--opaque-button-border-intensity)) / alp
146 |
147 |   /* Fallback for older browsers */
148 |   --primary-border: hsl(var(--primary));
149 |   --primary-border: hsl(from hsl(var(--primary)) h s calc(l + var(--opaque-button-border-intensity)) / alpha);
150 |
151 |   /* Fallback for older browsers */
152 |   --secondary-border: hsl(var(--secondary));
153 |   --secondary-border: hsl(from hsl(var(--secondary)) h s calc(l + var(--opaque-button-border-intensity)) / alpha);
154 |
155 |   /* Fallback for older browsers */
156 |   --muted-border: hsl(var(--muted));
157 |   --muted-border: hsl(from hsl(var(--muted)) h s calc(l + var(--opaque-button-border-intensity)) / alpha);
158 |
159 |   /* Fallback for older browsers */
160 |   --accent-border: hsl(var(--accent));
161 |   --accent-border: hsl(from hsl(var(--accent)) h s calc(l + var(--opaque-button-border-intensity)) / alpha);
162 |
163 |   /* Fallback for older browsers */
164 |   --destructive-border: hsl(var(--destructive));
165 |   --destructive-border: hsl(from hsl(var(--destructive)) h s calc(l + var(--opaque-button-border-intensity)) / alpha);
166 | }
167 |
168 | @layer base {
169 |   * {
170 |     @apply border-border;
171 |   }
172 |
173 |   body {
174 |     @apply font-sans antialiased bg-background text-foreground;
175 |   }
176 | }
177 |
178 | @keyframes gridMove {
179 |   0% { background-position: 0 0; }
180 |   100% { background-position: 60px 60px; }
181 | }
182 |
183 | @keyframes ringRotate {
184 |   from { transform: rotate(0deg); }
185 |   to { transform: rotate(360deg); }
186 | }
187 |
188 | @keyframes ringRotateReverse {
189 |   from { transform: rotate(360deg); }
190 |   to { transform: rotate(0deg); }
191 | }
192 |
193 | @keyframes glowPulse {
194 |   0%, 100% { box-shadow: 0 0 15px rgba(0,212,255,0.3), inset 0 0 15px rgba(0,212,255,0.1); }
195 |   50% { box-shadow: 0 0 25px rgba(0,212,255,0.5), inset 0 0 25px rgba(0,212,255,0.2); }
196 | }
197 |
198 | @keyframes fadeInUp {
199 |   from { opacity: 0; transform: translateY(12px); }
200 |   to { opacity: 1; transform: translateY(0); }
201 | }
202 |
203 | @keyframes barFill {
204 |   from { width: 0%; }
205 | }
206 |
207 | @keyframes blink {
208 |   0%, 100% { opacity: 1; }
209 |   50% { opacity: 0.3; }
210 | }
211 |
```

```
212 | @keyframes blinkFast {
213 |   0%, 100% { opacity: 1; }
214 |   50% { opacity: 0; }
215 | }
216 |
217 | @keyframes progressFill {
218 |   from { width: 0%; }
219 |   to { width: 100%; }
220 | }
221 |
222 | @keyframes textFade {
223 |   0%, 20% { opacity: 0; }
224 |   25%, 45% { opacity: 1; }
225 |   50%, 100% { opacity: 0; }
226 | }
227 |
228 | @keyframes splashFadeOut {
229 |   0% { opacity: 1; }
230 |   100% { opacity: 0; pointer-events: none; }
231 | }
232 |
233 | .dash-bg {
234 |   position: relative;
235 |   background-color: #020508;
236 | }
237 |
238 | .dash-bg::before {
239 |   content: '';
240 |   position: absolute;
241 |   inset: 0;
242 |   background-image:
243 |     linear-gradient(rgba(0,212,255,0.03) 1px, transparent 1px),
244 |     linear-gradient(90deg, rgba(0,212,255,0.03) 1px, transparent 1px);
245 |   background-size: 60px 60px;
246 |   animation: gridMove 20s linear infinite;
247 |   opacity: 0.15;
248 |   pointer-events: none;
249 |   z-index: 0;
250 | }
251 |
252 | .dash-bg::after {
253 |   content: '';
254 |   position: absolute;
255 |   inset: 0;
256 |   background: radial-gradient(ellipse at 50% 30%, rgba(0,212,255,0.04) 0%, transparent 70%);
257 |   pointer-events: none;
258 |   z-index: 0;
259 | }
260 |
261 | .dash-scanlines {
262 |   position: absolute;
263 |   inset: 0;
264 |   background: repeating-linear-gradient(
265 |     0deg,
266 |     transparent,
267 |     transparent 2px,
268 |     rgba(0,212,255,0.008) 2px,
269 |     rgba(0,212,255,0.008) 4px
270 |   );
271 |   pointer-events: none;
272 |   z-index: 0;
273 | }
274 |
275 | .dash-content {
276 |   position: relative;
277 |   z-index: 1;
278 | }
279 |
280 | .font-space { font-family: 'Space Mono', monospace; }
281 | .font-rajdhani { font-family: 'Rajdhani', sans-serif; }
282 | .font-exo { font-family: 'Exo 2', sans-serif; }
283 |
284 | .octagonal {
285 |   clip-path: polygon(15% 0%, 85% 0%, 100% 15%, 100% 85%, 85% 100%, 15% 100%, 0% 85%, 0% 15%);
286 | }
287 |
288 | .octagonal-inv {
289 |   clip-path: polygon(20px 0%, 100% 0%, 100% calc(100% - 20px), calc(100% - 20px) 100%, 0% 100%, 0% 20px);
290 | }
291 |
292 | .corner-cut-tl {
293 |   clip-path: polygon(0 0, calc(100% - 20px) 0, 100% 20px, 100% 100%, 20px 100%, 0 calc(100% - 20px));
294 | }
295 |
296 | .corner-cut-tr {
297 |   clip-path: polygon(20px 0, 100% 0, 100% calc(100% - 20px), calc(100% - 20px) 100%, 0 100%, 0 20px);
298 | }
299 |
300 | .dash-card {
301 |   background: #091018;
302 |   border: 1px solid #0d2035;
303 | }
304 |
305 | .dash-card-accent {
306 |   position: relative;
307 | }
308 |
309 | .dash-card-accent::before {
310 |   content: '';
311 |   position: absolute;
312 |   top: 0;
313 |   left: 0;
314 |   width: 20px;
315 |   height: 1px;
316 |   background: #00d4ff;
317 | }
318 |
```

```
.dash-card-accent::after {
  content: '';
  position: absolute;
  top: 0;
  left: 0;
  width: 1px;
  height: 20px;
  background: #00d4ff;
}

/**
 * Using the elevate system.
 * Automatic contrast adjustment.
 *
 * <element className="hover-elevate" />
 * <element className="active-elevate-2" />
 *
 * // Using the tailwind utility when a data attribute is "on"
 * <element className="toggle-elevate data-[state=on]:toggle-elevated" />
 * // Or manually controlling the toggle state
 * <element className="toggle-elevate toggle-elevated" />
 *
 * Elevation systems have to handle many states.
 * - not-hovered, vs. hovered vs. active  (three mutually exclusive states)
 * - toggled or not
 * - focused or not (this is not handled with these utilities)
 *
 * Even without handling focused or not, this is six possible combinations that
 * need to be distinguished from eachother visually.
 */
@layer utilities {

  /* Hide ugly search cancel button in Chrome until we can style it properly */
  input[type="search"]::-webkit-search-cancel-button {
    @apply hidden;
  }

  /* Placeholder styling for contentEditable div */
  [contenteditable][data-placeholder]:empty::before {
    content: attr(data-placeholder);
    color: hsl(var(--muted-foreground));
    pointer-events: none;
  }

  /* .no-default-hover-elevate/no-default-active-elevate is an escape hatch so consumers of
   * buttons/badges can remove the automatic brightness adjustment on interactions
   * and program their own. */
  .no-default-hover-elevate {}

  .no-default-active-elevate {}


  /**
   * Toggleable backgrounds go behind the content. Hoverable/active goes on top.
   * This way they can stack/compound. Both will overlap the parent's borders!
   * So borders will be automatically adjusted both on toggle, and hover/active,
   * and they will be compounded.
   */
  .toggle-elevate::before,
  .toggle-elevate-2::before {
    content: "";
    pointer-events: none;
    position: absolute;
    inset: 0px;
    /*border-radius: inherit;   match rounded corners */
    border-radius: inherit;
    z-index: -1;
    /* sits behind content but above backdrop */
  }

  .toggle-elevate.toggle-elevated::before {
    background-color: var(--elevate-2);
  }

  /* If there's a 1px border, adjust the inset so that it covers that parent's border */
  .border.toggle-elevate::before {
    inset: -1px;
  }

  /* Does not work on elements with overflow:hidden! */
  .hover-elevate:not(.no-default-hover-elevate),
  .active-elevate:not(.no-default-active-elevate),
  .hover-elevate-2:not(.no-default-hover-elevate),
  .active-elevate-2:not(.no-default-active-elevate) {
    position: relative;
    z-index: 0;
  }

  .hover-elevate:not(.no-default-hover-elevate)::after,
  .active-elevate:not(.no-default-active-elevate)::after,
  .hover-elevate-2:not(.no-default-hover-elevate)::after,
  .active-elevate-2:not(.no-default-active-elevate)::after {
    content: "";
    pointer-events: none;
    position: absolute;
    inset: 0px;
    /*border-radius: inherit;   match rounded corners */
    border-radius: inherit;
    z-index: 999;
    /* sits in front of content */
  }

  .hover-elevate:hover:not(.no-default-hover-elevate)::after,
  .active-elevate:active:not(.no-default-active-elevate)::after {
    background-color: var(--elevate-1);
  }

```

```
426 |   .hover-elevate-2:hover:not(.no-default-hover-elevate)::after,
427 |   .active-elevate-2:active:not(.no-default-active-elevate)::after {
428 |     background-color: var(--elevate-2);
429 |   }
430 |
431 |   /* If there's a 1px border, adjust the inset so that it covers that parent's border */
432 |   .border.hover-elevate:not(.no-hover-interaction-elevate)::after,
433 |   .border.active-elevate:not(.no-active-interaction-elevate)::after,
434 |   .border.hover-elevate-2:not(.no-hover-interaction-elevate)::after,
435 |   .border.active-elevate-2:not(.no-active-interaction-elevate)::after,
436 |   .border.hover-elevate:not(.no-hover-interaction-elevate)::after {
437 |     inset: -1px;
438 |   }
439 | }
440 |
441 | .dark-popup .leaflet-popup-content-wrapper {
442 |   background: transparent;
443 |   box-shadow: none;
444 |   border-radius: 4px;
445 |   padding: 0;
446 | }
447 | .dark-popup .leaflet-popup-content {
448 |   margin: 0;
449 | }
450 | .dark-popup .leaflet-popup-tip {
451 |   background: #0a1520;
452 |   border: 1px solid #0d2035;
453 | }
454 |
```

client/src/lib/queryClient.ts

58 lineas

```
1 | import { QueryClient, QueryFunction } from "@tanstack/react-query";
2 |
3 | async function throwIfResNotOk(res: Response) {
4 |   if (!res.ok) {
5 |     const text = (await res.text()) || res.statusText;
6 |     throw new Error(`${res.status}: ${text}`);
7 |   }
8 | }
9 |
10 | export async function apiRequest(
11 |   method: string,
12 |   url: string,
13 |   data?: unknown | undefined,
14 | ): Promise<Response> {
15 |   const res = await fetch(url, {
16 |     method,
17 |     headers: data ? { "Content-Type": "application/json" } : {},
18 |     body: data ? JSON.stringify(data) : undefined,
19 |     credentials: "include",
20 |   });
21 |
22 |   await throwIfResNotOk(res);
23 |   return res;
24 | }
25 |
26 | type UnauthorizedBehavior = "returnNull" | "throw";
27 | export const getQueryFn: <T>(options: {
28 |   on401: UnauthorizedBehavior;
29 | }) => QueryFunction<T> =
30 |   ({ on401: unauthorizedBehavior }) =>
31 |   async ({ queryKey }) => {
32 |     const res = await fetch(queryKey.join("/") as string, {
33 |       credentials: "include",
34 |     });
35 |
36 |     if (unauthorizedBehavior === "returnNull" && res.status === 401) {
37 |       return null;
38 |     }
39 |
40 |     await throwIfResNotOk(res);
41 |     return await res.json();
42 |   };
43 |
44 | export const queryClient = new QueryClient({
45 |   defaultOptions: {
46 |     queries: {
47 |       queryFn: getQueryFn({ on401: "throw" }),
48 |       refetchInterval: false,
49 |       refetchOnWindowFocus: false,
50 |       staleTime: Infinity,
51 |       retry: false,
52 |     },
53 |     mutations: {
54 |       retry: false,
55 |     },
56 |   },
57 | });
58 |
```

client/src/lib/utils.ts

7 lineas

```
1 | import { clsx, type ClassValue } from "clsx"
2 | import { twMerge } from "tailwind-merge"
3 |
4 | export function cn(...inputs: ClassValue[]) {
5 |   return twMerge(clsx(inputs))
6 | }
7 |
```

client/src/lib/fuel-utils.ts

118 lineas

```
1 | import type { Camion, Carga, Faena } from "@shared/schema";
2 |
3 | export const PRECIO = 1180;
4 |
5 | export interface CargaAnalizada extends Carga {
6 |   km: number;
7 |   dif: number;
8 |   rend: number | null;
9 |   estado: "CRITICO" | "ALERTA" | "OK";
10 | }
11 |
12 | export interface CamionStats extends Camion {
13 |   cargasAnalizadas: CargaAnalizada[];
14 |   criticos: number;
15 |   litDesv: number;
16 |   rendProm: number | null;
17 |   pctRal: number | null;
18 |   estado: "CRITICO" | "ALERTA" | "OK";
19 | }
20 |
21 | export function analizarCarga(c: Carga): CargaAnalizada {
22 |   const km = c.kmActual - c.kmAnterior;
23 |   const dif = c.litrosSurtidor - c.litrosEcu;
24 |   const rend = km > 0 && c.litrosEcu > 0 ? +(km / c.litrosEcu).toFixed(2) : null;
25 |   const estado: CargaAnalizada["estado"] = dif > 40 ? "CRITICO" : dif > 15 ? "ALERTA" : "OK";
26 |   return { ...c, km, dif, rend, estado };
27 | }
28 |
29 | export function statsCamion(cam: Camion, cargasList: Carga[]): CamionStats {
30 |   const cargasAnalizadas = cargasList.map(analizarCarga);
31 |   const criticos = cargasAnalizadas.filter(c => c.estado === "CRITICO").length;
32 |   const litDesv = cargasAnalizadas.filter(c => c.dif > 0).reduce((a, c) => a + c.dif, 0);
33 |   const rends = cargasAnalizadas.filter(c => c.rend).map(c => c.rend!);
34 |   const rendProm = rends.length ? +(rends.reduce((a, b) => a + b, 0) / rends.length).toFixed(2) : null;
35 |   const pctRal = cam.horasMotor ? Math.round((cam.horasRalenti! / cam.horasMotor) * 100) : null;
36 |   const estado: CamionStats["estado"] = criticos > 0 ? "CRITICO" : cargasAnalizadas.some(c => c.estado === "ALERTA") ? "
37 |   return { ...cam, cargasAnalizadas, criticos, litDesv, rendProm, pctRal, estado };
38 | }
39 |
40 | export const fN = (n: number | null | undefined, d = 0) =>
41 |   n == null ? "\u2014" : Number(n).toLocaleString("es-CL", { maximumFractionDigits: d });
42 | export const fL = (n: number | null | undefined) => n == null ? "\u2014" : `${fN(n)} L`;
43 | export const fK = (n: number | null | undefined) => n == null ? "\u2014" : `${fN(n)} km`;
44 | export const f$ = (n: number | null | undefined) => n == null ? "\u2014" : `$${fN(n)}`;
45 |
46 | export const statusColor = (s: string) =>
47 |   s === "CRITICO" ? "text-red-500" : s === "ALERTA" ? "text-amber-400" : "text-emerald-400";
48 |
49 | export const statusBg = (s: string) =>
50 |   s === "CRITICO" ? "bg-red-500/10 border-red-500/30 text-red-400" :
51 |   s === "ALERTA" ? "bg-amber-400/10 border-amber-400/30 text-amber-400" :
52 |   "bg-emerald-400/10 border-emerald-400/30 text-emerald-400";
53 |
54 | export const rendColor = (r: number | null, meta: number) =>
55 |   !r ? "text-muted-foreground" : r >= meta ? "text-emerald-400" : r >= meta * 0.9 ? "text-amber-400" : "text-red-400";
56 |
57 | export interface Percentiles {
58 |   p50: number;
59 |   p75: number;
60 |   p90: number;
61 |   totalCamiones: number;
62 |   fechaCalculo: string | null;
63 |   calibrando: boolean;
64 | }
65 |
66 | export type NivelPercentil = "EXCELENTE" | "BUENO" | "NORMAL" | "BAJO" | "CRITICO";
67 |
68 | export function getNivelPercentil(rend: number | null, p: Percentiles): NivelPercentil {
69 |   if (!rend || rend <= 0 || p.calibrando) return "BAJO";
70 |   if (rend >= p.p90) return "EXCELENTE";
71 |   if (rend >= p.p75) return "BUENO";
72 |   if (rend >= p.p50) return "NORMAL";
73 |   if (rend < p.p50 * 0.7) return "CRITICO";
74 |   return "BAJO";
75 | }
76 |
77 | export function rendColorPercentil(rend: number | null, p: Percentiles): string {
78 |   if (!rend || rend <= 0) return "text-muted-foreground";
79 |   if (p.calibrando) return rendColor(rend, 2.1);
80 |   const nivel = getNivelPercentil(rend, p);
81 |   if (nivel === "EXCELENTE") return "text-emerald-400";
82 |   if (nivel === "BUENO") return "text-emerald-400/70";
83 |   if (nivel === "NORMAL") return "text-blue-400";
84 |   if (nivel === "BAJO") return "text-amber-400";
85 |   return "text-red-400";
86 | }
87 |
88 | export function rendBgPercentil(rend: number | null, p: Percentiles): string {
89 |   if (!rend || rend <= 0) return "";
90 |   if (p.calibrando) return "";
91 |   const nivel = getNivelPercentil(rend, p);
92 |   if (nivel === "EXCELENTE") return "bg-emerald-400/10";
93 |   if (nivel === "BUENO") return "bg-emerald-400/5";
94 |   if (nivel === "NORMAL") return "bg-blue-400/5";
95 |   if (nivel === "BAJO") return "bg-amber-400/5";
96 |   return "bg-red-400/5";
97 | }
98 |
99 | export function getPercentilCamion(rend: number, allRends: number[]): number {
100 |   if (allRends.length === 0) return 0;
101 |   const below = allRends.filter(r => r < rend).length;
102 |   return Math.round((below / allRends.length) * 100);
103 | }
104 |
```

```
105 | export function getPercentilLabel(percentil: number): { text: string; color: string } {
106 |   if (percentil >= 90) return { text: `Top ${100 - percentil}% de la flota`, color: "text-emerald-400" };
107 |   if (percentil >= 75) return { text: `Top ${100 - percentil}% de la flota`, color: "text-emerald-400/70" };
108 |   if (percentil >= 50) return { text: `Top ${100 - percentil}% de la flota`, color: "text-blue-400" };
109 |   return { text: "Bajo el promedio", color: "text-amber-400" };
110 | }
111 |
112 | export function rendColorFlota(avgRend: number, p: Percentiles): string {
113 |   if (p.calibrando) return "#00C87A";
114 |   if (avgRend >= p.p75) return "#00C87A";
115 |   if (avgRend >= p.p50) return "#FFB020";
116 |   return "#FF2D4A";
117 | }
118 |
```

client/src/pages/hoy.tsx

657 lineas

```
1 | import { useState, useRef } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { RefreshCw, Loader2, ChevronRight, Info, Search, X, Brain, Fuel, Truck } from "lucide-react";
4 | import { LineChart, Line, XAxis, YAxis, CartesianGrid, Tooltip as ReTooltip, ResponsiveContainer } from "recharts";
5 | import { queryClient, apiRequest } from "@/lib/queryClient";
6 | import { Tooltip, TooltipContent, TooltipTrigger } from "@/components/ui/tooltip";
7 |
8 | interface KPIs {
9 |   camionesTotal: number;
10 |   camionesActivos: number;
11 |   rendPromedio: number;
12 |   rendCambioPct: number;
13 |   litrosPeriodo: number;
14 |   litrosCambioPct: number;
15 |   alertasCriticas: number;
16 |   excesosVel: number;
17 |   microSospechas: number;
18 | }
19 |
20 | interface ContratoEstado {
21 |   nombre: string;
22 |   color: string;
23 |   camionesTotal: number;
24 |   camionesActivos?: number;
25 |   rendPromedio: number;
26 |   alertasCriticas: number;
27 |   estado: "NORMAL" | "ATENCION" | "CRITICO";
28 | }
29 |
30 | interface TendenciaDay {
31 |   fecha: string;
32 |   litrosDia: number;
33 |   rendimientoDia: number;
34 | }
35 |
36 | interface Alerta {
37 |   tipo: string;
38 |   nivel: string;
39 |   titulo: string;
40 |   detalle: string;
41 |   hace: string;
42 |   gravedad: number;
43 | }
44 |
45 | interface DiagnosticoIA {
46 |   texto: string;
47 |   generadoHaceMin: number;
48 |   cached: boolean;
49 |   error?: string;
50 | }
51 |
52 | interface ContratoDetalle {
53 |   nombre: string;
54 |   color: string;
55 |   camionesTotal: number;
56 |   camionesOnline: number;
57 |   litrosHoy: number;
58 |   rendHoy: number;
59 |   cargasHoy: number;
60 |   alertasCriticas: number;
61 |   alertasAlerta: number;
62 |   camiones: {
63 |     patente: string;
64 |     modelo: string;
65 |     conductor: string;
66 |     online: boolean;
67 |     velocidad: number;
68 |     rendimiento: number;
69 |     litrosHoy: number;
70 |     estado: string;
71 |     cargasHoy: number;
72 |   }[];
73 | }
74 |
75 | type Periodo = "1d" | "7d" | "30d";
76 |
77 | const estadoColors: Record<string, string> = { NORMAL: "#00ff88", ATENCION: "#ffcc00", CRITICO: "#ff2244" };
78 | const estadoCamColors: Record<string, string> = { NORMAL: "#00ff88", ALERTA: "#ffcc00", CRITICO: "#ff2244" };
79 |
80 | const alertaIcons: Record<string, { icon: string; texto: string }> = {
81 |   desviacion: { icon: "\ud83d\udd34", texto: "Odometro alterado" },
82 |   micro_carga: { icon: "\ud83d\udd34", texto: "Posible robo combustible" },
83 |   exceso_velocidad: { icon: "\ud83d\udd34", texto: "Velocidad peligrosa" },
84 |   offline: { icon: "\ud83d\udfe1", texto: "Sin senal prolongada" },
85 |   rendimiento_bajo: { icon: "\ud83d\udfe1", texto: "Consumo anormal" },
86 | };
87 |
88 | function InfoTip({ text }: { text: string }) {
89 |   return (
90 |     <Tooltip>
91 |       <TooltipTrigger asChild>
92 |         <span className="inline-flex cursor-help ml-1 align-middle">
93 |           <Info className="w-3 h-3" style={{ color: "#3a6080" }} />
94 |         </span>
95 |       </TooltipTrigger>
96 |       <TooltipContent side="top" className="max-w-[220px] text-xs font-exo" style={{ background: "#1a2a3a", color: "#fff
97 |         {text}
98 |       </TooltipContent>
99 |     </Tooltip>
100 |   );
101 | }
102 |
103 | export default function Hoy({ onNavigateFlota, onOpenIA }: { onNavigateFlota?: (contrato?: string) => void; onOpenIA?: (
104 |   const [periodo, setPeriodo] = useState<Periodo>("7d");
```

```
105 |   const [lastRefresh, setLastRefresh] = useState(Date.now());
106 |   const alertasRef = useRef<HTMLDivElement>(null);
107 |   const [contratoDetalle, setContratoDetalle] = useState<string | null>(null);
108 |
109 |   const diasTendencia = periodo === "1d" ? 1 : periodo === "7d" ? 7 : 30;
110 |
111 |   async function safeFetch<T>(url: string): Promise<T> {
112 |     const r = await fetch(url, { credentials: "include" });
113 |     if (!r.ok) throw new Error(`${r.status}: ${await r.text()}`);
114 |     return r.json();
115 |   }
116 |
117 |   const { data: kpis, isLoading: loadingKpis } = useQuery<KPIs>({
118 |     queryKey: ["/api/ceo/kpis", periodo],
119 |     queryFn: () => safeFetch<KPIs>(`/api/ceo/kpis?periodo=${periodo}`),
120 |     refetchInterval: 120000,
121 |   });
122 |
123 |   const { data: contratos = [], isLoading: loadingContratos } = useQuery<ContratoEstado[]>({
124 |     queryKey: ["/api/ceo/contratos-estado"],
125 |     refetchInterval: 120000,
126 |   });
127 |
128 |   const { data: tendencia = [], isLoading: loadingTendencia } = useQuery<TendenciaDay[]>({
129 |     queryKey: ["/api/ceo/tendencia", diasTendencia],
130 |     queryFn: () => safeFetch<TendenciaDay[]>(`/api/ceo/tendencia?dias=${diasTendencia}`),
131 |     refetchInterval: 120000,
132 |   });
133 |
134 |   const { data: alertas = [], isLoading: loadingAlertas } = useQuery<Alerta[]>({
135 |     queryKey: ["/api/ceo/alertas"],
136 |     queryFn: () => safeFetch<Alerta[]>("/api/ceo/alertas?limite=5"),
137 |     refetchInterval: 120000,
138 |   });
139 |
140 |   const { data: diagnostico, isLoading: loadingDiag } = useQuery<DiagnosticoIA>({
141 |     queryKey: ["/api/ceo/diagnostico-ia"],
142 |     refetchInterval: 1800000,
143 |   });
144 |
145 |   const minAgo = Math.round((Date.now() - lastRefresh) / 60000);
146 |
147 |   function handleRefreshAll() {
148 |     setLastRefresh(Date.now());
149 |     queryClient.invalidateQueries({ queryKey: ["/api/ceo/kpis"] });
150 |     queryClient.invalidateQueries({ queryKey: ["/api/ceo/contratos-estado"] });
151 |     queryClient.invalidateQueries({ queryKey: ["/api/ceo/tendencia"] });
152 |     queryClient.invalidateQueries({ queryKey: ["/api/ceo/alertas"] });
153 |   }
154 |
155 |   function handleRefreshDiag() {
156 |     queryClient.invalidateQueries({ queryKey: ["/api/ceo/diagnostico-ia"] });
157 |     queryClient.fetchQuery({
158 |       queryKey: ["/api/ceo/diagnostico-ia"],
159 |       queryFn: () => fetch("/api/ceo/diagnostico-ia?force=true").then(r => { if (!r.ok) throw new Error("Error"); return
160 |     });
161 |   }
162 |
163 |   function scrollToAlertas() {
164 |     alertasRef.current?.scrollIntoView({ behavior: "smooth", block: "start" });
165 |   }
166 |
167 |   const estadoBadge = (() => {
168 |     const c = kpis?.alertasCriticas ?? 0;
169 |     if (c === 0) return { label: "OPERACION NORMAL", color: "#00ff88", blink: false };
170 |     if (c <= 3) return { label: "ATENCION REQUERIDA", color: "#ffcc00", blink: true };
171 |     return { label: "ALERTA ACTIVA", color: "#ff2244", blink: true };
172 |   })();
173 |
174 |   function parseDiagnostico(texto: string) {
175 |     const lines = texto.split("\n").filter(l => l.trim());
176 |     const result: { icon: string; label: string; text: string; color: string }[] = [];
177 |     for (const line of lines) {
178 |       if (line.startsWith("ESTADO:")) {
179 |         result.push({ icon: "\ud83d\udfe2", label: "ESTADO", text: line.replace("ESTADO:", "").trim(), color: "#00ff88"
180 |       } else if (line.startsWith("POSITIVO:")) {
181 |         result.push({ icon: "\u2705", label: "POSITIVO", text: line.replace("POSITIVO:", "").trim(), color: "#00ff88" })
182 |       } else if (line.startsWith("ATENCION:") || line.startsWith("ATENCI\u00d3N:")) {
183 |         result.push({ icon: "\u26a0\ufe0f", label: "ATENCION", text: line.replace(/ATENCI[O\u00d3]N:/, "").trim(), color
184 |       } else if (line.startsWith("ACCION:") || line.startsWith("ACCI\u00d3N:")) {
185 |         result.push({ icon: "\u25b6\ufe0f", label: "ACCION", text: line.replace(/ACCI[O\u00d3]N:/, "").trim(), color: "#
186 |       }
187 |     }
188 |     return result;
189 |   }
190 |
191 |   return (
192 |     <div className="space-y-5" data-testid="page-hoy">
193 |
194 |       {/* BLOQUE B — 4 KPIs */}
195 |       <div data-testid="hoy-kpis">
196 |         <div className="flex items-center justify-between mb-3">
197 |           <div />
198 |           <div className="flex items-center gap-3">
199 |             <div className="flex gap-1">
200 |               {(["1d", "7d", "30d"] as Periodo[]).map(p => (
201 |                 <button key={p} onClick={() => setPeriodo(p)}
202 |                   data-testid={`hoy-periodo-${p}`}
203 |                   className="font-space text-[10px] font-bold px-3 py-1 cursor-pointer transition-all border"
204 |                   style={{
205 |                     color: periodo === p ? "#020508" : "#3a6080",
206 |                     background: periodo === p ? "#00d4ff" : "transparent",
207 |                     borderColor: periodo === p ? "#00d4ff" : "#0d2035",
208 |                   }}>
209 |                   {p === "1d" ? "HOY" : p.toUpperCase()}
210 |                 </button>
211 |               ))}
```

```
212 |             </div>
213 |             <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>
214 |               Actualizado hace {minAgo < 1 ? "<1" : minAgo} min
215 |             </span>
216 |             <button onClick={handleRefreshAll} className="p-1.5 cursor-pointer transition-all hover:bg-[rgba(0,212,255,0
217 |               data-testid="hoy-refresh">
218 |               <RefreshCw className="w-3.5 h-3.5" style={{ color: "#00d4ff" }} />
219 |             </button>
220 |           </div>
221 |         </div>
222 |
223 |         <div className="grid grid-cols-2 lg:grid-cols-4 gap-4">
224 |           <KpiCard
225 |             label="ACTIVOS"
226 |             value={kpis ? String(kpis.camionesActivos) : "--"}
227 |             sub={kpis ? `de ${kpis.camionesTotal}` : ""}
228 |             progress={kpis ? Math.round((kpis.camionesActivos / Math.max(kpis.camionesTotal, 1)) * 100) : 0}
229 |             color="#00d4ff"
230 |             loading={loadingKpis}
231 |           />
232 |           <KpiCard
233 |             label="km/L"
234 |             value={kpis ? String(kpis.rendPromedio) : "--"}
235 |             sub={kpis ? `${kpis.rendCambioPct >= 0 ? "\u2191" : "\u2193"} ${kpis.rendCambioPct >= 0 ? "+" : ""}${kpis.re
236 |             color={kpis ? (kpis.rendCambioPct >= 0 ? "#00ff88" : "#ff2244") : "#3a6080"}
237 |             loading={loadingKpis}
238 |             tooltip="Kilometros recorridos por litro de combustible. Mayor = mas eficiente."
239 |           />
240 |           <KpiCard
241 |             label="LITROS"
242 |             value={kpis ? kpis.litrosPeriodo.toLocaleString() : "--"}
243 |             sub={kpis ? `${kpis.litrosCambioPct >= 0 ? "\u2191" : "\u2193"} ${kpis.litrosCambioPct >= 0 ? "+" : ""}${kpi
244 |             color="#ffcc00"
245 |             loading={loadingKpis}
246 |           />
247 |           <KpiCard
248 |             label="CRITICOS"
249 |             value={kpis ? String(kpis.alertasCriticas) : "--"}
250 |             sub={kpis?.alertasCriticas ? `${kpis.alertasCriticas} requieren accion` : "Sin alertas"}
251 |             color={kpis?.alertasCriticas ? "#ff2244" : "#00ff88"}
252 |             loading={loadingKpis}
253 |             onClick={kpis?.alertasCriticas ? scrollToAlertas : undefined}
254 |           />
255 |         </div>
256 |       </div>
257 |
258 |       {/* BLOQUE C — ESTADO POR CONTRATO */}
259 |       <div data-testid="hoy-contratos">
260 |         <div className="font-exo text-[9px] tracking-[0.2em] uppercase font-semibold mb-3" style={{ color: "#c8e8ff" }}>
261 |           TUS CONTRATOS
262 |         </div>
263 |         {loadingContratos ? (
264 |           <div className="flex items-center gap-2 py-6 justify-center">
265 |             <Loader2 className="w-4 h-4 animate-spin" style={{ color: "#00d4ff" }} />
266 |             <span className="font-exo text-[11px]" style={{ color: "#3a6080" }}>Cargando contratos...</span>
267 |           </div>
268 |         ) : (
269 |           <div className="flex gap-3 overflow-x-auto pb-2">
270 |             {contratos.map((c, i) => (
271 |               <div key={i} className="flex-shrink-0 min-w-[220px] p-4"
272 |                 style={{
273 |                   background: "#091018",
274 |                   border: "1px solid #0d2035",
275 |                   borderTopColor: estadoColors[c.estado] || "#3a6080",
276 |                   borderTopWidth: "3px",
277 |                 }}
278 |                 data-testid={`hoy-contrato-${i}`}>
279 |                 <div className="flex items-center justify-between mb-2">
280 |                   <div className="flex items-center gap-2">
281 |                     <span className="inline-block w-2 h-2 rounded-full" style={{ background: estadoColors[c.estado] }} /
282 |                     <span className="font-space text-[9px] font-bold tracking-[0.1em]" style={{ color: estadoColors[c.es
283 |                       {c.estado}
284 |                     </span>
285 |                   </div>
286 |                   <button onClick={() => setContratoDetalle(c.nombre)}
287 |                     className="octagonal p-1.5 cursor-pointer transition-all border hover:border-[#00d4ff] hover:shadow-
288 |                     style={{ borderColor: "#0d2035", background: "rgba(0,212,255,0.04)" }}
289 |                     title={`Ver detalle de ${c.nombre}`}
290 |                     data-testid={`hoy-contrato-lupa-${i}`}>
291 |                     <Search className="w-3.5 h-3.5" style={{ color: "#00d4ff" }} />
292 |                   </button>
293 |                 </div>
294 |                 <div className="font-space text-[13px] font-bold mb-1" style={{ color: "#c8e8ff" }}>{c.nombre}</div>
295 |                 <div className="font-exo text-[10px]" style={{ color: "#3a6080" }}>
296 |                   {c.camionesActivos != null ? `${c.camionesActivos}/` : ""}{c.camionesTotal} camiones
297 |                 </div>
298 |                 <div className="font-space text-[11px] font-bold mt-1" style={{ color: c.rendPromedio >= 2.5 ? "#00ff88"
299 |                   {c.rendPromedio > 0 ? `${c.rendPromedio} km/L` : "N/D"}
300 |                 </div>
301 |                 <div className="font-exo text-[10px] mt-1" style={{ color: c.alertasCriticas > 0 ? "#ff2244" : "#3a6080"
302 |                   {c.alertasCriticas} alertas
303 |                 </div>
304 |                 <button
305 |                   onClick={() => setContratoDetalle(c.nombre)}
306 |                   className="mt-3 w-full flex items-center justify-center gap-2 py-2.5 font-space text-[11px] font-bold
307 |                   style={{ background: "rgba(0,212,255,0.06)", borderColor: "rgba(0,212,255,0.25)", color: "#00d4ff" }}
308 |                   data-testid={`hoy-contrato-dia-${i}`}>
309 |                   DIA (24h)
310 |                 </button>
311 |                 <button
312 |                   onClick={() => onNavigateFlota?.(c.nombre)}
313 |                   className="mt-1.5 font-exo text-[10px] font-bold cursor-pointer transition-all hover:text-[#00d4ff] fl
314 |                   style={{ color: "#3a6080" }}
315 |                   data-testid={`hoy-contrato-detalle-${i}`}>
316 |                   Ver en flota <ChevronRight className="w-3 h-3" />
317 |                 </button>
318 |               </div>
```

```
319 |             ))}
320 |             {contratos.length === 0 && (
321 |               <div className="font-exo text-[11px] py-4" style={{ color: "#3a6080" }}>Sin contratos con camiones asignad
322 |             )}
323 |           </div>
324 |         )}
325 |       </div>
326 |
327 |       {/* BLOQUE D — REQUIERE ATENCION */}
328 |       <div ref={alertasRef} data-testid="hoy-alertas">
329 |         {loadingAlertas ? (
330 |           <div className="flex items-center gap-2 py-4 justify-center">
331 |             <Loader2 className="w-4 h-4 animate-spin" style={{ color: "#00d4ff" }} />
332 |           </div>
333 |         ) : alertas.length === 0 ? (
334 |           <div className="p-4 text-center" style={{ background: "#091018", border: "1px solid #0d2035" }}>
335 |             <span className="font-exo text-[13px]" style={{ color: "#00ff88" }}>Todo en orden hoy</span>
336 |           </div>
337 |         ) : (
338 |           <>
339 |             <div className="font-exo text-[9px] tracking-[0.2em] uppercase font-semibold mb-3" style={{ color: "#c8e8ff"
340 |               REQUIERE TU ATENCION
341 |             </div>
342 |             <div className="space-y-2">
343 |               {alertas.slice(0, 5).map((a, i) => {
344 |                 const info = alertaIcons[a.tipo] || { icon: a.nivel === "CRITICO" ? "\ud83d\udd34" : "\ud83d\udfe1", tex
345 |                 return (
346 |                   <div key={i} className="p-3 flex items-start gap-3 transition-all hover:bg-[rgba(0,212,255,0.03)]"
347 |                     style={{ background: "#091018", border: "1px solid #0d2035" }}
348 |                     data-testid={`hoy-alerta-${i}`}>
349 |                     <span className="text-[14px] mt-0.5 flex-shrink-0">{info.icon}</span>
350 |                     <div className="flex-1 min-w-0">
351 |                       <div className="font-space text-[11px] font-bold" style={{ color: "#c8e8ff" }}>{a.titulo}</div>
352 |                       <div className="font-exo text-[10px] mt-0.5" style={{ color: "#3a6080" }}>{a.detalle}</div>
353 |                       <div className="font-exo text-[9px] mt-1" style={{ color: "#1a3550" }}>{a.hace}</div>
354 |                     </div>
355 |                     <button
356 |                       onClick={() => onNavigateFlota?.()}
357 |                       className="font-exo text-[9px] font-bold px-2 py-1 cursor-pointer transition-all border hover:bord
358 |                       style={{ borderColor: "#0d2035", color: "#3a6080" }}
359 |                       data-testid={`hoy-alerta-investigar-${i}`}>
360 |                       Investigar <ChevronRight className="w-3 h-3 inline" />
361 |                     </button>
362 |                   </div>
363 |                 );
364 |               })}
365 |               {alertas.length > 5 && (
366 |                 <button onClick={scrollToAlertas}
367 |                   className="font-exo text-[10px] font-bold cursor-pointer px-3 py-2 w-full text-center transition-all h
368 |                   style={{ color: "#3a6080" }}
369 |                   data-testid="hoy-alertas-mas">
370 |                   Ver {alertas.length - 5} alertas mas <ChevronRight className="w-3 h-3 inline" />
371 |                 </button>
372 |               )}
373 |             </div>
374 |           </>
375 |         )}
376 |       </div>
377 |
378 |       {/* BLOQUE E — TENDENCIA + DIAGNOSTICO IA */}
379 |       <div className="grid grid-cols-1 lg:grid-cols-2 gap-4" data-testid="hoy-bottom">
380 |         <div className="p-4" style={{ background: "#091018", border: "1px solid #0d2035" }} data-testid="hoy-tendencia">
381 |           {loadingTendencia ? (
382 |             <div className="flex items-center justify-center py-12">
383 |               <Loader2 className="w-5 h-5 animate-spin" style={{ color: "#00d4ff" }} />
384 |             </div>
385 |           ) : (
386 |             <ResponsiveContainer width="100%" height={180}>
387 |               <LineChart data={tendencia} margin={{ top: 5, right: 30, left: 0, bottom: 5 }}>
388 |                 <CartesianGrid strokeDasharray="3 3" stroke="#0d2035" />
389 |                 <XAxis
390 |                   dataKey="fecha"
391 |                   tick={{ fill: "#3a6080", fontSize: 9, fontFamily: "Space Mono" }}
392 |                   tickFormatter={(v: string) => { const d = new Date(v); return `${d.getDate()}/${d.getMonth() + 1}`; }}
393 |                   stroke="#0d2035"
394 |                 />
395 |                 <YAxis yAxisId="left" tick={{ fill: "#3a6080", fontSize: 9 }} stroke="#0d2035" />
396 |                 <YAxis yAxisId="right" orientation="right" tick={{ fill: "#00ff88", fontSize: 9 }} stroke="#0d2035" />
397 |                 <ReTooltip
398 |                   contentStyle={{ background: "#091018", border: "1px solid #0d2035", fontFamily: "Exo 2", fontSize: 11,
399 |                   labelFormatter={(v: string) => new Date(v).toLocaleDateString("es-CL", { weekday: "short", day: "numer
400 |                 />
401 |                 <Line yAxisId="left" type="monotone" dataKey="litrosDia" name="Litros" stroke="#00d4ff" strokeWidth={2}
402 |                 <Line yAxisId="right" type="monotone" dataKey="rendimientoDia" name="km/L" stroke="#00ff88" strokeWidth=
403 |               </LineChart>
404 |             </ResponsiveContainer>
405 |           )}
406 |         </div>
407 |
408 |         <div className="p-4" style={{ background: "#091018", border: "1px solid #0d2035", borderLeft: "3px solid #00d4ff
409 |           data-testid="hoy-diagnostico">
410 |           <div className="flex items-center justify-between mb-3">
411 |             <span className="font-space text-[10px] font-bold tracking-[0.1em]" style={{ color: "#c8e8ff" }}>
412 |               DIAGNOSTICO IA
413 |             </span>
414 |             <div className="flex items-center gap-2">
415 |               {diagnostico && (
416 |                 <span className="font-exo text-[8px]" style={{ color: "#3a6080" }}>
417 |                   hace {diagnostico.generadoHaceMin < 1 ? "<1" : diagnostico.generadoHaceMin} min
418 |                 </span>
419 |               )}
420 |               <button onClick={handleRefreshDiag} className="p-1 cursor-pointer transition-all hover:bg-[rgba(0,212,255,
421 |                 data-testid="hoy-refresh-diag">
422 |                 <RefreshCw className="w-3 h-3" style={{ color: "#00d4ff" }} />
423 |               </button>
424 |             </div>
425 |           </div>
```

```
426 |
427 |           {loadingDiag ? (
428 |             <div className="flex items-center gap-2 py-6 justify-center">
429 |               <Loader2 className="w-4 h-4 animate-spin" style={{ color: "#00d4ff" }} />
430 |               <span className="font-exo text-[10px]" style={{ color: "#3a6080" }}>Generando diagnostico...</span>
431 |             </div>
432 |           ) : diagnostico?.texto ? (
433 |             <div className="space-y-2">
434 |               {parseDiagnostico(diagnostico.texto).map((line, i) => (
435 |                 <div key={i} className="flex items-start gap-2 py-1">
436 |                   <span className="text-[12px] flex-shrink-0">{line.icon}</span>
437 |                   <div className="min-w-0">
438 |                     <span className="font-space text-[9px] font-bold tracking-[0.1em] mr-1" style={{ color: line.color }
439 |                       {line.label}:
440 |                     </span>
441 |                     <span className="font-exo text-[11px]" style={{ color: "#c8e8ff" }}>{line.text}</span>
442 |                   </div>
443 |                 </div>
444 |               ))}
445 |               {onOpenIA && (
446 |                 <button onClick={onOpenIA}
447 |                   className="mt-2 font-exo text-[10px] font-bold cursor-pointer transition-all hover:text-[#00d4ff] flex
448 |                   style={{ color: "#3a6080" }}
449 |                   data-testid="hoy-consultar-ia">
450 |                   Consultar mas <ChevronRight className="w-3 h-3" />
451 |                 </button>
452 |               )}
453 |             </div>
454 |           ) : (
455 |             <div className="font-exo text-[11px] py-4" style={{ color: "#3a6080" }}>
456 |               No se pudo generar el diagnostico.
457 |             </div>
458 |           )}
459 |         </div>
460 |       </div>
461 |
462 |       {contratoDetalle && (
463 |         <ContratoDetalleModal nombre={contratoDetalle} onClose={() => setContratoDetalle(null)} />
464 |       )}
465 |
466 |     </div>
467 |   );
468 | }
469 |
470 | function ContratoDetalleModal({ nombre, onClose }: { nombre: string; onClose: () => void }) {
471 |   const { data: detalle, isLoading } = useQuery<ContratoDetalle>({
472 |     queryKey: ["/api/ceo/contrato-detalle", nombre],
473 |     queryFn: () => fetch(`/api/ceo/contrato-detalle?nombre=${encodeURIComponent(nombre)}`).then(r => {
474 |       if (!r.ok) throw new Error("Error cargando detalle");
475 |       return r.json();
476 |     }),
477 |   });
478 |
479 |   const [iaResumen, setIaResumen] = useState<string | null>(null);
480 |   const [iaLoading, setIaLoading] = useState(false);
481 |
482 |   const handleIA = async () => {
483 |     setIaLoading(true);
484 |     try {
485 |       const res = await apiRequest("POST", "/api/ceo/contrato-ia-resumen", { nombre });
486 |       const data = await res.json();
487 |       setIaResumen(data.resumen);
488 |     } catch {
489 |       setIaResumen("No se pudo generar el resumen IA.");
490 |     } finally {
491 |       setIaLoading(false);
492 |     }
493 |   };
494 |
495 |   return (
496 |     <div className="fixed inset-0 z-50 flex items-center justify-center" onClick={onClose}>
497 |       <div className="absolute inset-0" style={{ background: "rgba(2,5,8,0.85)", backdropFilter: "blur(8px)" }} />
498 |       <div className="relative w-full max-w-[700px] max-h-[85vh] overflow-y-auto m-4 p-0"
499 |         style={{ background: "#091018", border: "1px solid #0d2035" }}
500 |         onClick={e => e.stopPropagation()}
501 |         data-testid="modal-contrato-detalle">
502 |
503 |         {isLoading || !detalle ? (
504 |           <div className="flex items-center justify-center py-20">
505 |             <Loader2 className="w-6 h-6 animate-spin" style={{ color: "#00d4ff" }} />
506 |           </div>
507 |         ) : (
508 |           <>
509 |             <div className="sticky top-0 z-10 p-4 flex items-center justify-between"
510 |               style={{ background: "#091018", borderBottom: `3px solid ${detalle.color || "#00d4ff"}` }}>
511 |               <div className="flex items-center gap-3">
512 |                 <Search className="w-4 h-4" style={{ color: detalle.color || "#00d4ff" }} />
513 |                 <div>
514 |                   <div className="font-space text-[14px] font-bold tracking-[0.1em]" style={{ color: "#c8e8ff" }}>
515 |                     {detalle.nombre}
516 |                   </div>
517 |                   <div className="font-exo text-[9px] tracking-[0.1em] uppercase" style={{ color: "#3a6080" }}>
518 |                     Detalle contrato / Ultimas 24 horas
519 |                   </div>
520 |                 </div>
521 |               </div>
522 |               <button onClick={onClose} className="p-2 cursor-pointer hover:bg-[rgba(255,255,255,0.05)] transition-all"
523 |                 data-testid="btn-cerrar-detalle">
524 |                 <X className="w-4 h-4" style={{ color: "#3a6080" }} />
525 |               </button>
526 |             </div>
527 |
528 |             <div className="p-4">
529 |               <div className="grid grid-cols-2 sm:grid-cols-4 gap-3 mb-4">
530 |                 <MiniKpi label="CAMIONES" value={`${detalle.camionesOnline}/${detalle.camionesTotal}`} color="#00d4ff"
531 |                   sub="en linea" icon={<Truck className="w-3 h-3" />} />
532 |                 <MiniKpi label="LITROS HOY" value={detalle.litrosHoy.toLocaleString()} color="#ffcc00"
```

```
533 |                    sub={`${detalle.cargasHoy} cargas`} icon={<Fuel className="w-3 h-3" />} />
534 |                  <MiniKpi label="REND. HOY" value={detalle.rendHoy > 0 ? `${detalle.rendHoy}` : "N/D"} color="#00ff88"
535 |                    sub="km/L" />
536 |                  <MiniKpi label="ALERTAS" value={String(detalle.alertasCriticas)} color={detalle.alertasCriticas > 0 ? "#
537 |                    sub={detalle.alertasAlerta > 0 ? `+${detalle.alertasAlerta} alerta` : "Todo bien"} />
538 |                </div>
539 |
540 |                <button onClick={handleIA} disabled={iaLoading}
541 |                  className="w-full mb-4 flex items-center justify-center gap-2 py-3 font-space text-[11px] font-bold trac
542 |                  style={{ background: "rgba(0,212,255,0.06)", borderColor: "rgba(0,212,255,0.3)", color: "#00d4ff" }}
543 |                  data-testid="btn-ia-resumen-contrato">
544 |                  {iaLoading ? (
545 |                    <><Loader2 className="w-3.5 h-3.5 animate-spin" /> ANALIZANDO CON IA...</>
546 |                  ) : (
547 |                    <><Brain className="w-3.5 h-3.5" /> RESUMEN IA DEL DIA</>
548 |                  )}
549 |                </button>
550 |
551 |                {iaResumen && (
552 |                  <div className="mb-4 p-3" style={{ background: "#0a1520", border: "1px solid #0d2035", borderLeft: "3px
553 |                    data-testid="ia-resumen-texto">
554 |                    <div className="font-space text-[8px] font-bold tracking-[0.15em] mb-1.5" style={{ color: "#00d4ff" }}
555 |                      SOTRA IA
556 |                    </div>
557 |                    <div className="font-rajdhani text-[12px] leading-relaxed whitespace-pre-wrap" style={{ color: "#c8e8f
558 |                      {iaResumen}
559 |                    </div>
560 |                  </div>
561 |                )}
562 |
563 |                <div className="font-exo text-[9px] tracking-[0.2em] uppercase font-semibold mb-2" style={{ color: "#c8e8f
564 |                  CAMIONES ({detalle.camiones.length})
565 |                </div>
566 |
567 |                <div className="space-y-1.5">
568 |                  {detalle.camiones.map((cam, i) => (
569 |                    <div key={i} className="p-3 flex items-center gap-3"
570 |                      style={{ background: "#0a1520", border: "1px solid #0d2035" }}
571 |                      data-testid={`contrato-camion-${i}`}>
572 |                      <div className="flex items-center gap-1.5 w-[60px]">
573 |                        <span className="w-1.5 h-1.5 rounded-full flex-shrink-0"
574 |                          style={{ background: cam.online ? "#00ff88" : "#3a6080", boxShadow: cam.online ? "0 0 4px #00ff8
575 |                        <span className="font-space text-[11px] font-bold" style={{ color: "#c8e8ff" }}>{cam.patente}</spa
576 |                      </div>
577 |                      <div className="flex-1 min-w-0">
578 |                        <span className="font-exo text-[10px]" style={{ color: "#3a6080" }}>{cam.conductor}</span>
579 |                      </div>
580 |                      <div className="flex items-center gap-3 flex-shrink-0">
581 |                        {cam.online && cam.velocidad > 0 && (
582 |                          <span className="font-space text-[10px] font-bold" style={{ color: cam.velocidad > 90 ? "#ff2244
583 |                            {cam.velocidad} km/h
584 |                          </span>
585 |                        )}
586 |                        <span className="font-space text-[10px] font-bold" style={{ color: cam.rendimiento > 0 ? (cam.rend
587 |                          {cam.rendimiento > 0 ? `${cam.rendimiento} km/L` : "--"}
588 |                        </span>
589 |                        {cam.litrosHoy > 0 && (
590 |                          <span className="font-exo text-[9px]" style={{ color: "#ffcc00" }}>
591 |                            {cam.litrosHoy}L
592 |                          </span>
593 |                        )}
594 |                        <span className="font-space text-[8px] font-bold px-1.5 py-0.5"
595 |                          style={{ color: estadoCamColors[cam.estado] || "#3a6080", background: `${estadoCamColors[cam.est
596 |                          {cam.estado}
597 |                        </span>
598 |                      </div>
599 |                    </div>
600 |                  ))}
601 |                  {detalle.camiones.length === 0 && (
602 |                    <div className="font-exo text-[11px] py-4 text-center" style={{ color: "#3a6080" }}>
603 |                      Sin camiones asignados a este contrato
604 |                    </div>
605 |                  )}
606 |                </div>
607 |              </div>
608 |            </>
609 |          )}
610 |        </div>
611 |      </div>
612 |    );
613 | }
614 |
615 | function MiniKpi({ label, value, color, sub, icon }: { label: string; value: string; color: string; sub: string; icon?:
616 |   return (
617 |     <div className="p-3" style={{ background: "#0a1520", border: "1px solid #0d2035" }}>
618 |       <div className="flex items-center gap-1 mb-1">
619 |         {icon && <span style={{ color: "#3a6080" }}>{icon}</span>}
620 |         <span className="font-exo text-[7px] tracking-[0.2em] uppercase" style={{ color: "#3a6080" }}>{label}</span>
621 |       </div>
622 |       <div class="font-space text-[18px] font-bold leading-none" style={{ color }}>{value}</div>
623 |       <div class="font-exo text-[9px] mt-1" style={{ color: "#3a6080" }}>{sub}</div>
624 |     </div>
625 |   );
626 | }
627 |
628 | function KpiCard({ label, value, sub, color, progress, loading, tooltip, onClick }: {
629 |   label: string; value: string; sub: string; color: string; progress?: number; loading: boolean; tooltip?: string; onCli
630 | }) {
631 |   return (
632 |     <div className={`p-4 ${onClick ? "cursor-pointer hover:bg-[rgba(0,212,255,0.03)]" : ""}`}
633 |       style={{ background: "#091018", border: "1px solid #0d2035" }}
634 |       onClick={onClick}>
635 |       <div className="flex items-center gap-1 mb-2">
636 |         <span className="font-exo text-[8px] tracking-[0.2em] uppercase" style={{ color: "#3a6080" }}>{label}</span>
637 |         {tooltip && <InfoTip text={tooltip} />}
638 |       </div>
639 |       {loading ? (
```

```
640 |         <Loader2 className="w-5 h-5 animate-spin" style={{ color: "#3a6080" }} />
641 |       ) : (
642 |         <>
643 |           <div className="font-space text-[28px] font-bold leading-none" style={{ color }}>{value}</div>
644 |           <div className="font-exo text-[10px] mt-1.5" style={{ color: "#3a6080" }}>{sub}</div>
645 |           {progress != null && (
646 |             <div className="mt-2 h-1.5 overflow-hidden" style={{ background: "#0d2035" }}>
647 |               <div className="h-full transition-all duration-1000" style={{ width: `${progress}%`, background: color }}
648 |             </div>
649 |           )}
650 |         </>
651 |       )}
652 |     </div>
653 |   );
654 | }
655 |
656 | export { InfoTip };
657 |
```

client/src/pages/flota.tsx

747 lineas

```
1 | import { useState, useMemo, useRef, useEffect } from "react";
2 | import Volvo from "@/pages/volvo";
3 | import Camiones from "@/pages/camiones";
4 | import RankingConductores from "@/pages/ranking-conductores";
5 | import SigetraFusion from "@/pages/sigetra-fusion";
6 | import MicroCargas from "@/pages/micro-cargas";
7 | import Errores from "@/pages/errores";
8 | import { useQuery } from "@tanstack/react-query";
9 | import { ChevronDown, ChevronRight, Info, RefreshCw, Fuel, Users, TrendingDown, Truck, MapPin } from "lucide-react";
10 | import { Tooltip, TooltipContent, TooltipTrigger } from "@/components/ui/tooltip";
11 |
12 | type FlotaSub = "envivo" | "conductores" | "combustible";
13 |
14 | const FLOTA_TABS: { id: FlotaSub; label: string }[] = [
15 |   { id: "envivo", label: "EN VIVO" },
16 |   { id: "conductores", label: "CONDUCTORES" },
17 |   { id: "combustible", label: "COMBUSTIBLE" },
18 | ];
19 |
20 | function InfoTip({ text }: { text: string }) {
21 |   return (
22 |     <Tooltip>
23 |       <TooltipTrigger asChild>
24 |         <span className="inline-flex cursor-help ml-1 align-middle">
25 |           <Info className="w-3 h-3" style={{ color: "#3a6080" }} />
26 |         </span>
27 |       </TooltipTrigger>
28 |       <TooltipContent side="top" className="max-w-[220px] text-xs font-exo" style={{ background: "#1a2a3a", color: "#fff
29 |         {text}
30 |       </TooltipContent>
31 |     </Tooltip>
32 |   );
33 | }
34 |
35 | function AccordionSection({ title, defaultOpen, children, tooltip }: { title: string; defaultOpen?: boolean; children: R
36 |   const [open, setOpen] = useState(defaultOpen ?? false);
37 |   return (
38 |     <div style={{ border: "1px solid #0d2035" }} className="mb-3">
39 |       <button
40 |         onClick={() => setOpen(!open)}
41 |         className="w-full flex items-center justify-between p-3 cursor-pointer transition-all hover:bg-[rgba(0,212,255,0
42 |         style={{ background: "#091018" }}
43 |         data-testid={`accordion-${title.toLowerCase().replace(/\s+/g, "-")}`}>
44 |         <div className="flex items-center gap-2">
45 |           <span className="font-space text-[11px] font-bold tracking-[0.1em]" style={{ color: "#c8e8ff" }}>{title}</span
46 |           {tooltip && <InfoTip text={tooltip} />}
47 |         </div>
48 |         {open ? <ChevronDown className="w-4 h-4" style={{ color: "#3a6080" }} /> : <ChevronRight className="w-4 h-4" sty
49 |       </button>
50 |       {open && <div className="p-4" style={{ borderTop: "1px solid #0d2035" }}>{children}</div>}
51 |     </div>
52 |   );
53 | }
54 |
55 | function RendimientoTable() {
56 |   const { data: fusion = [] } = useQuery<any[]>({ queryKey: ["/api/datos/fusion"] });
57 |
58 |   const trucks = fusion.map((t: any) => {
59 |     const cargas = t.cargas || [];
60 |     const rendVals = cargas.map((c: any) => c.rendimiento).filter((r: any) => r > 0 && r <= 20);
61 |     const rendProm = rendVals.length > 0 ? +(rendVals.reduce((a: number, b: number) => a + b, 0) / rendVals.length).toFi
62 |     return { patente: t.patenteReal || t.fleetNum, faena: t.faenaSigetra || "Sin faena", rend: rendProm, cargas: cargas.
63 |   }).filter((t: any) => t.cargas > 0).sort((a: any, b: any) => a.rend - b.rend);
64 |
65 |   const allRend = trucks.map((t: any) => t.rend).filter((r: number) => r > 0).sort((a: number, b: number) => a - b);
66 |   const p50 = allRend.length > 0 ? allRend[Math.floor(allRend.length * 0.5)] : 0;
67 |   const p75 = allRend.length > 0 ? allRend[Math.floor(allRend.length * 0.75)] : 0;
68 |   const p90 = allRend.length > 0 ? allRend[Math.floor(allRend.length * 0.9)] : 0;
69 |
70 |   function getPercentilColor(rend: number) {
71 |     if (rend >= p90) return "#00ff88";
72 |     if (rend >= p75) return "#00d4ff";
73 |     if (rend >= p50) return "#ffcc00";
74 |     return "#ff2244";
75 |   }
76 |
77 |   return (
78 |     <div>
79 |       <div className="flex items-center gap-4 mb-3">
80 |         <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>
81 |           P50: {p50.toFixed(2)} | P75: {p75.toFixed(2)} | P90: {p90.toFixed(2)} km/L
82 |         </span>
83 |         <InfoTip text="P90 = el 90% de los camiones rinde menos que este. Significa que esta entre los mejores." />
84 |       </div>
85 |       <div className="overflow-x-auto max-h-[400px] overflow-y-auto">
86 |         <table className="w-full text-left" style={{ borderCollapse: "collapse" }}>
87 |           <thead>
88 |             <tr style={{ borderBottom: "1px solid #0d2035" }}>
89 |               <th className="py-2 px-3 font-exo text-[9px] tracking-wider uppercase" style={{ color: "#3a6080" }}>Camion
90 |               <th className="py-2 px-3 font-exo text-[9px] tracking-wider uppercase" style={{ color: "#3a6080" }}>Faena<
91 |               <th className="py-2 px-3 font-exo text-[9px] tracking-wider uppercase text-right" style={{ color: "#3a6080
92 |               <th className="py-2 px-3 font-exo text-[9px] tracking-wider uppercase text-right" style={{ color: "#3a6080
93 |               <th className="py-2 px-3 font-exo text-[9px] tracking-wider uppercase" style={{ color: "#3a6080" }}>Nivel<
94 |             </tr>
95 |           </thead>
96 |           <tbody>
97 |             {trucks.slice(0, 50).map((t: any, i: number) => (
98 |               <tr key={i} style={{ borderBottom: "1px solid rgba(13,32,53,0.5)" }}
99 |                 className="hover:bg-[rgba(0,212,255,0.02)] transition-colors"
100 |                 data-testid={`rend-row-${i}`}>
101 |                 <td className="py-2 px-3 font-space text-[11px] font-bold" style={{ color: "#c8e8ff" }}>{t.patente}</td>
102 |                 <td className="py-2 px-3 font-exo text-[10px]" style={{ color: "#3a6080" }}>{t.faena}</td>
103 |                 <td className="py-2 px-3 font-space text-[11px] font-bold text-right" style={{ color: getPercentilColor(
104 |                 <td className="py-2 px-3 font-exo text-[10px] text-right" style={{ color: "#3a6080" }}>{t.cargas}</td>
```

```
                <td className="py-2 px-3">
                  <span className="font-space text-[8px] font-bold px-2 py-0.5" style={{
                    color: getPercentilColor(t.rend),
                    border: `1px solid ${getPercentilColor(t.rend)}40`,
                    background: `${getPercentilColor(t.rend)}10`,
                  }}>
                    {t.rend >= p90 ? "P90+" : t.rend >= p75 ? "P75+" : t.rend >= p50 ? "P50+" : "<P50"}
                  </span>
                </td>
              </tr>
            ))}
          </tbody>
        </table>
      </div>
    </div>
  );
}

const FUEL_STATIONS: Record<string, { lat: number; lng: number; ciudad: string }> = {
  "QUILICURA": { lat: -33.3583, lng: -70.7250, ciudad: "Santiago" },
  "PANAM. NORTE KM 86.1": { lat: -33.2480, lng: -70.7170, ciudad: "Lampa" },
  "CARRETERA PANAMERICANA NORTE KM 90": { lat: -33.2350, lng: -70.7100, ciudad: "Lampa" },
  "CARRETERA PANAMERICANA NORTE KM. 10": { lat: -33.3890, lng: -70.6650, ciudad: "Renca" },
  "PANAM. NORTE KM.108": { lat: -33.1850, lng: -70.7280, ciudad: "Tiltil" },
  "RUTA 60-C 4, SECTOR 3 ESQUINAS": { lat: -32.8350, lng: -70.6120, ciudad: "Los Andes" },
  "LOS ANGELES 3": { lat: -37.4695, lng: -72.3538, ciudad: "Los Angeles" },
  "PANAMERICANA NORTE, CRUCE CAMINO HU": { lat: -33.32, lng: -70.73, ciudad: "Huechuraba" },
  "RUTA 5 SUR KM 906": { lat: -40.57, lng: -73.13, ciudad: "Osorno" },
  "CARRETERA PANAMERICANA NORTE KM 455": { lat: -30.40, lng: -70.98, ciudad: "La Serena" },
  "CARRETERA LONGITUDINAL SUR KM 275,3": { lat: -35.84, lng: -71.69, ciudad: "Linares" },
};

function AngloMapView({ camiones, conductores, subfaena }: { camiones: any[]; conductores: any[]; subfaena: string }) {
  const mapRef = useRef<HTMLDivElement>(null);
  const mapInstance = useRef<any>(null);
  const markersRef = useRef<any[]>([]);
  const [selectedTruck, setSelectedTruck] = useState<any>(null);
  const [leafletReady, setLeafletReady] = useState(false);

  useEffect(() => {
    const existing = document.querySelector('link[href*="leaflet"]');
    if (!existing) {
      const link = document.createElement("link");
      link.rel = "stylesheet";
      link.href = "https://unpkg.com/leaflet@1.9.4/dist/leaflet.css";
      document.head.appendChild(link);
    }
    const check = () => {
      if ((window as any).L) setLeafletReady(true);
      else setTimeout(check, 100);
    };
    check();
  }, []);

  useEffect(() => {
    if (!leafletReady || !mapRef.current) return;
    const L = (window as any).L;
    if (mapInstance.current) {
      mapInstance.current.remove();
      mapInstance.current = null;
    }
    const map = L.map(mapRef.current, {
      zoomControl: true,
      attributionControl: false,
    }).setView([-33.35, -70.72], 10);
    L.tileLayer("https://{s}.basemaps.cartocdn.com/dark_all/{z}/{x}/{y}{r}.png", {
      maxZoom: 18,
    }).addTo(map);
    mapInstance.current = map;
    return () => {
      if (mapInstance.current) {
        mapInstance.current.remove();
        mapInstance.current = null;
      }
    };
  }, [leafletReady, subfaena]);

  useEffect(() => {
    if (!mapInstance.current || !leafletReady) return;
    const L = (window as any).L;
    const map = mapInstance.current;

    markersRef.current.forEach(m => m.remove());
    markersRef.current = [];

    const stationAgg: Record<string, { cargas: number; litros: number; patentes: Set<string>; lat: number; lng: number;
    const truckLastStation: Record<string, string> = {};
    for (const cond of conductores) {
      if (!cond.cargasDetalle) continue;
      for (const cg of cond.cargasDetalle) {
        const lugar = cg.lugar || "";
        const st = FUEL_STATIONS[lugar];
        if (!st) continue;
        if (!stationAgg[lugar]) {
          stationAgg[lugar] = { cargas: 0, litros: 0, patentes: new Set(), lat: st.lat, lng: st.lng, ciudad: st.ciudad }
        }
        stationAgg[lugar].cargas++;
        stationAgg[lugar].litros += cg.litros || 0;
        if (cg.patente) {
          stationAgg[lugar].patentes.add(cg.patente);
          if (!truckLastStation[cg.patente]) truckLastStation[cg.patente] = lugar;
        }
      }
    }

    const trucksWithRend = camiones.filter(c => c.rendimientoSigetra > 0 && c.rendimientoSigetra < 100);
    const rendValues = trucksWithRend.map(c => c.rendimientoSigetra).sort((a: number, b: number) => a - b);
```

```
212 |     const p25 = rendValues.length > 0 ? rendValues[Math.floor(rendValues.length * 0.25)] : 2.5;
213 |     const p75 = rendValues.length > 0 ? rendValues[Math.floor(rendValues.length * 0.75)] : 3.5;
214 |
215 |     for (const [nombre, d] of Object.entries(stationAgg)) {
216 |       const size = Math.min(30, Math.max(14, Math.sqrt(d.cargas) * 6));
217 |       const fuelIcon = L.divIcon({
218 |         className: "",
219 |         html: `<div style="
220 |           width:${size}px;height:${size}px;
221 |           background:radial-gradient(circle, #ff660080, #ff660020);
222 |           border:2px solid #ff6600;
223 |           border-radius:50%;
224 |           display:flex;align-items:center;justify-content:center;
225 |           box-shadow:0 0 8px #ff660040;
226 |         "><svg width="${size*0.5}" height="${size*0.5}" viewBox="0 0 24 24" fill="none" stroke="#ff6600" stroke-width="2
227 |         iconSize: [size, size],
228 |         iconAnchor: [size / 2, size / 2],
229 |       });
230 |       const marker = L.marker([d.lat, d.lng], { icon: fuelIcon }).addTo(map);
231 |       marker.bindPopup(`
232 |         <div style="background:#0a1520;color:#c8e8ff;border:1px solid #ff6600;border-radius:4px;padding:10px;min-width:1
233 |           <div style="font-weight:bold;color:#ff6600;font-size:12px;margin-bottom:6px;">${nombre}</div>
234 |           <div style="font-size:10px;color:#3a6080;margin-bottom:4px;">${d.ciudad}</div>
235 |           <div style="display:grid;grid-template-columns:1fr 1fr;gap:4px;font-size:11px;">
236 |             <span style="color:#3a6080;">Cargas:</span><span style="color:#00d4ff;font-weight:bold;">${d.cargas}</span>
237 |             <span style="color:#3a6080;">Litros:</span><span style="color:#00d4ff;font-weight:bold;">${Math.round(d.litr
238 |             <span style="color:#3a6080;">Camiones:</span><span style="color:#FF6B35;font-weight:bold;">${d.patentes.size
239 |           </div>
240 |         </div>
241 |       `, { className: "dark-popup" });
242 |       markersRef.current.push(marker);
243 |     }
244 |
245 |     const stationCoords = Object.values(stationAgg).map(s => [s.lat, s.lng] as [number, number]);
246 |     const truckPositions: Array<{ patente: string; lat: number; lng: number; rend: number; litros: number }> = [];
247 |
248 |     for (const c of trucksWithRend) {
249 |       const lastStation = truckLastStation[c.patente];
250 |       if (!lastStation) continue;
251 |       const st = FUEL_STATIONS[lastStation];
252 |       if (!st) continue;
253 |       const jitter = () => (Math.random() - 0.5) * 0.015;
254 |       truckPositions.push({
255 |         patente: c.patente,
256 |         lat: st.lat + jitter(),
257 |         lng: st.lng + jitter(),
258 |         rend: c.rendimientoSigetra,
259 |         litros: c.litrosSigetra,
260 |       });
261 |     }
262 |
263 |     for (const t of truckPositions) {
264 |       const color = t.rend >= p75 ? "#00c97a" : t.rend >= p25 ? "#ffcc00" : "#ff2244";
265 |       const truckIcon = L.divIcon({
266 |         className: "",
267 |         html: `<div style="
268 |           width:10px;height:10px;
269 |           background:${color};
270 |           border:1.5px solid ${color};
271 |           border-radius:2px;
272 |           box-shadow:0 0 4px ${color}60;
273 |           cursor:pointer;
274 |         " title="${t.patente}: ${t.rend.toFixed(2)} km/L"></div>`,
275 |         iconSize: [10, 10],
276 |         iconAnchor: [5, 5],
277 |       });
278 |       const marker = L.marker([t.lat, t.lng], { icon: truckIcon }).addTo(map);
279 |       marker.on("click", () => setSelectedTruck(t));
280 |       marker.bindPopup(`
281 |         <div style="background:#0a1520;color:#c8e8ff;border:1px solid ${color};border-radius:4px;padding:8px;min-width:1
282 |           <div style="font-weight:bold;color:${color};font-size:13px;margin-bottom:4px;">${t.patente}</div>
283 |           <div style="font-size:11px;">
284 |             <span style="color:#3a6080;">Rend:</span> <span style="color:${color};font-weight:bold;">${t.rend.toFixed(2)
285 |           </div>
286 |           <div style="font-size:11px;">
287 |             <span style="color:#3a6080;">Litros:</span> <span style="color:#00d4ff;">${Math.round(t.litros).toLocaleStri
288 |           </div>
289 |         </div>
290 |       `, { className: "dark-popup" });
291 |       markersRef.current.push(marker);
292 |     }
293 |
294 |     const allCoords = [...stationCoords, ...truckPositions.map(t => [t.lat, t.lng] as [number, number])];
295 |     if (allCoords.length > 1) {
296 |       map.fitBounds(L.latLngBounds(allCoords).pad(0.15));
297 |     }
298 |   }, [camiones, conductores, leafletReady, subfaena]);
299 |
300 |   const stats = useMemo(() => {
301 |     const withRend = camiones.filter(c => c.rendimientoSigetra > 0 && c.rendimientoSigetra < 100);
302 |     const criticos = withRend.filter(c => c.rendimientoSigetra < 2.5).length;
303 |     const normales = withRend.filter(c => c.rendimientoSigetra >= 2.5 && c.rendimientoSigetra < 3.5).length;
304 |     const buenos = withRend.filter(c => c.rendimientoSigetra >= 3.5).length;
305 |     const stationSet = new Set<string>();
306 |     conductores.forEach((cond: any) => cond.cargasDetalle?.forEach((cg: any) => { if (cg.lugar && FUEL_STATIONS[cg.lugar
307 |     return { criticos, normales, buenos, estaciones: stationSet.size };
308 |   }, [camiones, conductores]);
309 |
310 |   return (
311 |     <div data-testid="anglo-map-view">
312 |       <div class="flex items-center gap-3 mb-3">
313 |         <div class="flex items-center gap-2">
314 |           <div class="w-2.5 h-2.5 rounded-sm" style={{ background: "#ff2244" }} />
315 |           <span class="font-exo text-[9px]" style={{ color: "#3a6080" }}>Critico &lt;2.5 ({stats.criticos})</span>
316 |         </div>
317 |         <div class="flex items-center gap-2">
318 |           <div class="w-2.5 h-2.5 rounded-sm" style={{ background: "#ffcc00" }} />
```

```
319 |           <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>Normal ({stats.normales})</span>
320 |         </div>
321 |         <div className="flex items-center gap-2">
322 |           <div className="w-2.5 h-2.5 rounded-sm" style={{ background: "#00c97a" }} />
323 |           <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>Bueno &gt;3.5 ({stats.buenos})</span>
324 |         </div>
325 |         <div className="flex items-center gap-2 ml-3">
326 |           <div className="w-3 h-3 rounded-full" style={{ border: "2px solid #ff6600", background: "#ff660030" }} />
327 |           <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>Estaciones ({stats.estaciones})</span>
328 |         </div>
329 |       </div>
330 |       <div className="relative rounded overflow-hidden" style={{ height: "500px", border: "1px solid #0d2035" }}>
331 |         <div ref={mapRef} className="absolute inset-0" data-testid="anglo-map-container" />
332 |       </div>
333 |     </div>
334 |   );
335 | }
336 |
337 | function ContractAngloPanel() {
338 |   const [activeSubfaena, setActiveSubfaena] = useState<string>("todas");
339 |   const [activeTab, setActiveTab] = useState<"resumen" | "conductores" | "cruce" | "mapa">("resumen");
340 |
341 |   const subfaenaParam = activeSubfaena === "todas" ? "" : activeSubfaena;
342 |   const { data: cruceData, isLoading: cruceLoading } = useQuery<any>({
343 |     queryKey: ["/api/geo/cruce-mensual", "anglo", subfaenaParam],
344 |     queryFn: async () => {
345 |       const params = new URLSearchParams({ contrato: "ANGLO" });
346 |       if (subfaenaParam) params.set("subfaena", subfaenaParam);
347 |       const res = await fetch(`/api/geo/cruce-mensual?${params}`);
348 |       return res.json();
349 |     },
350 |   });
351 |
352 |   const { data: conductoresData, isLoading: condLoading } = useQuery<any>({
353 |     queryKey: ["/api/geo/conductores", "anglo", subfaenaParam],
354 |     queryFn: async () => {
355 |       const params = new URLSearchParams({ contrato: "ANGLO" });
356 |       if (subfaenaParam) params.set("subfaena", subfaenaParam);
357 |       const res = await fetch(`/api/geo/conductores?${params}`);
358 |       return res.json();
359 |     },
360 |   });
361 |
362 |   const { data: contracts } = useQuery<any[]>({ queryKey: ["/api/contratos"] });
363 |   const angloContract = contracts?.find(c => c.name === "ANGLO");
364 |   const subfaenas = angloContract?.faenas || [];
365 |
366 |   const isLoading = cruceLoading || condLoading;
367 |
368 |   const resumen = useMemo(() => {
369 |     if (!cruceData?.camiones) return null;
370 |     const totalLitros = cruceData.totalSigetra || 0;
371 |     const totalCamiones = cruceData.totalCamiones || 0;
372 |     const withRend = cruceData.camiones.filter((c: any) => c.rendimientoSigetra > 0 && c.rendimientoSigetra < 100);
373 |     const rendProm = withRend.length > 0
374 |       ? withRend.reduce((s: number, c: any) => s + c.rendimientoSigetra, 0) / withRend.length
375 |       : 0;
376 |     const criticos = withRend.filter((c: any) => c.rendimientoSigetra < 2.5).length;
377 |     const bajos = withRend.filter((c: any) => c.rendimientoSigetra >= 2.5 && c.rendimientoSigetra < 3.5).length;
378 |     return { totalLitros, totalCamiones, rendProm, criticos, bajos, totalConductores: conductoresData?.totalConductores
379 |   }, [cruceData, conductoresData]);
380 |
381 |   const [sortCond, setSortCond] = useState<"litros" | "rend_asc" | "rend_desc">("litros");
382 |   const [searchCond, setSearchCond] = useState("");
383 |   const [expandedCond, setExpandedCond] = useState<string | null>(null);
384 |
385 |   const filteredConductores = useMemo(() => {
386 |     if (!conductoresData?.conductores) return [];
387 |     let list = conductoresData.conductores;
388 |     if (searchCond) {
389 |       const term = searchCond.toLowerCase();
390 |       list = list.filter((c: any) => c.nombre.toLowerCase().includes(term) || c.camiones.some((p: string) => p.includes(
391 |     }
392 |     if (sortCond === "rend_asc") list = [...list].sort((a: any, b: any) => (a.rendimiento || 999) - (b.rendimiento || 99
393 |     else if (sortCond === "rend_desc") list = [...list].sort((a: any, b: any) => (b.rendimiento || 0) - (a.rendimiento |
394 |     return list;
395 |   }, [conductoresData, searchCond, sortCond]);
396 |
397 |   const [sortCruce, setSortCruce] = useState<"patente" | "rend_asc" | "diff">("patente");
398 |   const sortedCruce = useMemo(() => {
399 |     if (!cruceData?.camiones) return [];
400 |     let list = [...cruceData.camiones];
401 |     if (sortCruce === "rend_asc") list.sort((a: any, b: any) => (a.rendimientoSigetra || 999) - (b.rendimientoSigetra ||
402 |     else if (sortCruce === "diff") list.sort((a: any, b: any) => Math.abs(b.diferencia || 0) - Math.abs(a.diferencia ||
403 |     return list;
404 |   }, [cruceData, sortCruce]);
405 |
406 |   return (
407 |     <div data-testid="anglo-panel">
408 |       <div className="flex items-center gap-2 mb-4">
409 |         <div className="font-space text-[14px] font-bold tracking-wider" style={{ color: "#FF6B35" }}>ANGLOAMERICAN</div
410 |         <div className="font-exo text-[10px] px-2 py-0.5 rounded" style={{ background: "#FF6B3515", border: "1px solid #
411 |           {angloContract?.camiones || 0} camiones
412 |         </div>
413 |       </div>
414 |
415 |       <div className="flex gap-1 mb-4">
416 |         <button onClick={() => setActiveSubfaena("todas")}
417 |           data-testid="btn-subfaena-todas"
418 |           className="px-3 py-1.5 font-exo text-[10px] font-bold cursor-pointer transition-all rounded"
419 |           style={{
420 |             background: activeSubfaena === "todas" ? "#FF6B3520" : "#0d203530",
421 |             border: `1px solid ${activeSubfaena === "todas" ? "#FF6B35" : "#0d2035"}`,
422 |             color: activeSubfaena === "todas" ? "#FF6B35" : "#3a6080",
423 |           }}>
424 |           TODAS
425 |         </button>
```

```
426 |          {subfaenas.map((sf: any) => {
427 |            const shortName = sf.nombre.replace("ANGLO-", "");
428 |            const colors: Record<string, string> = { "CARGAS VARIAS": "#FF6B35", "CAL": "#00C49A", "COCU": "#1A8FFF" };
429 |            const c = colors[shortName] || "#FF6B35";
430 |            return (
431 |              <button key={sf.id} onClick={() => setActiveSubfaena(sf.nombre)}
432 |                data-testid={`btn-subfaena-${sf.id}`}
433 |                className="px-3 py-1.5 font-exo text-[10px] font-bold cursor-pointer transition-all rounded"
434 |                style={{
435 |                  background: activeSubfaena === sf.nombre ? `${c}20` : "#0d203530",
436 |                  border: `1px solid ${activeSubfaena === sf.nombre ? c : "#0d2035"}`,
437 |                  color: activeSubfaena === sf.nombre ? c : "#3a6080",
438 |                }}>
439 |                {shortName}
440 |              </button>
441 |            );
442 |          })}
443 |        </div>
444 |
445 |        {isLoading ? (
446 |          <div className="flex items-center justify-center py-12">
447 |            <RefreshCw className="w-5 h-5 animate-spin" style={{ color: "#FF6B35" }} />
448 |          </div>
449 |        ) : (
450 |          <>
451 |            {resumen && (
452 |              <div className="grid grid-cols-2 md:grid-cols-4 gap-3 mb-5">
453 |                {[
454 |                  { label: "CAMIONES", value: resumen.totalCamiones, icon: Truck, color: "#FF6B35" },
455 |                  { label: "LITROS MES", value: resumen.totalLitros.toLocaleString() + " L", icon: Fuel, color: "#00d4ff"
456 |                  { label: "REND PROMEDIO", value: resumen.rendProm > 0 ? resumen.rendProm.toFixed(2) + " km/L" : "N/D", i
457 |                  { label: "CONDUCTORES", value: resumen.totalConductores, icon: Users, color: "#ffcc00" },
458 |                ].map((kpi, i) => (
459 |                  <div key={i} className="p-3 rounded" style={{ background: "#091018", border: "1px solid #0d2035" }}>
460 |                    <div className="flex items-center gap-2 mb-1">
461 |                      <kpi.icon className="w-3.5 h-3.5" style={{ color: kpi.color }} />
462 |                      <span className="font-exo text-[9px] tracking-wider uppercase" style={{ color: "#3a6080" }}>{kpi.lab
463 |                    </div>
464 |                    <div className="font-space text-[18px] font-bold" style={{ color: kpi.color }}>{kpi.value}</div>
465 |                    {kpi.label === "REND PROMEDIO" && resumen.criticos > 0 && (
466 |                      <div class="font-exo text-[9px] mt-1" style={{ color: "#ff2244" }}>
467 |                        {resumen.criticos} criticos (&lt;2.5)
468 |                      </div>
469 |                    )}
470 |                  </div>
471 |                ))}
472 |              </div>
473 |            )}
474 |
475 |            <div className="flex gap-1 mb-4">
476 |              {[
477 |                { id: "resumen" as const, label: "CRUCE CAMIONES" },
478 |                { id: "conductores" as const, label: "CONDUCTORES" },
479 |                { id: "mapa" as const, label: "MAPA" },
480 |              ].map(t => (
481 |                <button key={t.id} onClick={() => setActiveTab(t.id)}
482 |                  data-testid={`anglo-tab-${t.id}`}
483 |                  className="px-4 py-2 font-exo text-[10px] font-bold tracking-[0.15em] cursor-pointer transition-all bord
484 |                  style={{
485 |                    borderColor: activeTab === t.id ? "#FF6B35" : "transparent",
486 |                    color: activeTab === t.id ? "#FF6B35" : "#3a6080",
487 |                  }}>
488 |                  {t.label}
489 |                </button>
490 |              ))}
491 |            </div>
492 |
493 |            {activeTab === "resumen" && (
494 |              <div>
495 |                <div className="flex items-center gap-2 mb-3">
496 |                  {[
497 |                    { id: "patente" as const, label: "Por patente" },
498 |                    { id: "rend_asc" as const, label: "Peor rend" },
499 |                    { id: "diff" as const, label: "Mayor dif" },
500 |                  ].map(s => (
501 |                    <button key={s.id} onClick={() => setSortCruce(s.id)}
502 |                      data-testid={`btn-sort-cruce-${s.id}`}
503 |                      className="font-exo text-[10px] font-bold px-2.5 py-1 rounded cursor-pointer transition-all"
504 |                      style={{
505 |                        background: sortCruce === s.id ? "#FF6B3515" : "#0d203530",
506 |                        border: `1px solid ${sortCruce === s.id ? "#FF6B35" : "#0d2035"}`,
507 |                        color: sortCruce === s.id ? "#FF6B35" : "#3a6080",
508 |                      }}>
509 |                      {s.label}
510 |                    </button>
511 |                  ))}
512 |                </div>
513 |                <div className="rounded overflow-hidden" style={{ border: "1px solid #0d2035" }}>
514 |                  <div className="overflow-x-auto max-h-[500px] overflow-y-auto">
515 |                    <table class="w-full">
516 |                      <thead>
517 |                        <tr style={{ background: "#0d203540" }}>
518 |                          {["Cam", "Conductores", "Cargas", "Litros Sig", "Km Odo", "Rend km/L", "Litros ECU", "Dif L"].ma
519 |                            <th key={h} class="py-2 px-2 font-exo text-[9px] tracking-wider uppercase text-left" style
520 |                          ))}
521 |                        </tr>
522 |                      </thead>
523 |                      <tbody>
524 |                        {sortedCruce.map((c: any) => (
525 |                          <tr key={c.patente} style={{ borderBottom: "1px solid rgba(13,32,53,0.5)" }}
526 |                            class="hover:bg-[rgba(255,107,53,0.03)] transition-colors"
527 |                            data-testid={`anglo-cruce-${c.patente}`}>
528 |                            <td class="py-2 px-2 font-space text-[11px] font-bold" style={{ color: "#FF6B35" }}>{c.pat
529 |                            <td class="py-2 px-2 font-exo text-[10px]" style={{ color: "#c8e8ff" }}>
530 |                              {c.conductores?.slice(0, 2).join(", ") || "-"}
531 |                            </td>
532 |                            <td class="py-2 px-2 font-space text-[10px] text-center" style={{ color: "#3a6080" }}>{c.c
```

```
533 |                          <td className="py-2 px-2 font-space text-[11px] font-bold" style={{ color: "#00d4ff" }}>
534 |                            {c.litrosSigetra > 0 ? c.litrosSigetra.toLocaleString() : "-"}
535 |                          </td>
536 |                          <td className="py-2 px-2 font-space text-[10px]" style={{ color: "#c8e8ff" }}>
537 |                            {c.kmOdometro > 0 ? c.kmOdometro.toLocaleString() : "-"}
538 |                          </td>
539 |                          <td className="py-2 px-2 font-space text-[11px] font-bold" style={{
540 |                            color: c.rendimientoSigetra >= 3.5 ? "#00c97a" : c.rendimientoSigetra >= 2.5 ? "#ffcc00" : c
541 |                          }}>
542 |                            {c.rendimientoSigetra > 0 ? c.rendimientoSigetra.toFixed(2) : "-"}
543 |                          </td>
544 |                          <td className="py-2 px-2 font-space text-[10px]" style={{ color: "#3a6080" }}>
545 |                            {c.litrosEcu > 0 ? c.litrosEcu.toLocaleString() : "-"}
546 |                          </td>
547 |                          <td className="py-2 px-2 font-space text-[11px] font-bold" style={{
548 |                            color: Math.abs(c.diferencia) > 500 ? "#ff2244" : Math.abs(c.diferencia) > 100 ? "#ffcc00" :
549 |                          }}>
550 |                            {c.diferencia !== 0 ? (c.diferencia > 0 ? "+" : "") + c.diferencia.toLocaleString() : "-"}
551 |                          </td>
552 |                        </tr>
553 |                      ))}
554 |                    </tbody>
555 |                  </table>
556 |                </div>
557 |              </div>
558 |            </div>
559 |          )}
560 |
561 |          {activeTab === "conductores" && (
562 |            <div>
563 |              <div className="flex items-center justify-between mb-3">
564 |                <div className="flex items-center gap-2">
565 |                  <div className="font-exo text-[10px] px-2 py-0.5 rounded" style={{ background: "#FF6B3515", border: "1
566 |                    {conductoresData?.totalConductores || 0} conductores
567 |                  </div>
568 |                  <div className="flex gap-1">
569 |                    {([
570 |                      { id: "litros" as const, label: "Mas litros" },
571 |                      { id: "rend_asc" as const, label: "Peor rend" },
572 |                      { id: "rend_desc" as const, label: "Mejor rend" },
573 |                    ]).map(s => (
574 |                      <button key={s.id} onClick={() => setSortCond(s.id)}
575 |                        data-testid={`btn-sort-cond-${s.id}`}
576 |                        className="font-exo text-[10px] font-bold px-2.5 py-1 rounded cursor-pointer transition-all"
577 |                        style={{
578 |                          background: sortCond === s.id ? (s.id === "rend_asc" ? "#ff224420" : s.id === "rend_desc" ? "#
579 |                          border: `1px solid ${sortCond === s.id ? (s.id === "rend_asc" ? "#ff2244" : s.id === "rend_des
580 |                          color: sortCond === s.id ? (s.id === "rend_asc" ? "#ff2244" : s.id === "rend_desc" ? "#00c97a"
581 |                        }}>
582 |                        {s.label}
583 |                      </button>
584 |                    ))}
585 |                  </div>
586 |                </div>
587 |                <input type="text" value={searchCond} onChange={e => setSearchCond(e.target.value)}
588 |                  placeholder="Buscar conductor..." data-testid="input-search-anglo-conductor"
589 |                  className="pl-3 pr-3 py-1.5 rounded font-exo text-[11px]"
590 |                  style={{ background: "#0d203550", border: "1px solid #0d2035", color: "#c8e8ff", width: "200px" }} />
591 |              </div>
592 |              <div className="rounded overflow-hidden max-h-[500px] overflow-y-auto" style={{ border: "1px solid #0d2035
593 |                <table className="w-full">
594 |                  <thead>
595 |                    <tr style={{ background: "#0d203540" }}>
596 |                      {["Conductor", "Camiones", "Cargas", "Litros", "Km", "Rend km/L"].map(h => (
597 |                        <th key={h} className="py-2 px-2 font-exo text-[9px] tracking-wider uppercase text-left sticky t
598 |                      ))}
599 |                    </tr>
600 |                  </thead>
601 |                  <tbody>
602 |                    {filteredConductores.map((c: any) => (
603 |                      <tr key={c.nombre} style={{ borderBottom: "1px solid rgba(13,32,53,0.5)" }}
604 |                        className="hover:bg-[rgba(255,107,53,0.03)] transition-colors cursor-pointer"
605 |                        onClick={() => setExpandedCond(expandedCond === c.nombre ? null : c.nombre)}
606 |                        data-testid={`anglo-cond-${c.nombre}`}>
607 |                        <td className="py-2 px-2 font-exo text-[11px] font-bold" style={{ color: "#c8e8ff" }}>{c.nombre}
608 |                        <td className="py-2 px-2">
609 |                          <div className="flex gap-1 flex-wrap">
610 |                            {c.camiones.map((p: string) => (
611 |                              <span key={p} className="font-space text-[9px] px-1.5 py-0.5 rounded" style={{ background:
612 |                            ))}
613 |                          </div>
614 |                        </td>
615 |                        <td className="py-2 px-2 font-space text-[10px]" style={{ color: "#3a6080" }}>{c.cargas}</td>
616 |                        <td className="py-2 px-2 font-space text-[11px] font-bold" style={{ color: "#00d4ff" }}>{c.litro
617 |                        <td className="py-2 px-2 font-space text-[10px]" style={{ color: "#c8e8ff" }}>{c.kmTotales > 0 ?
618 |                        <td className="py-2 px-2 font-space text-[11px] font-bold" style={{
619 |                          color: c.rendimiento >= 3.5 ? "#00c97a" : c.rendimiento > 0 ? "#ff2244" : "#3a6080",
620 |                        }}>
621 |                          {c.rendimiento > 0 ? c.rendimiento.toFixed(2) : "-"}
622 |                        </td>
623 |                      </tr>
624 |                    ))}
625 |                  </tbody>
626 |                </table>
627 |              </div>
628 |            </div>
629 |          )}
630 |
631 |          {activeTab === "mapa" && (
632 |            <AngloMapView
633 |              camiones={cruceData?.camiones || []}
634 |              conductores={conductoresData?.conductores || []}
635 |              subfaena={activeSubfaena}
636 |            />
637 |          )}
638 |        </>
639 |      )}
```

```
640 |     </div>
641 |   );
642 | }
643 |
644 | export default function Flota({ initialSub }: { initialSub?: string }) {
645 |   const [activeSub, setActiveSub] = useState<FlotaSub>((initialSub as FlotaSub) || "envivo");
646 |   const [vistaEnVivo, setVistaEnVivo] = useState<"telemetria" | "camiones">("telemetria");
647 |   const [activeContract, setActiveContract] = useState<string>("CENCOSUD");
648 |
649 |   const { data: contracts } = useQuery<any[]>({ queryKey: ["/api/contratos"] });
650 |   const { data: microData } = useQuery<any>({ queryKey: ["/api/datos/micro-cargas"] });
651 |   const microBadge = (microData?.totals?.criticos || 0) + (microData?.totals?.sospechosos || 0);
652 |
653 |   const hasAnglo = contracts?.some(c => c.name === "ANGLO");
654 |
655 |   return (
656 |     <div data-testid="page-flota">
657 |       {hasAnglo && (
658 |         <div className="flex items-center gap-1 mb-3 pb-2" style={{ borderBottom: "1px solid #0d2035" }}>
659 |           <span className="font-exo text-[9px] tracking-wider uppercase mr-2" style={{ color: "#3a6080" }}>CONTRATO:</sp
660 |           {(contracts || []).map(c => {
661 |             const colors: Record<string, string> = { CENCOSUD: "#00d4ff", ANGLO: "#FF6B35" };
662 |             const col = colors[c.name] || "#00d4ff";
663 |             return (
664 |               <button key={c.name} onClick={() => setActiveContract(c.name)}
665 |                 data-testid={`contract-tab-${c.name}`}
666 |                 className="px-4 py-1.5 font-space text-[11px] font-bold tracking-wider cursor-pointer transition-all rou
667 |                 style={{
668 |                   background: activeContract === c.name ? `${col}15` : "transparent",
669 |                   border: `1px solid ${activeContract === c.name ? col : "transparent"}`,
670 |                   color: activeContract === c.name ? col : "#3a6080",
671 |                 }}>
672 |                 {c.name}
673 |                 <span className="ml-1.5 font-exo text-[9px]" style={{ color: "#3a6080" }}>({c.camiones})</span>
674 |               </button>
675 |             );
676 |           })}
677 |         </div>
678 |       )}
679 |
680 |       {activeContract === "ANGLO" ? (
681 |         <ContractAngloPanel />
682 |       ) : (
683 |         <>
684 |           <div className="flex items-center gap-1 mb-4 pb-2" style={{ borderBottom: "1px solid #0d2035" }}>
685 |             {FLOTA_TABS.map(t => (
686 |               <button key={t.id} onClick={() => setActiveSub(t.id)}
687 |                 data-testid={`flota-tab-${t.id}`}
688 |                 className={`px-4 py-2 font-exo text-[10px] font-bold tracking-[0.15em] cursor-pointer transition-all bor
689 |                   activeSub === t.id
690 |                     ? "border-[#00d4ff] text-[#00d4ff]"
691 |                     : "border-transparent text-[#3a6080] hover:text-[#c8e8ff]"
692 |                 }`}>
693 |                 {t.label}
694 |                 {t.id === "combustible" && microBadge > 0 && (
695 |                   <span className="ml-1.5 font-space text-[8px] font-bold px-1.5 py-0.5 rounded-full min-w-[18px] text-c
696 |                     style={{ background: "#ff2244", color: "#020508" }}>
697 |                     {microBadge}
698 |                   </span>
699 |                 )}
700 |               </button>
701 |             ))}
702 |           </div>
703 |
704 |           {activeSub === "envivo" && (
705 |             <div>
706 |               <div className="flex gap-1 mb-3">
707 |                 <button onClick={() => setVistaEnVivo("telemetria")}
708 |                   data-testid="envivo-toggle-telemetria"
709 |                   className={`px-3 py-1 font-exo text-[9px] font-bold cursor-pointer border ${
710 |                     vistaEnVivo === "telemetria" ? "bg-[rgba(0,212,255,0.1)] text-[#00d4ff] border-[rgba(0,212,255,0.3)]
711 |                   }`}>LISTA</button>
712 |                 <button onClick={() => setVistaEnVivo("camiones")}
713 |                   data-testid="envivo-toggle-camiones"
714 |                   className={`px-3 py-1 font-exo text-[9px] font-bold cursor-pointer border ${
715 |                     vistaEnVivo === "camiones" ? "bg-[rgba(0,212,255,0.1)] text-[#00d4ff] border-[rgba(0,212,255,0.3)]"
716 |                   }`}>FICHA CAMIONES</button>
717 |               </div>
718 |               {vistaEnVivo === "telemetria" ? <Volvo /> : <Camiones />}
719 |             </div>
720 |           )}
721 |
722 |           {activeSub === "conductores" && <RankingConductores />}
723 |
724 |           {activeSub === "combustible" && (
725 |             <div className="space-y-0">
726 |               <AccordionSection title="CRUCE SIGETRA-VOLVO" defaultOpen={true}
727 |                 tooltip="Comparacion entre lo que cargo el camion segun Sigetra y lo que consumio segun el GPS.">
728 |                 <SigetraFusion />
729 |               </AccordionSection>
730 |               <AccordionSection title="CARGAS SOSPECHOSAS"
731 |                 tooltip="Carga de poco combustible con senales que podrian indicar desvio. Requiere verificacion.">
732 |                 <MicroCargas />
733 |               </AccordionSection>
734 |               <AccordionSection title="RENDIMIENTO">
735 |                 <RendimientoTable />
736 |               </AccordionSection>
737 |               <AccordionSection title="CALIDAD DE DATOS">
738 |                 <Errores />
739 |               </AccordionSection>
740 |             </div>
741 |           )}
742 |         </>
743 |       )}
744 |     </div>
745 |   );
746 | }
```

747 |

client/src/pages/geovalidator.tsx

1956 lineas

```
1 | import { useState, useEffect, useMemo, useRef, useCallback, Fragment } from "react";
2 | import { useQuery, useMutation } from "@tanstack/react-query";
3 | import { queryClient, apiRequest } from "@/lib/queryClient";
4 | import { Map, CheckCircle, RefreshCw, MapPin, Clock, Gauge, Truck, ChevronDown, ChevronUp, Eye, Check, X, AlertTriangle,
5 |
6 | type GeoTab = "mapa" | "viajes" | "conductores" | "validacion" | "ia" | "recopilacion";
7 |
8 | interface CamionLive {
9 |   camionId: number;
10 |   patente: string;
11 |   modelo: string;
12 |   conductor: string | null;
13 |   lat: number | null;
14 |   lng: number | null;
15 |   velocidad: number;
16 |   rumbo: number;
17 |   timestamp: string | null;
18 |   estado: string;
19 |   ageMinutes: number;
20 |   fuelLevel: number | null;
21 | }
22 |
23 | interface GeoViaje {
24 |   id: number;
25 |   camionId: number;
26 |   patente: string;
27 |   contrato: string;
28 |   origenNombre: string | null;
29 |   destinoNombre: string | null;
30 |   origenTimestamp: string | null;
31 |   destinoTimestamp: string | null;
32 |   kmGps: string | null;
33 |   duracionMinutos: number | null;
34 |   velocidadMaxima: string | null;
35 |   velocidadPromedio: string | null;
36 |   validacionEstado: string;
37 |   validacionDetalle: any;
38 |   sigetraKmDeltaPct: string | null;
39 |   sigetraLitros: string | null;
40 |   sigetraSurtidorEnRuta: boolean | null;
41 |   validadoManualmente: boolean;
42 |   notas: string | null;
43 | }
44 |
45 | interface GeoBase {
46 |   id: number;
47 |   nombre: string;
48 |   lat: string;
49 |   lng: string;
50 |   radioMetros: number;
51 |   contrato: string;
52 | }
53 |
54 | const estadoColors: Record<string, string> = {
55 |   VALIDADO: "#00c97a",
56 |   REVISAR: "#ffcc00",
57 |   ANOMALIA: "#ff2244",
58 |   PENDIENTE: "#3a6080",
59 | };
60 |
61 | const estadoLabels: Record<string, string> = {
62 |   VALIDADO: "VALIDADO",
63 |   REVISAR: "REVISAR",
64 |   ANOMALIA: "ANOMALIA",
65 |   PENDIENTE: "PENDIENTE",
66 | };
67 |
68 | function EstadoBadge({ estado }: { estado: string }) {
69 |   const color = estadoColors[estado] || "#3a6080";
70 |   return (
71 |     <span className="font-exo text-[10px] font-bold px-2 py-0.5 rounded" style={{
72 |       background: color + "20",
73 |       border: `1px solid ${color}`,
74 |       color,
75 |     }} data-testid={`badge-estado-${estado}`}>
76 |       {estadoLabels[estado] || estado}
77 |     </span>
78 |   );
79 | }
80 |
81 | function CamionStatusDot({ estado }: { estado: string }) {
82 |   const colors: Record<string, string> = {
83 |     EN_MOVIMIENTO: "#00c97a",
84 |     DETENIDO_RECIENTE: "#ffcc00",
85 |     DETENIDO: "#ff2244",
86 |     "SIN_SEÑAL": "#3a6080",
87 |   };
88 |   return <div className="w-2 h-2 rounded-full" style={{ background: colors[estado] || "#3a6080" }} />;
89 | }
90 |
91 | function MapaEnVivo() {
92 |   const mapRef = useRef<HTMLDivElement>(null);
93 |   const mapInstance = useRef<any>(null);
94 |   const markersRef = useRef<any[]>([]);
95 |   const fuelMarkersRef = useRef<any[]>([]);
96 |   const [filter, setFilter] = useState("todos");
97 |   const [selectedCamion, setSelectedCamion] = useState<number | null>(null);
98 |   const [panelOpen, setPanelOpen] = useState(true);
99 |   const [showFuelStations, setShowFuelStations] = useState(true);
100 |   const [selectedEstacion, setSelectedEstacion] = useState<string | null>(null);
101 |
102 |   const { data: camiones, refetch } = useQuery<CamionLive[]>({
103 |     queryKey: ["/api/geo/camiones-live"],
104 |     refetchInterval: 60000,
```

```
105 |   });
106 |
107 |   const { data: bases } = useQuery<GeoBase[]>({
108 |     queryKey: ["/api/geo/bases"],
109 |   });
110 |
111 |   const { data: fuelData } = useQuery<any>({
112 |     queryKey: ["/api/geo/cargas-combustible"],
113 |   });
114 |
115 |   const { data: cruceMensual } = useQuery<any>({
116 |     queryKey: ["/api/geo/cruce-mensual"],
117 |   });
118 |
119 |   const ingestMutation = useMutation({
120 |     mutationFn: () => apiRequest("POST", "/api/geo/ingest-volvo"),
121 |     onSuccess: () => { refetch(); },
122 |   });
123 |
124 |   const filtered = useMemo(() => {
125 |     if (!camiones) return [];
126 |     if (filter === "todos") return camiones;
127 |     if (filter === "movimiento") return camiones.filter(c => c.estado === "EN_MOVIMIENTO");
128 |     if (filter === "detenidos") return camiones.filter(c => c.estado === "DETENIDO" || c.estado === "DETENIDO_RECIENTE")
129 |     if (filter === "sinsenal") return camiones.filter(c => c.estado === "SIN_SEÑAL");
130 |     return camiones;
131 |   }, [camiones, filter]);
132 |
133 |   useEffect(() => {
134 |     if (!mapRef.current || mapInstance.current) return;
135 |
136 |     const loadLeaflet = async () => {
137 |       const L = (window as any).L;
138 |       if (!L) {
139 |         const link = document.createElement("link");
140 |         link.rel = "stylesheet";
141 |         link.href = "https://unpkg.com/leaflet@1.9.4/dist/leaflet.css";
142 |         document.head.appendChild(link);
143 |
144 |         const script = document.createElement("script");
145 |         script.src = "https://unpkg.com/leaflet@1.9.4/dist/leaflet.js";
146 |         script.onload = () => initMap();
147 |         document.body.appendChild(script);
148 |       } else {
149 |         initMap();
150 |       }
151 |     };
152 |
153 |     const initMap = () => {
154 |       const L = (window as any).L;
155 |       if (!L || !mapRef.current) return;
156 |       const map = L.map(mapRef.current).setView([-33.45, -70.65], 6);
157 |       L.tileLayer("https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png", {
158 |         attribution: "&copy; OpenStreetMap",
159 |       }).addTo(map);
160 |       mapInstance.current = map;
161 |     };
162 |
163 |     loadLeaflet();
164 |   }, []);
165 |
166 |   useEffect(() => {
167 |     const L = (window as any).L;
168 |     if (!L || !mapInstance.current || !camiones) return;
169 |
170 |     markersRef.current.forEach(m => m.remove());
171 |     markersRef.current = [];
172 |
173 |     const statusColors: Record<string, string> = {
174 |       EN_MOVIMIENTO: "#00c97a",
175 |       DETENIDO_RECIENTE: "#ffcc00",
176 |       DETENIDO: "#ff2244",
177 |       "SIN_SEÑAL": "#3a6080",
178 |     };
179 |
180 |     for (const c of filtered) {
181 |       if (!c.lat || !c.lng) continue;
182 |       const color = statusColors[c.estado] || "#3a6080";
183 |       const icon = L.divIcon({
184 |         html: `<div style="width:24px;height:24px;background:${color};border:2px solid #020508;border-radius:50%;display
185 |         className: "",
186 |         iconSize: [24, 24],
187 |         iconAnchor: [12, 12],
188 |       });
189 |       const marker = L.marker([c.lat, c.lng], { icon })
190 |         .addTo(mapInstance.current)
191 |         .bindPopup(`
192 |           <div style="font-family:monospace;font-size:12px;min-width:180px">
193 |             <b>${c.patente}</b> ${c.conductor || ""}<br/>
194 |             ${c.velocidad} km/h · ${c.estado.replace(/_/g, " ")}<br/>
195 |             ${c.timestamp ? new Date(c.timestamp).toLocaleString("es-CL") : "Sin senal"}
196 |           </div>
197 |         `);
198 |       marker.on("click", () => setSelectedCamion(c.camionId));
199 |       markersRef.current.push(marker);
200 |     }
201 |
202 |     if (bases) {
203 |       for (const b of bases) {
204 |         const bLat = parseFloat(b.lat);
205 |         const bLng = parseFloat(b.lng);
206 |         const circle = L.circle([bLat, bLng], {
207 |           radius: b.radioMetros,
208 |           color: "#00d4ff",
209 |           fillColor: "#00d4ff",
210 |           fillOpacity: 0.05,
211 |           weight: 1,
```

```
212 |         }).addTo(mapInstance.current);
213 |         markersRef.current.push(circle);
214 |         const baseMarker = L.marker([bLat, bLng], {
215 |           icon: L.divIcon({
216 |             html: `<div style="width:8px;height:8px;background:#00d4ff;border-radius:2px;border:1px solid #020508"></div
217 |             className: "",
218 |             iconSize: [8, 8],
219 |             iconAnchor: [4, 4],
220 |           }),
221 |         }).addTo(mapInstance.current).bindTooltip(b.nombre, { permanent: false });
222 |         markersRef.current.push(baseMarker);
223 |       }
224 |     }
225 |
226 |     fuelMarkersRef.current.forEach(m => m.remove());
227 |     fuelMarkersRef.current = [];
228 |     if (showFuelStations && fuelData?.estaciones) {
229 |       for (const est of fuelData.estaciones) {
230 |         if (!est.lat || !est.lng) continue;
231 |         const fuelIcon = L.divIcon({
232 |           html: `<div style="width:28px;height:28px;background:#ff6600;border:2px solid #020508;border-radius:4px;displa
233 |           className: "",
234 |           iconSize: [28, 28],
235 |           iconAnchor: [14, 14],
236 |         });
237 |         const fuelMarker = L.marker([est.lat, est.lng], { icon: fuelIcon })
238 |           .addTo(mapInstance.current)
239 |           .bindPopup(`
240 |             <div style="font-family:monospace;font-size:12px;min-width:200px">
241 |               <b style="color:#ff6600">${est.nombre}</b><br/>
242 |               <span style="color:#666">${est.ciudad}</span><br/>
243 |               <hr style="border-color:#eee;margin:4px 0"/>
244 |               <b>${est.cargas}</b> cargas -- <b>${est.litros.toLocaleString("es-CL")}</b> L<br/>
245 |               ${est.camiones} camiones -- Ult: ${est.ultimaCarga ? new Date(est.ultimaCarga).toLocaleDateString("es-CL")
246 |             </div>
247 |           `);
248 |         fuelMarker.on("click", () => setSelectedEstacion(est.nombre));
249 |         fuelMarkersRef.current.push(fuelMarker);
250 |       }
251 |     }
252 |   }, [filtered, bases, fuelData, showFuelStations]);
253 |
254 |   const counts = useMemo(() => {
255 |     if (!camiones) return { total: 0, mov: 0, det: 0, sin: 0 };
256 |     return {
257 |       total: camiones.length,
258 |       mov: camiones.filter(c => c.estado === "EN_MOVIMIENTO").length,
259 |       det: camiones.filter(c => c.estado === "DETENIDO" || c.estado === "DETENIDO_RECIENTE").length,
260 |       sin: camiones.filter(c => c.estado === "SIN_SEÑAL").length,
261 |     };
262 |   }, [camiones]);
263 |
264 |   const estacionDetail = useMemo(() => {
265 |     if (!selectedEstacion || !fuelData?.cargas) return null;
266 |     const cargas = fuelData.cargas.filter((c: any) => c.lugar === selectedEstacion);
267 |     if (!cargas.length) return null;
268 |     const byTruck: Record<string, { patente: string; conductor: string; cargas: any[]; totalLitros: number }> = {};
269 |     for (const c of cargas) {
270 |       const key = String(c.numVeh);
271 |       if (!byTruck[key]) byTruck[key] = { patente: key, conductor: c.conductor || "--", cargas: [], totalLitros: 0 };
272 |       byTruck[key].cargas.push(c);
273 |       byTruck[key].totalLitros += c.litros;
274 |     }
275 |     const trucks = Object.values(byTruck).sort((a, b) => b.totalLitros - a.totalLitros);
276 |     const estacion = fuelData.estaciones?.find((e: any) => e.nombre === selectedEstacion);
277 |     return { nombre: selectedEstacion, ciudad: estacion?.ciudad || "", totalCargas: cargas.length, totalLitros: cargas.r
278 |   }, [selectedEstacion, fuelData]);
279 |
280 |   const [expandedTruck, setExpandedTruck] = useState<string | null>(null);
281 |
282 |   return (
283 |     <div className="relative" style={{ height: "calc(100vh - 120px)" }} data-testid="geo-mapa">
284 |       <div ref={mapRef} className="absolute inset-0 z-0 rounded" style={{ border: "1px solid #0d2035" }} />
285 |
286 |       <div className="absolute top-3 left-3 z-10 flex gap-2">
287 |         {[
288 |           { id: "todos", label: `Todos (${counts.total})` },
289 |           { id: "movimiento", label: `Mov (${counts.mov})`, color: "#00c97a" },
290 |           { id: "detenidos", label: `Det (${counts.det})`, color: "#ffcc00" },
291 |           { id: "sinsenal", label: `Sin (${counts.sin})`, color: "#3a6080" },
292 |         ].map(f => (
293 |           <button key={f.id} onClick={() => setFilter(f.id)}
294 |             data-testid={`geo-filter-${f.id}`}
295 |             className="font-exo text-[10px] font-bold px-3 py-1.5 rounded cursor-pointer transition-all"
296 |             style={{
297 |               background: filter === f.id ? (f.color || "#00d4ff") + "30" : "#020508cc",
298 |               border: `1px solid ${filter === f.id ? (f.color || "#00d4ff") : "#0d2035"}`,
299 |               color: filter === f.id ? (f.color || "#00d4ff") : "#3a6080",
300 |             }}>
301 |             {f.label}
302 |           </button>
303 |         ))}
304 |         <button onClick={() => ingestMutation.mutate()} disabled={ingestMutation.isPending}
305 |           data-testid="btn-ingest-volvo"
306 |           className="font-exo text-[10px] font-bold px-3 py-1.5 rounded cursor-pointer"
307 |           style={{ background: "#00d4ff20", border: "1px solid #00d4ff", color: "#00d4ff" }}>
308 |           <RefreshCw className={`w-3 h-3 inline mr-1 ${ingestMutation.isPending ? "animate-spin" : ""}`} />
309 |           Ingest GPS
310 |         </button>
311 |         <button onClick={() => setShowFuelStations(!showFuelStations)}
312 |           data-testid="btn-toggle-fuel"
313 |           className="font-exo text-[10px] font-bold px-3 py-1.5 rounded cursor-pointer"
314 |           style={{
315 |             background: showFuelStations ? "#ff660030" : "#020508cc",
316 |             border: `1px solid ${showFuelStations ? "#ff6600" : "#0d2035"}`,
317 |             color: showFuelStations ? "#ff6600" : "#3a6080",
318 |           }}>
```

```
319 |           <Fuel className="w-3 h-3 inline mr-1" />
320 |           Estaciones ({fuelData?.estaciones?.filter((e: any) => e.lat)?.length || 0})
321 |         </button>
322 |       </div>
323 |
324 |       {panelOpen && (
325 |         <div className="absolute top-3 right-3 bottom-3 w-72 z-10 overflow-y-auto rounded" style={{
326 |           background: "rgba(2,5,8,0.92)",
327 |           border: "1px solid #0d2035",
328 |           backdropFilter: "blur(8px)",
329 |         }}>
330 |           <div className="p-3 sticky top-0" style={{ background: "rgba(2,5,8,0.95)", borderBottom: "1px solid #0d2035" }
331 |             <div className="font-space text-[11px] font-bold tracking-wider" style={{ color: "#00d4ff" }}>
332 |               CAMIONES ACTIVOS
333 |             </div>
334 |           </div>
335 |           <div className="p-2 space-y-1">
336 |             {filtered?.map(c => (
337 |               <div key={c.camionId}
338 |                 onClick={() => {
339 |                   setSelectedCamion(c.camionId);
340 |                   if (c.lat && c.lng && mapInstance.current) {
341 |                     mapInstance.current.flyTo([c.lat, c.lng], 12, { duration: 0.5 });
342 |                   }
343 |                 }}
344 |                 className="flex items-center gap-2 p-2 rounded cursor-pointer transition-all"
345 |                 style={{
346 |                   background: selectedCamion === c.camionId ? "#00d4ff10" : "transparent",
347 |                   border: `1px solid ${selectedCamion === c.camionId ? "#00d4ff40" : "transparent"}`,
348 |                 }}
349 |                 data-testid={`geo-camion-${c.patente}`}>
350 |                 <CamionStatusDot estado={c.estado} />
351 |                 <div className="flex-1 min-w-0">
352 |                   <div className="font-space text-[11px] font-bold" style={{ color: "#c8e8ff" }}>{c.patente}</div>
353 |                   <div className="font-exo text-[9px] truncate" style={{ color: "#3a6080" }}>
354 |                     {c.velocidad > 0 ? `${c.velocidad} km/h` : c.estado === "SIN_SEÑAL" ? `Sin senal ${c.ageMinutes}min`
355 |                     {c.conductor ? ` · ${c.conductor}` : ""}
356 |                   </div>
357 |                 </div>
358 |               </div>
359 |             ))}
360 |           </div>
361 |         </div>
362 |       )}
363 |
364 |       <button onClick={() => setPanelOpen(!panelOpen)}
365 |         className="absolute top-3 z-10 font-exo text-[10px] px-2 py-1 rounded cursor-pointer"
366 |         style={{
367 |           right: panelOpen ? "19rem" : "0.75rem",
368 |           background: "#020508cc",
369 |           border: "1px solid #0d2035",
370 |           color: "#3a6080",
371 |         }}>
372 |         {panelOpen ? ">" : "<"}
373 |       </button>
374 |
375 |       {estacionDetail && (
376 |         <div className="absolute bottom-3 left-3 z-20 overflow-y-auto rounded" style={{
377 |           background: "rgba(2,5,8,0.95)",
378 |           border: "1px solid #ff6600",
379 |           backdropFilter: "blur(8px)",
380 |           maxHeight: "60%",
381 |           width: "380px",
382 |         }} data-testid="fuel-station-detail">
383 |           <div className="p-3 flex items-center justify-between sticky top-0" style={{ background: "rgba(2,5,8,0.98)", b
384 |             <div>
385 |               <div className="font-space text-[12px] font-bold tracking-wider" style={{ color: "#ff6600" }}>
386 |                 {estacionDetail.nombre}
387 |               </div>
388 |               <div className="font-exo text-[10px]" style={{ color: "#3a6080" }}>
389 |                 {estacionDetail.ciudad} -- {estacionDetail.totalCargas} cargas -- {Math.round(estacionDetail.totalLitros
390 |               </div>
391 |             </div>
392 |             <button onClick={() => setSelectedEstacion(null)} className="cursor-pointer" style={{ color: "#3a6080" }} da
393 |               <X className="w-4 h-4" />
394 |             </button>
395 |           </div>
396 |           <div className="p-2 space-y-1">
397 |             {estacionDetail.trucks.map(truck => (
398 |               <div key={truck.patente}>
399 |                 <button
400 |                   onClick={() => setExpandedTruck(expandedTruck === truck.patente ? null : truck.patente)}
401 |                   className="w-full flex items-center gap-2 p-2 rounded cursor-pointer transition-all"
402 |                   style={{
403 |                     background: expandedTruck === truck.patente ? "#ff660015" : "transparent",
404 |                     border: `1px solid ${expandedTruck === truck.patente ? "#ff660040" : "#0d203580"}`,
405 |                   }}
406 |                   data-testid={`btn-fuel-truck-${truck.patente}`}
407 |                 >
408 |                   <Truck className="w-3.5 h-3.5 flex-shrink-0" style={{ color: "#ff6600" }} />
409 |                   <div className="flex-1 text-left min-w-0">
410 |                     <span className="font-space text-[11px] font-bold" style={{ color: "#c8e8ff" }}>{truck.patente}</spa
411 |                     <span className="font-exo text-[10px] ml-2" style={{ color: "#3a6080" }}>{truck.conductor}</span>
412 |                   </div>
413 |                   <div className="text-right flex-shrink-0">
414 |                     <span className="font-space text-[11px] font-bold" style={{ color: "#ff6600" }}>{truck.cargas.length
415 |                     <span className="font-exo text-[9px] ml-1" style={{ color: "#3a6080" }}>cargas</span>
416 |                     <span className="font-space text-[10px] ml-2" style={{ color: "#c8e8ff" }}>{Math.round(truck.totalLi
417 |                   </div>
418 |                   {expandedTruck === truck.patente ? <ChevronUp className="w-3 h-3 flex-shrink-0" style={{ color: "#3a60
419 |                 </button>
420 |                 {expandedTruck === truck.patente && (
421 |                   <div className="ml-6 mt-1 mb-2 space-y-1" data-testid={`fuel-detail-${truck.patente}`}>
422 |                     {(() => {
423 |                       const cm = cruceMensual?.camiones?.find((c: any) => c.patente === truck.patente);
424 |                       if (!cm) return null;
425 |                       const diffColor = Math.abs(cm.diferencia) <= 50 ? "#00c97a" : cm.diferencia > 0 ? "#ffcc00" : "#ff
```

```
426 |                       return (
427 |                         <div className="p-2.5 rounded mb-2" style={{ background: "#00d4ff08", border: "1px solid #00d4ff
428 |                           <div className="font-space text-[10px] font-bold tracking-wider mb-2" style={{ color: "#00d4ff
429 |                           <div className="grid grid-cols-2 gap-x-4 gap-y-1.5">
430 |                             <div className="flex justify-between">
431 |                               <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>Cargas Sigetra</span>
432 |                               <span className="font-space text-[10px] font-bold" style={{ color: "#ff6600" }}>{cm.litros
433 |                             </div>
434 |                             <div className="flex justify-between">
435 |                               <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>Consumo ECU Volvo</span
436 |                               <span className="font-space text-[10px] font-bold" style={{ color: "#00d4ff" }}>{cm.litros
437 |                             </div>
438 |                             <div className="flex justify-between">
439 |                               <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>N cargas</span>
440 |                               <span className="font-space text-[10px]" style={{ color: "#c8e8ff" }}>{cm.cargasSigetra}</
441 |                             </div>
442 |                             <div className="flex justify-between">
443 |                               <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>N viajes ECU</span>
444 |                               <span className="font-space text-[10px]" style={{ color: "#c8e8ff" }}>{cm.viajesEcu}</span
445 |                             </div>
446 |                             <div className="flex justify-between">
447 |                               <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>Km odometro</span>
448 |                               <span className="font-space text-[10px] font-bold" style={{ color: "#c8e8ff" }}>{cm.kmOdom
449 |                             </div>
450 |                             <div className="flex justify-between">
451 |                               <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>Km viajes ECU</span>
452 |                               <span className="font-space text-[10px]" style={{ color: "#c8e8ff" }}>{cm.kmEcu > 0 ? cm.k
453 |                             </div>
454 |                             {cm.odoInicio > 0 && (
455 |                               <div className="col-span-2 flex justify-between">
456 |                                 <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>Odo inicio / fin</spa
457 |                                 <span className="font-space text-[10px]" style={{ color: "#3a6080" }}>{cm.odoInicio.toLo
458 |                               </div>
459 |                             )}
460 |                             <div className="flex justify-between">
461 |                               <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>Rend Sigetra</span>
462 |                               <span className="font-space text-[10px] font-bold" style={{ color: cm.rendimientoSigetra >
463 |                                 {cm.rendimientoSigetra > 0 ? `${cm.rendimientoSigetra.toFixed(2)} km/L` : "--"}
464 |                               </span>
465 |                             </div>
466 |                             <div className="flex justify-between">
467 |                               <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>Rend ECU Volvo</span>
468 |                               <span className="font-space text-[10px] font-bold" style={{ color: cm.rendimientoEcu >= 3.
469 |                                 {cm.rendimientoEcu > 0 ? `${cm.rendimientoEcu.toFixed(2)} km/L` : "--"}
470 |                               </span>
471 |                             </div>
472 |                           </div>
473 |                           <div className="mt-2 pt-2 flex justify-between items-center" style={{ borderTop: "1px solid #0
474 |                             <span className="font-exo text-[9px] font-bold" style={{ color: "#3a6080" }}>DIFERENCIA (Sig
475 |                             <div className="text-right">
476 |                               <span className="font-space text-[11px] font-bold" style={{ color: diffColor }}>
477 |                                 {cm.diferencia > 0 ? "+" : ""}{cm.diferencia.toLocaleString("es-CL")} L
478 |                               </span>
479 |                               {cm.diferenciaPct !== 0 && (
480 |                                 <span className="font-exo text-[9px] ml-1" style={{ color: diffColor }}>
481 |                                   ({cm.diferenciaPct > 0 ? "+" : ""}{cm.diferenciaPct}%)
482 |                                 </span>
483 |                               )}
484 |                             </div>
485 |                           </div>
486 |                         </div>
487 |                       );
488 |                     })()}
489 |                     {truck.cargas.sort((a: any, b: any) => new Date(b.fecha).getTime() - new Date(a.fecha).getTime()).ma
490 |                       <div key={i} className="p-2 rounded" style={{ background: "#0d203530", border: "1px solid #0d2035"
491 |                         <div className="flex justify-between items-center">
492 |                           <span className="font-exo text-[10px] font-bold" style={{ color: "#c8e8ff" }}>
493 |                             {new Date(c.fecha).toLocaleDateString("es-CL")} {new Date(c.fecha).toLocaleTimeString("es-CL
494 |                           </span>
495 |                           <span className="font-space text-[11px] font-bold" style={{ color: "#ff6600" }}>{c.litros.toFi
496 |                         </div>
497 |                         <div className="flex gap-3 mt-1">
498 |                           <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>Odo: {c.odometro?.toLocaleS
499 |                           <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>Km rec: {c.kmRecorrido?.toL
500 |                           <span className="font-exo text-[9px]" style={{ color: c.rendimiento >= 3.5 ? "#00c97a" : c.ren
501 |                             Rend: {c.rendimiento > 0 ? `${c.rendimiento.toFixed(2)} km/L` : "--"}
502 |                           </span>
503 |                         </div>
504 |                       </div>
505 |                     ))}
506 |                   </div>
507 |                 )}
508 |               </div>
509 |             ))}
510 |           </div>
511 |         </div>
512 |       )}
513 |     </div>
514 |   );
515 | }
516 |
517 | interface HistorialCamion {
518 |   camion: {
519 |     id: number;
520 |     patente: string;
521 |     modelo: string;
522 |     conductor: string | null;
523 |     contrato: string;
524 |     tiene_gps: boolean;
525 |     total_puntos_gps: number;
526 |     primer_punto: string | null;
527 |     ultimo_punto: string | null;
528 |   };
529 |   viajes: {
530 |     id: number;
531 |     fecha: string | null;
532 |     origen_nombre: string | null;
```

```
533 |     destino_nombre: string | null;
534 |     km_gps: number;
535 |     km_odometro: number | null;
536 |     duracion_horas: number;
537 |     duracion_minutos: number;
538 |     velocidad_maxima: number;
539 |     velocidad_promedio: number;
540 |     tiempo_detenido_min: number;
541 |     paradas: any[];
542 |     validacion_estado: string;
543 |     sigetra_match: { encontrado: boolean; litros?: number; km_delta_pct?: number };
544 |     origen_lat: number;
545 |     origen_lng: number;
546 |     destino_lat: number;
547 |     destino_lng: number;
548 |   }[];
549 |   sigetra: { total_cargas: number; total_litros: number; proveedores: string[] } | null;
550 |   resumen: {
551 |     total_viajes: number;
552 |     total_km_gps: number;
553 |     km_promedio_por_viaje: number;
554 |     destino_mas_frecuente: string | null;
555 |     viajes_validados: number;
556 |     viajes_anomalia: number;
557 |   };
558 |   mensaje: string | null;
559 | }
560 |
561 | interface ResumenFlota {
562 |   kpis: { total_camiones: number; con_gps: number; sin_gps: number; total_viajes: number; total_km: number; ocultos_sin_
563 |   camiones: {
564 |     patente: string; camionId: number; conductor: string | null; modelo: string;
565 |     tiene_gps: boolean; puntos_gps: number; total_viajes: number; total_km: number;
566 |     destino_top: string | null; validados: number; anomalias: number;
567 |     sigetra_cargas: number; sigetra_litros: number;
568 |   }[];
569 |   camiones_sin_gps: { patente: string; camionId: number; modelo: string; conductor: string | null }[];
570 | }
571 |
572 | function ViajeDetalleModal({ viaje, onClose }: { viaje: HistorialCamion["viajes"][0]; onClose: () => void }) {
573 |   const mapRef = useRef<HTMLDivElement>(null);
574 |   const mapInstanceRef = useRef<any>(null);
575 |   const [leafletReady, setLeafletReady] = useState(false);
576 |
577 |   useEffect(() => {
578 |     if ((window as any).L) { setLeafletReady(true); return; }
579 |     const link = document.createElement("link");
580 |     link.rel = "stylesheet";
581 |     link.href = "https://unpkg.com/leaflet@1.9.4/dist/leaflet.css";
582 |     document.head.appendChild(link);
583 |     const script = document.createElement("script");
584 |     script.src = "https://unpkg.com/leaflet@1.9.4/dist/leaflet.js";
585 |     script.onload = () => setLeafletReady(true);
586 |     document.head.appendChild(script);
587 |   }, []);
588 |
589 |   useEffect(() => {
590 |     if (!mapRef.current || mapInstanceRef.current || !leafletReady) return;
591 |     const L = (window as any).L;
592 |     if (!L) return;
593 |
594 |     const map = L.map(mapRef.current, {
595 |       center: [(viaje.origen_lat + viaje.destino_lat) / 2, (viaje.origen_lng + viaje.destino_lng) / 2],
596 |       zoom: 6, zoomControl: false,
597 |       attributionControl: false,
598 |     });
599 |     L.tileLayer("https://{s}.basemaps.cartocdn.com/dark_all/{z}/{x}/{y}{r}.png").addTo(map);
600 |
601 |     L.circleMarker([viaje.origen_lat, viaje.origen_lng], { radius: 8, color: "#00c97a", fillColor: "#00c97a", fillOpacit
602 |       .bindTooltip(viaje.origen_nombre || "Origen", { permanent: false }).addTo(map);
603 |     L.circleMarker([viaje.destino_lat, viaje.destino_lng], { radius: 8, color: "#ff2244", fillColor: "#ff2244", fillOpac
604 |       .bindTooltip(viaje.destino_nombre || "Destino", { permanent: false }).addTo(map);
605 |
606 |     if (viaje.paradas && Array.isArray(viaje.paradas)) {
607 |       viaje.paradas.forEach((p: any) => {
608 |         if (p.lat && p.lng) {
609 |           L.circleMarker([p.lat, p.lng], { radius: 6, color: "#ffcc00", fillColor: "#ffcc00", fillOpacity: 0.8, weight:
610 |             .bindTooltip(`${p.nombre || "Parada"} (${p.minutos}min)`, { permanent: false }).addTo(map);
611 |         }
612 |       });
613 |     }
614 |
615 |     L.polyline([[viaje.origen_lat, viaje.origen_lng], [viaje.destino_lat, viaje.destino_lng]], {
616 |       color: "#00d4ff", weight: 2, opacity: 0.5, dashArray: "8,8",
617 |     }).addTo(map);
618 |
619 |     const bounds = L.latLngBounds([[viaje.origen_lat, viaje.origen_lng], [viaje.destino_lat, viaje.destino_lng]]);
620 |     map.fitBounds(bounds, { padding: [30, 30] });
621 |
622 |     mapInstanceRef.current = map;
623 |     return () => { map.remove(); mapInstanceRef.current = null; };
624 |   }, [viaje, leafletReady]);
625 |
626 |   const validacionItems = [];
627 |   if (viaje.sigetra_match.encontrado) {
628 |     const deltaPct = viaje.sigetra_match.km_delta_pct || 0;
629 |     validacionItems.push({ label: `Km coinciden (${deltaPct.toFixed(1)}%)`, ok: deltaPct < 5 });
630 |     validacionItems.push({ label: "Sigetra encontrado", ok: true });
631 |   }
632 |   if (viaje.km_gps > 30) validacionItems.push({ label: "Viaje real (>30km)", ok: true });
633 |
634 |   return (
635 |     <div className="fixed inset-0 z-50 flex items-center justify-center" style={{ background: "rgba(2,5,8,0.85)" }}
636 |       onClick={onClose} data-testid="modal-viaje-detalle">
637 |       <div className="w-[95vw] max-w-[1200px] max-h-[85vh] overflow-auto rounded-lg"
638 |         style={{ background: "#091018", border: "1px solid #0d2035" }}
639 |         onClick={e => e.stopPropagation()}>
```

```
640 |         <div className="flex items-center justify-between px-5 py-3" style={{ borderBottom: "1px solid #0d2035" }}>
641 |           <div className="font-space text-[14px] font-bold" style={{ color: "#00d4ff" }}>
642 |             {viaje.fecha} -- {viaje.origen_nombre || "?"} !' {viaje.destino_nombre || "?"}
643 |           </div>
644 |           <button onClick={onClose} className="cursor-pointer" data-testid="btn-cerrar-modal">
645 |             <X className="w-4 h-4" style={{ color: "#3a6080" }} />
646 |           </button>
647 |         </div>
648 |
649 |         <div className="grid grid-cols-3 gap-0" style={{ minHeight: "400px" }}>
650 |           <div ref={mapRef} className="col-span-1" style={{ minHeight: "400px", background: "#020508" }} data-testid="mo
651 |
652 |           <div className="col-span-1 px-4 py-4 overflow-auto" style={{ borderLeft: "1px solid #0d2035", borderRight: "1p
653 |             <div className="font-exo text-[9px] tracking-[0.15em] uppercase mb-3" style={{ color: "#3a6080" }}>TIMELINE<
654 |             <div className="space-y-3">
655 |               <div className="flex gap-2">
656 |                 <div className="w-1.5 h-1.5 rounded-full mt-1.5 flex-shrink-0" style={{ background: "#00c97a" }} />
657 |                 <div>
658 |                   <div className="font-exo text-[10px]" style={{ color: "#3a6080" }}>SALIDA</div>
659 |                   <div className="font-exo text-[11px]" style={{ color: "#c8e8ff" }}>{viaje.origen_nombre || "Origen"}</
660 |                 </div>
661 |               </div>
662 |               {viaje.paradas && Array.isArray(viaje.paradas) && viaje.paradas.map((p: any, i: number) => (
663 |                 <div key={i} className="flex gap-2">
664 |                   <div className="w-1.5 h-1.5 rounded-full mt-1.5 flex-shrink-0" style={{ background: "#ffcc00" }} />
665 |                   <div>
666 |                     <div className="font-exo text-[10px]" style={{ color: "#ffcc00" }}>PARADA {p.minutos}min</div>
667 |                     <div className="font-exo text-[11px]" style={{ color: "#c8e8ff" }}>{p.nombre || "Parada"}</div>
668 |                   </div>
669 |                 </div>
670 |               ))}
671 |               <div className="flex gap-2">
672 |                 <div className="w-1.5 h-1.5 rounded-full mt-1.5 flex-shrink-0" style={{ background: "#ff2244" }} />
673 |                 <div>
674 |                   <div className="font-exo text-[10px]" style={{ color: "#3a6080" }}>LLEGADA</div>
675 |                   <div className="font-exo text-[11px]" style={{ color: "#c8e8ff" }}>{viaje.destino_nombre || "Destino"}
676 |                 </div>
677 |               </div>
678 |             </div>
679 |           </div>
680 |
681 |           <div className="col-span-1 px-4 py-4 overflow-auto" style={{ maxHeight: "500px" }}>
682 |             <div className="font-exo text-[9px] tracking-[0.15em] uppercase mb-3" style={{ color: "#3a6080" }}>METRICAS<
683 |             <div className="space-y-2">
684 |               {[
685 |                 { label: "KM GPS", value: `${viaje.km_gps.toFixed(1)}`, color: "#00d4ff" },
686 |                 ...(viaje.km_odometro ? [{ label: "KM ODOMETRO", value: `${viaje.km_odometro.toFixed(1)}`, color: "#c8e8
687 |                 { label: "DURACION", value: `${viaje.duracion_horas}h`, color: "#c8e8ff" },
688 |                 { label: "VEL. MAXIMA", value: `${viaje.velocidad_maxima} km/h`, color: "#ffcc00" },
689 |                 { label: "VEL. PROMEDIO", value: `${viaje.velocidad_promedio} km/h`, color: "#c8e8ff" },
690 |                 { label: "TIEMPO DETENIDO", value: `${Math.floor(viaje.tiempo_detenido_min / 60)}h ${viaje.tiempo_deteni
691 |               ].map(m => (
692 |                 <div key={m.label} className="flex justify-between items-center py-1" style={{ borderBottom: "1px solid
693 |                   <span className="font-exo text-[9px] tracking-[0.1em]" style={{ color: "#3a6080" }}>{m.label}</span>
694 |                   <span className="font-space text-[12px] font-bold" style={{ color: m.color }}>{m.value}</span>
695 |                 </div>
696 |               ))}
697 |             </div>
698 |
699 |             {viaje.sigetra_match.encontrado && (
700 |               <div className="mt-4">
701 |                 <div className="font-exo text-[9px] tracking-[0.15em] uppercase mb-2" style={{ color: "#3a6080" }}>SIGET
702 |                 <div className="flex justify-between py-1" style={{ borderBottom: "1px solid #0d203530" }}>
703 |                   <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>LITROS</span>
704 |                   <span className="font-space text-[12px] font-bold" style={{ color: "#00c97a" }}>{viaje.sigetra_match.l
705 |                 </div>
706 |                 {viaje.km_gps > 0 && viaje.sigetra_match.litros && (
707 |                   <div className="flex justify-between py-1">
708 |                     <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>RENDIMIENTO</span>
709 |                     <span className="font-space text-[12px] font-bold" style={{ color: "#c8e8ff" }}>
710 |                       {(viaje.km_gps / viaje.sigetra_match.litros).toFixed(2)} km/L
711 |                     </span>
712 |                   </div>
713 |                 )}
714 |               </div>
715 |             )}
716 |
717 |             {validacionItems.length > 0 && (
718 |               <div className="mt-4">
719 |                 <div className="font-exo text-[9px] tracking-[0.15em] uppercase mb-2" style={{ color: "#3a6080" }}>VALID
720 |                 {validacionItems.map((item, i) => (
721 |                   <div key={i} className="flex items-center gap-2 py-0.5">
722 |                     {item.ok ?
723 |                       <Check className="w-3 h-3" style={{ color: "#00c97a" }} /> :
724 |                       <X className="w-3 h-3" style={{ color: "#ff2244" }} />
725 |                     }
726 |                     <span className="font-exo text-[10px]" style={{ color: item.ok ? "#00c97a" : "#ff2244" }}>{item.labe
727 |                   </div>
728 |                 ))}
729 |                 <div className="mt-2">
730 |                   <EstadoBadge estado={viaje.validacion_estado} />
731 |                 </div>
732 |               </div>
733 |             )}
734 |           </div>
735 |         </div>
736 |       </div>
737 |     </div>
738 |   );
739 | }
740 |
741 | function PerfilCamion({ historial, onBack, onVerDetalle }: {
742 |   historial: HistorialCamion;
743 |   onBack: () => void;
744 |   onVerDetalle: (viaje: HistorialCamion["viajes"][0]) => void;
745 | }) {
746 |   const { camion, viajes, sigetra, resumen, mensaje } = historial;
```

```
747 |
748 |   return (
749 |     <div data-testid="perfil-camion">
750 |       <button onClick={onBack} className="flex items-center gap-1 mb-3 cursor-pointer font-exo text-[11px]" style={{ col
751 |         data-testid="btn-volver-flota">
752 |         <ArrowLeft className="w-3 h-3" /> Volver a flota
753 |       </button>
754 |
755 |       <div className="dash-card px-5 py-4 mb-4">
756 |         <div className="flex items-center justify-between mb-3">
757 |           <div className="flex items-center gap-3">
758 |             <Truck className="w-5 h-5" style={{ color: "#00d4ff" }} />
759 |             <span className="font-space text-[16px] font-bold" style={{ color: "#00d4ff" }}>{camion.patente}</span>
760 |             <span className="font-exo text-[10px] font-bold px-2 py-0.5 rounded" style={{
761 |               background: "#00c97a15", border: "1px solid #00c97a", color: "#00c97a",
762 |             }}>CENCOSUD</span>
763 |           </div>
764 |           <div className="flex items-center gap-2">
765 |             {camion.tiene_gps ? (
766 |               <span className="font-exo text-[10px]" style={{ color: "#00c97a" }}>
767 |                 <Check className="w-3 h-3 inline mr-1" />{camion.total_puntos_gps} puntos GPS
768 |               </span>
769 |             ) : (
770 |               <span className="font-exo text-[10px]" style={{ color: "#ff2244" }}>
771 |                 <X className="w-3 h-3 inline mr-1" />Sin GPS
772 |               </span>
773 |             )}
774 |           </div>
775 |         </div>
776 |         {camion.conductor && (
777 |           <div className="font-exo text-[11px]" style={{ color: "#3a6080" }}>Conductor: {camion.conductor}</div>
778 |         )}
779 |         {camion.tiene_gps && camion.primer_punto && (
780 |           <div className="font-exo text-[10px] mt-1" style={{ color: "#3a6080" }}>
781 |             GPS desde {new Date(camion.primer_punto).toLocaleDateString("es-CL")}
782 |           </div>
783 |         )}
784 |       </div>
785 |
786 |       {!camion.tiene_gps ? (
787 |         <div className="dash-card px-5 py-6 text-center">
788 |           <AlertTriangle className="w-8 h-8 mx-auto mb-3" style={{ color: "#ff2244" }} />
789 |           <div className="font-space text-[13px] font-bold" style={{ color: "#ff2244" }}>
790 |             Sin datos GPS desde 01-03-2026
791 |           </div>
792 |           {sigetra && sigetra.total_cargas > 0 ? (
793 |             <div className="mt-4 dash-card px-4 py-3 text-left" style={{ background: "#0a1520" }}>
794 |               <div className="font-exo text-[10px] font-bold tracking-[0.15em] mb-2" style={{ color: "#3a6080" }}>
795 |                 DATOS SIGETRA DISPONIBLES
796 |               </div>
797 |               <div className="font-exo text-[11px]" style={{ color: "#c8e8ff" }}>
798 |                 {sigetra.total_cargas} cargas -- {sigetra.total_litros}L
799 |               </div>
800 |               {sigetra.proveedores.length > 0 && (
801 |                 <div className="font-exo text-[10px] mt-1" style={{ color: "#3a6080" }}>
802 |                   Surtidores: {sigetra.proveedores.join(", ")}
803 |                 </div>
804 |               )}
805 |               <div className="font-exo text-[10px] mt-2" style={{ color: "#3a6080" }}>
806 |                 Sin GPS no es posible reconstruir los viajes automaticamente.
807 |               </div>
808 |             </div>
809 |           ) : (
810 |             <div className="font-exo text-[11px] mt-2" style={{ color: "#3a6080" }}>
811 |               {mensaje || "No hay datos disponibles para este camion en el periodo."}
812 |             </div>
813 |           )}
814 |         </div>
815 |       ) : (
816 |         <>
817 |           <div className="dash-card px-5 py-3 mb-4">
818 |             <div className="font-exo text-[9px] tracking-[0.15em] uppercase mb-2" style={{ color: "#3a6080" }}>
819 |               RESUMEN DEL PERIODO
820 |             </div>
821 |             <div className="grid grid-cols-4 gap-4">
822 |               {[
823 |                 { label: "VIAJES", value: resumen.total_viajes, color: "#00d4ff" },
824 |                 { label: "KM TOTAL", value: resumen.total_km_gps.toLocaleString(), color: "#00c97a" },
825 |                 { label: "DESTINO TOP", value: resumen.destino_mas_frecuente || "--", color: "#c8e8ff", small: true },
826 |                 { label: "ANOMALIAS", value: resumen.viajes_anomalia, color: resumen.viajes_anomalia > 0 ? "#ff2244" : "
827 |               ].map(k => (
828 |                 <div key={k.label}>
829 |                   <div className="font-exo text-[8px] tracking-[0.1em]" style={{ color: "#3a6080" }}>{k.label}</div>
830 |                   <div className={`font-space font-bold ${k.small ? "text-[11px]" : "text-[18px]"}`} style={{ color: k.c
831 |                 </div>
832 |               ))}
833 |             </div>
834 |           </div>
835 |
836 |           {viajes.length === 0 ? (
837 |             <div className="dash-card px-5 py-6 text-center">
838 |               <Route className="w-6 h-6 mx-auto mb-2" style={{ color: "#3a6080" }} />
839 |               <div className="font-exo text-[11px]" style={{ color: "#3a6080" }}>
840 |                 {"GPS disponible pero sin viajes que cumplan criterios minimos (>30km, >1h)"}
841 |               </div>
842 |             </div>
843 |           ) : (
844 |             <div className="space-y-2" data-testid="lista-viajes">
845 |               {viajes.map(v => (
846 |                 <div key={v.id} className="dash-card px-4 py-3" data-testid={`viaje-card-${v.id}`}>
847 |                   <div className="flex items-center justify-between">
848 |                     <div className="flex items-center gap-3">
849 |                       <div className="font-space text-[12px] font-bold" style={{ color: "#3a6080" }}>
850 |                         {v.fecha ? new Date(v.fecha + "T12:00:00").toLocaleDateString("es-CL", { day: "2-digit", month:
851 |                       </div>
852 |                       <div className="font-exo text-[12px]" style={{ color: "#c8e8ff" }}>
853 |                         {v.origen_nombre || "?"} !’ {v.destino_nombre || "?"}
```

```
854 |                       </div>
855 |                     </div>
856 |                     <EstadoBadge estado={v.validacion_estado} />
857 |                   </div>
858 |                   <div className="flex items-center gap-4 mt-2">
859 |                     <span className="font-space text-[11px]" style={{ color: "#00d4ff" }}>{v.km_gps.toFixed(0)} km</span
860 |                     <span className="font-exo text-[10px]" style={{ color: "#3a6080" }}>{v.duracion_horas}h</span>
861 |                     {v.velocidad_maxima > 0 && (
862 |                       <span className="font-exo text-[10px]" style={{ color: v.velocidad_maxima > 90 ? "#ffcc00" : "#3a6
863 |                         max {v.velocidad_maxima} km/h
864 |                       </span>
865 |                     )}
866 |                     {v.sigetra_match.encontrado && (
867 |                       <span className="font-exo text-[10px]" style={{ color: "#00c97a" }}>
868 |                         <Fuel className="w-3 h-3 inline mr-0.5" />{v.sigetra_match.litros}L
869 |                       </span>
870 |                     )}
871 |                     <div className="flex-1" />
872 |                     <button onClick={() => onVerDetalle(v)}
873 |                       className="font-exo text-[9px] font-bold px-2 py-1 rounded cursor-pointer"
874 |                       style={{ background: "#00d4ff15", border: "1px solid #00d4ff40", color: "#00d4ff" }}
875 |                       data-testid={`btn-ver-detalle-${v.id}`}>
876 |                       <Eye className="w-3 h-3 inline mr-1" />Ver detalle
877 |                     </button>
878 |                   </div>
879 |                 </div>
880 |               ))}
881 |             </div>
882 |           )}
883 |         </>
884 |       )}
885 |     </div>
886 |   );
887 | }
888 |
889 | function ViajesCerrados() {
890 |   const [busqueda, setBusqueda] = useState("");
891 |   const [patenteSeleccionada, setPatenteSeleccionada] = useState<string | null>(null);
892 |   const [expandedViaje, setExpandedViaje] = useState<number | null>(null);
893 |
894 |   const { data: viajesData, isLoading } = useQuery<any>({
895 |     queryKey: ["/api/geo/viajes-tms", patenteSeleccionada],
896 |     queryFn: async () => {
897 |       const url = patenteSeleccionada
898 |         ? `/api/geo/viajes-tms?patente=${patenteSeleccionada}`
899 |         : "/api/geo/viajes-tms";
900 |       const r = await fetch(url);
901 |       return r.json();
902 |     },
903 |   });
904 |
905 |   const patentesFiltradas = useMemo(() => {
906 |     if (!viajesData?.patentes) return [];
907 |     if (!busqueda) return viajesData.patentes;
908 |     return viajesData.patentes.filter((p: string) => p.includes(busqueda));
909 |   }, [viajesData?.patentes, busqueda]);
910 |
911 |   const viajes = viajesData?.viajes || [];
912 |   const totalKm = viajes.reduce((s: number, v: any) => s + (v.kmRecorridos || 0), 0);
913 |   const totalLitros = viajes.reduce((s: number, v: any) => s + (v.litrosEcu || 0), 0);
914 |
915 |   const formatFecha = (f: string | null) => {
916 |     if (!f) return "--";
917 |     return new Date(f).toLocaleString("es-CL", { day: "2-digit", month: "2-digit", hour: "2-digit", minute: "2-digit" })
918 |   };
919 |
920 |   const formatDuracion = (salida: string | null, llegada: string | null) => {
921 |     if (!salida || !llegada) return "--";
922 |     const ms = new Date(llegada).getTime() - new Date(salida).getTime();
923 |     const h = Math.floor(ms / 3600000);
924 |     const m = Math.floor((ms % 3600000) / 60000);
925 |     return `${h}h ${m}m`;
926 |   };
927 |
928 |   return (
929 |     <div data-testid="geo-viajes-cerrados">
930 |       <div className="flex items-center justify-between mb-4">
931 |         <div>
932 |           <div className="font-rajdhani text-sm font-bold" style={{ color: "#c8e8ff" }}>
933 |             Viajes Cerrados
934 |           </div>
935 |           <div className="font-exo text-[10px]" style={{ color: "#3a6080" }}>
936 |             {viajes.length} viajes -- {Math.round(totalKm).toLocaleString("es-CL")} km -- {Math.round(totalLitros).toLoc
937 |           </div>
938 |         </div>
939 |       </div>
940 |
941 |       <div className="flex gap-3 mb-4">
942 |         <div className="flex-1 relative">
943 |           <Search className="absolute left-3 top-1/2 -translate-y-1/2 w-3.5 h-3.5" style={{ color: "#3a6080" }} />
944 |           <input
945 |             type="text"
946 |             placeholder="Buscar por patente..."
947 |             value={busqueda}
948 |             onChange={(e) => setBusqueda(e.target.value)}
949 |             data-testid="input-buscar-patente"
950 |             className="w-full font-space text-[11px] pl-8 pr-3 py-2 rounded outline-none"
951 |             style={{ background: "#0a1520", border: "1px solid #0d2035", color: "#c8e8ff" }}
952 |           />
953 |         </div>
954 |         {patenteSeleccionada && (
955 |           <button onClick={() => setPatenteSeleccionada(null)}
956 |             data-testid="btn-limpiar-filtro"
957 |             className="font-exo text-[10px] font-bold px-3 py-2 rounded cursor-pointer"
958 |             style={{ background: "#ff224420", border: "1px solid #ff2244", color: "#ff2244" }}>
959 |             <X className="w-3 h-3 inline mr-1" />
960 |             Limpiar
```

```
961 |           </button>
962 |         )}
963 |       </div>
964 |
965 |       {!patenteSeleccionada && (
966 |         <div className="flex flex-wrap gap-1.5 mb-4">
967 |           {patentesFiltradas.map((p: string) => (
968 |             <button key={p} onClick={() => { setPatenteSeleccionada(p); setBusqueda(""); }}
969 |               data-testid={`btn-patente-${p}`}
970 |               className="font-space text-[10px] font-bold px-3 py-1.5 rounded cursor-pointer transition-all hover:scale-
971 |               style={{ background: "#00d4ff15", border: "1px solid #00d4ff40", color: "#00d4ff" }}>
972 |               <Truck className="w-3 h-3 inline mr-1" />{p}
973 |             </button>
974 |           ))}
975 |           {patentesFiltradas.length === 0 && busqueda && (
976 |             <div className="font-exo text-[10px] py-2" style={{ color: "#3a6080" }}>
977 |               Sin resultados para "{busqueda}"
978 |             </div>
979 |           )}
980 |         </div>
981 |       )}
982 |
983 |       {isLoading ? (
984 |         <div className="text-center py-8">
985 |           <RefreshCw className="w-5 h-5 animate-spin mx-auto" style={{ color: "#3a6080" }} />
986 |         </div>
987 |       ) : patenteSeleccionada ? (
988 |         <>
989 |           <div className="font-exo text-[11px] font-bold mb-3" style={{ color: "#00d4ff" }}>
990 |             <Truck className="w-4 h-4 inline mr-1" />
991 |             Camion {patenteSeleccionada} -- {viajes.length} viajes
992 |           </div>
993 |
994 |           <div className="dash-card overflow-hidden" data-testid="tabla-viajes-cerrados">
995 |             <table className="w-full">
996 |               <thead>
997 |                 <tr style={{ borderBottom: "1px solid #0d2035" }}>
998 |                   {["CODIGO", "SALIDA", "LLEGADA", "DURACION", "KM", "LITROS ECU", "REND.", "ESTADO"].map(h => (
999 |                     <th key={h} className="font-exo text-[9px] tracking-[0.15em] text-left px-3 py-2" style={{ color: "#
1000 |                   ))}
1001 |                 </tr>
1002 |               </thead>
1003 |               <tbody>
1004 |                 {viajes.map((v: any) => {
1005 |                   const isExpanded = expandedViaje === v.id;
1006 |                   const diffLitros = v.diferenciaLitros;
1007 |                   const hasDiff = diffLitros !== null && diffLitros !== undefined;
1008 |                   const diffColor = hasDiff ? (diffLitros > 15 ? "#ff2244" : diffLitros < -15 ? "#ffcc00" : "#00c97a") :
1009 |                   const diffLabel = hasDiff ? (diffLitros > 15 ? "EXCESO SIGETRA" : diffLitros < -15 ? "EXCESO ECU" : "O
1010 |                   return (
1011 |                     <Fragment key={v.id}>
1012 |                       <tr style={{ borderBottom: isExpanded ? "none" : "1px solid #0d203530", cursor: "pointer", backgro
1013 |                         onClick={() => setExpandedViaje(isExpanded ? null : v.id)}
1014 |                         data-testid={`fila-viaje-${v.id}`}>
1015 |                         <td className="font-space text-[10px] font-bold px-3 py-2" style={{ color: "#00d4ff" }}>
1016 |                           {isExpanded ? <ChevronUp className="w-3 h-3 inline mr-1" /> : <ChevronDown className="w-3 h-3
1017 |                           {v.codigo}
1018 |                         </td>
1019 |                         <td className="font-space text-[10px] px-3 py-2" style={{ color: "#c8e8ff" }}>{formatFecha(v.fec
1020 |                         <td class="font-space text-[10px] px-3 py-2" style={{ color: "#c8e8ff" }}>{formatFecha(v.fec
1021 |                         <td class="font-space text-[10px] px-3 py-2" style={{ color: "#c8e8ff" }}>{formatDuracion(v.
1022 |                         <td class="font-space text-[10px] font-bold px-3 py-2" style={{ color: "#c8e8ff" }}>{Math.ro
1023 |                         <td class="font-space text-[10px] px-3 py-2" style={{ color: "#ffcc00" }}>{v.litrosEcu ? `${
1024 |                         <td class="font-space text-[10px] font-bold px-3 py-2" style={{ color: v.rendimiento && v.re
1025 |                           {v.rendimiento ? `${v.rendimiento.toFixed(2)} km/L` : "--"}
1026 |                         </td>
1027 |                         <td class="px-3 py-2">
1028 |                           <span class="font-exo text-[9px] font-bold px-2 py-0.5 rounded" style={{
1029 |                             background: v.estado === "COMPLETADO" ? "#00c97a20" : "#ffcc0020",
1030 |                             color: v.estado === "COMPLETADO" ? "#00c97a" : "#ffcc00",
1031 |                           }}>
1032 |                             {v.estado}
1033 |                           </span>
1034 |                         </td>
1035 |                       </tr>
1036 |                       {isExpanded && (
1037 |                         <tr data-testid={`detalle-viaje-${v.id}`}>
1038 |                           <td colSpan={8} style={{ background: "#0a1520", borderBottom: "1px solid #0d2035" }}>
1039 |                             <div class="px-4 py-4">
1040 |                               <div class="grid grid-cols-2 gap-4 mb-4">
1041 |                                 <div>
1042 |                                   <div class="font-exo text-[9px] tracking-[0.15em] uppercase mb-3" style={{ color:
1043 |                                   <div class="space-y-2">
1044 |                                     <div class="flex justify-between">
1045 |                                       <span class="font-exo text-[10px]" style={{ color: "#3a6080" }}>Codigo</span>
1046 |                                       <span class="font-space text-[10px] font-bold" style={{ color: "#00d4ff" }}>{v
1047 |                                     </div>
1048 |                                     <div class="flex justify-between">
1049 |                                       <span class="font-exo text-[10px]" style={{ color: "#3a6080" }}>Patente</span>
1050 |                                       <span class="font-space text-[10px]" style={{ color: "#c8e8ff" }}>{v.patente}<
1051 |                                     </div>
1052 |                                     <div class="flex justify-between">
1053 |                                       <span class="font-exo text-[10px]" style={{ color: "#3a6080" }}>Conductor</spa
1054 |                                       <span class="font-space text-[10px]" style={{ color: "#c8e8ff" }}>{v.conductor
1055 |                                     </div>
1056 |                                     <div class="flex justify-between">
1057 |                                       <span class="font-exo text-[10px]" style={{ color: "#3a6080" }}>Fecha salida</
1058 |                                       <span class="font-space text-[10px]" style={{ color: "#c8e8ff" }}>{v.fechaSali
1059 |                                     </div>
1060 |                                     <div class="flex justify-between">
1061 |                                       <span class="font-exo text-[10px]" style={{ color: "#3a6080" }}>Fecha llegada<
1062 |                                       <span class="font-space text-[10px]" style={{ color: "#c8e8ff" }}>{v.fechaLleg
1063 |                                     </div>
1064 |                                     <div class="flex justify-between">
1065 |                                       <span class="font-exo text-[10px]" style={{ color: "#3a6080" }}>Duracion</span
1066 |                                       <span class="font-space text-[10px] font-bold" style={{ color: "#c8e8ff" }}>{f
1067 |                                     </div>
```

```
1068 |                                    {v.origenNombre && (
1069 |                                      <div className="flex justify-between">
1070 |                                        <span className="font-exo text-[10px]" style={{ color: "#3a6080" }}>Origen</span
1071 |                                        <span className="font-space text-[10px]" style={{ color: "#c8e8ff" }}>{v.origenN
1072 |                                      </div>
1073 |                                    )}
1074 |                                    {v.destinoNombre && (
1075 |                                      <div className="flex justify-between">
1076 |                                        <span className="font-exo text-[10px]" style={{ color: "#3a6080" }}>Destino</spa
1077 |                                        <span className="font-space text-[10px]" style={{ color: "#c8e8ff" }}>{v.destino
1078 |                                      </div>
1079 |                                    )}
1080 |                                  </div>
1081 |                                </div>
1082 |
1083 |                                <div>
1084 |                                  <div className="font-exo text-[9px] tracking-[0.15em] uppercase mb-3" style={{ color:
1085 |                                  <div className="space-y-2">
1086 |                                    <div className="flex justify-between">
1087 |                                      <span className="font-exo text-[10px]" style={{ color: "#3a6080" }}>Km inicio</spa
1088 |                                      <span className="font-space text-[10px]" style={{ color: "#c8e8ff" }}>{v.kmInicio
1089 |                                    </div>
1090 |                                    <div className="flex justify-between">
1091 |                                      <span className="font-exo text-[10px]" style={{ color: "#3a6080" }}>Km cierre</spa
1092 |                                      <span className="font-space text-[10px]" style={{ color: "#c8e8ff" }}>{v.kmCierre
1093 |                                    </div>
1094 |                                    <div className="flex justify-between">
1095 |                                      <span className="font-exo text-[10px]" style={{ color: "#3a6080" }}>Km recorridos<
1096 |                                      <span className="font-space text-[10px] font-bold" style={{ color: "#00d4ff" }}>{M
1097 |                                    </div>
1098 |                                    <div className="flex justify-between">
1099 |                                      <span className="font-exo text-[10px]" style={{ color: "#3a6080" }}>Litros Sigetra
1100 |                                      <span className="font-space text-[10px]" style={{ color: v.litrosSigetra ? "#c8e8f
1101 |                                    </div>
1102 |                                    <div className="flex justify-between">
1103 |                                      <span className="font-exo text-[10px]" style={{ color: "#3a6080" }}>Litros ECU Vol
1104 |                                      <span className="font-space text-[10px] font-bold" style={{ color: "#ffcc00" }}>{v
1105 |                                    </div>
1106 |                                    {hasDiff && (
1107 |                                      <div className="flex justify-between items-center">
1108 |                                        <span className="font-exo text-[10px]" style={{ color: "#3a6080" }}>Diferencia</
1109 |                                        <div className="flex items-center gap-2">
1110 |                                          <span className="font-space text-[10px] font-bold" style={{ color: diffColor }
1111 |                                            {diffLitros > 0 ? "+" : ""}{diffLitros.toFixed(1)} L
1112 |                                          </span>
1113 |                                          <span className="font-exo text-[8px] font-bold px-1.5 py-0.5 rounded" style={{
1114 |                                            {diffLabel}
1115 |                                          </span>
1116 |                                        </div>
1117 |                                      </div>
1118 |                                    )}
1119 |                                    <div className="flex justify-between">
1120 |                                      <span className="font-exo text-[10px]" style={{ color: "#3a6080" }}>Rendimiento</s
1121 |                                      <span className="font-space text-[10px] font-bold" style={{ color: v.rendimiento &
1122 |                                        {v.rendimiento ? `${v.rendimiento.toFixed(2)} km/L` : "--"}
1123 |                                      </span>
1124 |                                    </div>
1125 |                                  </div>
1126 |                                </div>
1127 |                              </div>
1128 |
1129 |                              {v.notas && (
1130 |                                <div className="mt-2 px-3 py-2 rounded" style={{ background: "#020508", border: "1px sol
1131 |                                  <div className="font-exo text-[9px] tracking-[0.15em] uppercase mb-1" style={{ color:
1132 |                                  <div className="font-exo text-[10px]" style={{ color: "#c8e8ff" }}>{v.notas}</div>
1133 |                                </div>
1134 |                              )}
1135 |
1136 |                              {v.detectadoPorIa && (
1137 |                                <div className="mt-2 flex items-center gap-1">
1138 |                                  <Cpu className="w-3 h-3" style={{ color: "#00d4ff" }} />
1139 |                                  <span className="font-exo text-[9px]" style={{ color: "#00d4ff" }}>Viaje detectado aut
1140 |                                </div>
1141 |                              )}
1142 |                            </div>
1143 |                          </td>
1144 |                        </tr>
1145 |                      )}
1146 |                    </Fragment>
1147 |                  );
1148 |                })}
1149 |              </tbody>
1150 |            </table>
1151 |          </div>
1152 |
1153 |          {viajes.length > 0 && (
1154 |            <div className="mt-3 flex gap-4">
1155 |              <div className="dash-card px-4 py-3 flex-1">
1156 |                <div className="font-exo text-[9px] tracking-[0.15em]" style={{ color: "#3a6080" }}>KM TOTAL</div>
1157 |                <div className="font-space text-lg font-bold" style={{ color: "#c8e8ff" }}>{Math.round(totalKm).toLocale
1158 |              </div>
1159 |              <div className="dash-card px-4 py-3 flex-1">
1160 |                <div className="font-exo text-[9px] tracking-[0.15em]" style={{ color: "#3a6080" }}>LITROS ECU</div>
1161 |                <div className="font-space text-lg font-bold" style={{ color: "#ffcc00" }}>{Math.round(totalLitros).toLo
1162 |              </div>
1163 |              <div className="dash-card px-4 py-3 flex-1">
1164 |                <div className="font-exo text-[9px] tracking-[0.15em]" style={{ color: "#3a6080" }}>REND. PROM</div>
1165 |                <div className="font-space text-lg font-bold" style={{ color: totalLitros > 0 && totalKm / totalLitros >
1166 |                  {totalLitros > 0 ? (totalKm / totalLitros).toFixed(2) : "--"}
1167 |                </div>
1168 |              </div>
1169 |            </div>
1170 |          )}
1171 |        </>
1172 |      ) : (
1173 |        <div className="dash-card px-5 py-8 text-center">
1174 |          <Search className="w-6 h-6 mx-auto mb-2" style={{ color: "#3a6080" }} />
```

```
1175 |           <div className="font-exo text-[11px] mb-1" style={{ color: "#c8e8ff" }}>
1176 |             Selecciona un camion
1177 |           </div>
1178 |           <div className="font-exo text-[10px]" style={{ color: "#3a6080" }}>
1179 |             Busca por patente para ver todos los viajes cerrados
1180 |           </div>
1181 |         </div>
1182 |       )}
1183 |     </div>
1184 |   );
1185 | }
1186 |
1187 | function ValidacionSigetra() {
1188 |   const [dias, setDias] = useState(7);
1189 |
1190 |   const { data: stats } = useQuery<any>({
1191 |     queryKey: ["/api/geo/validacion/stats", dias],
1192 |     queryFn: () => fetch(`/api/geo/validacion/stats?dias=${dias}`).then(r => r.json()),
1193 |   });
1194 |
1195 |   const { data: viajes } = useQuery<GeoViaje[]>({
1196 |     queryKey: ["/api/geo/viajes"],
1197 |   });
1198 |
1199 |   const aprobarMutation = useMutation({
1200 |     mutationFn: () => apiRequest("POST", "/api/geo/viajes/aprobar-masivo"),
1201 |     onSuccess: () => {
1202 |       queryClient.invalidateQueries({ queryKey: ["/api/geo/viajes"] });
1203 |       queryClient.invalidateQueries({ queryKey: ["/api/geo/validacion/stats"] });
1204 |     },
1205 |   });
1206 |
1207 |   const validarUno = useMutation({
1208 |     mutationFn: ({ id, estado }: { id: number; estado: string }) =>
1209 |       apiRequest("POST", `/api/geo/viajes/${id}/validar`, { estado }),
1210 |     onSuccess: () => {
1211 |       queryClient.invalidateQueries({ queryKey: ["/api/geo/viajes"] });
1212 |       queryClient.invalidateQueries({ queryKey: ["/api/geo/validacion/stats"] });
1213 |     },
1214 |   });
1215 |
1216 |   const porEstado = useMemo(() => {
1217 |     const map: Record<string, { count: number; pct: number }> = {
1218 |       VALIDADO: { count: 0, pct: 0 },
1219 |       REVISAR: { count: 0, pct: 0 },
1220 |       ANOMALIA: { count: 0, pct: 0 },
1221 |       PENDIENTE: { count: 0, pct: 0 },
1222 |     };
1223 |     if (stats?.porEstado) {
1224 |       for (const s of stats.porEstado) {
1225 |         map[s.estado] = { count: s.count, pct: s.pct };
1226 |       }
1227 |     }
1228 |     return map;
1229 |   }, [stats]);
1230 |
1231 |   const sorted = useMemo(() => {
1232 |     if (!viajes) return [];
1233 |     const order: Record<string, number> = { ANOMALIA: 0, REVISAR: 1, PENDIENTE: 2, VALIDADO: 3 };
1234 |     return [...viajes].sort((a, b) => (order[a.validacionEstado] ?? 9) - (order[b.validacionEstado] ?? 9));
1235 |   }, [viajes]);
1236 |
1237 |   return (
1238 |     <div data-testid="geo-validacion">
1239 |       <div className="flex items-center justify-between mb-4">
1240 |         <div>
1241 |           <h2 className="font-space text-[16px] font-bold tracking-[0.12em]" style={{ color: "#c8e8ff" }}>
1242 |             VALIDACION GPS vs SIGETRA
1243 |           </h2>
1244 |           <p className="font-exo text-[11px]" style={{ color: "#3a6080" }}>
1245 |             Cruzando kilometraje real con cargas registradas
1246 |           </p>
1247 |         </div>
1248 |         <div className="flex items-center gap-2">
1249 |           <select value={dias} onChange={e => setDias(parseInt(e.target.value))}
1250 |             data-testid="select-periodo"
1251 |             className="font-exo text-[11px] px-3 py-2 rounded"
1252 |             style={{ background: "#091018", border: "1px solid #0d2035", color: "#c8e8ff" }}>
1253 |             <option value={7}>Ultimos 7 dias</option>
1254 |             <option value={14}>Ultimos 14 dias</option>
1255 |             <option value={30}>Ultimos 30 dias</option>
1256 |           </select>
1257 |           <button onClick={() => aprobarMutation.mutate()} disabled={aprobarMutation.isPending}
1258 |             data-testid="btn-aprobar-masivo"
1259 |             className="font-exo text-[10px] font-bold px-3 py-2 rounded cursor-pointer"
1260 |             style={{ background: "#00c97a20", border: "1px solid #00c97a", color: "#00c97a" }}>
1261 |             <Check className="w-3 h-3 inline mr-1" />
1262 |             Aprobar validados automaticos
1263 |           </button>
1264 |         </div>
1265 |       </div>
1266 |
1267 |       <div className="grid grid-cols-4 gap-3 mb-4">
1268 |         {[
1269 |           { label: "VALIDADOS", count: porEstado.VALIDADO.count, pct: porEstado.VALIDADO.pct, color: "#00c97a" },
1270 |           { label: "REVISAR", count: porEstado.REVISAR.count, pct: porEstado.REVISAR.pct, color: "#ffcc00" },
1271 |           { label: "ANOMALIAS", count: porEstado.ANOMALIA.count, pct: porEstado.ANOMALIA.pct, color: "#ff2244" },
1272 |           { label: "PENDIENTES", count: porEstado.PENDIENTE.count, pct: porEstado.PENDIENTE.pct, color: "#3a6080" },
1273 |         ].map(k => (
1274 |           <div key={k.label} className="dash-card px-4 py-3" data-testid={`val-kpi-${k.label.toLowerCase()}`}>
1275 |             <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>{k.label}</div
1276 |             <div className="font-space text-2xl font-bold" style={{ color: k.color }}>{k.count}</div>
1277 |             <div className="font-exo text-[9px]" style={{ color: "#3a6080" }}>{k.pct}%</div>
1278 |           </div>
1279 |         ))}
1280 |       </div>
1281 |
```

```
1282 |       <div className="mb-4 dash-card p-3">
1283 |         <div className="h-3 rounded-full overflow-hidden flex" style={{ background: "#0d2035" }}>
1284 |           {porEstado.VALIDADO.pct > 0 && (
1285 |             <div style={{ width: `${porEstado.VALIDADO.pct}%`, background: "#00c97a" }} />
1286 |           )}
1287 |           {porEstado.REVISAR.pct > 0 && (
1288 |             <div style={{ width: `${porEstado.REVISAR.pct}%`, background: "#ffcc00" }} />
1289 |           )}
1290 |           {porEstado.ANOMALIA.pct > 0 && (
1291 |             <div style={{ width: `${porEstado.ANOMALIA.pct}%`, background: "#ff2244" }} />
1292 |           )}
1293 |           {porEstado.PENDIENTE.pct > 0 && (
1294 |             <div style={{ width: `${porEstado.PENDIENTE.pct}%`, background: "#3a6080" }} />
1295 |           )}
1296 |         </div>
1297 |       </div>
1298 |
1299 |       <div className="space-y-2">
1300 |         {sorted.map(v => (
1301 |           <div key={v.id} className="dash-card p-4" data-testid={`val-card-${v.id}`}
1302 |             style={{ borderLeft: `3px solid ${estadoColors[v.validacionEstado] || "#3a6080"}` }}>
1303 |             <div className="flex items-start justify-between mb-2">
1304 |               <div className="flex items-center gap-3">
1305 |                 <EstadoBadge estado={v.validacionEstado} />
1306 |                 <span className="font-space text-[12px] font-bold" style={{ color: "#00d4ff" }}>{v.patente}</span>
1307 |                 <span className="font-exo text-[10px]" style={{ color: "#3a6080" }}>
1308 |                   {v.origenTimestamp ? new Date(v.origenTimestamp).toLocaleDateString("es-CL") : ""}
1309 |                 </span>
1310 |               </div>
1311 |               {!v.validadoManualmente && v.validacionEstado !== "PENDIENTE" && (
1312 |                 <div className="flex gap-1">
1313 |                   <button onClick={() => validarUno.mutate({ id: v.id, estado: "VALIDADO" })}
1314 |                     data-testid={`btn-aprobar-${v.id}`}
1315 |                     className="font-exo text-[9px] px-2 py-1 rounded cursor-pointer"
1316 |                     style={{ background: "#00c97a20", border: "1px solid #00c97a", color: "#00c97a" }}>
1317 |                     Aprobar
1318 |                   </button>
1319 |                   <button onClick={() => validarUno.mutate({ id: v.id, estado: "ANOMALIA" })}
1320 |                     data-testid={`btn-rechazar-${v.id}`}
1321 |                     className="font-exo text-[9px] px-2 py-1 rounded cursor-pointer"
1322 |                     style={{ background: "#ff224420", border: "1px solid #ff2244", color: "#ff2244" }}>
1323 |                     Rechazar
1324 |                   </button>
1325 |                 </div>
1326 |               )}
1327 |             </div>
1328 |
1329 |             <div className="grid grid-cols-2 gap-4">
1330 |               <div>
1331 |                 <div className="font-exo text-[10px]" style={{ color: "#3a6080" }}>GPS detectado:</div>
1332 |                 <div className="font-exo text-[11px]" style={{ color: "#c8e8ff" }}>
1333 |                   {v.origenNombre || "?"} ›' {v.destinoNombre || "?"} · {v.kmGps ? `${Math.round(parseFloat(v.kmGps))} km
1334 |                 </div>
1335 |               </div>
1336 |               <div>
1337 |                 <div className="font-exo text-[10px]" style={{ color: "#3a6080" }}>Sigetra:</div>
1338 |                 <div className="font-exo text-[11px]" style={{ color: "#c8e8ff" }}>
1339 |                   {v.sigetraLitros ? `${parseFloat(v.sigetraLitros)}L cargados` : "Sin carga encontrada"}
1340 |                   {v.sigetraKmDeltaPct ? ` · Delta km: ${parseFloat(v.sigetraKmDeltaPct)}%` : ""}
1341 |                 </div>
1342 |               </div>
1343 |             </div>
1344 |
1345 |             {v.validacionDetalle && (
1346 |               <div className="mt-2 flex gap-3">
1347 |                 {v.validacionDetalle.kmStatus && (
1348 |                   <span className="font-exo text-[9px] flex items-center gap-1" style={{
1349 |                     color: v.validacionDetalle.kmStatus === "OK" ? "#00c97a" : v.validacionDetalle.kmStatus === "REVISAR
1350 |                   }}>
1351 |                     {v.validacionDetalle.kmStatus === "OK" ? <Check className="w-3 h-3" /> : <AlertTriangle className="w
1352 |                     Km: {v.validacionDetalle.kmStatus} {v.validacionDetalle.kmDeltaPct != null ? `(${v.validacionDetalle
1353 |                   </span>
1354 |                 )}
1355 |                 {v.validacionDetalle.timingStatus && (
1356 |                   <span className="font-exo text-[9px] flex items-center gap-1" style={{
1357 |                     color: v.validacionDetalle.timingStatus === "OK" ? "#00c97a" : "#ffcc00"
1358 |                   }}>
1359 |                     {v.validacionDetalle.timingStatus === "OK" ? <Check className="w-3 h-3" /> : <AlertTriangle classNam
1360 |                     Timing: {v.validacionDetalle.timingStatus}
1361 |                   </span>
1362 |                 )}
1363 |                 {v.validacionDetalle.surtidorStatus && (
1364 |                   <span className="font-exo text-[9px] flex items-center gap-1" style={{
1365 |                     color: v.validacionDetalle.surtidorStatus === "OK" ? "#00c97a" : "#ffcc00"
1366 |                   }}>
1367 |                     {v.validacionDetalle.surtidorStatus === "OK" ? <Check className="w-3 h-3" /> : <AlertTriangle classN
1368 |                     Surtidor: {v.validacionDetalle.surtidorStatus}
1369 |                   </span>
1370 |                 )}
1371 |               </div>
1372 |             )}
1373 |           </div>
1374 |         ))}
1375 |
1376 |         {sorted.length === 0 && (
1377 |           <div className="text-center py-12 dash-card">
1378 |             <CheckCircle className="w-8 h-8 mx-auto mb-3" style={{ color: "#3a6080" }} />
1379 |             <div className="font-space text-[13px] font-bold" style={{ color: "#3a6080" }}>Sin viajes para validar</div>
1380 |             <div className="font-exo text-[11px] mt-1" style={{ color: "#3a6080" }}>
1381 |               Los viajes se detectan automaticamente desde datos GPS
1382 |             </div>
1383 |           </div>
1384 |         )}
1385 |       </div>
1386 |     </div>
1387 |   );
1388 | }
```

```
1389 |
1390 | function AnalisisIATab() {
1391 |   const [analizando, setAnalizando] = useState(false);
1392 |   const [progreso, setProgreso] = useState<{ paso: string; progreso: number; total: number; detalles?: string } | null>(
1393 |
1394 |   const { data: analisis, isLoading } = useQuery<any>({
1395 |     queryKey: ["/api/geo/analisis-ia"],
1396 |   });
1397 |
1398 |   const generarMutation = useMutation({
1399 |     mutationFn: () => apiRequest("POST", "/api/geo/analisis-ia/generar"),
1400 |     onSuccess: () => {
1401 |       queryClient.invalidateQueries({ queryKey: ["/api/geo/analisis-ia"] });
1402 |     },
1403 |   });
1404 |
1405 |   const handleAnalizarHistorico = async () => {
1406 |     setAnalizando(true);
1407 |     setProgreso({ paso: "Iniciando...", progreso: 0, total: 1 });
1408 |     try {
1409 |       const res = await fetch("/api/geo/analizar-historico", { method: "POST" });
1410 |       const reader = res.body?.getReader();
1411 |       const decoder = new TextDecoder();
1412 |       if (reader) {
1413 |         let done = false;
1414 |         while (!done) {
1415 |           const { value, done: d } = await reader.read();
1416 |           done = d;
1417 |           if (value) {
1418 |             const text = decoder.decode(value);
1419 |             const lines = text.split("\n").filter(l => l.startsWith("data: "));
1420 |             for (const line of lines) {
1421 |               try {
1422 |                 const data = JSON.parse(line.replace("data: ", ""));
1423 |                 setProgreso(data);
1424 |               } catch {}
1425 |             }
1426 |           }
1427 |         }
1428 |       }
1429 |     } catch (e) {
1430 |       console.error("Error:", e);
1431 |     } finally {
1432 |       setAnalizando(false);
1433 |       queryClient.invalidateQueries({ queryKey: ["/api/geo/lugares"] });
1434 |       queryClient.invalidateQueries({ queryKey: ["/api/geo/lugares/stats"] });
1435 |     }
1436 |   };
1437 |
1438 |   const secciones = useMemo(() => {
1439 |     if (!analisis?.resumenTexto) return [];
1440 |     const text = analisis.resumenTexto as string;
1441 |     const titles = ["PATRONES DE RUTA:", "LOCALES CENCOSUD DETECTADOS:", "TIEMPOS DE PERMANENCIA:", "ANOMALIAS DETECTADA
1442 |     const parts: { titulo: string; contenido: string }[] = [];
1443 |     let rest = text;
1444 |
1445 |     for (let i = 0; i < titles.length; i++) {
1446 |       const idx = rest.indexOf(titles[i]);
1447 |       if (idx >= 0) {
1448 |         const nextIdx = i + 1 < titles.length ? rest.indexOf(titles[i + 1]) : rest.length;
1449 |         const contenido = rest.substring(idx + titles[i].length, nextIdx > idx ? nextIdx : rest.length).trim();
1450 |         parts.push({ titulo: titles[i].replace(":", ""), contenido });
1451 |         if (nextIdx > idx) rest = rest.substring(nextIdx);
1452 |       }
1453 |     }
1454 |     if (parts.length === 0 && text.length > 0) {
1455 |       parts.push({ titulo: "ANALISIS", contenido: text });
1456 |     }
1457 |     return parts;
1458 |   }, [analisis]);
1459 |
1460 |   const haceMinutos = analisis?.generadoAt
1461 |     ? Math.round((Date.now() - new Date(analisis.generadoAt).getTime()) / 60000)
1462 |     : null;
1463 |
1464 |   const seccionColors = ["#00d4ff", "#00c97a", "#ffcc00", "#ff2244", "#c8e8ff"];
1465 |
1466 |   return (
1467 |     <div data-testid="geo-ia">
1468 |       <div className="flex items-center justify-between mb-4">
1469 |         <div>
1470 |           <h2 className="font-space text-[16px] font-bold tracking-[0.12em]" style={{ color: "#c8e8ff" }}>
1471 |             ANALISIS INTELIGENTE DE OPERACION
1472 |           </h2>
1473 |           <p className="font-exo text-[11px]" style={{ color: "#3a6080" }}>
1474 |             Basado en GPS historico desde 01-03-2026
1475 |           </p>
1476 |         </div>
1477 |         <div className="flex items-center gap-3">
1478 |           {haceMinutos !== null && (
1479 |             <span className="font-exo text-[10px]" style={{ color: "#3a6080" }}>
1480 |               Ultimo analisis: {haceMinutos < 60 ? `hace ${haceMinutos}min` : haceMinutos < 1440 ? `hace ${Math.floor(ha
1481 |             </span>
1482 |           )}
1483 |           <button onClick={() => generarMutation.mutate()}
1484 |             disabled={generarMutation.isPending}
1485 |             className="font-exo text-[10px] font-bold px-3 py-2 rounded cursor-pointer"
1486 |             style={{ background: "#00d4ff20", border: "1px solid #00d4ff", color: "#00d4ff" }}
1487 |             data-testid="btn-generar-ia">
1488 |             <Cpu className={`w-3 h-3 inline mr-1 ${generarMutation.isPending ? "animate-spin" : ""}`} />
1489 |             {generarMutation.isPending ? "Generando..." : "Generar nuevo analisis"}
1490 |           </button>
1491 |         </div>
1492 |       </div>
1493 |
1494 |       <div className="dash-card p-4 mb-4" style={{ borderLeft: "3px solid #00d4ff" }}>
1495 |         <div className="flex items-center justify-between mb-3">
```

```
1496 |           <span className="font-space text-[12px] font-bold tracking-wider" style={{ color: "#00d4ff" }}>
1497 |             ANALIZAR HISTORIAL COMPLETO
1498 |           </span>
1499 |         </div>
1500 |         <p className="font-exo text-[11px] mb-3" style={{ color: "#3a6080" }}>
1501 |           Procesa todos los puntos GPS desde 01-03-2026, detecta paradas, identifica locales CENCOSUD via OpenStreetMap,
1502 |         </p>
1503 |         <button onClick={handleAnalizarHistorico}
1504 |           disabled={analizando}
1505 |           className="font-exo text-[11px] font-bold px-4 py-2 rounded cursor-pointer"
1506 |           style={{ background: analizando ? "#3a608020" : "#00c97a20", border: `1px solid ${analizando ? "#3a6080" : "#0
1507 |           data-testid="btn-analizar-historico">
1508 |           <RefreshCw className={`w-3 h-3 inline mr-1 ${analizando ? "animate-spin" : ""}`} />
1509 |           {analizando ? "Analizando..." : "Analizar historial desde 01-03-2026"}
1510 |         </button>
1511 |         {progreso && (
1512 |           <div className="mt-3">
1513 |             <div className="flex items-center justify-between mb-1">
1514 |               <span className="font-exo text-[10px]" style={{ color: "#c8e8ff" }}>{progreso.paso}</span>
1515 |               <span className="font-space text-[10px]" style={{ color: "#3a6080" }}>{progreso.progreso}/{progreso.total}
1516 |             </div>
1517 |             <div className="h-2 rounded-full overflow-hidden" style={{ background: "#0d2035" }}>
1518 |               <div className="h-full rounded-full transition-all" style={{
1519 |                 width: `${progreso.total > 0 ? (progreso.progreso / progreso.total) * 100 : 0}%`,
1520 |                 background: "#00c97a",
1521 |               }} />
1522 |             </div>
1523 |             {progreso.detalles && (
1524 |               <span className="font-exo text-[9px] mt-1 block" style={{ color: "#3a6080" }}>{progreso.detalles}</span>
1525 |             )}
1526 |           </div>
1527 |         )}
1528 |       </div>
1529 |
1530 |       {generarMutation.isPending && (
1531 |         <div className="text-center py-12 dash-card mb-4">
1532 |           <Cpu className="w-8 h-8 animate-spin mx-auto mb-3" style={{ color: "#00d4ff" }} />
1533 |           <div className="font-space text-[13px] font-bold" style={{ color: "#00d4ff" }}>Generando analisis con IA...</d
1534 |           <div className="font-exo text-[11px] mt-1" style={{ color: "#3a6080" }}>Procesando datos de viajes, lugares y
1535 |         </div>
1536 |       )}
1537 |
1538 |       {secciones.length > 0 ? (
1539 |         <div className="space-y-3">
1540 |           {secciones.map((s, i) => (
1541 |             <div key={i} className="dash-card p-4" style={{ borderLeft: `3px solid ${seccionColors[i % seccionColors.len
1542 |               data-testid={`ia-seccion-${i}`}>
1543 |               <div className="font-space text-[12px] font-bold tracking-wider mb-2" style={{ color: seccionColors[i % se
1544 |                 {s.titulo}
1545 |               </div>
1546 |               <div className="font-exo text-[11px] leading-relaxed whitespace-pre-line" style={{ color: "#c8e8ff" }}>
1547 |                 {s.contenido}
1548 |               </div>
1549 |             </div>
1550 |           ))}
1551 |         </div>
1552 |       ) : !generarMutation.isPending && (
1553 |         <div className="text-center py-12 dash-card">
1554 |           <Cpu className="w-8 h-8 mx-auto mb-3" style={{ color: "#3a6080" }} />
1555 |           <div className="font-space text-[13px] font-bold" style={{ color: "#3a6080" }}>Sin analisis generado</div>
1556 |           <div class="font-exo text-[11px] mt-1" style={{ color: "#3a6080" }}>
1557 |             Presiona "Generar nuevo analisis" para crear un reporte con IA
1558 |           </div>
1559 |         </div>
1560 |       )}
1561 |
1562 |       {analisis?.resultadoJson && (
1563 |         <div className="mt-4 grid grid-cols-3 gap-3">
1564 |           <div className="dash-card px-4 py-3">
1565 |             <div class="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>VIAJES ANALIZA
1566 |             <div class="font-space text-xl font-bold" style={{ color: "#00d4ff" }}>{analisis.resultadoJson.totalViaj
1567 |           </div>
1568 |           <div class="dash-card px-4 py-3">
1569 |             <div class="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>KM GPS TOTAL</
1570 |             <div class="font-space text-xl font-bold" style={{ color: "#00c97a" }}>{(analisis.resultadoJson.totalKmG
1571 |           </div>
1572 |           <div class="dash-card px-4 py-3">
1573 |             <div class="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>CAMIONES</div>
1574 |             <div class="font-space text-xl font-bold" style={{ color: "#c8e8ff" }}>{analisis.resultadoJson.camionesA
1575 |           </div>
1576 |         </div>
1577 |       )}
1578 |     </div>
1579 |   );
1580 | }
1581 |
1582 | function ConductoresTab() {
1583 |   const { data: conductoresData, isLoading } = useQuery<any>({
1584 |     queryKey: ["/api/geo/conductores"],
1585 |   });
1586 |
1587 |   const [expandedConductor, setExpandedConductor] = useState<string | null>(null);
1588 |   const [searchTerm, setSearchTerm] = useState("");
1589 |   const [sortBy, setSortBy] = useState<"litros" | "rend_asc" | "rend_desc">("litros");
1590 |
1591 |   const filtered = useMemo(() => {
1592 |     if (!conductoresData?.conductores) return [];
1593 |     let list = conductoresData.conductores;
1594 |     if (searchTerm) {
1595 |       const term = searchTerm.toLowerCase();
1596 |       list = list.filter((c: any) =>
1597 |         c.nombre.toLowerCase().includes(term) || c.camiones.some((p: string) => p.includes(term))
1598 |       );
1599 |     }
1600 |     if (sortBy === "rend_asc") {
1601 |       list = [...list].sort((a: any, b: any) => (a.rendimiento || 999) - (b.rendimiento || 999));
1602 |     } else if (sortBy === "rend_desc") {
```

```
      list = [...list].sort((a: any, b: any) => (b.rendimiento || 0) - (a.rendimiento || 0));
    }
    return list;
  }, [conductoresData, searchTerm, sortBy]);

  if (isLoading) return (
    <div className="flex items-center justify-center py-20">
      <RefreshCw className="w-6 h-6 animate-spin" style={{ color: "#00d4ff" }} />
    </div>
  );

  return (
    <div data-testid="conductores-tab">
      <div className="flex items-center justify-between mb-4">
        <div className="flex items-center gap-3">
          <div className="font-space text-[13px] font-bold tracking-wider" style={{ color: "#00d4ff" }}>
            CONDUCTORES CENCOSUD
          </div>
          <div className="font-exo text-[10px] px-2 py-0.5 rounded" style={{ background: "#00d4ff15", border: "1px solid
            {conductoresData?.totalConductores || 0} conductores
          </div>
          <div className="font-exo text-[10px]" style={{ color: "#3a6080" }}>
            Desde {conductoresData?.desde || "01-03-2026"}
          </div>
        </div>
        <div className="flex items-center gap-2">
          <div className="flex gap-1">
            {([
              { id: "litros" as const, label: "Mas litros" },
              { id: "rend_asc" as const, label: "Peor rend" },
              { id: "rend_desc" as const, label: "Mejor rend" },
            ]).map(s => (
              <button key={s.id} onClick={() => setSortBy(s.id)}
                data-testid={`btn-sort-${s.id}`}
                className="font-exo text-[10px] font-bold px-2.5 py-1 rounded cursor-pointer transition-all"
                style={{
                  background: sortBy === s.id ? (s.id === "rend_asc" ? "#ff224420" : s.id === "rend_desc" ? "#00c97a20"
                  border: `1px solid ${sortBy === s.id ? (s.id === "rend_asc" ? "#ff2244" : s.id === "rend_desc" ? "#00c
                  color: sortBy === s.id ? (s.id === "rend_asc" ? "#ff2244" : s.id === "rend_desc" ? "#00c97a" : "#00d4f
                }}>
                {s.label}
              </button>
            ))}
          </div>
          <div className="relative">
            <Search className="absolute left-2 top-1/2 -translate-y-1/2 w-3.5 h-3.5" style={{ color: "#3a6080" }} />
            <input
              type="text"
              value={searchTerm}
              onChange={e => setSearchTerm(e.target.value)}
              placeholder="Buscar conductor o patente..."
              data-testid="input-search-conductor"
              className="pl-7 pr-3 py-1.5 rounded font-exo text-[11px]"
              style={{ background: "#0d203550", border: "1px solid #0d2035", color: "#c8e8ff", width: "220px" }}
            />
          </div>
        </div>
      </div>

      <div className="rounded overflow-hidden" style={{ border: "1px solid #0d2035" }}>
        <table className="w-full">
          <thead>
            <tr style={{ background: "#0d203540" }}>
              {["CONDUCTOR", "CAMIONES", "CARGAS", "LITROS TOTAL", "KM TOTAL", "REND km/L", "ULT. CARGA"].map(h => (
                <th key={h} className="px-3 py-2 text-left font-space text-[10px] font-bold tracking-wider" style={{ col
                  {h}
                </th>
              ))}
            </tr>
          </thead>
          <tbody>
            {filtered.map((cond: any) => (
              <Fragment key={cond.nombre}>
                <tr
                  onClick={() => setExpandedConductor(expandedConductor === cond.nombre ? null : cond.nombre)}
                  className="cursor-pointer transition-all"
                  style={{
                    background: expandedConductor === cond.nombre ? "#00d4ff08" : "transparent",
                    borderBottom: "1px solid #0d203540",
                  }}
                  data-testid={`row-conductor-${cond.nombre.replace(/[^a-zA-Z]/g, "").slice(0, 15)}`}
                >
                  <td className="px-3 py-2">
                    <div className="flex items-center gap-2">
                      <Users className="w-3.5 h-3.5 flex-shrink-0" style={{ color: "#00d4ff" }} />
                      <span className="font-space text-[11px] font-bold" style={{ color: "#c8e8ff" }}>{cond.nombre}</spa
                      {expandedConductor === cond.nombre ? <ChevronUp className="w-3 h-3" style={{ color: "#3a6080" }} /
                    </div>
                  </td>
                  <td className="px-3 py-2">
                    <div className="flex gap-1">
                      {cond.camiones.map((p: string) => (
                        <span key={p} className="font-space text-[10px] px-1.5 py-0.5 rounded" style={{ background: "#ff
                      ))}
                    </div>
                  </td>
                  <td className="px-3 py-2 font-space text-[11px]" style={{ color: "#c8e8ff" }}>{cond.cargas}</td>
                  <td className="px-3 py-2 font-space text-[11px] font-bold" style={{ color: "#ff6600" }}>{Math.round(co
                  <td className="px-3 py-2 font-space text-[11px]" style={{ color: "#c8e8ff" }}>{cond.kmTotales > 0 ? co
                  <td className="px-3 py-2 font-space text-[11px] font-bold" style={{ color: cond.rendimiento >= 3.5 ? "
                    {cond.rendimiento > 0 ? cond.rendimiento.toFixed(2) : "--"}
                  </td>
                  <td className="px-3 py-2 font-exo text-[10px]" style={{ color: "#3a6080" }}>
                    {cond.ultimaCarga ? new Date(cond.ultimaCarga).toLocaleDateString("es-CL") : "--"}
                  </td>
                </tr>
                {expandedConductor === cond.nombre && (
```

```
                    <tr>
                      <td colSpan={7} className="px-3 py-2" style={{ background: "#0d203515" }}>
                        <div className="ml-6 space-y-1 max-h-[300px] overflow-y-auto">
                          <div className="font-space text-[10px] font-bold tracking-wider mb-2" style={{ color: "#00d4ff"
                            DETALLE DE CARGAS ({cond.cargasDetalle.length})
                          </div>
                          {cond.cargasDetalle.map((c: any, i: number) => (
                            <div key={i} className="flex items-center gap-3 p-1.5 rounded" style={{ background: "#0d203530
                              <span className="font-exo text-[10px] w-24 flex-shrink-0" style={{ color: "#c8e8ff" }}>
                                {new Date(c.fecha).toLocaleDateString("es-CL")} {new Date(c.fecha).toLocaleTimeString("es-
                              </span>
                              <span className="font-space text-[10px] font-bold w-14 text-right flex-shrink-0" style={{ co
                              <span className="font-space text-[10px] w-12 flex-shrink-0 px-1.5 py-0.5 rounded text-center
                              <span className="font-exo text-[9px] flex-1 truncate" style={{ color: "#3a6080" }}>{c.lugar}
                              <span className="font-exo text-[9px] w-16 text-right flex-shrink-0" style={{ color: "#3a6080
                              <span className="font-space text-[10px] w-16 text-right flex-shrink-0 font-bold" style={{ co
                                {c.rend > 0 && c.rend < 100 ? `${c.rend.toFixed(2)} km/L` : "--"}
                              </span>
                            </div>
                          ))}
                        </div>
                      </td>
                    </tr>
                  )}
                </Fragment>
              ))}
            </tbody>
          </table>
        </div>
      </div>
    );
  }

  function RecopilacionTab() {
    const { data: stats, isLoading } = useQuery<any>({
      queryKey: ["/api/geo/stats"],
      refetchInterval: 30000,
    });

    const ingestMutation = useMutation({
      mutationFn: () => apiRequest("POST", "/api/geo/ingest-volvo"),
      onSuccess: () => {
        queryClient.invalidateQueries({ queryKey: ["/api/geo/stats"] });
      },
    });

    if (isLoading) {
      return (
        <div className="flex items-center justify-center py-20">
          <RefreshCw className="w-5 h-5 animate-spin" style={{ color: "#3a6080" }} />
        </div>
      );
    }

    const kpis = [
      { label: "PUNTOS GPS TOTAL", value: stats?.totalPuntos?.toLocaleString() || "0", color: "#00d4ff" },
      { label: "CAMIONES RASTREADOS", value: stats?.totalCamiones || 0, color: "#00c97a" },
      { label: "PUNTOS HOY", value: stats?.puntosHoy?.toLocaleString() || "0", color: "#ffcc00" },
      { label: "PUNTOS 7 DIAS", value: stats?.puntos7d?.toLocaleString() || "0", color: "#ff6600" },
    ];

    return (
      <div data-testid="recopilacion-tab">
        <div className="flex items-center justify-between mb-4">
          <div className="font-exo text-[13px] font-bold tracking-wider" style={{ color: "#c8e8ff" }}>
            RECOPILACION DE DATOS GPS
          </div>
          <button
            onClick={() => ingestMutation.mutate()}
            disabled={ingestMutation.isPending}
            data-testid="button-ingest-manual"
            className="flex items-center gap-2 px-4 py-2 font-exo text-[11px] font-bold tracking-wider rounded cursor-poin
            style={{
              background: "rgba(0,212,255,0.12)",
              border: "1px solid #00d4ff",
              color: "#00d4ff",
            }}
          >
            <RefreshCw className={`w-3.5 h-3.5 ${ingestMutation.isPending ? "animate-spin" : ""}`} />
            {ingestMutation.isPending ? "INGESTING..." : "INGESTAR AHORA"}
          </button>
        </div>

        <div className="grid grid-cols-4 gap-3 mb-6">
          {kpis.map((k, i) => (
            <div key={i} className="rounded-lg p-4" style={{
              background: "rgba(6,13,20,0.6)",
              border: `1px solid ${k.color}30`,
            }} data-testid={`kpi-recopilacion-${i}`}>
              <div class="font-exo text-[10px] font-bold tracking-wider mb-1" style={{ color: "#3a6080" }}>
                {k.label}
              </div>
              <div class="font-space text-[24px] font-bold" style={{ color: k.color }}>
                {k.value}
              </div>
            </div>
          ))}
        </div>

        <div class="grid grid-cols-2 gap-4 mb-6">
          <div class="rounded-lg p-4" style={{
            background: "rgba(6,13,20,0.6)",
            border: "1px solid #0d2035",
          }}>
            <div class="font-exo text-[11px] font-bold tracking-wider mb-1" style={{ color: "#3a6080" }}>
              PRIMER PUNTO REGISTRADO
            </div>
```

```
1817 |           <div className="font-space text-[14px]" style={{ color: "#c8e8ff" }}>
1818 |             {stats?.primerPunto ? new Date(stats.primerPunto).toLocaleString("es-CL") : "---"}
1819 |           </div>
1820 |         </div>
1821 |         <div className="rounded-lg p-4" style={{
1822 |           background: "rgba(6,13,20,0.6)",
1823 |           border: "1px solid #0d2035",
1824 |         }}>
1825 |           <div className="font-exo text-[11px] font-bold tracking-wider mb-1" style={{ color: "#3a6080" }}>
1826 |             ULTIMO PUNTO REGISTRADO
1827 |           </div>
1828 |           <div className="font-space text-[14px]" style={{ color: "#c8e8ff" }}>
1829 |             {stats?.ultimoPunto ? new Date(stats.ultimoPunto).toLocaleString("es-CL") : "---"}
1830 |           </div>
1831 |         </div>
1832 |       </div>
1833 |
1834 |       <div className="rounded-lg p-4" style={{
1835 |         background: "rgba(6,13,20,0.6)",
1836 |         border: "1px solid #0d2035",
1837 |       }}>
1838 |         <div className="flex items-center gap-2 mb-3">
1839 |           <Activity className="w-4 h-4" style={{ color: "#00d4ff" }} />
1840 |           <div className="font-exo text-[12px] font-bold tracking-wider" style={{ color: "#c8e8ff" }}>
1841 |             HISTORIAL DIARIO (ULTIMOS 14 DIAS)
1842 |           </div>
1843 |         </div>
1844 |         <div className="font-exo text-[10px] tracking-wider mb-2" style={{ color: "#3a6080" }}>
1845 |           INGESTION AUTOMATICA CADA 5 MINUTOS - TODOS LOS CONTRATOS
1846 |         </div>
1847 |
1848 |         {stats?.porDia?.length > 0 ? (
1849 |           <div className="space-y-1">
1850 |             <div className="grid grid-cols-[120px_1fr_80px_80px] gap-2 py-1 border-b" style={{ borderColor: "#0d2035" }}
1851 |               <div className="font-exo text-[10px] font-bold tracking-wider" style={{ color: "#3a6080" }}>FECHA</div>
1852 |               <div className="font-exo text-[10px] font-bold tracking-wider" style={{ color: "#3a6080" }}>VOLUMEN</div>
1853 |               <div className="font-exo text-[10px] font-bold tracking-wider text-right" style={{ color: "#3a6080" }}>PUN
1854 |               <div className="font-exo text-[10px] font-bold tracking-wider text-right" style={{ color: "#3a6080" }}>CAM
1855 |             </div>
1856 |             {stats.porDia.map((d: any, i: number) => {
1857 |               const maxPuntos = Math.max(...stats.porDia.map((x: any) => x.puntos));
1858 |               const pct = maxPuntos > 0 ? (d.puntos / maxPuntos) * 100 : 0;
1859 |               return (
1860 |                 <div key={i} className="grid grid-cols-[120px_1fr_80px_80px] gap-2 py-1.5 items-center" data-testid={`ro
1861 |                   <div className="font-space text-[12px]" style={{ color: "#c8e8ff" }}>
1862 |                     {new Date(d.dia).toLocaleDateString("es-CL", { weekday: "short", day: "2-digit", month: "short" })}
1863 |                   </div>
1864 |                   <div className="h-3 rounded-full overflow-hidden" style={{ background: "#0d2035" }}>
1865 |                     <div className="h-full rounded-full transition-all" style={{
1866 |                       width: `${pct}%`,
1867 |                       background: `linear-gradient(90deg, #00d4ff, #00c97a)`,
1868 |                     }} />
1869 |                   </div>
1870 |                   <div className="font-space text-[12px] text-right" style={{ color: "#00d4ff" }}>
1871 |                     {d.puntos.toLocaleString()}
1872 |                   </div>
1873 |                   <div className="font-space text-[12px] text-right" style={{ color: "#00c97a" }}>
1874 |                     {d.camiones}
1875 |                   </div>
1876 |                 </div>
1877 |               );
1878 |             })}
1879 |           </div>
1880 |         ) : (
1881 |           <div className="text-center py-8">
1882 |             <Database className="w-8 h-8 mx-auto mb-2" style={{ color: "#3a6080" }} />
1883 |             <div className="font-exo text-[12px]" style={{ color: "#3a6080" }}>
1884 |               SIN DATOS AUN - LA INGESTION AUTOMATICA COMENZARA A ACUMULAR PUNTOS GPS
1885 |             </div>
1886 |           </div>
1887 |         )}
1888 |       </div>
1889 |
1890 |       <div className="mt-4 rounded-lg p-4" style={{
1891 |         background: "rgba(0,212,255,0.05)",
1892 |         border: "1px solid rgba(0,212,255,0.15)",
1893 |       }}>
1894 |         <div className="font-exo text-[11px] font-bold tracking-wider mb-2" style={{ color: "#00d4ff" }}>
1895 |           COMO FUNCIONA LA RECOPILACION
1896 |         </div>
1897 |         <div className="font-rajdhani text-[13px] leading-relaxed space-y-1" style={{ color: "#6a90aa" }}>
1898 |           <p>Cada 5 minutos se consulta Volvo Connect por la posicion GPS de todos los camiones con VIN registrado.</p>
1899 |           <p>Los puntos se almacenan con coordenadas, velocidad, rumbo y odometro para reconstruir viajes.</p>
1900 |           <p>Con suficientes datos se podran detectar patrones: rutas habituales, paradas sospechosas, desvios y tiempos
1901 |           <p>Mientras mas datos se acumulen, mejor sera la deteccion de anomalias en el comportamiento de cada camion.</
1902 |         </div>
1903 |       </div>
1904 |     </div>
1905 |   );
1906 | }
1907 |
1908 | export default function GeoValidator() {
1909 |   const [tab, setTab] = useState<GeoTab>("mapa");
1910 |
1911 |   const subtabs: { id: GeoTab; label: string; icon: typeof Map }[] = [
1912 |     { id: "mapa", label: "MAPA", icon: Map },
1913 |     { id: "recopilacion", label: "RECOPILACION", icon: Database },
1914 |     { id: "viajes", label: "VIAJES CERRADOS", icon: Route },
1915 |     { id: "conductores", label: "CONDUCTORES", icon: Users },
1916 |     { id: "validacion", label: "VALIDACION", icon: CheckCircle },
1917 |     { id: "ia", label: "IA", icon: Cpu },
1918 |   ];
1919 |
1920 |   return (
1921 |     <div data-testid="geovalidator">
1922 |       <div className="flex items-center justify-between mb-4">
1923 |         <div className="flex items-center gap-3">
```

```
1924 |           <div className="font-space text-[18px] font-bold tracking-[0.2em]" style={{ color: "#00d4ff" }}>
1925 |             GEOVALIDATOR
1926 |           </div>
1927 |         </div>
1928 |         <div className="flex gap-1">
1929 |           {subtabs.map(t => {
1930 |             const Icon = t.icon;
1931 |             return (
1932 |               <button key={t.id} onClick={() => setTab(t.id)}
1933 |                 data-testid={`geo-tab-${t.id}`}
1934 |                 className={`flex items-center gap-1.5 px-3 py-2 font-exo text-[10px] font-bold tracking-[0.12em] cursor-
1935 |                   tab === t.id
1936 |                     ? "border-[#00d4ff] text-[#00d4ff]"
1937 |                     : "border-transparent text-[#3a6080] hover:text-[#c8e8ff]"
1938 |                 }`}>
1939 |                 <Icon className="w-3.5 h-3.5" />
1940 |                 {t.label}
1941 |               </button>
1942 |             );
1943 |           })}
1944 |         </div>
1945 |       </div>
1946 |
1947 |       {tab === "mapa" && <MapaEnVivo />}
1948 |       {tab === "recopilacion" && <RecopilacionTab />}
1949 |       {tab === "viajes" && <ViajesCerrados />}
1950 |       {tab === "conductores" && <ConductoresTab />}
1951 |       {tab === "validacion" && <ValidacionSigetra />}
1952 |       {tab === "ia" && <AnalisisIATab />}
1953 |     </div>
1954 |   );
1955 | }
1956 |
```

client/src/pages/ia-tab.tsx

225 lineas

```
1 | import { useState, useRef, useEffect } from "react";
2 | import { apiRequest } from "@/lib/queryClient";
3 | import { Loader2, Send, RotateCcw, Brain } from "lucide-react";
4 | import Inteligencia from "@/pages/inteligencia";
5 | import RankingConductores from "@/pages/ranking-conductores";
6 | import Tarifas from "@/pages/tarifas";
7 |
8 | type IASub = "sotra-ia" | "inteligencia" | "ranking" | "tarifas";
9 |
10 | const IA_TABS: { id: IASub; label: string }[] = [
11 |   { id: "sotra-ia", label: "SOTRA IA" },
12 |   { id: "inteligencia", label: "INTELIGENCIA" },
13 |   { id: "ranking", label: "RANKING CONDUCTORES" },
14 |   { id: "tarifas", label: "TARIFAS" },
15 | ];
16 |
17 | interface ChatMessage {
18 |   role: "user" | "assistant";
19 |   content: string;
20 | }
21 |
22 | const SUGERENCIAS = [
23 |   "Como esta la flota hoy?",
24 |   "Que camion tiene el peor rendimiento?",
25 |   "Que faena consume mas combustible?",
26 |   "Hay alertas criticas activas?",
27 |   "Resumen ejecutivo de la semana",
28 |   "Patrones sospechosos detectados?",
29 | ];
30 |
31 | function SotraIAChat() {
32 |   const [messages, setMessages] = useState<ChatMessage[]>([]);
33 |   const [input, setInput] = useState("");
34 |   const [loading, setLoading] = useState(false);
35 |   const [inputFocused, setInputFocused] = useState(false);
36 |   const messagesEndRef = useRef<HTMLDivElement>(null);
37 |   const inputRef = useRef<HTMLInputElement>(null);
38 |
39 |   const scrollToBottom = () => {
40 |     messagesEndRef.current?.scrollIntoView({ behavior: "smooth" });
41 |   };
42 |
43 |   useEffect(() => {
44 |     scrollToBottom();
45 |   }, [messages]);
46 |
47 |   const sendMessage = async (text: string) => {
48 |     if (!text.trim() || loading) return;
49 |
50 |     const userMsg: ChatMessage = { role: "user", content: text.trim() };
51 |     const newMessages = [...messages, userMsg];
52 |     setMessages(newMessages);
53 |     setInput("");
54 |     setLoading(true);
55 |
56 |     try {
57 |       const res = await apiRequest("POST", "/api/ia/chat", { messages: newMessages });
58 |       const data = await res.json();
59 |       setMessages(prev => [...prev, { role: "assistant", content: data.respuesta }]);
60 |     } catch (e: any) {
61 |       setMessages(prev => [...prev, { role: "assistant", content: `Error: ${e.message}` }]);
62 |     } finally {
63 |       setLoading(false);
64 |       inputRef.current?.focus();
65 |     }
66 |   };
67 |
68 |   const handleNewConversation = () => {
69 |     setMessages([]);
70 |     setInput("");
71 |     inputRef.current?.focus();
72 |   };
73 |
74 |   return (
75 |     <div className="flex flex-col" style={{ height: "calc(100vh - 180px)", minHeight: "500px" }} data-testid="sotra-ia-c
76 |       <div className="flex items-center justify-between mb-3">
77 |         <div className="flex items-center gap-2">
78 |           <Brain className="w-4 h-4" style={{ color: "#00d4ff" }} />
79 |           <span className="font-space text-[12px] font-bold tracking-[0.1em]" style={{ color: "#c8e8ff" }}>
80 |             SOTRA IA
81 |           </span>
82 |           <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>
83 |             Asistente inteligente de flota
84 |           </span>
85 |         </div>
86 |         <button onClick={handleNewConversation}
87 |           className="flex items-center gap-1.5 px-3 py-1.5 font-exo text-[10px] font-bold cursor-pointer transition-all
88 |           style={{ borderColor: "#0d2035", color: "#3a6080" }}
89 |           data-testid="btn-nueva-conversacion">
90 |           <RotateCcw className="w-3 h-3" />
91 |           Nueva conversacion
92 |         </button>
93 |       </div>
94 |
95 |       <div className="flex-1 overflow-y-auto pr-1 space-y-3 mb-3" style={{ scrollbarWidth: "thin", scrollbarColor: "#0d2
96 |         {messages.length === 0 && !loading && (
97 |           <div className="flex flex-col items-center justify-center py-12">
98 |             <div className="octagonal w-16 h-16 flex items-center justify-center mb-4" style={{
99 |               background: "radial-gradient(circle, rgba(0,212,255,0.06) 0%, transparent 70%)",
100 |               border: "1px solid rgba(0,212,255,0.2)",
101 |             }}>
102 |               <Brain className="w-7 h-7" style={{ color: "#00d4ff" }} />
103 |             </div>
104 |             <div className="font-space text-[14px] font-bold tracking-[0.15em] mb-1" style={{ color: "#c8e8ff" }}>
```

```
105 |               SOTRA IA
106 |             </div>
107 |             <div className="font-exo text-[11px] mb-6" style={{ color: "#3a6080" }}>
108 |               Pregunta lo que necesites sobre tu flota
109 |             </div>
110 |             <div className="flex flex-wrap gap-2 justify-center max-w-lg">
111 |               {SUGERENCIAS.map((s, i) => (
112 |                 <button key={i} onClick={() => sendMessage(s)}
113 |                   className="px-3 py-2 font-exo text-[10px] border cursor-pointer transition-all hover:border-[#00d4ff]
114 |                   style={{ borderColor: "#0d2035", color: "#3a6080" }}
115 |                   data-testid={`sugerencia-${i}`}>
116 |                   {s}
117 |                 </button>
118 |               ))}
119 |             </div>
120 |           </div>
121 |         )}
122 |
123 |         {messages.map((msg, i) => (
124 |           <div key={i} className={`flex ${msg.role === "user" ? "justify-end" : "justify-start"}`}
125 |             data-testid={`chat-msg-${i}`}>
126 |             <div className={`max-w-[80%] p-3 ${msg.role === "user" ? "" : ""}`}
127 |               style={msg.role === "user"
128 |                 ? { background: "#0d2035", border: "1px solid rgba(13,32,53,0.8)" }
129 |                 : { background: "#091018", border: "1px solid #0d2035", borderLeft: "3px solid #00d4ff" }
130 |               }>
131 |               {msg.role === "assistant" && (
132 |                 <div className="font-space text-[8px] font-bold tracking-[0.15em] mb-1.5" style={{ color: "#00d4ff" }}>
133 |                   SOTRA IA
134 |                 </div>
135 |               )}
136 |               <div className="font-rajdhani text-[13px] leading-relaxed whitespace-pre-wrap"
137 |                 style={{ color: msg.role === "user" ? "#c8e8ff" : "#c8e8ff" }}>
138 |                 {msg.content}
139 |               </div>
140 |             </div>
141 |           </div>
142 |         ))}
143 |
144 |         {loading && (
145 |           <div className="flex justify-start" data-testid="chat-typing">
146 |             <div className="p-3" style={{ background: "#091018", border: "1px solid #0d2035", borderLeft: "3px solid #00
147 |               <div className="font-space text-[8px] font-bold tracking-[0.15em] mb-1.5" style={{ color: "#00d4ff" }}>
148 |                 SOTRA IA
149 |               </div>
150 |               <div className="flex items-center gap-1">
151 |                 <span class="font-exo text-[11px]" style={{ color: "#3a6080" }}>
152 |                   SOTRA IA esta escribiendo
153 |                 </span>
154 |                 <span class="flex gap-0.5">
155 |                   <span class="w-1 h-1 rounded-full animate-bounce" style={{ background: "#00d4ff", animationDelay:
156 |                   <span class="w-1 h-1 rounded-full animate-bounce" style={{ background: "#00d4ff", animationDelay:
157 |                   <span class="w-1 h-1 rounded-full animate-bounce" style={{ background: "#00d4ff", animationDelay:
158 |                 </span>
159 |               </div>
160 |             </div>
161 |           </div>
162 |         )}
163 |
164 |         <div ref={messagesEndRef} />
165 |       </div>
166 |
167 |       <div className="flex gap-2 pt-2" style={{ borderTop: "1px solid #0d2035" }}>
168 |         <input
169 |           ref={inputRef}
170 |           value={input}
171 |           onChange={e => setInput(e.target.value)}
172 |           onKeyDown={e => e.key === "Enter" && !e.shiftKey && sendMessage(input)}
173 |           onFocus={() => setInputFocused(true)}
174 |           onBlur={() => setInputFocused(false)}
175 |           placeholder="Escribe tu pregunta..."
176 |           className="flex-1 px-4 py-3 font-rajdhani text-sm bg-transparent border outline-none transition-all"
177 |           style={{
178 |             borderColor: inputFocused ? "#00d4ff" : "#0d2035",
179 |             color: "#c8e8ff",
180 |             boxShadow: inputFocused ? "0 0 12px rgba(0,212,255,0.15)" : undefined,
181 |           }}
182 |           data-testid="input-chat"
183 |           disabled={loading}
184 |         />
185 |         <button
186 |           onClick={() => sendMessage(input)}
187 |           disabled={loading || !input.trim()}
188 |           className="octagonal px-5 py-3 font-space text-sm font-bold cursor-pointer disabled:opacity-40 transition-all"
189 |           style={{ background: "rgba(0,212,255,0.1)", border: "1px solid rgba(0,212,255,0.3)", color: "#00d4ff" }}
190 |           data-testid="btn-enviar-chat"
191 |         >
192 |           {loading ? <Loader2 className="w-4 h-4 animate-spin" /> : <Send className="w-4 h-4" />}
193 |         </button>
194 |       </div>
195 |     </div>
196 |   );
197 | }
198 |
199 | export default function IATab() {
200 |   const [activeSub, setActiveSub] = useState<IASub>("sotra-ia");
201 |
202 |   return (
203 |     <div data-testid="page-ia">
204 |       <div className="flex items-center gap-1 mb-4 pb-2" style={{ borderBottom: "1px solid #0d2035" }}>
205 |         {IA_TABS.map(t => (
206 |           <button key={t.id} onClick={() => setActiveSub(t.id)}
207 |             data-testid={`ia-tab-${t.id}`}
208 |             className={`px-4 py-2 font-exo text-[10px] font-bold tracking-[0.15em] cursor-pointer transition-all border-
209 |               activeSub === t.id
210 |                 ? "border-[#00d4ff] text-[#00d4ff]"
211 |                 : "border-transparent text-[#3a6080] hover:text-[#c8e8ff]"
```

```
212 |             }`}>
213 |             {t.label}
214 |           </button>
215 |         ))}
216 |       </div>
217 |
218 |       {activeSub === "sotra-ia" && <SotraIAChat />}
219 |       {activeSub === "inteligencia" && <Inteligencia />}
220 |       {activeSub === "ranking" && <RankingConductores />}
221 |       {activeSub === "tarifas" && <Tarifas />}
222 |     </div>
223 |   );
224 | }
225 |
```

client/src/pages/configuration.tsx

505 lineas

```
1 | import { useState } from "react";
2 | import { useQuery, useMutation } from "@tanstack/react-query";
3 | import { Button } from "@/components/ui/button";
4 | import { Input } from "@/components/ui/input";
5 | import { Skeleton } from "@/components/ui/skeleton";
6 | import { StatusTag, getStatusVariant } from "@/components/status-tag";
7 | import { FaenaModal } from "@/components/faena-modal";
8 | import { CamionModal } from "@/components/camion-modal";
9 | import { statsCamion, type CamionStats } from "@/lib/fuel-utils";
10 | import { apiRequest, queryClient } from "@/lib/queryClient";
11 | import { useToast } from "@/hooks/use-toast";
12 | import { Plus, Pencil, X, Check, Upload, RefreshCw, Wifi, WifiOff, AlertTriangle, Eye, CheckCircle, Loader2, Truck, Chev
13 | import type { Faena, Camion, Carga } from "@shared/schema";
14 | import ExcesosVelocidad from "@/pages/excesos-velocidad";
15 | import PesoTara from "@/pages/peso-tara";
16 | import Errores from "@/pages/errores";
17 |
18 | interface AsignacionPreview {
19 |   total: number;
20 |   porAsignar: number;
21 |   yaAsignados: number;
22 |   sinDatos: number;
23 |   detalle: {
24 |     camionId: number;
25 |     patente: string;
26 |     clienteSigetra: string | null;
27 |     litrosSigetra: number;
28 |     faenaActualId: number;
29 |     faenaActualNombre: string;
30 |     faenaPropuesta: string | null;
31 |     confianza: "alta" | "media" | "sin_datos";
32 |     accion: "ASIGNAR" | "YA_ASIGNADO" | "SIN_DATOS";
33 |   }[];
34 | }
35 |
36 | interface AsignacionResult {
37 |   procesados: number;
38 |   asignados: number;
39 |   sin_datos_sigetra: number;
40 |   faenas_creadas: number;
41 |   detalle: { patente: string; cliente_detectado: string | null; faena_asignada: string | null; accion: string }[];
42 | }
43 |
44 | export default function Configuration() {
45 |   const { toast } = useToast();
46 |   const [faenaModal, setFaenaModal] = useState(false);
47 |   const [editFaena, setEditFaena] = useState<Faena | null>(null);
48 |   const [camionModal, setCamionModal] = useState(false);
49 |   const [editCamion, setEditCamion] = useState<Camion | null>(null);
50 |   const [asignacionPreview, setAsignacionPreview] = useState<AsignacionPreview | null>(null);
51 |   const [asignacionResult, setAsignacionResult] = useState<AsignacionResult | null>(null);
52 |
53 |   const { data: faenas = [], isLoading: loadingF } = useQuery<Faena[]>({ queryKey: ["/api/faenas"] });
54 |   const { data: camiones = [], isLoading: loadingC } = useQuery<Camion[]>({ queryKey: ["/api/camiones"] });
55 |   const { data: cargas = [], isLoading: loadingCg } = useQuery<Carga[]>({ queryKey: ["/api/cargas"] });
56 |   const { data: volvoStatus } = useQuery<{ user: string; configured: boolean; status: string; message: string }>({ query
57 |
58 |   const isLoading = loadingF || loadingC || loadingCg;
59 |
60 |   const deleteFaena = useMutation({
61 |     mutationFn: async (id: number) => { await apiRequest("DELETE", `/api/faenas/${id}`); },
62 |     onSuccess: () => {
63 |       queryClient.invalidateQueries({ queryKey: ["/api/faenas"] });
64 |       toast({ title: "Faena eliminada" });
65 |     },
66 |   });
67 |
68 |   const deleteCamion = useMutation({
69 |     mutationFn: async (id: number) => { await apiRequest("DELETE", `/api/camiones/${id}`); },
70 |     onSuccess: () => {
71 |       queryClient.invalidateQueries({ queryKey: ["/api/camiones"] });
72 |       queryClient.invalidateQueries({ queryKey: ["/api/cargas"] });
73 |       toast({ title: "Camion eliminado" });
74 |     },
75 |   });
76 |
77 |   const syncVolvo = useMutation({
78 |     mutationFn: async () => {
79 |       const res = await apiRequest("POST", "/api/volvo/sync");
80 |       return res.json();
81 |     },
82 |     onSuccess: (data: any) => {
83 |       queryClient.invalidateQueries({ queryKey: ["/api/camiones"] });
84 |       toast({ title: `Sincronizacion completa: ${data.synced} camiones` });
85 |     },
86 |     onError: (error: any) => {
87 |       toast({ title: "Error de sincronizacion", description: error.message, variant: "destructive" });
88 |     },
89 |   });
90 |
91 |   const previewAsignacion = useMutation({
92 |     mutationFn: async () => {
93 |       const res = await fetch("/api/camiones/preview-asignacion");
94 |       if (!res.ok) throw new Error("Error al obtener preview");
95 |       return res.json() as Promise<AsignacionPreview>;
96 |     },
97 |     onSuccess: (data) => {
98 |       setAsignacionPreview(data);
99 |       setAsignacionResult(null);
100 |     },
101 |     onError: (err: any) => toast({ title: "Error", description: err.message, variant: "destructive" }),
102 |   });
103 |
104 |   const confirmarAsignacion = useMutation({
```

```
105 |     mutationFn: async () => {
106 |       const res = await apiRequest("POST", "/api/camiones/asignar-faenas-sigetra");
107 |       return res.json() as Promise<AsignacionResult>;
108 |     },
109 |     onSuccess: (data) => {
110 |       setAsignacionResult(data);
111 |       setAsignacionPreview(null);
112 |       queryClient.invalidateQueries({ queryKey: ["/api/camiones"] });
113 |       queryClient.invalidateQueries({ queryKey: ["/api/faenas"] });
114 |       toast({ title: `Asignacion completada: ${data.asignados} camiones asignados, ${data.faenas_creadas} faenas creadas
115 |     },
116 |     onError: (err: any) => toast({ title: "Error", description: err.message, variant: "destructive" }),
117 |   });
118 |
119 |   if (isLoading) {
120 |     return (
121 |       <div className="grid grid-cols-2 gap-5">
122 |         <Skeleton className="h-64" />
123 |         <Skeleton className="h-64" />
124 |       </div>
125 |     );
126 |   }
127 |
128 |   return (
129 |     <div className="space-y-5" data-testid="config-view">
130 |       <div className="grid grid-cols-2 gap-5">
131 |         <div>
132 |           <div className="flex justify-between items-center mb-3.5 pb-2.5 border-b border-border">
133 |             <div className="text-[13px] font-mono font-bold text-foreground tracking-[0.15em] uppercase">
134 |               Faenas / Clientes
135 |             </div>
136 |             <Button size="sm" onClick={() => { setEditFaena(null); setFaenaModal(true); }}
137 |               data-testid="button-new-faena">
138 |               <Plus className="w-3 h-3 mr-1" /> Nueva
139 |             </Button>
140 |           </div>
141 |           <div className="bg-card border border-border">
142 |             {faenas.length === 0 && (
143 |               <div className="py-8 text-center text-muted-foreground font-mono text-xs" data-testid="text-no-faenas">
144 |                 Sin faenas. Agrega la primera.
145 |               </div>
146 |             )}
147 |             {faenas.map((f, i) => (
148 |               <div key={f.id}
149 |                 className={`flex justify-between items-center px-4 py-3 ${i < faenas.length - 1 ? "border-b border-borde
150 |                 data-testid={`faena-row-${f.id}`}>
151 |                 <div className="flex items-center gap-2.5">
152 |                   <div className="w-3 h-3 rounded-full flex-shrink-0" style={{ backgroundColor: f.color }} />
153 |                   <div>
154 |                     <div className="text-[13px] text-foreground font-semibold">{f.nombre}</div>
155 |                     <div className="text-[10px] font-mono text-muted-foreground mt-0.5">
156 |                       {camiones.filter(c => c.faenaId === f.id).length} camiones
157 |                     </div>
158 |                   </div>
159 |                 </div>
160 |                 <div className="flex gap-1.5">
161 |                   <Button variant="outline" size="sm" className="h-7"
162 |                     onClick={() => { setEditFaena(f); setFaenaModal(true); }}
163 |                     data-testid={`button-edit-faena-${f.id}`}>
164 |                     <Pencil className="w-3 h-3" />
165 |                   </Button>
166 |                   <Button variant="outline" size="sm" className="h-7 text-red-400 border-red-400/30"
167 |                     onClick={() => { if (confirm("Eliminar faena?")) deleteFaena.mutate(f.id); }}
168 |                     data-testid={`button-delete-faena-${f.id}`}>
169 |                     <X className="w-3 h-3" />
170 |                   </Button>
171 |                 </div>
172 |               </div>
173 |             ))}
174 |           </div>
175 |         </div>
176 |
177 |         <div>
178 |           <div className="flex justify-between items-center mb-3.5 pb-2.5 border-b border-border">
179 |             <div className="text-[13px] font-mono font-bold text-foreground tracking-[0.15em] uppercase">
180 |               Camiones
181 |             </div>
182 |             <Button size="sm" onClick={() => { setEditCamion(null); setCamionModal(true); }}
183 |               data-testid="button-new-camion">
184 |               <Plus className="w-3 h-3 mr-1" /> Nuevo
185 |             </Button>
186 |           </div>
187 |           <div className="bg-card border border-border">
188 |             {camiones.length === 0 && (
189 |               <div className="py-8 text-center text-muted-foreground font-mono text-xs" data-testid="text-no-camiones">
190 |                 Sin camiones.
191 |               </div>
192 |             )}
193 |             {camiones.map((cam, i) => {
194 |               const faena = faenas.find(f => f.id === cam.faenaId);
195 |               const cams = cargas.filter(c => c.camionId === cam.id);
196 |               const s = statsCamion(cam, cams);
197 |               return (
198 |                 <div key={cam.id}
199 |                   className={`flex justify-between items-center px-4 py-3 ${
200 |                     i < camiones.length - 1 ? "border-b border-border/20" : ""
201 |                   } ${s.estado === "CRITICO" ? "bg-red-500/[0.03]" : ""}`}
202 |                   data-testid={`camion-row-${cam.id}`}>
203 |                   <div className="flex items-center gap-2.5">
204 |                     <div className="w-2.5 h-2.5 rounded-full flex-shrink-0"
205 |                       style={{ backgroundColor: faena?.color || "#5A7A9A" }} />
206 |                     <div>
207 |                       <div className="flex items-center gap-2">
208 |                         <span className="text-sm font-mono font-bold text-primary">{cam.patente}</span>
209 |                         <StatusTag label={s.estado} variant={getStatusVariant(s.estado)} />
210 |                       </div>
211 |                       <div className="text-[10px] text-muted-foreground mt-0.5">
```

```
212 |                        {cam.modelo} · {faena?.nombre || "Sin faena"} · meta {cam.metaKmL} km/L
213 |                      </div>
214 |                    </div>
215 |                  </div>
216 |                  <div className="flex items-center gap-2">
217 |                    <div className={`text-[10px] font-mono mr-2 ${cam.syncOk ? "text-emerald-400" : "text-red-400"}`}>
218 |                      {cam.syncOk ? "Sync OK" : "Sin sync"}
219 |                    </div>
220 |                    <Button variant="outline" size="sm" className="h-7"
221 |                      onClick={() => { setEditCamion(cam); setCamionModal(true); }}
222 |                      data-testid={`button-edit-camion-${cam.id}`}>
223 |                      <Pencil className="w-3 h-3" />
224 |                    </Button>
225 |                    <Button variant="outline" size="sm" className="h-7 text-red-400 border-red-400/30"
226 |                      onClick={() => { if (confirm("Eliminar camion?")) deleteCamion.mutate(cam.id); }}
227 |                      data-testid={`button-delete-camion-${cam.id}`}>
228 |                      <X className="w-3 h-3" />
229 |                    </Button>
230 |                  </div>
231 |                </div>
232 |              );
233 |            })}
234 |          </div>
235 |        </div>
236 |      </div>
237 |
238 |      <div className="bg-card border border-border p-5" data-testid="section-asignacion-faenas">
239 |        <div className="flex justify-between items-center mb-4">
240 |          <div className="text-[13px] font-mono font-bold text-foreground tracking-[0.15em] uppercase">
241 |            Asignacion de Faenas por Sigetra
242 |          </div>
243 |          <div className="flex gap-2">
244 |            <Button variant="outline" size="sm"
245 |              disabled={previewAsignacion.isPending}
246 |              onClick={() => previewAsignacion.mutate()}
247 |              data-testid="button-preview-asignacion">
248 |              {previewAsignacion.isPending ? <Loader2 className="w-3.5 h-3.5 mr-1.5 animate-spin" /> : <Eye className="w
249 |              Previsualizar
250 |            </Button>
251 |            <Button size="sm"
252 |              className="bg-emerald-500 text-black font-mono disabled:opacity-50"
253 |              disabled={confirmarAsignacion.isPending || (!asignacionPreview || asignacionPreview.porAsignar === 0)}
254 |              onClick={() => { if (confirm("Confirmar asignacion de faenas basada en historial Sigetra?")) confirmarAsig
255 |              data-testid="button-confirmar-asignacion">
256 |              {confirmarAsignacion.isPending ? <Loader2 className="w-3.5 h-3.5 mr-1.5 animate-spin" /> : <CheckCircle cl
257 |              Confirmar y Asignar
258 |            </Button>
259 |          </div>
260 |        </div>
261 |
262 |        <div className="text-[11px] font-mono text-muted-foreground mb-3">
263 |          Detecta el cliente principal de cada camion usando el historial de cargas en Sigetra y asigna automaticamente
264 |        </div>
265 |
266 |        {asignacionResult && (
267 |          <div className="bg-emerald-500/10 border border-emerald-500/30 p-4 mb-4" data-testid="asignacion-result">
268 |            <div className="text-[12px] font-mono font-bold text-emerald-400 mb-2">Asignacion completada</div>
269 |            <div className="grid grid-cols-4 gap-3 text-[11px] font-mono">
270 |              <div><span className="text-muted-foreground">Procesados:</span> <span className="text-foreground font-bold
271 |              <div><span className="text-muted-foreground">Asignados:</span> <span className="text-emerald-400 font-bold
272 |              <div><span className="text-muted-foreground">Faenas creadas:</span> <span className="text-primary font-bol
273 |              <div><span className="text-muted-foreground">Sin datos:</span> <span className="text-muted-foreground font
274 |            </div>
275 |          </div>
276 |        )}
277 |
278 |        {asignacionPreview && (
279 |          <div data-testid="asignacion-preview">
280 |            <div className="grid grid-cols-4 gap-3 mb-3">
281 |              <div className="bg-background border border-border p-2.5 text-center">
282 |                <div className="text-lg font-mono font-bold text-primary">{asignacionPreview.total}</div>
283 |                <div className="text-[9px] font-mono text-muted-foreground tracking-wider">TOTAL</div>
284 |              </div>
285 |              <div className="bg-background border border-border p-2.5 text-center">
286 |                <div className="text-lg font-mono font-bold text-emerald-400">{asignacionPreview.porAsignar}</div>
287 |                <div className="text-[9px] font-mono text-muted-foreground tracking-wider">POR ASIGNAR</div>
288 |              </div>
289 |              <div className="bg-background border border-border p-2.5 text-center">
290 |                <div className="text-lg font-mono font-bold text-primary">{asignacionPreview.yaAsignados}</div>
291 |                <div className="text-[9px] font-mono text-muted-foreground tracking-wider">YA ASIGNADOS</div>
292 |              </div>
293 |              <div className="bg-background border border-border p-2.5 text-center">
294 |                <div className="text-lg font-mono font-bold text-muted-foreground">{asignacionPreview.sinDatos}</div>
295 |                <div className="text-[9px] font-mono text-muted-foreground tracking-wider">SIN DATOS</div>
296 |              </div>
297 |            </div>
298 |
299 |            <div className="border border-border max-h-[400px] overflow-y-auto">
300 |              <table className="w-full text-[11px] font-mono">
301 |                <thead className="bg-background sticky top-0">
302 |                  <tr className="text-[9px] text-muted-foreground tracking-wider">
303 |                    <th className="text-left px-3 py-2">PATENTE</th>
304 |                    <th className="text-left px-3 py-2">CLIENTE SIGETRA</th>
305 |                    <th className="text-right px-3 py-2">LITROS</th>
306 |                    <th className="text-left px-3 py-2">FAENA ACTUAL</th>
307 |                    <th className="text-left px-3 py-2">FAENA PROPUESTA</th>
308 |                    <th class="text-center px-3 py-2">ACCION</th>
309 |                  </tr>
310 |                </thead>
311 |                <tbody>
312 |                  {asignacionPreview.detalle.map((row, i) => (
313 |                    <tr key={row.camionId}
314 |                      className={`border-t border-border/30 ${row.accion === "ASIGNAR" ? "bg-emerald-500/[0.03]" : ""}`}
315 |                      data-testid={`asignacion-row-${row.patente}`}>
316 |                      <td className="px-3 py-1.5 font-bold text-primary">{row.patente}</td>
317 |                      <td className="px-3 py-1.5" style={{ color: row.clienteSigetra ? '#c8e8ff' : '#3a6080' }}>
318 |                        {row.clienteSigetra || "\u2014"}
```

```
                      </td>
                      <td className="px-3 py-1.5 text-right text-muted-foreground">
                        {row.litrosSigetra > 0 ? row.litrosSigetra.toLocaleString() + " L" : "\u2014"}
                      </td>
                      <td className="px-3 py-1.5 text-muted-foreground">{row.faenaActualNombre}</td>
                      <td className="px-3 py-1.5" style={{
                        color: row.confianza === "alta" ? "#00ff88" : row.confianza === "media" ? "#ffcc00" : "#3a6080"
                      }}>
                        {row.faenaPropuesta || "\u2014"}
                      </td>
                      <td className="px-3 py-1.5 text-center">
                        <span className={`inline-block px-2 py-0.5 text-[9px] font-bold tracking-wider ${
                          row.accion === "ASIGNAR" ? "text-emerald-400 bg-emerald-500/10 border border-emerald-500/30" :
                          row.accion === "YA_ASIGNADO" ? "text-primary bg-primary/10 border border-primary/30" :
                          "text-muted-foreground bg-muted/10 border border-border"
                        }`}>
                          {row.accion === "ASIGNAR" ? "ASIGNAR" : row.accion === "YA_ASIGNADO" ? "OK" : "SIN DATOS"}
                        </span>
                      </td>
                    </tr>
                  ))}
                </tbody>
              </table>
            </div>
          </div>
        )}

        {!asignacionPreview && !asignacionResult && (
          <div className="bg-background border border-border py-6 text-center">
            <Truck className="w-6 h-6 mx-auto mb-2 text-muted-foreground/40" />
            <div className="text-[11px] font-mono text-muted-foreground">
              Presiona Previsualizar para ver la asignacion propuesta antes de confirmar
            </div>
          </div>
        )}
      </div>

      <div className="bg-card border border-border p-5">
        <div className="flex justify-between items-center mb-4">
          <div className="text-[13px] font-mono font-bold text-foreground tracking-[0.15em] uppercase">
            Volvo rFMS API
          </div>
          {volvoStatus && (
            <div className={`flex items-center gap-2 text-[11px] font-mono ${
              volvoStatus.status === "connected" ? "text-emerald-400" :
              volvoStatus.status === "forbidden" ? "text-amber-400" :
              "text-red-400"
            }`} data-testid="text-volvo-api-status">
              {volvoStatus.status === "connected" ? <Wifi className="w-3.5 h-3.5" /> :
               volvoStatus.status === "forbidden" ? <AlertTriangle className="w-3.5 h-3.5" /> :
               <WifiOff className="w-3.5 h-3.5" />}
              {volvoStatus.status === "connected" ? "Conectado" :
               volvoStatus.status === "forbidden" ? "Pendiente activacion" :
               volvoStatus.status === "unauthorized" ? "No autorizado" :
               volvoStatus.status === "not_configured" ? "No configurado" : "Error"}
            </div>
          )}
        </div>
        <div className="grid grid-cols-2 gap-4">
          <div class="space-y-3">
            <div>
              <label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] uppercase block mb-1.5">
                Usuario API
              </label>
              <Input value={volvoStatus?.user || ""} readOnly className="font-mono bg-background" data-testid="input-vol
            </div>
            <div>
              <label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] uppercase block mb-1.5">
                Contrasena
              </label>
              <Input type="password" value="********" readOnly className="font-mono bg-background" data-testid="input-vo
            </div>

            {volvoStatus?.status === "forbidden" && (
              <div className="bg-amber-500/10 border border-amber-500/30 p-3 text-[11px] font-mono text-amber-300 leadin
                <AlertTriangle className="w-4 h-4 inline mr-1.5 -mt-0.5" />
                Credenciales validas pero acceso API no activado. Contactar a Volvo para activar acceso rFMS en la cuent
              </div>
            )}
            {volvoStatus?.status === "connected" && (
              <div className="flex items-center gap-2 text-[11px] font-mono text-emerald-400">
                <span className="w-2 h-2 rounded-full bg-emerald-400" />
                API rFMS activa - api.volvotrucks.com
              </div>
            )}

            <Button
              className="w-full bg-emerald-500 text-black font-mono disabled:opacity-50"
              disabled={volvoStatus?.status !== "connected" || syncVolvo.isPending}
              onClick={() => syncVolvo.mutate()}
              data-testid="button-sync-volvo">
              {syncVolvo.isPending ? (
                <><RefreshCw className="w-3.5 h-3.5 mr-1.5 animate-spin" /> Sincronizando...</>
              ) : (
                <><RefreshCw className="w-3.5 h-3.5 mr-1.5" /> Sincronizar flota</>
              )}
            </Button>
            {volvoStatus?.status !== "connected" && (
              <p class="text-[10px] text-muted-foreground/60 font-mono">
                La sincronizacion se habilitara cuando Volvo active el acceso API.
              </p>
            )}
          </div>
          <div class="bg-background border border-border p-4">
            <div class="text-[9px] font-mono text-primary tracking-[0.15em] mb-3">
              DATOS QUE SE SINCRONIZAN POR CAMION (rFMS v2.1)
            </div>
```

```
426 |             {[
427 |               "VIN e identificacion del vehiculo",
428 |               "Odometro (km totales ECU)",
429 |               "Consumo acumulado (litros ECU)",
430 |               "Horas motor encendido",
431 |               "Nivel de combustible",
432 |               "Posicion GPS (latitud/longitud)",
433 |               "Velocidad (tacografo y rueda)",
434 |               "Tipo de emision y ejes",
435 |             ].map((d, i) => (
436 |               <div key={i} className={`text-[11px] font-mono text-muted-foreground py-1.5 ${
437 |                 i < 7 ? "border-b border-border/20" : ""
438 |               }`}>
439 |                 <span className="text-emerald-400 mr-1.5">
440 |                   <Check className="w-3 h-3 inline" />
441 |                 </span>
442 |                 {d}
443 |               </div>
444 |             ))}
445 |           </div>
446 |         </div>
447 |       </div>
448 |
449 |       <div className="bg-card border border-border p-5">
450 |         <div className="text-[13px] font-mono font-bold text-foreground tracking-[0.15em] uppercase mb-3.5">
451 |           Importar Tarjetas Shell / Copec (Excel Sigetra)
452 |         </div>
453 |         <div className="bg-background border-2 border-dashed border-border py-8 text-center cursor-pointer transition-co
454 |           data-testid="upload-zone">
455 |           <Upload className="w-7 h-7 mx-auto mb-2 text-muted-foreground" />
456 |           <div className="text-[13px] font-mono text-muted-foreground">
457 |             Arrastra el Excel aqui o haz clic para subir
458 |           </div>
459 |           <div className="text-[10px] font-mono text-muted-foreground/60 mt-1.5">
460 |             Columnas: Patente · Fecha · Litros · Km · Proveedor
461 |           </div>
462 |         </div>
463 |       </div>
464 |
465 |       <ConfigAccordion title="EXCESOS DE VELOCIDAD">
466 |         <ExcesosVelocidad />
467 |       </ConfigAccordion>
468 |
469 |       <ConfigAccordion title="PESO Y TARA">
470 |         <PesoTara />
471 |       </ConfigAccordion>
472 |
473 |       <ConfigAccordion title="CALIDAD DE DATOS SIGETRA">
474 |         <Errores />
475 |       </ConfigAccordion>
476 |
477 |       {faenaModal && (
478 |         <FaenaModal faena={editFaena} open={faenaModal}
479 |           onClose={() => { setFaenaModal(false); setEditFaena(null); }} />
480 |       )}
481 |       {camionModal && (
482 |         <CamionModal faenas={faenas} camion={editCamion} open={camionModal}
483 |           onClose={() => { setCamionModal(false); setEditCamion(null); }} />
484 |       )}
485 |     </div>
486 |   );
487 | }
488 |
489 | function ConfigAccordion({ title, children }: { title: string; children: React.ReactNode }) {
490 |   const [open, setOpen] = useState(false);
491 |   return (
492 |     <div style={{ border: "1px solid #0d2035" }} className="mb-3">
493 |       <button
494 |         onClick={() => setOpen(!open)}
495 |         className="w-full flex items-center justify-between p-4 cursor-pointer transition-all hover:bg-[rgba(0,212,255,0
496 |         style={{ background: "#091018" }}
497 |         data-testid={`config-section-${title.toLowerCase().replace(/\s+/g, "-")}`}>
498 |         <span className="font-space text-[11px] font-bold tracking-[0.15em]" style={{ color: "#c8e8ff" }}>{title}</span>
499 |         {open ? <ChevronDown className="w-4 h-4" style={{ color: "#3a6080" }} /> : <ChevronRight className="w-4 h-4" sty
500 |       </button>
501 |       {open && <div className="p-4" style={{ borderTop: "1px solid #0d2035" }}>{children}</div>}
502 |     </div>
503 |   );
504 | }
505 |
```

client/src/pages/dashboard.tsx

619 lineas

```
1 | import { useState, useMemo, lazy, Suspense } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { Skeleton } from "@/components/ui/skeleton";
4 | import { statsCamion, fN, type CamionStats, type Percentiles } from "@/lib/fuel-utils";
5 | import { X, Truck, Loader2, Fuel, ChevronRight, Calendar } from "lucide-react";
6 | import type { Faena, Camion, Carga } from "@shared/schema";
7 |
8 | const VisionCEO = lazy(() => import("./vision-ceo"));
9 |
10 | interface HeroData {
11 |   totalCamiones: number;
12 |   totalFaenas: number;
13 |   litros30d: number;
14 |   km30d: number;
15 |   cargas30d: number;
16 |   rendPromedio: number;
17 |   camionesConVin: number;
18 |   camionesOnline: number;
19 |   conductores: number;
20 | }
21 |
22 | interface FaenaRow {
23 |   id: number;
24 |   nombre: string;
25 |   color: string;
26 |   camiones: number;
27 |   litros30d: number;
28 |   rendPromedio: number;
29 |   criticos: number;
30 |   alertas: number;
31 |   estado: string;
32 | }
33 |
34 | interface SistemaData {
35 |   volvo: { status: string; vehiculos: number; configured: boolean; health: number; error?: string };
36 |   sigetra: { status: string; connected: boolean; user: string; health: number };
37 |   ia: { status: string; configured: boolean; model: string; health: number };
38 |   ultimoSync: string;
39 |   healthGeneral: number;
40 | }
41 |
42 | function AnimatedRings() {
43 |   return (
44 |     <div className="relative w-[160px] h-[160px] mx-auto">
45 |       <div className="absolute inset-0 rounded-full" style={{ border: '1px solid rgba(0,212,255,0.1)', animation: 'ringR
46 |       <div className="absolute inset-[12px] rounded-full" style={{ border: '1px solid rgba(0,212,255,0.15)', animation:
47 |       <div className="absolute inset-[24px] rounded-full" style={{ border: '1px dashed rgba(0,212,255,0.2)', animation:
48 |       <div className="absolute inset-0 flex items-center justify-center">
49 |         <div className="octagonal w-[90px] h-[90px] flex items-center justify-center" style={{
50 |           background: 'radial-gradient(circle, rgba(0,212,255,0.08) 0%, transparent 70%)',
51 |           border: '1px solid rgba(0,212,255,0.3)',
52 |           animation: 'glowPulse 3s ease-in-out infinite'
53 |         }}>
54 |           <span className="text-4xl">&#9981;</span>
55 |         </div>
56 |       </div>
57 |     </div>
58 |   );
59 | }
60 |
61 | function PercentileBar({ label, value, color, maxVal }: { label: string; value: number; color: string; maxVal: number })
62 |   const pct = maxVal > 0 ? Math.min((value / maxVal) * 100, 100) : 0;
63 |   return (
64 |     <div className="flex items-center gap-3">
65 |       <span className="font-exo text-[9px] tracking-wider w-7" style={{ color: '#3a6080' }}>{label}</span>
66 |       <div className="flex-1 h-[6px] rounded-sm overflow-hidden" style={{ background: '#0d2035' }}>
67 |         <div className="h-full rounded-sm transition-all duration-1000" style={{ width: `${pct}%`, background: color, an
68 |       </div>
69 |       <span className="font-space text-[11px] font-bold w-16 text-right" style={{ color }}>{value > 0 ? `${value.toFixed
70 |     </div>
71 |   );
72 | }
73 |
74 | type FaenaDetailData = {
75 |   faena: Faena;
76 |   camiones: (Camion & { litros30d: number; rendimiento: number; cargas: number; viajes: number })[];
77 |   contrato: { id: number; nombre: string; totalViajes: number | null } | null;
78 |   viajes: { id: number; codigo: string; camionPatente: string; fechaSalida: string; estado: string; km: number; rendimie
79 |   resumen: { totalCamiones: number; litros30d: number; rendPromedio: number; criticos: number; alertas: number; totalVia
80 | };
81 |
82 | function formatFecha(d: string | null | undefined) {
83 |   if (!d) return "\u2014";
84 |   return new Date(d).toLocaleDateString("es-CL", { day: "2-digit", month: "short", hour: "2-digit", minute: "2-digit" })
85 | }
86 |
87 | function FaenaDetailModal({ faenaId, onClose }: { faenaId: number; onClose: () => void }) {
88 |   const { data, isLoading, isError, refetch } = useQuery<FaenaDetailData>({
89 |     queryKey: ["/api/dashboard/faena", faenaId],
90 |   });
91 |
92 |   if (isLoading) {
93 |     return (
94 |       <div className="fixed inset-0 z-50 flex items-center justify-center" style={{ background: 'rgba(2,5,8,0.88)' }}
95 |         onClick={onClose}>
96 |         <div className="dash-card p-8 max-w-lg w-full text-center" onClick={e => e.stopPropagation()}>
97 |           <Loader2 className="w-8 h-8 animate-spin mx-auto" style={{ color: '#00d4ff' }} />
98 |           <div className="font-exo text-[11px] mt-3" style={{ color: '#3a6080' }}>Cargando faena...</div>
99 |           <button onClick={onClose} className="mt-4 font-exo text-[10px] px-3 py-1 cursor-pointer border transition-all
100 |             style={{ color: '#3a6080', borderColor: '#0d2035' }}>Cancelar</button>
101 |         </div>
102 |       </div>
103 |     );
104 |   }
```

```
105 |
106 |   if (isError || !data) {
107 |     return (
108 |       <div className="fixed inset-0 z-50 flex items-center justify-center" style={{ background: 'rgba(2,5,8,0.88)' }}
109 |         onClick={onClose}>
110 |         <div className="dash-card p-8 max-w-lg w-full text-center" onClick={e => e.stopPropagation()}>
111 |           <div className="font-space text-[13px] font-bold" style={{ color: '#ff2244' }}>Error al cargar faena</div>
112 |           <div className="font-exo text-[11px] mt-2" style={{ color: '#3a6080' }}>No se pudo obtener la informacion de e
113 |           <div className="flex gap-3 justify-center mt-4">
114 |             <button onClick={() => refetch()} className="font-exo text-[10px] px-3 py-1 cursor-pointer border transition
115 |               style={{ color: '#00d4ff', borderColor: 'rgba(0,212,255,0.2)' }}>Reintentar</button>
116 |             <button onClick={onClose} className="font-exo text-[10px] px-3 py-1 cursor-pointer border transition-all hov
117 |               style={{ color: '#3a6080', borderColor: '#0d2035' }}>Cerrar</button>
118 |           </div>
119 |         </div>
120 |       </div>
121 |     );
122 |   }
123 |
124 |   const { faena, camiones, contrato, viajes, resumen } = data;
125 |   const estado = resumen.criticos > resumen.totalCamiones * 0.3 ? 'critica' : (resumen.alertas > resumen.totalCamiones *
126 |   const estadoColor = estado === 'critica' ? '#ff2244' : estado === 'alerta' ? '#ffcc00' : '#00ff88';
127 |
128 |   return (
129 |     <div className="fixed inset-0 z-50 flex items-start justify-center pt-6 pb-6 overflow-y-auto" style={{ background: '
130 |       onClick={e => { if (e.target === e.currentTarget) onClose(); }}
131 |       data-testid="faena-detail-modal">
132 |       <div className="dash-card w-full max-w-3xl mx-4" style={{ border: `1px solid ${faena.color || 'rgba(0,212,255,0.15
133 |
134 |         <div className="flex items-center justify-between p-4" style={{ borderBottom: '1px solid #0d2035' }}>
135 |           <div className="flex items-center gap-3">
136 |             <div className="w-4 h-4 rounded-full flex-shrink-0" style={{ background: faena.color, boxShadow: `0 0 8px ${
137 |             <div>
138 |               <div className="font-space text-lg font-bold tracking-[0.1em]" style={{ color: '#c8e8ff' }}>{faena.nombre}
139 |               <div className="flex items-center gap-2 mt-0.5">
140 |                 <span className="font-space text-[9px] font-bold px-2 py-0.5 border uppercase" style={{
141 |                   color: estadoColor, borderColor: `${estadoColor}40`, background: `${estadoColor}08`
142 |                 }}>{estado}</span>
143 |                 {contrato && (
144 |                   <span className="font-exo text-[10px]" style={{ color: '#3a6080' }}>Contrato TMS: {contrato.nombre}</s
145 |                 )}
146 |               </div>
147 |             </div>
148 |           </div>
149 |           <button onClick={onClose} className="p-2 cursor-pointer transition-all hover:bg-[rgba(255,255,255,0.05)] round
150 |             data-testid="btn-close-faena-modal">
151 |             <X className="w-4 h-4" style={{ color: '#3a6080' }} />
152 |           </button>
153 |         </div>
154 |
155 |         <div className="grid grid-cols-2 sm:grid-cols-4 gap-3 p-4" style={{ borderBottom: '1px solid #0d2035' }}>
156 |           <div>
157 |             <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>CAMIONES</div>
158 |             <div className="font-space text-xl font-bold" style={{ color: '#00d4ff' }}>{resumen.totalCamiones}</div>
159 |           </div>
160 |           <div>
161 |             <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>LITROS 30D</di
162 |             <div className="font-space text-xl font-bold" style={{ color: '#c8e8ff' }}>{resumen.litros30d.toLocaleString
163 |           </div>
164 |           <div>
165 |             <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>RENDIMIENTO</d
166 |             <div className="font-space text-xl font-bold" style={{ color: resumen.rendPromedio >= 3.5 ? '#00ff88' : resu
167 |               {resumen.rendPromedio > 0 ? resumen.rendPromedio : '\u2014'}
168 |             </div>
169 |             <div className="font-exo text-[8px]" style={{ color: '#3a6080' }}>km/L promedio</div>
170 |           </div>
171 |           <div>
172 |             <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>CARGAS</div>
173 |             <div className="font-space text-xl font-bold" style={{ color: '#c8e8ff' }}>{resumen.cargas}</div>
174 |             <div className="font-exo text-[8px]" style={{ color: '#3a6080' }}>ultimos 30 dias</div>
175 |           </div>
176 |           {resumen.criticos > 0 && (
177 |             <div>
178 |               <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>CRITICOS</di
179 |               <div className="font-space text-xl font-bold" style={{ color: '#ff2244' }}>{resumen.criticos}</div>
180 |               <div className="font-exo text-[8px]" style={{ color: '#3a6080' }}>rend &lt; 1.5 km/L</div>
181 |             </div>
182 |           )}
183 |           {resumen.alertas > 0 && (
184 |             <div>
185 |               <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>ALERTAS</div
186 |               <div className="font-space text-xl font-bold" style={{ color: '#ffcc00' }}>{resumen.alertas}</div>
187 |             </div>
188 |           )}
189 |           {resumen.totalViajes > 0 && (
190 |             <div>
191 |               <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>VIAJES TMS</
192 |               <div className="font-space text-xl font-bold" style={{ color: '#00d4ff' }}>{resumen.totalViajes}</div>
193 |               <div className="font-exo text-[8px]" style={{ color: '#3a6080' }}>{resumen.totalKmViajes.toLocaleString()}
194 |             </div>
195 |           )}
196 |         </div>
197 |
198 |         <div className="p-4 space-y-2" style={{ borderBottom: '1px solid #0d2035' }}>
199 |           <div className="font-exo text-[9px] tracking-[0.2em] uppercase font-semibold" style={{ color: '#c8e8ff' }}>
200 |             CAMIONES ({camiones.length})
201 |           </div>
202 |           <div className="max-h-[300px] overflow-y-auto space-y-0">
203 |             <div className="flex items-center gap-3 px-3 py-1.5 sticky top-0" style={{ background: '#020508' }}>
204 |               <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080', width: '90px'
205 |               <span className="font-exo text-[8px] tracking-[0.15em] uppercase flex-1" style={{ color: '#3a6080' }}>MODE
206 |               <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: '#3a6080', wi
207 |               <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: '#3a6080', wi
208 |               <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: '#3a6080', wi
209 |               <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: '#3a6080', wi
210 |             </div>
211 |             {camiones.map(cam => (
```

```
212 |               <div key={cam.id} className="flex items-center gap-3 px-3 py-2 transition-all hover:bg-[rgba(0,212,255,0.0
213 |                 style={{ borderBottom: '1px solid #0d2035' }}
214 |                 data-testid={`faena-camion-${cam.id}`}>
215 |                 <div className="flex items-center gap-2" style={{ width: '90px' }}>
216 |                   <Truck className="w-3 h-3 flex-shrink-0" style={{ color: cam.syncOk ? '#00ff88' : '#3a6080' }} />
217 |                   <span className="font-space text-[11px] font-bold" style={{ color: '#c8e8ff' }}>{cam.patente}</span>
218 |                 </div>
219 |                 <span className="font-exo text-[10px] flex-1 truncate" style={{ color: '#3a6080' }}>{cam.modelo}</span>
220 |                 <span className="font-space text-[10px] text-right" style={{ color: cam.litros30d > 0 ? '#c8e8ff' : '#3a
221 |                   {cam.litros30d > 0 ? cam.litros30d.toLocaleString() : '\u2014'}
222 |                 </span>
223 |                 <span className="font-space text-[10px] text-right" style={{
224 |                   color: cam.rendimiento >= 3.5 ? '#00ff88' : cam.rendimiento >= 2.0 ? '#c8e8ff' : cam.rendimiento > 0 ?
225 |                   width: '55px'
226 |                 }}>
227 |                   {cam.rendimiento > 0 ? cam.rendimiento : '\u2014'}
228 |                 </span>
229 |                 <span className="font-space text-[10px] text-right" style={{ color: '#3a6080', width: '50px' }}>
230 |                   {cam.cargas > 0 ? cam.cargas : '\u2014'}
231 |                 </span>
232 |                 <span className="text-right" style={{ width: '50px' }}>
233 |                   {cam.syncOk
234 |                     ? <span className="inline-block w-2 h-2 rounded-full" style={{ background: '#00ff88', boxShadow: '0
235 |                     : <span className="inline-block w-2 h-2 rounded-full" style={{ background: '#3a6080' }} />
236 |                   }
237 |                 </span>
238 |               </div>
239 |             ))}
240 |             {camiones.length === 0 && (
241 |               <div className="py-6 text-center font-exo text-[11px]" style={{ color: '#3a6080' }}>Sin camiones asignados
242 |             )}
243 |           </div>
244 |         </div>
245 |
246 |         {viajes.length > 0 && (
247 |           <div className="p-4 space-y-2">
248 |             <div className="font-exo text-[9px] tracking-[0.2em] uppercase font-semibold" style={{ color: '#00d4ff' }}>
249 |               ULTIMOS VIAJES ({viajes.length})
250 |             </div>
251 |             <div className="max-h-[200px] overflow-y-auto space-y-0">
252 |               {viajes.map(v => (
253 |                 <div key={v.id} className="flex items-center gap-3 px-3 py-2 transition-all hover:bg-[rgba(0,212,255,0.0
254 |                   style={{ borderBottom: '1px solid #0d2035' }}
255 |                   data-testid={`faena-viaje-${v.id}`}>
256 |                   <span className="font-space text-[10px] font-bold" style={{ color: '#c8e8ff', width: '110px' }}>{v.cod
257 |                   <span className="font-space text-[10px]" style={{ color: '#00d4ff', width: '50px' }}>{v.camionPatente}
258 |                   <span className="font-exo text-[9px]" style={{ color: '#3a6080', width: '100px' }}>{formatFecha(v.fech
259 |                   <span className="font-space text-[10px] text-right" style={{ color: '#c8e8ff', width: '55px' }}>{v.km
260 |                   <span className="font-space text-[10px] text-right" style={{
261 |                     color: v.rendimiento >= 3.5 ? '#00ff88' : v.rendimiento > 0 ? '#c8e8ff' : '#3a6080', width: '50px'
262 |                   }}>
263 |                     {v.rendimiento > 0 ? v.rendimiento : '\u2014'}
264 |                   </span>
265 |                   <span className="font-space text-[10px] text-right" style={{ color: '#3a6080', width: '45px' }}>{v.lit
266 |                 </div>
267 |               ))}
268 |             </div>
269 |           </div>
270 |         )}
271 |       </div>
272 |     </div>
273 |   );
274 | }
275 |
276 | function StatusDot({ status, fast }: { status: 'ok' | 'warn' | 'crit'; fast?: boolean }) {
277 |   const colors = { ok: '#00ff88', warn: '#ffcc00', crit: '#ff2244' };
278 |   return (
279 |     <span className="inline-block w-2 h-2 rounded-full flex-shrink-0" style={{
280 |       background: colors[status],
281 |       boxShadow: `0 0 6px ${colors[status]}`,
282 |       animation: fast ? 'blinkFast 0.6s infinite' : status !== 'ok' ? 'blink 2s infinite' : undefined
283 |     }} />
284 |   );
285 | }
286 |
287 | function HealthBar({ label, value, color }: { label: string; value: number; color: string }) {
288 |   return (
289 |     <div>
290 |       <div className="flex justify-between mb-1">
291 |         <span className="font-exo text-[9px] tracking-wider" style={{ color: '#3a6080' }}>{label}</span>
292 |         <span className="font-space text-[10px] font-bold" style={{ color }}>{value}%</span>
293 |       </div>
294 |       <div className="h-1.5 rounded-sm overflow-hidden" style={{ background: '#0d2035' }}>
295 |         <div className="h-full rounded-sm transition-all duration-1000" style={{ width: `${value}%`, background: color }
296 |       </div>
297 |     </div>
298 |   );
299 | }
300 |
301 | function getTurno(): string {
302 |   const h = new Date().getHours();
303 |   if (h >= 6 && h < 14) return "AM";
304 |   if (h >= 14 && h < 22) return "PM";
305 |   return "NOCHE";
306 | }
307 |
308 | export default function Dashboard() {
309 |   const [showCEO, setShowCEO] = useState(false);
310 |   const [selectedFaenaId, setSelectedFaenaId] = useState<number | null>(null);
311 |   const { data: faenas = [] } = useQuery<Faena[]>({ queryKey: ["/api/faenas"] });
312 |   const { data: camionesRaw = [], isLoading: loadingCamiones } = useQuery<Camion[]>({ queryKey: ["/api/camiones"] });
313 |   const { data: cargasRaw = [], isLoading: loadingCargas } = useQuery<Carga[]>({ queryKey: ["/api/cargas"] });
314 |   const { data: percentiles } = useQuery<Percentiles>({ queryKey: ["/api/flota/percentiles"] });
315 |   const { data: heroData } = useQuery<HeroData>({ queryKey: ["/api/dashboard/hero"], refetchInterval: 120000 });
316 |   const { data: faenasData } = useQuery<FaenaRow[]>({ queryKey: ["/api/dashboard/faenas"], refetchInterval: 120000 });
317 |   const { data: sistemaData } = useQuery<SistemaData>({ queryKey: ["/api/dashboard/sistema"], refetchInterval: 60000 });
318 |   const { data: iaResumen } = useQuery<{ resumen: string; cached?: boolean }>({ queryKey: ["/api/ia/resumen-dashboard"],
```

```
319 |
320 |   const isLoading = loadingCamiones || loadingCargas;
321 |
322 |   const camStats: CamionStats[] = useMemo(() =>
323 |     camionesRaw.map(cam => {
324 |       const cams = cargasRaw.filter(c => c.camionId === cam.id);
325 |       return statsCamion(cam, cams);
326 |     }), [camionesRaw, cargasRaw]);
327 |
328 |   const criticos = camStats.filter(c => c.estado === "CRITICO");
329 |   const alertas = camStats.filter(c => c.estado === "ALERTA");
330 |   const rendValues = camStats.map(c => c.rendProm).filter((r): r is number => r != null);
331 |   const rendFlota = rendValues.length > 0 ? +(rendValues.reduce((a, b) => a + b, 0) / rendValues.length).toFixed(2) : nu
332 |   const turno = getTurno();
333 |   const p90Max = percentiles ? Math.max(percentiles.p90, rendFlota || 0) * 1.1 : 3;
334 |
335 |   if (isLoading) {
336 |     return (
337 |       <div className="space-y-5" data-testid="dashboard-loading">
338 |         <div className="grid grid-cols-2 gap-4">
339 |           <Skeleton className="h-64" />
340 |           <Skeleton className="h-64" />
341 |         </div>
342 |         <div className="grid grid-cols-5 gap-3">
343 |           {Array.from({ length: 5 }).map((_, i) => <Skeleton key={i} className="h-24" />)}
344 |         </div>
345 |       </div>
346 |     );
347 |   }
348 |
349 |   return (
350 |     <div className="dash-bg min-h-screen -mx-5 -mt-5 px-5 pt-5 pb-10" data-testid="dashboard-view">
351 |       <div className="dash-scanlines" />
352 |       <div className="dash-content max-w-[1400px] mx-auto space-y-5">
353 |
354 |         <div className="flex justify-end mb-[-8px]">
355 |           <button
356 |             onClick={() => setShowCEO(true)}
357 |             data-testid="btn-vision-ceo"
358 |             className="font-bold text-[13px] px-5 py-2.5 rounded-lg cursor-pointer transition-all hover:brightness-110 h
359 |             style={{
360 |               background: "linear-gradient(135deg, #00d4ff, #0055aa)",
361 |               color: "#ffffff",
362 |               fontFamily: "Helvetica, Arial, sans-serif",
363 |               border: "none",
364 |               boxShadow: "0 2px 12px rgba(0,212,255,0.3)",
365 |             }}
366 |           >
367 |             {"\ud83d\udc54"} VISION CEO
368 |           </button>
369 |         </div>
370 |
371 |         <div className="grid grid-cols-1 lg:grid-cols-2 gap-4" data-testid="hero-section">
372 |           <div className="corner-cut-tl dash-card relative overflow-hidden" data-testid="hero-logo-panel">
373 |             <div className="absolute top-0 left-0 w-12 h-[1px]" style={{ background: '#00d4ff' }} />
374 |             <div className="absolute top-0 left-0 w-[1px] h-12" style={{ background: '#00d4ff' }} />
375 |             <div className="absolute bottom-0 right-0 w-12 h-[1px]" style={{ background: '#00d4ff' }} />
376 |             <div className="absolute bottom-0 right-0 w-[1px] h-12" style={{ background: '#00d4ff' }} />
377 |
378 |             <div className="p-6 flex flex-col items-center justify-center min-h-[280px]">
379 |               <AnimatedRings />
380 |               <div className="mt-4 text-center">
381 |                 <div className="font-space text-[28px] font-bold tracking-[0.4em]" style={{ color: '#00d4ff', textShadow
382 |                   SOTRASER
383 |                 </div>
384 |                 <div className="font-exo text-[11px] font-extralight tracking-[0.35em] mt-1" style={{ color: '#3a6080' }
385 |                   CONTROL OPERACIONAL DE FLOTA
386 |                 </div>
387 |                 <div className="font-space text-[9px] mt-2" style={{ color: '#1a3550' }}>
388 |                   VOLVO rFMS &middot; SIGETRA &middot; ANTHROPIC AI
389 |                 </div>
390 |               </div>
391 |             </div>
392 |
393 |             <div className="border-t px-6 py-3 grid grid-cols-3 gap-4" style={{ borderColor: '#0d2035' }}>
394 |               <div className="text-center">
395 |                 <div className="font-exo text-[8px] tracking-[0.2em] mb-1" style={{ color: '#3a6080' }}>LITROS 30D</div>
396 |                 <div className="font-space text-base font-bold" style={{ color: '#ffcc00' }} data-testid="text-hero-litr
397 |                   {heroData ? fN(Math.round(heroData.litros30d)) : '\u2014'}
398 |                 </div>
399 |               </div>
400 |               <div className="text-center border-x" style={{ borderColor: '#0d2035' }}>
401 |                 <div className="font-exo text-[8px] tracking-[0.2em] mb-1" style={{ color: '#3a6080' }}>KM TOTAL</div>
402 |                 <div className="font-space text-base font-bold" style={{ color: '#c8e8ff' }} data-testid="text-hero-km">
403 |                   {heroData ? fN(Math.round(heroData.km30d)) : '\u2014'}
404 |                 </div>
405 |               </div>
406 |               <div className="text-center">
407 |                 <div className="font-exo text-[8px] tracking-[0.2em] mb-1" style={{ color: '#3a6080' }}>CARGAS</div>
408 |                 <div class="font-space text-base font-bold" style={{ color: '#c8e8ff' }} data-testid="text-hero-carg
409 |                   {heroData ? fN(heroData.cargas30d) : '\u2014'}
410 |                 </div>
411 |               </div>
412 |             </div>
413 |           </div>
414 |
415 |           <div className="corner-cut-tr dash-card relative overflow-hidden" data-testid="hero-ia-panel">
416 |             <div class="p-5">
417 |               <div class="flex items-center justify-between mb-4">
418 |                 <div class="flex items-center gap-2">
419 |                   <div class="w-7 h-7 flex items-center justify-center border" style={{ borderColor: '#0d2035', anim
420 |                     <span class="text-sm">&#129302;</span>
421 |                   </div>
422 |                   <span class="font-exo text-[9px] tracking-[0.2em] font-light" style={{ color: '#3a6080' }}>
423 |                     INTELIGENCIA ARTIFICIAL &middot; DIAGNOSTICO
424 |                   </span>
425 |                 </div>
```

```
426 |                <div className="flex items-center gap-1.5">
427 |                  <StatusDot status="ok" />
428 |                  <span className="font-exo text-[9px] tracking-wider" style={{ color: '#00ff88' }}>ACTIVO</span>
429 |                </div>
430 |              </div>
431 |
432 |              <div className="mb-4 pl-3.5" style={{ borderLeft: '2px solid #00d4ff' }}>
433 |                <p className="font-exo text-[14px] font-light leading-relaxed" style={{ color: '#c8e8ff', lineHeight: '1
434 |                  {iaResumen?.resumen || "Cargando diagnostico de flota..."}
435 |                </p>
436 |              </div>
437 |
438 |              <div className="mb-4">
439 |                <div className="font-exo text-[8px] tracking-[0.2em] uppercase mb-3" style={{ color: '#3a6080' }}>
440 |                  REFERENCIA PERCENTIL &middot; FLOTA REAL
441 |                </div>
442 |                <div className="space-y-2">
443 |                  <PercentileBar label="P90" value={percentiles?.p90 || 0} color="#00ff88" maxVal={p90Max} />
444 |                  <PercentileBar label="P75" value={percentiles?.p75 || 0} color="#00d4ff" maxVal={p90Max} />
445 |                  <PercentileBar label="P50" value={percentiles?.p50 || 0} color="#3a6080" maxVal={p90Max} />
446 |                </div>
447 |              </div>
448 |
449 |              <div className="flex flex-wrap gap-2" data-testid="hero-tags">
450 |                <span className="px-2.5 py-1 font-space text-[9px] font-bold border" style={{
451 |                  color: criticos.length > 0 ? '#ff2244' : '#00ff88',
452 |                  borderColor: criticos.length > 0 ? 'rgba(255,34,68,0.4)' : 'rgba(0,255,136,0.3)',
453 |                  background: criticos.length > 0 ? 'rgba(255,34,68,0.08)' : 'rgba(0,255,136,0.05)'
454 |                }} data-testid="tag-criticos">
455 |                  {criticos.length} CRITICOS
456 |                </span>
457 |                <span className="px-2.5 py-1 font-space text-[9px] font-bold border" style={{
458 |                  color: alertas.length > 0 ? '#ffcc00' : '#00ff88',
459 |                  borderColor: alertas.length > 0 ? 'rgba(255,204,0,0.4)' : 'rgba(0,255,136,0.3)',
460 |                  background: alertas.length > 0 ? 'rgba(255,204,0,0.08)' : 'rgba(0,255,136,0.05)'
461 |                }} data-testid="tag-alertas">
462 |                  {alertas.length} ALERTAS
463 |                </span>
464 |                <span className="px-2.5 py-1 font-space text-[9px] font-bold border" style={{
465 |                  color: '#ffcc00', borderColor: 'rgba(255,204,0,0.3)', background: 'rgba(255,204,0,0.05)'
466 |                }}>
467 |                  {heroData?.cargas30d || 0} CARGAS 30D
468 |                </span>
469 |                <span className="px-2.5 py-1 font-space text-[9px] border" style={{ color: '#3a6080', borderColor: '#0d2
470 |                  ACTUALIZADO {new Date().toLocaleTimeString('es-CL', { hour: '2-digit', minute: '2-digit' })}
471 |                </span>
472 |              </div>
473 |            </div>
474 |          </div>
475 |        </div>
476 |
477 |        <div className="grid grid-cols-2 sm:grid-cols-3 lg:grid-cols-5 gap-3" data-testid="kpi-grid">
478 |          {[
479 |            { label: 'CAMIONES ACTIVOS', value: camStats.length, accent: '#00d4ff', sub: `${heroData?.camionesOnline ||
480 |            { label: 'CRITICOS', value: criticos.length, accent: '#ff2244', sub: criticos.length ? `${criticos.length} r
481 |            { label: 'ALERTAS', value: alertas.length, accent: '#ffcc00', sub: alertas.length ? `${alertas.length} en ob
482 |            { label: 'REND. FLOTA', value: rendFlota != null ? rendFlota : '\u2014', accent: '#00ff88', sub: `KM/L · ${h
483 |            { label: 'CARGAS 30D', value: heroData?.cargas30d || 0, accent: '#3a6080', sub: `${heroData?.totalFaenas ||
484 |          ].map((kpi, i) => (
485 |            <div
486 |              key={kpi.label}
487 |              className="dash-card dash-card-accent relative p-4 transition-all"
488 |              style={{ borderTopWidth: 2, borderTopColor: kpi.accent, animationDelay: `${i * 0.08}s`, animation: 'fadeIn
489 |              data-testid={`kpi-${kpi.label.toLowerCase().replace(/[\s.]/g, "-")}`}
490 |            >
491 |              <div className="font-exo text-[8px] tracking-[0.2em] uppercase mb-2" style={{ color: '#3a6080' }}>
492 |                {kpi.label}
493 |              </div>
494 |              <div className="font-space text-2xl font-bold" style={{ color: kpi.accent }} data-testid={`kpi-value-${i}`
495 |                {kpi.value}
496 |              </div>
497 |              <div className="font-exo text-[9px] mt-1" style={{ color: '#3a6080' }}>{kpi.sub}</div>
498 |            </div>
499 |          ))}
500 |        </div>
501 |
502 |        <div className="grid grid-cols-1 lg:grid-cols-[2fr_1fr] gap-4">
503 |          <div className="dash-card" data-testid="panel-faenas">
504 |            <div className="flex items-center justify-between p-4 border-b" style={{ borderColor: '#0d2035' }}>
505 |              <span className="font-exo text-[10px] tracking-[0.2em] font-semibold uppercase" style={{ color: '#c8e8ff'
506 |                FAENAS ACTIVAS
507 |              </span>
508 |            </div>
509 |            <div className="divide-y" style={{ borderColor: '#0d2035' }}>
510 |              {(faenasData || []).filter(f => f.camiones > 0).map((f, i) => {
511 |                const dotStatus = f.estado === 'critica' ? 'crit' : f.estado === 'alerta' ? 'warn' : 'ok';
512 |                return (
513 |                  <div key={f.nombre} className="flex items-center gap-3 px-4 py-2.5 transition-colors hover:bg-[rgba(0,
514 |                    onClick={() => setSelectedFaenaId(f.id)}>
515 |                    <StatusDot status={dotStatus} fast={dotStatus === 'crit'} />
516 |                    <span className="font-rajdhani text-[13px] font-medium flex-1 truncate" style={{ color: f.color || '
517 |                      {f.nombre}
518 |                    </span>
519 |                    <span className="font-space text-[10px]" style={{ color: '#3a6080' }}>{f.camiones} cam</span>
520 |                    <span className="font-space text-[11px] font-bold w-16 text-right" style={{ color: f.rendPromedio >=
521 |                      {f.rendPromedio > 0 ? f.rendPromedio.toFixed(2) : '\u2014'} km/L
522 |                    </span>
523 |                    <span className="font-space text-[9px] font-bold px-2 py-0.5 border uppercase" style={{
524 |                      color: f.estado === 'critica' ? '#ff2244' : f.estado === 'alerta' ? '#ffcc00' : '#00ff88',
525 |                      borderColor: f.estado === 'critica' ? 'rgba(255,34,68,0.3)' : f.estado === 'alerta' ? 'rgba(255,20
526 |                      background: f.estado === 'critica' ? 'rgba(255,34,68,0.06)' : f.estado === 'alerta' ? 'rgba(255,20
527 |                    }}>
528 |                      {f.estado}
529 |                    </span>
530 |                    <ChevronRight className="w-3.5 h-3.5 flex-shrink-0" style={{ color: '#3a6080' }} />
531 |                  </div>
532 |                );
```

```
533 |               })}
534 |               {!faenasData && (
535 |                 <div className="px-4 py-6 text-center font-exo text-[11px]" style={{ color: '#3a6080' }}>
536 |                   Cargando faenas...
537 |                 </div>
538 |               )}
539 |               {faenasData && faenasData.filter(f => f.camiones > 0).length === 0 && (
540 |                 <div className="px-4 py-6 text-center font-exo text-[11px]" style={{ color: '#3a6080' }}>
541 |                   Sin faenas activas con camiones asignados
542 |                 </div>
543 |               )}
544 |             </div>
545 |           </div>
546 |
547 |           <div className="dash-card" data-testid="panel-sistema">
548 |             <div className="p-4 border-b" style={{ borderColor: '#0d2035' }}>
549 |               <span className="font-exo text-[10px] tracking-[0.2em] font-semibold uppercase" style={{ color: '#c8e8ff'
550 |                 SISTEMA
551 |               </span>
552 |             </div>
553 |             <div className="p-4 space-y-3">
554 |               <div className="space-y-2">
555 |                 {[
556 |                   { name: 'VOLVO rFMS', ok: sistemaData?.volvo?.status === 'connected', label: 'CONECTADO' },
557 |                   { name: 'SIGETRA', ok: sistemaData?.sigetra?.connected ?? false, label: 'CONECTADO' },
558 |                   { name: 'ANTHROPIC IA', ok: sistemaData?.ia?.status === 'available', label: 'ACTIVO' },
559 |                 ].map(s => (
560 |                   <div key={s.name} className="flex items-center gap-2">
561 |                     <StatusDot status={s.ok ? 'ok' : 'crit'} />
562 |                     <span className="font-space text-[10px] font-bold flex-1" style={{ color: '#c8e8ff' }}>{s.name}</spa
563 |                     <span className="font-exo text-[9px]" style={{ color: s.ok ? '#00ff88' : '#ff2244' }}>{s.ok ? s.labe
564 |                   </div>
565 |                 ))}
566 |               </div>
567 |
568 |               <div className="border-t pt-3 space-y-2.5" style={{ borderColor: '#0d2035' }}>
569 |                 <HealthBar label="Volvo rFMS" value={sistemaData?.volvo?.health || 0}
570 |                   color={(sistemaData?.volvo?.health || 0) > 70 ? '#00ff88' : (sistemaData?.volvo?.health || 0) > 50 ? '
571 |                 <HealthBar label="Sigetra" value={sistemaData?.sigetra?.health || 0} color="#00d4ff" />
572 |                 <HealthBar label="IA Claude" value={sistemaData?.ia?.health || 0}
573 |                   color={(sistemaData?.ia?.health || 0) > 80 ? '#00ff88' : (sistemaData?.ia?.health || 0) > 60 ? '#ffcc0
574 |               </div>
575 |
576 |               <div className="border-t pt-3 space-y-1" style={{ borderColor: '#0d2035' }}>
577 |                 <div className="flex justify-between">
578 |                   <span className="font-space text-[9px]" style={{ color: '#1a3550' }}>ULTIMO SYNC</span>
579 |                   <span className="font-space text-[9px]" style={{ color: '#3a6080' }}>{sistemaData?.ultimoSync || '\u20
580 |                 </div>
581 |                 <div className="flex justify-between">
582 |                   <span className="font-space text-[9px]" style={{ color: '#1a3550' }}>DATOS DESDE</span>
583 |                   <span className="font-space text-[9px]" style={{ color: '#3a6080' }}>01-03-2026</span>
584 |                 </div>
585 |                 <div className="flex justify-between">
586 |                   <span className="font-space text-[9px]" style={{ color: '#1a3550' }}>PERCENTIL P90</span>
587 |                   <span className="font-space text-[9px] font-bold" style={{ color: '#00ff88' }}>{percentiles?.p90?.toFi
588 |                 </div>
589 |               </div>
590 |             </div>
591 |           </div>
592 |         </div>
593 |
594 |         <div className="text-[9px] font-space tracking-[0.15em]" style={{ color: '#3a6080' }} data-testid="text-system-n
595 |           Sistema de medicion activo desde 01-03-2026 &middot; Turno actual: {turno}
596 |           {percentiles && !percentiles.calibrando && (
597 |             <span> &middot; Ref. P90: {percentiles.p90.toFixed(2)} km/L</span>
598 |           )}
599 |         </div>
600 |
601 |       </div>
602 |
603 |       {selectedFaenaId !== null && (
604 |         <FaenaDetailModal faenaId={selectedFaenaId} onClose={() => setSelectedFaenaId(null)} />
605 |       )}
606 |
607 |       {showCEO && (
608 |         <Suspense fallback={
609 |           <div className="fixed inset-0 z-[100] flex items-center justify-center" style={{ background: "rgba(2,5,8,0.98)
610 |             <Loader2 className="w-8 h-8 animate-spin" style={{ color: "#00d4ff" }} />
611 |           </div>
612 |         }>
613 |           <VisionCEO onClose={() => setShowCEO(false)} />
614 |         </Suspense>
615 |       )}
616 |     </div>
617 |   );
618 | }
619 |
```

client/src/pages/desviaciones.tsx

918 lineas

```
1 | import { useState, useMemo, useCallback } from "react";
2 | import { useQuery, useMutation } from "@tanstack/react-query";
3 | import { Input } from "@/components/ui/input";
4 | import { Skeleton } from "@/components/ui/skeleton";
5 | import {
6 |   AlertTriangle, Search, Shield, ShieldAlert, ShieldX,
7 |   ArrowRightLeft, X, CheckCircle2, Circle, Gauge, MapPin, BarChart3
8 | } from "lucide-react";
9 | import { apiRequest, queryClient } from "@/lib/queryClient";
10 | import { getErrorFleetNums, isErrorCarga } from "@/pages/errores";
11 |
12 | interface FusionCarga {
13 |   numGuia: number;
14 |   fecha: string;
15 |   litros: number;
16 |   odometro: number | null;
17 |   kmRecorrido: number | null;
18 |   rendimiento: number | null;
19 |   lugar: string | null;
20 |   tarjeta: string | null;
21 |   conductor: string | null;
22 | }
23 |
24 | interface FusionTruck {
25 |   fleetNum: string;
26 |   patenteReal: string | null;
27 |   vin: string | null;
28 |   modeloVolvo: string;
29 |   faenaSigetra: string | null;
30 |   conductorSigetra: string | null;
31 |   totalLitrosSigetra: number;
32 |   totalCargas: number;
33 |   rendPromedio: number;
34 |   odometroSigetra: number | null;
35 |   odometroVolvo: number | null;
36 |   deltaOdometro: number | null;
37 |   fuelLevelVolvo: number | null;
38 |   totalFuelUsedVolvo: number | null;
39 |   litrosVolvoPeriodo: number | null;
40 |   engineHoursVolvo: number | null;
41 |   gpsVolvo: { latitude: number | null; longitude: number | null } | null;
42 |   alertLevel: "ok" | "alerta" | "critico";
43 |   cargas: FusionCarga[];
44 | }
45 |
46 | interface FusionResponse {
47 |   from: string;
48 |   to: string;
49 |   totalTrucksMatched: number;
50 |   totalLitros: number;
51 |   totalCargas: number;
52 |   alertCount: number;
53 |   trucks: FusionTruck[];
54 | }
55 |
56 | interface DesviacionCheck {
57 |   id: number;
58 |   fleetNum: string;
59 |   tipo: string;
60 |   gestionado: boolean;
61 |   gestionadoAt: string | null;
62 |   nota: string | null;
63 | }
64 |
65 | interface KmDeviation {
66 |   truck: FusionTruck;
67 |   absDelta: number;
68 |   severity: "critico" | "alerta";
69 |   faena: string;
70 | }
71 |
72 | interface RendDeviation {
73 |   truck: FusionTruck;
74 |   faena: string;
75 |   rendimiento: number;
76 |   faenaAvg: number;
77 |   pctDesv: number;
78 |   direction: "sobre" | "bajo";
79 | }
80 |
81 | function fN(n: number): string {
82 |   return n.toLocaleString("es-CL");
83 | }
84 |
85 | function formatDateTime(iso: string): string {
86 |   if (!iso) return "\u2014";
87 |   const d = new Date(iso);
88 |   return d.toLocaleDateString("es-CL", { day: "2-digit", month: "2-digit", hour: "2-digit", minute: "2-digit" });
89 | }
90 |
91 | function getDateRange7d(): { from: string; to: string } {
92 |   const to = new Date();
93 |   const from = new Date(Date.now() - 7 * 24 * 60 * 60 * 1000);
94 |   return { from: from.toISOString().slice(0, 10), to: to.toISOString().slice(0, 10) };
95 | }
96 |
97 | export default function Desviaciones() {
98 |   const [busq, setBusq] = useState("");
99 |   const [selectedTruck, setSelectedTruck] = useState<string | null>(null);
100 |   const [severityFilter, setSeverityFilter] = useState<"all" | "critico" | "alerta">("all");
101 |   const [showGestionado, setShowGestionado] = useState(false);
102 |   const [devMode, setDevMode] = useState<"km" | "rend">("km");
103 |
104 |   const range = useMemo(() => getDateRange7d(), []);
```

```
105 |
106 |   const { data: fusion, isLoading, isError } = useQuery<FusionResponse>({
107 |     queryKey: [`/api/sigetra/fusion?from=${range.from}&to=${range.to}`],
108 |     refetchInterval: 600000,
109 |   });
110 |
111 |   const { data: checks = [] } = useQuery<DesviacionCheck[]>({
112 |     queryKey: ["/api/desviaciones/checks"],
113 |   });
114 |
115 |   const checkMutation = useMutation({
116 |     mutationFn: async (params: { fleetNum: string; tipo: string; gestionado: boolean }) => {
117 |       return apiRequest("POST", "/api/desviaciones/check", params);
118 |     },
119 |     onSuccess: () => {
120 |       queryClient.invalidateQueries({ queryKey: ["/api/desviaciones/checks"] });
121 |     },
122 |   });
123 |
124 |   const isGestionado = useCallback((fleetNum: string, tipo: string) => {
125 |     return checks.some(c => c.fleetNum === fleetNum && c.tipo === tipo && c.gestionado);
126 |   }, [checks]);
127 |
128 |   const getCheckInfo = useCallback((fleetNum: string, tipo: string) => {
129 |     return checks.find(c => c.fleetNum === fleetNum && c.tipo === tipo && c.gestionado);
130 |   }, [checks]);
131 |
132 |   const errorNums = useMemo(() => {
133 |     if (!fusion?.trucks) return new Set<string>();
134 |     return getErrorFleetNums(fusion.trucks);
135 |   }, [fusion]);
136 |
137 |   const kmDeviations = useMemo(():KmDeviation[] => {
138 |     if (!fusion?.trucks) return [];
139 |     const results: KmDeviation[] = [];
140 |     for (const t of fusion.trucks) {
141 |       if (errorNums.has(t.fleetNum)) continue;
142 |       if (t.deltaOdometro == null) continue;
143 |       const absDelta = Math.abs(t.deltaOdometro);
144 |       if (absDelta <= 1000) continue;
145 |       const severity = absDelta > 5000 ? "critico" as const : "alerta" as const;
146 |       results.push({ truck: t, absDelta, severity, faena: t.faenaSigetra || "Sin faena" });
147 |     }
148 |     results.sort((a, b) => {
149 |       const sevOrder = a.severity === "critico" ? 0 : 1;
150 |       const sevOrderB = b.severity === "critico" ? 0 : 1;
151 |       if (sevOrder !== sevOrderB) return sevOrder - sevOrderB;
152 |       return b.absDelta - a.absDelta;
153 |     });
154 |     return results;
155 |   }, [fusion, errorNums]);
156 |
157 |   const rendDeviations = useMemo(():RendDeviation[] => {
158 |     if (!fusion?.trucks) return [];
159 |     const cleanTrucks = fusion.trucks
160 |       .filter(t => !errorNums.has(t.fleetNum))
161 |       .map(t => {
162 |         const validCargas = t.cargas.filter(c => !isErrorCarga(c));
163 |         const totalLitros = validCargas.reduce((s, c) => s + c.litros, 0);
164 |         const totalKm = validCargas.reduce((s, c) => s + (c.kmRecorrido != null && c.kmRecorrido > 0 ? c.kmRecorrido : 0
165 |         const rendPromedio = totalLitros > 0 ? totalKm / totalLitros : 0;
166 |         return { ...t, cargas: validCargas, totalLitrosSigetra: totalLitros, totalCargas: validCargas.length, rendPromed
167 |       });
168 |
169 |     type CleanTruck = typeof cleanTrucks[number];
170 |     const byFaena = new Map<string, CleanTruck[]>();
171 |     for (const t of cleanTrucks) {
172 |       const faena = t.faenaSigetra || "Sin Faena";
173 |       const existing = byFaena.get(faena) || [];
174 |       existing.push(t);
175 |       byFaena.set(faena, existing);
176 |     }
177 |
178 |     const results: RendDeviation[] = [];
179 |     const faenaEntries = Array.from(byFaena.entries());
180 |     for (const [faena, trucks] of faenaEntries) {
181 |       const withRend = trucks.filter((t: CleanTruck) => t.rendPromedio > 0);
182 |       if (withRend.length < 2) continue;
183 |       const avg = withRend.reduce((s: number, t: CleanTruck) => s + t.rendPromedio, 0) / withRend.length;
184 |       if (avg <= 0) continue;
185 |       for (const t of withRend) {
186 |         const pctDesv = ((t.rendPromedio - avg) / avg) * 100;
187 |         if (Math.abs(pctDesv) > 15) {
188 |           results.push({
189 |             truck: t,
190 |             faena,
191 |             rendimiento: t.rendPromedio,
192 |             faenaAvg: avg,
193 |             pctDesv,
194 |             direction: pctDesv > 0 ? "sobre" : "bajo",
195 |           });
196 |         }
197 |       }
198 |     }
199 |     results.sort((a, b) => Math.abs(b.pctDesv) - Math.abs(a.pctDesv));
200 |     return results;
201 |   }, [fusion, errorNums]);
202 |
203 |   const filteredKm = useMemo(() => {
204 |     let list = kmDeviations;
205 |
206 |     if (severityFilter !== "all") {
207 |       list = list.filter(d => d.severity === severityFilter);
208 |     }
209 |
210 |     if (showGestionado) {
211 |       list = list.filter(d => isGestionado(d.truck.fleetNum, "ODO_KM"));
```

```
212 |     } else {
213 |       list = list.filter(d => !isGestionado(d.truck.fleetNum, "ODO_KM"));
214 |     }
215 |
216 |     if (busq.trim()) {
217 |       const q = busq.toLowerCase();
218 |       list = list.filter(d =>
219 |         d.truck.fleetNum.includes(q) ||
220 |         d.truck.patenteReal?.toLowerCase().includes(q) ||
221 |         d.truck.vin?.toLowerCase().includes(q) ||
222 |         d.faena.toLowerCase().includes(q) ||
223 |         d.truck.conductorSigetra?.toLowerCase().includes(q)
224 |       );
225 |     }
226 |
227 |     return list;
228 |   }, [kmDeviations, busq, severityFilter, showGestionado, isGestionado]);
229 |
230 |   const filteredRend = useMemo(() => {
231 |     let list = rendDeviations;
232 |
233 |     if (severityFilter === "critico") {
234 |       list = list.filter(d => Math.abs(d.pctDesv) > 30);
235 |     } else if (severityFilter === "alerta") {
236 |       list = list.filter(d => Math.abs(d.pctDesv) <= 30);
237 |     }
238 |
239 |     if (showGestionado) {
240 |       list = list.filter(d => isGestionado(d.truck.fleetNum, "REND_DESV"));
241 |     } else {
242 |       list = list.filter(d => !isGestionado(d.truck.fleetNum, "REND_DESV"));
243 |     }
244 |
245 |     if (busq.trim()) {
246 |       const q = busq.toLowerCase();
247 |       list = list.filter(d =>
248 |         d.truck.fleetNum.includes(q) ||
249 |         d.truck.patenteReal?.toLowerCase().includes(q) ||
250 |         d.faena.toLowerCase().includes(q) ||
251 |         d.truck.conductorSigetra?.toLowerCase().includes(q)
252 |       );
253 |     }
254 |
255 |     return list;
256 |   }, [rendDeviations, busq, severityFilter, showGestionado, isGestionado]);
257 |
258 |   const faenaGroups = useMemo(() => {
259 |     const groups = new Map<string, KmDeviation[]>();
260 |     for (const d of filteredKm) {
261 |       const existing = groups.get(d.faena) || [];
262 |       existing.push(d);
263 |       groups.set(d.faena, existing);
264 |     }
265 |     return Array.from(groups.entries()).sort((a, b) => {
266 |       const aCrit = a[1].filter(d => d.severity === "critico").length;
267 |       const bCrit = b[1].filter(d => d.severity === "critico").length;
268 |       if (aCrit !== bCrit) return bCrit - aCrit;
269 |       return b[1].length - a[1].length;
270 |     });
271 |   }, [filteredKm]);
272 |
273 |   const rendFaenaGroups = useMemo(() => {
274 |     const groups = new Map<string, RendDeviation[]>();
275 |     for (const d of filteredRend) {
276 |       const existing = groups.get(d.faena) || [];
277 |       existing.push(d);
278 |       groups.set(d.faena, existing);
279 |     }
280 |     return Array.from(groups.entries()).sort((a, b) => b[1].length - a[1].length);
281 |   }, [filteredRend]);
282 |
283 |   const stats = useMemo(() => {
284 |     if (devMode === "km") {
285 |       const total = kmDeviations.length;
286 |       const criticos = kmDeviations.filter(d => d.severity === "critico").length;
287 |       const alertas = kmDeviations.filter(d => d.severity === "alerta").length;
288 |       const gestionados = kmDeviations.filter(d => isGestionado(d.truck.fleetNum, "ODO_KM")).length;
289 |       const pendientes = total - gestionados;
290 |       return { total, criticos, alertas, gestionados, pendientes };
291 |     } else {
292 |       const total = rendDeviations.length;
293 |       const criticos = rendDeviations.filter(d => Math.abs(d.pctDesv) > 30).length;
294 |       const alertas = total - criticos;
295 |       const gestionados = rendDeviations.filter(d => isGestionado(d.truck.fleetNum, "REND_DESV")).length;
296 |       const pendientes = total - gestionados;
297 |       return { total, criticos, alertas, gestionados, pendientes };
298 |     }
299 |   }, [kmDeviations, rendDeviations, isGestionado, devMode]);
300 |
301 |   const selectedKm = devMode === "km" && selectedTruck ? kmDeviations.find(d => d.truck.fleetNum === selectedTruck) : nu
302 |   const selectedRend = devMode === "rend" && selectedTruck ? rendDeviations.find(d => d.truck.fleetNum === selectedTruck
303 |
304 |   const handleToggleCheck = (fleetNum: string, currentlyChecked: boolean) => {
305 |     const tipo = devMode === "km" ? "ODO_KM" : "REND_DESV";
306 |     checkMutation.mutate({ fleetNum, tipo, gestionado: !currentlyChecked });
307 |   };
308 |
309 |   if (isLoading) {
310 |     return (
311 |       <div className="space-y-4" data-testid="desviaciones-loading">
312 |         <Skeleton className="h-16 w-full" />
313 |         <div className="grid grid-cols-4 gap-3">
314 |           {Array.from({ length: 4 }).map((_, i) => <Skeleton key={i} className="h-20" />)}
315 |         </div>
316 |         <Skeleton className="h-96 w-full" />
317 |       </div>
318 |     );
```

```
319 |    }
320 |
321 |    return (
322 |      <div className="space-y-4" data-testid="desviaciones-page">
323 |        <div className="flex items-center gap-3 flex-wrap">
324 |          <div className="flex items-center gap-2">
325 |            <ShieldAlert className="w-5 h-5 text-red-400" />
326 |            <h1 className="text-sm font-mono font-bold tracking-[0.15em] uppercase text-foreground" data-testid="text-desv
327 |              Desviaciones
328 |            </h1>
329 |          </div>
330 |          <div className="flex items-center gap-1">
331 |            <button onClick={() => { setDevMode("km"); setSelectedTruck(null); setSeverityFilter("all"); setShowGestionado
332 |              data-testid="btn-mode-km"
333 |              className={`px-3 py-1 text-[10px] font-mono font-bold border cursor-pointer transition-colors ${
334 |                devMode === "km" ? "bg-primary/20 border-primary text-primary" : "bg-card border-border text-muted-foregro
335 |              }`}>
336 |              KILOMETRAJE
337 |            </button>
338 |            <button onClick={() => { setDevMode("rend"); setSelectedTruck(null); setSeverityFilter("all"); setShowGestiona
339 |              data-testid="btn-mode-rend"
340 |              className={`px-3 py-1 text-[10px] font-mono font-bold border cursor-pointer transition-colors ${
341 |                devMode === "rend" ? "bg-primary/20 border-primary text-primary" : "bg-card border-border text-muted-foreg
342 |              }`}>
343 |              RENDIMIENTO
344 |            </button>
345 |          </div>
346 |          <span className="text-[10px] font-mono text-muted-foreground">
347 |            {devMode === "km" ? "Diferencias odometro Sigetra vs Volvo ECU" : "Camiones con rendimiento >15% sobre/bajo pr
348 |          </span>
349 |        </div>
350 |
351 |        <div className="grid grid-cols-2 sm:grid-cols-5 gap-3">
352 |          <button onClick={() => { setSeverityFilter("all"); setShowGestionado(false); }}
353 |            data-testid="btn-filter-all"
354 |            className={`bg-card border p-3 text-left cursor-pointer transition-colors ${severityFilter === "all" && !showG
355 |            <div className="flex items-center gap-2 mb-1">
356 |              <Shield className="w-4 h-4 text-primary" />
357 |              <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">PENDIENTES</span>
358 |            </div>
359 |            <div className="text-2xl font-mono font-bold text-primary" data-testid="text-desv-pendientes">{stats.pendiente
360 |            <div className="text-[10px] font-mono text-muted-foreground">por gestionar</div>
361 |          </button>
362 |          <button onClick={() => { setSeverityFilter("critico"); setShowGestionado(false); }}
363 |            data-testid="btn-filter-critico"
364 |            className={`bg-card border p-3 text-left cursor-pointer transition-colors ${severityFilter === "critico" ? "bo
365 |            <div className="flex items-center gap-2 mb-1">
366 |              <ShieldX className="w-4 h-4 text-red-400" />
367 |              <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">CRITICOS</span>
368 |            </div>
369 |            <div className="text-2xl font-mono font-bold text-red-400" data-testid="text-desv-criticos">{stats.criticos}</
370 |            <div className="text-[10px] font-mono text-red-400/70">{devMode === "km" ? ">5.000 km dif" : ">30% desv"}</div
371 |          </button>
372 |          <button onClick={() => { setSeverityFilter("alerta"); setShowGestionado(false); }}
373 |            data-testid="btn-filter-alerta"
374 |            className={`bg-card border p-3 text-left cursor-pointer transition-colors ${severityFilter === "alerta" ? "bor
375 |            <div className="flex items-center gap-2 mb-1">
376 |              <AlertTriangle className="w-4 h-4 text-amber-400" />
377 |              <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">ALERTAS</span>
378 |            </div>
379 |            <div className="text-2xl font-mono font-bold text-amber-400" data-testid="text-desv-alertas">{stats.alertas}</
380 |            <div className="text-[10px] font-mono text-amber-400/70">{devMode === "km" ? "1.000-5.000 km" : "15-30% desv"}
381 |          </button>
382 |          <button onClick={() => { setSeverityFilter("all"); setShowGestionado(true); }}
383 |            data-testid="btn-filter-gestionados"
384 |            className={`bg-card border p-3 text-left cursor-pointer transition-colors ${showGestionado ? "border-emerald-5
385 |            <div className="flex items-center gap-2 mb-1">
386 |              <CheckCircle2 className="w-4 h-4 text-emerald-400" />
387 |              <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">GESTIONADOS</span>
388 |            </div>
389 |            <div className="text-2xl font-mono font-bold text-emerald-400" data-testid="text-desv-gestionados">{stats.gest
390 |            <div className="text-[10px] font-mono text-emerald-400/70">revisados</div>
391 |          </button>
392 |          <div className="bg-card border border-border p-3">
393 |            <div className="flex items-center gap-2 mb-1">
394 |              <Gauge className="w-4 h-4 text-blue-400" />
395 |              <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">TOTAL</span>
396 |            </div>
397 |            <div className="text-2xl font-mono font-bold text-blue-400" data-testid="text-desv-total">{stats.total}</div>
398 |            <div className="text-[10px] font-mono text-muted-foreground">{devMode === "km" ? "desviaciones km" : "desv. re
399 |          </div>
400 |        </div>
401 |
402 |        {isError && (
403 |          <div className="bg-red-500/10 border border-red-500/30 p-3 flex items-center gap-2" data-testid="desv-error">
404 |            <AlertTriangle className="w-4 h-4 text-red-400" />
405 |            <span className="text-[11px] font-mono text-red-400">
406 |              Error al obtener datos de fusion. Verifique la conexion con Sigetra y Volvo.
407 |            </span>
408 |          </div>
409 |        )}
410 |
411 |        {!isError && (
412 |          <>
413 |            <div className="flex items-center gap-2">
414 |              <div className="relative flex-1 max-w-xs">
415 |                <Search className="absolute left-2.5 top-1/2 -translate-y-1/2 w-3.5 h-3.5 text-muted-foreground" />
416 |                <Input value={busq} onChange={e => setBusq(e.target.value)}
417 |                  placeholder="Buscar tracto, patente, faena..."
418 |                  className="pl-8 font-mono text-xs bg-card"
419 |                  data-testid="input-desv-search" />
420 |              </div>
421 |              <span className="text-[10px] font-mono text-muted-foreground">
422 |                {devMode === "km" ? filteredKm.length : filteredRend.length} desviaciones
423 |              </span>
424 |            </div>
425 |
```

```
426 |           <div className="flex gap-4">
427 |             <div className={`border border-border bg-card overflow-hidden ${(selectedKm || selectedRend) ? "w-[45%]" : "
428 |               <div className="bg-background border-b border-border px-3 py-2 flex items-center justify-between">
429 |                 <span className="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">
430 |                   {devMode === "km" ? "DESVIACIONES KILOMETRAJE POR FAENA" : "DESVIACIONES RENDIMIENTO POR FAENA"}
431 |                 </span>
432 |                 <span className="text-[8px] font-mono text-muted-foreground">
433 |                   7 dias
434 |                 </span>
435 |               </div>
436 |
437 |               <div className="max-h-[600px] overflow-y-auto">
438 |                 {devMode === "km" && faenaGroups.map(([faena, devs]) => {
439 |                   const critCount = devs.filter(d => d.severity === "critico").length;
440 |                   const gestCount = devs.filter(d => isGestionado(d.truck.fleetNum, "ODO_KM")).length;
441 |
442 |                   return (
443 |                     <div key={faena}>
444 |                       <div className="bg-background/50 border-b border-border px-3 py-2 flex items-center justify-betwee
445 |                         <div className="flex items-center gap-2">
446 |                           <MapPin className="w-3 h-3 text-primary" />
447 |                           <span className="text-[10px] font-mono font-bold text-foreground">{faena}</span>
448 |                           <span className="text-[9px] font-mono text-muted-foreground">({devs.length} camiones)</span>
449 |                         </div>
450 |                         <div className="flex items-center gap-2">
451 |                           {critCount > 0 && (
452 |                             <span className="text-[8px] font-mono px-1.5 py-0.5 bg-red-500/10 border border-red-500/30 t
453 |                               {critCount} CRIT
454 |                             </span>
455 |                           )}
456 |                           {gestCount > 0 && (
457 |                             <span className="text-[8px] font-mono px-1.5 py-0.5 bg-emerald-500/10 border border-emerald-
458 |                               {gestCount} OK
459 |                             </span>
460 |                           )}
461 |                         </div>
462 |                       </div>
463 |
464 |                       {devs.map(d => {
465 |                         const t = d.truck;
466 |                         const checked = isGestionado(t.fleetNum, "ODO_KM");
467 |                         const isActive = selectedTruck === t.fleetNum;
468 |
469 |                         return (
470 |                           <div key={t.fleetNum}
471 |                             className={`px-3 py-2.5 border-b border-border/50 cursor-pointer transition-colors hover:bg-
472 |                             style={{ borderLeftWidth: 3, borderLeftColor: d.severity === "critico" ? "#FF2D4A" : "#FFB02
473 |                             data-testid={`row-desv-${t.fleetNum}`}
474 |                           >
475 |                             <div className="flex items-center gap-2">
476 |                               <button
477 |                                 onClick={(e) => { e.stopPropagation(); handleToggleCheck(t.fleetNum, checked); }}
478 |                                 className="flex-shrink-0 cursor-pointer"
479 |                                 data-testid={`btn-check-${t.fleetNum}`}
480 |                               >
481 |                                 {checked
482 |                                   ? <CheckCircle2 className="w-4.5 h-4.5 text-emerald-400" />
483 |                                   : <Circle className="w-4.5 h-4.5 text-muted-foreground hover:text-primary" />
484 |                                 }
485 |                               </button>
486 |
487 |                               <div className="flex-1 min-w-0" onClick={() => setSelectedTruck(isActive ? null : t.fleetN
488 |                                 <div className="flex items-center justify-between mb-1">
489 |                                   <div className="flex items-center gap-2">
490 |                                     <span className={`text-sm font-mono font-bold ${checked ? "text-muted-foreground lin
491 |                                     <span className="text-[10px] font-mono text-muted-foreground">{t.patenteReal || ""}<
492 |                                   </div>
493 |                                   <div className="flex items-center gap-1.5">
494 |                                     <span className={`w-2 h-2 rounded-full ${d.severity === "critico" ? "bg-red-500 anim
495 |                                     <span className={`text-[9px] font-mono font-bold ${d.severity === "critico" ? "text-
496 |                                       {d.severity === "critico" ? "CRITICO" : "ALERTA"}
497 |                                     </span>
498 |                                   </div>
499 |                                 </div>
500 |                                 <div className="flex items-center gap-3 text-[10px] font-mono">
501 |                                   <span className="text-muted-foreground">{t.conductorSigetra || "\u2014"}</span>
502 |                                   <span className="text-muted-foreground">{t.totalCargas} cargas</span>
503 |                                   <span className={`font-bold ${d.severity === "critico" ? "text-red-400" : "text-amber-
504 |                                     {t.deltaOdometro! > 0 ? "+" : ""}{fN(t.deltaOdometro!)} km
505 |                                   </span>
506 |                                 </div>
507 |                                 {checked && (() => {
508 |                                   const info = getCheckInfo(t.fleetNum, "ODO_KM");
509 |                                   return info?.gestionadoAt ? (
510 |                                     <div className="text-[9px] font-mono text-emerald-400/70 mt-0.5">
511 |                                       Gestionado: {info.gestionadoAt}
512 |                                     </div>
513 |                                   ) : null;
514 |                                 })()}
515 |                               </div>
516 |                             </div>
517 |                           </div>
518 |                         );
519 |                       })}
520 |                     </div>
521 |                   );
522 |                 })}
523 |
524 |                 {devMode === "rend" && rendFaenaGroups.map(([faena, devs]) => {
525 |                   const sobreCount = devs.filter(d => d.direction === "sobre").length;
526 |                   const bajoCount = devs.filter(d => d.direction === "bajo").length;
527 |                   const avg = devs[0]?.faenaAvg || 0;
528 |
529 |                   return (
530 |                     <div key={faena}>
531 |                       <div className="bg-background/50 border-b border-border px-3 py-2 flex items-center justify-betwee
532 |                         <div className="flex items-center gap-2">
```

```
533 |                          <MapPin className="w-3 h-3 text-primary" />
534 |                          <span className="text-[10px] font-mono font-bold text-foreground">{faena}</span>
535 |                          <span className="text-[9px] font-mono text-muted-foreground">({devs.length} camiones)</span>
536 |                          <span className="text-[9px] font-mono text-muted-foreground">prom: {avg.toFixed(2)} km/L</span
537 |                        </div>
538 |                        <div className="flex items-center gap-2">
539 |                          {sobreCount > 0 && (
540 |                            <span className="text-[8px] font-mono px-1.5 py-0.5 bg-emerald-500/10 border border-emerald-
541 |                              {sobreCount} SOBRE
542 |                            </span>
543 |                          )}
544 |                          {bajoCount > 0 && (
545 |                            <span className="text-[8px] font-mono px-1.5 py-0.5 bg-red-500/10 border border-red-500/30 t
546 |                              {bajoCount} BAJO
547 |                            </span>
548 |                          )}
549 |                        </div>
550 |                      </div>
551 |
552 |                      {devs.map(d => {
553 |                        const t = d.truck;
554 |                        const checked = isGestionado(t.fleetNum, "REND_DESV");
555 |                        const isActive = selectedTruck === t.fleetNum;
556 |                        const isCritico = Math.abs(d.pctDesv) > 30;
557 |
558 |                        return (
559 |                          <div key={t.fleetNum}
560 |                            className={`px-3 py-2.5 border-b border-border/50 cursor-pointer transition-colors hover:bg-
561 |                            style={{ borderLeftWidth: 3, borderLeftColor: d.direction === "bajo" ? (isCritico ? "#FF2D4A
562 |                            data-testid={`row-rend-desv-${t.fleetNum}`}
563 |                          >
564 |                            <div className="flex items-center gap-2">
565 |                              <button
566 |                                onClick={(e) => { e.stopPropagation(); handleToggleCheck(t.fleetNum, checked); }}
567 |                                className="flex-shrink-0 cursor-pointer"
568 |                                data-testid={`btn-rend-check-${t.fleetNum}`}
569 |                              >
570 |                                {checked
571 |                                  ? <CheckCircle2 className="w-4.5 h-4.5 text-emerald-400" />
572 |                                  : <Circle className="w-4.5 h-4.5 text-muted-foreground hover:text-primary" />
573 |                                }
574 |                              </button>
575 |
576 |                              <div className="flex-1 min-w-0" onClick={() => setSelectedTruck(isActive ? null : t.fleetN
577 |                                <div className="flex items-center justify-between mb-1">
578 |                                  <div className="flex items-center gap-2">
579 |                                    <span className={`text-sm font-mono font-bold ${checked ? "text-muted-foreground lin
580 |                                    <span className="text-[10px] font-mono text-muted-foreground">{t.patenteReal || ""}<
581 |                                  </div>
582 |                                  <div className="flex items-center gap-1.5">
583 |                                    <span className={`w-2 h-2 rounded-full ${d.direction === "bajo" ? (isCritico ? "bg-r
584 |                                    <span className={`text-[9px] font-mono font-bold ${d.direction === "bajo" ? (isCriti
585 |                                      {d.direction === "sobre" ? "SOBRE PROMEDIO" : isCritico ? "CRITICO BAJO" : "BAJO P
586 |                                    </span>
587 |                                  </div>
588 |                                </div>
589 |                                <div className="flex items-center gap-3 text-[10px] font-mono">
590 |                                  <span className="text-muted-foreground">{t.conductorSigetra || "\u2014"}</span>
591 |                                  <span className="text-foreground font-bold">{d.rendimiento.toFixed(2)} km/L</span>
592 |                                  <span className={`font-bold ${d.direction === "bajo" ? "text-red-400" : "text-emerald-
593 |                                    {d.pctDesv > 0 ? "+" : ""}{d.pctDesv.toFixed(1)}%
594 |                                  </span>
595 |                                  <span className="text-muted-foreground">vs {d.faenaAvg.toFixed(2)} prom</span>
596 |                                </div>
597 |                                {checked && (() => {
598 |                                  const info = getCheckInfo(t.fleetNum, "REND_DESV");
599 |                                  return info?.gestionadoAt ? (
600 |                                    <div className="text-[9px] font-mono text-emerald-400/70 mt-0.5">
601 |                                      Gestionado: {info.gestionadoAt}
602 |                                    </div>
603 |                                  ) : null;
604 |                                })()}
605 |                              </div>
606 |                            </div>
607 |                          </div>
608 |                        );
609 |                      })}
610 |                    </div>
611 |                  );
612 |                })}
613 |
614 |                {devMode === "km" && filteredKm.length === 0 && (
615 |                  <div className="p-8 text-center text-[11px] font-mono text-muted-foreground" data-testid="desv-empty">
616 |                    {busq ? "Sin resultados para la busqueda" : showGestionado ? "No hay desviaciones gestionadas" : "No
617 |                  </div>
618 |                )}
619 |
620 |                {devMode === "rend" && filteredRend.length === 0 && (
621 |                  <div className="p-8 text-center text-[11px] font-mono text-muted-foreground" data-testid="desv-rend-em
622 |                    {busq ? "Sin resultados para la busqueda" : showGestionado ? "No hay desviaciones gestionadas" : "No
623 |                  </div>
624 |                )}
625 |              </div>
626 |            </div>
627 |
628 |            {selectedKm && (
629 |              <div className="w-[55%] border border-border bg-card overflow-hidden" data-testid="desv-detail-panel">
630 |                <div className="bg-background border-b border-border px-4 py-2.5 flex items-center justify-between">
631 |                  <div class="flex items-center gap-2">
632 |                    <ShieldAlert className={`w-4 h-4 ${selectedKm.severity === "critico" ? "text-red-400" : "text-amber-
633 |                    <span class="text-xs font-mono font-bold text-foreground">
634 |                      TRACTO {selectedKm.truck.fleetNum}
635 |                    </span>
636 |                    <span class="text-[10px] font-mono text-muted-foreground">
637 |                      {selectedKm.truck.patenteReal}
638 |                    </span>
639 |                  </div>
```

```
640 |                    <button onClick={() => setSelectedTruck(null)} className="p-1 hover:bg-primary/10 cursor-pointer" data
641 |                      <X className="w-4 h-4 text-muted-foreground" />
642 |                    </button>
643 |                  </div>
644 |
645 |                  <div className="max-h-[570px] overflow-y-auto">
646 |                    <div className="p-4 border-b border-border/50">
647 |                      <div className="grid grid-cols-3 gap-4">
648 |                        <div>
649 |                          <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-1.5">CAMION</div
650 |                          <div className="space-y-1 text-[11px] font-mono">
651 |                            <div><span className="text-muted-foreground">Interno:</span> <span className="text-foreground
652 |                            <div><span className="text-muted-foreground">Patente:</span> <span className="text-foreground"
653 |                            <div><span className="text-muted-foreground">VIN:</span> <span className="text-foreground text
654 |                            <div><span className="text-muted-foreground">Modelo:</span> <span className="text-foreground">
655 |                            <div><span className="text-muted-foreground">Faena:</span> <span className="text-foreground">{
656 |                          </div>
657 |                        </div>
658 |                        <div>
659 |                          <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-1.5">SIGETRA</di
660 |                          <div className="space-y-1 text-[11px] font-mono">
661 |                            <div><span className="text-muted-foreground">Odometro:</span> <span className="text-amber-400
662 |                            <div><span className="text-muted-foreground">Total litros:</span> <span className="text-foregr
663 |                            <div><span className="text-muted-foreground">Cargas:</span> <span className="text-foreground">
664 |                            <div><span className="text-muted-foreground">Rendimiento:</span> <span className="text-foregro
665 |                            <div><span className="text-muted-foreground">Conductor:</span> <span className="text-foregroun
666 |                          </div>
667 |                        </div>
668 |                        <div>
669 |                          <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-1.5">VOLVO ECU</
670 |                          <div className="space-y-1 text-[11px] font-mono">
671 |                            <div><span className="text-muted-foreground">Odometro:</span> <span className="text-blue-400 f
672 |                            <div><span className="text-muted-foreground">Nivel tanque:</span> <span className={`font-bold
673 |                            <div><span className="text-muted-foreground">Fuel ECU (periodo):</span> <span className="text-
674 |                            <div><span className="text-muted-foreground">Horas motor:</span> <span className="text-foregro
675 |                            <div><span className="text-muted-foreground">GPS:</span> <span className="text-foreground">{se
676 |                          </div>
677 |                        </div>
678 |                      </div>
679 |                    </div>
680 |
681 |                    <div className="p-4 border-b border-border/50">
682 |                      <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-2">COMPARACION ODOME
683 |                      <div className="flex items-center gap-3">
684 |                        <div className="flex-1">
685 |                          <div className="text-[9px] font-mono text-muted-foreground mb-1">SIGETRA</div>
686 |                          <div className="text-lg font-mono font-bold text-amber-400">{fN(selectedKm.truck.odometroSigetra
687 |                        </div>
688 |                        <div className="flex flex-col items-center gap-1">
689 |                          <ArrowRightLeft className={`w-5 h-5 ${selectedKm.severity === "critico" ? "text-red-400" : "text
690 |                          <div className={`text-xs font-mono font-bold ${selectedKm.severity === "critico" ? "text-red-400
691 |                            {selectedKm.truck.deltaOdometro! > 0 ? "+" : ""}{fN(selectedKm.truck.deltaOdometro!)} km
692 |                          </div>
693 |                        </div>
694 |                        <div className="flex-1 text-right">
695 |                          <div className="text-[9px] font-mono text-muted-foreground mb-1">VOLVO ECU</div>
696 |                          <div className="text-lg font-mono font-bold text-blue-400">{fN(selectedKm.truck.odometroVolvo!)}
697 |                        </div>
698 |                      </div>
699 |                      <div className={`mt-3 p-2 border text-[10px] font-mono ${
700 |                        selectedKm.severity === "critico" ? "bg-red-500/10 border-red-500/30 text-red-400" : "bg-amber-500
701 |                      }`}>
702 |                        {selectedKm.severity === "critico"
703 |                          ? `Desviacion critica de ${fN(selectedKm.absDelta)} km. Diferencia superior a 5.000 km indica po
704 |                          : `Desviacion moderada de ${fN(selectedKm.absDelta)} km. Puede deberse a retraso en actualizacio
705 |                        }
706 |                      </div>
707 |                    </div>
708 |
709 |                    <div className="p-4 border-b border-border/50">
710 |                      <div className="flex items-center justify-between mb-2">
711 |                        <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em]">GESTION</div>
712 |                      </div>
713 |                      {(() => {
714 |                        const checked = isGestionado(selectedKm.truck.fleetNum, "ODO_KM");
715 |                        const info = getCheckInfo(selectedKm.truck.fleetNum, "ODO_KM");
716 |                        return (
717 |                          <button
718 |                            onClick={() => handleToggleCheck(selectedKm.truck.fleetNum, checked)}
719 |                            data-testid={`btn-detail-check-${selectedKm.truck.fleetNum}`}
720 |                            className={`w-full flex items-center gap-3 p-3 border cursor-pointer transition-colors ${
721 |                              checked
722 |                                ? "bg-emerald-500/10 border-emerald-500/30"
723 |                                : "bg-card border-border hover:border-primary/30"
724 |                            }`}
725 |                          >
726 |                            {checked
727 |                              ? <CheckCircle2 className="w-5 h-5 text-emerald-400" />
728 |                              : <Circle className="w-5 h-5 text-muted-foreground" />
729 |                            }
730 |                            <div className="text-left">
731 |                              <div className={`text-xs font-mono font-bold ${checked ? "text-emerald-400" : "text-foregrou
732 |                                {checked ? "GESTIONADO" : "MARCAR COMO GESTIONADO"}
733 |                              </div>
734 |                              {checked && info?.gestionadoAt && (
735 |                                <div className="text-[9px] font-mono text-emerald-400/70">{info.gestionadoAt}</div>
736 |                              )}
737 |                              {!checked && (
738 |                                <div className="text-[9px] font-mono text-muted-foreground">Click para marcar como revisad
739 |                              )}
740 |                            </div>
741 |                          </button>
742 |                        );
743 |                      })()}
744 |                    </div>
745 |
746 |                    <div className="p-4">
```

```
747 |                    <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-2">
748 |                      HISTORIAL DE CARGAS ({selectedKm.truck.cargas.length})
749 |                    </div>
750 |                    <div className="border border-border/50 max-h-[200px] overflow-y-auto">
751 |                      <div className="grid grid-cols-[1fr_70px_80px_80px_70px_1fr] gap-0 bg-card/50 px-2 py-1.5 border-b
752 |                        <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">FECHA</span>
753 |                        <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">LITROS</span>
754 |                        <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">ODOMETRO</span>
755 |                        <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">KM REC.</span>
756 |                        <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">REND.</span>
757 |                        <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">CONDUCTOR</span>
758 |                      </div>
759 |                      {selectedKm.truck.cargas.map((c, i) => (
760 |                        <div key={i} className="grid grid-cols-[1fr_70px_80px_80px_70px_1fr] gap-0 px-2 py-1.5 border-b
761 |                          <span className="text-muted-foreground">{formatDateTime(c.fecha)}</span>
762 |                          <span className="text-amber-400 font-bold">{c.litros.toFixed(1)}</span>
763 |                          <span className="text-foreground">{c.odometro != null ? fN(c.odometro) : "\u2014"}</span>
764 |                          <span className={`${c.kmRecorrido != null && c.kmRecorrido < 0 ? "text-red-400" : "text-foregr
765 |                          <span className="text-foreground">{c.rendimiento != null ? c.rendimiento.toFixed(1) : "\u2014"
766 |                          <span className="text-muted-foreground truncate">{c.conductor || "\u2014"}</span>
767 |                        </div>
768 |                      ))}
769 |                    </div>
770 |                  </div>
771 |                </div>
772 |              </div>
773 |            )}
774 |
775 |            {selectedRend && (
776 |              <div className="w-[55%] border border-border bg-card overflow-hidden" data-testid="desv-rend-detail-panel"
777 |                <div className="bg-background border-b border-border px-4 py-2.5 flex items-center justify-between">
778 |                  <div className="flex items-center gap-2">
779 |                    <BarChart3 className={`w-4 h-4 ${selectedRend.direction === "bajo" ? "text-red-400" : "text-emerald-
780 |                    <span className="text-xs font-mono font-bold text-foreground">
781 |                      TRACTO {selectedRend.truck.fleetNum}
782 |                    </span>
783 |                    <span className="text-[10px] font-mono text-muted-foreground">
784 |                      {selectedRend.truck.patenteReal}
785 |                    </span>
786 |                  </div>
787 |                  <button onClick={() => setSelectedTruck(null)} className="p-1 hover:bg-primary/10 cursor-pointer" data
788 |                    <X className="w-4 h-4 text-muted-foreground" />
789 |                  </button>
790 |                </div>
791 |
792 |                <div className="max-h-[570px] overflow-y-auto">
793 |                  <div className="p-4 border-b border-border/50">
794 |                    <div class="grid grid-cols-2 gap-4">
795 |                      <div>
796 |                        <div class="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-1.5">CAMION</div
797 |                        <div class="space-y-1 text-[11px] font-mono">
798 |                          <div><span class="text-muted-foreground">Interno:</span> <span class="text-foreground
799 |                          <div><span class="text-muted-foreground">Patente:</span> <span class="text-foreground"
800 |                          <div><span class="text-muted-foreground">Faena:</span> <span class="text-foreground">{
801 |                          <div><span class="text-muted-foreground">Conductor:</span> <span class="text-foregroun
802 |                          <div><span class="text-muted-foreground">Cargas:</span> <span class="text-foreground">
803 |                        </div>
804 |                      </div>
805 |                      <div>
806 |                        <div class="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-1.5">RENDIMIENTO
807 |                        <div class="space-y-1 text-[11px] font-mono">
808 |                          <div><span class="text-muted-foreground">Rendimiento:</span> <span class={`font-bold $
809 |                          <div><span class="text-muted-foreground">Promedio faena:</span> <span class="text-fore
810 |                          <div><span class="text-muted-foreground">Desviacion:</span> <span class={`font-bold ${
811 |                          <div><span class="text-muted-foreground">Litros periodo:</span> <span class="text-ambe
812 |                        </div>
813 |                      </div>
814 |                    </div>
815 |                  </div>
816 |
817 |                  <div class="p-4 border-b border-border/50">
818 |                    <div class="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-2">COMPARACION RENDI
819 |                    <div class="flex items-center gap-3">
820 |                      <div class="flex-1">
821 |                        <div class="text-[9px] font-mono text-muted-foreground mb-1">CAMION</div>
822 |                        <div class={`text-lg font-mono font-bold ${selectedRend.direction === "bajo" ? "text-red-400
823 |                      </div>
824 |                      <div class="flex flex-col items-center gap-1">
825 |                        <ArrowRightLeft class={`w-5 h-5 ${selectedRend.direction === "bajo" ? "text-red-400" : "text
826 |                        <div class={`text-xs font-mono font-bold ${selectedRend.direction === "bajo" ? "text-red-400
827 |                          {selectedRend.pctDesv > 0 ? "+" : ""}{selectedRend.pctDesv.toFixed(1)}%
828 |                        </div>
829 |                      </div>
830 |                      <div class="flex-1 text-right">
831 |                        <div class="text-[9px] font-mono text-muted-foreground mb-1">PROM. FAENA</div>
832 |                        <div class="text-lg font-mono font-bold text-foreground">{selectedRend.faenaAvg.toFixed(2)}
833 |                      </div>
834 |                    </div>
835 |                    <div class={`mt-3 p-2 border text-[10px] font-mono ${
836 |                      selectedRend.direction === "bajo"
837 |                        ? (Math.abs(selectedRend.pctDesv) > 30 ? "bg-red-500/10 border-red-500/30 text-red-400" : "bg-am
838 |                        : "bg-emerald-500/10 border-emerald-500/30 text-emerald-400"
839 |                    }`}>
840 |                      {selectedRend.direction === "bajo"
841 |                        ? `Rendimiento ${Math.abs(selectedRend.pctDesv).toFixed(1)}% por debajo del promedio de la faena
842 |                        : `Rendimiento ${selectedRend.pctDesv.toFixed(1)}% sobre el promedio de la faena (${selectedRend
843 |                      }
844 |                    </div>
845 |                  </div>
846 |
847 |                  <div class="p-4 border-b border-border/50">
848 |                    <div class="flex items-center justify-between mb-2">
849 |                      <div class="text-[8px] font-mono text-muted-foreground tracking-[0.15em]">GESTION</div>
850 |                    </div>
851 |                    {(() => {
852 |                      const checked = isGestionado(selectedRend.truck.fleetNum, "REND_DESV");
853 |                      const info = getCheckInfo(selectedRend.truck.fleetNum, "REND_DESV");
```

```
854 |                       return (
855 |                         <button
856 |                           onClick={() => handleToggleCheck(selectedRend.truck.fleetNum, checked)}
857 |                           data-testid={`btn-detail-rend-check-${selectedRend.truck.fleetNum}`}
858 |                           className={`w-full flex items-center gap-3 p-3 border cursor-pointer transition-colors ${
859 |                             checked
860 |                               ? "bg-emerald-500/10 border-emerald-500/30"
861 |                               : "bg-card border-border hover:border-primary/30"
862 |                           }`}
863 |                         >
864 |                           {checked
865 |                             ? <CheckCircle2 className="w-5 h-5 text-emerald-400" />
866 |                             : <Circle className="w-5 h-5 text-muted-foreground" />
867 |                           }
868 |                           <div className="text-left">
869 |                             <div className={`text-xs font-mono font-bold ${checked ? "text-emerald-400" : "text-foregrou
870 |                               {checked ? "GESTIONADO" : "MARCAR COMO GESTIONADO"}
871 |                             </div>
872 |                             {checked && info?.gestionadoAt && (
873 |                               <div className="text-[9px] font-mono text-emerald-400/70">{info.gestionadoAt}</div>
874 |                             )}
875 |                             {!checked && (
876 |                               <div className="text-[9px] font-mono text-muted-foreground">Click para marcar como revisad
877 |                             )}
878 |                           </div>
879 |                         </button>
880 |                       );
881 |                     })()}
882 |                   </div>
883 |
884 |                   <div className="p-4">
885 |                     <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-2">
886 |                       HISTORIAL DE CARGAS ({selectedRend.truck.cargas.length})
887 |                     </div>
888 |                     <div className="border border-border/50 max-h-[200px] overflow-y-auto">
889 |                       <div className="grid grid-cols-[1fr_70px_80px_80px_70px_1fr] gap-0 bg-card/50 px-2 py-1.5 border-b
890 |                         <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">FECHA</span>
891 |                         <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">LITROS</span>
892 |                         <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">ODOMETRO</span>
893 |                         <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">KM REC.</span>
894 |                         <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">REND.</span>
895 |                         <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">CONDUCTOR</span>
896 |                       </div>
897 |                       {selectedRend.truck.cargas.map((c, i) => (
898 |                         <div key={i} className="grid grid-cols-[1fr_70px_80px_80px_70px_1fr] gap-0 px-2 py-1.5 border-b
899 |                           <span className="text-muted-foreground">{formatDateTime(c.fecha)}</span>
900 |                           <span className="text-amber-400 font-bold">{c.litros.toFixed(1)}</span>
901 |                           <span className="text-foreground">{c.odometro != null ? fN(c.odometro) : "\u2014"}</span>
902 |                           <span className={`${c.kmRecorrido != null && c.kmRecorrido < 0 ? "text-red-400" : "text-foregr
903 |                           <span className={`font-bold ${c.rendimiento != null && c.rendimiento > 0 ? (c.rendimiento >= s
904 |                           <span className="text-muted-foreground truncate">{c.conductor || "\u2014"}</span>
905 |                         </div>
906 |                       ))}
907 |                     </div>
908 |                   </div>
909 |                 </div>
910 |               </div>
911 |             )}
912 |           </div>
913 |         </>
914 |       )}
915 |     </div>
916 |   );
917 | }
918 |
```

client/src/pages/volvo.tsx

679 lineas

```
1 | import { useState, useMemo, useEffect, useRef } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { Skeleton } from "@/components/ui/skeleton";
4 | import { Input } from "@/components/ui/input";
5 | import { Button } from "@/components/ui/button";
6 | import { Card } from "@/components/ui/card";
7 | import { Badge } from "@/components/ui/badge";
8 | import { VolvoTruckModal } from "@/components/volvo-truck-modal";
9 | import { fN } from "@/lib/fuel-utils";
10 | import {
11 |   Truck, MapPin, Gauge, Clock, Fuel, Thermometer, Search,
12 |   Radio, Wifi, WifiOff, AlertTriangle, List, Map as MapIcon,
13 |   AlertCircle, Link2Off, Shield
14 | } from "lucide-react";
15 | import type { Faena, Camion } from "@shared/schema";
16 | import L from "leaflet";
17 | import "leaflet/dist/leaflet.css";
18 |
19 | interface UnifiedStatus {
20 |   vin: string;
21 |   createdDateTime: string | null;
22 |   gps: {
23 |     latitude: number | null;
24 |     longitude: number | null;
25 |     altitude: number | null;
26 |     heading: number | null;
27 |     speed: number | null;
28 |     positionDateTime: string | null;
29 |   };
30 |   fuelLevel: number | null;
31 |   engineHours: number | null;
32 |   totalDistance: number | null;
33 |   totalFuelUsed: number | null;
34 |   engineSpeed: number | null;
35 |   wheelBasedSpeed: number | null;
36 |   ambientTemperature: number | null;
37 |   driverId: string | null;
38 |   driverWorkingState: string | null;
39 |   catalystFuelLevel: number | null;
40 |   grossWeight: number | null;
41 | }
42 |
43 | function createColorIcon(color: string): L.DivIcon {
44 |   return L.divIcon({
45 |     className: "",
46 |     iconSize: [28, 28],
47 |     iconAnchor: [14, 14],
48 |     popupAnchor: [0, -16],
49 |     html: `<div style="width:28px;height:28px;border-radius:50%;background:${color};border:3px solid white;box-shadow:0
50 |       <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="white" stroke-width="2.5" stroke-linecap="roun
51 |     </div>`,
52 |   });
53 | }
54 |
55 | const greenIcon = createColorIcon("#22c55e");
56 | const yellowIcon = createColorIcon("#eab308");
57 | const greyIcon = createColorIcon("#6b7280");
58 |
59 | function TruckMap({ camiones, statusMap, faenas, onSelectVin }: {
60 |   camiones: Camion[];
61 |   statusMap: Map<string, UnifiedStatus>;
62 |   faenas: Faena[];
63 |   onSelectVin: (vin: string) => void;
64 | }) {
65 |   const mapRef = useRef<HTMLDivElement>(null);
66 |   const mapInstanceRef = useRef<L.Map | null>(null);
67 |
68 |   useEffect(() => {
69 |     if (!mapRef.current) return;
70 |
71 |     if (mapInstanceRef.current) {
72 |       mapInstanceRef.current.remove();
73 |       mapInstanceRef.current = null;
74 |     }
75 |
76 |     const map = L.map(mapRef.current, {
77 |       center: [-33.45, -70.65],
78 |       zoom: 6,
79 |       zoomControl: true,
80 |     });
81 |
82 |     L.tileLayer("https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png", {
83 |       attribution: '&copy; OpenStreetMap',
84 |       maxZoom: 18,
85 |     }).addTo(map);
86 |
87 |     mapInstanceRef.current = map;
88 |
89 |     const bounds: L.LatLngExpression[] = [];
90 |
91 |     camiones.forEach(cam => {
92 |       if (!cam.vin) return;
93 |       const vs = statusMap.get(cam.vin);
94 |       const gps = vs?.gps;
95 |       const hasGps = gps && gps.latitude != null && gps.longitude != null;
96 |
97 |       if (!hasGps) return;
98 |
99 |       const lat = gps!.latitude!;
100 |       const lng = gps!.longitude!;
101 |       bounds.push([lat, lng]);
102 |
103 |       const isOnline = !!vs;
104 |       const hasData = vs && (vs.fuelLevel != null || vs.totalDistance != null);
```

```
105 |       let icon = greyIcon;
106 |       if (isOnline && hasData) icon = greenIcon;
107 |       else if (isOnline) icon = yellowIcon;
108 |
109 |       const faena = faenas.find(f => f.id === cam.faenaId);
110 |       const distKm = vs?.totalDistance ? Math.round(vs.totalDistance / 1000) : cam.odometro;
111 |       const fuelLvl = vs?.fuelLevel;
112 |       const speed = vs?.wheelBasedSpeed;
113 |
114 |       const popupContent = `
115 |         <div style="font-family:monospace;font-size:11px;min-width:180px;">
116 |           <div style="font-weight:bold;font-size:13px;margin-bottom:4px;color:#1a8fff;">${cam.patente}</div>
117 |           <div style="font-size:10px;color:#888;margin-bottom:6px;">${cam.modelo || ""}</div>
118 |           ${faena ? `<div style="font-size:10px;margin-bottom:4px;"><span style="display:inline-block;width:8px;height:8
119 |           ${cam.conductor ? `<div style="font-size:10px;margin-bottom:2px;">Conductor: ${cam.conductor}</div>` : ""}
120 |           ${distKm ? `<div style="font-size:10px;margin-bottom:2px;">Km: ${fN(distKm)}</div>` : ""}
121 |           ${fuelLvl != null ? `<div style="font-size:10px;margin-bottom:2px;">Combustible: ${fuelLvl}%</div>` : ""}
122 |           ${speed != null && speed > 0 ? `<div style="font-size:10px;margin-bottom:2px;">Velocidad: ${speed.toFixed(0)}
123 |           <div style="margin-top:6px;"><a href="#" onclick="window.__volvoSelectVin__('${cam.vin}');return false;" style
124 |         </div>
125 |       `;
126 |
127 |       L.marker([lat, lng], { icon })
128 |         .addTo(map)
129 |         .bindPopup(popupContent);
130 |     });
131 |
132 |     if (bounds.length > 0) {
133 |       map.fitBounds(bounds as L.LatLngBoundsExpression, { padding: [40, 40], maxZoom: 12 });
134 |     }
135 |
136 |     (window as any).__volvoSelectVin__ = (vin: string) => {
137 |       onSelectVin(vin);
138 |     };
139 |
140 |     return () => {
141 |       if (mapInstanceRef.current) {
142 |         mapInstanceRef.current.remove();
143 |         mapInstanceRef.current = null;
144 |       }
145 |       delete (window as any).__volvoSelectVin__;
146 |     };
147 |   }, [camiones, statusMap, faenas, onSelectVin]);
148 |
149 |   return (
150 |     <div
151 |       ref={mapRef}
152 |       className="w-full border border-border"
153 |       style={{ height: "500px" }}
154 |       data-testid="volvo-map"
155 |     />
156 |   );
157 | }
158 |
159 | function DiagnosticsSection({ camiones, statusMap, faenas, onSelectVin }: {
160 |   camiones: Camion[];
161 |   statusMap: Map<string, UnifiedStatus>;
162 |   faenas: Faena[];
163 |   onSelectVin: (vin: string) => void;
164 | }) {
165 |   const noVin = camiones.filter(c => c.syncOk && !c.vin);
166 |
167 |   const noDataVins = useMemo(() => {
168 |     const now = Date.now();
169 |     const h48 = 48 * 60 * 60 * 1000;
170 |     return camiones.filter(c => {
171 |       if (!c.vin || !c.syncOk) return false;
172 |       const vs = statusMap.get(c.vin);
173 |       if (!vs) return true;
174 |       if (vs.createdDateTime) {
175 |         const dt = new Date(vs.createdDateTime).getTime();
176 |         return now - dt > h48;
177 |       }
178 |       return false;
179 |     });
180 |   }, [camiones, statusMap]);
181 |
182 |   const unassigned = camiones.filter(c => c.syncOk && (!c.faenaId || c.faenaId === 0));
183 |
184 |   const inactive = useMemo(() => {
185 |     return camiones.filter(c => {
186 |       if (!c.vin || !c.syncOk) return false;
187 |       const vs = statusMap.get(c.vin);
188 |       if (!vs) return true;
189 |       const speed = vs.wheelBasedSpeed;
190 |       const engineSpeed = vs.engineSpeed;
191 |       return (speed == null || speed === 0) && (engineSpeed == null || engineSpeed === 0);
192 |     });
193 |   }, [camiones, statusMap]);
194 |
195 |   const hasIssues = noVin.length > 0 || noDataVins.length > 0 || unassigned.length > 0 || inactive.length > 0;
196 |
197 |   if (!hasIssues) return null;
198 |
199 |   return (
200 |     <div className="space-y-3" data-testid="volvo-diagnostics">
201 |       <div className="flex items-center gap-2 pb-2 border-b border-border">
202 |         <AlertCircle className="w-4 h-4 text-amber-400" />
203 |         <span className="text-[11px] font-mono font-bold tracking-[0.15em] uppercase text-amber-400">
204 |           Diagnosticos
205 |         </span>
206 |       </div>
207 |
208 |       <div className="grid grid-cols-1 md:grid-cols-2 gap-3">
209 |         {noVin.length > 0 && (
210 |           <Card className="p-3" data-testid="diag-no-vin">
211 |             <div className="flex items-center gap-2 mb-2">
```

```
212 |               <Link2Off className="w-4 h-4 text-muted-foreground" />
213 |               <span className="text-[10px] font-mono font-bold tracking-[0.15em] uppercase text-muted-foreground">
214 |                 Sin VIN asignado
215 |               </span>
216 |               <Badge variant="secondary" className="text-[9px]">{noVin.length}</Badge>
217 |             </div>
218 |             <div className="space-y-1">
219 |               {noVin.map(c => (
220 |                 <div key={c.id} className="text-[11px] font-mono text-muted-foreground" data-testid={`diag-novin-${c.id}
221 |                   {c.patente} — {c.modelo}
222 |                 </div>
223 |               ))}
224 |             </div>
225 |           </Card>
226 |         )}
227 |
228 |         {noDataVins.length > 0 && (
229 |           <Card className="p-3" data-testid="diag-no-data-48h">
230 |             <div className="flex items-center gap-2 mb-2">
231 |               <AlertTriangle className="w-4 h-4 text-amber-400" />
232 |               <span className="text-[10px] font-mono font-bold tracking-[0.15em] uppercase text-amber-400">
233 |                 Sin datos rFMS 48h
234 |               </span>
235 |               <Badge variant="secondary" className="text-[9px]">{noDataVins.length}</Badge>
236 |             </div>
237 |             <div className="space-y-1">
238 |               {noDataVins.map(c => (
239 |                 <div key={c.id} className="text-[11px] font-mono text-muted-foreground" data-testid={`diag-nodata-${c.id
240 |                   {c.patente} — VIN: {c.vin}
241 |                 </div>
242 |               ))}
243 |             </div>
244 |           </Card>
245 |         )}
246 |
247 |         {inactive.length > 0 && (
248 |           <Card className="p-3" data-testid="diag-inactive">
249 |             <div className="flex items-center gap-2 mb-2">
250 |               <Shield className="w-4 h-4 text-muted-foreground/60" />
251 |               <span className="text-[10px] font-mono font-bold tracking-[0.15em] uppercase text-muted-foreground">
252 |                 Inactivos
253 |               </span>
254 |               <Badge variant="secondary" className="text-[9px]">{inactive.length}</Badge>
255 |             </div>
256 |             <div className="space-y-1">
257 |               {inactive.slice(0, 10).map(c => (
258 |                 <div
259 |                   key={c.id}
260 |                   className="text-[11px] font-mono text-muted-foreground cursor-pointer hover:text-primary"
261 |                   onClick={() => c.vin && onSelectVin(c.vin)}
262 |                   data-testid={`diag-inactive-${c.id}`}
263 |                 >
264 |                   {c.patente}
265 |                 </div>
266 |               ))}
267 |               {inactive.length > 10 && (
268 |                 <div className="text-[10px] font-mono text-muted-foreground/60">
269 |                   y {inactive.length - 10} mas...
270 |                 </div>
271 |               )}
272 |             </div>
273 |           </Card>
274 |         )}
275 |
276 |         {unassigned.length > 0 && (
277 |           <Card className="p-3" data-testid="diag-unassigned">
278 |             <div className="flex items-center gap-2 mb-2">
279 |               <MapPin className="w-4 h-4 text-amber-400" />
280 |               <span className="text-[10px] font-mono font-bold tracking-[0.15em] uppercase text-amber-400">
281 |                 Sin faena asignada
282 |               </span>
283 |               <Badge variant="secondary" className="text-[9px]">{unassigned.length}</Badge>
284 |             </div>
285 |             <div className="space-y-1">
286 |               {unassigned.map(c => (
287 |                 <div key={c.id} className="flex items-center justify-between" data-testid={`diag-unassigned-${c.id}`}>
288 |                   <span className="text-[11px] font-mono text-muted-foreground">{c.patente}</span>
289 |                 </div>
290 |               ))}
291 |             </div>
292 |           </Card>
293 |         )}
294 |       </div>
295 |     </div>
296 |   );
297 | }
298 |
299 | function TruckCard({ cam, vs, faenaColor, onClick }: {
300 |   cam: Camion;
301 |   vs: UnifiedStatus | null;
302 |   faenaColor: string;
303 |   onClick: () => void;
304 | }) {
305 |   const gps = vs?.gps;
306 |   const distKm = vs?.totalDistance ? Math.round(vs.totalDistance / 1000) : cam.odometro;
307 |   const engineH = vs?.engineHours ? Math.round(vs.engineHours * 10) / 10 : cam.horasMotor;
308 |   const fuelLvl = vs?.fuelLevel;
309 |   const speed = vs?.wheelBasedSpeed;
310 |   const temp = vs?.ambientTemperature;
311 |   const hasLiveData = !!vs;
312 |
313 |   return (
314 |     <div
315 |       onClick={onClick}
316 |       className="bg-card border border-border p-3 cursor-pointer transition-all hover:border-primary/40 hover:bg-primary
317 |       style={{ borderLeftWidth: 3, borderLeftColor: faenaColor }}
318 |       data-testid={`volvo-truck-card-${cam.id}`}
```

```
319 |     >
320 |       <div className="flex justify-between items-start mb-2">
321 |         <div>
322 |           <div className="text-sm font-mono font-bold text-primary">{cam.patente}</div>
323 |           <div className="text-[10px] font-mono text-muted-foreground">{cam.modelo}</div>
324 |         </div>
325 |         <div className={`flex items-center gap-1 text-[9px] font-mono ${hasLiveData ? "text-emerald-400" : "text-muted-f
326 |           {hasLiveData ? <Wifi className="w-3 h-3" /> : <WifiOff className="w-3 h-3" />}
327 |           {hasLiveData ? "LIVE" : "OFF"}
328 |         </div>
329 |       </div>
330 |
331 |       <div className="grid grid-cols-2 gap-x-3 gap-y-1.5 text-[10px] font-mono">
332 |         <div className="flex items-center gap-1.5 text-muted-foreground">
333 |           <Gauge className="w-3 h-3 flex-shrink-0" />
334 |           <span>{distKm ? `${fN(distKm)} km` : "\u2014"}</span>
335 |         </div>
336 |         <div className="flex items-center gap-1.5 text-muted-foreground">
337 |           <Clock className="w-3 h-3 flex-shrink-0" />
338 |           <span>{engineH ? `${fN(engineH, 1)} h` : "\u2014"}</span>
339 |         </div>
340 |         {fuelLvl != null && (
341 |           <div className="flex items-center gap-1.5">
342 |             <Fuel className="w-3 h-3 flex-shrink-0 text-amber-400" />
343 |             <span className={fuelLvl < 20 ? "text-red-400" : fuelLvl < 40 ? "text-amber-400" : "text-emerald-400"}>
344 |               {fuelLvl}%
345 |             </span>
346 |           </div>
347 |         )}
348 |         {speed != null && speed > 0 && (
349 |           <div className="flex items-center gap-1.5 text-blue-400">
350 |             <MapPin className="w-3 h-3 flex-shrink-0" />
351 |             <span>{speed.toFixed(0)} km/h</span>
352 |           </div>
353 |         )}
354 |         {gps && gps.latitude != null && (
355 |           <div className="col-span-2 flex items-center gap-1.5 text-muted-foreground/70">
356 |             <MapPin className="w-3 h-3 flex-shrink-0" />
357 |             <span className="truncate">{gps.latitude?.toFixed(4)}, {gps.longitude?.toFixed(4)}</span>
358 |           </div>
359 |         )}
360 |         {temp != null && (
361 |           <div className="flex items-center gap-1.5 text-muted-foreground">
362 |             <Thermometer className="w-3 h-3 flex-shrink-0" />
363 |             <span>{temp.toFixed(0)}C</span>
364 |           </div>
365 |         )}
366 |       </div>
367 |     </div>
368 |   );
369 | }
370 |
371 | export default function Volvo() {
372 |   const [faenaFil, setFaenaFil] = useState<string>("all");
373 |   const [busq, setBusq] = useState("");
374 |   const [selectedVin, setSelectedVin] = useState<string | null>(null);
375 |   const [viewMode, setViewMode] = useState<"lista" | "mapa">("lista");
376 |
377 |   const { data: faenas = [], isLoading: loadingF } = useQuery<Faena[]>({ queryKey: ["/api/faenas"] });
378 |   const { data: camiones = [], isLoading: loadingC } = useQuery<Camion[]>({ queryKey: ["/api/camiones"] });
379 |   const { data: fleetStatus = [], isLoading: loadingS, isError: errorS } = useQuery<UnifiedStatus[]>({
380 |     queryKey: ["/api/volvo/fleet-status"],
381 |     refetchInterval: 300000,
382 |   });
383 |
384 |   const isLoading = loadingF || loadingC || loadingS;
385 |
386 |   const statusMap = useMemo(() => {
387 |     const map = new Map<string, UnifiedStatus>();
388 |     fleetStatus.forEach(s => map.set(s.vin, s));
389 |     return map;
390 |   }, [fleetStatus]);
391 |
392 |   const filtered = useMemo(() => {
393 |     let list = camiones.filter(c => c.syncOk);
394 |     if (faenaFil !== "all") list = list.filter(c => c.faenaId === parseInt(faenaFil));
395 |     if (busq) list = list.filter(c =>
396 |       c.patente.toLowerCase().includes(busq.toLowerCase()) ||
397 |       c.vin?.toLowerCase().includes(busq.toLowerCase()) ||
398 |       c.conductor?.toLowerCase().includes(busq.toLowerCase())
399 |     );
400 |     return list;
401 |   }, [camiones, faenaFil, busq]);
402 |
403 |   const { onlineTrucks, offlineTrucks } = useMemo(() => {
404 |     const online: Camion[] = [];
405 |     const offline: Camion[] = [];
406 |     for (const c of filtered) {
407 |       if (c.vin && statusMap.has(c.vin)) {
408 |         online.push(c);
409 |       } else {
410 |         offline.push(c);
411 |       }
412 |     }
413 |     return { onlineTrucks: online, offlineTrucks: offline };
414 |   }, [filtered, statusMap]);
415 |
416 |   const groupByFaena = (trucks: Camion[]) => {
417 |     const groups: { faena: Faena | null; trucks: Camion[] }[] = [];
418 |     const faenaMap = new Map<number, Camion[]>();
419 |
420 |     trucks.forEach(c => {
421 |       const key = c.faenaId || 0;
422 |       if (!faenaMap.has(key)) faenaMap.set(key, []);
423 |       faenaMap.get(key)!.push(c);
424 |     });
425 |
```

```
426 |     faenaMap.forEach((trks, faenaId) => {
427 |       const faena = faenas.find(f => f.id === faenaId) || null;
428 |       groups.push({ faena, trucks: trks });
429 |     });
430 |
431 |     groups.sort((a, b) => {
432 |       if (!a.faena) return 1;
433 |       if (!b.faena) return -1;
434 |       return a.faena.nombre.localeCompare(b.faena.nombre);
435 |     });
436 |
437 |     return groups;
438 |   };
439 |
440 |   const onlineGrouped = useMemo(() => groupByFaena(onlineTrucks), [onlineTrucks, faenas]);
441 |   const offlineGrouped = useMemo(() => groupByFaena(offlineTrucks), [offlineTrucks, faenas]);
442 |
443 |   const selectedCamion = selectedVin ? camiones.find(c => c.vin === selectedVin) : null;
444 |   const selectedStatus = selectedVin ? statusMap.get(selectedVin) || null : null;
445 |   const totalWithGps = fleetStatus.filter(s => s.gps.latitude != null).length;
446 |
447 |   const handleSelectVin = useMemo(() => (vin: string) => setSelectedVin(vin), []);
448 |
449 |   if (isLoading) {
450 |     return (
451 |       <div className="space-y-5" data-testid="volvo-loading">
452 |         <div className="grid grid-cols-4 gap-3">
453 |           {Array.from({ length: 4 }).map((_, i) => <Skeleton key={i} className="h-20" />)}
454 |         </div>
455 |         <Skeleton className="h-96" />
456 |       </div>
457 |     );
458 |   }
459 |
460 |   const renderFaenaGroup = (group: { faena: Faena | null; trucks: Camion[] }) => (
461 |     <div key={group.faena?.id ?? 0}>
462 |       <div className="flex items-center gap-2 mb-2.5 pb-2 border-b border-border">
463 |         {group.faena ? (
464 |           <>
465 |             <div className="w-3 h-3 rounded-full" style={{ backgroundColor: group.faena.color }} />
466 |             <span className="text-[13px] font-mono font-bold tracking-[0.1em] uppercase" style={{ color: group.faena.col
467 |               {group.faena.nombre}
468 |             </span>
469 |           </>
470 |         ) : (
471 |           <>
472 |             <div className="w-3 h-3 rounded-full bg-muted-foreground/40" />
473 |             <span className="text-[13px] font-mono font-bold tracking-[0.1em] uppercase text-muted-foreground">
474 |               Sin Asignar
475 |             </span>
476 |           </>
477 |         )}
478 |         <span className="text-[10px] font-mono text-muted-foreground ml-1">
479 |           ({group.trucks.length} camiones)
480 |         </span>
481 |       </div>
482 |
483 |       <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 xl:grid-cols-4 gap-2.5 mb-5">
484 |         {group.trucks.map(cam => (
485 |           <TruckCard
486 |             key={cam.id}
487 |             cam={cam}
488 |             vs={cam.vin ? statusMap.get(cam.vin) || null : null}
489 |             faenaColor={group.faena?.color || "#5A7A9A"}
490 |             onClick={() => cam.vin && setSelectedVin(cam.vin)}
491 |           />
492 |         ))}
493 |       </div>
494 |     </div>
495 |   );
496 |
497 |   return (
498 |     <div className="space-y-5" data-testid="volvo-view">
499 |       <div className="grid grid-cols-4 gap-3" data-testid="volvo-kpi-grid">
500 |         <div className="bg-card border border-border p-3.5">
501 |           <div className="flex items-center gap-2 mb-1.5">
502 |             <Truck className="w-4 h-4 text-primary" />
503 |             <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">FLOTA rFMS</span>
504 |           </div>
505 |           <div className="text-2xl font-mono font-bold text-primary" data-testid="text-fleet-count">{filtered.length}</d
506 |           <div class="text-[10px] font-mono text-muted-foreground">{camiones.filter(c => c.syncOk).length} sincroniz
507 |         </div>
508 |         <div className="bg-card border border-border p-3.5">
509 |           <div className="flex items-center gap-2 mb-1.5">
510 |             <Wifi className="w-4 h-4 text-emerald-400" />
511 |             <span class="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">ONLINE</span>
512 |           </div>
513 |           <div class="text-2xl font-mono font-bold text-emerald-400" data-testid="text-online-count">{onlineTrucks.l
514 |           <div class="text-[10px] font-mono text-muted-foreground">reportando telemetria</div>
515 |         </div>
516 |         <div class="bg-card border border-border p-3.5">
517 |           <div class="flex items-center gap-2 mb-1.5">
518 |             <WifiOff class="w-4 h-4 text-muted-foreground/50" />
519 |             <span class="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">OFFLINE</span>
520 |           </div>
521 |           <div class="text-2xl font-mono font-bold text-muted-foreground" data-testid="text-offline-count">{offlineT
522 |           <div class="text-[10px] font-mono text-muted-foreground">sin telemetria</div>
523 |         </div>
524 |         <div class="bg-card border border-border p-3.5">
525 |           <div class="flex items-center gap-2 mb-1.5">
526 |             <Radio class="w-4 h-4 text-blue-400" />
527 |             <span class="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">CON GPS</span>
528 |           </div>
529 |           <div class="text-2xl font-mono font-bold text-blue-400" data-testid="text-gps-count">{totalWithGps}</div>
530 |           <div class="text-[10px] font-mono text-muted-foreground">posicion disponible</div>
531 |         </div>
532 |       </div>
```

```
533 |
534 |       {errorS && (
535 |         <div className="bg-red-500/10 border border-red-500/30 p-3 flex items-center gap-2" data-testid="volvo-error-ban
536 |           <AlertTriangle className="w-4 h-4 text-red-400 flex-shrink-0" />
537 |           <span className="text-[11px] font-mono text-red-400">Error al obtener telemetria Volvo. Los datos pueden estar
538 |         </div>
539 |       )}
540 |
541 |       <div className="flex gap-2 items-center flex-wrap" data-testid="volvo-filters">
542 |         <div className="flex border border-border overflow-visible">
543 |           <Button
544 |             variant={viewMode === "lista" ? "default" : "ghost"}
545 |             size="sm"
546 |             onClick={() => setViewMode("lista")}
547 |             className="gap-1.5 text-[10px] font-mono font-bold tracking-[0.1em] rounded-none"
548 |             data-testid="btn-view-lista"
549 |           >
550 |             <List className="w-3.5 h-3.5" />
551 |             LISTA
552 |           </Button>
553 |           <Button
554 |             variant={viewMode === "mapa" ? "default" : "ghost"}
555 |             size="sm"
556 |             onClick={() => setViewMode("mapa")}
557 |             className="gap-1.5 text-[10px] font-mono font-bold tracking-[0.1em] rounded-none"
558 |             data-testid="btn-view-mapa"
559 |           >
560 |             <MapIcon className="w-3.5 h-3.5" />
561 |             MAPA
562 |           </Button>
563 |         </div>
564 |
565 |         <div className="relative">
566 |           <Search className="absolute left-2.5 top-1/2 -translate-y-1/2 w-3.5 h-3.5 text-muted-foreground" />
567 |           <Input value={busq} onChange={e => setBusq(e.target.value)}
568 |             placeholder="Buscar patente, VIN o conductor..."
569 |             className="pl-8 w-64 font-mono text-xs bg-card"
570 |             data-testid="input-volvo-search" />
571 |         </div>
572 |         {[{ id: "all", nombre: "Todas las faenas", color: undefined } as const, ...faenas,
573 |           { id: 0, nombre: "Sin Asignar", color: "#5A7A9A" } as any].map((f: any) => (
574 |           <button
575 |             key={f.id}
576 |             onClick={() => setFaenaFil(String(f.id))}
577 |             data-testid={`volvo-filter-faena-${f.id}`}
578 |             className={`px-3 py-1.5 text-[10px] font-mono cursor-pointer border whitespace-nowrap transition-all ${
579 |               String(faenaFil) === String(f.id)
580 |                 ? "border-primary/50 text-primary bg-primary/10"
581 |                 : "border-border text-muted-foreground bg-transparent"
582 |             }`}
583 |             style={
584 |               f.id !== "all" && String(faenaFil) === String(f.id) && f.color
585 |                 ? { borderColor: `${f.color}80`, color: f.color, backgroundColor: `${f.color}15` }
586 |                 : undefined
587 |             }
588 |           >
589 |             {f.nombre}
590 |             <span className="ml-1 opacity-60">
591 |               ({f.id === "all"
592 |                 ? filtered.length
593 |                 : f.id === 0
594 |                   ? camiones.filter(c => c.syncOk && (c.faenaId === 0 || !c.faenaId)).length
595 |                   : camiones.filter(c => c.syncOk && c.faenaId === f.id).length})
596 |             </span>
597 |           </button>
598 |         ))}
599 |       </div>
600 |
601 |       {viewMode === "mapa" ? (
602 |         <div className="space-y-3">
603 |           <div className="flex items-center gap-4 text-[10px] font-mono text-muted-foreground">
604 |             <div className="flex items-center gap-1.5">
605 |               <div className="w-3 h-3 rounded-full bg-emerald-500" />
606 |               Online
607 |             </div>
608 |             <div className="flex items-center gap-1.5">
609 |               <div className="w-3 h-3 rounded-full bg-yellow-500" />
610 |               Offline
611 |             </div>
612 |             <div className="flex items-center gap-1.5">
613 |               <div className="w-3 h-3 rounded-full bg-gray-500" />
614 |               Sin datos
615 |             </div>
616 |           </div>
617 |           <TruckMap
618 |             camiones={filtered}
619 |             statusMap={statusMap}
620 |             faenas={faenas}
621 |             onSelectVin={handleSelectVin}
622 |           />
623 |         </div>
624 |       ) : (
625 |         <>
626 |           {onlineTrucks.length > 0 && (
627 |             <div data-testid="volvo-online-section">
628 |               <div className="flex items-center gap-2 mb-3 pb-2 border-b-2 border-emerald-500/30">
629 |                 <Wifi className="w-4 h-4 text-emerald-400" />
630 |                 <span className="text-[11px] font-mono font-bold text-emerald-400 tracking-[0.15em] uppercase">
631 |                   Online / Trabajando
632 |                 </span>
633 |                 <span className="text-[10px] font-mono text-emerald-400/60">({onlineTrucks.length})</span>
634 |               </div>
635 |               {onlineGrouped.map(renderFaenaGroup)}
636 |             </div>
637 |           )}
638 |
639 |           {offlineTrucks.length > 0 && (
```

```
640 |             <div data-testid="volvo-offline-section">
641 |               <div className="flex items-center gap-2 mb-3 pb-2 border-b-2 border-muted-foreground/20">
642 |                 <WifiOff className="w-4 h-4 text-muted-foreground/50" />
643 |                 <span className="text-[11px] font-mono font-bold text-muted-foreground/60 tracking-[0.15em] uppercase">
644 |                   Offline
645 |                 </span>
646 |                 <span className="text-[10px] font-mono text-muted-foreground/40">({offlineTrucks.length})</span>
647 |               </div>
648 |               {offlineGrouped.map(renderFaenaGroup)}
649 |             </div>
650 |           )}
651 |         </>
652 |       )}
653 |
654 |       {filtered.length === 0 && (
655 |         <div className="py-16 text-center text-muted-foreground font-mono text-xs" data-testid="text-no-volvo-trucks">
656 |           Sin camiones sincronizados con Volvo. Sincroniza la flota desde Configuracion.
657 |         </div>
658 |       )}
659 |
660 |       <DiagnosticsSection
661 |         camiones={camiones}
662 |         statusMap={statusMap}
663 |         faenas={faenas}
664 |         onSelectVin={handleSelectVin}
665 |       />
666 |
667 |       {selectedCamion && selectedVin && (
668 |         <VolvoTruckModal
669 |           camion={selectedCamion}
670 |           status={selectedStatus || null}
671 |           faena={faenas.find(f => f.id === selectedCamion.faenaId) || null}
672 |           open={!!selectedVin}
673 |           onClose={() => setSelectedVin(null)}
674 |         />
675 |       )}
676 |     </div>
677 |   );
678 | }
679 |
```

client/src/pages/combustibles-reales-ia.tsx

513 lineas

```
1 | import { useState } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { apiRequest } from "@/lib/queryClient";
4 | import { Search, Loader2, ChevronDown, ChevronUp, AlertTriangle, CheckCircle, TrendingDown, TrendingUp, Route, Fuel, X,
5 |
6 | interface ViajeDetail {
7 |   id: number;
8 |   codigo: string;
9 |   fecha: string;
10 |   kmRecorridos: number;
11 |   litrosSigetra: number;
12 |   litrosEcu: number;
13 |   rendimiento: number;
14 |   desviacionPct: number;
15 |   esAnomalia: boolean;
16 | }
17 |
18 | interface RouteCluster {
19 |   routeId: string;
20 |   camionId: number;
21 |   patente: string;
22 |   faena: string;
23 |   origenNombre: string;
24 |   destinoNombre: string;
25 |   viajes: ViajeDetail[];
26 |   totalViajes: number;
27 |   promedioRendimiento: number;
28 |   promedioKm: number;
29 |   promedioLitros: number;
30 |   desviacionMax: number;
31 |   anomalias: number;
32 | }
33 |
34 | interface Resumen {
35 |   totalRutas: number;
36 |   totalViajes: number;
37 |   totalAnomalias: number;
38 |   rutasConAnomalias: number;
39 |   camionesAnalizados: number;
40 |   toleranciaKm: number;
41 |   umbralAnomaliaPct: number;
42 |   minimoViajes: number;
43 | }
44 |
45 | interface CombustiblesData {
46 |   rutas: RouteCluster[];
47 |   resumen: Resumen;
48 | }
49 |
50 | const PREGUNTAS = [
51 |   "Que camion tiene mas anomalias en sus rutas?",
52 |   "Hay sospechas de robo de combustible?",
53 |   "Que ruta consume mas combustible?",
54 |   "Diferencia surtidor vs ECU por ruta?",
55 |   "Camiones con peor rendimiento constante?",
56 |   "Resumen de patrones sospechosos",
57 | ];
58 |
59 | function ConsultaPanel({ onClose }: { onClose: () => void }) {
60 |   const [pregunta, setPregunta] = useState("");
61 |   const [loading, setLoading] = useState(false);
62 |   const [respuesta, setRespuesta] = useState<string | null>(null);
63 |
64 |   const handleAsk = async (q: string) => {
65 |     const query = q || pregunta;
66 |     if (!query.trim()) return;
67 |     setPregunta(query);
68 |     setLoading(true);
69 |     setRespuesta(null);
70 |     try {
71 |       const res = await apiRequest("POST", "/api/ia/combustibles-reales/consulta", { pregunta: query });
72 |       const data = await res.json();
73 |       setRespuesta(data.respuesta);
74 |     } catch {
75 |       setRespuesta("Error al consultar. Intenta nuevamente.");
76 |     } finally {
77 |       setLoading(false);
78 |     }
79 |   };
80 |
81 |   return (
82 |     <div className="fixed inset-0 z-50 flex items-center justify-center" style={{ background: "rgba(0,0,0,0.7)" }}>
83 |       <div
84 |         className="relative w-full max-w-2xl mx-4 rounded border overflow-hidden"
85 |         style={{ background: "#0a1929", borderColor: "#0d2035", maxHeight: "85vh" }}
86 |         data-testid="modal-consulta-combustibles"
87 |       >
88 |         <div className="flex items-center justify-between p-4 border-b" style={{ borderColor: "#0d2035" }}>
89 |           <div className="flex items-center gap-2">
90 |             <Search className="w-4 h-4" style={{ color: "#00d4ff" }} />
91 |             <span className="font-space text-sm font-bold" style={{ color: "#00d4ff" }}>CONSULTAR COMBUSTIBLES REALES</s
92 |           </div>
93 |           <button onClick={onClose} className="p-1 hover:opacity-70" data-testid="btn-close-consulta-combustibles">
94 |             <X className="w-4 h-4" style={{ color: "#3a6080" }} />
95 |           </button>
96 |         </div>
97 |
98 |         <div className="p-4 space-y-4 overflow-y-auto" style={{ maxHeight: "calc(85vh - 140px)" }}>
99 |           <div className="flex flex-wrap gap-2">
100 |             {PREGUNTAS.map((p, i) => (
101 |               <button
102 |                 key={i}
103 |                 onClick={() => handleAsk(p)}
104 |                 disabled={loading}
```

```
105 |                 className="px-3 py-1.5 text-xs font-rajdhani rounded border transition-all hover:border-[#00d4ff]"
106 |                 style={{ borderColor: "#0d2035", color: "#3a6080", background: "rgba(0,212,255,0.05)" }}
107 |                 data-testid={`btn-pregunta-sugerida-${i}`}
108 |               >
109 |                 {p}
110 |               </button>
111 |             ))}
112 |           </div>
113 |
114 |           <div className="flex gap-2">
115 |             <input
116 |               value={pregunta}
117 |               onChange={e => setPregunta(e.target.value)}
118 |               onKeyDown={e => e.key === "Enter" && handleAsk("")}
119 |               placeholder="Escribe tu pregunta sobre combustibles..."
120 |               className="flex-1 px-3 py-2 rounded border text-sm font-rajdhani"
121 |               style={{ background: "#020508", borderColor: "#0d2035", color: "#c8e8ff" }}
122 |               disabled={loading}
123 |               data-testid="input-pregunta-combustibles"
124 |             />
125 |             <button
126 |               onClick={() => handleAsk("")}
127 |               disabled={loading || !pregunta.trim()}
128 |               className="px-4 py-2 rounded border font-space text-xs font-bold transition-all"
129 |               style={{
130 |                 background: loading ? "#0d2035" : "rgba(0,212,255,0.15)",
131 |                 borderColor: "#00d4ff",
132 |                 color: "#00d4ff",
133 |                 opacity: loading || !pregunta.trim() ? 0.4 : 1,
134 |               }}
135 |               data-testid="btn-enviar-pregunta-combustibles"
136 |             >
137 |               {loading ? <Loader2 className="w-4 h-4 animate-spin" /> : <Send className="w-4 h-4" />}
138 |             </button>
139 |           </div>
140 |
141 |           {loading && (
142 |             <div className="flex items-center gap-3 p-4 rounded border" style={{ borderColor: "#0d2035", background: "rg
143 |               <Loader2 className="w-5 h-5 animate-spin" style={{ color: "#00d4ff" }} />
144 |               <span className="text-sm font-rajdhani" style={{ color: "#3a6080" }}>Analizando datos de combustible por r
145 |             </div>
146 |           )}
147 |
148 |           {respuesta && (
149 |             <div
150 |               className="p-4 rounded border text-sm font-rajdhani leading-relaxed whitespace-pre-wrap"
151 |               style={{ borderColor: "#0d2035", background: "rgba(0,212,255,0.03)", color: "#c8e8ff" }}
152 |               data-testid="text-respuesta-combustibles"
153 |             >
154 |               {respuesta}
155 |             </div>
156 |           )}
157 |         </div>
158 |       </div>
159 |     </div>
160 |   );
161 | }
162 |
163 | function RouteCard({ cluster, isOpen, onToggle }: { cluster: RouteCluster; isOpen: boolean; onToggle: () => void }) {
164 |   const hasAnomalias = cluster.anomalias > 0;
165 |
166 |   return (
167 |     <div
168 |       className="rounded border overflow-hidden"
169 |       style={{ borderColor: hasAnomalias ? "rgba(255,34,68,0.3)" : "#0d2035", background: "#0a1929" }}
170 |       data-testid={`card-ruta-${cluster.routeId}`}
171 |     >
172 |       <button
173 |         onClick={onToggle}
174 |         className="w-full flex items-center justify-between p-4 text-left transition-all hover:bg-[rgba(0,212,255,0.03)]
175 |         data-testid={`btn-toggle-ruta-${cluster.routeId}`}
176 |       >
177 |         <div className="flex items-center gap-4 flex-1 min-w-0">
178 |           <div
179 |             className="w-10 h-10 rounded flex items-center justify-center flex-shrink-0"
180 |             style={{
181 |               background: hasAnomalias ? "rgba(255,34,68,0.15)" : "rgba(0,255,136,0.1)",
182 |               border: `1px solid ${hasAnomalias ? "rgba(255,34,68,0.3)" : "rgba(0,255,136,0.2)"}`,
183 |             }}
184 |           >
185 |             {hasAnomalias ? (
186 |               <AlertTriangle class=name="w-5 h-5" style={{ color: "#ff2244" }} />
187 |             ) : (
188 |               <CheckCircle class=name="w-5 h-5" style={{ color: "#00ff88" }} />
189 |             )}
190 |           </div>
191 |
192 |           <div class=name="flex-1 min-w-0">
193 |             <div class=name="flex items-center gap-2 flex-wrap">
194 |               <span class=name="font-space text-sm font-bold" style={{ color: "#00d4ff" }} data-testid={`text-patente-rut
195 |                 {cluster.patente}
196 |               </span>
197 |               <span class=name="text-xs font-rajdhani px-2 py-0.5 rounded" style={{ background: "rgba(0,212,255,0.1)", co
198 |                 {cluster.faena}
199 |               </span>
200 |             </div>
201 |             <div class=name="flex items-center gap-1 mt-1">
202 |               <Route class=name="w-3 h-3 flex-shrink-0" style={{ color: "#3a6080" }} />
203 |               <span class=name="text-xs font-rajdhani truncate" style={{ color: "#3a6080" }}>
204 |                 {cluster.origenNombre} &rarr; {cluster.destinoNombre}
205 |               </span>
206 |             </div>
207 |           </div>
208 |
209 |           <div class=name="hidden sm:flex items-center gap-6 flex-shrink-0">
210 |             <div class=name="text-center">
211 |               <div class=name="font-space text-lg font-bold" style={{ color: "#c8e8ff" }}>{cluster.totalViajes}</div>
```

```
212 |               <div className="text-[10px] font-rajdhani" style={{ color: "#3a6080" }}>VIAJES</div>
213 |             </div>
214 |             <div className="text-center">
215 |               <div className="font-space text-lg font-bold" style={{ color: "#00d4ff" }}>{cluster.promedioRendimiento}</
216 |               <div className="text-[10px] font-rajdhani" style={{ color: "#3a6080" }}>KM/L PROM</div>
217 |             </div>
218 |             <div className="text-center">
219 |               <div className="font-space text-lg font-bold" style={{ color: hasAnomalias ? "#ff2244" : "#00ff88" }}>
220 |                 {cluster.anomalias}
221 |               </div>
222 |               <div className="text-[10px] font-rajdhani" style={{ color: "#3a6080" }}>ANOMALIAS</div>
223 |             </div>
224 |             <div className="text-center">
225 |               <div className="font-space text-lg font-bold" style={{ color: cluster.desviacionMax > 20 ? "#ffcc00" : "#c
226 |                 {cluster.desviacionMax}%
227 |               </div>
228 |               <div className="text-[10px] font-rajdhani" style={{ color: "#3a6080" }}>DESV MAX</div>
229 |             </div>
230 |           </div>
231 |         </div>
232 |
233 |         {isOpen ? (
234 |           <ChevronUp className="w-5 h-5 flex-shrink-0 ml-2" style={{ color: "#3a6080" }} />
235 |         ) : (
236 |           <ChevronDown className="w-5 h-5 flex-shrink-0 ml-2" style={{ color: "#3a6080" }} />
237 |         )}
238 |       </button>
239 |
240 |       {isOpen && (
241 |         <div className="border-t px-4 pb-4" style={{ borderColor: "#0d2035" }}>
242 |           <div className="sm:hidden flex gap-4 py-3 border-b" style={{ borderColor: "#0d2035" }}>
243 |             <div className="text-center flex-1">
244 |               <div className="font-space text-sm font-bold" style={{ color: "#c8e8ff" }}>{cluster.totalViajes}</div>
245 |               <div className="text-[9px] font-rajdhani" style={{ color: "#3a6080" }}>VIAJES</div>
246 |             </div>
247 |             <div className="text-center flex-1">
248 |               <div className="font-space text-sm font-bold" style={{ color: "#00d4ff" }}>{cluster.promedioRendimiento}</
249 |               <div className="text-[9px] font-rajdhani" style={{ color: "#3a6080" }}>KM/L</div>
250 |             </div>
251 |             <div className="text-center flex-1">
252 |               <div className="font-space text-sm font-bold" style={{ color: hasAnomalias ? "#ff2244" : "#00ff88" }}>{clu
253 |               <div className="text-[9px] font-rajdhani" style={{ color: "#3a6080" }}>ANOMALIAS</div>
254 |             </div>
255 |           </div>
256 |
257 |           <div className="mt-3 flex gap-4 flex-wrap mb-3">
258 |             <div className="flex items-center gap-2 px-3 py-2 rounded border" style={{ borderColor: "#0d2035", backgroun
259 |               <Fuel className="w-4 h-4" style={{ color: "#00d4ff" }} />
260 |               <div>
261 |                 <div className="text-[10px] font-rajdhani" style={{ color: "#3a6080" }}>PROMEDIO LITROS</div>
262 |                 <div className="font-space text-sm font-bold" style={{ color: "#c8e8ff" }}>{cluster.promedioLitros} L</d
263 |               </div>
264 |             </div>
265 |             <div className="flex items-center gap-2 px-3 py-2 rounded border" style={{ borderColor: "#0d2035", backgroun
266 |               <Route className="w-4 h-4" style={{ color: "#00d4ff" }} />
267 |               <div>
268 |                 <div className="text-[10px] font-rajdhani" style={{ color: "#3a6080" }}>PROMEDIO KM</div>
269 |                 <div className="font-space text-sm font-bold" style={{ color: "#c8e8ff" }}>{cluster.promedioKm} km</div>
270 |               </div>
271 |             </div>
272 |           </div>
273 |
274 |           <div className="overflow-x-auto">
275 |             <table className="w-full text-xs font-rajdhani" data-testid={`table-viajes-${cluster.routeId}`}>
276 |               <thead>
277 |                 <tr style={{ borderBottom: "1px solid #0d2035" }}>
278 |                   <th className="text-left py-2 px-2 font-space text-[10px] font-bold" style={{ color: "#3a6080" }}>FECH
279 |                   <th className="text-right py-2 px-2 font-space text-[10px] font-bold" style={{ color: "#3a6080" }}>KM<
280 |                   <th className="text-right py-2 px-2 font-space text-[10px] font-bold" style={{ color: "#3a6080" }}>LT
281 |                   <th className="text-right py-2 px-2 font-space text-[10px] font-bold" style={{ color: "#3a6080" }}>LT
282 |                   <th className="text-right py-2 px-2 font-space text-[10px] font-bold" style={{ color: "#3a6080" }}>KM/
283 |                   <th className="text-right py-2 px-2 font-space text-[10px] font-bold" style={{ color: "#3a6080" }}>DES
284 |                   <th className="text-center py-2 px-2 font-space text-[10px] font-bold" style={{ color: "#3a6080" }}>ES
285 |                 </tr>
286 |               </thead>
287 |               <tbody>
288 |                 {cluster.viajes.map(v => (
289 |                   <tr
290 |                     key={v.id}
291 |                     className="transition-colors"
292 |                     style={{
293 |                       borderBottom: "1px solid rgba(13,32,53,0.5)",
294 |                       background: v.esAnomalia ? "rgba(255,34,68,0.05)" : "transparent",
295 |                     }}
296 |                     data-testid={`row-viaje-${v.id}`}
297 |                   >
298 |                     <td className="py-2 px-2" style={{ color: "#c8e8ff" }}>{v.fecha}</td>
299 |                     <td className="py-2 px-2 text-right font-space" style={{ color: "#c8e8ff" }}>{v.kmRecorridos}</td>
300 |                     <td className="py-2 px-2 text-right font-space" style={{ color: "#ffcc00" }}>{v.litrosSigetra || "—"
301 |                     <td className="py-2 px-2 text-right font-space" style={{ color: "#00d4ff" }}>{v.litrosEcu || "—"}</t
302 |                     <td className="py-2 px-2 text-right font-space font-bold" style={{ color: "#c8e8ff" }}>{v.rendimient
303 |                     <td className="py-2 px-2 text-right font-space font-bold" style={{
304 |                       color: v.esAnomalia
305 |                         ? (v.desviacionPct < 0 ? "#ff2244" : "#ffcc00")
306 |                         : "#00ff88"
307 |                     }}>
308 |                       <span className="flex items-center justify-end gap-1">
309 |                         {v.desviacionPct > 0 ? (
310 |                           <TrendingUp className="w-3 h-3" />
311 |                         ) : (
312 |                           <TrendingDown className="w-3 h-3" />
313 |                         )}
314 |                         {v.desviacionPct > 0 ? "+" : ""}{v.desviacionPct}%
315 |                       </span>
316 |                     </td>
317 |                     <td className="py-2 px-2 text-center">
318 |                       {v.esAnomalia ? (
```

```
319 |                         <span className="px-2 py-0.5 rounded text-[10px] font-space font-bold" style={{ background: "rgb
320 |                           ANOMALIA
321 |                         </span>
322 |                       ) : (
323 |                         <span className="px-2 py-0.5 rounded text-[10px] font-space font-bold" style={{ background: "rgb
324 |                           NORMAL
325 |                         </span>
326 |                       )}
327 |                     </td>
328 |                   </tr>
329 |                 ))}
330 |               </tbody>
331 |             </table>
332 |           </div>
333 |
334 |           <div className="mt-3 p-3 rounded border" style={{ borderColor: "#0d2035", background: "rgba(0,212,255,0.03)" }
335 |             <div className="text-[10px] font-space font-bold mb-1" style={{ color: "#3a6080" }}>BASELINE PROPIO</div>
336 |             <div className="text-xs font-rajdhani" style={{ color: "#c8e8ff" }}>
337 |               Basado en {cluster.totalViajes} viajes del mismo camion en la misma ruta.
338 |               Rendimiento promedio: <span className="font-bold" style={{ color: "#00d4ff" }}>{cluster.promedioRendimient
339 |               {cluster.anomalias > 0 ? (
340 |                 <span style={{ color: "#ff2244" }}> Se detectaron {cluster.anomalias} viaje(s) con desviacion superior a
341 |               ) : (
342 |                 <span style={{ color: "#00ff88" }}> Sin anomalias detectadas.</span>
343 |               )}
344 |             </div>
345 |           </div>
346 |         </div>
347 |       )}
348 |     </div>
349 |   );
350 | }
351 |
352 | export default function CombustiblesRealesIA() {
353 |   const [showConsulta, setShowConsulta] = useState(false);
354 |   const [openRoutes, setOpenRoutes] = useState<Set<string>>(new Set());
355 |   const [filterFaena, setFilterFaena] = useState<string>("todas");
356 |   const [filterAnomalias, setFilterAnomalias] = useState(false);
357 |
358 |   const { data, isLoading, error } = useQuery<CombustiblesData>({
359 |     queryKey: ["/api/ia/combustibles-reales"],
360 |   });
361 |
362 |   const toggleRoute = (routeId: string) => {
363 |     setOpenRoutes(prev => {
364 |       const next = new Set(prev);
365 |       if (next.has(routeId)) next.delete(routeId);
366 |       else next.add(routeId);
367 |       return next;
368 |     });
369 |   };
370 |
371 |   const faenas = data ? [...new Set(data.rutas.map(r => r.faena))].sort() : [];
372 |   const filteredRutas = data?.rutas.filter(r => {
373 |     if (filterFaena !== "todas" && r.faena !== filterFaena) return false;
374 |     if (filterAnomalias && r.anomalias === 0) return false;
375 |     return true;
376 |   }) || [];
377 |
378 |   const noData = !isLoading && data && data.resumen.totalRutas === 0;
379 |
380 |   return (
381 |     <div className="space-y-6" data-testid="page-combustibles-reales">
382 |       <div className="flex flex-col sm:flex-row items-start sm:items-center justify-between gap-4">
383 |         <div>
384 |           <h1 className="font-space text-xl font-bold tracking-wider" style={{ color: "#00d4ff" }} data-testid="title-co
385 |             COMBUSTIBLES REALES IA
386 |           </h1>
387 |           <p className="text-sm font-rajdhani mt-1" style={{ color: "#3a6080" }}>
388 |             Comparacion de consumo real por ruta &mdash; mismo camion, misma ruta, minimo 3 viajes
389 |           </p>
390 |         </div>
391 |         <button
392 |           onClick={() => setShowConsulta(true)}
393 |           className="flex items-center gap-2 px-4 py-2 rounded border font-space text-xs font-bold transition-all hover:
394 |           style={{ borderColor: "#0d2035", color: "#00d4ff", background: "rgba(0,212,255,0.05)" }}
395 |           data-testid="btn-consultar-combustibles"
396 |         >
397 |           <Search className="w-4 h-4" />
398 |           CONSULTAR IA
399 |         </button>
400 |       </div>
401 |
402 |       {isLoading && (
403 |         <div className="flex flex-col items-center justify-center py-20 gap-4">
404 |           <Loader2 className="w-8 h-8 animate-spin" style={{ color: "#00d4ff" }} />
405 |           <span class="font-rajdhani text-sm" style={{ color: "#3a6080" }}>Analizando rutas y agrupando viajes por G
406 |         </div>
407 |       )}
408 |
409 |       {error && (
410 |         <div class="p-4 rounded border text-sm font-rajdhani" style={{ borderColor: "rgba(255,34,68,0.3)", backgroun
411 |           Error al cargar datos: {(error as Error).message}
412 |         </div>
413 |       )}
414 |
415 |       {noData && (
416 |         <div class="flex flex-col items-center justify-center py-20 gap-4 rounded border" style={{ borderColor: "#0d
417 |           <div class="w-16 h-16 rounded-full flex items-center justify-center" style={{ background: "rgba(0,212,255,
418 |             <Route class="w-8 h-8" style={{ color: "#00d4ff" }} />
419 |           </div>
420 |           <div class="text-center space-y-2">
421 |             <h3 class="font-space text-sm font-bold" style={{ color: "#c8e8ff" }}>Analisis pendiente</h3>
422 |             <p class="text-sm font-rajdhani max-w-md" style={{ color: "#3a6080" }}>
423 |               Se necesitan viajes TMS detectados con coordenadas GPS para comparar rutas.
424 |               Ve al tab TMS, selecciona un contrato y presiona "Detectar Viajes" para generar el historial.
425 |               Se requieren minimo 3 viajes del mismo camion en la misma ruta.
```

```
426 |             </p>
427 |           </div>
428 |         </div>
429 |       )}
430 |
431 |       {data && data.resumen.totalRutas > 0 && (
432 |         <>
433 |           <div className="grid grid-cols-2 sm:grid-cols-4 gap-3" data-testid="kpi-combustibles">
434 |             <div className="p-4 rounded border" style={{ borderColor: "#0d2035", background: "#0a1929" }}>
435 |               <div className="text-[10px] font-space font-bold" style={{ color: "#3a6080" }}>RUTAS ANALIZADAS</div>
436 |               <div className="font-space text-2xl font-bold mt-1" style={{ color: "#00d4ff" }} data-testid="kpi-total-ru
437 |             </div>
438 |             <div className="p-4 rounded border" style={{ borderColor: "#0d2035", background: "#0a1929" }}>
439 |               <div className="text-[10px] font-space font-bold" style={{ color: "#3a6080" }}>TOTAL VIAJES</div>
440 |               <div className="font-space text-2xl font-bold mt-1" style={{ color: "#c8e8ff" }} data-testid="kpi-total-vi
441 |             </div>
442 |             <div className="p-4 rounded border" style={{ borderColor: data.resumen.totalAnomalias > 0 ? "rgba(255,34,68,
443 |               <div className="text-[10px] font-space font-bold" style={{ color: "#3a6080" }}>ANOMALIAS</div>
444 |               <div className="font-space text-2xl font-bold mt-1" style={{ color: data.resumen.totalAnomalias > 0 ? "#ff
445 |                 {data.resumen.totalAnomalias}
446 |               </div>
447 |             </div>
448 |             <div className="p-4 rounded border" style={{ borderColor: "#0d2035", background: "#0a1929" }}>
449 |               <div className="text-[10px] font-space font-bold" style={{ color: "#3a6080" }}>CAMIONES</div>
450 |               <div className="font-space text-2xl font-bold mt-1" style={{ color: "#c8e8ff" }} data-testid="kpi-camiones
451 |             </div>
452 |           </div>
453 |
454 |           <div className="flex flex-wrap items-center gap-3">
455 |             <select
456 |               value={filterFaena}
457 |               onChange={e => setFilterFaena(e.target.value)}
458 |               className="px-3 py-2 rounded border text-sm font-rajdhani"
459 |               style={{ background: "#020508", borderColor: "#0d2035", color: "#c8e8ff" }}
460 |               data-testid="select-filter-faena"
461 |             >
462 |               <option value="todas">Todas las faenas</option>
463 |               {faenas.map(f => (
464 |                 <option key={f} value={f}>{f}</option>
465 |               ))}
466 |             </select>
467 |
468 |             <button
469 |               onClick={() => setFilterAnomalias(!filterAnomalias)}
470 |               className="px-3 py-2 rounded border text-xs font-space font-bold transition-all"
471 |               style={{
472 |                 borderColor: filterAnomalias ? "rgba(255,34,68,0.5)" : "#0d2035",
473 |                 background: filterAnomalias ? "rgba(255,34,68,0.1)" : "transparent",
474 |                 color: filterAnomalias ? "#ff2244" : "#3a6080",
475 |               }}
476 |               data-testid="btn-filter-anomalias"
477 |             >
478 |               <span className="flex items-center gap-1.5">
479 |                 <AlertTriangle className="w-3 h-3" />
480 |                 SOLO ANOMALIAS
481 |               </span>
482 |             </button>
483 |
484 |             <span className="text-xs font-rajdhani ml-auto" style={{ color: "#3a6080" }}>
485 |               {filteredRutas.length} ruta(s) &mdash; tolerancia {data.resumen.toleranciaKm}km, umbral {data.resumen.umbr
486 |             </span>
487 |           </div>
488 |
489 |           <div className="space-y-3" data-testid="list-rutas">
490 |             {filteredRutas.map(cluster => (
491 |               <RouteCard
492 |                 key={cluster.routeId}
493 |                 cluster={cluster}
494 |                 isOpen={openRoutes.has(cluster.routeId)}
495 |                 onToggle={() => toggleRoute(cluster.routeId)}
496 |               />
497 |             ))}
498 |             {filteredRutas.length === 0 && (
499 |               <div className="text-center py-10">
500 |                 <span className="font-rajdhani text-sm" style={{ color: "#3a6080" }}>
501 |                   No hay rutas que coincidan con los filtros seleccionados
502 |                 </span>
503 |               </div>
504 |             )}
505 |           </div>
506 |         </>
507 |       )}
508 |
509 |       {showConsulta && <ConsultaPanel onClose={() => setShowConsulta(false)} />}
510 |     </div>
511 |   );
512 | }
513 |
```

client/src/pages/sigetra-fusion.tsx

631 lineas

```
1 | import { useState, useMemo } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { Input } from "@/components/ui/input";
4 | import { Skeleton } from "@/components/ui/skeleton";
5 | import {
6 |   Truck, Fuel, Search, Wifi, WifiOff, AlertTriangle, Check,
7 |   ArrowRightLeft, Database, ChevronDown, ChevronUp, Calendar,
8 |   MapPin
9 | } from "lucide-react";
10 |
11 | interface FusionCarga {
12 |   numGuia: number;
13 |   fecha: string;
14 |   litros: number;
15 |   odometro: number | null;
16 |   kmRecorrido: number | null;
17 |   rendimiento: number | null;
18 |   lugar: string | null;
19 |   tarjeta: string | null;
20 |   conductor: string | null;
21 | }
22 |
23 | interface FusionTruck {
24 |   fleetNum: string;
25 |   patenteReal: string | null;
26 |   vin: string | null;
27 |   modeloVolvo: string;
28 |   faenaSigetra: string | null;
29 |   conductorSigetra: string | null;
30 |   totalLitrosSigetra: number;
31 |   totalCargas: number;
32 |   rendPromedio: number;
33 |   odometroSigetra: number | null;
34 |   odometroVolvo: number | null;
35 |   deltaOdometro: number | null;
36 |   fuelLevelVolvo: number | null;
37 |   totalFuelUsedVolvo: number | null;
38 |   litrosVolvoPeriodo: number | null;
39 |   litrosSigetraViajes: number;
40 |   viajesMatched: number;
41 |   viajesSinEcu: number;
42 |   engineHoursVolvo: number | null;
43 |   gpsVolvo: { latitude: number | null; longitude: number | null } | null;
44 |   alertLevel: "ok" | "alerta" | "critico";
45 |   cargas: FusionCarga[];
46 | }
47 |
48 | interface MatchBreakdown {
49 |   viajesMatched: number;
50 |   viajesSinEcu: number;
51 |   camionesOffline: number;
52 | }
53 |
54 | interface FusionResponse {
55 |   from: string;
56 |   to: string;
57 |   totalTrucksMatched: number;
58 |   totalLitros: number;
59 |   totalCargas: number;
60 |   alertCount: number;
61 |   matchBreakdown: MatchBreakdown;
62 |   trucks: FusionTruck[];
63 | }
64 |
65 | interface ConnectionStatus {
66 |   connected?: boolean;
67 |   status?: string;
68 |   configured?: boolean;
69 |   message: string;
70 |   user: string;
71 | }
72 |
73 | interface FaenaGroup {
74 |   faena: string;
75 |   trucks: FusionTruck[];
76 |   totalLitros: number;
77 |   totalCargas: number;
78 |   alertCount: number;
79 | }
80 |
81 | function isConnected(s: ConnectionStatus | undefined): boolean {
82 |   return s?.connected === true || s?.status === "connected";
83 | }
84 |
85 | function fN(n: number): string {
86 |   return n.toLocaleString("es-CL");
87 | }
88 |
89 | function formatDate(iso: string): string {
90 |   if (!iso) return "\u2014";
91 |   const d = new Date(iso);
92 |   return d.toLocaleDateString("es-CL", { day: "2-digit", month: "2-digit", year: "numeric" });
93 | }
94 |
95 | function formatDateTime(iso: string): string {
96 |   if (!iso) return "\u2014";
97 |   const d = new Date(iso);
98 |   return d.toLocaleDateString("es-CL", { day: "2-digit", month: "2-digit", hour: "2-digit", minute: "2-digit" });
99 | }
100 |
101 | function getDateRange(days: number): { from: string; to: string } {
102 |   const to = new Date();
103 |   const from = new Date(Date.now() - days * 24 * 60 * 60 * 1000);
104 |   return { from: from.toISOString().slice(0, 10), to: to.toISOString().slice(0, 10) };
```

```
105 | }
106 |
107 | function computeCuadratura(t: FusionTruck): number | null {
108 |   const volvo = t.litrosVolvoPeriodo;
109 |   const sigetra = t.litrosSigetraViajes > 0 ? t.litrosSigetraViajes : t.totalLitrosSigetra;
110 |   if (volvo == null || volvo <= 0 || sigetra <= 0) return null;
111 |   return (Math.min(volvo, sigetra) / Math.max(volvo, sigetra)) * 100;
112 | }
113 |
114 | function groupByFaena(trucks: FusionTruck[]): FaenaGroup[] {
115 |   const map: Record<string, FusionTruck[]> = {};
116 |   for (const t of trucks) {
117 |     const key = t.faenaSigetra || "Sin Faena";
118 |     if (!map[key]) map[key] = [];
119 |     map[key].push(t);
120 |   }
121 |   const groups: FaenaGroup[] = Object.keys(map).map((faena: string) => {
122 |     const grpTrucks = map[faena];
123 |     return {
124 |       faena,
125 |       trucks: grpTrucks,
126 |       totalLitros: grpTrucks.reduce((s: number, t: FusionTruck) => s + t.totalLitrosSigetra, 0),
127 |       totalCargas: grpTrucks.reduce((s: number, t: FusionTruck) => s + t.totalCargas, 0),
128 |       alertCount: grpTrucks.filter((t: FusionTruck) => t.alertLevel !== "ok").length,
129 |     };
130 |   });
131 |   groups.sort((a: FaenaGroup, b: FaenaGroup) => {
132 |     if (a.faena === "Sin Faena") return 1;
133 |     if (b.faena === "Sin Faena") return -1;
134 |     return a.faena.localeCompare(b.faena);
135 |   });
136 |   return groups;
137 | }
138 |
139 | export default function SigetraFusion() {
140 |   const [days, setDays] = useState(7);
141 |   const [busq, setBusq] = useState("");
142 |   const [expandedRow, setExpandedRow] = useState<string | null>(null);
143 |   const [collapsedFaenas, setCollapsedFaenas] = useState<Set<string>>(new Set());
144 |
145 |   const range = useMemo(() => getDateRange(days), [days]);
146 |
147 |   const { data: sigetraStatus, isLoading: loadingSigetra } = useQuery<ConnectionStatus>({
148 |     queryKey: ["/api/sigetra/status"],
149 |   });
150 |
151 |   const { data: volvoStatus, isLoading: loadingVolvo } = useQuery<ConnectionStatus>({
152 |     queryKey: ["/api/volvo/status"],
153 |   });
154 |
155 |   const { data: fusion, isLoading: loadingFusion, isError: errorFusion } = useQuery<FusionResponse>({
156 |     queryKey: [`/api/sigetra/fusion?from=${range.from}&to=${range.to}`],
157 |     enabled: sigetraStatus?.connected === true,
158 |     refetchInterval: 600000,
159 |   });
160 |
161 |   const cuadraturaTrucks = useMemo(() => {
162 |     if (!fusion?.trucks) return [];
163 |     return fusion.trucks;
164 |   }, [fusion]);
165 |
166 |   const filtered = useMemo(() => {
167 |     if (!cuadraturaTrucks.length) return [];
168 |     if (!busq.trim()) return cuadraturaTrucks;
169 |     const q = busq.toLowerCase();
170 |     return cuadraturaTrucks.filter(t =>
171 |       t.fleetNum.includes(q) ||
172 |       t.patenteReal?.toLowerCase().includes(q) ||
173 |       t.vin?.toLowerCase().includes(q) ||
174 |       t.faenaSigetra?.toLowerCase().includes(q) ||
175 |       t.conductorSigetra?.toLowerCase().includes(q)
176 |     );
177 |   }, [cuadraturaTrucks, busq]);
178 |
179 |   const faenaGroups = useMemo(() => groupByFaena(filtered), [filtered]);
180 |
181 |   const toggleFaena = (faena: string) => {
182 |     setCollapsedFaenas(prev => {
183 |       const next = new Set(prev);
184 |       if (next.has(faena)) next.delete(faena);
185 |       else next.add(faena);
186 |       return next;
187 |     });
188 |   };
189 |
190 |   const isLoading = loadingSigetra || loadingVolvo;
191 |
192 |   if (isLoading) {
193 |     return (
194 |       <div className="space-y-4" data-testid="fusion-loading">
195 |         <Skeleton className="h-20 w-full" />
196 |         <div className="grid grid-cols-4 gap-3">
197 |           {Array.from({ length: 4 }).map((_, i) => <Skeleton key={i} className="h-24" />)}
198 |         </div>
199 |         <Skeleton className="h-96 w-full" />
200 |       </div>
201 |     );
202 |   }
203 |
204 |   return (
205 |     <div className="space-y-4" data-testid="fusion-page">
206 |       <div className="flex items-center gap-3 flex-wrap">
207 |         <div className="flex items-center gap-2">
208 |           <ArrowRightLeft className="w-5 h-5 text-primary" />
209 |           <h1 className="text-sm font-mono font-bold tracking-[0.15em] uppercase text-foreground" data-testid="text-fusi
210 |             Conexion Sigetra-Volvo
211 |           </h1>
```

```
212 |          </div>
213 |          <div className="flex-1" />
214 |          <div className="flex items-center gap-3">
215 |            <div className={`flex items-center gap-1.5 px-3 py-1.5 border text-[10px] font-mono ${
216 |              sigetraStatus?.connected
217 |                ? "bg-emerald-500/10 border-emerald-500/30 text-emerald-400"
218 |                : "bg-red-500/10 border-red-500/30 text-red-400"
219 |            }`} data-testid="status-sigetra">
220 |              {sigetraStatus?.connected ? <Wifi className="w-3 h-3" /> : <WifiOff className="w-3 h-3" />}
221 |              SIGETRA: {sigetraStatus?.connected ? "CONECTADO" : "DESCONECTADO"}
222 |            </div>
223 |            <div className={`flex items-center gap-1.5 px-3 py-1.5 border text-[10px] font-mono ${
224 |              isConnected(volvoStatus)
225 |                ? "bg-emerald-500/10 border-emerald-500/30 text-emerald-400"
226 |                : "bg-red-500/10 border-red-500/30 text-red-400"
227 |            }`} data-testid="status-volvo">
228 |              {isConnected(volvoStatus) ? <Wifi className="w-3 h-3" /> : <WifiOff className="w-3 h-3" />}
229 |              VOLVO rFMS: {isConnected(volvoStatus) ? "CONECTADO" : "DESCONECTADO"}
230 |            </div>
231 |          </div>
232 |        </div>
233 |
234 |        {sigetraStatus?.connected && isConnected(volvoStatus) && (
235 |          <div className="bg-emerald-500/5 border border-emerald-500/20 p-3 flex items-center gap-2" data-testid="fusion-s
236 |            <Check className="w-4 h-4 text-emerald-400" />
237 |            <span className="text-[11px] font-mono text-emerald-400">
238 |              Ambas fuentes de datos conectadas. Cruce automatico activo.
239 |            </span>
240 |          </div>
241 |        )}
242 |
243 |        {!sigetraStatus?.connected && (
244 |          <div className="bg-red-500/10 border border-red-500/30 p-4" data-testid="fusion-sigetra-error">
245 |            <div className="flex items-center gap-2 mb-2">
246 |              <WifiOff className="w-4 h-4 text-red-400" />
247 |              <span className="text-xs font-mono font-bold text-red-400">SIGETRA NO CONECTADO</span>
248 |            </div>
249 |            <p className="text-[11px] font-mono text-red-400/80">
250 |              {sigetraStatus?.message || "No se pudo conectar con Sigetra Web. Verifique las credenciales."}
251 |            </p>
252 |          </div>
253 |        )}
254 |
255 |        <div className="flex items-center gap-2 flex-wrap">
256 |          <Calendar className="w-3.5 h-3.5 text-muted-foreground" />
257 |          <span className="text-[10px] font-mono text-muted-foreground tracking-[0.15em]">PERIODO:</span>
258 |          {[3, 7, 14, 30].map(d => (
259 |            <button key={d} onClick={() => setDays(d)}
260 |              data-testid={`btn-days-${d}`}
261 |              className={`px-3 py-1 text-[10px] font-mono font-bold border cursor-pointer transition-colors ${
262 |                days === d
263 |                  ? "bg-primary/20 border-primary text-primary"
264 |                  : "bg-card border-border text-muted-foreground hover:text-foreground"
265 |              }`}>
266 |              {d}D
267 |            </button>
268 |          ))}
269 |          <span className="text-[10px] font-mono text-muted-foreground ml-2">
270 |            {formatDate(range.from)} — {formatDate(range.to)}
271 |          </span>
272 |        </div>
273 |
274 |        {fusion && (() => {
275 |          const allTrucks = fusion.trucks;
276 |          const trucksWithBoth = allTrucks.filter(t => t.litrosVolvoPeriodo != null && t.litrosVolvoPeriodo > 0 && t.total
277 |          const soloSigetra = allTrucks.filter(t => t.totalLitrosSigetra > 0 && (t.litrosVolvoPeriodo == null || t.litrosV
278 |          const soloVolvo = allTrucks.filter(t => (t.litrosVolvoPeriodo != null && t.litrosVolvoPeriodo > 0) && t.totalLit
279 |          const sinDatos = allTrucks.filter(t => (t.litrosVolvoPeriodo == null || t.litrosVolvoPeriodo <= 0) && t.totalLit
280 |          const totalLitrosVolvo = trucksWithBoth.reduce((s, t) => s + (t.litrosVolvoPeriodo ?? 0), 0);
281 |          const totalLitrosSigetraMatched = trucksWithBoth.reduce((s, t) => s + (t.litrosSigetraViajes || t.totalLitrosSig
282 |          const bd = fusion.matchBreakdown;
283 |
284 |          const breakdownRows = [
285 |            { label: "CRUZADOS METODO VIAJE", desc: "Viajes con ECU match", count: bd.viajesMatched, color: "text-emerald-
286 |            { label: "SIN ECU DISPONIBLE", desc: "Viajes sin snapshot ECU", count: bd.viajesSinEcu, color: "text-amber-400
287 |            { label: "CAMIONES OFFLINE", desc: "Sin VIN asignado", count: bd.camionesOffline, color: "text-muted-foregroun
288 |          ];
289 |
290 |          const sourceRows = [
291 |            { label: "CRUCE COMPLETO", desc: "ECU + Sigetra", count: trucksWithBoth.length, color: "text-emerald-400", dot
292 |            { label: "SOLO SIGETRA", desc: "Sin datos ECU", count: soloSigetra.length, color: "text-amber-400", dotColor:
293 |            { label: "SOLO VOLVO", desc: "Sin datos Sigetra", count: soloVolvo.length, color: "text-amber-400", dotColor:
294 |            { label: "SIN DATOS", desc: "Sin fuentes", count: sinDatos.length, color: "text-muted-foreground", dotColor: "
295 |            { label: "TOTAL FLOTA", desc: "Camiones con registros", count: allTrucks.length, color: "text-blue-400", dotCo
296 |          ];
297 |
298 |          return (
299 |            <>
300 |              <div className="grid grid-cols-2 sm:grid-cols-4 gap-3">
301 |                <div className="bg-card border border-border p-3.5">
302 |                  <div className="flex items-center gap-2 mb-1.5">
303 |                    <Truck className="w-4 h-4 text-primary" />
304 |                    <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">CRUZADOS</span>
305 |                  </div>
306 |                  <div className="text-2xl font-mono font-bold text-primary" data-testid="text-fusion-matched">{trucksWith
307 |                  <div className="text-[10px] font-mono text-muted-foreground">camiones con cruce viaje</div>
308 |                </div>
309 |                <div className="bg-card border border-border p-3.5">
310 |                  <div className="flex items-center gap-2 mb-1.5">
311 |                    <Fuel className="w-4 h-4 text-amber-400" />
312 |                    <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">LITROS SIGETRA</span>
313 |                  </div>
314 |                  <div className="text-2xl font-mono font-bold text-amber-400" data-testid="text-fusion-litros">{fN(fusion
315 |                  <div className="text-[10px] font-mono text-muted-foreground">total periodo</div>
316 |                </div>
317 |                <div className="bg-card border border-border p-3.5">
318 |                  <div className="flex items-center gap-2 mb-1.5">
```

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319 |                    <Database className="w-4 h-4 text-blue-400" />
320 |                    <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">CARGAS</span>
321 |                  </div>
322 |                  <div className="text-2xl font-mono font-bold text-blue-400" data-testid="text-fusion-cargas">{fN(fusion.
323 |                  <div className="text-[10px] font-mono text-muted-foreground">registros combustible</div>
324 |                </div>
325 |                <div className="bg-card border border-border p-3.5">
326 |                  <div className="flex items-center gap-2 mb-1.5">
327 |                    <ArrowRightLeft className="w-4 h-4 text-primary" />
328 |                    <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">CRUCE SIGETRA-VOLVO</spa
329 |                  </div>
330 |                  <div className="text-lg font-mono font-bold text-primary" data-testid="text-match-flota">{trucksWithBoth
331 |                  <div className="text-[10px] font-mono text-muted-foreground leading-relaxed mt-1">
332 |                    Sistema de cruce en calibracion.<br />
333 |                    {trucksWithBoth.length} camiones con datos de ambas fuentes.<br />
334 |                    Precision mejora con mas historial.
335 |                  </div>
336 |                </div>
337 |              </div>
338 |
339 |              <div className="grid grid-cols-1 sm:grid-cols-2 gap-3">
340 |                <div className="bg-card border border-border p-3.5" data-testid="breakdown-table">
341 |                  <div className="flex items-center gap-2 mb-3">
342 |                    <ArrowRightLeft className="w-4 h-4 text-primary" />
343 |                    <span className="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.2em]">MATCH POR METO
344 |                  </div>
345 |                  <div className="space-y-0">
346 |                    {breakdownRows.map(row => (
347 |                      <div key={row.label} className="flex items-center gap-3 px-2 py-2" data-testid={`breakdown-row-${row
348 |                        <span className={`w-2 h-2 rounded-full flex-shrink-0 ${row.dotColor}`} />
349 |                        <div className="flex-1 min-w-0">
350 |                          <span className={`text-[11px] font-mono font-bold ${row.color}`}>{row.label}</span>
351 |                          <span className="text-[10px] font-mono text-muted-foreground ml-2">{row.desc}</span>
352 |                        </div>
353 |                        <span className={`text-sm font-mono font-bold ${row.color}`}>{row.count}</span>
354 |                      </div>
355 |                    ))}
356 |                  </div>
357 |                  <div className="mt-3 pt-3 border-t border-border">
358 |                    <div className="flex items-center gap-2 mb-2">
359 |                      <Database className="w-3.5 h-3.5 text-muted-foreground" />
360 |                      <span className="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.2em]">FUENTES DE D
361 |                    </div>
362 |                    {sourceRows.map(row => (
363 |                      <div key={row.label} className={`flex items-center gap-3 px-2 py-1.5 ${row.label === "TOTAL FLOTA" ?
364 |                        <span className={`w-1.5 h-1.5 rounded-full flex-shrink-0 ${row.dotColor}`} />
365 |                        <div className="flex-1 min-w-0">
366 |                          <span className={`text-[10px] font-mono ${row.color}`}>{row.label}</span>
367 |                        </div>
368 |                        <span className={`text-[11px] font-mono font-bold ${row.color}`}>{row.count}</span>
369 |                      </div>
370 |                    ))}
371 |                  </div>
372 |                </div>
373 |
374 |                <div className="grid grid-cols-1 gap-3">
375 |                  <div className="bg-card border border-border p-3.5">
376 |                    <div className="flex items-center gap-2 mb-1.5">
377 |                      <Fuel className="w-4 h-4 text-blue-400" />
378 |                      <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">LITROS VOLVO (VIAJES)<
379 |                    </div>
380 |                    <div className="text-2xl font-mono font-bold text-blue-400" data-testid="text-litros-volvo">{fN(Math.r
381 |                    <div className="text-[10px] font-mono text-muted-foreground">ECU delta por viaje ({trucksWithBoth.leng
382 |                  </div>
383 |                  <div className="bg-card border border-border p-3.5">
384 |                    <div className="flex items-center gap-2 mb-1.5">
385 |                      <Fuel className="w-4 h-4 text-amber-400" />
386 |                      <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">LITROS SIGETRA (VIAJES
387 |                    </div>
388 |                    <div className="text-2xl font-mono font-bold text-amber-400" data-testid="text-litros-sigetra-viajes">
389 |                    <div class="text-[10px] font-mono text-muted-foreground">solo viajes con cruce ECU</div>
390 |                  </div>
391 |                </div>
392 |              </div>
393 |
394 |              <div className="bg-blue-500/5 border border-blue-500/20 p-2.5 flex items-center gap-2" data-testid="note-dat
395 |                <Database class="w-3.5 h-3.5 text-blue-400 flex-shrink-0" />
396 |                <span class="text-[10px] font-mono text-blue-400">Datos analizados desde 01-03-2026. Metodo: cruce por
397 |              </div>
398 |            </>
399 |          );
400 |        })()}
401 |
402 |        {loadingFusion && (
403 |          <div class="space-y-2">
404 |            {Array.from({ length: 5 }).map((_, i) => <Skeleton key={i} class="h-14 w-full" />)}
405 |          </div>
406 |        )}
407 |
408 |        {errorFusion && (
409 |          <div class="bg-red-500/10 border border-red-500/30 p-3 flex items-center gap-2" data-testid="fusion-error">
410 |            <AlertTriangle class="w-4 h-4 text-red-400" />
411 |            <span class="text-[11px] font-mono text-red-400">Error al obtener datos de fusion. Intente nuevamente.</sp
412 |          </div>
413 |        )}
414 |
415 |        {fusion && (
416 |          <>
417 |            <div class="flex items-center gap-2">
418 |              <div class="relative flex-1 max-w-xs">
419 |                <Search class="absolute left-2.5 top-1/2 -translate-y-1/2 w-3.5 h-3.5 text-muted-foreground" />
420 |                <Input value={busq} onChange={e => setBusq(e.target.value)}
421 |                  placeholder="Buscar patente, VIN, faena..."
422 |                  class="pl-8 font-mono text-xs bg-card"
423 |                  data-testid="input-fusion-search" />
424 |              </div>
425 |              <span class="text-[10px] font-mono text-muted-foreground">{filtered.length} resultados en {faenaGroups.l
```

```
426 |           </div>
427 |
428 |           <div className="space-y-3">
429 |             {faenaGroups.map(group => {
430 |               const isCollapsed = collapsedFaenas.has(group.faena);
431 |               return (
432 |                 <div key={group.faena} className="border border-border bg-card overflow-hidden" data-testid={`faena-grou
433 |                   <div
434 |                     className="flex items-center gap-3 px-3 py-2.5 bg-background border-b border-border cursor-pointer t
435 |                     onClick={() => toggleFaena(group.faena)}
436 |                     data-testid={`btn-faena-toggle-${group.faena}`}
437 |                   >
438 |                     <div className="flex items-center gap-2">
439 |                       <MapPin className="w-3.5 h-3.5 text-primary" />
440 |                       <span className="text-xs font-mono font-bold text-foreground tracking-[0.1em]">{group.faena}</span
441 |                     </div>
442 |                     <div className="flex items-center gap-3 flex-wrap">
443 |                       <span className="text-[10px] font-mono text-muted-foreground">
444 |                         <Truck className="w-3 h-3 inline mr-1" />{group.trucks.length} camiones
445 |                       </span>
446 |                       <span className="text-[10px] font-mono text-amber-400">
447 |                         <Fuel className="w-3 h-3 inline mr-1" />{fN(Math.round(group.totalLitros))} L
448 |                       </span>
449 |                       <span className="text-[10px] font-mono text-blue-400">
450 |                         {group.totalCargas} cargas
451 |                       </span>
452 |                       {group.alertCount > 0 && (
453 |                         <span className="text-[10px] font-mono text-red-400">
454 |                           <AlertTriangle className="w-3 h-3 inline mr-1" />{group.alertCount} alertas
455 |                         </span>
456 |                       )}
457 |                     </div>
458 |                     <div className="flex-1" />
459 |                     {isCollapsed ? <ChevronDown className="w-4 h-4 text-muted-foreground" /> : <ChevronUp className="w-4
460 |                   </div>
461 |
462 |                   {!isCollapsed && (
463 |                     <>
464 |                       <div className="grid grid-cols-[60px_80px_100px_80px_80px_80px_80px_70px_60px] gap-0 bg-background
465 |                         <span className="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">N.INT</
466 |                         <span className="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">PATENTE
467 |                         <span className="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">LT SIGE
468 |                         <span className="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">CARGAS<
469 |                         <span className="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">REND</s
470 |                         <span className="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">ODO.SIG
471 |                         <span className="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">ODO.VOL
472 |                         <span className="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">CUAD.</
473 |                         <span className="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">ESTADO<
474 |                       </div>
475 |
476 |                       <div className="max-h-[500px] overflow-y-auto">
477 |                         {group.trucks.map(t => (
478 |                           <div key={t.fleetNum}>
479 |                             <div
480 |                               className={`grid grid-cols-[60px_80px_100px_80px_80px_80px_80px_70px_60px] gap-0 px-3 py-2
481 |                                 t.alertLevel === "critico" ? "bg-red-500/5" : t.alertLevel === "alerta" ? "bg-amber-500/
482 |                               }`}
483 |                               style={{ borderLeftWidth: 3, borderLeftColor: t.alertLevel === "critico" ? "#FF2D4A" : t.a
484 |                               onClick={() => setExpandedRow(expandedRow === t.fleetNum ? null : t.fleetNum)}
485 |                               data-testid={`row-fusion-${t.fleetNum}`}
486 |                             >
487 |                               <span className="text-xs font-mono font-bold text-foreground">{t.fleetNum}</span>
488 |                               <span className="text-[10px] font-mono text-muted-foreground">{t.patenteReal || "\u2014"}<
489 |                               <span className="text-xs font-mono font-bold text-amber-400">{fN(Math.round(t.totalLitrosS
490 |                               <span className="text-[10px] font-mono text-muted-foreground">{t.totalCargas}</span>
491 |                               <span className="text-[10px] font-mono text-foreground">{t.rendPromedio > 0 ? `${t.rendPro
492 |                               <span className="text-[10px] font-mono text-foreground">{t.odometroSigetra != null ? fN(t.
493 |                               <span className="text-[10px] font-mono text-blue-400">{t.odometroVolvo != null ? fN(t.odom
494 |                               <span className={`text-[10px] font-mono font-bold ${(() => { const c = computeCuadratura(t
495 |                               <div className="flex items-center gap-1">
496 |                                 <span className={`w-2 h-2 rounded-full ${
497 |                                   t.alertLevel === "critico" ? "bg-red-500" : t.alertLevel === "alerta" ? "bg-amber-500"
498 |                                 }`} />
499 |                                 {expandedRow === t.fleetNum ? <ChevronUp className="w-3 h-3 text-muted-foreground" /> :
500 |                               </div>
501 |                             </div>
502 |
503 |                             {expandedRow === t.fleetNum && (
504 |                               <div className="bg-background border-b border-border p-4">
505 |                                 <div className="grid grid-cols-3 gap-4 mb-4">
506 |                                   <div>
507 |                                     <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-1">I
508 |                                     <div className="space-y-1 text-[11px] font-mono">
509 |                                       <div><span className="text-muted-foreground">Interno:</span> <span className="text
510 |                                       <div><span className="text-muted-foreground">Patente:</span> <span className="text
511 |                                       <div><span className="text-muted-foreground">VIN:</span> <span className="text-for
512 |                                       <div><span className="text-muted-foreground">Modelo:</span> <span className="text-
513 |                                       <div><span className="text-muted-foreground">Conductor:</span> <span className="te
514 |                                     </div>
515 |                                   </div>
516 |                                   <div>
517 |                                     <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-1">D
518 |                                     <div className="space-y-1 text-[11px] font-mono">
519 |                                       <div><span className="text-muted-foreground">Total litros:</span> <span className=
520 |                                       <div><span className="text-muted-foreground">Cargas:</span> <span className="text-
521 |                                       <div><span className="text-muted-foreground">Rendimiento:</span> <span className="
522 |                                       <div><span className="text-muted-foreground">Odometro:</span> <span className="tex
523 |                                       <div><span className="text-muted-foreground">Faena:</span> <span className="text-f
524 |                                     </div>
525 |                                   </div>
526 |                                   <div>
527 |                                     <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-1">D
528 |                                     <div className="space-y-1 text-[11px] font-mono">
529 |                                       <div><span className="text-muted-foreground">Odometro ECU:</span> <span className=
530 |                                       <div><span className="text-muted-foreground">Nivel tanque:</span> <span className=
531 |                                       <div><span className="text-muted-foreground">Fuel ECU (periodo):</span> <span clas
532 |                                       <div><span className="text-muted-foreground">Fuel ECU (acum. vida):</span> <span c
```

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533 |                                    <div><span className="text-muted-foreground">Horas motor:</span> <span className="
534 |                                    <div><span className="text-muted-foreground">GPS:</span> <span className="text-for
535 |                                  </div>
536 |                                </div>
537 |                              </div>
538 |
539 |                              {(() => {
540 |                                const cuad = computeCuadratura(t);
541 |                                const sigetraComp = t.litrosSigetraViajes > 0 ? t.litrosSigetraViajes : t.totalLitrosS
542 |                                const hasComparison = t.litrosVolvoPeriodo != null && t.litrosVolvoPeriodo > 0 && sige
543 |                                return hasComparison ? (
544 |                                  <div className={`border p-3 mb-3 ${cuad != null && cuad >= 95 ? "bg-emerald-500/10 b
545 |                                    <div className="flex items-center gap-2 mb-2">
546 |                                      <ArrowRightLeft className={`w-3.5 h-3.5 ${cuad != null && cuad >= 95 ? "text-eme
547 |                                      <span className={`text-[10px] font-mono font-bold tracking-[0.15em] ${cuad != nu
548 |                                        CUADRATURA VOLVO vs SIGETRA
549 |                                      </span>
550 |                                      {t.viajesMatched > 0 && (
551 |                                        <span className="text-[9px] font-mono text-muted-foreground ml-2">({t.viajesMa
552 |                                      )}
553 |                                    </div>
554 |                                    <div className="grid grid-cols-3 gap-3">
555 |                                      <div>
556 |                                        <div className="text-[8px] font-mono text-muted-foreground mb-0.5">LITROS VOLV
557 |                                        <div className="text-sm font-mono font-bold text-blue-400">{fN(t.litrosVolvoPe
558 |                                      </div>
559 |                                      <div>
560 |                                        <div className="text-[8px] font-mono text-muted-foreground mb-0.5">LITROS SIGE
561 |                                        <div className="text-sm font-mono font-bold text-amber-400">{fN(Math.round(sig
562 |                                      </div>
563 |                                      <div>
564 |                                        <div className="text-[8px] font-mono text-muted-foreground mb-0.5">CUADRATURA<
565 |                                        <div className={`text-sm font-mono font-bold ${cuad != null && cuad >= 95 ? "t
566 |                                      </div>
567 |                                    </div>
568 |                                    <div className="mt-2 text-[10px] font-mono text-muted-foreground">
569 |                                      Diferencia: {fN(Math.abs((t.litrosVolvoPeriodo ?? 0) - sigetraComp))} L ({(t.lit
570 |                                    </div>
571 |                                  </div>
572 |                                ) : null;
573 |                              })()}
574 |
575 |                              {t.deltaOdometro != null && (
576 |                                <div className={`border p-2 mb-3 flex items-center gap-2 ${
577 |                                  t.alertLevel === "critico" ? "bg-red-500/10 border-red-500/30" : t.alertLevel === "a
578 |                                }`}>
579 |                                  <ArrowRightLeft className={`w-3.5 h-3.5 ${t.alertLevel === "critico" ? "text-red-400
580 |                                  <span className={`text-[11px] font-mono font-bold ${t.alertLevel === "critico" ? "te
581 |                                    Delta odometro: {fN(t.deltaOdometro)} km (Sigetra - Volvo)
582 |                                  </span>
583 |                                </div>
584 |                              )}
585 |
586 |                              <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-2">HISTO
587 |                              <div className="border border-border/50 max-h-[200px] overflow-y-auto">
588 |                                <div className="grid grid-cols-[1fr_70px_80px_80px_70px_1fr_70px] gap-0 bg-card/50 px-
589 |                                  <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">FECHA<
590 |                                  <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">LITROS
591 |                                  <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">ODOMET
592 |                                  <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">KM REC
593 |                                  <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">REND.<
594 |                                  <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">LUGAR<
595 |                                  <span className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">TARJET
596 |                                </div>
597 |                                {t.cargas.map((c, i) => (
598 |                                  <div key={i} className="grid grid-cols-[1fr_70px_80px_80px_70px_1fr_70px] gap-0 px-2
599 |                                    <span className="text-muted-foreground">{formatDateTime(c.fecha)}</span>
600 |                                    <span className="text-amber-400 font-bold">{c.litros.toFixed(1)}</span>
601 |                                    <span className="text-foreground">{c.odometro != null ? fN(c.odometro) : "\u2014"}
602 |                                    <span className="text-foreground">{c.kmRecorrido != null ? fN(c.kmRecorrido) : "\u
603 |                                    <span className="text-foreground">{c.rendimiento != null ? c.rendimiento.toFixed(1
604 |                                    <span className="text-muted-foreground truncate">{c.lugar || "\u2014"}</span>
605 |                                    <span className="text-muted-foreground">{c.tarjeta || "\u2014"}</span>
606 |                                  </div>
607 |                                ))}
608 |                              </div>
609 |                            </div>
610 |                          )}
611 |                        </div>
612 |                      ))}
613 |                    </div>
614 |                  </>
615 |                )}
616 |              </div>
617 |            );
618 |          })}
619 |
620 |          {faenaGroups.length === 0 && !loadingFusion && (
621 |            <div className="p-8 text-center text-[11px] font-mono text-muted-foreground border border-border bg-card"
622 |              {busq ? "Sin resultados para la busqueda" : "No se encontraron camiones con datos cruzados en el periodo
623 |            </div>
624 |          )}
625 |        </div>
626 |      </>
627 |    )}
628 |  </div>
629 | );
630 | }
631 |
```

client/src/pages/ranking-conductores.tsx

607 lineas

```
1 | import { useState, useMemo } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { Skeleton } from "@/components/ui/skeleton";
4 | import { Card, CardContent, CardHeader, CardTitle } from "@/components/ui/card";
5 | import {
6 |   Star, MapPin, Users, Fuel, TrendingUp, ChevronDown, ChevronUp,
7 |   Trophy, Medal, AlertTriangle, Filter
8 | } from "lucide-react";
9 | import { getErrorFleetNums, isErrorCarga } from "@/pages/errores";
10 | import { getPercentilCamion } from "@/lib/fuel-utils";
11 |
12 | interface FusionCarga {
13 |   numGuia: number;
14 |   fecha: string;
15 |   litros: number;
16 |   odometro: number | null;
17 |   kmRecorrido: number | null;
18 |   rendimiento: number | null;
19 |   lugar: string | null;
20 |   tarjeta: string | null;
21 |   conductor: string | null;
22 | }
23 |
24 | interface FusionTruck {
25 |   fleetNum: string;
26 |   patenteReal: string | null;
27 |   vin: string | null;
28 |   modeloVolvo: string;
29 |   faenaSigetra: string | null;
30 |   conductorSigetra: string | null;
31 |   totalLitrosSigetra: number;
32 |   totalCargas: number;
33 |   rendPromedio: number;
34 |   odometroSigetra: number | null;
35 |   odometroVolvo: number | null;
36 |   deltaOdometro: number | null;
37 |   fuelLevelVolvo: number | null;
38 |   totalFuelUsedVolvo: number | null;
39 |   litrosVolvoPeriodo: number | null;
40 |   engineHoursVolvo: number | null;
41 |   gpsVolvo: { latitude: number | null; longitude: number | null } | null;
42 |   alertLevel: "ok" | "alerta" | "critico";
43 |   cargas: FusionCarga[];
44 | }
45 |
46 | interface FusionResponse {
47 |   from: string;
48 |   to: string;
49 |   totalTrucksMatched: number;
50 |   totalLitros: number;
51 |   totalCargas: number;
52 |   alertCount: number;
53 |   trucks: FusionTruck[];
54 | }
55 |
56 | interface DriverStats {
57 |   conductor: string;
58 |   faena: string;
59 |   totalLitros: number;
60 |   totalKm: number;
61 |   rendimiento: number;
62 |   litros100km: number;
63 |   totalCargas: number;
64 |   camiones: string[];
65 |   diasPeriodo: number;
66 |   percentil: number;
67 | }
68 |
69 | interface FaenaDriverGroup {
70 |   faena: string;
71 |   drivers: DriverStats[];
72 |   avgRendimiento: number;
73 |   totalLitros: number;
74 |   totalKm: number;
75 |   bestDriver: DriverStats | null;
76 |   worstDriver: DriverStats | null;
77 | }
78 |
79 | const DATA_START_DATE = "2026-03-01";
80 |
81 | function fN(n: number): string {
82 |   return n.toLocaleString("es-CL");
83 | }
84 |
85 | function getDateRange(days: number): { from: string; to: string } {
86 |   const to = new Date();
87 |   const from = new Date(Date.now() - days * 24 * 60 * 60 * 1000);
88 |   const startMin = new Date(DATA_START_DATE);
89 |   const effectiveFrom = from < startMin ? startMin : from;
90 |   return { from: effectiveFrom.toISOString().slice(0, 10), to: to.toISOString().slice(0, 10) };
91 | }
92 |
93 | function RendBar({ value, max, color }: { value: number; max: number; color: string }) {
94 |   const pct = max > 0 ? Math.min((value / max) * 100, 100) : 0;
95 |   return (
96 |     <div className="h-2 w-full bg-muted rounded-md overflow-hidden">
97 |       <div className={`h-full rounded-md ${color}`} style={{ width: `${pct}%` }} />
98 |     </div>
99 |   );
100 | }
101 |
102 | function FaenaRankingCard({ group, expanded, onToggle }: {
103 |   group: FaenaDriverGroup;
104 |   expanded: boolean;
```

```
105 |   onToggle: () => void;
106 | }) {
107 |   return (
108 |     <Card className="overflow-visible" data-testid={`card-ranking-faena-${group.faena}`}>
109 |       <CardHeader className="flex flex-row items-center justify-between gap-2 space-y-0 pb-2 cursor-pointer" onClick={on
110 |         <div className="flex items-center gap-3 flex-wrap min-w-0">
111 |           <MapPin className="w-4 h-4 text-primary flex-shrink-0" />
112 |           <CardTitle className="text-sm font-mono font-bold tracking-[0.1em] uppercase truncate" data-testid={`text-rank
113 |             {group.faena}
114 |           </CardTitle>
115 |           <span className="text-[10px] font-mono text-muted-foreground">{group.drivers.length} conductores</span>
116 |         </div>
117 |         <div className="flex items-center gap-4 flex-wrap flex-shrink-0">
118 |           <div className="text-right">
119 |             <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em]">REND. PROMEDIO</div>
120 |             <div className={`text-lg font-mono font-bold ${
121 |               group.avgRendimiento >= 1.8 ? "text-emerald-400" : group.avgRendimiento >= 1.2 ? "text-amber-400" : "text-
122 |             }`} data-testid={`text-ranking-avg-rend-${group.faena}`}>
123 |               {group.avgRendimiento > 0 ? `${group.avgRendimiento.toFixed(2)} km/L` : "\u2014"}
124 |             </div>
125 |           </div>
126 |           {expanded ? <ChevronUp className="w-4 h-4 text-muted-foreground" /> : <ChevronDown className="w-4 h-4 text-mut
127 |         </div>
128 |       </CardHeader>
129 |       <CardContent className="pt-0">
130 |         <div className="grid grid-cols-2 sm:grid-cols-4 gap-4 mb-3">
131 |           <div>
132 |             <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-0.5">CONDUCTORES</div>
133 |             <div className="text-sm font-mono font-bold text-foreground">{group.drivers.length}</div>
134 |           </div>
135 |           <div>
136 |             <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-0.5">L/100KM</div>
137 |             <div className="text-sm font-mono font-bold text-cyan-400">
138 |               {group.totalKm > 0 ? ((group.totalLitros / group.totalKm) * 100).toFixed(1) : "\u2014"}
139 |             </div>
140 |           </div>
141 |           <div>
142 |             <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-0.5">LITROS TOTAL</div>
143 |             <div className="text-sm font-mono font-bold text-amber-400">{fN(Math.round(group.totalLitros))} L</div>
144 |           </div>
145 |           <div>
146 |             <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-0.5">KM TOTAL</div>
147 |             <div className="text-sm font-mono font-bold text-foreground">{fN(Math.round(group.totalKm))} km</div>
148 |           </div>
149 |         </div>
150 |
151 |         {group.drivers.length > 0 && (
152 |           <div className="mb-3">
153 |             <div class="text-[8px] font-mono text-amber-400 tracking-[0.2em] uppercase mb-2 font-bold">TOP 3 CONDUCT
154 |             <div class="grid grid-cols-1 sm:grid-cols-3 gap-2">
155 |               {group.drivers.slice(0, 3).map((d, idx) => {
156 |                 const podiumStyles = [
157 |                   { bg: "bg-amber-500/15 border-amber-500/40", label: "text-amber-300", icon: <Trophy className="w-5 h-5
158 |                   { bg: "bg-gray-500/10 border-gray-400/30", label: "text-gray-300", icon: <Medal className="w-5 h-5 tex
159 |                   { bg: "bg-orange-500/10 border-orange-700/30", label: "text-orange-300", icon: <Medal className="w-5 h
160 |                 ];
161 |                 const style = podiumStyles[idx];
162 |                 return (
163 |                   <div key={d.conductor} className={`border px-3 py-2.5 ${style.bg}`} data-testid={`podium-${group.faena
164 |                     <div className="flex items-center gap-2 mb-1.5">
165 |                       {style.icon}
166 |                       <span className="text-[9px] font-mono font-bold text-muted-foreground tracking-[0.15em]">{style.po
167 |                     </div>
168 |                     <div className={`text-sm font-mono font-bold truncate ${style.label}`} data-testid={`text-podium-nam
169 |                       {d.conductor}
170 |                     </div>
171 |                     <div className="flex items-center justify-between mt-1.5">
172 |                       <div>
173 |                         <div className={`text-lg font-mono font-bold ${
174 |                           d.percentil >= 75 ? "text-emerald-400" : d.percentil >= 50 ? "text-blue-400" : d.percentil >=
175 |                         }`}>
176 |                           {d.rendimiento.toFixed(2)}
177 |                           <span className="text-[9px] text-muted-foreground ml-1">km/L</span>
178 |                         </div>
179 |                         <div className="text-[9px] font-mono text-cyan-400/70">
180 |                           {d.litros100km.toFixed(1)} L/100km
181 |                         </div>
182 |                       </div>
183 |                       <span className={`text-sm font-mono font-bold ${
184 |                         d.percentil >= 90 ? "text-emerald-400" : d.percentil >= 75 ? "text-emerald-400/70" : d.percentil
185 |                       }`}>P{d.percentil}</span>
186 |                     </div>
187 |                     <div className="grid grid-cols-3 gap-1 mt-1.5 pt-1.5 border-t border-border/40">
188 |                       <div>
189 |                         <div className="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">LITROS</div>
190 |                         <div class="text-[10px] font-mono font-bold text-foreground">{fN(Math.round(d.totalLitros))}
191 |                       </div>
192 |                       <div>
193 |                         <div class="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">KM</div>
194 |                         <div class="text-[10px] font-mono font-bold text-foreground">{fN(Math.round(d.totalKm))}</di
195 |                       </div>
196 |                       <div>
197 |                         <div class="text-[7px] font-mono text-muted-foreground tracking-[0.1em]">CARGAS</div>
198 |                         <div class="text-[10px] font-mono font-bold text-foreground">{d.totalCargas}</div>
199 |                       </div>
200 |                     </div>
201 |                     <div class="text-[9px] font-mono text-muted-foreground mt-1 truncate">
202 |                       Cam: {d.camiones.join(", ")}
203 |                     </div>
204 |                   </div>
205 |                 );
206 |               })}
207 |             </div>
208 |           </div>
209 |         )}
210 |
211 |         {expanded && (
```

```
212 |           <div className="mt-3 space-y-0">
213 |             <div className="grid grid-cols-[2.5rem_1fr_5rem_5rem_4.5rem_5rem_5rem_4rem] gap-1.5 px-2 pb-1.5 border-b bor
214 |               <span className="text-[8px] font-mono text-muted-foreground tracking-[0.15em]">#</span>
215 |               <span className="text-[8px] font-mono text-muted-foreground tracking-[0.15em]">CONDUCTOR</span>
216 |               <span className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] text-right">PERCENTIL</span>
217 |               <span className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] text-right">REND.</span>
218 |               <span className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] text-right">L/100KM</span>
219 |               <span className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] text-right">LITROS</span>
220 |               <span className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] text-right">KM</span>
221 |               <span className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] text-right">CARGAS</span>
222 |             </div>
223 |             {group.drivers.map((d, idx) => {
224 |               const isBest = idx === 0;
225 |               const isWorst = idx === group.drivers.length - 1 && group.drivers.length > 1;
226 |               const rowBg = isBest ? "bg-emerald-500/5" : isWorst ? "bg-red-500/5" : "";
227 |
228 |               return (
229 |                 <div
230 |                   key={`${d.conductor}-${d.faena}`}
231 |                   data-testid={`row-driver-${idx}`}
232 |                   className={`grid grid-cols-[2.5rem_1fr_5rem_5rem_4.5rem_5rem_5rem_4rem] gap-1.5 px-2 py-1.5 items-cent
233 |                 >
234 |                   <span className="text-[11px] font-mono font-bold text-muted-foreground">
235 |                     {idx === 0 && <Trophy className="w-3.5 h-3.5 text-amber-400 inline" />}
236 |                     {idx === 1 && <Medal className="w-3.5 h-3.5 text-gray-400 inline" />}
237 |                     {idx === 2 && <Medal className="w-3.5 h-3.5 text-amber-700 inline" />}
238 |                     {idx > 2 && `${idx + 1}`}
239 |                   </span>
240 |                   <div className="min-w-0">
241 |                     <div className="text-[11px] font-mono font-bold text-foreground truncate" data-testid={`text-driver-
242 |                       {d.conductor}
243 |                     </div>
244 |                     <div className="text-[9px] font-mono text-muted-foreground truncate">
245 |                       {d.camiones.length > 0 ? `Cam: ${d.camiones.join(", ")}` : ""}
246 |                     </div>
247 |                   </div>
248 |                   <div className="text-right">
249 |                     <span className={`text-[11px] font-mono font-bold ${
250 |                       d.percentil >= 90 ? "text-emerald-400" : d.percentil >= 75 ? "text-emerald-400/70" : d.percentil >
251 |                     }`} data-testid={`text-driver-percentil-${idx}`}>
252 |                       P{d.percentil}
253 |                     </span>
254 |                     <div className={`text-[8px] font-mono ${
255 |                       d.percentil >= 90 ? "text-emerald-400/70" : d.percentil >= 75 ? "text-blue-400/70" : d.percentil >
256 |                     }`}>
257 |                       {d.percentil >= 90 ? `Top ${100 - d.percentil}%` : d.percentil >= 50 ? `Top ${100 - d.percentil}%`
258 |                     </div>
259 |                   </div>
260 |                   <div className="text-right">
261 |                     <span className={`text-[11px] font-mono font-bold ${
262 |                       d.percentil >= 75 ? "text-emerald-400" : d.percentil >= 50 ? "text-blue-400" : d.percentil >= 25 ?
263 |                     }`}>
264 |                       {d.rendimiento.toFixed(2)}
265 |                     </span>
266 |                     <span className="text-[9px] font-mono text-muted-foreground ml-0.5">km/L</span>
267 |                   </div>
268 |                   <div className="text-right text-[11px] font-mono font-bold text-cyan-400">{d.litros100km.toFixed(1)}</
269 |                   <div className="text-right text-[11px] font-mono text-foreground">{fN(Math.round(d.totalLitros))}</div
270 |                   <div className="text-right text-[11px] font-mono text-foreground">{fN(Math.round(d.totalKm))}</div>
271 |                   <div className="text-right text-[11px] font-mono text-muted-foreground">{d.totalCargas}</div>
272 |                 </div>
273 |               );
274 |             })}
275 |           </div>
276 |         )}
277 |       </CardContent>
278 |     </Card>
279 |   );
280 | }
281 |
282 | export default function RankingConductores() {
283 |   const [days, setDays] = useState(7);
284 |   const [expandedFaenas, setExpandedFaenas] = useState<Set<string>>(new Set());
285 |   const [sortBy, setSortBy] = useState<"rendimiento" | "litros" | "conductores">("rendimiento");
286 |   const [faenaFilter, setFaenaFilter] = useState<string>("todas");
287 |
288 |   const range = useMemo(() => getDateRange(days), [days]);
289 |   const { data, isLoading, isError } = useQuery<FusionResponse>({
290 |     queryKey: [`/api/sigetra/fusion?from=${range.from}&to=${range.to}`],
291 |   });
292 |
293 |   const allFaenas = useMemo(() => {
294 |     if (!data) return [];
295 |     const faenaSet = new Set<string>();
296 |     for (const truck of data.trucks) {
297 |       const faena = truck.faenaSigetra || "Sin Faena";
298 |       faenaSet.add(faena);
299 |     }
300 |     return Array.from(faenaSet).sort();
301 |   }, [data]);
302 |
303 |   const driverRankings = useMemo(() => {
304 |     if (!data) return { groups: [], trucksTotal: 0, trucksFiltered: 0 };
305 |     const errorNums = getErrorFleetNums(data.trucks as any);
306 |
307 |     let trucksTotal = 0;
308 |     let trucksFiltered = 0;
309 |
310 |     const driverMap = new Map<string, {
311 |       conductor: string;
312 |       faena: string;
313 |       totalLitros: number;
314 |       totalKm: number;
315 |       totalCargas: number;
316 |       camiones: Set<string>;
317 |       validCargas: number;
318 |     }>();
```

```
319 |
320 |     for (const truck of data.trucks) {
321 |       if (errorNums.has(truck.fleetNum)) continue;
322 |       trucksTotal++;
323 |
324 |       if (truck.litrosVolvoPeriodo != null && truck.litrosVolvoPeriodo > 0 && truck.totalLitrosSigetra > 0) {
325 |         const avg = (truck.litrosVolvoPeriodo + truck.totalLitrosSigetra) / 2;
326 |         const pctDiff = Math.abs(truck.litrosVolvoPeriodo - truck.totalLitrosSigetra) / avg * 100;
327 |         if (pctDiff > 5) { trucksFiltered++; continue; }
328 |       }
329 |
330 |       const faena = truck.faenaSigetra || "Sin Faena";
331 |
332 |       for (const c of truck.cargas) {
333 |         if (!c.conductor || c.conductor.trim() === "") continue;
334 |         if (isErrorCarga(c as any)) continue;
335 |         if (c.litros == null || c.litros <= 0) continue;
336 |         if (c.kmRecorrido == null || c.kmRecorrido <= 0) continue;
337 |         if (c.rendimiento == null || c.rendimiento <= 0 || c.rendimiento > 15) continue;
338 |
339 |         const fecha = c.fecha ? c.fecha.slice(0, 10) : null;
340 |         if (fecha && fecha < DATA_START_DATE) continue;
341 |
342 |         const conductorNorm = c.conductor.trim();
343 |         const key = `${conductorNorm.toLowerCase()}|||${faena}`;
344 |         const existing = driverMap.get(key) || {
345 |           conductor: conductorNorm,
346 |           faena,
347 |           totalLitros: 0,
348 |           totalKm: 0,
349 |           totalCargas: 0,
350 |           camiones: new Set<string>(),
351 |           validCargas: 0,
352 |         };
353 |         existing.totalLitros += c.litros;
354 |         existing.totalKm += c.kmRecorrido;
355 |         existing.totalCargas += 1;
356 |         existing.camiones.add(truck.fleetNum);
357 |         existing.validCargas += 1;
358 |         driverMap.set(key, existing);
359 |       }
360 |     }
361 |
362 |     const byFaena = new Map<string, DriverStats[]>();
363 |     const diasPeriodo = days;
364 |     for (const [, d] of driverMap) {
365 |       if (d.validCargas < 2) continue;
366 |       const rend = d.totalKm / d.totalLitros;
367 |       const l100 = d.totalKm > 0 ? (d.totalLitros / d.totalKm) * 100 : 0;
368 |       const stat: DriverStats = {
369 |         conductor: d.conductor,
370 |         faena: d.faena,
371 |         totalLitros: d.totalLitros,
372 |         totalKm: d.totalKm,
373 |         rendimiento: rend,
374 |         litros100km: l100,
375 |         totalCargas: d.totalCargas,
376 |         camiones: Array.from(d.camiones),
377 |         diasPeriodo,
378 |         percentil: 0,
379 |       };
380 |       const arr = byFaena.get(d.faena) || [];
381 |       arr.push(stat);
382 |       byFaena.set(d.faena, arr);
383 |     }
384 |
385 |     const allDriversFlat: DriverStats[] = [];
386 |     for (const [, arr] of byFaena) {
387 |       allDriversFlat.push(...arr);
388 |     }
389 |     const allRend = allDriversFlat.map(d => d.rendimiento).filter(v => v > 0);
390 |
391 |     for (const d of allDriversFlat) {
392 |       d.percentil = getPercentilCamion(d.rendimiento, allRend);
393 |     }
394 |
395 |     const groups: FaenaDriverGroup[] = [];
396 |     for (const [faena, drivers] of Array.from(byFaena.entries())) {
397 |       if (faenaFilter !== "todas" && faena !== faenaFilter) continue;
398 |       drivers.sort((a, b) => b.rendimiento - a.rendimiento);
399 |
400 |       const rends = drivers.map(d => d.rendimiento).filter(r => r > 0);
401 |       const avg = rends.length > 0 ? rends.reduce((a, b) => a + b, 0) / rends.length : 0;
402 |       const totalLitros = drivers.reduce((s, d) => s + d.totalLitros, 0);
403 |       const totalKm = drivers.reduce((s, d) => s + d.totalKm, 0);
404 |
405 |       groups.push({
406 |         faena,
407 |         drivers,
408 |         avgRendimiento: avg,
409 |         totalLitros,
410 |         totalKm,
411 |         bestDriver: drivers[0] || null,
412 |         worstDriver: drivers[drivers.length - 1] || null,
413 |       });
414 |     }
415 |
416 |     if (sortBy === "rendimiento") groups.sort((a, b) => b.avgRendimiento - a.avgRendimiento);
417 |     else if (sortBy === "litros") groups.sort((a, b) => b.totalLitros - a.totalLitros);
418 |     else groups.sort((a, b) => b.drivers.length - a.drivers.length);
419 |
420 |     return { groups, trucksTotal, trucksFiltered };
421 |   }, [data, sortBy, days, faenaFilter]);
422 |
423 |   const rankingGroups = driverRankings.groups;
424 |   const { trucksTotal, trucksFiltered } = driverRankings;
425 |
```

```
426 |   const globalKpis = useMemo(() => {
427 |     const totalDrivers = rankingGroups.reduce((s, g) => s + g.drivers.length, 0);
428 |     const totalLitros = rankingGroups.reduce((s, g) => s + g.totalLitros, 0);
429 |     const totalKm = rankingGroups.reduce((s, g) => s + g.totalKm, 0);
430 |     const totalCargas = rankingGroups.reduce((s, g) => s + g.drivers.reduce((sc, d) => sc + d.totalCargas, 0), 0);
431 |     const avgL100 = totalKm > 0 ? (totalLitros / totalKm) * 100 : 0;
432 |     const allRends = rankingGroups.flatMap(g => g.drivers.map(d => d.rendimiento));
433 |     const avgRend = allRends.length > 0 ? allRends.reduce((a, b) => a + b, 0) / allRends.length : 0;
434 |     const topP90 = rankingGroups.flatMap(g => g.drivers).filter(d => d.percentil >= 90).length;
435 |     return { totalDrivers, totalLitros, totalKm, totalCargas, avgL100, avgRend, topP90, faenas: rankingGroups.length };
436 |   }, [rankingGroups]);
437 |
438 |   function toggleFaena(f: string) {
439 |     setExpandedFaenas(prev => {
440 |       const n = new Set(prev);
441 |       n.has(f) ? n.delete(f) : n.add(f);
442 |       return n;
443 |     });
444 |   }
445 |
446 |   if (isLoading) {
447 |     return (
448 |       <div className="space-y-4">
449 |         <Skeleton className="h-8 w-64" />
450 |         <Skeleton className="h-24 w-full" />
451 |         <Skeleton className="h-40 w-full" />
452 |         <Skeleton className="h-40 w-full" />
453 |       </div>
454 |     );
455 |   }
456 |
457 |   return (
458 |     <div className="space-y-4">
459 |       <div className="flex items-center justify-between flex-wrap gap-3">
460 |         <div>
461 |           <h2 className="text-lg font-mono font-bold tracking-[0.15em] uppercase" data-testid="text-ranking-title">
462 |             Ranking Conductores
463 |           </h2>
464 |           <p className="text-[10px] font-mono text-muted-foreground tracking-[0.1em]">
465 |             Eficiencia de combustible por conductor y faena &mdash; datos desde 01-03-2026
466 |             {trucksFiltered > 0 && (
467 |               <span className="ml-2 text-amber-400">
468 |                 ({trucksFiltered} de {trucksTotal} camiones excluidos por dif. Volvo/Sigetra &gt;5%)
469 |               </span>
470 |             )}
471 |           </p>
472 |         </div>
473 |         <div className="flex items-center gap-2">
474 |           {[3, 7, 14, 30].map(d => (
475 |             <button key={d} onClick={() => setDays(d)}
476 |               data-testid={`btn-days-${d}`}
477 |               className={`px-3 py-1.5 text-[10px] font-mono font-bold tracking-[0.1em] border transition-colors cursor-p
478 |                 days === d ? "bg-primary text-primary-foreground border-primary" : "bg-card text-muted-foreground border
479 |               }`}>
480 |               {d}D
481 |             </button>
482 |           ))}
483 |         </div>
484 |       </div>
485 |
486 |       <div className="flex items-center gap-3 flex-wrap">
487 |         <div className="flex items-center gap-2 bg-card border border-border px-3 py-1.5">
488 |           <Filter className="w-3.5 h-3.5 text-primary" />
489 |           <span className="text-[10px] font-mono text-muted-foreground tracking-[0.1em]">FAENA:</span>
490 |           <select
491 |             value={faenaFilter}
492 |             onChange={e => setFaenaFilter(e.target.value)}
493 |             data-testid="select-faena-filter"
494 |             className="bg-muted border border-border px-2 py-0.5 text-[11px] font-mono text-foreground outline-none focu
495 |           >
496 |             <option value="todas">Todas las faenas</option>
497 |             {allFaenas.map(f => (
498 |               <option key={f} value={f}>{f}</option>
499 |             ))}
500 |           </select>
501 |         </div>
502 |
503 |         <div className="flex items-center gap-1.5">
504 |           {(["rendimiento", "litros", "conductores"] as const).map(s => (
505 |             <button key={s} onClick={() => setSortBy(s)}
506 |               data-testid={`btn-sort-${s}`}
507 |               className={`px-2.5 py-1 text-[9px] font-mono font-bold tracking-[0.1em] uppercase border transition-colors
508 |                 sortBy === s ? "bg-primary/20 text-primary border-primary/50" : "bg-card text-muted-foreground border-bo
509 |               }`}>
510 |               {s === "rendimiento" ? "Rendimiento" : s === "litros" ? "Litros" : "Conductores"}
511 |             </button>
512 |           ))}
513 |         </div>
514 |       </div>
515 |
516 |       <div className="grid grid-cols-2 sm:grid-cols-4 lg:grid-cols-7 gap-3">
517 |         <div className="bg-card border border-border p-3">
518 |           <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-1">FAENAS</div>
519 |           <div className="text-xl font-mono font-bold text-foreground" data-testid="text-kpi-faenas">{globalKpis.faenas}
520 |         </div>
521 |         <div className="bg-card border border-border p-3">
522 |           <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-1">CONDUCTORES</div>
523 |           <div className="text-xl font-mono font-bold text-foreground" data-testid="text-kpi-conductores">{globalKpis.to
524 |         </div>
525 |         <div className="bg-card border border-border p-3">
526 |           <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-1">REND. GLOBAL</div>
527 |           <div className={`text-xl font-mono font-bold ${
528 |             globalKpis.avgRend >= 1.8 ? "text-emerald-400" : globalKpis.avgRend >= 1.2 ? "text-amber-400" : "text-red-40
529 |           }`} data-testid="text-kpi-rend-global">
530 |             {globalKpis.avgRend > 0 ? `${globalKpis.avgRend.toFixed(2)}` : "\u2014"}
531 |           </div>
532 |           <div className="text-[9px] font-mono text-muted-foreground">km/L</div>
```

```
533 |         </div>
534 |         <div className="bg-card border border-border p-3">
535 |           <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-1">L/100KM</div>
536 |           <div className="text-xl font-mono font-bold text-cyan-400" data-testid="text-kpi-l100">
537 |             {globalKpis.avgL100 > 0 ? globalKpis.avgL100.toFixed(1) : "\u2014"}
538 |           </div>
539 |           <div className="text-[9px] font-mono text-muted-foreground">litros/100km</div>
540 |         </div>
541 |         <div className="bg-card border border-border p-3">
542 |           <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-1">LITROS TOTAL</div>
543 |           <div className="text-xl font-mono font-bold text-amber-400" data-testid="text-kpi-litros">{fN(Math.round(globa
544 |           <div className="text-[9px] font-mono text-muted-foreground">litros</div>
545 |         </div>
546 |         <div className="bg-card border border-border p-3">
547 |           <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-1">KM TOTAL</div>
548 |           <div className="text-xl font-mono font-bold text-foreground" data-testid="text-kpi-km">{fN(Math.round(globalKp
549 |           <div className="text-[9px] font-mono text-muted-foreground">km</div>
550 |         </div>
551 |         <div className="bg-card border border-border p-3">
552 |           <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-1">CARGAS</div>
553 |           <div className="text-xl font-mono font-bold text-foreground" data-testid="text-kpi-cargas">{fN(globalKpis.tota
554 |           <div className="text-[9px] font-mono text-muted-foreground">total cargas</div>
555 |         </div>
556 |       </div>
557 |
558 |       {isError ? (
559 |         <Card>
560 |           <CardContent className="py-10 text-center">
561 |             <AlertTriangle className="w-10 h-10 text-red-400 mx-auto mb-3" />
562 |             <p className="text-sm font-mono text-red-400">Error al cargar datos de Sigetra Fusion</p>
563 |             <p className="text-[10px] font-mono text-muted-foreground mt-1">Verifique la conexion con el portal Sigetra<
564 |           </CardContent>
565 |         </Card>
566 |       ) : rankingGroups.length === 0 ? (
567 |         <Card>
568 |           <CardContent className="py-10 text-center">
569 |             <Users className="w-10 h-10 text-muted-foreground mx-auto mb-3" />
570 |             <p className="text-sm font-mono text-muted-foreground">Sin datos de conductores en el periodo seleccionado</
571 |           </CardContent>
572 |         </Card>
573 |       ) : (
574 |         <div className="space-y-3">
575 |           {rankingGroups.map(g => (
576 |             <FaenaRankingCard
577 |               key={g.faena}
578 |               group={g}
579 |
580 |               expanded={expandedFaenas.has(g.faena)}
581 |               onToggle={() => toggleFaena(g.faena)}
582 |             />
583 |           ))}
584 |         </div>
585 |       )}
586 |
587 |       <div className="bg-amber-500/10 border border-amber-500/30 px-4 py-3">
588 |         <div className="flex items-start gap-2">
589 |           <Star className="w-4 h-4 text-amber-400 flex-shrink-0 mt-0.5" />
590 |           <div className="text-[10px] font-mono text-amber-200/80 leading-relaxed">
591 |             <span className="font-bold text-amber-300">Ranking por percentil de rendimiento (km/L):</span>{" "}
592 |             Cada conductor se ubica en un percentil respecto al rendimiento global de todos los conductores de la flota
593 |             <span className="text-emerald-300">P90+ = Top 10%</span> (excelente),{" "}
594 |             <span className="text-emerald-400/70">P75-P89 = Top 25%</span> (bueno),{" "}
595 |             <span className="text-blue-400">P50-P74 = Sobre promedio</span>,{" "}
596 |             <span className="text-amber-400">P25-P49 = Bajo promedio</span>,{" "}
597 |             <span className="text-red-400">Bajo P25 = Bajo rendimiento</span>.{" "}
598 |             <span className="font-bold text-emerald-300">Filtro de calidad:</span> solo se incluyen camiones donde la di
599 |             Se excluyen conductores con &lt;2 cargas validas y rendimiento &gt;15 km/L o &le;0.{" "}
600 |             <span className="font-bold text-amber-300">Sistema de medicion activo desde 01-03-2026.</span>
601 |           </div>
602 |         </div>
603 |       </div>
604 |     </div>
605 |   );
606 | }
607 |
```

client/src/pages/tarifas.tsx

598 lineas

```
1 | import { useState, useMemo } from "react";
2 | import { useQuery, useMutation } from "@tanstack/react-query";
3 | import { queryClient, apiRequest } from "@/lib/queryClient";
4 | import { Button } from "@/components/ui/button";
5 | import { Input } from "@/components/ui/input";
6 | import { Skeleton } from "@/components/ui/skeleton";
7 | import { KpiCard } from "@/components/kpi-card";
8 | import { useToast } from "@/hooks/use-toast";
9 | import { MapPin, Route, Plus, Pencil, Trash2, Check, X, Filter, DollarSign, TrendingUp, Fuel } from "lucide-react";
10 | import type { PuntoRuta, TarifaRuta } from "@shared/schema";
11 |
12 | type SubTab = "puntos" | "tarifas";
13 |
14 | function PuntosConfirmados() {
15 |   const { data: puntos = [], isLoading } = useQuery<PuntoRuta[]>({ queryKey: ["/api/tarifas/puntos-confirmados"] });
16 |   const [tipoFilter, setTipoFilter] = useState<string>("todos");
17 |   const [editId, setEditId] = useState<number | null>(null);
18 |   const [editName, setEditName] = useState("");
19 |   const { toast } = useToast();
20 |
21 |   const updateMutation = useMutation({
22 |     mutationFn: async ({ id, nombreInferido }: { id: number; nombreInferido: string }) => {
23 |       await apiRequest("PUT", `/api/puntos-ruta/${id}`, { nombreInferido });
24 |     },
25 |     onSuccess: () => {
26 |       queryClient.invalidateQueries({ queryKey: ["/api/tarifas/puntos-confirmados"] });
27 |       setEditId(null);
28 |       toast({ title: "Punto actualizado" });
29 |     },
30 |     onError: (e: Error) => {
31 |       toast({ title: "Error", description: e.message, variant: "destructive" });
32 |     },
33 |   });
34 |
35 |   const tipos = useMemo(() => {
36 |     const set = new Set(puntos.map(p => p.tipo || "SIN TIPO"));
37 |     return ["todos", ...Array.from(set)];
38 |   }, [puntos]);
39 |
40 |   const filtered = useMemo(() => {
41 |     if (tipoFilter === "todos") return puntos;
42 |     return puntos.filter(p => (p.tipo || "SIN TIPO") === tipoFilter);
43 |   }, [puntos, tipoFilter]);
44 |
45 |   const grouped = useMemo(() => {
46 |     const map = new Map<string, PuntoRuta[]>();
47 |     for (const p of filtered) {
48 |       const key = p.tipo || "SIN TIPO";
49 |       const arr = map.get(key) || [];
50 |       arr.push(p);
51 |       map.set(key, arr);
52 |     }
53 |     return map;
54 |   }, [filtered]);
55 |
56 |   const tipoColor: Record<string, string> = {
57 |     CARGA: "#FFB020",
58 |     ENTREGA: "#00C87A",
59 |     COMBUSTIBLE: "#1A8FFF",
60 |     DESCANSO: "#A855F7",
61 |     "SIN TIPO": "#5A7A9A",
62 |   };
63 |
64 |   if (isLoading) {
65 |     return (
66 |       <div className="space-y-3" data-testid="puntos-loading">
67 |         {Array.from({ length: 3 }).map((_, i) => <Skeleton key={i} className="h-16" />)}
68 |       </div>
69 |     );
70 |   }
71 |
72 |   return (
73 |     <div className="space-y-4" data-testid="puntos-view">
74 |       <div className="flex items-center gap-2 flex-wrap" data-testid="puntos-tipo-filters">
75 |         <Filter className="w-3.5 h-3.5 text-muted-foreground" />
76 |         {tipos.map(t => (
77 |           <button
78 |             key={t}
79 |             onClick={() => setTipoFilter(t)}
80 |             data-testid={`filter-tipo-${t.toLowerCase().replace(/\s/g, "-")}`}
81 |             className={`px-3 py-1.5 text-[10px] font-mono tracking-[0.1em] cursor-pointer border uppercase transition-al
82 |               tipoFilter === t
83 |                 ? "border-primary/50 text-primary bg-primary/10"
84 |                 : "border-border text-muted-foreground"
85 |             }`}
86 |           >
87 |             {t === "todos" ? "Todos" : t}
88 |             {t !== "todos" && (
89 |               <span className="ml-1 opacity-60">
90 |                 ({puntos.filter(p => (p.tipo || "SIN TIPO") === t).length})
91 |               </span>
92 |             )}
93 |           </button>
94 |         ))}
95 |         <div className="ml-auto text-[11px] font-mono text-muted-foreground">
96 |           {filtered.length} puntos confirmados
97 |         </div>
98 |       </div>
99 |
100 |       {filtered.length === 0 && (
101 |         <div className="py-12 text-center text-muted-foreground font-mono text-xs" data-testid="text-no-puntos">
102 |           No hay puntos confirmados. Detecta y confirma puntos desde la ficha de cada camion.
103 |         </div>
104 |       )}
```

```
105 |
106 |       {Array.from(grouped.entries()).map(([tipo, pts]) => (
107 |         <div key={tipo} data-testid={`grupo-tipo-${tipo.toLowerCase().replace(/\s/g, "-")}`}>
108 |           <div className="flex items-center gap-2 mb-2">
109 |             <div className="w-2.5 h-2.5 rounded-full" style={{ backgroundColor: tipoColor[tipo] || "#5A7A9A" }} />
110 |             <span className="text-[10px] font-mono font-bold tracking-[0.15em]" style={{ color: tipoColor[tipo] || "#5A7
111 |               {tipo}
112 |             </span>
113 |             <span className="text-[10px] font-mono text-muted-foreground">({pts.length})</span>
114 |           </div>
115 |           <div className="bg-card border border-border">
116 |             <div className="grid grid-cols-[1fr_200px_120px_100px_80px] px-4 py-2 border-b border-border bg-muted/30 tex
117 |               <div>NOMBRE</div><div>COORDENADAS</div><div>DURACION</div><div>FECHA</div><div>ACCIONES</div>
118 |             </div>
119 |             {pts.map((p, i) => (
120 |               <div
121 |                 key={p.id}
122 |                 className={`grid grid-cols-[1fr_200px_120px_100px_80px] px-4 py-3 items-center ${
123 |                   i < pts.length - 1 ? "border-b border-border/20" : ""
124 |                 }`}
125 |                 data-testid={`punto-row-${p.id}`}
126 |               >
127 |                 <div>
128 |                   {editId === p.id ? (
129 |                     <div className="flex items-center gap-1.5">
130 |                       <Input
131 |                         value={editName}
132 |                         onChange={e => setEditName(e.target.value)}
133 |                         className="h-7 text-xs font-mono w-48"
134 |                         data-testid={`input-edit-punto-${p.id}`}
135 |                         autoFocus
136 |                       />
137 |                       <Button
138 |                         size="icon"
139 |                         variant="ghost"
140 |                         onClick={() => updateMutation.mutate({ id: p.id, nombreInferido: editName })}
141 |                         disabled={updateMutation.isPending}
142 |                         data-testid={`btn-save-punto-${p.id}`}
143 |                       >
144 |                         <Check className="w-3.5 h-3.5 text-emerald-400" />
145 |                       </Button>
146 |                       <Button size="icon" variant="ghost" onClick={() => setEditId(null)} data-testid={`btn-cancel-edit-
147 |                         <X className="w-3.5 h-3.5 text-muted-foreground" />
148 |                       </Button>
149 |                     </div>
150 |                   ) : (
151 |                     <span className="text-xs font-mono text-foreground">{p.nombreInferido || "Sin nombre"}</span>
152 |                   )}
153 |                 </div>
154 |                 <div className="text-[10px] font-mono text-muted-foreground">
155 |                   {p.lat}, {p.lng}
156 |                 </div>
157 |                 <div className="text-[10px] font-mono text-muted-foreground">
158 |                   {p.duracionMin != null ? `${p.duracionMin} min` : "--"}
159 |                 </div>
160 |                 <div className="text-[10px] font-mono text-muted-foreground">
161 |                   {p.fecha ? new Date(p.fecha).toLocaleDateString("es-CL") : "--"}
162 |                 </div>
163 |                 <div>
164 |                   <Button
165 |                     size="icon"
166 |                     variant="ghost"
167 |                     onClick={() => { setEditId(p.id); setEditName(p.nombreInferido || ""); }}
168 |                     data-testid={`btn-edit-punto-${p.id}`}
169 |                   >
170 |                     <Pencil className="w-3.5 h-3.5" />
171 |                   </Button>
172 |                 </div>
173 |               </div>
174 |             ))}
175 |           </div>
176 |         </div>
177 |       ))}
178 |     </div>
179 |   );
180 | }
181 |
182 | function TarifasPorRuta() {
183 |   const { data: tarifas = [], isLoading } = useQuery<TarifaRuta[]>({ queryKey: ["/api/tarifas"] });
184 |   const { data: puntosConfirmados = [] } = useQuery<PuntoRuta[]>({ queryKey: ["/api/tarifas/puntos-confirmados"] });
185 |   const [showForm, setShowForm] = useState(false);
186 |   const [editingTarifa, setEditingTarifa] = useState<TarifaRuta | null>(null);
187 |   const { toast } = useToast();
188 |
189 |   const [formData, setFormData] = useState({
190 |     nombreRuta: "",
191 |     origenNombre: "",
192 |     destinoNombre: "",
193 |     distanciaKm: "",
194 |     litrosPromedio: "",
195 |     tiempoHoras: "",
196 |     tarifaClp: "",
197 |     tarifaUsd: "",
198 |     notas: "",
199 |     puntoOrigenId: "",
200 |     puntoDestinoId: "",
201 |   });
202 |
203 |   const activeTarifas = useMemo(() => tarifas.filter(t => t.activa !== false), [tarifas]);
204 |
205 |   const kpis = useMemo(() => {
206 |     const total = activeTarifas.length;
207 |     const mostExpensive = activeTarifas.reduce((max, t) => {
208 |       const val = t.tarifaClp ? parseFloat(String(t.tarifaClp)) : 0;
209 |       return val > max.val ? { val, nombre: t.nombreRuta } : max;
210 |     }, { val: 0, nombre: "--" });
211 |     const litrosValues = activeTarifas.map(t => t.litrosPromedio ? parseFloat(String(t.litrosPromedio)) : 0).filter(v =>
```

```
212 |     const avgLitros = litrosValues.length > 0 ? (litrosValues.reduce((a, b) => a + b, 0) / litrosValues.length).toFixed(
213 |     const distValues = activeTarifas.map(t => t.distanciaKm ? parseFloat(String(t.distanciaKm)) : 0).filter(v => v > 0);
214 |     const mostFrequent = activeTarifas.length > 0 ? activeTarifas[0].nombreRuta : "--";
215 |     return { total, mostExpensive: mostExpensive.nombre, avgLitros, mostFrequent };
216 |   }, [activeTarifas]);
217 |
218 |   const createMutation = useMutation({
219 |     mutationFn: async (data: typeof formData) => {
220 |       const body = {
221 |         nombreRuta: data.nombreRuta,
222 |         origenNombre: data.origenNombre || null,
223 |         destinoNombre: data.destinoNombre || null,
224 |         distanciaKm: data.distanciaKm || null,
225 |         litrosPromedio: data.litrosPromedio || null,
226 |         tiempoHoras: data.tiempoHoras || null,
227 |         tarifaClp: data.tarifaClp || null,
228 |         tarifaUsd: data.tarifaUsd || null,
229 |         notas: data.notas || null,
230 |         puntoOrigenId: data.puntoOrigenId ? parseInt(data.puntoOrigenId) : null,
231 |         puntoDestinoId: data.puntoDestinoId ? parseInt(data.puntoDestinoId) : null,
232 |         activa: true,
233 |       };
234 |       await apiRequest("POST", "/api/tarifas", body);
235 |     },
236 |     onSuccess: () => {
237 |       queryClient.invalidateQueries({ queryKey: ["/api/tarifas"] });
238 |       resetForm();
239 |       toast({ title: "Tarifa creada" });
240 |     },
241 |     onError: (e: Error) => {
242 |       toast({ title: "Error", description: e.message, variant: "destructive" });
243 |     },
244 |   });
245 |
246 |   const updateMutation = useMutation({
247 |     mutationFn: async ({ id, data }: { id: number; data: typeof formData }) => {
248 |       const body = {
249 |         nombreRuta: data.nombreRuta,
250 |         origenNombre: data.origenNombre || null,
251 |         destinoNombre: data.destinoNombre || null,
252 |         distanciaKm: data.distanciaKm || null,
253 |         litrosPromedio: data.litrosPromedio || null,
254 |         tiempoHoras: data.tiempoHoras || null,
255 |         tarifaClp: data.tarifaClp || null,
256 |         tarifaUsd: data.tarifaUsd || null,
257 |         notas: data.notas || null,
258 |         puntoOrigenId: data.puntoOrigenId ? parseInt(data.puntoOrigenId) : null,
259 |         puntoDestinoId: data.puntoDestinoId ? parseInt(data.puntoDestinoId) : null,
260 |       };
261 |       await apiRequest("PUT", `/api/tarifas/${id}`, body);
262 |     },
263 |     onSuccess: () => {
264 |       queryClient.invalidateQueries({ queryKey: ["/api/tarifas"] });
265 |       resetForm();
266 |       toast({ title: "Tarifa actualizada" });
267 |     },
268 |     onError: (e: Error) => {
269 |       toast({ title: "Error", description: e.message, variant: "destructive" });
270 |     },
271 |   });
272 |
273 |   const deleteMutation = useMutation({
274 |     mutationFn: async (id: number) => {
275 |       await apiRequest("DELETE", `/api/tarifas/${id}`);
276 |     },
277 |     onSuccess: () => {
278 |       queryClient.invalidateQueries({ queryKey: ["/api/tarifas"] });
279 |       toast({ title: "Tarifa eliminada" });
280 |     },
281 |     onError: (e: Error) => {
282 |       toast({ title: "Error", description: e.message, variant: "destructive" });
283 |     },
284 |   });
285 |
286 |   function resetForm() {
287 |     setFormData({
288 |       nombreRuta: "", origenNombre: "", destinoNombre: "",
289 |       distanciaKm: "", litrosPromedio: "", tiempoHoras: "",
290 |       tarifaClp: "", tarifaUsd: "", notas: "",
291 |       puntoOrigenId: "", puntoDestinoId: "",
292 |     });
293 |     setShowForm(false);
294 |     setEditingTarifa(null);
295 |   }
296 |
297 |   function startEdit(t: TarifaRuta) {
298 |     setEditingTarifa(t);
299 |     setFormData({
300 |       nombreRuta: t.nombreRuta,
301 |       origenNombre: t.origenNombre || "",
302 |       destinoNombre: t.destinoNombre || "",
303 |       distanciaKm: t.distanciaKm ? String(t.distanciaKm) : "",
304 |       litrosPromedio: t.litrosPromedio ? String(t.litrosPromedio) : "",
305 |       tiempoHoras: t.tiempoHoras ? String(t.tiempoHoras) : "",
306 |       tarifaClp: t.tarifaClp ? String(t.tarifaClp) : "",
307 |       tarifaUsd: t.tarifaUsd ? String(t.tarifaUsd) : "",
308 |       notas: t.notas || "",
309 |       puntoOrigenId: t.puntoOrigenId ? String(t.puntoOrigenId) : "",
310 |       puntoDestinoId: t.puntoDestinoId ? String(t.puntoDestinoId) : "",
311 |     });
312 |     setShowForm(true);
313 |   }
314 |
315 |   function handleSubmit() {
316 |     if (!formData.nombreRuta.trim()) {
317 |       toast({ title: "Error", description: "Nombre de ruta requerido", variant: "destructive" });
318 |       return;
```

```
319 |     }
320 |     if (editingTarifa) {
321 |       updateMutation.mutate({ id: editingTarifa.id, data: formData });
322 |     } else {
323 |       createMutation.mutate(formData);
324 |     }
325 |   }
326 |
327 |   if (isLoading) {
328 |     return (
329 |       <div className="space-y-3" data-testid="tarifas-loading">
330 |         <div className="grid grid-cols-4 gap-3">
331 |           {Array.from({ length: 4 }).map((_, i) => <Skeleton key={i} className="h-24" />)}
332 |         </div>
333 |         <Skeleton className="h-48" />
334 |       </div>
335 |     );
336 |   }
337 |
338 |   return (
339 |     <div className="space-y-4" data-testid="tarifas-ruta-view">
340 |       <div className="grid grid-cols-4 gap-3" data-testid="tarifas-kpis">
341 |         <KpiCard label="Total Rutas" value={kpis.total} color="#1A8FFF"
342 |           icon={<Route className="w-5 h-5" />} />
343 |         <KpiCard label="Mas Costosa" value={kpis.mostExpensive} color="#FF2D4A"
344 |           icon={<DollarSign className="w-5 h-5" />} />
345 |         <KpiCard label="Mas Frecuente" value={kpis.mostFrequent} color="#FFB020"
346 |           icon={<TrendingUp className="w-5 h-5" />} />
347 |         <KpiCard label="Litros Promedio" value={kpis.avgLitros} unit="L" color="#00C87A"
348 |           icon={<Fuel className="w-5 h-5" />} />
349 |       </div>
350 |
351 |       <div className="flex items-center justify-between">
352 |         <div className="text-[11px] font-mono text-muted-foreground">
353 |           {activeTarifas.length} tarifas activas
354 |         </div>
355 |         <Button
356 |           variant="outline"
357 |           size="sm"
358 |           onClick={() => { resetForm(); setShowForm(true); }}
359 |           data-testid="btn-nueva-tarifa"
360 |         >
361 |           <Plus className="w-3.5 h-3.5 mr-1" />
362 |           Nueva Tarifa
363 |         </Button>
364 |       </div>
365 |
366 |       {showForm && (
367 |         <div className="bg-card border border-border p-4 space-y-3" data-testid="tarifa-form">
368 |           <div className="text-[9px] font-mono text-muted-foreground tracking-[0.2em] uppercase mb-2">
369 |             {editingTarifa ? "EDITAR TARIFA" : "NUEVA TARIFA"}
370 |           </div>
371 |           <div className="grid grid-cols-3 gap-3">
372 |             <div>
373 |               <label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] block mb-1">NOMBRE RUTA *</
374 |               <Input
375 |                 value={formData.nombreRuta}
376 |                 onChange={e => setFormData({ ...formData, nombreRuta: e.target.value })}
377 |                 placeholder="Ej: Faena Norte - Puerto"
378 |                 className="text-xs font-mono"
379 |                 data-testid="input-nombre-ruta"
380 |               />
381 |             </div>
382 |             <div>
383 |               <label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] block mb-1">ORIGEN</label>
384 |               <Input
385 |                 value={formData.origenNombre}
386 |                 onChange={e => setFormData({ ...formData, origenNombre: e.target.value })}
387 |                 placeholder="Nombre origen"
388 |                 className="text-xs font-mono"
389 |                 data-testid="input-origen"
390 |               />
391 |             </div>
392 |             <div>
393 |               <label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] block mb-1">DESTINO</label>
394 |               <Input
395 |                 value={formData.destinoNombre}
396 |                 onChange={e => setFormData({ ...formData, destinoNombre: e.target.value })}
397 |                 placeholder="Nombre destino"
398 |                 className="text-xs font-mono"
399 |                 data-testid="input-destino"
400 |               />
401 |             </div>
402 |           </div>
403 |           <div className="grid grid-cols-5 gap-3">
404 |             <div>
405 |               <label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] block mb-1">DISTANCIA KM</l
406 |               <Input
407 |                 value={formData.distanciaKm}
408 |                 onChange={e => setFormData({ ...formData, distanciaKm: e.target.value })}
409 |                 placeholder="0"
410 |                 type="number"
411 |                 className="text-xs font-mono"
412 |                 data-testid="input-distancia"
413 |               />
414 |             </div>
415 |             <div>
416 |               <label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] block mb-1">LITROS PROM.</l
417 |               <Input
418 |                 value={formData.litrosPromedio}
419 |                 onChange={e => setFormData({ ...formData, litrosPromedio: e.target.value })}
420 |                 placeholder="0"
421 |                 type="number"
422 |                 className="text-xs font-mono"
423 |                 data-testid="input-litros-prom"
424 |               />
425 |             </div>
```

```
426 |             <div>
427 |               <label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] block mb-1">TIEMPO HRS</lab
428 |               <Input
429 |                 value={formData.tiempoHoras}
430 |                 onChange={e => setFormData({ ...formData, tiempoHoras: e.target.value })}
431 |                 placeholder="0"
432 |                 type="number"
433 |                 className="text-xs font-mono"
434 |                 data-testid="input-tiempo"
435 |               />
436 |             </div>
437 |             <div>
438 |               <label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] block mb-1">TARIFA CLP</lab
439 |               <Input
440 |                 value={formData.tarifaClp}
441 |                 onChange={e => setFormData({ ...formData, tarifaClp: e.target.value })}
442 |                 placeholder="0"
443 |                 type="number"
444 |                 className="text-xs font-mono"
445 |                 data-testid="input-tarifa-clp"
446 |               />
447 |             </div>
448 |             <div>
449 |               <label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] block mb-1">TARIFA USD</lab
450 |               <Input
451 |                 value={formData.tarifaUsd}
452 |                 onChange={e => setFormData({ ...formData, tarifaUsd: e.target.value })}
453 |                 placeholder="0"
454 |                 type="number"
455 |                 className="text-xs font-mono"
456 |                 data-testid="input-tarifa-usd"
457 |               />
458 |             </div>
459 |           </div>
460 |           <div>
461 |             <label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] block mb-1">NOTAS</label>
462 |             <Input
463 |               value={formData.notas}
464 |               onChange={e => setFormData({ ...formData, notas: e.target.value })}
465 |               placeholder="Notas adicionales..."
466 |               className="text-xs font-mono"
467 |               data-testid="input-notas"
468 |             />
469 |           </div>
470 |           {puntosConfirmados.length > 0 && (
471 |             <div className="grid grid-cols-2 gap-3">
472 |               <div>
473 |                 <label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] block mb-1">PUNTO ORIGEN
474 |                 <select
475 |                   value={formData.puntoOrigenId}
476 |                   onChange={e => setFormData({ ...formData, puntoOrigenId: e.target.value })}
477 |                   className="w-full h-9 px-3 text-xs font-mono bg-background border border-border text-foreground"
478 |                   data-testid="select-punto-origen"
479 |                 >
480 |                   <option value="">Sin punto asociado</option>
481 |                   {puntosConfirmados.map(p => (
482 |                     <option key={p.id} value={String(p.id)}>{p.nombreInferido || `Punto #${p.id}`} ({p.tipo})</option>
483 |                   ))}
484 |                 </select>
485 |               </div>
486 |               <div>
487 |                 <label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] block mb-1">PUNTO DESTINO
488 |                 <select
489 |                   value={formData.puntoDestinoId}
490 |                   onChange={e => setFormData({ ...formData, puntoDestinoId: e.target.value })}
491 |                   className="w-full h-9 px-3 text-xs font-mono bg-background border border-border text-foreground"
492 |                   data-testid="select-punto-destino"
493 |                 >
494 |                   <option value="">Sin punto asociado</option>
495 |                   {puntosConfirmados.map(p => (
496 |                     <option key={p.id} value={String(p.id)}>{p.nombreInferido || `Punto #${p.id}`} ({p.tipo})</option>
497 |                   ))}
498 |                 </select>
499 |               </div>
500 |             </div>
501 |           )}
502 |           <div className="flex gap-2 justify-end">
503 |             <Button variant="outline" size="sm" onClick={resetForm} data-testid="btn-cancelar-tarifa">
504 |               Cancelar
505 |             </Button>
506 |             <Button
507 |               size="sm"
508 |               onClick={handleSubmit}
509 |               disabled={createMutation.isPending || updateMutation.isPending}
510 |               data-testid="btn-guardar-tarifa"
511 |             >
512 |               {(createMutation.isPending || updateMutation.isPending) ? "Guardando..." : editingTarifa ? "Actualizar" :
513 |             </Button>
514 |           </div>
515 |         </div>
516 |       )}
517 |
518 |       {activeTarifas.length === 0 && !showForm && (
519 |         <div className="py-12 text-center text-muted-foreground font-mono text-xs" data-testid="text-no-tarifas">
520 |           No hay tarifas registradas. Crea la primera tarifa para comenzar.
521 |         </div>
522 |       )}
523 |
524 |       {activeTarifas.length > 0 && (
525 |         <div className="bg-card border border-border" data-testid="tarifas-table">
526 |           <div className="grid grid-cols-[1fr_120px_120px_100px_100px_100px_80px] px-4 py-2 border-b border-border bg-mu
527 |             <div>RUTA</div><div>ORIGEN</div><div>DESTINO</div><div>DISTANCIA</div><div>LITROS</div><div>TARIFA CLP</div>
528 |           </div>
529 |           {activeTarifas.map((t, i) => (
530 |             <div
531 |               key={t.id}
532 |               className={`grid grid-cols-[1fr_120px_120px_100px_100px_100px_80px] px-4 py-3 items-center ${
```

```
533 |                 i < activeTarifas.length - 1 ? "border-b border-border/20" : ""
534 |               }`}
535 |               data-testid={`tarifa-row-${t.id}`}
536 |             >
537 |               <div>
538 |                 <div className="text-xs font-mono font-bold text-foreground">{t.nombreRuta}</div>
539 |                 {t.notas && <div className="text-[9px] text-muted-foreground mt-0.5">{t.notas}</div>}
540 |               </div>
541 |               <div className="text-[10px] font-mono text-muted-foreground">{t.origenNombre || "--"}</div>
542 |               <div className="text-[10px] font-mono text-muted-foreground">{t.destinoNombre || "--"}</div>
543 |               <div className="text-[10px] font-mono text-foreground">{t.distanciaKm ? `${t.distanciaKm} km` : "--"}</div
544 |               <div className="text-[10px] font-mono text-foreground">{t.litrosPromedio ? `${t.litrosPromedio} L` : "--"}
545 |               <div className="text-[10px] font-mono text-foreground">{t.tarifaClp ? `$${parseInt(String(t.tarifaClp)).to
546 |               <div className="flex gap-1" onClick={e => e.stopPropagation()}>
547 |                 <Button size="icon" variant="ghost" onClick={() => startEdit(t)} data-testid={`btn-edit-tarifa-${t.id}`}
548 |                   <Pencil className="w-3.5 h-3.5" />
549 |                 </Button>
550 |                 <Button
551 |                   size="icon"
552 |                   variant="ghost"
553 |                   onClick={() => deleteMutation.mutate(t.id)}
554 |                   disabled={deleteMutation.isPending}
555 |                   data-testid={`btn-delete-tarifa-${t.id}`}
556 |                 >
557 |                   <Trash2 className="w-3.5 h-3.5 text-red-400" />
558 |                 </Button>
559 |               </div>
560 |             </div>
561 |           ))}
562 |         </div>
563 |       )}
564 |     </div>
565 |   );
566 | }
567 |
568 | export default function Tarifas() {
569 |   const [subTab, setSubTab] = useState<SubTab>("puntos");
570 |
571 |   return (
572 |     <div className="space-y-5" data-testid="tarifas-page">
573 |       <div className="flex gap-1" data-testid="tarifas-subtabs">
574 |         {([
575 |           { id: "puntos" as SubTab, label: "Puntos Confirmados", icon: MapPin },
576 |           { id: "tarifas" as SubTab, label: "Tarifas por Ruta", icon: Route },
577 |         ].map(t => (
578 |           <button
579 |             key={t.id}
580 |             onClick={() => setSubTab(t.id)}
581 |             data-testid={`subtab-${t.id}`}
582 |             className={`flex items-center gap-1.5 px-4 py-2 text-[11px] font-mono font-bold tracking-[0.1em] uppercase c
583 |               subTab === t.id
584 |                 ? "border-primary text-primary"
585 |                 : "border-transparent text-muted-foreground"
586 |             }`}
587 |           >
588 |             <t.icon className="w-3.5 h-3.5" />
589 |             {t.label}
590 |           </button>
591 |         ))}
592 |       </div>
593 |
594 |       {subTab === "puntos" ? <PuntosConfirmados /> : <TarifasPorRuta />}
595 |     </div>
596 |   );
597 | }
598 |
```

client/src/pages/performance-ia.tsx

597 lineas

```
1 | import { useState, useMemo } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { Skeleton } from "@/components/ui/skeleton";
4 | import { Card, CardContent, CardHeader, CardTitle } from "@/components/ui/card";
5 | import {
6 |   Cpu, TrendingUp, TrendingDown, Fuel, Truck, ChevronDown, ChevronUp,
7 |   BarChart3, AlertTriangle, Info, MapPin, Target, Route, Navigation
8 | } from "lucide-react";
9 | import { getErrorFleetNums, isErrorCarga } from "@/pages/errores";
10 | import { type Percentiles, getNivelPercentil, rendColorFlota } from "@/lib/fuel-utils";
11 | import RutasContrato from "@/pages/rutas-contrato";
12 | import CombustibleIA from "@/pages/combustible-ia";
13 | import AnalizadorRutas from "@/pages/analizador-rutas";
14 |
15 | interface FusionCarga {
16 |   numGuia: number;
17 |   fecha: string;
18 |   litros: number;
19 |   odometro: number | null;
20 |   kmRecorrido: number | null;
21 |   rendimiento: number | null;
22 |   lugar: string | null;
23 |   tarjeta: string | null;
24 |   conductor: string | null;
25 | }
26 |
27 | interface FusionTruck {
28 |   fleetNum: string;
29 |   patenteReal: string | null;
30 |   vin: string | null;
31 |   modeloVolvo: string;
32 |   faenaSigetra: string | null;
33 |   conductorSigetra: string | null;
34 |   totalLitrosSigetra: number;
35 |   totalCargas: number;
36 |   rendPromedio: number;
37 |   odometroSigetra: number | null;
38 |   odometroVolvo: number | null;
39 |   deltaOdometro: number | null;
40 |   fuelLevelVolvo: number | null;
41 |   totalFuelUsedVolvo: number | null;
42 |   litrosVolvoPeriodo: number | null;
43 |   engineHoursVolvo: number | null;
44 |   gpsVolvo: { latitude: number | null; longitude: number | null } | null;
45 |   alertLevel: "ok" | "alerta" | "critico";
46 |   cargas: FusionCarga[];
47 | }
48 |
49 | interface FusionResponse {
50 |   from: string;
51 |   to: string;
52 |   totalTrucksMatched: number;
53 |   totalLitros: number;
54 |   totalCargas: number;
55 |   alertCount: number;
56 |   trucks: FusionTruck[];
57 | }
58 |
59 | interface FaenaAnalysis {
60 |   faena: string;
61 |   trucks: FusionTruck[];
62 |   truckCount: number;
63 |   avgRendimiento: number;
64 |   optimo: number;
65 |   totalLitros: number;
66 |   totalLitrosVolvo: number;
67 |   totalLitrosVolvoPeriodo: number;
68 |   totalCargas: number;
69 |   bestTruck: FusionTruck | null;
70 |   worstTruck: FusionTruck | null;
71 |   rendRange: { min: number; max: number };
72 | }
73 |
74 | function calcPercentile(values: number[], pct: number): number {
75 |   if (values.length === 0) return 0;
76 |   const sorted = [...values].sort((a, b) => a - b);
77 |   const idx = (pct / 100) * (sorted.length - 1);
78 |   const lower = Math.floor(idx);
79 |   const upper = Math.ceil(idx);
80 |   if (lower === upper) return sorted[lower];
81 |   return sorted[lower] + (sorted[upper] - sorted[lower]) * (idx - lower);
82 | }
83 |
84 | function fN(n: number): string {
85 |   return n.toLocaleString("es-CL");
86 | }
87 |
88 | const DATA_START_DATE = "2026-03-01";
89 |
90 | function getDateRange(days: number): { from: string; to: string } {
91 |   const to = new Date();
92 |   const from = new Date(Date.now() - days * 24 * 60 * 60 * 1000);
93 |   const minDate = new Date(DATA_START_DATE + "T00:00:00");
94 |   const effectiveFrom = from < minDate ? minDate : from;
95 |   return { from: effectiveFrom.toISOString().slice(0, 10), to: to.toISOString().slice(0, 10) };
96 | }
97 |
98 | function RendBar({ value, max, color }: { value: number; max: number; color: string }) {
99 |   const pct = max > 0 ? Math.min((value / max) * 100, 100) : 0;
100 |   return (
101 |     <div className="h-2 w-full bg-muted rounded-md overflow-hidden">
102 |       <div className={`h-full rounded-md ${color}`} style={{ width: `${pct}%` }} />
103 |     </div>
104 |   );
```

```
105 | }
106 |
107 | function FaenaCard({ analysis, globalMax, expanded, onToggle }: {
108 |   analysis: FaenaAnalysis;
109 |   globalMax: number;
110 |   expanded: boolean;
111 |   onToggle: () => void;
112 | }) {
113 |   const trucksWithRend = analysis.trucks.filter(t => t.rendPromedio > 0).sort((a, b) => b.rendPromedio - a.rendPromedio)
114 |
115 |   return (
116 |     <Card className="overflow-visible" data-testid={`card-faena-${analysis.faena}`}>
117 |       <CardHeader className="flex flex-row items-center justify-between gap-2 space-y-0 pb-2 cursor-pointer" onClick={on
118 |         <div className="flex items-center gap-3 flex-wrap min-w-0">
119 |           <MapPin className="w-4 h-4 text-primary flex-shrink-0" />
120 |           <CardTitle className="text-sm font-mono font-bold tracking-[0.1em] uppercase truncate" data-testid={`text-faen
121 |             {analysis.faena}
122 |           </CardTitle>
123 |           <span className="text-[10px] font-mono text-muted-foreground">{analysis.truckCount} camiones</span>
124 |         </div>
125 |         <div className="flex items-center gap-4 flex-wrap flex-shrink-0">
126 |           <div className="text-right">
127 |             <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em]">REND. PROMEDIO</div>
128 |             <div className={`text-lg font-mono font-bold ${
129 |               analysis.avgRendimiento >= 1.8 ? "text-emerald-400" : analysis.avgRendimiento >= 1.2 ? "text-amber-400" :
130 |             }`} data-testid={`text-avg-rend-${analysis.faena}`}>
131 |               {analysis.avgRendimiento > 0 ? `${analysis.avgRendimiento.toFixed(2)} km/L` : "\u2014"}
132 |             </div>
133 |           </div>
134 |           {expanded ? <ChevronUp className="w-4 h-4 text-muted-foreground" /> : <ChevronDown className="w-4 h-4 text-mut
135 |         </div>
136 |       </CardHeader>
137 |       <CardContent className="pt-0">
138 |         <div className="grid grid-cols-3 sm:grid-cols-5 gap-4 mb-3">
139 |           <div>
140 |             <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-0.5">OPTIMO RUTA</div>
141 |             <div className="text-sm font-mono font-bold text-cyan-400" data-testid={`text-optimo-${analysis.faena}`}>
142 |               {analysis.optimo > 0 ? `${analysis.optimo.toFixed(2)} km/L` : "\u2014"}
143 |             </div>
144 |             <div className="text-[9px] font-mono text-muted-foreground">P75 referencia</div>
145 |           </div>
146 |           <div>
147 |             <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-0.5">LITROS SIGETRA</div>
148 |             <div className="text-sm font-mono font-bold text-amber-400" data-testid={`text-litros-sigetra-${analysis.fae
149 |             <div className="text-[9px] font-mono text-muted-foreground">periodo</div>
150 |           </div>
151 |           <div>
152 |             <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-0.5">LITROS VOLVO ECU</div>
153 |             <div className="text-sm font-mono font-bold text-blue-400" data-testid={`text-litros-volvo-${analysis.faena}
154 |               {analysis.totalLitrosVolvoPeriodo > 0 ? `${fN(Math.round(analysis.totalLitrosVolvoPeriodo))} L` : "\u2014"
155 |             </div>
156 |             <div className="text-[9px] font-mono text-muted-foreground">{analysis.totalLitrosVolvoPeriodo > 0 ? "periodo
157 |           </div>
158 |           <div>
159 |             <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-0.5">CARGAS</div>
160 |             <div className="text-sm font-mono font-bold text-foreground" data-testid={`text-cargas-${analysis.faena}`}>{
161 |           </div>
162 |           <div>
163 |             <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-0.5">RANGO REND.</div>
164 |             <div className="text-sm font-mono font-bold text-foreground" data-testid={`text-range-${analysis.faena}`}>
165 |               {analysis.rendRange.min > 0 ? `${analysis.rendRange.min.toFixed(1)} - ${analysis.rendRange.max.toFixed(1)}
166 |             </div>
167 |           </div>
168 |         </div>
169 |
170 |         {analysis.bestTruck && analysis.worstTruck && (
171 |           <div className="grid grid-cols-2 gap-3 mb-3">
172 |             <div className="bg-emerald-500/5 border border-emerald-500/20 p-2.5">
173 |               <div className="flex items-center gap-1.5 mb-1">
174 |                 <TrendingUp className="w-3 h-3 text-emerald-400" />
175 |                 <span className="text-[8px] font-mono text-emerald-400 tracking-[0.15em]">MEJOR RENDIMIENTO</span>
176 |               </div>
177 |               <div class="text-xs font-mono font-bold text-foreground" data-testid={`text-best-truck-${analysis.faen
178 |                 N.{analysis.bestTruck.fleetNum} {analysis.bestTruck.patenteReal ? `(${analysis.bestTruck.patenteReal})`
179 |               </div>
180 |               <div class="text-sm font-mono font-bold text-emerald-400">
181 |                 {analysis.bestTruck.rendPromedio.toFixed(2)} km/L
182 |               </div>
183 |             </div>
184 |             <div class="bg-red-500/5 border border-red-500/20 p-2.5">
185 |               <div class="flex items-center gap-1.5 mb-1">
186 |                 <TrendingDown class="w-3 h-3 text-red-400" />
187 |                 <span class="text-[8px] font-mono text-red-400 tracking-[0.15em]">MENOR RENDIMIENTO</span>
188 |               </div>
189 |               <div class="text-xs font-mono font-bold text-foreground" data-testid={`text-worst-truck-${analysis.fae
190 |                 N.{analysis.worstTruck.fleetNum} {analysis.worstTruck.patenteReal ? `(${analysis.worstTruck.patenteReal}
191 |               </div>
192 |               <div class="text-sm font-mono font-bold text-red-400">
193 |                 {analysis.worstTruck.rendPromedio.toFixed(2)} km/L
194 |               </div>
195 |             </div>
196 |           </div>
197 |         )}
198 |
199 |         <RendBar value={analysis.avgRendimiento} max={globalMax} color="bg-primary" />
200 |
201 |         {expanded && trucksWithRend.length > 0 && (
202 |           <div className="mt-4 border border-border bg-background">
203 |             <div class="grid grid-cols-[55px_70px_1fr_95px_70px_80px_80px_55px] gap-0 bg-card px-3 py-2 border-b bor
204 |               <span class="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">N.INT</span>
205 |               <span class="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">PATENTE</span>
206 |               <span class="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">CONDUCTOR</span>
207 |               <span class="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">REND.</span>
208 |               <span class="text-[8px] font-mono font-bold text-cyan-400/70 tracking-[0.15em]">vs OPT</span>
209 |               <span class="text-[8px] font-mono font-bold text-amber-400/70 tracking-[0.15em]">L.SIGETRA</span>
210 |               <span class="text-[8px] font-mono font-bold text-blue-400/70 tracking-[0.15em]">L.VOLVO ECU</span>
211 |               <span class="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">CARG.</span>
```

```
212 |             </div>
213 |             {trucksWithRend.map((t, idx) => {
214 |               const isTop = idx === 0 && trucksWithRend.length > 1;
215 |               const isBottom = idx === trucksWithRend.length - 1 && trucksWithRend.length > 1;
216 |               const vsOptPct = analysis.optimo > 0 ? ((t.rendPromedio - analysis.optimo) / analysis.optimo) * 100 : 0;
217 |               return (
218 |                 <div
219 |                   key={t.fleetNum}
220 |                   className={`grid grid-cols-[55px_70px_1fr_95px_70px_80px_80px_55px] gap-0 px-3 py-2 border-b border-bo
221 |                     isTop ? "bg-emerald-500/5" : isBottom ? "bg-red-500/5" : ""
222 |                   }`}
223 |                   data-testid={`row-perf-${t.fleetNum}`}
224 |                 >
225 |                   <span className="text-xs font-mono font-bold text-foreground">{t.fleetNum}</span>
226 |                   <span className="text-[10px] font-mono text-muted-foreground">{t.patenteReal || "\u2014"}</span>
227 |                   <span className="text-[10px] font-mono text-foreground truncate">{t.conductorSigetra || "\u2014"}</spa
228 |                   <div className="flex items-center gap-2">
229 |                     <span className={`text-xs font-mono font-bold ${
230 |                       t.rendPromedio >= 1.8 ? "text-emerald-400" : t.rendPromedio >= 1.2 ? "text-amber-400" : "text-red-
231 |                     }`}>
232 |                       {t.rendPromedio.toFixed(2)}
233 |                     </span>
234 |                     <div className="flex-1">
235 |                       <RendBar value={t.rendPromedio} max={globalMax} color={
236 |                         t.rendPromedio >= 1.8 ? "bg-emerald-500" : t.rendPromedio >= 1.2 ? "bg-amber-500" : "bg-red-500"
237 |                       } />
238 |                     </div>
239 |                   </div>
240 |                   <span className={`text-[10px] font-mono font-bold ${vsOptPct >= 0 ? "text-emerald-400" : vsOptPct >= -
241 |                     {vsOptPct >= 0 ? "+" : ""}{vsOptPct.toFixed(0)}%
242 |                   </span>
243 |                   <span className="text-[10px] font-mono text-amber-400">{fN(Math.round(t.totalLitrosSigetra))} L</span>
244 |                   <span className="text-[10px] font-mono text-blue-400">{t.litrosVolvoPeriodo != null ? `${fN(Math.round
245 |                   <span className="text-[10px] font-mono text-muted-foreground">{t.totalCargas}</span>
246 |                 </div>
247 |               );
248 |             })}
249 |           </div>
250 |         )}
251 |
252 |         {expanded && trucksWithRend.length === 0 && (
253 |           <div className="mt-4 p-4 text-center text-[11px] font-mono text-muted-foreground border border-border">
254 |             Sin datos de rendimiento para los camiones de esta faena
255 |           </div>
256 |         )}
257 |       </CardContent>
258 |     </Card>
259 |   );
260 | }
261 |
262 | export default function PerformanceIA() {
263 |   const [subTab, setSubTab] = useState<"rendimiento" | "rutas" | "combustible" | "analizador">("rendimiento");
264 |   const [days, setDays] = useState(7);
265 |   const [expandedFaena, setExpandedFaena] = useState<string | null>(null);
266 |   const [sortBy, setSortBy] = useState<"rend" | "litros" | "trucks">("rend");
267 |
268 |   const range = useMemo(() => getDateRange(days), [days]);
269 |
270 |   const { data: fusion, isLoading, isError } = useQuery<FusionResponse>({
271 |     queryKey: [`/api/sigetra/fusion?from=${range.from}&to=${range.to}`],
272 |     refetchInterval: 600000,
273 |   });
274 |
275 |   const errorNums = useMemo(() => {
276 |     if (!fusion?.trucks) return new Set<string>();
277 |     return getErrorFleetNums(fusion.trucks);
278 |   }, [fusion]);
279 |
280 |   const cleanTrucks = useMemo(() => {
281 |     if (!fusion?.trucks) return [];
282 |     return fusion.trucks
283 |       .filter(t => !errorNums.has(t.fleetNum))
284 |       .map(t => {
285 |         const validCargas = t.cargas.filter(c => !isErrorCarga(c));
286 |         const totalLitros = validCargas.reduce((s, c) => s + c.litros, 0);
287 |         const totalKm = validCargas.reduce((s, c) => s + (c.kmRecorrido != null && c.kmRecorrido > 0 ? c.kmRecorrido : 0
288 |         const rendPromedio = totalLitros > 0 ? totalKm / totalLitros : 0;
289 |         return { ...t, cargas: validCargas, totalLitrosSigetra: totalLitros, totalCargas: validCargas.length, rendPromed
290 |       });
291 |   }, [fusion, errorNums]);
292 |
293 |   const faenaAnalysis = useMemo(() => {
294 |     if (cleanTrucks.length === 0) return [];
295 |
296 |     const byFaena = new Map<string, FusionTruck[]>();
297 |     for (const t of cleanTrucks) {
298 |       const faena = t.faenaSigetra || "Sin Faena";
299 |       const existing = byFaena.get(faena) || [];
300 |       existing.push(t);
301 |       byFaena.set(faena, existing);
302 |     }
303 |
304 |     const analyses: FaenaAnalysis[] = [];
305 |     const entries = Array.from(byFaena.entries());
306 |     for (const [faena, trucks] of entries) {
307 |       const trucksWithRend = trucks.filter((t: FusionTruck) => t.rendPromedio > 0);
308 |       const avgRendimiento = trucksWithRend.length > 0
309 |         ? trucksWithRend.reduce((s: number, t: FusionTruck) => s + t.rendPromedio, 0) / trucksWithRend.length
310 |         : 0;
311 |       const totalLitros = trucks.reduce((s: number, t: FusionTruck) => s + t.totalLitrosSigetra, 0);
312 |       const totalLitrosVolvo = trucks.reduce((s: number, t: FusionTruck) => s + (t.totalFuelUsedVolvo || 0), 0);
313 |       const totalLitrosVolvoPeriodo = trucks.reduce((s: number, t: FusionTruck) => s + (t.litrosVolvoPeriodo || 0), 0);
314 |       const totalCargas = trucks.reduce((s: number, t: FusionTruck) => s + t.totalCargas, 0);
315 |
316 |       const sorted = [...trucksWithRend].sort((a, b) => b.rendPromedio - a.rendPromedio);
317 |       const bestTruck = sorted[0] || null;
318 |       const worstTruck = sorted.length > 1 ? sorted[sorted.length - 1] : null;
```

```
319 |
320 |       const rendValues = trucksWithRend.map((t: FusionTruck) => t.rendPromedio);
321 |       const rendRange = rendValues.length > 0
322 |         ? { min: Math.min(...rendValues), max: Math.max(...rendValues) }
323 |         : { min: 0, max: 0 };
324 |
325 |       const optimo = rendValues.length >= 3 ? calcPercentile(rendValues, 75) : (rendValues.length > 0 ? Math.max(...rend
326 |
327 |       analyses.push({
328 |         faena,
329 |         trucks,
330 |         truckCount: trucks.length,
331 |         avgRendimiento,
332 |         optimo,
333 |         totalLitros,
334 |         totalLitrosVolvo,
335 |         totalLitrosVolvoPeriodo,
336 |         totalCargas,
337 |         bestTruck,
338 |         worstTruck,
339 |         rendRange,
340 |       });
341 |     }
342 |
343 |     if (sortBy === "rend") analyses.sort((a, b) => b.avgRendimiento - a.avgRendimiento);
344 |     else if (sortBy === "litros") analyses.sort((a, b) => b.totalLitros - a.totalLitros);
345 |     else analyses.sort((a, b) => b.truckCount - a.truckCount);
346 |
347 |     return analyses;
348 |   }, [cleanTrucks, sortBy]);
349 |
350 |   const globalMax = useMemo(() => {
351 |     return faenaAnalysis.reduce((max, f) => {
352 |       const fMax = f.trucks.reduce((m, t) => Math.max(m, t.rendPromedio), 0);
353 |       return Math.max(max, fMax);
354 |     }, 0);
355 |   }, [faenaAnalysis]);
356 |
357 |   const globalStats = useMemo(() => {
358 |     if (cleanTrucks.length === 0) return { avgRend: 0, totalFaenas: 0, totalTrucks: 0, totalLitrosSigetra: 0, totalLitro
359 |     const withRend = faenaAnalysis.filter(f => f.avgRendimiento > 0);
360 |     const sortedByRend = [...withRend].sort((a, b) => b.avgRendimiento - a.avgRendimiento);
361 |     const totalLitrosSigetra = faenaAnalysis.reduce((s, f) => s + f.totalLitros, 0);
362 |     const totalLitrosVolvo = faenaAnalysis.reduce((s, f) => s + f.totalLitrosVolvo, 0);
363 |     const totalLitrosVolvoPeriodo = faenaAnalysis.reduce((s, f) => s + f.totalLitrosVolvoPeriodo, 0);
364 |     return {
365 |       avgRend: withRend.length > 0 ? withRend.reduce((s, f) => s + f.avgRendimiento, 0) / withRend.length : 0,
366 |       totalFaenas: faenaAnalysis.length,
367 |       totalTrucks: cleanTrucks.length,
368 |       totalLitrosSigetra,
369 |       totalLitrosVolvo,
370 |       totalLitrosVolvoPeriodo,
371 |       bestFaena: sortedByRend[0]?.faena || "",
372 |       worstFaena: sortedByRend.length > 1 ? sortedByRend[sortedByRend.length - 1]?.faena || "" : "",
373 |     };
374 |   }, [cleanTrucks, faenaAnalysis]);
375 |
376 |   if (isLoading) {
377 |     return (
378 |       <div className="space-y-4" data-testid="performance-loading">
379 |         <Skeleton className="h-20 w-full" />
380 |         <div className="grid grid-cols-4 gap-3">
381 |           {Array.from({ length: 4 }).map((_, i) => <Skeleton key={i} className="h-24" />)}
382 |         </div>
383 |         <Skeleton className="h-96 w-full" />
384 |       </div>
385 |     );
386 |   }
387 |
388 |   const subTabs: { key: typeof subTab; label: string; icon: typeof Cpu; activeClass: string }[] = [
389 |     { key: "rendimiento", label: "Rendimiento", icon: Cpu, activeClass: "border-primary text-primary" },
390 |     { key: "combustible", label: "Combustible", icon: Fuel, activeClass: "border-amber-400 text-amber-400" },
391 |     { key: "rutas", label: "Rutas Contrato", icon: Route, activeClass: "border-cyan-400 text-cyan-400" },
392 |     { key: "analizador", label: "Analizador Rutas", icon: Navigation, activeClass: "border-violet-400 text-violet-400" }
393 |   ];
394 |
395 |   const renderTabBar = () => (
396 |     <>
397 |       <div className="flex items-center gap-3 flex-wrap">
398 |         <div className="flex items-center gap-2">
399 |           <Cpu className="w-5 h-5 text-primary" />
400 |           <h1 className="text-sm font-mono font-bold tracking-[0.15em] uppercase text-foreground" data-testid="text-perf
401 |             Sotra IA
402 |           </h1>
403 |         </div>
404 |       </div>
405 |       <div className="flex items-center gap-1 border-b border-border overflow-x-auto mb-1">
406 |         {subTabs.map(tab => (
407 |           <button key={tab.key} onClick={() => setSubTab(tab.key)} data-testid={`subtab-${tab.key}`}
408 |             className={`px-4 py-2 text-[10px] font-mono font-bold tracking-[0.15em] uppercase border-b-2 cursor-pointer
409 |               subTab === tab.key ? tab.activeClass : "border-transparent text-muted-foreground hover:text-foreground"
410 |             }`}>
411 |             <tab.icon className="w-3.5 h-3.5 inline mr-1.5 -mt-0.5" />
412 |             {tab.label}
413 |           </button>
414 |         ))}
415 |       </div>
416 |     </>
417 |   );
418 |
419 |   if (subTab === "rutas") {
420 |     return (
421 |       <div className="space-y-4" data-testid="performance-page">
422 |         {renderTabBar()}
423 |         <RutasContrato />
424 |       </div>
425 |     );
```

```
426 |   }
427 |
428 |   if (subTab === "combustible") {
429 |     return (
430 |       <div className="space-y-4" data-testid="performance-page">
431 |         {renderTabBar()}
432 |         <CombustibleIA />
433 |       </div>
434 |     );
435 |   }
436 |
437 |   if (subTab === "analizador") {
438 |     return (
439 |       <div className="space-y-4" data-testid="performance-page">
440 |         {renderTabBar()}
441 |         <AnalizadorRutas />
442 |       </div>
443 |     );
444 |   }
445 |
446 |   return (
447 |     <div className="space-y-4" data-testid="performance-page">
448 |       {renderTabBar()}
449 |
450 |       <div className="flex items-center gap-2 flex-wrap">
451 |         <span className="text-[10px] font-mono text-muted-foreground tracking-[0.15em]">PERIODO:</span>
452 |         {[3, 7, 14, 30, 60].map(d => (
453 |           <button key={d} onClick={() => setDays(d)}
454 |             data-testid={`btn-perf-days-${d}`}
455 |             className={`px-3 py-1 text-[10px] font-mono font-bold border cursor-pointer transition-colors ${
456 |               days === d
457 |                 ? "bg-primary/20 border-primary text-primary"
458 |                 : "bg-card border-border text-muted-foreground hover:text-foreground"
459 |             }`}>
460 |             {d}D
461 |           </button>
462 |         ))}
463 |         <div className="flex-1" />
464 |         <span className="text-[10px] font-mono text-muted-foreground tracking-[0.15em]">ORDENAR:</span>
465 |         {([["rend", "RENDIMIENTO"], ["litros", "LITROS"], ["trucks", "CAMIONES"]] as const).map(([key, label]) => (
466 |           <button key={key} onClick={() => setSortBy(key)}
467 |             data-testid={`btn-sort-${key}`}
468 |             className={`px-3 py-1 text-[10px] font-mono font-bold border cursor-pointer transition-colors ${
469 |               sortBy === key
470 |                 ? "bg-primary/20 border-primary text-primary"
471 |                 : "bg-card border-border text-muted-foreground hover:text-foreground"
472 |             }`}>
473 |             {label}
474 |           </button>
475 |         ))}
476 |       </div>
477 |
478 |       {isError && (
479 |         <div className="bg-red-500/10 border border-red-500/30 p-3 flex items-center gap-2" data-testid="performance-err
480 |           <AlertTriangle className="w-4 h-4 text-red-400" />
481 |           <span className="text-[11px] font-mono text-red-400">Error al obtener datos de fusion. Intente nuevamente.</sp
482 |         </div>
483 |       )}
484 |
485 |       {fusion && (
486 |         <>
487 |           <div className="grid grid-cols-3 sm:grid-cols-5 gap-3">
488 |             <div className="bg-card border border-border p-3.5">
489 |               <div className="flex items-center gap-2 mb-1.5">
490 |                 <MapPin className="w-4 h-4 text-primary" />
491 |                 <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">FAENAS</span>
492 |               </div>
493 |               <div className="text-2xl font-mono font-bold text-primary" data-testid="text-perf-faenas">{globalStats.tot
494 |               <div className="text-[10px] font-mono text-muted-foreground">con datos</div>
495 |             </div>
496 |             <div className="bg-card border border-border p-3.5">
497 |               <div className="flex items-center gap-2 mb-1.5">
498 |                 <Truck className="w-4 h-4 text-blue-400" />
499 |                 <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">CAMIONES</span>
500 |               </div>
501 |               <div className="text-2xl font-mono font-bold text-blue-400" data-testid="text-perf-trucks">{globalStats.to
502 |               <div className="text-[10px] font-mono text-muted-foreground">en periodo</div>
503 |             </div>
504 |             <div className="bg-card border border-border p-3.5">
505 |               <div className="flex items-center gap-2 mb-1.5">
506 |                 <BarChart3 className="w-4 h-4 text-emerald-400" />
507 |                 <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">REND. GLOBAL</span>
508 |               </div>
509 |               <div className="text-2xl font-mono font-bold text-emerald-400" data-testid="text-perf-global-rend">
510 |                 {globalStats.avgRend > 0 ? `${globalStats.avgRend.toFixed(2)}` : "\u2014"}
511 |               </div>
512 |               <div className="text-[10px] font-mono text-muted-foreground">km/L promedio</div>
513 |             </div>
514 |             <div className="bg-card border border-border p-3.5">
515 |               <div className="flex items-center gap-2 mb-1.5">
516 |                 <Fuel className="w-4 h-4 text-amber-400" />
517 |                 <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">L. SIGETRA</span>
518 |               </div>
519 |               <div className="text-2xl font-mono font-bold text-amber-400" data-testid="text-perf-litros-sigetra">{fN(Ma
520 |               <div className="text-[10px] font-mono text-muted-foreground">litros tarjeta (periodo)</div>
521 |             </div>
522 |             <div className="bg-card border border-border p-3.5">
523 |               <div className="flex items-center gap-2 mb-1.5">
524 |                 <Fuel className="w-4 h-4 text-blue-400" />
525 |                 <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">L. VOLVO ECU</span>
526 |               </div>
527 |               <div className="text-2xl font-mono font-bold text-blue-400" data-testid="text-perf-litros-volvo">
528 |                 {globalStats.totalLitrosVolvoPeriodo > 0 ? fN(Math.round(globalStats.totalLitrosVolvoPeriodo)) : "\u2014
529 |               </div>
530 |               <div className="text-[10px] font-mono text-muted-foreground">
531 |                 {globalStats.totalLitrosVolvoPeriodo > 0 ? "litros ECU (periodo)" : "sin historial ECU aun"}
532 |               </div>
```

```
533 |             </div>
534 |           </div>
535 |
536 |           {globalStats.bestFaena && (
537 |             <div className="grid grid-cols-2 gap-3">
538 |               <div className="bg-emerald-500/5 border border-emerald-500/20 p-3 flex items-center gap-3">
539 |                 <TrendingUp className="w-5 h-5 text-emerald-400 flex-shrink-0" />
540 |                 <div>
541 |                   <div className="text-[8px] font-mono text-emerald-400 tracking-[0.15em]">MEJOR FAENA</div>
542 |                   <div className="text-sm font-mono font-bold text-foreground" data-testid="text-best-faena">{globalStat
543 |                 </div>
544 |               </div>
545 |               {globalStats.worstFaena && (
546 |                 <div className="bg-red-500/5 border border-red-500/20 p-3 flex items-center gap-3">
547 |                   <TrendingDown className="w-5 h-5 text-red-400 flex-shrink-0" />
548 |                   <div>
549 |                     <div className="text-[8px] font-mono text-red-400 tracking-[0.15em]">FAENA CON MENOR REND.</div>
550 |                     <div className="text-sm font-mono font-bold text-foreground" data-testid="text-worst-faena">{globalS
551 |                   </div>
552 |                 </div>
553 |               )}
554 |             </div>
555 |           )}
556 |
557 |           <div className="bg-blue-500/5 border border-blue-500/20 p-3 flex items-start gap-2" data-testid="note-rfms">
558 |             <Info className="w-4 h-4 text-blue-400 flex-shrink-0 mt-0.5" />
559 |             <div className="text-[11px] font-mono text-blue-400/80">
560 |               <span className="font-bold text-blue-400">Nota:</span> Los litros Volvo ECU se calculan como delta del tot
561 |               Si un camion muestra "\u2014", es porque aun no hay historial ECU suficiente (se requieren al menos 2 lect
562 |               Algunos camiones no reportan datos Volvo porque su suscripcion rFMS esta inactiva o el hardware no envia t
563 |             </div>
564 |           </div>
565 |
566 |           <div className="bg-cyan-500/5 border border-cyan-500/20 p-3 flex items-start gap-2" data-testid="note-optimo">
567 |             <Target className="w-4 h-4 text-cyan-400 flex-shrink-0 mt-0.5" />
568 |             <div className="text-[11px] font-mono text-cyan-400/80">
569 |               <span className="font-bold text-cyan-400">Optimo por ruta:</span> Se calcula como el percentil 75 (P75) de
570 |               Camiones sobre el optimo son referencia de eficiencia. Camiones bajo -15% del optimo requieren atencion.
571 |               La columna "vs OPT" muestra la desviacion porcentual de cada camion respecto al optimo de su faena.
572 |             </div>
573 |           </div>
574 |
575 |           <div className="space-y-3">
576 |             {faenaAnalysis.map(analysis => (
577 |               <FaenaCard
578 |                 key={analysis.faena}
579 |                 analysis={analysis}
580 |                 globalMax={globalMax}
581 |                 expanded={expandedFaena === analysis.faena}
582 |                 onToggle={() => setExpandedFaena(expandedFaena === analysis.faena ? null : analysis.faena)}
583 |               />
584 |             ))}
585 |           </div>
586 |
587 |           {faenaAnalysis.length === 0 && (
588 |             <div className="p-8 text-center text-[11px] font-mono text-muted-foreground border border-border bg-card" da
589 |               No se encontraron datos de rendimiento por faena en el periodo seleccionado
590 |             </div>
591 |           )}
592 |         </>
593 |       )}
594 |     </div>
595 |   );
596 | }
597 |
```

client/src/pages/analizador-rutas.tsx

587 lineas

```
1 | import { useState, useMemo } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { Skeleton } from "@/components/ui/skeleton";
4 | import { Card, CardContent, CardHeader, CardTitle } from "@/components/ui/card";
5 | import {
6 |   Navigation, TrendingUp, TrendingDown, Fuel, Truck, ChevronDown, ChevronUp,
7 |   BarChart3, AlertTriangle, Info, MapPin, Target
8 | } from "lucide-react";
9 | import { getErrorFleetNums, isErrorCarga } from "@/pages/errores";
10 |
11 | interface FusionCarga {
12 |   numGuia: number;
13 |   fecha: string;
14 |   litros: number;
15 |   odometro: number | null;
16 |   kmRecorrido: number | null;
17 |   rendimiento: number | null;
18 |   lugar: string | null;
19 |   tarjeta: string | null;
20 |   conductor: string | null;
21 | }
22 |
23 | interface FusionTruck {
24 |   fleetNum: string;
25 |   patenteReal: string | null;
26 |   vin: string | null;
27 |   modeloVolvo: string;
28 |   faenaSigetra: string | null;
29 |   conductorSigetra: string | null;
30 |   totalLitrosSigetra: number;
31 |   totalCargas: number;
32 |   rendPromedio: number;
33 |   odometroSigetra: number | null;
34 |   odometroVolvo: number | null;
35 |   deltaOdometro: number | null;
36 |   fuelLevelVolvo: number | null;
37 |   totalFuelUsedVolvo: number | null;
38 |   litrosVolvoPeriodo: number | null;
39 |   engineHoursVolvo: number | null;
40 |   gpsVolvo: { latitude: number | null; longitude: number | null } | null;
41 |   alertLevel: "ok" | "alerta" | "critico";
42 |   cargas: FusionCarga[];
43 | }
44 |
45 | interface FusionResponse {
46 |   from: string;
47 |   to: string;
48 |   totalTrucksMatched: number;
49 |   totalLitros: number;
50 |   totalCargas: number;
51 |   alertCount: number;
52 |   trucks: FusionTruck[];
53 | }
54 |
55 | interface RouteDefinition {
56 |   id: string;
57 |   name: string;
58 |   keywords: string[][];
59 |   distanciaRef: number;
60 | }
61 |
62 | const KNOWN_ROUTES: RouteDefinition[] = [
63 |   {
64 |     id: "stgo-concepcion",
65 |     name: "Santiago - Concepcion",
66 |     keywords: [
67 |       ["quilicura", "santiago", "pudahuel", "evc vespucio", "evc colorado"],
68 |       ["concepcion", "ruta 5 sur km 484", "ruta 5 sur km 518", "longitudinal sur km 275", "carretera panamericana sur km
69 |     ],
70 |     distanciaRef: 500,
71 |   },
72 |   {
73 |     id: "stgo-losangeles",
74 |     name: "Santiago - Los Angeles",
75 |     keywords: [
76 |       ["quilicura", "santiago", "pudahuel", "evc vespucio", "evc colorado"],
77 |       ["los angeles", "longitudinal sur 5 km. 409", "ruta 5 km 409"],
78 |     ],
79 |     distanciaRef: 520,
80 |   },
81 |   {
82 |     id: "stgo-coquimbo",
83 |     name: "Santiago - Coquimbo / La Serena",
84 |     keywords: [
85 |       ["quilicura", "santiago", "pudahuel", "evc vespucio", "evc colorado"],
86 |       ["coquimbo", "la serena", "ruta 5 norte km 838", "panamericana norte km 455", "carretera panamericana norte km 455
87 |     ],
88 |     distanciaRef: 470,
89 |   },
90 |   {
91 |     id: "stgo-antofagasta",
92 |     name: "Santiago - Antofagasta",
93 |     keywords: [
94 |       ["quilicura", "santiago", "pudahuel", "evc vespucio", "evc colorado"],
95 |       ["antofagasta", "ruta 5 norte km 1392", "ruta 5 norte km 1398", "ruta 5 norte km 1351", "panamericana norte km 135
96 |     ],
97 |     distanciaRef: 1350,
98 |   },
99 |   {
100 |     id: "stgo-temuco",
101 |     name: "Santiago - Temuco / Araucania",
102 |     keywords: [
103 |       ["quilicura", "santiago", "pudahuel", "evc vespucio", "evc colorado"],
104 |       ["temuco", "ruta 5 sur km 904", "ruta 5 sur km 906", "trafun", "ruta 5 km 786"],
```

```
105 |     ],
106 |     distanciaRef: 680,
107 |   },
108 |   {
109 |     id: "losangeles-temuco",
110 |     name: "Los Angeles - Temuco",
111 |     keywords: [
112 |       ["los angeles"],
113 |       ["temuco", "ruta 5 sur km 904", "ruta 5 sur km 906", "trafun"],
114 |     ],
115 |     distanciaRef: 300,
116 |   },
117 | ];
118 |
119 | function matchesZone(lugar: string, keywords: string[]): boolean {
120 |   const lug = lugar.toLowerCase().trim();
121 |   return keywords.some(kw => lug.includes(kw));
122 | }
123 |
124 | interface RouteTrip {
125 |   fleetNum: string;
126 |   patente: string | null;
127 |   conductor: string | null;
128 |   fecha: string;
129 |   lugarOrigen: string;
130 |   lugarDestino: string;
131 |   litros: number;
132 |   km: number;
133 |   rendimiento: number;
134 | }
135 |
136 | interface RouteAnalysis {
137 |   route: RouteDefinition;
138 |   trips: RouteTrip[];
139 |   avgRendimiento: number;
140 |   p75Rendimiento: number;
141 |   totalLitros: number;
142 |   totalKm: number;
143 |   bestTrip: RouteTrip | null;
144 |   worstTrip: RouteTrip | null;
145 |   truckCount: number;
146 | }
147 |
148 | function calcPercentile(values: number[], pct: number): number {
149 |   if (values.length === 0) return 0;
150 |   const sorted = [...values].sort((a, b) => a - b);
151 |   const idx = (pct / 100) * (sorted.length - 1);
152 |   const lower = Math.floor(idx);
153 |   const upper = Math.ceil(idx);
154 |   if (lower === upper) return sorted[lower];
155 |   return sorted[lower] + (sorted[upper] - sorted[lower]) * (idx - lower);
156 | }
157 |
158 | function fN(n: number): string {
159 |   return n.toLocaleString("es-CL");
160 | }
161 |
162 | const DATA_START_DATE = "2026-03-01";
163 |
164 | function getDateRange(days: number): { from: string; to: string } {
165 |   const to = new Date();
166 |   const from = new Date(Date.now() - days * 24 * 60 * 60 * 1000);
167 |   const minDate = new Date(DATA_START_DATE + "T00:00:00");
168 |   const effectiveFrom = from < minDate ? minDate : from;
169 |   return { from: effectiveFrom.toISOString().slice(0, 10), to: to.toISOString().slice(0, 10) };
170 | }
171 |
172 | function RendBar({ value, max, color }: { value: number; max: number; color: string }) {
173 |   const pct = max > 0 ? Math.min((value / max) * 100, 100) : 0;
174 |   return (
175 |     <div className="h-2 w-full bg-muted rounded-md overflow-hidden">
176 |       <div className={`h-full rounded-md ${color}`} style={{ width: `${pct}%` }} />
177 |     </div>
178 |   );
179 | }
180 |
181 | export default function AnalizadorRutas() {
182 |   const [days, setDays] = useState(30);
183 |   const [expandedRoute, setExpandedRoute] = useState<string | null>(null);
184 |
185 |   const range = useMemo(() => getDateRange(days), [days]);
186 |
187 |   const { data: fusion, isLoading, isError } = useQuery<FusionResponse>({
188 |     queryKey: [`/api/sigetra/fusion?from=${range.from}&to=${range.to}`],
189 |     refetchInterval: 600000,
190 |   });
191 |
192 |   const errorNums = useMemo(() => {
193 |     if (!fusion?.trucks) return new Set<string>();
194 |     return getErrorFleetNums(fusion.trucks);
195 |   }, [fusion]);
196 |
197 |   const routeAnalyses = useMemo(() => {
198 |     if (!fusion?.trucks) return [];
199 |
200 |     const cleanTrucks = fusion.trucks
201 |       .filter(t => !errorNums.has(t.fleetNum))
202 |       .map(t => ({
203 |         ...t,
204 |         cargas: t.cargas.filter(c => !isErrorCarga(c) && c.litros > 0),
205 |       }));
206 |
207 |     const analyses: RouteAnalysis[] = [];
208 |
209 |     KNOWN_ROUTES.forEach(route => {
210 |       const trips: RouteTrip[] = [];
211 |
```

```
212 |       cleanTrucks.forEach(truck => {
213 |         const sortedCargas = [...truck.cargas].sort((a, b) => (a.fecha || "").localeCompare(b.fecha || ""));
214 |
215 |         for (let i = 0; i < sortedCargas.length; i++) {
216 |           const carga = sortedCargas[i];
217 |           if (!carga.lugar) continue;
218 |
219 |           const isInOrigin = matchesZone(carga.lugar, route.keywords[0]);
220 |           const isInDest = matchesZone(carga.lugar, route.keywords[1]);
221 |
222 |           if (!isInDest && !isInOrigin) continue;
223 |
224 |           const minKm = route.distanciaRef * 0.3;
225 |           const maxKm = route.distanciaRef * 3;
226 |
227 |           if (isInDest && i > 0) {
228 |             const prev = sortedCargas[i - 1];
229 |             if (prev.lugar && matchesZone(prev.lugar, route.keywords[0])) {
230 |               const km = carga.kmRecorrido != null && carga.kmRecorrido > 0 ? carga.kmRecorrido : 0;
231 |               const litros = carga.litros;
232 |               if (litros > 0 && km >= minKm && km <= maxKm) {
233 |                 const rend = km / litros;
234 |                 if (rend > 0.5 && rend < 10) {
235 |                   trips.push({
236 |                     fleetNum: truck.fleetNum,
237 |                     patente: truck.patenteReal,
238 |                     conductor: carga.conductor || truck.conductorSigetra,
239 |                     fecha: carga.fecha,
240 |                     lugarOrigen: prev.lugar,
241 |                     lugarDestino: carga.lugar,
242 |                     litros,
243 |                     km,
244 |                     rendimiento: rend,
245 |                   });
246 |                 }
247 |               }
248 |             }
249 |           }
250 |
251 |           if (isInOrigin && i > 0) {
252 |             const prev = sortedCargas[i - 1];
253 |             if (prev.lugar && matchesZone(prev.lugar, route.keywords[1])) {
254 |               const km = carga.kmRecorrido != null && carga.kmRecorrido > 0 ? carga.kmRecorrido : 0;
255 |               const litros = carga.litros;
256 |               if (litros > 0 && km >= minKm && km <= maxKm) {
257 |                 const rend = km / litros;
258 |                 if (rend > 0.5 && rend < 10) {
259 |                   trips.push({
260 |                     fleetNum: truck.fleetNum,
261 |                     patente: truck.patenteReal,
262 |                     conductor: carga.conductor || truck.conductorSigetra,
263 |                     fecha: carga.fecha,
264 |                     lugarOrigen: prev.lugar,
265 |                     lugarDestino: carga.lugar,
266 |                     litros,
267 |                     km,
268 |                     rendimiento: rend,
269 |                   });
270 |                 }
271 |               }
272 |             }
273 |           }
274 |         }
275 |       });
276 |
277 |       if (trips.length === 0) return;
278 |
279 |       const rendValues = trips.map(t => t.rendimiento);
280 |       const avgRendimiento = rendValues.length > 0 ? rendValues.reduce((s, v) => s + v, 0) / rendValues.length : 0;
281 |       const p75Rendimiento = calcPercentile(rendValues, 75);
282 |       const totalLitros = trips.reduce((s, t) => s + t.litros, 0);
283 |       const totalKm = trips.reduce((s, t) => s + t.km, 0);
284 |       const uniqueTrucks = new Set(trips.map(t => t.fleetNum));
285 |
286 |       const sortedByRend = [...trips].sort((a, b) => b.rendimiento - a.rendimiento);
287 |       const bestTrip = sortedByRend[0] || null;
288 |       const worstTrip = sortedByRend.length > 1 ? sortedByRend[sortedByRend.length - 1] : null;
289 |
290 |       analyses.push({
291 |         route,
292 |         trips,
293 |         avgRendimiento,
294 |         p75Rendimiento,
295 |         totalLitros,
296 |         totalKm,
297 |         bestTrip,
298 |         worstTrip,
299 |         truckCount: uniqueTrucks.size,
300 |       });
301 |     });
302 |
303 |     analyses.sort((a, b) => b.trips.length - a.trips.length);
304 |     return analyses;
305 |   }, [fusion, errorNums]);
306 |
307 |   const globalMax = useMemo(() => {
308 |     return routeAnalyses.reduce((max, r) => Math.max(max, r.avgRendimiento, r.p75Rendimiento), 0);
309 |   }, [routeAnalyses]);
310 |
311 |   const totalTrips = routeAnalyses.reduce((s, r) => s + r.trips.length, 0);
312 |
313 |   if (isLoading) {
314 |     return (
315 |       <div className="space-y-4" data-testid="rutas-analizador-loading">
316 |         <Skeleton className="h-20 w-full" />
317 |         <div className="grid grid-cols-3 gap-3">
318 |           {Array.from({ length: 3 }).map((_, i) => <Skeleton key={i} className="h-24" />)}
```

```
319 |         </div>
320 |         <Skeleton className="h-96 w-full" />
321 |       </div>
322 |     );
323 |   }
324 |
325 |   return (
326 |     <div className="space-y-4" data-testid="analizador-rutas-page">
327 |       <div className="flex items-center gap-3 flex-wrap">
328 |         <div className="flex items-center gap-2">
329 |           <Navigation className="w-5 h-5 text-violet-400" />
330 |           <h1 className="text-sm font-mono font-bold tracking-[0.15em] uppercase text-foreground" data-testid="text-anal
331 |             Analizador de Rutas
332 |           </h1>
333 |         </div>
334 |         <span className="text-[10px] font-mono text-muted-foreground">Rendimiento ideal por ruta de carretera</span>
335 |       </div>
336 |
337 |       <div className="flex items-center gap-2 flex-wrap">
338 |         <span className="text-[10px] font-mono text-muted-foreground tracking-[0.15em]">PERIODO:</span>
339 |         {[14, 30, 60].map(d => (
340 |           <button key={d} onClick={() => setDays(d)}
341 |             data-testid={`btn-analizador-days-${d}`}
342 |             className={`px-3 py-1 text-[10px] font-mono font-bold border cursor-pointer transition-colors ${
343 |               days === d ? "bg-violet-500/20 border-violet-500 text-violet-400" : "bg-card border-border text-muted-fore
344 |             }`}>
345 |             {d}D
346 |           </button>
347 |         ))}
348 |       </div>
349 |
350 |       {isError && (
351 |         <div className="bg-red-500/10 border border-red-500/30 p-3 flex items-center gap-2" data-testid="analizador-erro
352 |           <AlertTriangle className="w-4 h-4 text-red-400" />
353 |           <span className="text-[11px] font-mono text-red-400">Error al obtener datos.</span>
354 |         </div>
355 |       )}
356 |
357 |       {fusion && (
358 |         <>
359 |           <div className="grid grid-cols-3 gap-3">
360 |             <div className="bg-card border border-border p-3.5">
361 |               <div className="flex items-center gap-2 mb-1.5">
362 |                 <Navigation className="w-4 h-4 text-violet-400" />
363 |                 <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">RUTAS DETECTADAS</span>
364 |               </div>
365 |               <div className="text-2xl font-mono font-bold text-violet-400" data-testid="text-rutas-detectadas">{routeAn
366 |               <div className="text-[10px] font-mono text-muted-foreground">de {KNOWN_ROUTES.length} configuradas</div>
367 |             </div>
368 |             <div className="bg-card border border-border p-3.5">
369 |               <div className="flex items-center gap-2 mb-1.5">
370 |                 <Truck className="w-4 h-4 text-blue-400" />
371 |                 <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">VIAJES TOTALES</span>
372 |               </div>
373 |               <div className="text-2xl font-mono font-bold text-blue-400" data-testid="text-viajes-total">{totalTrips}</
374 |               <div className="text-[10px] font-mono text-muted-foreground">tramos identificados</div>
375 |             </div>
376 |             <div className="bg-card border border-border p-3.5">
377 |               <div className="flex items-center gap-2 mb-1.5">
378 |                 <BarChart3 className="w-4 h-4 text-emerald-400" />
379 |                 <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">MEJOR RUTA</span>
380 |               </div>
381 |               {routeAnalyses.length > 0 ? (
382 |                 <>
383 |                   <div className="text-sm font-mono font-bold text-foreground truncate" data-testid="text-mejor-ruta">
384 |                     {[...routeAnalyses].sort((a, b) => b.avgRendimiento - a.avgRendimiento)[0].route.name}
385 |                   </div>
386 |                   <div className="text-lg font-mono font-bold text-emerald-400">
387 |                     {[...routeAnalyses].sort((a, b) => b.avgRendimiento - a.avgRendimiento)[0].avgRendimiento.toFixed(2)
388 |                   </div>
389 |                 </>
390 |               ) : (
391 |                 <div className="text-lg font-mono font-bold text-muted-foreground">{"\u2014"}</div>
392 |               )}
393 |             </div>
394 |           </div>
395 |
396 |           <div className="bg-violet-500/5 border border-violet-500/20 p-3 flex items-start gap-2" data-testid="note-anal
397 |             <Info className="w-4 h-4 text-violet-400 flex-shrink-0 mt-0.5" />
398 |             <div className="text-[11px] font-mono text-violet-400/80">
399 |               <span class="font-bold text-violet-400">Analizador de Rutas:</span> Identifica viajes entre ciudades a
400 |               Cuando un camion carga en una zona de origen (ej: Quilicura/Santiago) y su siguiente carga es en una zona
401 |               se detecta como un viaje en esa ruta. El rendimiento ideal (P75) muestra la eficiencia alcanzable.
402 |               Rutas configuradas: {KNOWN_ROUTES.map(r => r.name).join(", ")}.
403 |             </div>
404 |           </div>
405 |
406 |           <div className="space-y-3">
407 |             {routeAnalyses.map(analysis => {
408 |               const isExpanded = expandedRoute === analysis.route.id;
409 |               const sortedTrips = [...analysis.trips].sort((a, b) => b.rendimiento - a.rendimiento);
410 |
411 |               return (
412 |                 <Card key={analysis.route.id} className="overflow-visible" data-testid={`card-route-${analysis.route.id}
413 |                   <CardHeader
414 |                     className="flex flex-row items-center justify-between gap-2 space-y-0 pb-2 cursor-pointer"
415 |                     onClick={() => setExpandedRoute(isExpanded ? null : analysis.route.id)}
416 |                     data-testid={`btn-expand-route-${analysis.route.id}`}
417 |                   >
418 |                     <div className="flex items-center gap-3 flex-wrap min-w-0">
419 |                       <Navigation className="w-4 h-4 text-violet-400 flex-shrink-0" />
420 |                       <CardTitle className="text-sm font-mono font-bold tracking-[0.1em] uppercase truncate" data-testid
421 |                         {analysis.route.name}
422 |                       </CardTitle>
423 |                       <span className="text-[10px] font-mono text-muted-foreground">
424 |                         {analysis.trips.length} viajes · {analysis.truckCount} camiones · ~{fN(analysis.route.distanciaR
425 |                       </span>
```

```
                    </div>
                    <div className="flex items-center gap-4 flex-wrap flex-shrink-0">
                      <div className="text-right">
                        <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em]">REND. PROMEDIO</di
                        <div className={`text-lg font-mono font-bold ${
                          analysis.avgRendimiento >= 1.8 ? "text-emerald-400" : analysis.avgRendimiento >= 1.2 ? "text-a
                        }`} data-testid={`text-route-rend-${analysis.route.id}`}>
                          {analysis.avgRendimiento.toFixed(2)} km/L
                        </div>
                      </div>
                      {isExpanded ? <ChevronUp className="w-4 h-4 text-muted-foreground" /> : <ChevronDown className="w-
                    </div>
                  </CardHeader>
                  <CardContent className="pt-0">
                    <div className="grid grid-cols-5 gap-3 mb-3">
                      <div>
                        <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-0.5">OPTIMO (P75
                        <div className="text-sm font-mono font-bold text-cyan-400" data-testid={`text-route-p75-${analys
                          {analysis.p75Rendimiento.toFixed(2)} km/L
                        </div>
                      </div>
                      <div>
                        <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-0.5">L/100KM</di
                        <div className="text-sm font-mono font-bold text-foreground">
                          {analysis.avgRendimiento > 0 ? (100 / analysis.avgRendimiento).toFixed(1) : "\u2014"}
                        </div>
                      </div>
                      <div>
                        <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-0.5">LITROS TOTA
                        <div className="text-sm font-mono font-bold text-amber-400">{fN(Math.round(analysis.totalLitros)
                      </div>
                      <div>
                        <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-0.5">KM TOTAL</d
                        <div className="text-sm font-mono font-bold text-blue-400">{fN(Math.round(analysis.totalKm))}</d
                      </div>
                      <div>
                        <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-0.5">VIAJES</div
                        <div className="text-sm font-mono font-bold text-foreground">{analysis.trips.length}</div>
                      </div>
                    </div>

                    {analysis.bestTrip && analysis.worstTrip && (
                      <div className="grid grid-cols-2 gap-3 mb-3">
                        <div className="bg-emerald-500/5 border border-emerald-500/20 p-2.5">
                          <div className="flex items-center gap-1.5 mb-1">
                            <TrendingUp className="w-3 h-3 text-emerald-400" />
                            <span className="text-[8px] font-mono text-emerald-400 tracking-[0.15em]">MEJOR VIAJE</span>
                          </div>
                          <div class="text-xs font-mono font-bold text-foreground" data-testid={`text-best-trip-${an
                            N.{analysis.bestTrip.fleetNum} ({analysis.bestTrip.fecha?.slice(0, 10)})
                          </div>
                          <div class="text-sm font-mono font-bold text-emerald-400">
                            {analysis.bestTrip.rendimiento.toFixed(2)} km/L
                          </div>
                        </div>
                        <div className="bg-red-500/5 border border-red-500/20 p-2.5">
                          <div className="flex items-center gap-1.5 mb-1">
                            <TrendingDown className="w-3 h-3 text-red-400" />
                            <span className="text-[8px] font-mono text-red-400 tracking-[0.15em]">PEOR VIAJE</span>
                          </div>
                          <div class="text-xs font-mono font-bold text-foreground" data-testid={`text-worst-trip-${a
                            N.{analysis.worstTrip.fleetNum} ({analysis.worstTrip.fecha?.slice(0, 10)})
                          </div>
                          <div class="text-sm font-mono font-bold text-red-400">
                            {analysis.worstTrip.rendimiento.toFixed(2)} km/L
                          </div>
                        </div>
                      </div>
                    )}

                    <RendBar value={analysis.avgRendimiento} max={globalMax} color="bg-violet-500" />

                    {isExpanded && (
                      <div className="mt-4 border border-border bg-background">
                        <div className="grid grid-cols-[3.5rem_4rem_1fr_1fr_5rem_5rem_5rem_4.5rem] gap-1.5 bg-card px-3
                          <span class="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">N.INT
                          <span class="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">FECHA
                          <span class="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">ORIGE
                          <span class="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">DESTI
                          <span class="text-[8px] font-mono font-bold text-amber-400/70 tracking-[0.15em] text-right
                          <span class="text-[8px] font-mono font-bold text-blue-400/70 tracking-[0.15em] text-right"
                          <span class="text-[8px] font-mono font-bold text-emerald-400/70 tracking-[0.15em] text-rig
                          <span class="text-[8px] font-mono font-bold text-cyan-400/70 tracking-[0.15em] text-right"
                        </div>
                        <div class="divide-y divide-border/30 max-h-[400px] overflow-y-auto">
                          {sortedTrips.map((trip, idx) => {
                            const vsP75 = analysis.p75Rendimiento > 0 ? ((trip.rendimiento - analysis.p75Rendimiento) /
                            const isTop = idx === 0 && sortedTrips.length > 1;
                            const isBottom = idx === sortedTrips.length - 1 && sortedTrips.length > 1;
                            return (
                              <div
                                key={`${trip.fleetNum}-${trip.fecha}-${idx}`}
                                className={`grid grid-cols-[3.5rem_4rem_1fr_1fr_5rem_5rem_5rem_4.5rem] gap-1.5 px-3 py-2
                                  isTop ? "bg-emerald-500/5" : isBottom ? "bg-red-500/5" : ""
                                }`}
                                data-testid={`row-trip-${analysis.route.id}-${idx}`}
                              >
                                <span class="text-xs font-mono font-bold text-foreground">{trip.fleetNum}</span>
                                <span class="text-[9px] font-mono text-muted-foreground">{trip.fecha?.slice(5, 10)}<
                                <span class="text-[9px] font-mono text-foreground truncate">{trip.lugarOrigen?.subst
                                <span class="text-[9px] font-mono text-foreground truncate">{trip.lugarDestino?.subs
                                <span class="text-[10px] font-mono text-amber-400 text-right">{fN(Math.round(trip.li
                                <span class="text-[10px] font-mono text-blue-400 text-right">{fN(Math.round(trip.km)
                                <span class={`text-xs font-mono font-bold text-right ${
                                  trip.rendimiento >= 1.8 ? "text-emerald-400" : trip.rendimiento >= 1.2 ? "text-amber-4
                                }`}>
                                  {trip.rendimiento.toFixed(2)}
```

```
                                </span>
                                <span className={`text-[10px] font-mono font-bold text-right ${vsP75 >= 0 ? "text-emeral
                                  {vsP75 >= 0 ? "+" : ""}{vsP75.toFixed(0)}%
                                </span>
                              </div>
                            );
                          })}
                        </div>
                      </div>
                    )}
                  </CardContent>
                </Card>
              );
            })}
          </div>

          {routeAnalyses.length === 0 && (
            <div className="p-8 text-center border border-border bg-card" data-testid="analizador-empty">
              <div className="text-[11px] font-mono text-muted-foreground mb-2">
                No se detectaron viajes en las rutas configuradas para el periodo seleccionado.
              </div>
              <div className="text-[10px] font-mono text-muted-foreground/60">
                Las rutas se detectan cuando un camion carga combustible en una zona de origen y su siguiente carga es e
                Intente ampliar el periodo o verifique que existen cargas en las zonas configuradas.
              </div>
            </div>
          )}

          <div className="bg-background border border-border p-3" data-testid="rutas-config-info">
            <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] uppercase mb-2">Rutas configura
            <div class="grid grid-cols-2 sm:grid-cols-3 gap-2">
              {KNOWN_ROUTES.map(route => {
                const analysis = routeAnalyses.find(a => a.route.id === route.id);
                return (
                  <div key={route.id} className={`p-2 border ${analysis ? "border-violet-500/30 bg-violet-500/5" : "bord
                    <div className="text-[10px] font-mono font-bold text-foreground">{route.name}</div>
                    <div className="text-[9px] font-mono text-muted-foreground">~{fN(route.distanciaRef)} km</div>
                    {analysis ? (
                      <div className="text-[10px] font-mono text-violet-400 font-bold mt-0.5">
                        {analysis.trips.length} viajes · {analysis.avgRendimiento.toFixed(2)} km/L
                      </div>
                    ) : (
                      <div className="text-[9px] font-mono text-muted-foreground/50 mt-0.5">Sin datos</div>
                    )}
                  </div>
                );
              })}
            </div>
          </div>
        </>
      )}
    </div>
  );
}
```

client/src/pages/errores.tsx

560 lineas

```
1 | import { useState, useMemo } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { Input } from "@/components/ui/input";
4 | import { Button } from "@/components/ui/button";
5 | import { Skeleton } from "@/components/ui/skeleton";
6 | import {
7 |   AlertOctagon, Search, Bug, ChevronDown, ChevronUp, Gauge, Fuel,
8 |   MapPin, ArrowDownUp, Download
9 | } from "lucide-react";
10 |
11 | interface FusionCarga {
12 |   numGuia: number;
13 |   fecha: string;
14 |   litros: number;
15 |   odometro: number | null;
16 |   kmRecorrido: number | null;
17 |   rendimiento: number | null;
18 |   lugar: string | null;
19 |   tarjeta: string | null;
20 |   conductor: string | null;
21 | }
22 |
23 | interface FusionTruck {
24 |   fleetNum: string;
25 |   patenteReal: string | null;
26 |   vin: string | null;
27 |   modeloVolvo: string;
28 |   faenaSigetra: string | null;
29 |   conductorSigetra: string | null;
30 |   totalLitrosSigetra: number;
31 |   totalCargas: number;
32 |   rendPromedio: number;
33 |   odometroSigetra: number | null;
34 |   odometroVolvo: number | null;
35 |   deltaOdometro: number | null;
36 |   fuelLevelVolvo: number | null;
37 |   totalFuelUsedVolvo: number | null;
38 |   litrosVolvoPeriodo: number | null;
39 |   engineHoursVolvo: number | null;
40 |   gpsVolvo: { latitude: number | null; longitude: number | null } | null;
41 |   alertLevel: "ok" | "alerta" | "critico";
42 |   cargas: FusionCarga[];
43 | }
44 |
45 | interface FusionResponse {
46 |   from: string;
47 |   to: string;
48 |   totalTrucksMatched: number;
49 |   totalLitros: number;
50 |   totalCargas: number;
51 |   alertCount: number;
52 |   trucks: FusionTruck[];
53 | }
54 |
55 | type ErrorType = "REND_IMPOSIBLE" | "KM_NEGATIVO" | "ODO_SALTO" | "MICRO_CARGA" | "REND_NEGATIVO" | "KM_EXTREMO" | "CUAD
56 |
57 | interface ErrorRecord {
58 |   truck: FusionTruck;
59 |   carga: FusionCarga | null;
60 |   errorType: ErrorType;
61 |   errorLabel: string;
62 |   detail: string;
63 | }
64 |
65 | const ERROR_LABELS: Record<ErrorType, string> = {
66 |   REND_IMPOSIBLE: "Rendimiento imposible",
67 |   KM_NEGATIVO: "KM negativos",
68 |   ODO_SALTO: "Salto de odometro",
69 |   MICRO_CARGA: "Micro-carga sospechosa",
70 |   REND_NEGATIVO: "Rendimiento negativo",
71 |   KM_EXTREMO: "KM extremos",
72 |   CUADRATURA_BAJA: "Cuadratura baja",
73 | };
74 |
75 | const ERROR_COLORS: Record<ErrorType, string> = {
76 |   REND_IMPOSIBLE: "text-amber-400 bg-amber-500/10 border-amber-500/30",
77 |   KM_NEGATIVO: "text-red-400 bg-red-500/10 border-red-500/30",
78 |   ODO_SALTO: "text-red-400 bg-red-500/10 border-red-500/30",
79 |   MICRO_CARGA: "text-blue-400 bg-blue-500/10 border-blue-500/30",
80 |   REND_NEGATIVO: "text-red-400 bg-red-500/10 border-red-500/30",
81 |   KM_EXTREMO: "text-amber-400 bg-amber-500/10 border-amber-500/30",
82 |   CUADRATURA_BAJA: "text-amber-400 bg-amber-500/10 border-amber-500/30",
83 | };
84 |
85 | function fN(n: number): string {
86 |   return n.toLocaleString("es-CL");
87 | }
88 |
89 | function formatDateTime(iso: string): string {
90 |   if (!iso) return "\u2014";
91 |   const d = new Date(iso);
92 |   return d.toLocaleDateString("es-CL", { day: "2-digit", month: "2-digit", hour: "2-digit", minute: "2-digit" });
93 | }
94 |
95 | type PeriodOption = "7D" | "14D" | "30D";
96 |
97 | function getDateRange(period: PeriodOption): { from: string; to: string } {
98 |   const days = period === "7D" ? 7 : period === "14D" ? 14 : 30;
99 |   const to = new Date();
100 |   const from = new Date(Date.now() - days * 24 * 60 * 60 * 1000);
101 |   return { from: from.toISOString().slice(0, 10), to: to.toISOString().slice(0, 10) };
102 | }
103 |
104 | function detectErrors(t: FusionTruck): ErrorRecord[] {
```

```
105 |   const errors: ErrorRecord[] = [];
106 |
107 |   for (const c of t.cargas) {
108 |     if (c.rendimiento != null && c.rendimiento > 15) {
109 |       errors.push({
110 |         truck: t, carga: c, errorType: "REND_IMPOSIBLE",
111 |         errorLabel: ERROR_LABELS.REND_IMPOSIBLE,
112 |         detail: `Rendimiento de ${c.rendimiento.toFixed(1)} km/L es fisicamente imposible para un camion (max real ~3.5
113 |       });
114 |     }
115 |
116 |     if (c.rendimiento != null && c.rendimiento < 0) {
117 |       errors.push({
118 |         truck: t, carga: c, errorType: "REND_NEGATIVO",
119 |         errorLabel: ERROR_LABELS.REND_NEGATIVO,
120 |         detail: `Rendimiento negativo (${c.rendimiento.toFixed(1)} km/L) indica que el odometro retrocedio entre cargas.
121 |       });
122 |     }
123 |
124 |     if (c.kmRecorrido != null && c.kmRecorrido < 0) {
125 |       errors.push({
126 |         truck: t, carga: c, errorType: "KM_NEGATIVO",
127 |         errorLabel: ERROR_LABELS.KM_NEGATIVO,
128 |         detail: `KM recorridos negativos (${fN(c.kmRecorrido)} km). El odometro actual es menor que el anterior, lo que
129 |       });
130 |     }
131 |
132 |     if (c.kmRecorrido != null && c.kmRecorrido > 50000) {
133 |       errors.push({
134 |         truck: t, carga: c, errorType: "KM_EXTREMO",
135 |         errorLabel: ERROR_LABELS.KM_EXTREMO,
136 |         detail: `KM recorridos extremos (${fN(c.kmRecorrido)} km) entre cargas. Probablemente un error de digitacion en
137 |       });
138 |     }
139 |
140 |     if (c.litros > 0 && c.litros < 0.5 && c.kmRecorrido != null && c.kmRecorrido > 100) {
141 |       errors.push({
142 |         truck: t, carga: c, errorType: "MICRO_CARGA",
143 |         errorLabel: ERROR_LABELS.MICRO_CARGA,
144 |         detail: `Carga de ${c.litros.toFixed(2)}L con ${fN(c.kmRecorrido!)} km recorridos. Posible test de tarjeta o err
145 |       });
146 |     }
147 |   }
148 |
149 |   const odos = t.cargas
150 |     .filter(c => c.odometro != null && c.odometro > 0)
151 |     .sort((a, b) => new Date(a.fecha).getTime() - new Date(b.fecha).getTime())
152 |     .map(c => ({ odo: c.odometro!, fecha: c.fecha, carga: c }));
153 |
154 |   for (let i = 1; i < odos.length; i++) {
155 |     const prev = odos[i - 1];
156 |     const curr = odos[i];
157 |     const diff = Math.abs(curr.odo - prev.odo);
158 |     if (diff > 100000) {
159 |       errors.push({
160 |         truck: t, carga: curr.carga, errorType: "ODO_SALTO",
161 |         errorLabel: ERROR_LABELS.ODO_SALTO,
162 |         detail: `Salto de odometro de ${fN(diff)} km entre ${formatDateTime(prev.fecha)} (${fN(prev.odo)}) y ${formatDat
163 |       });
164 |     }
165 |   }
166 |
167 |   if (t.litrosVolvoPeriodo != null && t.litrosVolvoPeriodo > 0 && t.totalLitrosSigetra > 0) {
168 |     const volvo = t.litrosVolvoPeriodo;
169 |     const sigetra = t.totalLitrosSigetra;
170 |     const cuadratura = (Math.min(volvo, sigetra) / Math.max(volvo, sigetra)) * 100;
171 |     if (cuadratura < 95) {
172 |       const diff = Math.abs(volvo - sigetra);
173 |       errors.push({
174 |         truck: t, carga: null, errorType: "CUADRATURA_BAJA",
175 |         errorLabel: ERROR_LABELS.CUADRATURA_BAJA,
176 |         detail: `Cuadratura ${cuadratura.toFixed(1)}% — Volvo: ${fN(Math.round(volvo))} L vs Sigetra: ${fN(Math.round(si
177 |       });
178 |     }
179 |   }
180 |
181 |   return errors;
182 | }
183 |
184 | export function getErrorFleetNums(trucks: FusionTruck[]): Set<string> {
185 |   const errorNums = new Set<string>();
186 |   for (const t of trucks) {
187 |     const errors = detectErrors(t);
188 |     if (errors.length > 0) {
189 |       const hasCuadraturaBaja = errors.some(e => e.errorType === "CUADRATURA_BAJA");
190 |       const hasSerious = errors.some(e =>
191 |         e.errorType === "ODO_SALTO" || e.errorType === "KM_NEGATIVO" || e.errorType === "KM_EXTREMO"
192 |       );
193 |       const rendErrors = errors.filter(e => e.errorType === "REND_IMPOSIBLE");
194 |       const validCargas = t.cargas.length - rendErrors.length;
195 |       if (hasCuadraturaBaja || hasSerious || (rendErrors.length > 0 && validCargas === 0)) {
196 |         errorNums.add(t.fleetNum);
197 |       }
198 |     }
199 |   }
200 |   return errorNums;
201 | }
202 |
203 | export function isErrorCarga(c: FusionCarga): boolean {
204 |   if (c.rendimiento != null && (c.rendimiento > 15 || c.rendimiento < 0)) return true;
205 |   if (c.kmRecorrido != null && (c.kmRecorrido < 0 || c.kmRecorrido > 50000)) return true;
206 |   if (c.litros > 0 && c.litros < 0.5 && c.kmRecorrido != null && c.kmRecorrido > 100) return true;
207 |   return false;
208 | }
209 |
210 | function exportErrorsToCSV(errors: ErrorRecord[]) {
211 |   const header = "Patente,Fecha,Tipo Error,Valor,Descripcion\n";
```

```
212 |   const rows = errors.map(e => {
213 |     const patente = e.truck.patenteReal || e.truck.fleetNum;
214 |     const fecha = e.carga ? formatDateTime(e.carga.fecha) : "\u2014";
215 |     const tipoError = e.errorLabel;
216 |     let valor = "";
217 |     if (e.errorType === "REND_IMPOSIBLE" || e.errorType === "REND_NEGATIVO") {
218 |       valor = e.carga?.rendimiento != null ? `${e.carga.rendimiento.toFixed(1)} km/L` : "";
219 |     } else if (e.errorType === "KM_NEGATIVO" || e.errorType === "KM_EXTREMO") {
220 |       valor = e.carga?.kmRecorrido != null ? `${fN(e.carga.kmRecorrido)} km` : "";
221 |     } else if (e.errorType === "ODO_SALTO") {
222 |       valor = e.carga?.odometro != null ? `${fN(e.carga.odometro)} km` : "";
223 |     } else if (e.errorType === "MICRO_CARGA") {
224 |       valor = e.carga ? `${e.carga.litros.toFixed(2)} L` : "";
225 |     } else if (e.errorType === "CUADRATURA_BAJA") {
226 |       const v = e.truck.litrosVolvoPeriodo;
227 |       const s = e.truck.totalLitrosSigetra;
228 |       if (v != null && v > 0 && s > 0) {
229 |         valor = `${((Math.min(v, s) / Math.max(v, s)) * 100).toFixed(1)}%`;
230 |       }
231 |     }
232 |     const desc = e.detail.replace(/,/g, ";").replace(/\n/g, " ");
233 |     return `"${patente}","${fecha}","${tipoError}","${valor}","${desc}"`;
234 |   }).join("\n");
235 |
236 |   const blob = new Blob([header + rows], { type: "text/csv;charset=utf-8;" });
237 |   const url = URL.createObjectURL(blob);
238 |   const a = document.createElement("a");
239 |   a.href = url;
240 |   a.download = `errores_datos_${new Date().toISOString().slice(0, 10)}.csv`;
241 |   a.click();
242 |   URL.revokeObjectURL(url);
243 | }
244 |
245 | export default function Errores() {
246 |   const [busq, setBusq] = useState("");
247 |   const [expandedTruck, setExpandedTruck] = useState<string | null>(null);
248 |   const [filterType, setFilterType] = useState<ErrorType | "all" | "odo_group" | "CUADRATURA_BAJA">("all");
249 |   const [period, setPeriod] = useState<PeriodOption>("7D");
250 |
251 |   const range = useMemo(() => getDateRange(period), [period]);
252 |
253 |   const { data: fusion, isLoading } = useQuery<FusionResponse>({
254 |     queryKey: [`/api/sigetra/fusion?from=${range.from}&to=${range.to}`],
255 |     refetchInterval: 600000,
256 |   });
257 |
258 |   const allErrors = useMemo(() => {
259 |     if (!fusion?.trucks) return [];
260 |     const errors: ErrorRecord[] = [];
261 |     for (const t of fusion.trucks) {
262 |       errors.push(...detectErrors(t));
263 |     }
264 |     return errors;
265 |   }, [fusion]);
266 |
267 |   const errorsByTruck = useMemo(() => {
268 |     const map = new Map<string, ErrorRecord[]>();
269 |     for (const e of allErrors) {
270 |       const existing = map.get(e.truck.fleetNum) || [];
271 |       existing.push(e);
272 |       map.set(e.truck.fleetNum, existing);
273 |     }
274 |     return Array.from(map.entries())
275 |       .sort((a, b) => b[1].length - a[1].length);
276 |   }, [allErrors]);
277 |
278 |   const filtered = useMemo(() => {
279 |     let list = errorsByTruck;
280 |
281 |     if (filterType === "odo_group") {
282 |       const odoTypes: ErrorType[] = ["KM_NEGATIVO", "ODO_SALTO", "KM_EXTREMO"];
283 |       list = list.map(([num, errs]) => [num, errs.filter(e => odoTypes.includes(e.errorType))] as [string, ErrorRecord[]
284 |         .filter(([, errs]) => errs.length > 0);
285 |     } else if (filterType !== "all") {
286 |       list = list.map(([num, errs]) => [num, errs.filter(e => e.errorType === filterType)] as [string, ErrorRecord[]])
287 |         .filter(([, errs]) => errs.length > 0);
288 |     }
289 |
290 |     if (busq.trim()) {
291 |       const q = busq.toLowerCase();
292 |       list = list.filter(([num, errs]) =>
293 |         num.includes(q) ||
294 |         errs[0]?.truck.patenteReal?.toLowerCase().includes(q) ||
295 |         errs[0]?.truck.faenaSigetra?.toLowerCase().includes(q) ||
296 |         errs[0]?.truck.conductorSigetra?.toLowerCase().includes(q)
297 |       );
298 |     }
299 |
300 |     return list;
301 |   }, [errorsByTruck, busq, filterType]);
302 |
303 |   const stats = useMemo(() => {
304 |     const byType = new Map<ErrorType, number>();
305 |     for (const e of allErrors) {
306 |       byType.set(e.errorType, (byType.get(e.errorType) || 0) + 1);
307 |     }
308 |     return {
309 |       total: allErrors.length,
310 |       trucks: errorsByTruck.length,
311 |       byType,
312 |     };
313 |   }, [allErrors, errorsByTruck]);
314 |
315 |   if (isLoading) {
316 |     return (
317 |       <div className="space-y-4" data-testid="errores-loading">
318 |         <Skeleton className="h-16 w-full" />
```

```
319 |         <Skeleton className="h-96 w-full" />
320 |       </div>
321 |     );
322 |   }
323 |
324 |   return (
325 |     <div className="space-y-4" data-testid="errores-page">
326 |       <div className="flex items-center justify-between gap-3 flex-wrap">
327 |         <div className="flex items-center gap-3 flex-wrap">
328 |           <div className="flex items-center gap-2">
329 |             <Bug className="w-5 h-5 text-amber-400" />
330 |             <h1 className="text-sm font-mono font-bold tracking-[0.15em] uppercase text-foreground" data-testid="text-er
331 |               Errores de Datos
332 |             </h1>
333 |           </div>
334 |           <span className="text-[10px] font-mono text-muted-foreground">
335 |             Registros con datos imposibles o inconsistentes en Sigetra — ultimos {period === "7D" ? "7" : period === "14
336 |           </span>
337 |         </div>
338 |         <div className="flex items-center gap-2">
339 |           <div className="flex items-center border border-border rounded-md overflow-hidden">
340 |             {(["7D", "14D", "30D"] as PeriodOption[]).map(p => (
341 |               <button
342 |                 key={p}
343 |                 onClick={() => setPeriod(p)}
344 |                 data-testid={`btn-period-${p.toLowerCase()}`}
345 |                 className={`px-3 py-1.5 text-[10px] font-mono font-bold transition-colors ${period === p ? "bg-amber-500
346 |               >
347 |                 {p}
348 |               </button>
349 |             ))}
350 |           </div>
351 |           <Button
352 |             size="sm"
353 |             variant="outline"
354 |             onClick={() => exportErrorsToCSV(allErrors)}
355 |             disabled={allErrors.length === 0}
356 |             data-testid="btn-export-csv"
357 |           >
358 |             <Download className="w-3.5 h-3.5 mr-1.5" />
359 |             <span className="text-xs font-mono">CSV</span>
360 |           </Button>
361 |         </div>
362 |       </div>
363 |
364 |       <div className="bg-amber-500/5 border border-amber-500/20 p-3 flex items-center gap-2">
365 |         <AlertOctagon className="w-4 h-4 text-amber-400 flex-shrink-0" />
366 |         <span className="text-[10px] font-mono text-amber-400">
367 |           Estos registros tienen datos que no son fisicamente posibles (rendimiento &gt;15 km/L, odometros que retrocede
368 |         </span>
369 |       </div>
370 |
371 |       <div className="grid grid-cols-2 sm:grid-cols-5 gap-3">
372 |         <button onClick={() => setFilterType("all")}
373 |           data-testid="btn-err-filter-all"
374 |           className={`bg-card border p-3 text-left cursor-pointer transition-colors ${filterType === "all" ? "border-amb
375 |           <div className="flex items-center gap-2 mb-1">
376 |             <Bug className="w-4 h-4 text-amber-400" />
377 |             <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">TOTAL ERRORES</span>
378 |           </div>
379 |           <div className="text-2xl font-mono font-bold text-amber-400" data-testid="text-err-total">{stats.total}</div>
380 |           <div className="text-[10px] font-mono text-muted-foreground">{stats.trucks} camiones</div>
381 |         </button>
382 |         <button onClick={() => setFilterType("CUADRATURA_BAJA")}
383 |           data-testid="btn-err-filter-cuadratura"
384 |           className={`bg-card border p-3 text-left cursor-pointer transition-colors ${filterType === "CUADRATURA_BAJA" ?
385 |           <div className="flex items-center gap-2 mb-1">
386 |             <ArrowDownUp className="w-4 h-4 text-amber-400" />
387 |             <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">CUADRATURA BAJA</span>
388 |           </div>
389 |           <div className="text-2xl font-mono font-bold text-amber-400" data-testid="text-err-cuadratura">{stats.byType.g
390 |           <div class="text-[10px] font-mono text-muted-foreground">&lt;95% Volvo vs Sigetra</div>
391 |         </button>
392 |         <button onClick={() => setFilterType("REND_IMPOSIBLE")}
393 |           data-testid="btn-err-filter-rend"
394 |           className={`bg-card border p-3 text-left cursor-pointer transition-colors ${filterType === "REND_IMPOSIBLE" ?
395 |           <div class="flex items-center gap-2 mb-1">
396 |             <Gauge class="w-4 h-4 text-amber-400" />
397 |             <span class="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">REND. IMPOSIBLE</span>
398 |           </div>
399 |           <div class="text-2xl font-mono font-bold text-amber-400">{stats.byType.get("REND_IMPOSIBLE") || 0}</div>
400 |           <div class="text-[10px] font-mono text-muted-foreground">&gt;15 km/L</div>
401 |         </button>
402 |         <button onClick={() => setFilterType("odo_group")}
403 |           data-testid="btn-err-filter-km-neg"
404 |           class={`bg-card border p-3 text-left cursor-pointer transition-colors ${filterType === "odo_group" ? "bord
405 |           <div class="flex items-center gap-2 mb-1">
406 |             <ArrowDownUp class="w-4 h-4 text-red-400" />
407 |             <span class="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">ODOMETRO CORRUPTO</span>
408 |           </div>
409 |           <div class="text-2xl font-mono font-bold text-red-400">{(stats.byType.get("KM_NEGATIVO") || 0) + (stats.by
410 |           <div class="text-[10px] font-mono text-muted-foreground">km neg / salto / extremo</div>
411 |         </button>
412 |         <button onClick={() => setFilterType("MICRO_CARGA")}
413 |           data-testid="btn-err-filter-micro"
414 |           class={`bg-card border p-3 text-left cursor-pointer transition-colors ${filterType === "MICRO_CARGA" ? "bo
415 |           <div class="flex items-center gap-2 mb-1">
416 |             <Fuel class="w-4 h-4 text-blue-400" />
417 |             <span class="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">MICRO-CARGAS</span>
418 |           </div>
419 |           <div class="text-2xl font-mono font-bold text-blue-400">{stats.byType.get("MICRO_CARGA") || 0}</div>
420 |           <div class="text-[10px] font-mono text-muted-foreground">&lt;0.5L sospechosas</div>
421 |         </button>
422 |       </div>
423 |
424 |       <div class="flex items-center gap-2">
425 |         <div class="relative flex-1 max-w-xs">
```

```
426 |           <Search className="absolute left-2.5 top-1/2 -translate-y-1/2 w-3.5 h-3.5 text-muted-foreground" />
427 |           <Input value={busq} onChange={e => setBusq(e.target.value)}
428 |             placeholder="Buscar tracto, patente, faena..."
429 |             className="pl-8 font-mono text-xs bg-card"
430 |             data-testid="input-err-search" />
431 |         </div>
432 |         <span className="text-[10px] font-mono text-muted-foreground">{filtered.length} camiones</span>
433 |       </div>
434 |
435 |       <div className="border border-border bg-card overflow-hidden">
436 |         <div className="bg-background border-b border-border px-3 py-2">
437 |           <span className="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">
438 |             REGISTROS CON ERRORES DE DATOS
439 |           </span>
440 |         </div>
441 |
442 |         <div className="max-h-[600px] overflow-y-auto">
443 |           {filtered.map(([fleetNum, errors]) => {
444 |             const t = errors[0].truck;
445 |             const isExpanded = expandedTruck === fleetNum;
446 |
447 |             return (
448 |               <div key={fleetNum}>
449 |                 <div
450 |                   onClick={() => setExpandedTruck(isExpanded ? null : fleetNum)}
451 |                   className="px-3 py-2.5 border-b border-border/50 cursor-pointer transition-colors hover:bg-primary/5"
452 |                   style={{ borderLeftWidth: 3, borderLeftColor: "#FFB020" }}
453 |                   data-testid={`row-err-${fleetNum}`}
454 |                 >
455 |                   <div className="flex items-center justify-between mb-1.5">
456 |                     <div className="flex items-center gap-2">
457 |                       <span className="text-sm font-mono font-bold text-foreground">{fleetNum}</span>
458 |                       <span className="text-[10px] font-mono text-muted-foreground">{t.patenteReal || ""}</span>
459 |                       <span className="text-[10px] font-mono text-muted-foreground">{t.faenaSigetra || ""}</span>
460 |                     </div>
461 |                     <div className="flex items-center gap-2">
462 |                       <span className="text-[9px] font-mono font-bold text-amber-400">{errors.length} error{errors.lengt
463 |                       {isExpanded ? <ChevronUp className="w-3 h-3 text-muted-foreground" /> : <ChevronDown className="w-
464 |                     </div>
465 |                   </div>
466 |                   <div className="flex flex-wrap gap-1.5">
467 |                     {Array.from(new Set(errors.map(e => e.errorType))).map(type => (
468 |                       <span key={type} className={`text-[8px] font-mono px-1.5 py-0.5 border ${ERROR_COLORS[type]}`}>
469 |                         {ERROR_LABELS[type]}
470 |                       </span>
471 |                     ))}
472 |                   </div>
473 |                 </div>
474 |
475 |                 {isExpanded && (
476 |                   <div className="bg-background border-b border-border p-4 space-y-3">
477 |                     <div className="grid grid-cols-3 gap-4 mb-3">
478 |                       <div className="text-[11px] font-mono">
479 |                         <span className="text-muted-foreground">Modelo:</span> <span className="text-foreground">{t.mode
480 |                       </div>
481 |                       <div class="text-[11px] font-mono">
482 |                         <span class="text-muted-foreground">Conductor:</span> <span class="text-foreground">{t.c
483 |                       </div>
484 |                       <div class="text-[11px] font-mono">
485 |                         <span class="text-muted-foreground">Total cargas:</span> <span class="text-foreground">{
486 |                       </div>
487 |                     </div>
488 |
489 |                     {errors.map((e, i) => (
490 |                       <div key={i} className={`border p-3 ${ERROR_COLORS[e.errorType]}`}>
491 |                         <div class="flex items-center gap-2 mb-2 flex-wrap">
492 |                           <AlertOctagon class="w-3.5 h-3.5" />
493 |                           <span class="text-[10px] font-mono font-bold">{e.errorLabel}</span>
494 |                           <span class="text-[9px] font-mono opacity-70" data-testid={`text-err-patente-${i}`}>{e.tru
495 |                           {e.carga && <span class="text-[9px] font-mono opacity-70" data-testid={`text-err-fecha-${i
496 |                           {e.carga && <span class="text-[9px] font-mono opacity-70">Guia #{e.carga.numGuia}</span>}
497 |                         </div>
498 |                         {e.carga ? (
499 |                           <div class="grid grid-cols-[1fr_70px_80px_80px_70px] gap-0 mb-2 text-[10px] font-mono">
500 |                             <div>
501 |                               <span class="text-muted-foreground">Fecha: </span>
502 |                               <span>{formatDateTime(e.carga.fecha)}</span>
503 |                             </div>
504 |                             <div>
505 |                               <span class="text-muted-foreground">Litros: </span>
506 |                               <span class="text-amber-400 font-bold">{e.carga.litros.toFixed(2)}</span>
507 |                             </div>
508 |                             <div>
509 |                               <span class="text-muted-foreground">Odo: </span>
510 |                               <span>{e.carga.odometro != null ? fN(e.carga.odometro) : "\u2014"}</span>
511 |                             </div>
512 |                             <div>
513 |                               <span class="text-muted-foreground">KM: </span>
514 |                               <span class={e.carga.kmRecorrido != null && e.carga.kmRecorrido < 0 ? "text-red-400" :
515 |                             </div>
516 |                             <div>
517 |                               <span class="text-muted-foreground">Rend: </span>
518 |                               <span class={e.carga.rendimiento != null && (e.carga.rendimiento > 15 || e.carga.rendi
519 |                             </div>
520 |                           </div>
521 |                         ) : e.errorType === "CUADRATURA_BAJA" && (
522 |                           <div class="grid grid-cols-3 gap-4 mb-2 text-[10px] font-mono">
523 |                             <div>
524 |                               <span class="text-muted-foreground">Volvo: </span>
525 |                               <span class="text-amber-400 font-bold">{e.truck.litrosVolvoPeriodo != null ? fN(Math.r
526 |                             </div>
527 |                             <div>
528 |                               <span class="text-muted-foreground">Sigetra: </span>
529 |                               <span class="text-amber-400 font-bold">{fN(Math.round(e.truck.totalLitrosSigetra))} L<
530 |                             </div>
531 |                             <div>
532 |                               <span class="text-muted-foreground">Cuadratura: </span>
```

```
533 |                              <span className="text-red-400 font-bold">
534 |                                {e.truck.litrosVolvoPeriodo != null && e.truck.litrosVolvoPeriodo > 0 && e.truck.totalLi
535 |                                  ? ((Math.min(e.truck.litrosVolvoPeriodo, e.truck.totalLitrosSigetra) / Math.max(e.truc
536 |                                  : "\u2014"}%
537 |                              </span>
538 |                            </div>
539 |                          </div>
540 |                        )}
541 |                        <div className="text-[9px] font-mono opacity-80">{e.detail}</div>
542 |                      </div>
543 |                    ))}
544 |                  </div>
545 |                )}
546 |              </div>
547 |            );
548 |          })}
549 |
550 |          {filtered.length === 0 && (
551 |            <div className="p-8 text-center text-[11px] font-mono text-muted-foreground" data-testid="err-empty">
552 |              {busq ? "Sin resultados para la busqueda" : `No se detectaron errores de datos en los ultimos ${period ===
553 |            </div>
554 |          )}
555 |        </div>
556 |      </div>
557 |    </div>
558 |  );
559 | }
560 |
```

client/src/pages/rutas-contrato.tsx

530 lineas

```
1 | import { useState, useMemo } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { Skeleton } from "@/components/ui/skeleton";
4 | import { Card, CardContent, CardHeader, CardTitle } from "@/components/ui/card";
5 | import {
6 |   Route, TrendingUp, TrendingDown, Fuel, Truck, ChevronDown, ChevronUp,
7 |   MapPin, Target, AlertTriangle, Info
8 | } from "lucide-react";
9 | import { getErrorFleetNums, isErrorCarga } from "@/pages/errores";
10 |
11 | interface FusionCarga {
12 |   numGuia: number;
13 |   fecha: string;
14 |   litros: number;
15 |   odometro: number | null;
16 |   kmRecorrido: number | null;
17 |   rendimiento: number | null;
18 |   lugar: string | null;
19 |   tarjeta: string | null;
20 |   conductor: string | null;
21 | }
22 |
23 | interface FusionTruck {
24 |   fleetNum: string;
25 |   patenteReal: string | null;
26 |   vin: string | null;
27 |   modeloVolvo: string;
28 |   faenaSigetra: string | null;
29 |   conductorSigetra: string | null;
30 |   totalLitrosSigetra: number;
31 |   totalCargas: number;
32 |   rendPromedio: number;
33 |   odometroSigetra: number | null;
34 |   odometroVolvo: number | null;
35 |   deltaOdometro: number | null;
36 |   fuelLevelVolvo: number | null;
37 |   totalFuelUsedVolvo: number | null;
38 |   litrosVolvoPeriodo: number | null;
39 |   engineHoursVolvo: number | null;
40 |   gpsVolvo: { latitude: number | null; longitude: number | null } | null;
41 |   alertLevel: "ok" | "alerta" | "critico";
42 |   cargas: FusionCarga[];
43 | }
44 |
45 | interface FusionResponse {
46 |   from: string;
47 |   to: string;
48 |   totalTrucksMatched: number;
49 |   totalLitros: number;
50 |   totalCargas: number;
51 |   alertCount: number;
52 |   trucks: FusionTruck[];
53 | }
54 |
55 | interface SectorAnalysis {
56 |   sector: string;
57 |   totalLitros: number;
58 |   totalKm: number;
59 |   totalCargas: number;
60 |   rendimiento: number;
61 |   trucks: Set<string>;
62 |   conductores: Set<string>;
63 |   bestRend: { fleetNum: string; rend: number } | null;
64 |   worstRend: { fleetNum: string; rend: number } | null;
65 | }
66 |
67 | interface FaenaRouteAnalysis {
68 |   faena: string;
69 |   sectors: SectorAnalysis[];
70 |   globalRendimiento: number;
71 |   totalLitros: number;
72 |   totalKm: number;
73 |   totalCargas: number;
74 |   truckCount: number;
75 | }
76 |
77 | function fN(n: number): string {
78 |   return n.toLocaleString("es-CL");
79 | }
80 |
81 | function getDateRange(days: number): { from: string; to: string } {
82 |   const to = new Date();
83 |   const from = new Date(Date.now() - days * 24 * 60 * 60 * 1000);
84 |   return { from: from.toISOString().slice(0, 10), to: to.toISOString().slice(0, 10) };
85 | }
86 |
87 | function RendBar({ value, max, color }: { value: number; max: number; color: string }) {
88 |   const pct = max > 0 ? Math.min((value / max) * 100, 100) : 0;
89 |   return (
90 |     <div className="h-2 w-full bg-muted rounded-md overflow-hidden">
91 |       <div className={`h-full rounded-md ${color}`} style={{ width: `${pct}%` }} />
92 |     </div>
93 |   );
94 | }
95 |
96 | function rendColor(r: number): string {
97 |   if (r >= 1.8) return "text-emerald-400";
98 |   if (r >= 1.2) return "text-amber-400";
99 |   return "text-red-400";
100 | }
101 |
102 | function rendBgColor(r: number): string {
103 |   if (r >= 1.8) return "bg-emerald-500";
104 |   if (r >= 1.2) return "bg-amber-500";
```

```
105 |   return "bg-red-500";
106 | }
107 |
108 | function SectorCard({ sector, faenaRend, globalMax, rank, total }: {
109 |   sector: SectorAnalysis;
110 |   faenaRend: number;
111 |   globalMax: number;
112 |   rank: number;
113 |   total: number;
114 | }) {
115 |   const vsAvg = faenaRend > 0 ? ((sector.rendimiento - faenaRend) / faenaRend) * 100 : 0;
116 |   const isTop = rank === 1;
117 |   const isBottom = rank === total && total > 1;
118 |
119 |   return (
120 |     <div
121 |       className={`border border-border p-3 ${
122 |         isTop ? "bg-emerald-500/5 border-emerald-500/20" : isBottom ? "bg-red-500/5 border-red-500/20" : "bg-background"
123 |       }`}
124 |       data-testid={`sector-card-${sector.sector}`}
125 |     >
126 |       <div className="flex items-start justify-between gap-3 mb-2">
127 |         <div className="flex items-center gap-2 min-w-0 flex-1">
128 |           <div className={`w-6 h-6 rounded flex items-center justify-center text-[10px] font-mono font-bold ${
129 |             isTop ? "bg-emerald-500/20 text-emerald-400" : isBottom ? "bg-red-500/20 text-red-400" : "bg-muted text-mute
130 |           }`}>
131 |             #{rank}
132 |           </div>
133 |           <div className="min-w-0">
134 |             <div className="text-xs font-mono font-bold text-foreground truncate" data-testid={`text-sector-name-${secto
135 |               {sector.sector}
136 |             </div>
137 |             <div className="text-[9px] font-mono text-muted-foreground">
138 |               {sector.trucks.size} camiones · {sector.conductores.size} conductores · {sector.totalCargas} cargas
139 |             </div>
140 |           </div>
141 |         </div>
142 |         <div className="text-right flex-shrink-0">
143 |           <div className={`text-lg font-mono font-bold ${rendColor(sector.rendimiento)}`} data-testid={`text-sector-rend
144 |             {sector.rendimiento > 0 ? sector.rendimiento.toFixed(2) : "\u2014"}
145 |           </div>
146 |           <div className="text-[8px] font-mono text-muted-foreground">km/L</div>
147 |         </div>
148 |       </div>
149 |
150 |       <div className="mb-2">
151 |         <RendBar value={sector.rendimiento} max={globalMax} color={rendBgColor(sector.rendimiento)} />
152 |       </div>
153 |
154 |       <div className="grid grid-cols-4 gap-3 text-[10px] font-mono">
155 |         <div>
156 |           <span className="text-muted-foreground">Litros:</span>
157 |           <span className="ml-1 text-amber-400 font-bold">{fN(Math.round(sector.totalLitros))}</span>
158 |         </div>
159 |         <div>
160 |           <span className="text-muted-foreground">Km:</span>
161 |           <span className="ml-1 text-foreground font-bold">{fN(Math.round(sector.totalKm))}</span>
162 |         </div>
163 |         <div>
164 |           <span className="text-muted-foreground">vs Prom:</span>
165 |           <span className={`ml-1 font-bold ${vsAvg >= 0 ? "text-emerald-400" : vsAvg >= -15 ? "text-amber-400" : "text-r
166 |             {vsAvg >= 0 ? "+" : ""}{vsAvg.toFixed(0)}%
167 |           </span>
168 |         </div>
169 |         <div>
170 |           <span className="text-muted-foreground">L/100km:</span>
171 |           <span className="ml-1 text-foreground font-bold">
172 |             {sector.rendimiento > 0 ? (100 / sector.rendimiento).toFixed(1) : "\u2014"}
173 |           </span>
174 |         </div>
175 |       </div>
176 |
177 |       {(sector.bestRend || sector.worstRend) && (
178 |         <div className="grid grid-cols-2 gap-2 mt-2 pt-2 border-t border-border/30">
179 |           {sector.bestRend && (
180 |             <div className="flex items-center gap-1.5 text-[9px] font-mono">
181 |               <TrendingUp className="w-3 h-3 text-emerald-400 flex-shrink-0" />
182 |               <span className="text-muted-foreground">Mejor:</span>
183 |               <span className="text-emerald-400 font-bold">N.{sector.bestRend.fleetNum}</span>
184 |               <span class="text-emerald-400">{sector.bestRend.rend.toFixed(2)}</span>
185 |             </div>
186 |           )}
187 |           {sector.worstRend && (
188 |             <div className="flex items-center gap-1.5 text-[9]px] font-mono">
189 |               <TrendingDown className="w-3 h-3 text-red-400 flex-shrink-0" />
190 |               <span class="text-muted-foreground">Peor:</span>
191 |               <span class="text-red-400 font-bold">N.{sector.worstRend.fleetNum}</span>
192 |               <span class="text-red-400">{sector.worstRend.rend.toFixed(2)}</span>
193 |             </div>
194 |           )}
195 |         </div>
196 |       )}
197 |     </div>
198 |   );
199 | }
200 |
201 | export default function RutasContrato() {
202 |   const [days, setDays] = useState(14);
203 |   const [expandedFaena, setExpandedFaena] = useState<string | null>(null);
204 |
205 |   const range = useMemo(() => getDateRange(days), [days]);
206 |
207 |   const { data: fusion, isLoading, isError } = useQuery<FusionResponse>({
208 |     queryKey: [`/api/sigetra/fusion?from=${range.from}&to=${range.to}`],
209 |     refetchInterval: 600000,
210 |   });
211 |
```

```
212 |   const errorNums = useMemo(() => {
213 |     if (!fusion?.trucks) return new Set<string>();
214 |     return getErrorFleetNums(fusion.trucks);
215 |   }, [fusion]);
216 |
217 |   const faenaRoutes = useMemo(() => {
218 |     if (!fusion?.trucks) return [];
219 |
220 |     const cleanTrucks = fusion.trucks
221 |       .filter(t => !errorNums.has(t.fleetNum))
222 |       .map(t => {
223 |         const validCargas = t.cargas.filter(c => !isErrorCarga(c));
224 |         return { ...t, cargas: validCargas };
225 |       });
226 |
227 |     const byFaena = new Map<string, FusionTruck[]>();
228 |     cleanTrucks.forEach(t => {
229 |       const faena = t.faenaSigetra || "Sin Faena";
230 |       if (!byFaena.has(faena)) byFaena.set(faena, []);
231 |       byFaena.get(faena)!.push(t);
232 |     });
233 |
234 |     const analyses: FaenaRouteAnalysis[] = [];
235 |
236 |     Array.from(byFaena.entries()).forEach(([faena, trucks]) => {
237 |       const sectorMap = new Map<string, {
238 |         litros: number;
239 |         km: number;
240 |         cargas: number;
241 |         trucks: Set<string>;
242 |         conductores: Set<string>;
243 |         truckRends: { fleetNum: string; litros: number; km: number }[];
244 |       }>();
245 |
246 |       let faenaTotalLitros = 0;
247 |       let faenaTotalKm = 0;
248 |       let faenaTotalCargas = 0;
249 |
250 |       trucks.forEach(t => {
251 |         t.cargas.forEach(c => {
252 |           const sector = c.lugar?.trim() || "Sin Lugar";
253 |           if (c.litros <= 0) return;
254 |           const km = c.kmRecorrido != null && c.kmRecorrido > 0 ? c.kmRecorrido : 0;
255 |
256 |           if (!sectorMap.has(sector)) {
257 |             sectorMap.set(sector, { litros: 0, km: 0, cargas: 0, trucks: new Set(), conductores: new Set(), truckRends:
258 |           }
259 |
260 |           const s = sectorMap.get(sector)!;
261 |           s.litros += c.litros;
262 |           s.km += km;
263 |           s.cargas += 1;
264 |           s.trucks.add(t.fleetNum);
265 |           if (c.conductor) s.conductores.add(c.conductor);
266 |           s.truckRends.push({ fleetNum: t.fleetNum, litros: c.litros, km });
267 |
268 |           faenaTotalLitros += c.litros;
269 |           faenaTotalKm += km;
270 |           faenaTotalCargas += 1;
271 |         });
272 |       });
273 |
274 |       const sectors: SectorAnalysis[] = [];
275 |       Array.from(sectorMap.entries()).forEach(([sector, data]) => {
276 |         if (data.cargas < 2) return;
277 |
278 |         const rendimiento = data.litros > 0 ? data.km / data.litros : 0;
279 |
280 |         const truckAgg = new Map<string, { litros: number; km: number }>();
281 |         data.truckRends.forEach(r => {
282 |           if (!truckAgg.has(r.fleetNum)) truckAgg.set(r.fleetNum, { litros: 0, km: 0 });
283 |           const a = truckAgg.get(r.fleetNum)!;
284 |           a.litros += r.litros;
285 |           a.km += r.km;
286 |         });
287 |
288 |         let bestRend: { fleetNum: string; rend: number } | null = null;
289 |         let worstRend: { fleetNum: string; rend: number } | null = null;
290 |
291 |         Array.from(truckAgg.entries()).forEach(([fleetNum, agg]) => {
292 |           if (agg.litros <= 0 || agg.km <= 0) return;
293 |           const rend = agg.km / agg.litros;
294 |           if (rend <= 0 || rend > 15) return;
295 |           if (!bestRend || rend > bestRend.rend) bestRend = { fleetNum, rend };
296 |           if (!worstRend || rend < worstRend.rend) worstRend = { fleetNum, rend };
297 |         });
298 |
299 |         if (bestRend && worstRend && bestRend.fleetNum === worstRend.fleetNum) {
300 |           worstRend = null;
301 |         }
302 |
303 |         sectors.push({
304 |           sector,
305 |           totalLitros: data.litros,
306 |           totalKm: data.km,
307 |           totalCargas: data.cargas,
308 |           rendimiento,
309 |           trucks: data.trucks,
310 |           conductores: data.conductores,
311 |           bestRend,
312 |           worstRend,
313 |         });
314 |       });
315 |
316 |       sectors.sort((a, b) => b.rendimiento - a.rendimiento);
317 |
318 |       const faenaRend = faenaTotalLitros > 0 ? faenaTotalKm / faenaTotalLitros : 0;
```

```
319 |       const uniqueTrucks = new Set<string>(trucks.map(t => t.fleetNum));
320 |
321 |       analyses.push({
322 |         faena,
323 |         sectors,
324 |         globalRendimiento: faenaRend,
325 |         totalLitros: faenaTotalLitros,
326 |         totalKm: faenaTotalKm,
327 |         totalCargas: faenaTotalCargas,
328 |         truckCount: uniqueTrucks.size,
329 |       });
330 |     });
331 |
332 |     analyses.sort((a, b) => b.totalCargas - a.totalCargas);
333 |     return analyses;
334 |   }, [fusion, errorNums]);
335 |
336 |   const globalMax = useMemo(() => {
337 |     return faenaRoutes.reduce((max, f) => {
338 |       return f.sectors.reduce((m, s) => Math.max(m, s.rendimiento), max);
339 |     }, 0);
340 |   }, [faenaRoutes]);
341 |
342 |   const globalStats = useMemo(() => {
343 |     const totalSectors = faenaRoutes.reduce((s, f) => s + f.sectors.length, 0);
344 |     const allSectors = faenaRoutes.flatMap(f => f.sectors).filter(s => s.rendimiento > 0);
345 |     const bestSector = allSectors.length > 0 ? allSectors.reduce((best, s) => s.rendimiento > best.rendimiento ? s : bes
346 |     const worstSector = allSectors.length > 0 ? allSectors.reduce((worst, s) => s.rendimiento < worst.rendimiento ? s :
347 |     return { totalSectors, bestSector, worstSector };
348 |   }, [faenaRoutes]);
349 |
350 |   if (isLoading) {
351 |     return (
352 |       <div className="space-y-4" data-testid="rutas-loading">
353 |         <Skeleton className="h-20 w-full" />
354 |         <div className="grid grid-cols-3 gap-3">
355 |           {Array.from({ length: 3 }).map((_, i) => <Skeleton key={i} className="h-24" />)}
356 |         </div>
357 |         <Skeleton className="h-96 w-full" />
358 |       </div>
359 |     );
360 |   }
361 |
362 |   return (
363 |     <div className="space-y-4" data-testid="rutas-page">
364 |       <div className="flex items-center gap-3 flex-wrap">
365 |         <div className="flex items-center gap-2">
366 |           <Route className="w-5 h-5 text-cyan-400" />
367 |           <h1 className="text-sm font-mono font-bold tracking-[0.15em] uppercase text-foreground" data-testid="text-ruta
368 |             Rutas por Contrato
369 |           </h1>
370 |         </div>
371 |         <span className="text-[10px] font-mono text-muted-foreground">Analisis de consumo por sector dentro de cada faen
372 |       </div>
373 |
374 |       <div className="flex items-center gap-2 flex-wrap">
375 |         <span className="text-[10px] font-mono text-muted-foreground tracking-[0.15em]">PERIODO:</span>
376 |         {[7, 14, 30].map(d => (
377 |           <button key={d} onClick={() => setDays(d)}
378 |             data-testid={`btn-rutas-days-${d}`}
379 |             className={`px-3 py-1 text-[10px] font-mono font-bold border cursor-pointer transition-colors ${
380 |               days === d
381 |                 ? "bg-cyan-500/20 border-cyan-500 text-cyan-400"
382 |                 : "bg-card border-border text-muted-foreground hover:text-foreground"
383 |             }`}>
384 |             {d}D
385 |           </button>
386 |         ))}
387 |       </div>
388 |
389 |       {isError && (
390 |         <div className="bg-red-500/10 border border-red-500/30 p-3 flex items-center gap-2" data-testid="rutas-error">
391 |           <AlertTriangle className="w-4 h-4 text-red-400" />
392 |           <span className="text-[11px] font-mono text-red-400">Error al obtener datos de fusion.</span>
393 |         </div>
394 |       )}
395 |
396 |       {fusion && (
397 |         <>
398 |           <div className="grid grid-cols-3 gap-3">
399 |             <div className="bg-card border border-border p-3.5">
400 |               <div className="flex items-center gap-2 mb-1.5">
401 |                 <MapPin className="w-4 h-4 text-cyan-400" />
402 |                 <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">SECTORES</span>
403 |               </div>
404 |               <div className="text-2xl font-mono font-bold text-cyan-400" data-testid="text-rutas-sectors">{globalStats.
405 |               <div className="text-[10px] font-mono text-muted-foreground">con 2+ cargas</div>
406 |             </div>
407 |             {globalStats.bestSector && (
408 |               <div className="bg-emerald-500/5 border border-emerald-500/20 p-3.5">
409 |                 <div className="flex items-center gap-2 mb-1.5">
410 |                   <TrendingUp className="w-4 h-4 text-emerald-400" />
411 |                   <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">MEJOR SECTOR</span>
412 |                 </div>
413 |                 <div className="text-sm font-mono font-bold text-foreground truncate" data-testid="text-best-sector">
414 |                   {globalStats.bestSector.sector}
415 |                 </div>
416 |                 <div className="text-lg font-mono font-bold text-emerald-400">
417 |                   {globalStats.bestSector.rendimiento.toFixed(2)} km/L
418 |                 </div>
419 |               </div>
420 |             )}
421 |             {globalStats.worstSector && (
422 |               <div className="bg-red-500/5 border border-red-500/20 p-3.5">
423 |                 <div className="flex items-center gap-2 mb-1.5">
424 |                   <TrendingDown className="w-4 h-4 text-red-400" />
425 |                   <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">PEOR SECTOR</span>
```

```
426 |                  </div>
427 |                  <div className="text-sm font-mono font-bold text-foreground truncate" data-testid="text-worst-sector">
428 |                    {globalStats.worstSector.sector}
429 |                  </div>
430 |                  <div className="text-lg font-mono font-bold text-red-400">
431 |                    {globalStats.worstSector.rendimiento.toFixed(2)} km/L
432 |                  </div>
433 |                </div>
434 |              )}
435 |            </div>
436 |
437 |            <div className="bg-cyan-500/5 border border-cyan-500/20 p-3 flex items-start gap-2" data-testid="note-rutas">
438 |              <Info className="w-4 h-4 text-cyan-400 flex-shrink-0 mt-0.5" />
439 |              <div className="text-[11px] font-mono text-cyan-400/80">
440 |                <span className="font-bold text-cyan-400">Rutas por contrato:</span> Cada faena (contrato) contiene multip
441 |                El rendimiento por sector permite identificar que tramos de la ruta tienen mayor o menor eficiencia.
442 |                Solo se muestran sectores con 2 o mas cargas. Los porcentajes "vs Prom" comparan contra el promedio de la
443 |              </div>
444 |            </div>
445 |
446 |            <div className="space-y-3">
447 |              {faenaRoutes.map(fr => (
448 |                <Card key={fr.faena} className="overflow-visible" data-testid={`card-ruta-${fr.faena}`}>
449 |                  <CardHeader
450 |                    className="flex flex-row items-center justify-between gap-2 space-y-0 pb-2 cursor-pointer"
451 |                    onClick={() => setExpandedFaena(expandedFaena === fr.faena ? null : fr.faena)}
452 |                    data-testid={`btn-expand-ruta-${fr.faena}`}
453 |                  >
454 |                    <div className="flex items-center gap-3 flex-wrap min-w-0">
455 |                      <Target className="w-4 h-4 text-cyan-400 flex-shrink-0" />
456 |                      <CardTitle className="text-sm font-mono font-bold tracking-[0.1em] uppercase truncate" data-testid={
457 |                        {fr.faena}
458 |                      </CardTitle>
459 |                      <span className="text-[10px] font-mono text-muted-foreground">
460 |                        {fr.sectors.length} sectores · {fr.truckCount} camiones
461 |                      </span>
462 |                    </div>
463 |                    <div className="flex items-center gap-4 flex-wrap flex-shrink-0">
464 |                      <div className="text-right">
465 |                        <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em]">REND. FAENA</div>
466 |                        <div className={`text-lg font-mono font-bold ${rendColor(fr.globalRendimiento)}`} data-testid={`te
467 |                          {fr.globalRendimiento > 0 ? `${fr.globalRendimiento.toFixed(2)} km/L` : "\u2014"}
468 |                        </div>
469 |                      </div>
470 |                      {expandedFaena === fr.faena ? <ChevronUp className="w-4 h-4 text-muted-foreground" /> : <ChevronDown
471 |                    </div>
472 |                  </CardHeader>
473 |                  <CardContent className="pt-0">
474 |                    <div className="grid grid-cols-4 gap-3 mb-3 text-[10px] font-mono">
475 |                      <div>
476 |                        <span className="text-muted-foreground">Litros totales:</span>
477 |                        <span className="ml-1 text-amber-400 font-bold">{fN(Math.round(fr.totalLitros))}</span>
478 |                      </div>
479 |                      <div>
480 |                        <span className="text-muted-foreground">Km totales:</span>
481 |                        <span className="ml-1 text-foreground font-bold">{fN(Math.round(fr.totalKm))}</span>
482 |                      </div>
483 |                      <div>
484 |                        <span className="text-muted-foreground">Cargas:</span>
485 |                        <span className="ml-1 text-foreground font-bold">{fN(fr.totalCargas)}</span>
486 |                      </div>
487 |                      <div>
488 |                        <span className="text-muted-foreground">L/100km:</span>
489 |                        <span className="ml-1 text-foreground font-bold">
490 |                          {fr.globalRendimiento > 0 ? (100 / fr.globalRendimiento).toFixed(1) : "\u2014"}
491 |                        </span>
492 |                      </div>
493 |                    </div>
494 |
495 |                    {expandedFaena === fr.faena && (
496 |                      <div className="space-y-2 mt-3">
497 |                        {fr.sectors.length > 0 ? (
498 |                          fr.sectors.map((s, idx) => (
499 |                            <SectorCard
500 |                              key={s.sector}
501 |                              sector={s}
502 |                              faenaRend={fr.globalRendimiento}
503 |                              globalMax={globalMax}
504 |                              rank={idx + 1}
505 |                              total={fr.sectors.length}
506 |                            />
507 |                          ))
508 |                        ) : (
509 |                          <div className="p-4 text-center text-[11px] font-mono text-muted-foreground border border-border
510 |                            Sin sectores con datos suficientes (se requieren 2+ cargas por sector)
511 |                          </div>
512 |                        )}
513 |                      </div>
514 |                    )}
515 |                  </CardContent>
516 |                </Card>
517 |              ))}
518 |            </div>
519 |
520 |            {faenaRoutes.length === 0 && (
521 |              <div className="p-8 text-center text-[11px] font-mono text-muted-foreground border border-border bg-card" da
522 |                No se encontraron datos de rutas en el periodo seleccionado
523 |              </div>
524 |            )}
525 |          </>
526 |        )}
527 |      </div>
528 |    );
529 | }
530 |
```

client/src/pages/vision-ceo.tsx

428 lineas

```
1 | import { useState, useEffect } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { X, RefreshCw, Loader2 } from "lucide-react";
4 | import { LineChart, Line, XAxis, YAxis, CartesianGrid, Tooltip, ResponsiveContainer } from "recharts";
5 | import { apiRequest, queryClient } from "@/lib/queryClient";
6 |
7 | interface KPIs {
8 |   camionesTotal: number;
9 |   camionesActivos: number;
10 |   rendPromedio: number;
11 |   rendCambioPct: number;
12 |   litrosPeriodo: number;
13 |   litrosCambioPct: number;
14 |   alertasCriticas: number;
15 |   excesosVel: number;
16 |   microSospechas: number;
17 | }
18 |
19 | interface ContratoEstado {
20 |   nombre: string;
21 |   color: string;
22 |   camionesTotal: number;
23 |   rendPromedio: number;
24 |   alertasCriticas: number;
25 |   estado: "NORMAL" | "ATENCION" | "CRITICO";
26 | }
27 |
28 | interface TendenciaDay {
29 |   fecha: string;
30 |   litrosDia: number;
31 |   rendimientoDia: number;
32 | }
33 |
34 | interface Alerta {
35 |   tipo: string;
36 |   nivel: string;
37 |   titulo: string;
38 |   detalle: string;
39 |   hace: string;
40 |   gravedad: number;
41 | }
42 |
43 | interface DiagnosticoIA {
44 |   texto: string;
45 |   generadoHaceMin: number;
46 |   cached: boolean;
47 |   error?: string;
48 | }
49 |
50 | type Periodo = "1d" | "7d" | "30d";
51 |
52 | export default function VisionCEO({ onClose }: { onClose: () => void }) {
53 |   const [periodo, setPeriodo] = useState<Periodo>("7d");
54 |   const [lastRefresh, setLastRefresh] = useState(Date.now());
55 |
56 |   const periodoLabel = periodo === "1d" ? "HOY" : periodo === "7d" ? "7D" : "30D";
57 |   const diasTendencia = periodo === "1d" ? 1 : periodo === "7d" ? 7 : 30;
58 |
59 |   async function safeFetch<T>(url: string): Promise<T> {
60 |     const r = await fetch(url, { credentials: "include" });
61 |     if (!r.ok) throw new Error(`${r.status}: ${await r.text()}`);
62 |     return r.json();
63 |   }
64 |
65 |   const { data: kpis, isLoading: loadingKpis } = useQuery<KPIs>({
66 |     queryKey: ["/api/ceo/kpis", periodo],
67 |     queryFn: () => safeFetch<KPIs>(`/api/ceo/kpis?periodo=${periodo}`),
68 |     refetchInterval: 120000,
69 |   });
70 |
71 |   const { data: contratos = [], isLoading: loadingContratos } = useQuery<ContratoEstado[]>({
72 |     queryKey: ["/api/ceo/contratos-estado"],
73 |     refetchInterval: 120000,
74 |   });
75 |
76 |   const { data: tendencia = [], isLoading: loadingTendencia } = useQuery<TendenciaDay[]>({
77 |     queryKey: ["/api/ceo/tendencia", diasTendencia],
78 |     queryFn: () => safeFetch<TendenciaDay[]>(`/api/ceo/tendencia?dias=${diasTendencia}`),
79 |     refetchInterval: 120000,
80 |   });
81 |
82 |   const { data: alertas = [], isLoading: loadingAlertas } = useQuery<Alerta[]>({
83 |     queryKey: ["/api/ceo/alertas"],
84 |     queryFn: () => safeFetch<Alerta[]>("/api/ceo/alertas?limite=5"),
85 |     refetchInterval: 120000,
86 |   });
87 |
88 |   const { data: diagnostico, isLoading: loadingDiag, refetch: refetchDiag } = useQuery<DiagnosticoIA>({
89 |     queryKey: ["/api/ceo/diagnostico-ia"],
90 |     refetchInterval: 1800000,
91 |   });
92 |
93 |   const minAgo = Math.round((Date.now() - lastRefresh) / 60000);
94 |
95 |   function handleRefreshAll() {
96 |     setLastRefresh(Date.now());
97 |     queryClient.invalidateQueries({ queryKey: ["/api/ceo/kpis"] });
98 |     queryClient.invalidateQueries({ queryKey: ["/api/ceo/contratos-estado"] });
99 |     queryClient.invalidateQueries({ queryKey: ["/api/ceo/tendencia"] });
100 |     queryClient.invalidateQueries({ queryKey: ["/api/ceo/alertas"] });
101 |   }
102 |
103 |   function handleRefreshDiag() {
104 |     queryClient.invalidateQueries({ queryKey: ["/api/ceo/diagnostico-ia"] });
```

```
105 |     queryClient.fetchQuery({
106 |       queryKey: ["/api/ceo/diagnostico-ia"],
107 |       queryFn: () => fetch("/api/ceo/diagnostico-ia?force=true").then(r => { if (!r.ok) throw new Error("Error"); return
108 |     });
109 |   }
110 |
111 |   useEffect(() => {
112 |     document.body.style.overflow = "hidden";
113 |     return () => { document.body.style.overflow = ""; };
114 |   }, []);
115 |
116 |   const estadoBadge = (() => {
117 |     const c = kpis?.alertasCriticas ?? 0;
118 |     if (c === 0) return { label: "OPERACION NORMAL", color: "#00ff88", blink: false };
119 |     if (c <= 3) return { label: "ATENCION REQUERIDA", color: "#ffcc00", blink: false };
120 |     return { label: "ALERTA ACTIVA", color: "#ff2244", blink: true };
121 |   })();
122 |
123 |   const fechaHoy = new Date().toLocaleDateString("es-CL", {
124 |     weekday: "long", year: "numeric", month: "long", day: "numeric",
125 |   });
126 |
127 |   function parseDiagnostico(texto: string) {
128 |     const lines = texto.split("\n").filter(l => l.trim());
129 |     const result: { icon: string; label: string; text: string }[] = [];
130 |     for (const line of lines) {
131 |       if (line.startsWith("ESTADO:")) {
132 |         result.push({ icon: "\ud83d\udfe2", label: "ESTADO", text: line.replace("ESTADO:", "").trim() });
133 |       } else if (line.startsWith("POSITIVO:")) {
134 |         result.push({ icon: "\u2705", label: "POSITIVO", text: line.replace("POSITIVO:", "").trim() });
135 |       } else if (line.startsWith("ATENCION:") || line.startsWith("ATENCI\u00d3N:")) {
136 |         result.push({ icon: "\u26a0\ufe0f", label: "ATENCION", text: line.replace(/ATENCI[OÓ]N:/, "").trim() });
137 |       } else if (line.startsWith("ACCION:") || line.startsWith("ACCI\u00d3N:")) {
138 |         result.push({ icon: "\u25b6\ufe0f", label: "ACCION", text: line.replace(/ACCI[OÓ]N:/, "").trim() });
139 |       }
140 |     }
141 |     return result;
142 |   }
143 |
144 |   const estadoColors: Record<string, string> = { NORMAL: "#00ff88", ATENCION: "#ffcc00", CRITICO: "#ff2244" };
145 |
146 |   return (
147 |     <div className="fixed inset-0 z-[100] overflow-y-auto" style={{ background: "rgba(2,5,8,0.98)" }}
148 |       data-testid="ceo-panel">
149 |       <div className="max-w-[1400px] mx-auto px-5 py-5 space-y-5">
150 |
151 |         {/* ZONA 1 — HEADER */}
152 |         <div className="flex items-center justify-between flex-wrap gap-3" data-testid="ceo-header">
153 |           <div>
154 |             <div className="font-space text-[20px] font-bold tracking-[0.2em]" style={{ color: "#c8e8ff" }}>
155 |               SOTRASER &middot; VISION CEO
156 |             </div>
157 |             <div className="font-exo text-[11px] capitalize" style={{ color: "#3a6080" }}>{fechaHoy}</div>
158 |           </div>
159 |
160 |           <div className="flex items-center gap-2">
161 |             <span className="inline-block w-2.5 h-2.5 rounded-full" style={{
162 |               background: estadoBadge.color,
163 |               boxShadow: `0 0 8px ${estadoBadge.color}`,
164 |               animation: estadoBadge.blink ? "blinkFast 0.6s infinite" : undefined,
165 |             }} />
166 |             <span className="font-space text-[11px] font-bold tracking-[0.1em]" style={{ color: estadoBadge.color }}>
167 |               {estadoBadge.label}
168 |             </span>
169 |           </div>
170 |
171 |           <div className="flex items-center gap-3">
172 |             <div className="flex gap-1">
173 |               {(["1d", "7d", "30d"] as Periodo[]).map(p => (
174 |                 <button key={p} onClick={() => setPeriodo(p)}
175 |                   data-testid={`ceo-periodo-${p}`}
176 |                   className="font-space text-[10px] font-bold px-3 py-1 cursor-pointer transition-all border"
177 |                   style={{
178 |                     color: periodo === p ? "#020508" : "#3a6080",
179 |                     background: periodo === p ? "#00d4ff" : "transparent",
180 |                     borderColor: periodo === p ? "#00d4ff" : "#0d2035",
181 |                   }}>
182 |                   {p === "1d" ? "HOY" : p.toUpperCase()}
183 |                 </button>
184 |               ))}
185 |             </div>
186 |             <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>
187 |               Actualizado hace {minAgo < 1 ? "<1" : minAgo} min
188 |             </span>
189 |             <button onClick={handleRefreshAll} className="p-1.5 cursor-pointer transition-all hover:bg-[rgba(0,212,255,0
190 |               data-testid="ceo-refresh-all">
191 |               <RefreshCw className="w-3.5 h-3.5" style={{ color: "#00d4ff" }} />
192 |             </button>
193 |             <button onClick={onClose}
194 |               className="font-space text-[11px] font-bold px-4 py-1.5 cursor-pointer transition-all border hover:bg-[rgb
195 |               style={{ color: "#c8e8ff", borderColor: "#0d2035" }}
196 |               data-testid="ceo-close">
197 |               VOLVER AL DASHBOARD
198 |             </button>
199 |           </div>
200 |         </div>
201 |
202 |         {/* ZONA 2 — 4 KPIs GRANDES */}
203 |         <div className="grid grid-cols-2 lg:grid-cols-4 gap-4" data-testid="ceo-kpis">
204 |           <KpiCard
205 |             label="FLOTA ACTIVA"
206 |             value={kpis ? String(kpis.camionesActivos) : "--"}
207 |             sub={kpis ? `de ${kpis.camionesTotal} CENCOSUD` : ""}
208 |             progress={kpis ? Math.round((kpis.camionesActivos / Math.max(kpis.camionesTotal, 1)) * 100) : 0}
209 |             color="#00d4ff"
210 |             loading={loadingKpis}
211 |           />
```

```
212 |           <KpiCard
213 |             label="RENDIMIENTO FLOTA"
214 |             value={kpis ? String(kpis.rendPromedio) : "--"}
215 |             sub={kpis ? `${kpis.rendCambioPct >= 0 ? "\u2191" : "\u2193"} ${kpis.rendCambioPct >= 0 ? "+" : ""}${kpis.re
216 |             color={kpis ? (kpis.rendCambioPct >= 0 ? "#00ff88" : "#ff2244") : "#3a6080"}
217 |             unit="km/L"
218 |             loading={loadingKpis}
219 |           />
220 |           <KpiCard
221 |             label="LITROS PERIODO"
222 |             value={kpis ? kpis.litrosPeriodo.toLocaleString() : "--"}
223 |             sub={kpis ? `${kpis.litrosCambioPct >= 0 ? "\u2191" : "\u2193"} ${kpis.litrosCambioPct >= 0 ? "+" : ""}${kpi
224 |             color="#ffcc00"
225 |             loading={loadingKpis}
226 |           />
227 |           <KpiCard
228 |             label="ALERTAS ACTIVAS"
229 |             value={kpis ? String(kpis.alertasCriticas) : "--"}
230 |             sub={kpis?.alertasCriticas ? `${kpis.alertasCriticas} requieren accion hoy` : "Sin alertas criticas"}
231 |             color={kpis?.alertasCriticas ? "#ff2244" : "#00ff88"}
232 |             loading={loadingKpis}
233 |           />
234 |         </div>
235 |
236 |         {/* ZONA 3 — SEMAFORO POR CONTRATO */}
237 |         <div data-testid="ceo-contratos">
238 |           <div className="font-exo text-[9px] tracking-[0.2em] uppercase font-semibold mb-3" style={{ color: "#c8e8ff" }
239 |             ESTADO POR CONTRATO
240 |           </div>
241 |           {loadingContratos ? (
242 |             <div className="flex items-center gap-2 py-6 justify-center">
243 |               <Loader2 className="w-4 h-4 animate-spin" style={{ color: "#00d4ff" }} />
244 |               <span className="font-exo text-[11px]" style={{ color: "#3a6080" }}>Cargando contratos...</span>
245 |             </div>
246 |           ) : (
247 |             <div className="flex gap-3 overflow-x-auto pb-2">
248 |               {contratos.map((c, i) => (
249 |                 <div key={i} className="flex-shrink-0 min-w-[180px] p-4 rounded-sm" style={{
250 |                   background: "#091018",
251 |                   borderTop: `3px solid ${estadoColors[c.estado] || "#3a6080"}`,
252 |                   border: "1px solid #0d2035",
253 |                   borderTopColor: estadoColors[c.estado] || "#3a6080",
254 |                 }} data-testid={`ceo-contrato-${i}`}>
255 |                   <div className="flex items-center gap-2 mb-2">
256 |                     <span className="inline-block w-2 h-2 rounded-full" style={{ background: estadoColors[c.estado] }} /
257 |                     <span className="font-space text-[9px] font-bold tracking-[0.1em]" style={{ color: estadoColors[c.es
258 |                       {c.estado}
259 |                     </span>
260 |                   </div>
261 |                   <div className="font-space text-[13px] font-bold mb-1" style={{ color: "#c8e8ff" }}>{c.nombre}</div>
262 |                   <div className="font-exo text-[10px]" style={{ color: "#3a6080" }}>{c.camionesTotal} camiones</div>
263 |                   <div className="font-space text-[11px] font-bold mt-1" style={{ color: c.rendPromedio >= 2.5 ? "#00ff8
264 |                     {c.rendPromedio > 0 ? `${c.rendPromedio} km/L` : "N/D"}
265 |                   </div>
266 |                   <div className="font-exo text-[10px] mt-1" style={{ color: c.alertasCriticas > 0 ? "#ff2244" : "#3a608
267 |                     {c.alertasCriticas} alertas
268 |                   </div>
269 |                 </div>
270 |               ))}
271 |               {contratos.length === 0 && (
272 |                 <div className="font-exo text-[11px] py-4" style={{ color: "#3a6080" }}>Sin contratos con camiones asign
273 |               )}
274 |             </div>
275 |           )}
276 |         </div>
277 |
278 |         {/* ZONA 4+5 — TENDENCIA + ALERTAS */}
279 |         <div className="grid grid-cols-1 lg:grid-cols-[3fr_2fr] gap-4">
280 |           <div className="p-4 rounded-sm" style={{ background: "#091018", border: "1px solid #0d2035" }} data-testid="ce
281 |             <div className="font-exo text-[9px] tracking-[0.2em] uppercase font-semibold mb-3" style={{ color: "#c8e8ff"
282 |               TENDENCIA &middot; {periodoLabel}
283 |             </div>
284 |             {loadingTendencia ? (
285 |               <div className="flex items-center justify-center py-12">
286 |                 <Loader2 className="w-5 h-5 animate-spin" style={{ color: "#00d4ff" }} />
287 |               </div>
288 |             ) : (
289 |               <ResponsiveContainer width="100%" height={220}>
290 |                 <LineChart data={tendencia} margin={{ top: 5, right: 30, left: 0, bottom: 5 }}>
291 |                   <CartesianGrid strokeDasharray="3 3" stroke="#0d2035" />
292 |                   <XAxis
293 |                     dataKey="fecha"
294 |                     tick={{ fill: "#3a6080", fontSize: 9, fontFamily: "Space Mono" }}
295 |                     tickFormatter={(v: string) => { const d = new Date(v); return `${d.getDate()}/${d.getMonth() + 1}`;
296 |                     stroke="#0d2035"
297 |                   />
298 |                   <YAxis yAxisId="left" tick={{ fill: "#3a6080", fontSize: 9 }} stroke="#0d2035" />
299 |                   <YAxis yAxisId="right" orientation="right" tick={{ fill: "#00ff88", fontSize: 9 }} stroke="#0d2035" />
300 |                   <Tooltip
301 |                     contentStyle={{ background: "#091018", border: "1px solid #0d2035", fontFamily: "Exo 2", fontSize: 1
302 |                     labelFormatter={(v: string) => new Date(v).toLocaleDateString("es-CL", { weekday: "short", day: "num
303 |                   />
304 |                   <Line yAxisId="left" type="monotone" dataKey="litrosDia" name="Litros" stroke="#00d4ff" strokeWidth={2
305 |                   <Line yAxisId="right" type="monotone" dataKey="rendimientoDia" name="km/L" stroke="#00ff88" strokeWidt
306 |                 </LineChart>
307 |               </ResponsiveContainer>
308 |             )}
309 |           </div>
310 |
311 |           <div className="p-4 rounded-sm" style={{ background: "#091018", border: "1px solid #0d2035" }} data-testid="ce
312 |             <div className="font-exo text-[9px] tracking-[0.2em] uppercase font-semibold mb-3" style={{ color: "#c8e8ff"
313 |               REQUIERE ATENCION
314 |             </div>
315 |             {loadingAlertas ? (
316 |               <div className="flex items-center justify-center py-8">
317 |                 <Loader2 className="w-4 h-4 animate-spin" style={{ color: "#00d4ff" }} />
318 |               </div>
```

```
319 |             ) : alertas.length === 0 ? (
320 |               <div className="py-8 text-center">
321 |                 <div className="font-exo text-[13px]" style={{ color: "#00ff88" }}>Sin alertas criticas</div>
322 |               </div>
323 |             ) : (
324 |               <div className="space-y-2 max-h-[260px] overflow-y-auto">
325 |                 {alertas.map((a, i) => (
326 |                   <div key={i} className="p-3 rounded-sm transition-all hover:bg-[rgba(0,212,255,0.03)]"
327 |                     style={{ background: "rgba(2,5,8,0.5)", border: "1px solid #0d2035" }}
328 |                     data-testid={`ceo-alerta-${i}`}>
329 |                     <div className="flex items-start gap-2">
330 |                       <span className="text-[12px] mt-0.5 flex-shrink-0">
331 |                         {a.nivel === "CRITICO" ? "\ud83d\udd34" : "\ud83d\udfe1"}
332 |                       </span>
333 |                       <div className="flex-1 min-w-0">
334 |                         <div className="font-space text-[11px] font-bold" style={{ color: "#c8e8ff" }}>{a.titulo}</div>
335 |                         <div className="font-exo text-[10px] mt-0.5" style={{ color: "#3a6080" }}>{a.detalle}</div>
336 |                         <div className="font-exo text-[9px] mt-1" style={{ color: "#1a3550" }}>{a.hace}</div>
337 |                       </div>
338 |                     </div>
339 |                   </div>
340 |                 ))}
341 |               </div>
342 |             )}
343 |           </div>
344 |         </div>
345 |
346 |         {/* ZONA 6 — DIAGNOSTICO IA */}
347 |         <div className="p-5 rounded-sm" style={{ background: "#091018", border: "1px solid #0d2035", borderLeft: "3px so
348 |           data-testid="ceo-diagnostico">
349 |           <div className="flex items-center justify-between mb-4">
350 |             <div className="flex items-center gap-2">
351 |               <span className="text-[16px]">{"\ud83e\udd16"}</span>
352 |               <span className="font-space text-[11px] font-bold tracking-[0.1em]" style={{ color: "#c8e8ff" }}>
353 |                 DIAGNOSTICO EJECUTIVO &middot; IA
354 |               </span>
355 |             </div>
356 |             <div className="flex items-center gap-3">
357 |               {diagnostico && (
358 |                 <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>
359 |                   Actualizado hace {diagnostico.generadoHaceMin < 1 ? "<1" : diagnostico.generadoHaceMin} min
360 |                   {diagnostico.cached && " (cache)"}
361 |                 </span>
362 |               )}
363 |               <button onClick={handleRefreshDiag} className="flex items-center gap-1.5 px-2.5 py-1 cursor-pointer transi
364 |                 style={{ borderColor: "rgba(0,212,255,0.2)", color: "#00d4ff" }}
365 |                 data-testid="ceo-refresh-diag">
366 |                 <RefreshCw class=Name="w-3 h-3" />
367 |                 <span class=Name="font-exo text-[9px]">Actualizar diagnostico</span>
368 |               </button>
369 |             </div>
370 |           </div>
371 |
372 |           {loadingDiag ? (
373 |             <div class=Name="flex items-center gap-3 py-6 justify-center">
374 |               <Loader2 class=Name="w-5 h-5 animate-spin" style={{ color: "#00d4ff" }} />
375 |               <span class=Name="font-exo text-[11px]" style={{ color: "#3a6080" }}>Generando diagnostico con IA...</span>
376 |             </div>
377 |           ) : diagnostico?.texto ? (
378 |             <div class=Name="space-y-2">
379 |               {parseDiagnostico(diagnostico.texto).map((line, i) => (
380 |                 <div key={i} class=Name="flex items-start gap-3 py-1.5">
381 |                   <span class=Name="text-[14px] flex-shrink-0">{line.icon}</span>
382 |                   <div>
383 |                     <span class=Name="font-space text-[10px] font-bold tracking-[0.1em] mr-2" style={{ color: "#00d4ff" }
384 |                       {line.label}:
385 |                     </span>
386 |                     <span class=Name="font-exo text-[12px]" style={{ color: "#c8e8ff" }}>{line.text}</span>
387 |                   </div>
388 |                 </div>
389 |               ))}
390 |             </div>
391 |           ) : (
392 |             <div class=Name="font-exo text-[11px] py-4" style={{ color: "#3a6080" }}>
393 |               No se pudo generar el diagnostico. Intente nuevamente.
394 |             </div>
395 |           )}
396 |         </div>
397 |
398 |       </div>
399 |     </div>
400 |   );
401 | }
402 |
403 | function KpiCard({ label, value, sub, color, unit, progress, loading }: {
404 |   label: string; value: string; sub: string; color: string; unit?: string; progress?: number; loading: boolean;
405 | }) {
406 |   return (
407 |     <div class=Name="p-5 rounded-sm" style={{ background: "#091018", border: "1px solid #0d2035" }}>
408 |       <div class=Name="font-exo text-[8px] tracking-[0.2em] uppercase mb-3" style={{ color: "#3a6080" }}>{label}</div>
409 |       {loading ? (
410 |         <Loader2 class=Name="w-6 h-6 animate-spin" style={{ color: "#3a6080" }} />
411 |       ) : (
412 |         <>
413 |           <div class=Name="flex items-baseline gap-2">
414 |             <span class=Name="font-space text-[32px] font-bold leading-none" style={{ color }}>{value}</span>
415 |             {unit && <span class=Name="font-exo text-[12px]" style={{ color: "#3a6080" }}>{unit}</span>}
416 |           </div>
417 |           <div class=Name="font-exo text-[10px] mt-2" style={{ color: "#3a6080" }}>{sub}</div>
418 |           {progress != null && (
419 |             <div class=Name="mt-3 h-1.5 rounded-sm overflow-hidden" style={{ background: "#0d2035" }}>
420 |               <div class=Name="h-full rounded-sm transition-all duration-1000" style={{ width: `${progress}%`, background
421 |             </div>
422 |           )}
423 |         </>
424 |       )}
425 |     </div>
```

```
426 |   );
427 | }
428 |
```

client/src/pages/combustible-ia.tsx

419 lineas

```
1 | import { useState, useMemo } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { Skeleton } from "@/components/ui/skeleton";
4 | import {
5 |   Fuel, Truck, TrendingUp, TrendingDown, Calendar, BarChart3, AlertTriangle, ChevronDown, ChevronUp
6 | } from "lucide-react";
7 | import { getErrorFleetNums, isErrorCarga } from "@/pages/errores";
8 |
9 | interface FusionCarga {
10 |   numGuia: number;
11 |   fecha: string;
12 |   litros: number;
13 |   odometro: number | null;
14 |   kmRecorrido: number | null;
15 |   rendimiento: number | null;
16 |   lugar: string | null;
17 |   tarjeta: string | null;
18 |   conductor: string | null;
19 | }
20 |
21 | interface FusionTruck {
22 |   fleetNum: string;
23 |   patenteReal: string | null;
24 |   vin: string | null;
25 |   modeloVolvo: string;
26 |   faenaSigetra: string | null;
27 |   conductorSigetra: string | null;
28 |   totalLitrosSigetra: number;
29 |   totalCargas: number;
30 |   rendPromedio: number;
31 |   odometroSigetra: number | null;
32 |   odometroVolvo: number | null;
33 |   deltaOdometro: number | null;
34 |   fuelLevelVolvo: number | null;
35 |   totalFuelUsedVolvo: number | null;
36 |   litrosVolvoPeriodo: number | null;
37 |   engineHoursVolvo: number | null;
38 |   gpsVolvo: { latitude: number | null; longitude: number | null } | null;
39 |   alertLevel: "ok" | "alerta" | "critico";
40 |   cargas: FusionCarga[];
41 | }
42 |
43 | interface FusionResponse {
44 |   from: string;
45 |   to: string;
46 |   totalTrucksMatched: number;
47 |   totalLitros: number;
48 |   totalCargas: number;
49 |   alertCount: number;
50 |   trucks: FusionTruck[];
51 | }
52 |
53 | type Granularity = "diario" | "semanal" | "mensual";
54 |
55 | interface PeriodRow {
56 |   sortKey: string;
57 |   label: string;
58 |   litros: number;
59 |   km: number;
60 |   cargas: number;
61 |   trucks: Set<string>;
62 |   rendimiento: number;
63 | }
64 |
65 | function fN(n: number): string {
66 |   return n.toLocaleString("es-CL");
67 | }
68 |
69 | const DATA_START_DATE = "2026-03-01";
70 |
71 | function getDateRange(days: number): { from: string; to: string } {
72 |   const to = new Date();
73 |   const from = new Date(Date.now() - days * 24 * 60 * 60 * 1000);
74 |   const minDate = new Date(DATA_START_DATE + "T00:00:00");
75 |   const effectiveFrom = from < minDate ? minDate : from;
76 |   return { from: effectiveFrom.toISOString().slice(0, 10), to: to.toISOString().slice(0, 10) };
77 | }
78 |
79 | function getWeekKey(dateStr: string): string {
80 |   const d = new Date(dateStr);
81 |   const day = d.getDay();
82 |   const monday = new Date(d);
83 |   monday.setDate(d.getDate() - ((day + 6) % 7));
84 |   return monday.toISOString().slice(0, 10);
85 | }
86 |
87 | function getMonthKey(dateStr: string): string {
88 |   return dateStr.slice(0, 7);
89 | }
90 |
91 | function RendBar({ value, max, color }: { value: number; max: number; color: string }) {
92 |   const pct = max > 0 ? Math.min((value / max) * 100, 100) : 0;
93 |   return (
94 |     <div className="h-2 w-full bg-muted rounded-md overflow-hidden">
95 |       <div className={`h-full rounded-md ${color}`} style={{ width: `${pct}%` }} />
96 |     </div>
97 |   );
98 | }
99 |
100 | export default function CombustibleIA() {
101 |   const [days, setDays] = useState(30);
102 |   const [granularity, setGranularity] = useState<Granularity>("diario");
103 |   const [showTrucks, setShowTrucks] = useState(false);
104 |
```

```
  const range = useMemo(() => getDateRange(days), [days]);

  const { data: fusion, isLoading, isError } = useQuery<FusionResponse>({
    queryKey: [`/api/sigetra/fusion?from=${range.from}&to=${range.to}`],
    refetchInterval: 600000,
  });

  const errorNums = useMemo(() => {
    if (!fusion?.trucks) return new Set<string>();
    return getErrorFleetNums(fusion.trucks);
  }, [fusion]);

  const allCargas = useMemo(() => {
    if (!fusion?.trucks) return [];
    return fusion.trucks
      .filter(t => !errorNums.has(t.fleetNum))
      .flatMap(t =>
        t.cargas
          .filter(c => !isErrorCarga(c) && c.litros > 0)
          .map(c => ({ ...c, fleetNum: t.fleetNum, patenteReal: t.patenteReal }))
      );
  }, [fusion, errorNums]);

  const periodRows = useMemo(() => {
    if (allCargas.length === 0) return [];

    const groups = new Map<string, { litros: number; km: number; cargas: number; trucks: Set<string> }>();

    allCargas.forEach(c => {
      const dateStr = c.fecha?.slice(0, 10) || "";
      if (!dateStr || dateStr.length < 10 || isNaN(new Date(dateStr + "T12:00:00").getTime())) return;
      let key: string;
      if (granularity === "diario") key = dateStr;
      else if (granularity === "semanal") key = getWeekKey(dateStr);
      else key = getMonthKey(dateStr);

      if (!groups.has(key)) groups.set(key, { litros: 0, km: 0, cargas: 0, trucks: new Set() });
      const g = groups.get(key)!;
      g.litros += c.litros;
      g.km += (c.kmRecorrido != null && c.kmRecorrido > 0 ? c.kmRecorrido : 0);
      g.cargas += 1;
      g.trucks.add(c.fleetNum);
    });

    const rows: PeriodRow[] = Array.from(groups.entries()).map(([key, data]) => {
      let label: string;
      if (granularity === "diario") {
        const d = new Date(key + "T12:00:00");
        const dayNames = ["Dom", "Lun", "Mar", "Mie", "Jue", "Vie", "Sab"];
        label = `${dayNames[d.getDay()]} ${key.slice(8, 10)}/${key.slice(5, 7)}`;
      } else if (granularity === "semanal") {
        const end = new Date(key);
        end.setDate(end.getDate() + 6);
        label = `Sem ${key.slice(8, 10)}/${key.slice(5, 7)} - ${end.toISOString().slice(8, 10)}/${end.toISOString().slic
      } else {
        const monthNames = ["Ene", "Feb", "Mar", "Abr", "May", "Jun", "Jul", "Ago", "Sep", "Oct", "Nov", "Dic"];
        const m = parseInt(key.slice(5, 7)) - 1;
        label = `${monthNames[m]} ${key.slice(0, 4)}`;
      }

      return {
        sortKey: key,
        label,
        litros: data.litros,
        km: data.km,
        cargas: data.cargas,
        trucks: data.trucks,
        rendimiento: data.litros > 0 ? data.km / data.litros : 0,
      };
    });

    rows.sort((a, b) => a.sortKey.localeCompare(b.sortKey));

    return rows;
  }, [allCargas, granularity]);

  const truckSummary = useMemo(() => {
    if (!fusion?.trucks) return [];
    return fusion.trucks
      .filter(t => !errorNums.has(t.fleetNum))
      .map(t => {
        const validCargas = t.cargas.filter(c => !isErrorCarga(c) && c.litros > 0);
        const totalLitros = validCargas.reduce((s, c) => s + c.litros, 0);
        const totalKm = validCargas.reduce((s, c) => s + (c.kmRecorrido != null && c.kmRecorrido > 0 ? c.kmRecorrido : 0
        return {
          fleetNum: t.fleetNum,
          patente: t.patenteReal,
          faena: t.faenaSigetra,
          conductor: t.conductorSigetra,
          litros: totalLitros,
          km: totalKm,
          cargas: validCargas.length,
          rend: totalLitros > 0 ? totalKm / totalLitros : 0,
        };
      })
      .filter(t => t.litros > 0)
      .sort((a, b) => b.litros - a.litros);
  }, [fusion, errorNums]);

  const totals = useMemo(() => {
    const totalLitros = periodRows.reduce((s, r) => s + r.litros, 0);
    const totalKm = periodRows.reduce((s, r) => s + r.km, 0);
    const totalCargas = periodRows.reduce((s, r) => s + r.cargas, 0);
    const avgRendimiento = totalLitros > 0 ? totalKm / totalLitros : 0;
    const avgDailyLitros = periodRows.length > 0 ? totalLitros / periodRows.length : 0;
    const avgDailyKm = periodRows.length > 0 ? totalKm / periodRows.length : 0;
    const maxLitrosRow = periodRows.length > 0 ? periodRows.reduce((max, r) => r.litros > max.litros ? r : max, periodRo
```

```
212 |     const minLitrosRow = periodRows.length > 0 ? periodRows.reduce((min, r) => r.litros < min.litros ? r : min, periodRo
213 |     return { totalLitros, totalKm, totalCargas, avgRendimiento, avgDailyLitros, avgDailyKm, maxLitrosRow, minLitrosRow }
214 |   }, [periodRows]);
215 |
216 |   const maxLitros = useMemo(() => Math.max(...periodRows.map(r => r.litros), 1), [periodRows]);
217 |
218 |   if (isLoading) {
219 |     return (
220 |       <div className="space-y-4" data-testid="combustible-loading">
221 |         <Skeleton className="h-20 w-full" />
222 |         <div className="grid grid-cols-4 gap-3">
223 |           {Array.from({ length: 4 }).map((_, i) => <Skeleton key={i} className="h-24" />)}
224 |         </div>
225 |         <Skeleton className="h-96 w-full" />
226 |       </div>
227 |     );
228 |   }
229 |
230 |   return (
231 |     <div className="space-y-4" data-testid="combustible-page">
232 |       <div className="flex items-center gap-3 flex-wrap">
233 |         <div className="flex items-center gap-2">
234 |           <Fuel className="w-5 h-5 text-amber-400" />
235 |           <h1 className="text-sm font-mono font-bold tracking-[0.15em] uppercase text-foreground" data-testid="text-comb
236 |             Combustible & Kilometraje
237 |           </h1>
238 |         </div>
239 |         <span className="text-[10px] font-mono text-muted-foreground">Analisis de consumo y distancia</span>
240 |       </div>
241 |
242 |       <div className="flex items-center gap-2 flex-wrap">
243 |         <span className="text-[10px] font-mono text-muted-foreground tracking-[0.15em]">PERIODO:</span>
244 |         {[7, 14, 30, 60].map(d => (
245 |           <button key={d} onClick={() => setDays(d)}
246 |             data-testid={`btn-comb-days-${d}`}
247 |             className={`px-3 py-1 text-[10px] font-mono font-bold border cursor-pointer transition-colors ${
248 |               days === d ? "bg-amber-500/20 border-amber-500 text-amber-400" : "bg-card border-border text-muted-foregro
249 |             }`}>
250 |             {d}D
251 |           </button>
252 |         ))}
253 |         <div className="w-px h-5 bg-border mx-1" />
254 |         <span className="text-[10px] font-mono text-muted-foreground tracking-[0.15em]">VISTA:</span>
255 |         {(["diario", "semanal", "mensual"] as const).map(g => (
256 |           <button key={g} onClick={() => setGranularity(g)}
257 |             data-testid={`btn-gran-${g}`}
258 |             className={`px-3 py-1 text-[10px] font-mono font-bold border cursor-pointer transition-colors ${
259 |               granularity === g ? "bg-amber-500/20 border-amber-500 text-amber-400" : "bg-card border-border text-muted-
260 |             }`}>
261 |             {g.toUpperCase()}
262 |           </button>
263 |         ))}
264 |       </div>
265 |
266 |       {isError && (
267 |         <div className="bg-red-500/10 border border-red-500/30 p-3 flex items-center gap-2" data-testid="combustible-err
268 |           <AlertTriangle className="w-4 h-4 text-red-400" />
269 |           <span className="text-[11px] font-mono text-red-400">Error al obtener datos.</span>
270 |         </div>
271 |       )}
272 |
273 |       {fusion && (
274 |         <>
275 |           <div className="grid grid-cols-2 sm:grid-cols-4 gap-3">
276 |             <div className="bg-card border border-border p-3.5">
277 |               <div className="flex items-center gap-2 mb-1.5">
278 |                 <Fuel className="w-4 h-4 text-amber-400" />
279 |                 <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">LITROS TOTAL</span>
280 |               </div>
281 |               <div className="text-2xl font-mono font-bold text-amber-400" data-testid="text-comb-litros">{fN(Math.round
282 |               <div className="text-[10px] font-mono text-muted-foreground">en {days} dias</div>
283 |             </div>
284 |             <div className="bg-card border border-border p-3.5">
285 |               <div className="flex items-center gap-2 mb-1.5">
286 |                 <Truck className="w-4 h-4 text-blue-400" />
287 |                 <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">KM TOTAL</span>
288 |               </div>
289 |               <div className="text-2xl font-mono font-bold text-blue-400" data-testid="text-comb-km">{fN(Math.round(tota
290 |               <div className="text-[10px] font-mono text-muted-foreground">{fN(totals.totalCargas)} cargas</div>
291 |             </div>
292 |             <div className="bg-card border border-border p-3.5">
293 |               <div className="flex items-center gap-2 mb-1.5">
294 |                 <BarChart3 className="w-4 h-4 text-emerald-400" />
295 |                 <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">REND. PROMEDIO</span>
296 |               </div>
297 |               <div className="text-2xl font-mono font-bold text-emerald-400" data-testid="text-comb-rend">
298 |                 {totals.avgRendimiento > 0 ? totals.avgRendimiento.toFixed(2) : "\u2014"}
299 |               </div>
300 |               <div className="text-[10px] font-mono text-muted-foreground">km/L global</div>
301 |             </div>
302 |             <div className="bg-card border border-border p-3.5">
303 |               <div className="flex items-center gap-2 mb-1.5">
304 |                 <Calendar className="w-4 h-4 text-primary" />
305 |                 <span className="text-[8px] font-mono text-muted-foreground tracking-[0.2em]">PROMEDIO/{granularity ===
306 |               </div>
307 |               <div class="text-lg font-mono font-bold text-primary" data-testid="text-comb-avg">
308 |                 {fN(Math.round(totals.avgDailyLitros))} L
309 |               </div>
310 |               <div class="text-[10px] font-mono text-muted-foreground">{fN(Math.round(totals.avgDailyKm))} km</div>
311 |             </div>
312 |           </div>
313 |
314 |           {totals.maxLitrosRow && totals.minLitrosRow && periodRows.length > 1 && (
315 |             <div class="grid grid-cols-2 gap-3">
316 |               <div class="bg-red-500/5 border border-red-500/20 p-3 flex items-center gap-3">
317 |                 <TrendingUp class="w-5 h-5 text-red-400 flex-shrink-0" />
318 |                 <div>
```

```
                  <div className="text-[8px] font-mono text-red-400 tracking-[0.15em]">MAYOR CONSUMO</div>
                  <div className="text-sm font-mono font-bold text-foreground" data-testid="text-comb-max">{totals.maxLi
                  <div className="text-xs font-mono text-red-400">{fN(Math.round(totals.maxLitrosRow.litros))} L · {fN(M
                </div>
              </div>
              <div className="bg-emerald-500/5 border border-emerald-500/20 p-3 flex items-center gap-3">
                <TrendingDown className="w-5 h-5 text-emerald-400 flex-shrink-0" />
                <div>
                  <div className="text-[8px] font-mono text-emerald-400 tracking-[0.15em]">MENOR CONSUMO</div>
                  <div className="text-sm font-mono font-bold text-foreground" data-testid="text-comb-min">{totals.minLi
                  <div className="text-xs font-mono text-emerald-400">{fN(Math.round(totals.minLitrosRow.litros))} L · {
                </div>
              </div>
            </div>
          )}

          <div className="bg-background border border-border" data-testid="combustible-table">
            <div className="grid grid-cols-[1fr_6rem_6rem_4rem_4rem_5rem] gap-2 px-3 py-2 border-b border-border bg-card
              <span className="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">PERIODO</span>
              <span className="text-[8px] font-mono font-bold text-amber-400/70 tracking-[0.15em] text-right">LITROS</sp
              <span class="text-[8px] font-mono font-bold text-blue-400/70 tracking-[0.15em] text-right">KM</span>
              <span class="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em] text-right">CARG.<
              <span class="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em] text-right">CAM.</
              <span class="text-[8px] font-mono font-bold text-emerald-400/70 tracking-[0.15em] text-right">REND.</s
            </div>
            <div class="divide-y divide-border/30 max-h-[500px] overflow-y-auto">
              {periodRows.map((row, idx) => (
                <div key={idx} class="grid grid-cols-[1fr_6rem_6rem_4rem_4rem_5rem] gap-2 px-3 py-2 items-center" da
                  <div class="flex items-center gap-2">
                    <span class="text-xs font-mono font-bold text-foreground">{row.label}</span>
                    <div class="flex-1">
                      <RendBar value={row.litros} max={maxLitros} color="bg-amber-500/60" />
                    </div>
                  </div>
                  <span class="text-xs font-mono font-bold text-amber-400 text-right">{fN(Math.round(row.litros))}</
                  <span class="text-xs font-mono text-blue-400 text-right">{fN(Math.round(row.km))}</span>
                  <span class="text-[10px] font-mono text-muted-foreground text-right">{row.cargas}</span>
                  <span class="text-[10px] font-mono text-muted-foreground text-right">{row.trucks.size}</span>
                  <span class={`text-xs font-mono font-bold text-right ${
                    row.rendimiento >= 1.8 ? "text-emerald-400" : row.rendimiento >= 1.2 ? "text-amber-400" : row.rendim
                  }`}>
                    {row.rendimiento > 0 ? row.rendimiento.toFixed(2) : "\u2014"}
                  </span>
                </div>
              ))}
            </div>
          </div>

          <div>
            <button
              onClick={() => setShowTrucks(!showTrucks)}
              class="flex items-center gap-2 text-[10px] font-mono font-bold text-primary cursor-pointer mb-2"
              data-testid="btn-toggle-trucks"
            >
              {showTrucks ? <ChevronUp class="w-3.5 h-3.5" /> : <ChevronDown class="w-3.5 h-3.5" />}
              {showTrucks ? "Ocultar detalle por camion" : `Ver detalle por camion (${truckSummary.length})`}
            </button>

            {showTrucks && truckSummary.length > 0 && (
              <div class="bg-background border border-border" data-testid="truck-detail-table">
                <div class="grid grid-cols-[3.5rem_4.5rem_1fr_6rem_6rem_3.5rem_5rem] gap-2 px-3 py-2 border-b border
                  <span class="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">N.INT</span>
                  <span class="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">PATENTE</span
                  <span class="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em]">FAENA</span>
                  <span class="text-[8px] font-mono font-bold text-amber-400/70 tracking-[0.15em] text-right">LITROS
                  <span class="text-[8px] font-mono font-bold text-blue-400/70 tracking-[0.15em] text-right">KM</spa
                  <span class="text-[8px] font-mono font-bold text-muted-foreground tracking-[0.15em] text-right">CA
                  <span class="text-[8px] font-mono font-bold text-emerald-400/70 tracking-[0.15em] text-right">REND
                </div>
                <div class="divide-y divide-border/30 max-h-[400px] overflow-y-auto">
                  {truckSummary.map((t, idx) => (
                    <div key={t.fleetNum} class="grid grid-cols-[3.5rem_4.5rem_1fr_6rem_6rem_3.5rem_5rem] gap-2 px-3
                      <span class="text-xs font-mono font-bold text-foreground">{t.fleetNum}</span>
                      <span class="text-[10px] font-mono text-muted-foreground">{t.patente || "\u2014"}</span>
                      <span class="text-[10px] font-mono text-foreground truncate">{t.faena || "\u2014"}</span>
                      <span class="text-xs font-mono font-bold text-amber-400 text-right">{fN(Math.round(t.litros))}
                      <span class="text-xs font-mono text-blue-400 text-right">{fN(Math.round(t.km))}</span>
                      <span class="text-[10px] font-mono text-muted-foreground text-right">{t.cargas}</span>
                      <span class={`text-xs font-mono font-bold text-right ${
                        t.rend >= 1.8 ? "text-emerald-400" : t.rend >= 1.2 ? "text-amber-400" : t.rend > 0 ? "text-red-4
                      }`}>
                        {t.rend > 0 ? t.rend.toFixed(2) : "\u2014"}
                      </span>
                    </div>
                  ))}
                </div>
              </div>
            )}
          </div>

          {periodRows.length === 0 && (
            <div class="p-8 text-center text-[11px] font-mono text-muted-foreground border border-border bg-card" da
              Sin datos de combustible en el periodo seleccionado
            </div>
          )}
        </>
      )}
    </div>
  );
}
```

client/src/pages/inteligencia.tsx

436 lineas

```
1 | import { useState } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { apiRequest } from "@/lib/queryClient";
4 | import {
5 |   Brain,
6 |   Truck,
7 |   Loader2,
8 |   AlertTriangle,
9 |   Droplets,
10 |   ChevronDown,
11 |   ChevronUp,
12 |   ArrowRight,
13 |   Search,
14 |   Fuel,
15 | } from "lucide-react";
16 | import type { Faena } from "@shared/schema";
17 |
18 | interface CamionContratoResult {
19 |   patente: string;
20 |   modelo: string;
21 |   conductor: string;
22 |   metaKmL: number;
23 |   online: boolean;
24 |   rendPromedio: number | null;
25 |   totalLitros: number;
26 |   cargas: number;
27 |   estado: string;
28 |   weeklyData: { semana: string; litros: number; rendimiento: number | null; cargas: number }[];
29 | }
30 |
31 | interface ContratoIAResult {
32 |   estado_general: string;
33 |   resumen: string;
34 |   rendimiento_contrato: string;
35 |   camiones_problema: string[];
36 |   camion_destacado: string;
37 |   riesgos: string[];
38 |   recomendaciones: string[];
39 |   score_contrato: number;
40 | }
41 |
42 | interface ContratoAnalisis {
43 |   faena: string;
44 |   color: string;
45 |   camionesTotal: number;
46 |   camionesOnline: number;
47 |   totalLitros: number;
48 |   rendPromedio: number | null;
49 |   criticos: number;
50 |   camiones: CamionContratoResult[];
51 |   ia: ContratoIAResult | null;
52 | }
53 |
54 | const estadoCamColors: Record<string, string> = { NORMAL: "#00ff88", ALERTA: "#ffcc00", CRITICO: "#ff2244" };
55 |
56 | function ScoreGauge({ score }: { score: number }) {
57 |   const color = score >= 80 ? "#00ff88" : score >= 60 ? "#ffcc00" : score >= 40 ? "#FF8C00" : "#ff2244";
58 |   const circumference = 2 * Math.PI * 40;
59 |   const offset = circumference - (score / 100) * circumference;
60 |
61 |   return (
62 |     <div className="relative w-24 h-24 flex items-center justify-center" data-testid="gauge-score-contrato">
63 |       <svg className="absolute w-full h-full -rotate-90" viewBox="0 0 100 100">
64 |         <circle cx="50" cy="50" r="40" fill="none" stroke="#0d2035" strokeWidth="6" />
65 |         <circle cx="50" cy="50" r="40" fill="none" stroke={color} strokeWidth="6"
66 |           strokeDasharray={circumference} strokeDashoffset={offset} strokeLinecap="round" />
67 |       </svg>
68 |       <div className="text-center">
69 |         <div className="font-space text-[22px] font-bold" style={{ color }}>{score}</div>
70 |         <div className="font-exo text-[7px] tracking-[0.15em]" style={{ color: "#3a6080" }}>SALUD</div>
71 |       </div>
72 |     </div>
73 |   );
74 | }
75 |
76 | function CamionRow({ cam, expanded, onToggle }: { cam: CamionContratoResult; expanded: boolean; onToggle: () => void })
77 |   const rendColor = cam.rendPromedio !== null
78 |     ? (cam.rendPromedio >= cam.metaKmL ? "#00ff88" : cam.rendPromedio >= cam.metaKmL * 0.7 ? "#ffcc00" : "#ff2244")
79 |     : "#3a6080";
80 |
81 |   return (
82 |     <div style={{ border: "1px solid #0d2035" }} data-testid={`contrato-camion-${cam.patente}`}>
83 |       <button onClick={onToggle}
84 |         className="w-full p-3 flex items-center gap-3 cursor-pointer transition-all hover:bg-[rgba(0,212,255,0.03)]"
85 |         style={{ background: "#0a1520" }}
86 |         data-testid={`btn-expand-${cam.patente}`}>
87 |         <span className="w-1.5 h-1.5 rounded-full flex-shrink-0"
88 |           style={{ background: cam.online ? "#00ff88" : "#3a6080", boxShadow: cam.online ? "0 0 4px #00ff88" : undefined
89 |         <span className="font-space text-[12px] font-bold w-[70px]" style={{ color: "#c8e8ff" }}>{cam.patente}</span>
90 |         <span className="font-exo text-[10px] flex-1 min-w-0 truncate" style={{ color: "#3a6080" }}>{cam.conductor}</spa
91 |         <span className="font-space text-[11px] font-bold" style={{ color: rendColor }}>
92 |           {cam.rendPromedio !== null ? `${cam.rendPromedio} km/L` : "--"}
93 |         </span>
94 |         <span class="font-exo text-[9px]" style={{ color: "#ffcc00" }}>{cam.totalLitros}L</span>
95 |         <span class="font-space text-[8px] font-bold px-1.5 py-0.5"
96 |           style={{ color: estadoCamColors[cam.estado], background: `${estadoCamColors[cam.estado]}15`, border: `1px soli
97 |           {cam.estado}
98 |         </span>
99 |         {expanded ? <ChevronUp className="w-3 h-3 flex-shrink-0" style={{ color: "#3a6080" }} />
100 |           : <ChevronDown className="w-3 h-3 flex-shrink-0" style={{ color: "#3a6080" }} />}
101 |       </button>
102 |
103 |       {expanded && (
104 |         <div className="p-3 space-y-3" style={{ background: "#091018", borderTop: "1px solid #0d2035" }}>
```

```
105 |           <div className="grid grid-cols-4 gap-2">
106 |             <div className="p-2" style={{ background: "#0a1520", border: "1px solid #0d2035" }}>
107 |               <div className="font-exo text-[7px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>MODELO</div>
108 |               <div className="font-space text-[11px] font-bold" style={{ color: "#c8e8ff" }}>{cam.modelo}</div>
109 |             </div>
110 |             <div className="p-2" style={{ background: "#0a1520", border: "1px solid #0d2035" }}>
111 |               <div className="font-exo text-[7px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>META</div>
112 |               <div className="font-space text-[11px] font-bold" style={{ color: "#00d4ff" }}>{cam.metaKmL} km/L</div>
113 |             </div>
114 |             <div className="p-2" style={{ background: "#0a1520", border: "1px solid #0d2035" }}>
115 |               <div className="font-exo text-[7px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>CARGAS</div>
116 |               <div className="font-space text-[11px] font-bold" style={{ color: "#c8e8ff" }}>{cam.cargas}</div>
117 |             </div>
118 |             <div className="p-2" style={{ background: "#0a1520", border: "1px solid #0d2035" }}>
119 |               <div className="font-exo text-[7px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>LITROS</div>
120 |               <div className="font-space text-[11px] font-bold" style={{ color: "#ffcc00" }}>{cam.totalLitros.toLocaleSt
121 |             </div>
122 |           </div>
123 |
124 |           {cam.weeklyData.length > 0 && (
125 |             <div>
126 |               <div className="font-exo text-[8px] tracking-[0.15em] uppercase mb-2" style={{ color: "#3a6080" }}>
127 |                 RENDIMIENTO SEMANAL
128 |               </div>
129 |               <div className="flex items-end gap-1" style={{ height: 60 }}>
130 |                 {cam.weeklyData.map((w, i) => {
131 |                   const maxLt = Math.max(...cam.weeklyData.map(d => d.litros), 1);
132 |                   const h = Math.max((w.litros / maxLt) * 50, 2);
133 |                   const barColor = w.rendimiento !== null
134 |                     ? (w.rendimiento >= cam.metaKmL ? "#00ff88" : w.rendimiento >= cam.metaKmL * 0.7 ? "#ffcc00" : "#ff2
135 |                     : "#0d2035";
136 |                   return (
137 |                     <div key={i} className="flex flex-col items-center gap-0.5 flex-1" title={`${w.semana}: ${w.litros}L
138 |                       {w.rendimiento !== null && (
139 |                         <span className="font-space text-[7px]" style={{ color: barColor }}>{w.rendimiento}</span>
140 |                       )}
141 |                       <div style={{ height: h, background: barColor, width: "100%", minWidth: 8, opacity: 0.7 }} />
142 |                       <span className="font-exo text-[6px]" style={{ color: "#3a6080" }}>{w.semana}</span>
143 |                     </div>
144 |                   );
145 |                 })}
146 |               </div>
147 |             </div>
148 |           )}
149 |         </div>
150 |       )}
151 |     </div>
152 |   );
153 | }
154 |
155 | export default function Inteligencia() {
156 |   const [selectedFaenaId, setSelectedFaenaId] = useState<number | null>(null);
157 |   const [periodo, setPeriodo] = useState<string>("56");
158 |   const [result, setResult] = useState<ContratoAnalisis | null>(null);
159 |   const [loading, setLoading] = useState(false);
160 |   const [error, setError] = useState<string | null>(null);
161 |   const [expandedCamion, setExpandedCamion] = useState<string | null>(null);
162 |
163 |   const { data: faenas = [], isLoading: loadingFaenas } = useQuery<Faena[]>({ queryKey: ["/api/faenas"] });
164 |
165 |   const analyze = async () => {
166 |     if (!selectedFaenaId) return;
167 |     setLoading(true);
168 |     setError(null);
169 |     setResult(null);
170 |     try {
171 |       const res = await apiRequest("POST", "/api/ia/contrato", {
172 |         faenaId: selectedFaenaId,
173 |         dias: parseInt(periodo),
174 |       });
175 |       const data: ContratoAnalisis = await res.json();
176 |       setResult(data);
177 |     } catch (e: any) {
178 |       setError(e.message);
179 |     } finally {
180 |       setLoading(false);
181 |     }
182 |   };
183 |
184 |   if (loadingFaenas) {
185 |     return (
186 |       <div className="flex items-center gap-2 py-12 justify-center" data-testid="inteligencia-loading">
187 |         <Loader2 className="w-5 h-5 animate-spin" style={{ color: "#00d4ff" }} />
188 |         <span className="font-exo text-[11px]" style={{ color: "#3a6080" }}>Cargando contratos...</span>
189 |       </div>
190 |     );
191 |   }
192 |
193 |   return (
194 |     <div className="space-y-5" data-testid="inteligencia-view">
195 |       <div className="flex items-center gap-2">
196 |         <Brain className="w-4 h-4" style={{ color: "#00d4ff" }} />
197 |         <span className="font-space text-[12px] font-bold tracking-[0.15em]" style={{ color: "#c8e8ff" }}>
198 |           INTELIGENCIA POR CONTRATO
199 |         </span>
200 |         <span className="font-exo text-[9px]" style={{ color: "#3a6080" }}>
201 |           Analisis profundo con IA por contrato, detalle por camion
202 |         </span>
203 |       </div>
204 |
205 |       <div className="flex items-end gap-3 flex-wrap">
206 |         <div className="flex-1 min-w-[200px]">
207 |           <div className="font-exo text-[8px] tracking-[0.2em] uppercase mb-1.5" style={{ color: "#3a6080" }}>
208 |             SELECCIONAR CONTRATO
209 |           </div>
210 |           <div className="flex gap-2 flex-wrap">
211 |             {faenas.map(f => (
```

```
212 |               <button key={f.id} onClick={() => setSelectedFaenaId(f.id)}
213 |                 className="px-4 py-2.5 font-space text-[11px] font-bold cursor-pointer transition-all border"
214 |                 style={{
215 |                   background: selectedFaenaId === f.id ? `${f.color}15` : "rgba(9,16,24,0.8)",
216 |                   borderColor: selectedFaenaId === f.id ? f.color : "#0d2035",
217 |                   color: selectedFaenaId === f.id ? f.color : "#3a6080",
218 |                   boxShadow: selectedFaenaId === f.id ? `0 0 10px ${f.color}20` : undefined,
219 |                 }}
220 |                 data-testid={`btn-faena-${f.id}`}>
221 |                 {f.nombre}
222 |               </button>
223 |             ))}
224 |           </div>
225 |         </div>
226 |
227 |         <div>
228 |           <div className="font-exo text-[8px] tracking-[0.2em] uppercase mb-1.5" style={{ color: "#3a6080" }}>PERIODO</d
229 |           <div className="flex gap-1">
230 |             {[{ val: "28", label: "4 sem" }, { val: "56", label: "8 sem" }, { val: "84", label: "12 sem" }].map(p => (
231 |               <button key={p.val} onClick={() => setPeriodo(p.val)}
232 |                 className="px-3 py-2 font-space text-[10px] font-bold cursor-pointer transition-all border"
233 |                 style={{
234 |                   background: periodo === p.val ? "#00d4ff" : "transparent",
235 |                   color: periodo === p.val ? "#020508" : "#3a6080",
236 |                   borderColor: periodo === p.val ? "#00d4ff" : "#0d2035",
237 |                 }}
238 |                 data-testid={`btn-periodo-${p.val}`}>
239 |                 {p.label}
240 |               </button>
241 |             ))}
242 |           </div>
243 |         </div>
244 |
245 |         <button onClick={analyze} disabled={!selectedFaenaId || loading}
246 |           className="flex items-center gap-2 px-6 py-2.5 font-space text-[11px] font-bold tracking-[0.1em] cursor-pointe
247 |           style={{ background: "rgba(0,212,255,0.1)", borderColor: "rgba(0,212,255,0.3)", color: "#00d4ff" }}
248 |           data-testid="btn-analizar-contrato">
249 |           {loading ? (
250 |             <><Loader2 className="w-4 h-4 animate-spin" /> ANALIZANDO...</>
251 |           ) : (
252 |             <><Brain className="w-4 h-4" /> ANALIZAR CONTRATO</>
253 |           )}
254 |         </button>
255 |       </div>
256 |
257 |       {!result && !loading && !error && (
258 |         <div className="flex flex-col items-center justify-center py-16 text-center" data-testid="empty-state">
259 |           <div className="octagonal w-16 h-16 flex items-center justify-center mb-4"
260 |             style={{ background: "radial-gradient(circle, rgba(0,212,255,0.06) 0%, transparent 70%)", border: "1px solid
261 |             <Search className="w-7 h-7" style={{ color: "#0d2035" }} />
262 |           </div>
263 |           <div className="font-space text-[12px] font-bold mb-1" style={{ color: "#3a6080" }}>
264 |             Selecciona un contrato y presiona "ANALIZAR CONTRATO"
265 |           </div>
266 |           <div className="font-exo text-[10px]" style={{ color: "#1a3550" }}>
267 |             Analisis IA del contrato con detalle por cada camion asignado
268 |           </div>
269 |         </div>
270 |       )}
271 |
272 |       {loading && (
273 |         <div className="flex flex-col items-center justify-center py-16 gap-3" data-testid="loading-state">
274 |           <Loader2 className="w-8 h-8 animate-spin" style={{ color: "#00d4ff" }} />
275 |           <div class="font-space text-[12px]" style={{ color: "#c8e8ff" }}>Analizando contrato con IA...</div>
276 |           <div class="font-exo text-[10px]" style={{ color: "#3a6080" }}>Evaluando cada camion del contrato</div>
277 |         </div>
278 |       )}
279 |
280 |       {error && (
281 |         <div className="p-4" style={{ background: "#091018", border: "1px solid rgba(255,34,68,0.3)" }} data-testid="err
282 |           <div className="flex items-center gap-2 mb-2">
283 |             <AlertTriangle className="w-4 h-4" style={{ color: "#ff2244" }} />
284 |             <span class="font-space text-[11px] font-bold" style={{ color: "#ff2244" }}>Error en el analisis</span>
285 |           </div>
286 |           <div class="font-exo text-[10px]" style={{ color: "#c8e8ff" }}>{error}</div>
287 |           <button onClick={analyze}
288 |             class="mt-2 px-3 py-1.5 font-exo text-[10px] font-bold cursor-pointer border hover:border-[#00d4ff]"
289 |             style={{ borderColor: "#0d2035", color: "#3a6080" }}
290 |             data-testid="btn-retry">
291 |             Reintentar
292 |           </button>
293 |         </div>
294 |       )}
295 |
296 |       {result && result.ia && (
297 |         <div class="space-y-4" data-testid="resultado-contrato">
298 |           <div class="p-4" style={{ background: "#091018", border: "1px solid #0d2035", borderTop: `3px solid ${resu
299 |             <div class="flex items-start gap-4">
300 |               <ScoreGauge score={result.ia.score_contrato} />
301 |               <div class="flex-1 min-w-0">
302 |                 <div class="flex items-center gap-2 mb-1">
303 |                   <span class="font-space text-[14px] font-bold tracking-[0.1em]" style={{ color: "#c8e8ff" }}>
304 |                     {result.faena}
305 |                   </span>
306 |                   <span class="font-space text-[9px] font-bold px-2 py-0.5"
307 |                     style={{
308 |                       color: estadoCamColors[result.ia.estado_general] || "#00ff88",
309 |                       background: `${estadoCamColors[result.ia.estado_general] || "#00ff88"}15`,
310 |                       border: `1px solid ${estadoCamColors[result.ia.estado_general] || "#00ff88"}30`,
311 |                     }}>
312 |                     {result.ia.estado_general}
313 |                   </span>
314 |                 </div>
315 |                 <div class="font-rajdhani text-[12px] leading-relaxed mb-3" style={{ color: "#c8e8ff" }}>
316 |                   {result.ia.resumen}
317 |                 </div>
318 |                 <div class="grid grid-cols-2 sm:grid-cols-4 gap-2">
```

```
319 |                   <MiniStat label="CAMIONES" value={`${result.camionesOnline}/${result.camionesTotal}`} color="#00d4ff"
320 |                     icon={<Truck className="w-3 h-3" />} />
321 |                   <MiniStat label="LITROS TOTAL" value={result.totalLitros.toLocaleString()} color="#ffcc00"
322 |                     icon={<Fuel className="w-3 h-3" />} />
323 |                   <MiniStat label="REND. PROM" value={result.rendPromedio !== null ? `${result.rendPromedio} km/L` : "N/
324 |                     icon={<Droplets className="w-3 h-3" />} />
325 |                   <MiniStat label="CRITICOS" value={String(result.criticos)} color={result.criticos > 0 ? "#ff2244" : "#
326 |                     icon={<AlertTriangle className="w-3 h-3" />} />
327 |                 </div>
328 |               </div>
329 |             </div>
330 |           </div>
331 |
332 |           {result.ia.rendimiento_contrato && (
333 |             <div className="p-3" style={{ background: "#091018", border: "1px solid #0d2035", borderLeft: "3px solid #00
334 |               <div className="font-space text-[8px] font-bold tracking-[0.15em] mb-1" style={{ color: "#00d4ff" }}>RENDI
335 |               <div className="font-rajdhani text-[12px] leading-relaxed" style={{ color: "#c8e8ff" }}>
336 |                 {result.ia.rendimiento_contrato}
337 |               </div>
338 |             </div>
339 |           )}
340 |
341 |           <div className="grid grid-cols-1 lg:grid-cols-2 gap-3">
342 |             {result.ia.camiones_problema.length > 0 && (
343 |               <div className="p-3" style={{ background: "#091018", border: "1px solid rgba(255,34,68,0.2)" }}>
344 |                 <div className="font-space text-[8px] font-bold tracking-[0.15em] mb-2" style={{ color: "#ff2244" }}>
345 |                   CAMIONES CON PROBLEMAS
346 |                 </div>
347 |                 <div className="space-y-1">
348 |                   {result.ia.camiones_problema.map((p, i) => (
349 |                     <div key={i} className="flex items-start gap-2">
350 |                       <AlertTriangle className="w-3 h-3 mt-0.5 flex-shrink-0" style={{ color: "#ff2244" }} />
351 |                       <span className="font-exo text-[10px]" style={{ color: "#c8e8ff" }}>{p}</span>
352 |                     </div>
353 |                   ))}
354 |                 </div>
355 |               </div>
356 |             )}
357 |
358 |             {result.ia.riesgos.length > 0 && (
359 |               <div className="p-3" style={{ background: "#091018", border: "1px solid rgba(255,204,0,0.2)" }}>
360 |                 <div className="font-space text-[8px] font-bold tracking-[0.15em] mb-2" style={{ color: "#ffcc00" }}>
361 |                   RIESGOS DETECTADOS
362 |                 </div>
363 |                 <div className="space-y-1">
364 |                   {result.ia.riesgos.map((r, i) => (
365 |                     <div key={i} className="flex items-start gap-2">
366 |                       <span className="font-exo text-[10px]" style={{ color: "#c8e8ff" }}>- {r}</span>
367 |                     </div>
368 |                   ))}
369 |                 </div>
370 |               </div>
371 |             )}
372 |           </div>
373 |
374 |           {result.ia.camion_destacado && (
375 |             <div className="p-3" style={{ background: "#091018", border: "1px solid rgba(0,255,136,0.2)" }}>
376 |               <div className="font-space text-[8px] font-bold tracking-[0.15em] mb-1" style={{ color: "#00ff88" }}>
377 |                 CAMION DESTACADO
378 |               </div>
379 |               <div className="font-exo text-[10px]" style={{ color: "#c8e8ff" }}>{result.ia.camion_destacado}</div>
380 |             </div>
381 |           )}
382 |
383 |           {result.ia.recomendaciones.length > 0 && (
384 |             <div className="p-3" style={{ background: "#091018", border: "1px solid rgba(0,212,255,0.2)" }}>
385 |               <div className="font-space text-[8px] font-bold tracking-[0.15em] mb-2" style={{ color: "#00d4ff" }}>
386 |                 RECOMENDACIONES
387 |               </div>
388 |               <div className="space-y-1.5">
389 |                 {result.ia.recomendaciones.map((r, i) => (
390 |                   <div key={i} className="flex items-start gap-2">
391 |                     <ArrowRight className="w-3 h-3 mt-0.5 flex-shrink-0" style={{ color: "#00d4ff" }} />
392 |                     <span className="font-exo text-[10px]" style={{ color: "#c8e8ff" }}>{r}</span>
393 |                   </div>
394 |                 ))}
395 |               </div>
396 |             </div>
397 |           )}
398 |
399 |           <div>
400 |             <div className="font-exo text-[9px] tracking-[0.2em] uppercase font-semibold mb-2" style={{ color: "#c8e8ff"
401 |               DETALLE POR CAMION ({result.camiones.length})
402 |             </div>
403 |             <div className="space-y-1.5">
404 |               {result.camiones.map(cam => (
405 |                 <CamionRow key={cam.patente} cam={cam}
406 |                   expanded={expandedCamion === cam.patente}
407 |                   onToggle={() => setExpandedCamion(expandedCamion === cam.patente ? null : cam.patente)} />
408 |               ))}
409 |             </div>
410 |           </div>
411 |         </div>
412 |       )}
413 |
414 |       {result && !result.ia && (
415 |         <div className="p-4 text-center" style={{ background: "#091018", border: "1px solid #0d2035" }}>
416 |           <div className="font-exo text-[11px]" style={{ color: "#3a6080" }}>
417 |             Sin camiones asignados a este contrato
418 |           </div>
419 |         </div>
420 |       )}
421 |     </div>
422 |   );
423 | }
424 |
425 | function MiniStat({ label, value, color, icon }: { label: string; value: string; color: string; icon?: React.ReactNode }
```

```
426 |   return (
427 |     <div className="p-2" style={{ background: "#0a1520", border: "1px solid #0d2035" }}>
428 |       <div className="flex items-center gap-1 mb-0.5">
429 |         {icon && <span style={{ color: "#3a6080" }}>{icon}</span>}
430 |         <span className="font-exo text-[7px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>{label}</span>
431 |       </div>
432 |       <div className="font-space text-[13px] font-bold" style={{ color }}>{value}</div>
433 |     </div>
434 |   );
435 | }
436 |
```

client/src/pages/peso-tara.tsx

356 lineas

```
1 | import { useState } from "react";
2 | import { useQuery, useMutation } from "@tanstack/react-query";
3 | import { queryClient, apiRequest } from "@/lib/queryClient";
4 | import { Loader2, AlertTriangle, Weight, Plus, X, Truck, CheckCircle } from "lucide-react";
5 |
6 | interface PesoTaraRecord {
7 |   id: number;
8 |   patente: string;
9 |   modelo: string;
10 |   faena: string | null;
11 |   tipoVehiculo: string | null;
12 |   taraKg: number;
13 |   pesoMaximoKg: number | null;
14 |   capacidadCargaKg: number | null;
15 |   cargaEstimada: number | null;
16 |   pesoTotal: number | null;
17 |   pesoVolvoKg: number | null;
18 |   cumple: "CUMPLE" | "AL_LIMITE" | "EXCEDE" | "SIN_DATOS";
19 |   ultimaCarga: { fecha: string; litros: number; lugar: string } | null;
20 |   anioFabricacion: number | null;
21 |   configuracionEjes: string | null;
22 | }
23 |
24 | interface PesoTaraData {
25 |   registros: PesoTaraRecord[];
26 |   totals: {
27 |     conTara: number;
28 |     cumplen: number;
29 |     exceden: number;
30 |     alLimite: number;
31 |     sinDatos: number;
32 |     totalCamiones: number;
33 |   };
34 | }
35 |
36 | function CumpleBadge({ cumple }: { cumple: string }) {
37 |   const cfg: Record<string, { color: string; bg: string; border: string; label: string }> = {
38 |     CUMPLE: { color: "#00ff88", bg: "rgba(0,255,136,0.08)", border: "rgba(0,255,136,0.3)", label: "CUMPLE" },
39 |     AL_LIMITE: { color: "#ffcc00", bg: "rgba(255,204,0,0.08)", border: "rgba(255,204,0,0.3)", label: "AL LIMITE" },
40 |     EXCEDE: { color: "#ff2244", bg: "rgba(255,34,68,0.1)", border: "rgba(255,34,68,0.3)", label: "EXCEDE" },
41 |     SIN_DATOS: { color: "#3a6080", bg: "rgba(58,96,128,0.08)", border: "rgba(58,96,128,0.3)", label: "SIN DATOS" },
42 |   };
43 |   const c = cfg[cumple] || cfg.SIN_DATOS;
44 |   return (
45 |     <span className="font-space text-[8px] font-bold px-2 py-0.5 uppercase tracking-[0.1em]"
46 |       style={{ color: c.color, background: c.bg, border: `1px solid ${c.border}` }}>
47 |       {c.label}
48 |     </span>
49 |   );
50 | }
51 |
52 | function TaraModal({ onClose, camiones }: { onClose: () => void; camiones: { id: number; patente: string }[] }) {
53 |   const [selectedId, setSelectedId] = useState<number | null>(null);
54 |   const [taraKg, setTaraKg] = useState("");
55 |   const [pesoMaximoKg, setPesoMaximoKg] = useState("");
56 |   const [tipoVehiculo, setTipoVehiculo] = useState("");
57 |   const [anioFabricacion, setAnioFabricacion] = useState("");
58 |   const [configuracionEjes, setConfiguracionEjes] = useState("");
59 |
60 |   const mutation = useMutation({
61 |     mutationFn: async () => {
62 |       if (!selectedId || !taraKg) throw new Error("Seleccione camion y ingrese tara");
63 |       return apiRequest("PATCH", `/api/camiones/${selectedId}/tara`, {
64 |         taraKg: parseFloat(taraKg),
65 |         pesoMaximoKg: pesoMaximoKg ? parseFloat(pesoMaximoKg) : null,
66 |         tipoVehiculo: tipoVehiculo || null,
67 |         anioFabricacion: anioFabricacion ? parseInt(anioFabricacion) : null,
68 |         configuracionEjes: configuracionEjes || null,
69 |         capacidadCargaKg: pesoMaximoKg && taraKg ? parseFloat(pesoMaximoKg) - parseFloat(taraKg) : null,
70 |       });
71 |     },
72 |     onSuccess: () => {
73 |       queryClient.invalidateQueries({ queryKey: ["/api/datos/peso-tara"] });
74 |       onClose();
75 |     },
76 |   });
77 |
78 |   return (
79 |     <div className="fixed inset-0 z-50 flex items-center justify-center" style={{ background: "rgba(2,5,8,0.85)" }}
80 |       onClick={onClose}>
81 |       <div className="dash-card w-full max-w-lg mx-4 overflow-hidden" onClick={e => e.stopPropagation()}>
82 |         <div className="flex items-center justify-between px-5 py-3" style={{ borderBottom: "1px solid #0d2035" }}>
83 |           <h3 className="font-space text-[14px] font-bold tracking-[0.1em]" style={{ color: "#c8e8ff" }}>
84 |             Registrar Tara de Camion
85 |           </h3>
86 |           <button onClick={onClose} className="p-1 hover:opacity-70" data-testid="btn-close-tara-modal">
87 |             <X className="w-4 h-4" style={{ color: "#3a6080" }} />
88 |           </button>
89 |         </div>
90 |         <div className="px-5 py-4 space-y-4">
91 |           <div>
92 |             <label className="font-exo text-[9px] tracking-[0.15em] uppercase block mb-1" style={{ color: "#3a6080" }}>C
93 |             <select
94 |               className="w-full px-3 py-2 font-space text-[12px]"
95 |               style={{ background: "#020508", border: "1px solid #0d2035", color: "#c8e8ff" }}
96 |               value={selectedId || ""}
97 |               onChange={e => setSelectedId(parseInt(e.target.value))}
98 |               data-testid="select-camion-tara"
99 |             >
100 |               <option value="">Seleccionar patente...</option>
101 |               {camiones.map(c => (
102 |                 <option key={c.id} value={c.id}>{c.patente}</option>
103 |               ))}
104 |             </select>
```

```
105 |           </div>
106 |           <div className="grid grid-cols-2 gap-3">
107 |             <div>
108 |               <label className="font-exo text-[9px] tracking-[0.15em] uppercase block mb-1" style={{ color: "#3a6080" }}
109 |               <input type="number" placeholder="Ej: 8500"
110 |                 className="w-full px-3 py-2 font-space text-[12px]"
111 |                 style={{ background: "#020508", border: "1px solid #0d2035", color: "#c8e8ff" }}
112 |                 value={taraKg} onChange={e => setTaraKg(e.target.value)}
113 |                 data-testid="input-tara-kg" />
114 |             </div>
115 |             <div>
116 |               <label className="font-exo text-[9px] tracking-[0.15em] uppercase block mb-1" style={{ color: "#3a6080" }}
117 |               <input type="number" placeholder="Ej: 45000"
118 |                 className="w-full px-3 py-2 font-space text-[12px]"
119 |                 style={{ background: "#020508", border: "1px solid #0d2035", color: "#c8e8ff" }}
120 |                 value={pesoMaximoKg} onChange={e => setPesoMaximoKg(e.target.value)}
121 |                 data-testid="input-peso-max" />
122 |             </div>
123 |           </div>
124 |           <div>
125 |             <label className="font-exo text-[9px] tracking-[0.15em] uppercase block mb-1" style={{ color: "#3a6080" }}>T
126 |             <select
127 |               className="w-full px-3 py-2 font-space text-[12px]"
128 |               style={{ background: "#020508", border: "1px solid #0d2035", color: "#c8e8ff" }}
129 |               value={tipoVehiculo} onChange={e => setTipoVehiculo(e.target.value)}
130 |               data-testid="select-tipo-vehiculo"
131 |             >
132 |               <option value="">Seleccionar...</option>
133 |               <option value="CAMION RIGIDO">CAMION RIGIDO</option>
134 |               <option value="ARTICULADO">ARTICULADO</option>
135 |               <option value="TRACTO-CAMION">TRACTO-CAMION</option>
136 |               <option value="OTRO">OTRO</option>
137 |             </select>
138 |           </div>
139 |           <div class="grid grid-cols-2 gap-3">
140 |             <div>
141 |               <label class="font-exo text-[9px] tracking-[0.15em] uppercase block mb-1" style={{ color: "#3a6080" }}
142 |               <input type="number" placeholder="Ej: 2022"
143 |                 class="w-full px-3 py-2 font-space text-[12px]"
144 |                 style={{ background: "#020508", border: "1px solid #0d2035", color: "#c8e8ff" }}
145 |                 value={anioFabricacion} onChange={e => setAnioFabricacion(e.target.value)}
146 |                 data-testid="input-anio" />
147 |             </div>
148 |             <div>
149 |               <label class="font-exo text-[9px] tracking-[0.15em] uppercase block mb-1" style={{ color: "#3a6080" }}
150 |               <input type="text" placeholder="Ej: 6x4"
151 |                 class="w-full px-3 py-2 font-space text-[12px]"
152 |                 style={{ background: "#020508", border: "1px solid #0d2035", color: "#c8e8ff" }}
153 |                 value={configuracionEjes} onChange={e => setConfiguracionEjes(e.target.value)}
154 |                 data-testid="input-ejes" />
155 |             </div>
156 |           </div>
157 |           {mutation.isError && (
158 |             <div class="font-exo text-[10px]" style={{ color: "#ff2244" }}>
159 |               {(mutation.error as Error).message}
160 |             </div>
161 |           )}
162 |           <button
163 |             class="w-full py-2.5 font-space text-[11px] font-bold tracking-[0.1em] uppercase transition-all hover:op
164 |             style={{ background: "#00d4ff", color: "#020508" }}
165 |             onClick={() => mutation.mutate()}
166 |             disabled={mutation.isPending || !selectedId || !taraKg}
167 |             data-testid="btn-guardar-tara"
168 |           >
169 |             {mutation.isPending ? "GUARDANDO..." : "GUARDAR"}
170 |           </button>
171 |         </div>
172 |       </div>
173 |     </div>
174 |   );
175 | }
176 |
177 | export default function PesoTara() {
178 |   const [showModal, setShowModal] = useState(false);
179 |
180 |   const { data, isLoading, error } = useQuery<PesoTaraData>({
181 |     queryKey: ["/api/datos/peso-tara"],
182 |   });
183 |
184 |   const { data: allCamiones } = useQuery<any[]>({
185 |     queryKey: ["/api/camiones"],
186 |   });
187 |
188 |   if (isLoading) {
189 |     return (
190 |       <div class="flex items-center justify-center py-20" data-testid="peso-tara-loading">
191 |         <Loader2 class="w-8 h-8 animate-spin" style={{ color: "#00d4ff" }} />
192 |         <span class="ml-3 font-exo text-[12px]" style={{ color: "#3a6080" }}>Cargando datos de peso y tara...</span>
193 |       </div>
194 |     );
195 |   }
196 |
197 |   if (error) {
198 |     return (
199 |       <div class="dash-card px-8 py-12 text-center">
200 |         <AlertTriangle class="w-8 h-8 mx-auto mb-3" style={{ color: "#ff2244" }} />
201 |         <div class="font-space text-[14px] font-bold" style={{ color: "#ff2244" }}>Error al cargar datos</div>
202 |       </div>
203 |     );
204 |   }
205 |
206 |   if (!data) return null;
207 |   const { registros, totals } = data;
208 |
209 |   return (
210 |     <div class="space-y-5" data-testid="peso-tara-page">
211 |       <div class="flex items-center justify-between">
```

```
212 |          <div>
213 |            <h2 className="font-space text-[18px] font-bold tracking-[0.12em] uppercase" style={{ color: "#c8e8ff" }}>
214 |              Peso y Tara
215 |            </h2>
216 |            <p className="font-exo text-[11px] mt-1" style={{ color: "#3a6080" }}>
217 |              Solo mostrando {totals.conTara} camiones con datos de tara registrados de {totals.totalCamiones} CENCOSUD
218 |            </p>
219 |          </div>
220 |          <div className="flex items-center gap-3">
221 |            <span className="font-exo text-[9px] px-2 py-1" style={{ color: "#3a6080", border: "1px solid #0d2035" }}>
222 |              Para agregar camiones ir a Configuracion o usar boton +
223 |            </span>
224 |            <button
225 |              className="flex items-center gap-1.5 px-3 py-1.5 font-space text-[10px] font-bold tracking-[0.1em] uppercase
226 |              style={{ background: "#00d4ff", color: "#020508" }}
227 |              onClick={() => setShowModal(true)}
228 |              data-testid="btn-registrar-tara"
229 |            >
230 |              <Plus className="w-3.5 h-3.5" />
231 |              REGISTRAR TARA
232 |            </button>
233 |          </div>
234 |        </div>
235 |
236 |        <div className="grid grid-cols-4 gap-3">
237 |          <div className="dash-card px-4 py-3" data-testid="kpi-con-tara">
238 |            <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>CAMIONES CON TAR
239 |            <div className="font-space text-2xl font-bold" style={{ color: "#00d4ff" }}>{totals.conTara}</div>
240 |            <div className="font-exo text-[8px]" style={{ color: "#3a6080" }}>de {totals.totalCamiones} CENCOSUD</div>
241 |          </div>
242 |          <div className="dash-card px-4 py-3" data-testid="kpi-cumplen">
243 |            <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>CUMPLEN LIMITE</
244 |            <div className="font-space text-2xl font-bold" style={{ color: "#00ff88" }}>{totals.cumplen}</div>
245 |            <div className="font-exo text-[8px]" style={{ color: "#3a6080" }}>dentro de norma</div>
246 |          </div>
247 |          <div className="dash-card px-4 py-3" data-testid="kpi-exceden">
248 |            <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>EXCEDEN LIMITE</
249 |            <div className="font-space text-2xl font-bold" style={{ color: "#ff2244" }}>{totals.exceden}</div>
250 |            <div className="font-exo text-[8px]" style={{ color: "#3a6080" }}>sobre peso maximo</div>
251 |          </div>
252 |          <div className="dash-card px-4 py-3" data-testid="kpi-pendientes">
253 |            <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>PENDIENTE VERIFI
254 |            <div className="font-space text-2xl font-bold" style={{ color: "#3a6080" }}>{totals.sinDatos}</div>
255 |            <div className="font-exo text-[8px]" style={{ color: "#3a6080" }}>sin datos Volvo</div>
256 |          </div>
257 |        </div>
258 |
259 |        {registros.length === 0 ? (
260 |          <div className="dash-card px-8 py-12 text-center">
261 |            <Weight className="w-10 h-10 mx-auto mb-3" style={{ color: "#3a6080" }} />
262 |            <div className="font-space text-[14px] font-bold" style={{ color: "#3a6080" }}>Sin camiones con tara registrad
263 |            <div className="font-exo text-[11px] mt-2" style={{ color: "#3a6080" }}>
264 |              Registre datos de tara para comenzar a monitorear el peso de la flota.
265 |            </div>
266 |            <button
267 |              className="mt-4 px-4 py-2 font-space text-[10px] font-bold tracking-[0.1em] uppercase transition-all hover:o
268 |              style={{ background: "#00d4ff", color: "#020508" }}
269 |              onClick={() => setShowModal(true)}
270 |              data-testid="btn-registrar-tara-empty"
271 |            >
272 |              + REGISTRAR TARA
273 |            </button>
274 |          </div>
275 |        ) : (
276 |          <div className="dash-card overflow-hidden">
277 |            <div className="flex items-center gap-3 px-5 py-2" style={{ borderBottom: "1px solid #0d2035", background: "#0
278 |              <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080", width: "70px" }
279 |              <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080", width: "80px" }
280 |              <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: "#3a6080", widt
281 |              <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: "#3a6080", widt
282 |              <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: "#3a6080", widt
283 |              <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: "#3a6080", widt
284 |              <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: "#3a6080", widt
285 |              <span className="font-exo text-[8px] tracking-[0.15em] uppercase flex-1 text-right" style={{ color: "#3a6080
286 |            </div>
287 |            {registros.map(r => (
288 |              <div key={r.id} className="flex items-center gap-3 px-5 py-2.5 transition-all hover:bg-[rgba(0,212,255,0.03)
289 |                style={{ borderBottom: "1px solid #0d2035" }}
290 |                data-testid={`peso-row-${r.patente}`}>
291 |                <div className="flex items-center gap-1.5" style={{ width: "70px" }}>
292 |                  <Truck className="w-3 h-3 flex-shrink-0" style={{ color: "#3a6080" }} />
293 |                  <span className="font-space text-[11px] font-bold" style={{ color: "#c8e8ff" }}>{r.patente}</span>
294 |                </div>
295 |                <span className="font-exo text-[9px] truncate" style={{ color: "#3a6080", width: "80px" }}>{r.tipoVehiculo
296 |                <span className="font-space text-[11px] text-right" style={{ color: "#c8e8ff", width: "75px" }}>
297 |                  {r.taraKg.toLocaleString()}
298 |                </span>
299 |                <span className="font-space text-[11px] text-right" style={{ color: "#c8e8ff", width: "80px" }}>
300 |                  {r.pesoMaximoKg ? r.pesoMaximoKg.toLocaleString() : "---"}
301 |                </span>
302 |                <span className="font-space text-[11px] text-right" style={{
303 |                  color: r.cargaEstimada != null ? "#c8e8ff" : "#3a6080", width: "85px"
304 |                }}>
305 |                  {r.cargaEstimada != null ? `${r.cargaEstimada.toLocaleString()} kg` : "---"}
306 |                </span>
307 |                <span className="font-space text-[11px] font-bold text-right" style={{
308 |                  color: r.cumple === "EXCEDE" ? "#ff2244" : r.cumple === "AL_LIMITE" ? "#ffcc00" : r.cumple === "CUMPLE"
309 |                  width: "85px"
310 |                }}>
311 |                  {r.pesoTotal != null ? `${r.pesoTotal.toLocaleString()} kg` : "---"}
312 |                </span>
313 |                <span className="font-space text-[10px] text-right" style={{ color: "#3a6080", width: "80px" }}>
314 |                  {r.pesoVolvoKg != null ? `${r.pesoVolvoKg.toLocaleString()}` : "---"}
315 |                </span>
316 |                <div className="flex-1 text-right">
317 |                  <CumpleBadge cumple={r.cumple} />
318 |                </div>
```

```
319 |             </div>
320 |           ))}
321 |         </div>
322 |       )}
323 |
324 |       <div className="dash-card px-5 py-4">
325 |         <h3 className="font-space text-[12px] font-bold tracking-[0.1em] uppercase mb-3" style={{ color: "#c8e8ff" }}>
326 |           Registrar tara de camion
327 |         </h3>
328 |         <p className="font-exo text-[10px] mb-3" style={{ color: "#3a6080" }}>
329 |           Agregue datos de tara a mas camiones para monitorear el cumplimiento de peso.
330 |           La carga estimada se calcula automaticamente con datos de peso bruto de Volvo rFMS (GrossCombinationVehicleWei
331 |         </p>
332 |         <button
333 |           className="flex items-center gap-1.5 px-4 py-2 font-space text-[10px] font-bold tracking-[0.1em] uppercase tra
334 |           style={{ background: "#00d4ff", color: "#020508" }}
335 |           onClick={() => setShowModal(true)}
336 |           data-testid="btn-registrar-tara-bottom"
337 |         >
338 |           <Plus className="w-3.5 h-3.5" />
339 |           REGISTRAR TARA DE CAMION
340 |         </button>
341 |       </div>
342 |
343 |       <div className="text-[9px] font-space tracking-[0.15em]" style={{ color: "#3a6080" }}>
344 |         Peso bruto via Volvo rFMS GrossCombinationVehicleWeight &middot; CUMPLE: &lt;95% del maximo &middot; AL LIMITE:
345 |       </div>
346 |
347 |       {showModal && allCamiones && (
348 |         <TaraModal
349 |           onClose={() => setShowModal(false)}
350 |           camiones={allCamiones.map((c: any) => ({ id: c.id, patente: c.patente }))}
351 |         />
352 |       )}
353 |     </div>
354 |   );
355 | }
356 |
```

client/src/pages/excesos-velocidad.tsx

318 lineas

```
1 | import { useState } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { Loader2, AlertTriangle, Gauge, ChevronDown, ChevronRight, Truck } from "lucide-react";
4 |
5 | interface EventoVelocidad {
6 |   camionId: number;
7 |   patente: string;
8 |   modelo: string;
9 |   conductor: string | null;
10 |   faena: string | null;
11 |   velocidadMaxima: number;
12 |   fecha: string;
13 |   lat: number | null;
14 |   lng: number | null;
15 |   riesgo: "ALTO" | "MEDIO" | "BAJO";
16 | }
17 |
18 | interface CamionExceso {
19 |   patente: string;
20 |   modelo: string;
21 |   conductor: string | null;
22 |   faena: string | null;
23 |   velocidadMaxima: number;
24 |   totalExcesos: number;
25 |   ultimoExceso: string;
26 |   eventos: EventoVelocidad[];
27 |   riesgo: "ALTO" | "MEDIO" | "BAJO";
28 | }
29 |
30 | interface ExcesosData {
31 |   eventos: EventoVelocidad[];
32 |   porCamion: CamionExceso[];
33 |   totals: {
34 |     camionesConExceso: number;
35 |     eventosTotal: number;
36 |     masGrave: { patente: string; velocidad: number } | null;
37 |     hoy: number;
38 |     limiteKmh: number;
39 |   };
40 |   resumen: {
41 |     nuncaExcedieron: number;
42 |     ocasionales: number;
43 |     frecuentes: number;
44 |   };
45 |   desde: string;
46 | }
47 |
48 | function RiesgoBadge({ riesgo }: { riesgo: string }) {
49 |   const cfg: Record<string, { color: string; bg: string; border: string }> = {
50 |     ALTO: { color: "#ff2244", bg: "rgba(255,34,68,0.1)", border: "rgba(255,34,68,0.3)" },
51 |     MEDIO: { color: "#ffcc00", bg: "rgba(255,204,0,0.08)", border: "rgba(255,204,0,0.3)" },
52 |     BAJO: { color: "#00ff88", bg: "rgba(0,255,136,0.08)", border: "rgba(0,255,136,0.3)" },
53 |   };
54 |   const c = cfg[riesgo] || cfg.BAJO;
55 |   return (
56 |     <span className="font-space text-[8px] font-bold px-2 py-0.5 uppercase tracking-[0.1em]"
57 |       style={{ color: c.color, background: c.bg, border: `1px solid ${c.border}` }}>
58 |       {riesgo}
59 |     </span>
60 |   );
61 | }
62 |
63 | export default function ExcesosVelocidad() {
64 |   const [vista, setVista] = useState<"camion" | "evento">("camion");
65 |   const [expandedPatente, setExpandedPatente] = useState<string | null>(null);
66 |
67 |   const { data, isLoading, error } = useQuery<ExcesosData>({
68 |     queryKey: ["/api/datos/excesos-velocidad"],
69 |     refetchInterval: 60000,
70 |   });
71 |
72 |   if (isLoading) {
73 |     return (
74 |       <div className="flex items-center justify-center py-20" data-testid="excesos-loading">
75 |         <Loader2 className="w-8 h-8 animate-spin" style={{ color: "#00d4ff" }} />
76 |         <span className="ml-3 font-exo text-[12px]" style={{ color: "#3a6080" }}>Analizando excesos de velocidad...</spa
77 |       </div>
78 |     );
79 |   }
80 |
81 |   if (error) {
82 |     return (
83 |       <div className="dash-card px-8 py-12 text-center">
84 |         <AlertTriangle className="w-8 h-8 mx-auto mb-3" style={{ color: "#ff2244" }} />
85 |         <div className="font-space text-[14px] font-bold" style={{ color: "#ff2244" }}>Error al cargar datos</div>
86 |       </div>
87 |     );
88 |   }
89 |
90 |   if (!data) return null;
91 |   const { eventos, porCamion, totals, resumen } = data;
92 |
93 |   return (
94 |     <div className="space-y-5" data-testid="excesos-velocidad-page">
95 |       <div>
96 |         <h2 className="font-space text-[18px] font-bold tracking-[0.12em] uppercase" style={{ color: "#c8e8ff" }}>
97 |           Excesos de Velocidad
98 |         </h2>
99 |         <p className="font-exo text-[11px] mt-1" style={{ color: "#3a6080" }}>
100 |           Limite: {totals.limiteKmh} km/h &middot; Datos en tiempo real via Volvo rFMS
101 |         </p>
102 |       </div>
103 |
104 |       <div className="grid grid-cols-4 gap-3">
```

```
105 |          <div className="dash-card px-4 py-3" data-testid="kpi-camiones-exceso">
106 |            <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>CAMIONES CON EXC
107 |            <div className="font-space text-2xl font-bold" style={{ color: totals.camionesConExceso > 0 ? "#ff2244" : "#00
108 |              {totals.camionesConExceso}
109 |            </div>
110 |            <div className="font-exo text-[8px]" style={{ color: "#3a6080" }}>actualmente sobre limite</div>
111 |          </div>
112 |          <div className="dash-card px-4 py-3" data-testid="kpi-eventos-total">
113 |            <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>EVENTOS TOTALES<
114 |            <div className="font-space text-2xl font-bold" style={{ color: "#ffcc00" }}>{totals.eventosTotal}</div>
115 |            <div className="font-exo text-[8px]" style={{ color: "#3a6080" }}>detecciones activas</div>
116 |          </div>
117 |          <div className="dash-card px-4 py-3" data-testid="kpi-mas-grave">
118 |            <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>MAS GRAVE</div>
119 |            {totals.masGrave ? (
120 |              <>
121 |                <div className="font-space text-2xl font-bold" style={{ color: "#ff2244" }}>{totals.masGrave.velocidad} km
122 |                <div className="font-exo text-[8px]" style={{ color: "#3a6080" }}>{totals.masGrave.patente}</div>
123 |              </>
124 |            ) : (
125 |              <div className="font-space text-2xl font-bold" style={{ color: "#00ff88" }}>---</div>
126 |            )}
127 |          </div>
128 |          <div className="dash-card px-4 py-3" data-testid="kpi-hoy">
129 |            <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>HOY</div>
130 |            <div className="font-space text-2xl font-bold" style={{ color: totals.hoy > 0 ? "#ffcc00" : "#00ff88" }}>{tota
131 |            <div className="font-exo text-[8px]" style={{ color: "#3a6080" }}>eventos detectados</div>
132 |          </div>
133 |        </div>
134 |
135 |        <div className="flex items-center gap-2">
136 |          <button
137 |            className="font-space text-[10px] font-bold tracking-[0.1em] uppercase px-4 py-1.5 transition-all"
138 |            style={{
139 |              color: vista === "camion" ? "#00d4ff" : "#3a6080",
140 |              background: vista === "camion" ? "rgba(0,212,255,0.1)" : "transparent",
141 |              border: `1px solid ${vista === "camion" ? "#00d4ff30" : "#0d2035"}`,
142 |            }}
143 |            onClick={() => setVista("camion")}
144 |            data-testid="btn-vista-camion"
145 |          >
146 |            POR CAMION
147 |          </button>
148 |          <button
149 |            className="font-space text-[10px] font-bold tracking-[0.1em] uppercase px-4 py-1.5 transition-all"
150 |            style={{
151 |              color: vista === "evento" ? "#00d4ff" : "#3a6080",
152 |              background: vista === "evento" ? "rgba(0,212,255,0.1)" : "transparent",
153 |              border: `1px solid ${vista === "evento" ? "#00d4ff30" : "#0d2035"}`,
154 |            }}
155 |            onClick={() => setVista("evento")}
156 |            data-testid="btn-vista-evento"
157 |          >
158 |            POR EVENTO
159 |          </button>
160 |        </div>
161 |
162 |        {vista === "camion" ? (
163 |          <div className="dash-card overflow-hidden">
164 |            {porCamion.length === 0 ? (
165 |              <div className="px-8 py-12 text-center">
166 |                <Gauge className="w-10 h-10 mx-auto mb-3" style={{ color: "#00ff88" }} />
167 |                <div className="font-space text-[14px] font-bold" style={{ color: "#00ff88" }}>Ningun camion excede el lim
168 |                <div class="font-exo text-[11px] mt-2" style={{ color: "#3a6080" }}>
169 |                  Todos los camiones CENCOSUD con GPS activo circulan bajo {totals.limiteKmh} km/h
170 |                </div>
171 |              </div>
172 |            ) : (
173 |              <>
174 |                <div className="flex items-center gap-3 px-5 py-2" style={{ borderBottom: "1px solid #0d2035", background:
175 |                  <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080", width: "70p
176 |                  <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080", width: "120
177 |                  <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080", width: "120
178 |                  <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: "#3a6080",
179 |                  <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: "#3a6080",
180 |                  <span className="font-exo text-[8px] tracking-[0.15em] uppercase flex-1 text-right" style={{ color: "#3a
181 |                </div>
182 |                {porCamion.map(c => (
183 |                  <div key={c.patente}>
184 |                    <div
185 |                      className="flex items-center gap-3 px-5 py-2.5 cursor-pointer transition-all hover:bg-[rgba(0,212,25
186 |                      style={{ borderBottom: "1px solid #0d2035" }}
187 |                      onClick={() => setExpandedPatente(expandedPatente === c.patente ? null : c.patente)}
188 |                      data-testid={`exceso-camion-${c.patente}`}
189 |                    >
190 |                      <div className="flex items-center gap-1.5" style={{ width: "70px" }}>
191 |                        <Truck className="w-3 h-3 flex-shrink-0" style={{ color: c.riesgo === "ALTO" ? "#ff2244" : c.riesg
192 |                        <span className="font-space text-[11px] font-bold" style={{ color: "#c8e8ff" }}>{c.patente}</span>
193 |                      </div>
194 |                      <span className="font-exo text-[10px] truncate" style={{ color: "#c8e8ff", width: "120px" }}>{c.cond
195 |                      <span class="font-exo text-[10px] truncate" style={{ color: "#3a6080", width: "120px" }}>{c.faen
196 |                      <span className="font-space text-[12px] font-bold text-right" style={{
197 |                        color: c.velocidadMaxima > 120 ? "#ff2244" : c.velocidadMaxima > 100 ? "#ffcc00" : "#c8e8ff",
198 |                        width: "80px"
199 |                      }}>
200 |                        {c.velocidadMaxima} km/h
201 |                      </span>
202 |                      <span class="font-space text-[11px] text-right" style={{ color: "#c8e8ff", width: "70px" }}>{c.t
203 |                      <div class="flex-1 text-right flex items-center justify-end gap-2">
204 |                        <RiesgoBadge riesgo={c.riesgo} />
205 |                        {expandedPatente === c.patente
206 |                          ? <ChevronDown class="w-4 h-4" style={{ color: "#00d4ff" }} />
207 |                          : <ChevronRight class="w-4 h-4" style={{ color: "#3a6080" }} />}
208 |                      </div>
209 |                    </div>
210 |                    {expandedPatente === c.patente && (
211 |                      <div class="px-5 py-3" style={{ background: "rgba(0,212,255,0.02)", borderBottom: "1px solid #0d
```

```
212 |                       <div className="font-exo text-[8px] tracking-[0.15em] uppercase mb-2" style={{ color: "#3a6080" }}
213 |                       {c.eventos.map((e, i) => (
214 |                         <div key={i} className="flex items-center gap-3 py-1.5" style={{ borderBottom: "1px solid #0d203
215 |                           <span className="font-space text-[10px]" style={{ color: "#3a6080", width: "130px" }}>
216 |                             {new Date(e.fecha).toLocaleString("es-CL")}
217 |                           </span>
218 |                           <span className="font-space text-[12px] font-bold" style={{
219 |                             color: e.velocidadMaxima > 120 ? "#ff2244" : e.velocidadMaxima > 100 ? "#ffcc00" : "#c8e8ff"
220 |                           }}>
221 |                             {e.velocidadMaxima} km/h
222 |                           </span>
223 |                           {e.lat && e.lng && (
224 |                             <span className="font-space text-[8px]" style={{ color: "#3a6080" }}>
225 |                               {e.lat.toFixed(3)}, {e.lng.toFixed(3)}
226 |                             </span>
227 |                           )}
228 |                           <RiesgoBadge riesgo={e.riesgo} />
229 |                         </div>
230 |                       ))}
231 |                     </div>
232 |                   )}
233 |                 </div>
234 |               ))}
235 |             </>
236 |           )}
237 |         </div>
238 |       ) : (
239 |         <div className="dash-card overflow-hidden">
240 |           {eventos.length === 0 ? (
241 |             <div className="px-8 py-12 text-center">
242 |               <Gauge className="w-10 h-10 mx-auto mb-3" style={{ color: "#00ff88" }} />
243 |               <div className="font-space text-[14px] font-bold" style={{ color: "#00ff88" }}>Sin excesos activos</div>
244 |             </div>
245 |           ) : (
246 |             <>
247 |               <div className="flex items-center gap-3 px-5 py-2" style={{ borderBottom: "1px solid #0d2035", background:
248 |                 <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080", width: "130
249 |                 <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080", width: "70p
250 |                 <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080", width: "120
251 |                 <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: "#3a6080",
252 |                 <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080", width: "100
253 |                 <span className="font-exo text-[8px] tracking-[0.15em] uppercase flex-1 text-right" style={{ color: "#3a
254 |               </div>
255 |               {eventos.map((e, i) => (
256 |                 <div key={i} className="flex items-center gap-3 px-5 py-2.5"
257 |                   style={{ borderBottom: "1px solid #0d2035" }}
258 |                   data-testid={`exceso-evento-${i}`}>
259 |                   <span className="font-space text-[10px]" style={{ color: "#3a6080", width: "130px" }}>
260 |                     {new Date(e.fecha).toLocaleString("es-CL")}
261 |                   </span>
262 |                   <span className="font-space text-[11px] font-bold" style={{ color: "#c8e8ff", width: "70px" }}>{e.pate
263 |                   <span className="font-exo text-[10px] truncate" style={{ color: "#c8e8ff", width: "120px" }}>{e.conduc
264 |                   <span className="font-space text-[12px] font-bold text-right" style={{
265 |                     color: e.velocidadMaxima > 120 ? "#ff2244" : e.velocidadMaxima > 100 ? "#ffcc00" : "#c8e8ff",
266 |                     width: "80px"
267 |                   }}>
268 |                     {e.velocidadMaxima} km/h
269 |                   </span>
270 |                   <span className="font-space text-[8px]" style={{ color: "#3a6080", width: "100px" }}>
271 |                     {e.lat && e.lng ? `${e.lat.toFixed(3)}, ${e.lng.toFixed(3)}` : "---"}
272 |                   </span>
273 |                   <div className="flex-1 text-right">
274 |                     <RiesgoBadge riesgo={e.riesgo} />
275 |                   </div>
276 |                 </div>
277 |               ))}
278 |             </>
279 |           )}
280 |         </div>
281 |       )}
282 |
283 |       <div className="dash-card px-5 py-4">
284 |         <div className="font-exo text-[8px] tracking-[0.15em] uppercase mb-3" style={{ color: "#3a6080" }}>
285 |           RESUMEN DEL PERIODO &middot; Desde 01-03-2026
286 |         </div>
287 |         <div className="grid grid-cols-3 gap-4">
288 |           <div className="flex items-center gap-3">
289 |             <span className="inline-block w-3 h-3 rounded-full" style={{ background: "#00ff88" }} />
290 |             <div>
291 |               <span className="font-space text-[16px] font-bold" style={{ color: "#00ff88" }}>{resumen.nuncaExcedieron}<
292 |               <span className="font-exo text-[10px] ml-2" style={{ color: "#3a6080" }}>camiones nunca excedieron {totals
293 |             </div>
294 |           </div>
295 |           <div className="flex items-center gap-3">
296 |             <span className="inline-block w-3 h-3 rounded-full" style={{ background: "#ffcc00" }} />
297 |             <div>
298 |               <span className="font-space text-[16px] font-bold" style={{ color: "#ffcc00" }}>{resumen.ocasionales}</spa
299 |               <span className="font-exo text-[10px] ml-2" style={{ color: "#3a6080" }}>con excesos ocasionales</span>
300 |             </div>
301 |           </div>
302 |           <div className="flex items-center gap-3">
303 |             <span className="inline-block w-3 h-3 rounded-full" style={{ background: "#ff2244" }} />
304 |             <div>
305 |               <span className="font-space text-[16px] font-bold" style={{ color: "#ff2244" }}>{resumen.frecuentes}</span
306 |               <span className="font-exo text-[10px] ml-2" style={{ color: "#3a6080" }}>con excesos frecuentes</span>
307 |             </div>
308 |           </div>
309 |         </div>
310 |       </div>
311 |
312 |       <div className="text-[9px] font-space tracking-[0.15em]" style={{ color: "#3a6080" }}>
313 |         Datos en tiempo real via Volvo rFMS &middot; Limite configurable en parametros &middot; Auto-refresh cada 60s &m
314 |       </div>
315 |     </div>
316 |   );
317 | }
318 |
```

client/src/pages/faenas.tsx

294 lineas

```
1 | import { useState, useMemo } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { Loader2, ChevronDown, ChevronRight, Truck, Calendar, Fuel, TrendingUp } from "lucide-react";
4 |
5 | interface DiaData {
6 |   total: number;
7 |   km: number;
8 |   camiones: string[];
9 | }
10 |
11 | interface CamionResumen {
12 |   patente: string;
13 |   modelo: string;
14 |   viajes: number;
15 |   km: number;
16 |   litros: number;
17 |   rendimiento: number;
18 | }
19 |
20 | interface FaenaResumen {
21 |   id: number;
22 |   nombre: string;
23 |   color: string;
24 |   totalCamiones: number;
25 |   litros30d: number;
26 |   rendPromedio: number;
27 |   contrato: { id: number; nombre: string } | null;
28 |   totalViajes: number;
29 |   totalKm: number;
30 |   viajesPorDia: Record<string, DiaData>;
31 |   camionResumen: CamionResumen[];
32 | }
33 |
34 | function generarDiasDesdeMarzo(): string[] {
35 |   const dias: string[] = [];
36 |   const inicio = new Date("2026-03-01");
37 |   const hoy = new Date();
38 |   const d = new Date(inicio);
39 |   while (d <= hoy) {
40 |     dias.push(d.toISOString().split("T")[0]);
41 |     d.setDate(d.getDate() + 1);
42 |   }
43 |   return dias.reverse();
44 | }
45 |
46 | function formatDia(fecha: string): string {
47 |   const d = new Date(fecha + "T12:00:00");
48 |   const dia = d.getDate().toString().padStart(2, "0");
49 |   const meses = ["ENE", "FEB", "MAR", "ABR", "MAY", "JUN", "JUL", "AGO", "SEP", "OCT", "NOV", "DIC"];
50 |   const mes = meses[d.getMonth()];
51 |   const dow = ["DOM", "LUN", "MAR", "MIE", "JUE", "VIE", "SAB"][d.getDay()];
52 |   return `${dow} ${dia} ${mes}`;
53 | }
54 |
55 | function FaenaCard({ faena }: { faena: FaenaResumen }) {
56 |   const [expanded, setExpanded] = useState(false);
57 |   const [showCamiones, setShowCamiones] = useState(false);
58 |   const dias = useMemo(() => generarDiasDesdeMarzo(), []);
59 |
60 |   const diasConViajes = dias.filter(d => faena.viajesPorDia[d]);
61 |   const maxViajesDia = Math.max(1, ...Object.values(faena.viajesPorDia).map(v => v.total));
62 |
63 |   return (
64 |     <div className="dash-card overflow-hidden" style={{ border: `1px solid ${faena.color}20` }}>
65 |       <div className="flex items-center gap-3 px-5 py-4 cursor-pointer transition-all hover:bg-[rgba(0,212,255,0.03)]"
66 |         onClick={() => setExpanded(!expanded)}
67 |         data-testid={`faena-resumen-card-${faena.id}`}>
68 |         <div className="w-3 h-3 rounded-full flex-shrink-0" style={{ background: faena.color, boxShadow: `0 0 8px ${faen
69 |         {expanded ? <ChevronDown className="w-4 h-4 flex-shrink-0" style={{ color: '#00d4ff' }} /> : <ChevronRight class
70 |         <div className="flex-1 min-w-0">
71 |           <div class="font-space text-[14px] font-bold tracking-[0.08em]" style={{ color: '#c8e8ff' }}>{faena.nombre
72 |           <div class="flex items-center gap-3 mt-0.5">
73 |             {faena.contrato && (
74 |               <span class="font-exo text-[9px]" style={{ color: '#3a6080' }}>TMS: {faena.contrato.nombre}</span>
75 |             )}
76 |           </div>
77 |         </div>
78 |         <div class="flex items-center gap-5">
79 |           <div class="text-right">
80 |             <div class="font-space text-[16px] font-bold" style={{ color: '#00d4ff' }}>{faena.totalViajes}</div>
81 |             <div class="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>VIAJES</div>
82 |           </div>
83 |           <div class="text-right">
84 |             <div class="font-space text-[14px] font-bold" style={{ color: '#c8e8ff' }}>{faena.totalKm.toLocaleString
85 |             <div class="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>KM</div>
86 |           </div>
87 |           <div class="text-right">
88 |             <div class="font-space text-[14px] font-bold" style={{ color: '#c8e8ff' }}>{faena.litros30d.toLocaleStri
89 |             <div class="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>LITROS</div>
90 |           </div>
91 |           <div class="text-right">
92 |             <div class="font-space text-[14px] font-bold" style={{ color: faena.rendPromedio >= 3.5 ? '#00ff88' : fa
93 |               {faena.rendPromedio > 0 ? faena.rendPromedio : '\u2014'}
94 |             </div>
95 |             <div class="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>KM/L</div>
96 |           </div>
97 |           <div class="text-right">
98 |             <div class="font-space text-[14px] font-bold" style={{ color: '#c8e8ff' }}>{faena.totalCamiones}</div>
99 |             <div class="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>CAMIONES</div>
100 |           </div>
101 |         </div>
102 |       </div>
103 |
104 |       {expanded && (
```

```
105 |         <div style={{ borderTop: '1px solid #0d2035' }}>
106 |           <div className="flex gap-1 px-5 py-2" style={{ borderBottom: '1px solid #0d2035' }}>
107 |             <button
108 |               className={`font-exo text-[10px] tracking-[0.1em] px-3 py-1.5 cursor-pointer transition-all ${!showCamione
109 |               style={{
110 |                 color: !showCamiones ? '#00d4ff' : '#3a6080',
111 |                 background: !showCamiones ? 'rgba(0,212,255,0.08)' : 'transparent',
112 |                 border: `1px solid ${!showCamiones ? 'rgba(0,212,255,0.2)' : '#0d2035'}`
113 |               }}
114 |               onClick={() => setShowCamiones(false)}
115 |               data-testid={`btn-viajes-dia-${faena.id}`}>
116 |               <Calendar className="w-3 h-3 inline mr-1.5" />VIAJES POR DIA
117 |             </button>
118 |             <button
119 |               className={`font-exo text-[10px] tracking-[0.1em] px-3 py-1.5 cursor-pointer transition-all ${showCamiones
120 |               style={{
121 |                 color: showCamiones ? '#00d4ff' : '#3a6080',
122 |                 background: showCamiones ? 'rgba(0,212,255,0.08)' : 'transparent',
123 |                 border: `1px solid ${showCamiones ? 'rgba(0,212,255,0.2)' : '#0d2035'}`
124 |               }}
125 |               onClick={() => setShowCamiones(true)}
126 |               data-testid={`btn-camion-resumen-${faena.id}`}>
127 |               <Truck className="w-3 h-3 inline mr-1.5" />RESUMEN POR CAMION
128 |             </button>
129 |           </div>
130 |
131 |           {!showCamiones ? (
132 |             <div className="max-h-[400px] overflow-y-auto">
133 |               <div className="flex items-center gap-3 px-5 py-1.5 sticky top-0" style={{ background: '#020508', zIndex:
134 |                 <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080', width: '90p
135 |                 <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080', width: '55p
136 |                 <span className="font-exo text-[8px] tracking-[0.15em] uppercase flex-1" style={{ color: '#3a6080' }}>AC
137 |                 <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080', width: '70p
138 |                 <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080', width: '60p
139 |               </div>
140 |               {dias.map(dia => {
141 |                 const d = faena.viajesPorDia[dia];
142 |                 if (!d) return (
143 |                   <div key={dia} className="flex items-center gap-3 px-5 py-1.5" style={{ borderBottom: '1px solid #0d20
144 |                     <span className="font-space text-[10px]" style={{ color: '#1a3050', width: '90px' }}>{formatDia(dia)
145 |                     <span className="font-space text-[10px]" style={{ color: '#1a3050', width: '55px' }}>{'\u2014'}</spa
146 |                     <div className="flex-1 h-[4px] rounded-sm" style={{ background: '#0d2035' }} />
147 |                     <span style={{ width: '70px' }} />
148 |                     <span style={{ width: '60px' }} />
149 |                   </div>
150 |                 );
151 |                 const barPct = Math.min((d.total / maxViajesDia) * 100, 100);
152 |                 return (
153 |                   <div key={dia} className="flex items-center gap-3 px-5 py-2 transition-all hover:bg-[rgba(0,212,255,0.
154 |                     style={{ borderBottom: '1px solid #0d2035' }}
155 |                     data-testid={`dia-row-${dia}`}>
156 |                     <span className="font-space text-[10px] font-bold" style={{ color: '#c8e8ff', width: '90px' }}>{form
157 |                     <span className="font-space text-[12px] font-bold" style={{ color: '#00d4ff', width: '55px' }}>{d.to
158 |                     <div className="flex-1 h-[6px] rounded-sm overflow-hidden" style={{ background: '#0d2035' }}>
159 |                       <div className="h-full rounded-sm transition-all" style={{
160 |                         width: `${barPct}%`,
161 |                         background: `linear-gradient(90deg, ${faena.color}, #00d4ff)`,
162 |                         boxShadow: `0 0 4px ${faena.color}40`
163 |                       }} />
164 |                     </div>
165 |                     <span className="font-space text-[10px] text-right" style={{ color: '#c8e8ff', width: '70px' }}>{d.k
166 |                     <span className="font-space text-[10px] text-right" style={{ color: '#3a6080', width: '60px' }}>{d.c
167 |                   </div>
168 |                 );
169 |               })}
170 |             </div>
171 |           ) : (
172 |             <div className="max-h-[400px] overflow-y-auto">
173 |               <div className="flex items-center gap-3 px-5 py-1.5 sticky top-0" style={{ background: '#020508', zIndex:
174 |                 <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080', width: '70p
175 |                 <span className="font-exo text-[8px] tracking-[0.15em] uppercase flex-1" style={{ color: '#3a6080' }}>MO
176 |                 <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: '#3a6080',
177 |                 <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: '#3a6080',
178 |                 <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: '#3a6080',
179 |                 <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: '#3a6080',
180 |               </div>
181 |               {faena.camionResumen.map(cam => (
182 |                 <div key={cam.patente} className="flex items-center gap-3 px-5 py-2 transition-all hover:bg-[rgba(0,212,
183 |                   style={{ borderBottom: '1px solid #0d2035' }}
184 |                   data-testid={`camion-resumen-${cam.patente}`}>
185 |                   <div className="flex items-center gap-2" style={{ width: '70px' }}>
186 |                     <Truck className="w-3 h-3 flex-shrink-0" style={{ color: '#00d4ff' }} />
187 |                     <span className="font-space text-[11px] font-bold" style={{ color: '#c8e8ff' }}>{cam.patente}</span>
188 |                   </div>
189 |                   <span className="font-exo text-[10px] flex-1 truncate" style={{ color: '#3a6080' }}>{cam.modelo}</span
190 |                   <span className="font-space text-[11px] font-bold text-right" style={{ color: '#00d4ff', width: '55px'
191 |                   <span className="font-space text-[10px] text-right" style={{ color: '#c8e8ff', width: '70px' }}>{cam.k
192 |                   <span className="font-space text-[10px] text-right" style={{ color: '#c8e8ff', width: '70px' }}>{cam.l
193 |                   <span className="font-space text-[10px] font-bold text-right" style={{
194 |                     color: cam.rendimiento >= 3.5 ? '#00ff88' : cam.rendimiento >= 2.0 ? '#c8e8ff' : cam.rendimiento > 0
195 |                     width: '55px'
196 |                   }}>
197 |                     {cam.rendimiento > 0 ? cam.rendimiento : '\u2014'}
198 |                   </span>
199 |                 </div>
200 |               ))}
201 |               {faena.camionResumen.length === 0 && (
202 |                 <div className="px-5 py-6 text-center font-exo text-[11px]" style={{ color: '#3a6080' }}>
203 |                   Sin datos de camiones para esta faena
204 |                 </div>
205 |               )}
206 |             </div>
207 |           )}
208 |         </div>
209 |       )}
210 |     </div>
211 |   );
```

```
212 | }
213 |
214 | export default function Faenas() {
215 |   const { data, isLoading } = useQuery<FaenaResumen[]>({
216 |     queryKey: ["/api/faenas/resumen"],
217 |     refetchInterval: 120000,
218 |   });
219 |
220 |   const faenas = data || [];
221 |
222 |   const totales = useMemo(() => {
223 |     return {
224 |       faenas: faenas.length,
225 |       viajes: faenas.reduce((s, f) => s + f.totalViajes, 0),
226 |       km: Math.round(faenas.reduce((s, f) => s + f.totalKm, 0)),
227 |       litros: faenas.reduce((s, f) => s + f.litros30d, 0),
228 |       camiones: faenas.reduce((s, f) => s + f.totalCamiones, 0),
229 |     };
230 |   }, [faenas]);
231 |
232 |   if (isLoading) {
233 |     return (
234 |       <div className="flex items-center justify-center py-20" data-testid="faenas-loading">
235 |         <Loader2 className="w-8 h-8 animate-spin" style={{ color: '#00d4ff' }} />
236 |         <span className="ml-3 font-exo text-[12px]" style={{ color: '#3a6080' }}>Cargando resumenes de faenas...</span>
237 |       </div>
238 |     );
239 |   }
240 |
241 |   return (
242 |     <div className="space-y-5" data-testid="faenas-page">
243 |       <div className="flex items-center justify-between">
244 |         <div>
245 |           <h2 className="font-space text-[18px] font-bold tracking-[0.12em] uppercase" style={{ color: '#c8e8ff' }}>
246 |             Resumenes por Faena
247 |           </h2>
248 |           <p className="font-exo text-[11px] mt-1" style={{ color: '#3a6080' }}>
249 |             Viajes detectados desde 01-03-2026 agrupados por dia y camion
250 |           </p>
251 |         </div>
252 |       </div>
253 |
254 |       <div className="grid grid-cols-5 gap-3">
255 |         <div className="dash-card px-4 py-3">
256 |           <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>FAENAS</div>
257 |           <div className="font-space text-2xl font-bold" style={{ color: '#00d4ff' }}>{totales.faenas}</div>
258 |         </div>
259 |         <div className="dash-card px-4 py-3">
260 |           <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>VIAJES TOTALES</
261 |           <div className="font-space text-2xl font-bold" style={{ color: '#00d4ff' }}>{totales.viajes}</div>
262 |         </div>
263 |         <div className="dash-card px-4 py-3">
264 |           <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>KM TOTALES</div>
265 |           <div className="font-space text-2xl font-bold" style={{ color: '#c8e8ff' }}>{totales.km.toLocaleString()}</div
266 |         </div>
267 |         <div className="dash-card px-4 py-3">
268 |           <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>LITROS 30D</div>
269 |           <div className="font-space text-2xl font-bold" style={{ color: '#c8e8ff' }}>{totales.litros.toLocaleString()}<
270 |         </div>
271 |         <div className="dash-card px-4 py-3">
272 |           <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>CAMIONES</div>
273 |           <div className="font-space text-2xl font-bold" style={{ color: '#c8e8ff' }}>{totales.camiones}</div>
274 |         </div>
275 |       </div>
276 |
277 |       <div className="space-y-3">
278 |         {faenas.map(f => (
279 |           <FaenaCard key={f.id} faena={f} />
280 |         ))}
281 |       </div>
282 |
283 |       {faenas.length === 0 && (
284 |         <div className="dash-card px-8 py-12 text-center">
285 |           <div className="font-space text-[14px] font-bold" style={{ color: '#3a6080' }}>Sin faenas con datos</div>
286 |           <div className="font-exo text-[11px] mt-2" style={{ color: '#3a6080' }}>
287 |             No se encontraron faenas con camiones asignados
288 |           </div>
289 |         </div>
290 |       )}
291 |     </div>
292 |   );
293 | }
294 |
```

client/src/pages/faenas-movimiento.tsx

256 lineas

```
1 | import { useState } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { Loader2, ChevronDown, ChevronRight, Truck, Navigation, Fuel, Clock, MapPin, Radio } from "lucide-react";
4 |
5 | interface TruckLive {
6 |   id: number;
7 |   patente: string;
8 |   modelo: string;
9 |   vin: string;
10 |   lat: number;
11 |   lng: number;
12 |   speed: number;
13 |   heading: number | null;
14 |   gpsTime: string;
15 |   gpsAgeMin: number;
16 |   fuelLevel: number | null;
17 |   engineHours: number | null;
18 |   totalDistance: number | null;
19 |   driverWorkingState: string | null;
20 |   isMoving: boolean;
21 |   isRecent: boolean;
22 |   estado: "EN_RUTA" | "DETENIDO" | "INACTIVO";
23 | }
24 |
25 | interface FaenaLive {
26 |   faenaId: number;
27 |   nombre: string;
28 |   color: string;
29 |   trucks: TruckLive[];
30 |   enRuta: number;
31 |   detenidos: number;
32 |   inactivos: number;
33 |   totalOnline: number;
34 | }
35 |
36 | interface EnMovimientoData {
37 |   faenas: FaenaLive[];
38 |   totals: {
39 |     totalOnline: number;
40 |     enRuta: number;
41 |     detenidos: number;
42 |     inactivos: number;
43 |     faenasActivas: number;
44 |   };
45 | }
46 |
47 | function timeAgo(min: number): string {
48 |   if (min < 1) return "ahora";
49 |   if (min < 60) return `${min}min`;
50 |   const h = Math.floor(min / 60);
51 |   return `${h}h ${min % 60}m`;
52 | }
53 |
54 | function estadoBadge(estado: string) {
55 |   const cfg: Record<string, { color: string; bg: string; border: string; label: string }> = {
56 |     EN_RUTA: { color: "#00ff88", bg: "rgba(0,255,136,0.08)", border: "rgba(0,255,136,0.3)", label: "EN RUTA" },
57 |     DETENIDO: { color: "#ffcc00", bg: "rgba(255,204,0,0.08)", border: "rgba(255,204,0,0.3)", label: "DETENIDO" },
58 |     INACTIVO: { color: "#3a6080", bg: "rgba(58,96,128,0.08)", border: "rgba(58,96,128,0.3)", label: "INACTIVO" },
59 |   };
60 |   const c = cfg[estado] || cfg.INACTIVO;
61 |   return (
62 |     <span className="font-space text-[8px] font-bold px-2 py-0.5 uppercase tracking-[0.1em]"
63 |       style={{ color: c.color, background: c.bg, border: `1px solid ${c.border}` }}>
64 |       {c.label}
65 |     </span>
66 |   );
67 | }
68 |
69 | function FaenaLiveCard({ faena }: { faena: FaenaLive }) {
70 |   const [expanded, setExpanded] = useState(faena.enRuta > 0);
71 |
72 |   return (
73 |     <div className="dash-card overflow-hidden" style={{ border: `1px solid ${faena.color}20` }}>
74 |       <div className="flex items-center gap-3 px-5 py-4 cursor-pointer transition-all hover:bg-[rgba(0,212,255,0.03)]"
75 |         onClick={() => setExpanded(!expanded)}
76 |         data-testid={`faena-live-card-${faena.faenaId}`}>
77 |         {expanded ? <ChevronDown className="w-4 h-4 flex-shrink-0" style={{ color: '#00d4ff' }} /> : <ChevronRight class
78 |         <div className="w-3 h-3 rounded-full flex-shrink-0" style={{ background: faena.color, boxShadow: `0 0 8px ${faen
79 |         <div className="flex-1 min-w-0">
80 |           <div className="font-space text-[14px] font-bold tracking-[0.08em]" style={{ color: '#c8e8ff' }}>{faena.nombre
81 |         </div>
82 |         <div className="flex items-center gap-4">
83 |           {faena.enRuta > 0 && (
84 |             <div className="flex items-center gap-1.5">
85 |               <span className="inline-block w-2 h-2 rounded-full" style={{ background: '#00ff88', boxShadow: '0 0 6px #0
86 |               <span className="font-space text-[14px] font-bold" style={{ color: '#00ff88' }}>{faena.enRuta}</span>
87 |               <span className="font-exo text-[8px] tracking-[0.1em] uppercase" style={{ color: '#3a6080' }}>EN RUTA</spa
88 |             </div>
89 |           )}
90 |           {faena.detenidos > 0 && (
91 |             <div className="flex items-center gap-1.5">
92 |               <span className="inline-block w-2 h-2 rounded-full" style={{ background: '#ffcc00' }} />
93 |               <span className="font-space text-[13px] font-bold" style={{ color: '#ffcc00' }}>{faena.detenidos}</span>
94 |               <span className="font-exo text-[8px] tracking-[0.1em] uppercase" style={{ color: '#3a6080' }}>DET</span>
95 |             </div>
96 |           )}
97 |           {faena.inactivos > 0 && (
98 |             <div className="flex items-center gap-1.5">
99 |               <span className="inline-block w-2 h-2 rounded-full" style={{ background: '#3a6080' }} />
100 |               <span className="font-space text-[13px]" style={{ color: '#3a6080' }}>{faena.inactivos}</span>
101 |               <span className="font-exo text-[8px] tracking-[0.1em] uppercase" style={{ color: '#3a6080' }}>INACT</span>
102 |             </div>
103 |           )}
104 |           <div className="text-right ml-2">
```

```
105 |             <div className="font-space text-[12px]" style={{ color: '#3a6080' }}>{faena.totalOnline} online</div>
106 |           </div>
107 |         </div>
108 |       </div>
109 |
110 |       {expanded && (
111 |         <div style={{ borderTop: '1px solid #0d2035' }}>
112 |           <div className="flex items-center gap-3 px-5 py-1.5 sticky top-0" style={{ background: '#020508', zIndex: 1, b
113 |             <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080', width: '55px' }
114 |             <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080', width: '70px' }
115 |             <span className="font-exo text-[8px] tracking-[0.15em] uppercase flex-1" style={{ color: '#3a6080' }}>MODELO
116 |             <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: '#3a6080', widt
117 |             <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: '#3a6080', widt
118 |             <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: '#3a6080', widt
119 |             <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: '#3a6080', widt
120 |           </div>
121 |           {faena.trucks.map(truck => (
122 |             <div key={truck.id} className="flex items-center gap-3 px-5 py-2.5 transition-all hover:bg-[rgba(0,212,255,0
123 |               style={{ borderBottom: '1px solid #0d2035' }}
124 |               data-testid={`truck-live-${truck.patente}`}>
125 |               <div style={{ width: '55px' }}>{estadoBadge(truck.estado)}</div>
126 |               <div className="flex items-center gap-1.5" style={{ width: '70px' }}>
127 |                 <Truck className="w-3 h-3 flex-shrink-0" style={{
128 |                   color: truck.estado === "EN_RUTA" ? '#00ff88' : truck.estado === "DETENIDO" ? '#ffcc00' : '#3a6080'
129 |                 }} />
130 |                 <span className="font-space text-[11px] font-bold" style={{ color: '#c8e8ff' }}>{truck.patente}</span>
131 |               </div>
132 |               <span className="font-exo text-[10px] flex-1 truncate" style={{ color: '#3a6080' }}>{truck.modelo}</span>
133 |               <div className="text-right" style={{ width: '55px' }}>
134 |                 {truck.speed > 0 ? (
135 |                   <span className="font-space text-[11px] font-bold" style={{ color: '#00ff88' }}>
136 |                     {Math.round(truck.speed)} km/h
137 |                   </span>
138 |                 ) : (
139 |                   <span className="font-space text-[10px]" style={{ color: '#3a6080' }}>0</span>
140 |                 )}
141 |               </div>
142 |               <div className="text-right" style={{ width: '50px' }}>
143 |                 {truck.fuelLevel != null ? (
144 |                   <span className="font-space text-[10px]" style={{
145 |                     color: truck.fuelLevel > 50 ? '#00ff88' : truck.fuelLevel > 20 ? '#ffcc00' : '#ff2244'
146 |                   }}>
147 |                     {Math.round(truck.fuelLevel)}%
148 |                   </span>
149 |                 ) : (
150 |                   <span className="font-space text-[10px]" style={{ color: '#3a6080' }}>{'\u2014'}</span>
151 |                 )}
152 |               </div>
153 |               <div className="text-right" style={{ width: '70px' }}>
154 |                 <span className="font-exo text-[9px]" style={{
155 |                   color: truck.gpsAgeMin < 15 ? '#00ff88' : truck.gpsAgeMin < 60 ? '#c8e8ff' : '#ffcc00'
156 |                 }}>
157 |                   {timeAgo(truck.gpsAgeMin)}
158 |                 </span>
159 |               </div>
160 |               <div className="text-right" style={{ width: '80px' }}>
161 |                 <span className="font-space text-[8px]" style={{ color: '#3a6080' }}>
162 |                   {truck.lat.toFixed(3)}, {truck.lng.toFixed(3)}
163 |                 </span>
164 |               </div>
165 |             </div>
166 |           ))}
167 |         </div>
168 |       )}
169 |     </div>
170 |   );
171 | }
172 |
173 | export default function FaenasEnMovimiento() {
174 |   const { data, isLoading } = useQuery<EnMovimientoData>({
175 |     queryKey: ["/api/faenas/en-movimiento"],
176 |     refetchInterval: 60000,
177 |   });
178 |
179 |   if (isLoading || !data) {
180 |     return (
181 |       <div className="flex items-center justify-center py-20" data-testid="movimiento-loading">
182 |         <Loader2 className="w-8 h-8 animate-spin" style={{ color: '#00d4ff' }} />
183 |         <span className="ml-3 font-exo text-[12px]" style={{ color: '#3a6080' }}>Consultando estado de flota...</span>
184 |       </div>
185 |     );
186 |   }
187 |
188 |   const { faenas, totals } = data;
189 |
190 |   return (
191 |     <div className="space-y-5" data-testid="faenas-movimiento-page">
192 |       <div className="flex items-center justify-between">
193 |         <div>
194 |           <h2 className="font-space text-[18px] font-bold tracking-[0.12em] uppercase" style={{ color: '#c8e8ff' }}>
195 |             Faenas en Movimiento
196 |           </h2>
197 |           <p className="font-exo text-[11px] mt-1" style={{ color: '#3a6080' }}>
198 |             Estado en tiempo real de camiones por faena via Volvo rFMS
199 |           </p>
200 |         </div>
201 |         <div className="flex items-center gap-2">
202 |           <Radio className="w-3.5 h-3.5" style={{ color: '#00ff88', animation: 'blink 2s infinite' }} />
203 |           <span className="font-space text-[10px] font-bold" style={{ color: '#00ff88' }}>LIVE</span>
204 |         </div>
205 |       </div>
206 |
207 |       <div className="grid grid-cols-5 gap-3">
208 |         <div className="dash-card px-4 py-3">
209 |           <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>ONLINE</div>
210 |           <div className="font-space text-2xl font-bold" style={{ color: '#00d4ff' }}>{totals.totalOnline}</div>
211 |           <div className="font-exo text-[8px]" style={{ color: '#3a6080' }}>camiones con GPS</div>
```

```
212 |         </div>
213 |         <div className="dash-card px-4 py-3">
214 |           <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>EN RUTA</div>
215 |           <div className="font-space text-2xl font-bold" style={{ color: '#00ff88' }}>{totals.enRuta}</div>
216 |           <div className="font-exo text-[8px]" style={{ color: '#3a6080' }}>en movimiento</div>
217 |         </div>
218 |         <div className="dash-card px-4 py-3">
219 |           <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>DETENIDOS</div>
220 |           <div className="font-space text-2xl font-bold" style={{ color: '#ffcc00' }}>{totals.detenidos}</div>
221 |           <div className="font-exo text-[8px]" style={{ color: '#3a6080' }}>motor encendido</div>
222 |         </div>
223 |         <div className="dash-card px-4 py-3">
224 |           <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>INACTIVOS</div>
225 |           <div className="font-space text-2xl font-bold" style={{ color: '#3a6080' }}>{totals.inactivos}</div>
226 |           <div className="font-exo text-[8px]" style={{ color: '#3a6080' }}>sin actividad reciente</div>
227 |         </div>
228 |         <div className="dash-card px-4 py-3">
229 |           <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: '#3a6080' }}>FAENAS</div>
230 |           <div className="font-space text-2xl font-bold" style={{ color: '#c8e8ff' }}>{totals.faenasActivas}</div>
231 |           <div className="font-exo text-[8px]" style={{ color: '#3a6080' }}>con camiones online</div>
232 |         </div>
233 |       </div>
234 |
235 |       <div className="space-y-3">
236 |         {faenas.map(f => (
237 |           <FaenaLiveCard key={f.faenaId} faena={f} />
238 |         ))}
239 |       </div>
240 |
241 |       {faenas.length === 0 && (
242 |         <div className="dash-card px-8 py-12 text-center">
243 |           <div className="font-space text-[14px] font-bold" style={{ color: '#3a6080' }}>Sin camiones online</div>
244 |           <div className="font-exo text-[11px] mt-2" style={{ color: '#3a6080' }}>
245 |             No se detectaron camiones con GPS activo en las ultimas 6 horas
246 |           </div>
247 |         </div>
248 |       )}
249 |
250 |       <div className="text-[9px] font-space tracking-[0.15em]" style={{ color: '#3a6080' }}>
251 |         Datos via Volvo rFMS API &middot; Auto-refresh cada 60s &middot; EN RUTA: vel &gt; 2 km/h &middot; DETENIDO: GPS
252 |       </div>
253 |     </div>
254 |   );
255 | }
256 |
```

client/src/pages/camiones.tsx

246 lineas

```
1 | import { useState, useMemo } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { Input } from "@/components/ui/input";
4 | import { Button } from "@/components/ui/button";
5 | import { Skeleton } from "@/components/ui/skeleton";
6 | import { StatusTag, getStatusVariant } from "@/components/status-tag";
7 | import { FichaCamionModal } from "@/components/ficha-camion-modal";
8 | import { CargaModal } from "@/components/carga-modal";
9 | import { statsCamion, fN, fL, f$, PRECIO, rendColor, type CamionStats } from "@/lib/fuel-utils";
10 | import { AlertTriangle, Search, FileText, Plus } from "lucide-react";
11 | import type { Faena, Camion, Carga } from "@shared/schema";
12 |
13 | export default function Camiones() {
14 |   const [faenaFil, setFaenaFil] = useState<string>("all");
15 |   const [estadoFil, setEstadoFil] = useState("todos");
16 |   const [busq, setBusq] = useState("");
17 |   const [fichaId, setFichaId] = useState<number | null>(null);
18 |   const [cargaId, setCargaId] = useState<number | null>(null);
19 |
20 |   const { data: faenas = [], isLoading: loadingFaenas } = useQuery<Faena[]>({ queryKey: ["/api/faenas"] });
21 |   const { data: camionesRaw = [], isLoading: loadingCamiones } = useQuery<Camion[]>({ queryKey: ["/api/camiones"] });
22 |   const { data: cargasRaw = [], isLoading: loadingCargas } = useQuery<Carga[]>({ queryKey: ["/api/cargas"] });
23 |
24 |   const isLoading = loadingFaenas || loadingCamiones || loadingCargas;
25 |
26 |   const camStats: CamionStats[] = useMemo(() =>
27 |     camionesRaw.map(cam => {
28 |       const cams = cargasRaw.filter(c => c.camionId === cam.id);
29 |       return statsCamion(cam, cams);
30 |     }), [camionesRaw, cargasRaw]);
31 |
32 |   const filtrados = useMemo(() => {
33 |     let d = camStats;
34 |     if (faenaFil !== "all") d = d.filter(c => c.faenaId === parseInt(faenaFil));
35 |     if (estadoFil !== "todos") d = d.filter(c => c.estado === estadoFil.toUpperCase());
36 |     if (busq) d = d.filter(c =>
37 |       c.patente.toLowerCase().includes(busq.toLowerCase()) ||
38 |       c.conductor?.toLowerCase().includes(busq.toLowerCase())
39 |     );
40 |     const order: Record<string, number> = { CRITICO: 0, ALERTA: 1, OK: 2 };
41 |     return d.sort((a, b) => order[a.estado] - order[b.estado]);
42 |   }, [camStats, faenaFil, estadoFil, busq]);
43 |
44 |   const criticos = filtrados.filter(c => c.estado === "CRITICO");
45 |
46 |   const fichaStats = fichaId ? camStats.find(c => c.id === fichaId) : null;
47 |
48 |   if (isLoading) {
49 |     return (
50 |       <div className="space-y-5" data-testid="camiones-loading">
51 |         <div className="flex gap-2 flex-wrap">
52 |           {Array.from({ length: 3 }).map((_, i) => <Skeleton key={i} className="h-8 w-40" />)}
53 |         </div>
54 |         <Skeleton className="h-64" />
55 |       </div>
56 |     );
57 |   }
58 |
59 |   return (
60 |     <div className="space-y-5" data-testid="camiones-view">
61 |       <div className="flex gap-1.5 flex-wrap items-center" data-testid="faena-filters">
62 |         {[{ id: "all", nombre: "Todas las faenas", color: undefined } as const, ...faenas].map(f => (
63 |           <button
64 |             key={f.id}
65 |             onClick={() => setFaenaFil(String(f.id))}
66 |             data-testid={`filter-faena-${f.id}`}
67 |             className={`px-3.5 py-1.5 text-[11px] font-mono cursor-pointer border whitespace-nowrap transition-all ${
68 |               String(faenaFil) === String(f.id)
69 |                 ? "border-primary/50 text-primary bg-primary/10"
70 |                 : "border-border text-muted-foreground bg-transparent"
71 |             }`}
72 |             style={
73 |               f.id !== "all" && String(faenaFil) === String(f.id)
74 |                 ? { borderColor: `${f.color}80`, color: f.color, backgroundColor: `${f.color}15` }
75 |                 : undefined
76 |             }
77 |           >
78 |             {f.nombre}
79 |             {f.id !== "all" && (
80 |               <span className="ml-1.5 opacity-60">
81 |                 ({camionesRaw.filter(c => c.faenaId === f.id).length})
82 |               </span>
83 |             )}
84 |           </button>
85 |         ))}
86 |       </div>
87 |
88 |       {criticos.length > 0 && (
89 |         <div data-testid="criticos-section">
90 |           <div className="text-[9px] font-mono text-red-400 tracking-[0.2em] uppercase mb-2 flex items-center gap-2">
91 |             <span className="w-1.5 h-1.5 rounded-full bg-red-500 animate-pulse" />
92 |             ACCION INMEDIATA — POSIBLE ROBO O FRAUDE
93 |           </div>
94 |           {criticos.map(cam => {
95 |             const faena = faenas.find(f => f.id === cam.faenaId);
96 |             const criticCarga = cam.cargasAnalizadas.find(c => c.estado === "CRITICO");
97 |             return (
98 |               <div key={cam.id}
99 |                 className="bg-red-500/5 border border-red-500/25 p-3 px-4 mb-2 flex justify-between items-center"
100 |                 style={{ borderLeftWidth: 4, borderLeftColor: "#FF2D4A" }}
101 |                 data-testid={`critico-alert-${cam.id}`}>
102 |                 <div className="flex gap-5 items-center">
103 |                   <div>
104 |                     <div className="text-base font-mono font-bold text-primary">{cam.patente}</div>
```

```
105 |                     <div className="text-[10px] text-muted-foreground mt-0.5">
106 |                       {cam.modelo} · {faena && <span style={{ color: faena.color }}>{faena.nombre}</span>}
107 |                     </div>
108 |                   </div>
109 |                   <div className="text-center">
110 |                     <div className="text-[8px] font-mono text-muted-foreground mb-0.5">TARJETA</div>
111 |                     <div className="text-base font-mono text-foreground">{criticCarga?.litrosSurtidor}L</div>
112 |                   </div>
113 |                   <div className="text-xl text-red-400 font-bold">/=</div>
114 |                   <div className="text-center">
115 |                     <div className="text-[8px] font-mono text-muted-foreground mb-0.5">ECU VOLVO</div>
116 |                     <div className="text-base font-mono text-emerald-400">{criticCarga?.litrosEcu}L</div>
117 |                   </div>
118 |                   <div className="text-center">
119 |                     <div className="text-[8px] font-mono text-muted-foreground mb-0.5">DIFERENCIA TOTAL</div>
120 |                     <div className="text-lg font-mono font-bold text-red-400">+{cam.litDesv}L</div>
121 |                     <div className="text-[10px] text-red-400">{f$(cam.litDesv * PRECIO)}</div>
122 |                   </div>
123 |                 </div>
124 |                 <div className="flex gap-2">
125 |                   <Button variant="outline" size="sm" onClick={() => setCargaId(cam.id)} data-testid={`button-add-carga-
126 |                     <Plus className="w-3 h-3 mr-1" /> Carga
127 |                   </Button>
128 |                   <Button variant="destructive" size="sm" onClick={() => setFichaId(cam.id)} data-testid={`button-ver-fi
129 |                     VER FICHA
130 |                   </Button>
131 |                 </div>
132 |               </div>
133 |             );
134 |           })}
135 |         </div>
136 |       )}
137 |
138 |       <div className="flex gap-2 items-center flex-wrap" data-testid="search-filters">
139 |         <div className="relative">
140 |           <Search className="absolute left-2.5 top-1/2 -translate-y-1/2 w-3.5 h-3.5 text-muted-foreground" />
141 |           <Input value={busq} onChange={e => setBusq(e.target.value)}
142 |             placeholder="Buscar patente o conductor..."
143 |             className="pl-8 w-56 font-mono text-xs bg-card"
144 |             data-testid="input-search" />
145 |         </div>
146 |         {["todos", "CRITICO", "ALERTA", "OK"].map(f => (
147 |           <button key={f} onClick={() => setEstadoFil(f)}
148 |             data-testid={`filter-estado-${f.toLowerCase()}`}
149 |             className={`px-3.5 py-1.5 text-[10px] font-mono tracking-[0.1em] cursor-pointer border uppercase transition-
150 |               estadoFil === f
151 |                 ? f === "CRITICO" ? "border-red-500/50 text-red-400 bg-red-500/10"
152 |                 : f === "ALERTA" ? "border-amber-400/50 text-amber-400 bg-amber-400/10"
153 |                 : f === "OK" ? "border-emerald-400/50 text-emerald-400 bg-emerald-400/10"
154 |                 : "border-muted-foreground/30 text-muted-foreground bg-muted/30"
155 |                 : "border-border text-muted-foreground"
156 |             }`}>
157 |             {f === "todos" ? "Todos" : f}
158 |           </button>
159 |         ))}
160 |         <div className="ml-auto text-[11px] font-mono text-muted-foreground">
161 |           {filtrados.length} camiones
162 |         </div>
163 |       </div>
164 |
165 |       <div className="bg-card border border-border" data-testid="truck-table">
166 |         <div className="grid grid-cols-[76px_140px_1fr_80px_80px_80px_70px_70px_90px] px-4 py-2 border-b border-border b
167 |           <div>ESTADO</div><div>PATENTE</div><div>FAENA / MODELO</div><div>SURT.</div><div>ECU</div><div>DIF</div><div>R
168 |         </div>
169 |
170 |         {filtrados.length === 0 && (
171 |           <div className="py-12 text-center text-muted-foreground font-mono text-xs" data-testid="text-empty-table">
172 |             Sin camiones con este filtro.
173 |           </div>
174 |         )}
175 |
176 |         {filtrados.map((cam, i) => {
177 |           const faena = faenas.find(f => f.id === cam.faenaId);
178 |           const totS = cam.cargasAnalizadas.reduce((a, c) => a + c.litrosSurtidor, 0);
179 |           const totE = cam.cargasAnalizadas.reduce((a, c) => a + c.litrosEcu, 0);
180 |           return (
181 |             <div key={cam.id}
182 |               className={`grid grid-cols-[76px_140px_1fr_80px_80px_80px_70px_70px_90px] px-4 py-3 items-center cursor-po
183 |                 i < filtrados.length - 1 ? "border-b border-border/20" : ""
184 |               } ${cam.estado === "CRITICO" ? "bg-red-500/[0.03]" : ""}`}
185 |               onClick={() => setFichaId(cam.id)}
186 |               data-testid={`truck-row-${cam.id}`}>
187 |
188 |               <StatusTag label={cam.estado === "OK" ? "OK" : cam.estado} variant={getStatusVariant(cam.estado)} />
189 |
190 |               <div>
191 |                 <div className="text-sm font-mono font-bold text-primary">{cam.patente}</div>
192 |                 <div className={`text-[9px] font-mono mt-0.5 ${cam.syncOk ? "text-emerald-400" : "text-red-400"}`}>
193 |                   {cam.syncOk ? "sync OK" : "sin sync"}
194 |                 </div>
195 |               </div>
196 |
197 |               <div>
198 |                 {faena && (
199 |                   <div class="flex items-center gap-1.5 mb-0.5">
200 |                     <div class="w-2 h-2 rounded-full flex-shrink-0" style={{ backgroundColor: faena.color }} />
201 |                     <span class="text-[11px] text-muted-foreground">{faena.nombre}</span>
202 |                   </div>
203 |                 )}
204 |                 <div class="text-[11px] text-muted-foreground/60">{cam.modelo}</div>
205 |               </div>
206 |
207 |               <div class="text-xs font-mono text-foreground">{fL(totS)}</div>
208 |               <div class="text-xs font-mono text-emerald-400">{fL(totE)}</div>
209 |               <div class={`text-[13px] font-mono font-bold ${
210 |                 cam.litDesv > 0 ? cam.estado === "CRITICO" ? "text-red-400" : cam.estado === "ALERTA" ? "text-amber-400"
211 |               }`}>
```

```
212 |                 {cam.litDesv > 0 ? `+${cam.litDesv}L` : "\u2014"}
213 |               </div>
214 |               <div className={`text-[13px] font-mono font-bold ${rendColor(cam.rendProm, cam.metaKmL)}`}>
215 |                 {cam.rendProm ?? "\u2014"}
216 |               </div>
217 |               <div className={`text-xs font-mono ${cam.pctRal != null && cam.pctRal > 15 ? "text-amber-400" : "text-mute
218 |                 {cam.pctRal != null ? `${cam.pctRal}%` : "\u2014"}
219 |               </div>
220 |
221 |               <div className="flex gap-1.5" onClick={e => e.stopPropagation()}>
222 |                 <Button variant="outline" size="sm" className="h-7 px-2 text-[10px] font-mono"
223 |                   onClick={() => setCargaId(cam.id)} data-testid={`button-add-carga-row-${cam.id}`}>
224 |                   +L
225 |                 </Button>
226 |                 <Button variant="outline" size="sm" className="h-7 px-2 text-[10px] font-mono"
227 |                   onClick={() => setFichaId(cam.id)} data-testid={`button-ficha-row-${cam.id}`}>
228 |                   <FileText className="w-3 h-3" />
229 |                 </Button>
230 |               </div>
231 |             </div>
232 |           );
233 |         })}
234 |       </div>
235 |
236 |       {fichaStats && (
237 |         <FichaCamionModal cam={fichaStats} faenas={faenas} open={!!fichaId} onClose={() => setFichaId(null)} />
238 |       )}
239 |
240 |       {cargaId && (
241 |         <CargaModal camionId={cargaId} open={!!cargaId} onClose={() => setCargaId(null)} />
242 |       )}
243 |     </div>
244 |   );
245 | }
246 |
```

client/src/pages/micro-cargas.tsx

229 lineas

```
1 | import { useState } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { Loader2, ChevronDown, ChevronRight, AlertTriangle, MapPin, Fuel, Ban, FileDown, CheckCircle } from "lucide-reac
4 |
5 | interface MicroCargaRecord {
6 |   numGuia: number;
7 |   patente: string;
8 |   conductor: string | null;
9 |   fecha: string;
10 |   litros: number;
11 |   lugar: string | null;
12 |   faena: string | null;
13 |   signals: string[];
14 |   riesgo: "CRITICO" | "SOSPECHOSO";
15 |   fuelLevelActual: number | null;
16 | }
17 |
18 | interface MicroCargasData {
19 |   registros: MicroCargaRecord[];
20 |   totals: {
21 |     criticos: number;
22 |     sospechosos: number;
23 |     normales: number;
24 |     totalCargas: number;
25 |     cargasPequenas: number;
26 |   };
27 |   desde: string;
28 | }
29 |
30 | function SignalBadge({ signal }: { signal: string }) {
31 |   const cfg: Record<string, { icon: any; label: string; color: string; bg: string }> = {
32 |     MULTIPLE_DIA: { icon: MapPin, label: "MULTIPLE DIA", color: "#ffcc00", bg: "rgba(255,204,0,0.08)" },
33 |     SIN_MOVIMIENTO: { icon: Ban, label: "SIN MOVIMIENTO", color: "#ff2244", bg: "rgba(255,34,68,0.08)" },
34 |     TANQUE_LLENO: { icon: Fuel, label: "TANQUE LLENO", color: "#ff6b35", bg: "rgba(255,107,53,0.08)" },
35 |   };
36 |   const c = cfg[signal] || cfg.MULTIPLE_DIA;
37 |   const Icon = c.icon;
38 |   return (
39 |     <span className="inline-flex items-center gap-1 font-space text-[8px] font-bold px-2 py-0.5 uppercase tracking-[0.08
40 |       style={{ color: c.color, background: c.bg, border: `1px solid ${c.color}30` }}>
41 |       <Icon className="w-2.5 h-2.5" />
42 |       {c.label}
43 |     </span>
44 |   );
45 | }
46 |
47 | function ExpandedDetail({ record }: { record: MicroCargaRecord }) {
48 |   return (
49 |     <div className="px-5 py-4 space-y-3" style={{ background: "rgba(0,212,255,0.02)", borderTop: "1px solid #0d2035" }}>
50 |       <div className="grid grid-cols-4 gap-4">
51 |         <div>
52 |           <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>PATENTE</div>
53 |           <div className="font-space text-[13px] font-bold" style={{ color: "#c8e8ff" }}>{record.patente}</div>
54 |         </div>
55 |         <div>
56 |           <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>CONDUCTOR</div>
57 |           <div className="font-exo text-[11px]" style={{ color: "#c8e8ff" }}>{record.conductor || "Sin datos"}</div>
58 |         </div>
59 |         <div>
60 |           <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>LUGAR</div>
61 |           <div className="font-exo text-[11px]" style={{ color: "#c8e8ff" }}>{record.lugar || "Sin datos"}</div>
62 |         </div>
63 |         <div>
64 |           <div className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080" }}>NIVEL TANQUE ACT
65 |           <div className="font-space text-[13px] font-bold" style={{
66 |             color: record.fuelLevelActual != null
67 |               ? record.fuelLevelActual > 80 ? "#ff2244" : record.fuelLevelActual > 50 ? "#ffcc00" : "#00ff88"
68 |               : "#3a6080"
69 |           }}>
70 |             {record.fuelLevelActual != null ? `${Math.round(record.fuelLevelActual)}%` : "N/D"}
71 |           </div>
72 |         </div>
73 |       </div>
74 |       <div>
75 |         <div className="font-exo text-[8px] tracking-[0.15em] uppercase mb-1" style={{ color: "#3a6080" }}>SENALES DETEC
76 |         <div className="space-y-1">
77 |           {record.signals.includes("MULTIPLE_DIA") && (
78 |             <div className="font-exo text-[10px]" style={{ color: "#ffcc00" }}>
79 |               Multiples cargas pequenas en distintos puntos el mismo dia. Patron consistente con desvio de combustible.
80 |             </div>
81 |           )}
82 |           {record.signals.includes("SIN_MOVIMIENTO") && (
83 |             <div className="font-exo text-[10px]" style={{ color: "#ff2244" }}>
84 |               Carga registrada sin movimiento GPS detectado. El camion no muestra actividad cercana al momento de la car
85 |             </div>
86 |           )}
87 |           {record.signals.includes("TANQUE_LLENO") && (
88 |             <div className="font-exo text-[10px]" style={{ color: "#ff6b35" }}>
89 |               El nivel de combustible Volvo indica tanque sobre 80% al momento de consulta. Carga posiblemente no ingres
90 |             </div>
91 |           )}
92 |         </div>
93 |       </div>
94 |     </div>
95 |   );
96 | }
97 |
98 | export default function MicroCargas() {
99 |   const [expandedGuia, setExpandedGuia] = useState<number | null>(null);
100 |
101 |   const { data, isLoading, error } = useQuery<MicroCargasData>({
102 |     queryKey: ["/api/datos/micro-cargas"],
103 |   });
104 |
```

```
105 |   if (isLoading) {
106 |     return (
107 |       <div className="flex items-center justify-center py-20" data-testid="micro-cargas-loading">
108 |         <Loader2 className="w-8 h-8 animate-spin" style={{ color: "#00d4ff" }} />
109 |         <span className="ml-3 font-exo text-[12px]" style={{ color: "#3a6080" }}>Analizando cargas sospechosas...</span>
110 |       </div>
111 |     );
112 |   }
113 |
114 |   if (error) {
115 |     return (
116 |       <div className="dash-card px-8 py-12 text-center" data-testid="micro-cargas-error">
117 |         <AlertTriangle className="w-8 h-8 mx-auto mb-3" style={{ color: "#ff2244" }} />
118 |         <div className="font-space text-[14px] font-bold" style={{ color: "#ff2244" }}>Error al cargar datos</div>
119 |         <div className="font-exo text-[11px] mt-2" style={{ color: "#3a6080" }}>{(error as Error).message}</div>
120 |       </div>
121 |     );
122 |   }
123 |
124 |   if (!data) return null;
125 |   const { registros, totals } = data;
126 |
127 |   return (
128 |     <div className="space-y-5" data-testid="micro-cargas-page">
129 |       <div className="flex items-center justify-between">
130 |         <div>
131 |           <h2 className="font-space text-[18px] font-bold tracking-[0.12em] uppercase" style={{ color: "#c8e8ff" }}>
132 |             Micro-Cargas Sospechosas
133 |           </h2>
134 |           <p className="font-exo text-[11px] mt-1" style={{ color: "#3a6080" }}>
135 |             Analizando desde 01-03-2026 &middot; {totals.totalCargas.toLocaleString()} cargas totales &middot; {totals.c
136 |           </p>
137 |         </div>
138 |         <button className="flex items-center gap-2 px-3 py-1.5 font-space text-[10px] font-bold tracking-[0.1em] upperca
139 |           style={{ color: "#00d4ff", border: "1px solid #00d4ff30", background: "rgba(0,212,255,0.05)" }}
140 |           data-testid="btn-export-sospechosos">
141 |           <FileDown className="w-3.5 h-3.5" />
142 |           EXPORTAR SOSPECHOSOS
143 |         </button>
144 |       </div>
145 |
146 |       <div className="flex items-center gap-4">
147 |         {totals.criticos > 0 && (
148 |           <div className="flex items-center gap-2 px-3 py-2 dash-card" data-testid="badge-criticos">
149 |             <span className="inline-block w-3 h-3 rounded-full" style={{ background: "#ff2244", boxShadow: "0 0 8px #ff2
150 |             <span className="font-space text-[16px] font-bold" style={{ color: "#ff2244" }}>{totals.criticos}</span>
151 |             <span className="font-exo text-[10px] tracking-[0.1em] uppercase" style={{ color: "#ff2244" }}>CRITICOS</spa
152 |           </div>
153 |         )}
154 |         <div className="flex items-center gap-2 px-3 py-2 dash-card" data-testid="badge-sospechosos">
155 |           <span className="inline-block w-3 h-3 rounded-full" style={{ background: "#ffcc00" }} />
156 |           <span className="font-space text-[16px] font-bold" style={{ color: "#ffcc00" }}>{totals.sospechosos}</span>
157 |           <span className="font-exo text-[10px] tracking-[0.1em] uppercase" style={{ color: "#ffcc00" }}>SOSPECHOSOS</sp
158 |         </div>
159 |         <div className="flex items-center gap-2 px-3 py-2 dash-card" data-testid="badge-normales">
160 |           <span className="inline-block w-3 h-3 rounded-full" style={{ background: "#00ff88" }} />
161 |           <span className="font-space text-[16px] font-bold" style={{ color: "#00ff88" }}>{totals.normales}</span>
162 |           <span className="font-exo text-[10px] tracking-[0.1em] uppercase" style={{ color: "#3a6080" }}>NORMALES (ocult
163 |         </div>
164 |       </div>
165 |
166 |       {registros.length === 0 ? (
167 |         <div className="dash-card px-8 py-12 text-center" data-testid="micro-cargas-empty">
168 |           <CheckCircle className="w-10 h-10 mx-auto mb-3" style={{ color: "#00ff88" }} />
169 |           <div className="font-space text-[16px] font-bold" style={{ color: "#00ff88" }}>Sin cargas sospechosas detectad
170 |           <div className="font-exo text-[11px] mt-2" style={{ color: "#3a6080" }}>
171 |             Todas las cargas pequenas del periodo son consistentes con los datos GPS y niveles de tanque.
172 |           </div>
173 |         </div>
174 |       ) : (
175 |         <div className="dash-card overflow-hidden">
176 |           <div className="flex items-center gap-3 px-5 py-2" style={{ borderBottom: "1px solid #0d2035", background: "#0
177 |             <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080", width: "70px" }
178 |             <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080", width: "70px" }
179 |             <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080", width: "140px"
180 |             <span className="font-exo text-[8px] tracking-[0.15em] uppercase" style={{ color: "#3a6080", width: "130px"
181 |             <span className="font-exo text-[8px] tracking-[0.15em] uppercase text-right" style={{ color: "#3a6080", widt
182 |             <span className="font-exo text-[8px] tracking-[0.15em] uppercase flex-1" style={{ color: "#3a6080" }}>SENALE
183 |           </div>
184 |
185 |           {registros.map(r => (
186 |             <div key={r.numGuia}>
187 |               <div
188 |                 className="flex items-center gap-3 px-5 py-2.5 cursor-pointer transition-all hover:bg-[rgba(0,212,255,0.
189 |                 style={{ borderBottom: "1px solid #0d2035" }}
190 |                 onClick={() => setExpandedGuia(expandedGuia === r.numGuia ? null : r.numGuia)}
191 |                 data-testid={`micro-carga-row-${r.numGuia}`}
192 |               >
193 |                 <div style={{ width: "70px" }}>
194 |                   {r.riesgo === "CRITICO" ? (
195 |                     <span className="inline-flex items-center gap-1 font-space text-[9px] font-bold px-2 py-0.5"
196 |                       style={{ color: "#ff2244", background: "rgba(255,34,68,0.1)", border: "1px solid rgba(255,34,68,0.
197 |                       CRITICO
198 |                     </span>
199 |                   ) : (
200 |                     <span className="inline-flex items-center gap-1 font-space text-[9px] font-bold px-2 py-0.5"
201 |                       style={{ color: "#ffcc00", background: "rgba(255,204,0,0.08)", border: "1px solid rgba(255,204,0,0
202 |                       SOSPECHOSO
203 |                     </span>
204 |                   )}
205 |                 </div>
206 |                 <span className="font-space text-[11px] font-bold" style={{ color: "#c8e8ff", width: "70px" }}>{r.patent
207 |                 <span className="font-exo text-[10px] truncate" style={{ color: "#c8e8ff", width: "140px" }}>{r.conducto
208 |                 <span className="font-space text-[10px]" style={{ color: "#3a6080", width: "130px" }}>{r.fecha}</span>
209 |                 <span className="font-space text-[11px] font-bold text-right" style={{ color: "#ffcc00", width: "60px" }
210 |                 <div className="flex items-center gap-1.5 flex-1">
211 |                   {r.signals.map(s => <SignalBadge key={s} signal={s} />)}
```

```
212 |                 </div>
213 |                 {expandedGuia === r.numGuia
214 |                   ? <ChevronDown className="w-4 h-4 flex-shrink-0" style={{ color: "#00d4ff" }} />
215 |                   : <ChevronRight className="w-4 h-4 flex-shrink-0" style={{ color: "#3a6080" }} />}
216 |               </div>
217 |               {expandedGuia === r.numGuia && <ExpandedDetail record={r} />}
218 |             </div>
219 |           ))}
220 |         </div>
221 |       )}
222 |
223 |       <div className="text-[9px] font-space tracking-[0.15em]" style={{ color: "#3a6080" }}>
224 |         Senales: MULTIPLE DIA = 2+ cargas &lt;100L en distintos puntos mismo dia &middot; SIN MOVIMIENTO = sin GPS activ
225 |       </div>
226 |     </div>
227 |   );
228 | }
229 |
```

client/src/components/camion-modal.tsx

146 lineas

```
1 | import { useState } from "react";
2 | import { Dialog, DialogContent, DialogHeader, DialogTitle } from "@/components/ui/dialog";
3 | import { Button } from "@/components/ui/button";
4 | import { Input } from "@/components/ui/input";
5 | import { Label } from "@/components/ui/label";
6 | import { Select, SelectContent, SelectItem, SelectTrigger, SelectValue } from "@/components/ui/select";
7 | import { Separator } from "@/components/ui/separator";
8 | import { useMutation } from "@tanstack/react-query";
9 | import { apiRequest, queryClient } from "@/lib/queryClient";
10 | import { useToast } from "@/hooks/use-toast";
11 | import type { Camion, Faena } from "@shared/schema";
12 |
13 | interface CamionModalProps {
14 |   faenas: Faena[];
15 |   camion?: Camion | null;
16 |   open: boolean;
17 |   onClose: () => void;
18 | }
19 |
20 | export function CamionModal({ faenas, camion, open, onClose }: CamionModalProps) {
21 |   const { toast } = useToast();
22 |   const isEdit = !!camion;
23 |
24 |   const [form, setForm] = useState({
25 |     patente: camion?.patente || "",
26 |     modelo: camion?.modelo || "",
27 |     faenaId: camion?.faenaId?.toString() || (faenas[0]?.id?.toString() || ""),
28 |     metaKmL: camion?.metaKmL?.toString() || "2.1",
29 |     vin: camion?.vin || "",
30 |     conductor: camion?.conductor || "",
31 |   });
32 |
33 |   const set = (key: string, value: string) => setForm(prev => ({ ...prev, [key]: value }));
34 |
35 |   const mutation = useMutation({
36 |     mutationFn: async () => {
37 |       const data = {
38 |         patente: form.patente.toUpperCase().replace(/\s/g, ""),
39 |         modelo: form.modelo,
40 |         faenaId: parseInt(form.faenaId),
41 |         metaKmL: parseFloat(form.metaKmL) || 2.1,
42 |         vin: form.vin || null,
43 |         conductor: form.conductor || null,
44 |         syncOk: false,
45 |       };
46 |       if (isEdit) {
47 |         await apiRequest("PATCH", `/api/camiones/${camion!.id}`, data);
48 |       } else {
49 |         await apiRequest("POST", "/api/camiones", data);
50 |       }
51 |     },
52 |     onSuccess: () => {
53 |       queryClient.invalidateQueries({ queryKey: ["/api/camiones"] });
54 |       toast({ title: isEdit ? "Camion actualizado" : "Camion creado" });
55 |       onClose();
56 |     },
57 |     onError: (error: Error) => {
58 |       toast({ title: "Error", description: error.message, variant: "destructive" });
59 |     },
60 |   });
61 |
62 |   const canSubmit = form.patente && form.modelo && form.faenaId;
63 |
64 |   return (
65 |     <Dialog open={open} onOpenChange={() => onClose()}>
66 |       <DialogContent className="max-w-md bg-card border-border">
67 |         <DialogHeader>
68 |           <DialogTitle className="text-sm font-mono font-bold tracking-[0.15em] uppercase">
69 |             {isEdit ? "Editar Camion" : "Nuevo Camion"}
70 |           </DialogTitle>
71 |         </DialogHeader>
72 |
73 |         <div className="space-y-4">
74 |           <div className="grid grid-cols-2 gap-3">
75 |             <div>
76 |               <Label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] uppercase">Patente</Label>
77 |               <Input value={form.patente} onChange={e => set("patente", e.target.value.toUpperCase())}
78 |                 placeholder="DKPW78" className="font-mono bg-background" data-testid="input-patente" />
79 |             </div>
80 |             <div>
81 |               <Label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] uppercase">Modelo</Label>
82 |               <Input value={form.modelo} onChange={e => set("modelo", e.target.value)}
83 |                 placeholder="Volvo FH16 750" className="font-mono bg-background" data-testid="input-modelo" />
84 |             </div>
85 |           </div>
86 |
87 |           <div>
88 |             <Label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] uppercase">Faena</Label>
89 |             <Select value={form.faenaId} onValueChange={v => set("faenaId", v)}>
90 |               <SelectTrigger className="font-mono bg-background" data-testid="select-faena">
91 |                 <SelectValue />
92 |               </SelectTrigger>
93 |               <SelectContent>
94 |                 {faenas.map(f => (
95 |                   <SelectItem key={f.id} value={f.id.toString()}>{f.nombre}</SelectItem>
96 |                 ))}
97 |               </SelectContent>
98 |             </Select>
99 |           </div>
100 |
101 |           <div>
102 |             <Label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] uppercase">Meta rendimiento (
103 |             <Input type="number" value={form.metaKmL} onChange={e => set("metaKmL", e.target.value)}
104 |               placeholder="2.1" className="font-mono bg-background" data-testid="input-meta" />
```

```
105 |             <p className="text-[10px] text-muted-foreground mt-1 font-mono">
106 |               Se usa para detectar bajo rendimiento automaticamente
107 |             </p>
108 |           </div>
109 |
110 |           <div className="flex items-center gap-3 py-1">
111 |             <Separator className="flex-1" />
112 |             <span className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] uppercase">Datos Volvo Connect
113 |             <Separator className="flex-1" />
114 |           </div>
115 |
116 |           <div className="grid grid-cols-2 gap-3">
117 |             <div>
118 |               <Label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] uppercase">VIN (del camion)
119 |               <Input value={form.vin} onChange={e => set("vin", e.target.value)}
120 |                 placeholder="YV2RT40A5KB123403" className="font-mono bg-background" data-testid="input-vin" />
121 |             </div>
122 |             <div>
123 |               <Label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] uppercase">Conductor habitu
124 |               <Input value={form.conductor} onChange={e => set("conductor", e.target.value)}
125 |                 placeholder="P. Contreras" className="font-mono bg-background" data-testid="input-conductor" />
126 |             </div>
127 |           </div>
128 |
129 |           <div className="bg-primary/5 border border-primary/20 p-3 text-[11px] text-muted-foreground font-mono">
130 |             El odometro, horas motor, ralenti y velocidad se sincronizan automaticamente
131 |             desde Volvo Connect usando el VIN. El VIN es el ID unico del camion en Volvo.
132 |           </div>
133 |
134 |           <div className="flex gap-2 justify-end">
135 |             <Button variant="outline" onClick={onClose} data-testid="button-cancel-camion">Cancelar</Button>
136 |             <Button onClick={() => mutation.mutate()} disabled={!canSubmit || mutation.isPending}
137 |               data-testid="button-save-camion">
138 |               {mutation.isPending ? "Guardando..." : "Guardar camion"}
139 |             </Button>
140 |           </div>
141 |         </div>
142 |       </DialogContent>
143 |     </Dialog>
144 |   );
145 | }
146 |
```

client/src/components/carga-modal.tsx

118 lineas

```
1 | import { useState } from "react";
2 | import { Dialog, DialogContent, DialogHeader, DialogTitle } from "@/components/ui/dialog";
3 | import { Button } from "@/components/ui/button";
4 | import { Input } from "@/components/ui/input";
5 | import { Label } from "@/components/ui/label";
6 | import { Select, SelectContent, SelectItem, SelectTrigger, SelectValue } from "@/components/ui/select";
7 | import { useMutation } from "@tanstack/react-query";
8 | import { apiRequest, queryClient } from "@/lib/queryClient";
9 | import { useToast } from "@/hooks/use-toast";
10 |
11 | interface CargaModalProps {
12 |   camionId: number;
13 |   open: boolean;
14 |   onClose: () => void;
15 | }
16 |
17 | export function CargaModal({ camionId, open, onClose }: CargaModalProps) {
18 |   const { toast } = useToast();
19 |   const [form, setForm] = useState({
20 |     fecha: new Date().toISOString().slice(0, 10),
21 |     litrosSurtidor: "",
22 |     litrosEcu: "",
23 |     kmAnterior: "",
24 |     kmActual: "",
25 |     proveedor: "COPEC",
26 |   });
27 |
28 |   const set = (key: string, value: string) => setForm(prev => ({ ...prev, [key]: value }));
29 |
30 |   const mutation = useMutation({
31 |     mutationFn: async () => {
32 |       await apiRequest("POST", "/api/cargas", {
33 |         camionId,
34 |         fecha: form.fecha,
35 |         litrosSurtidor: parseFloat(form.litrosSurtidor),
36 |         litrosEcu: parseFloat(form.litrosEcu),
37 |         kmAnterior: parseInt(form.kmAnterior),
38 |         kmActual: parseInt(form.kmActual),
39 |         proveedor: form.proveedor,
40 |       });
41 |     },
42 |     onSuccess: () => {
43 |       queryClient.invalidateQueries({ queryKey: ["/api/cargas"] });
44 |       toast({ title: "Carga registrada correctamente" });
45 |       onClose();
46 |     },
47 |     onError: (error: Error) => {
48 |       toast({ title: "Error", description: error.message, variant: "destructive" });
49 |     },
50 |   });
51 |
52 |   const canSubmit = form.litrosSurtidor && form.litrosEcu && form.kmAnterior && form.kmActual;
53 |
54 |   return (
55 |     <Dialog open={open} onOpenChange={() => onClose()}>
56 |       <DialogContent className="max-w-md bg-card border-border">
57 |         <DialogHeader>
58 |           <DialogTitle className="text-sm font-mono font-bold tracking-[0.15em] uppercase">
59 |             Registrar Carga
60 |           </DialogTitle>
61 |         </DialogHeader>
62 |
63 |         <div className="space-y-4">
64 |           <div>
65 |             <Label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] uppercase">Fecha</Label>
66 |             <Input type="date" value={form.fecha} onChange={e => set("fecha", e.target.value)}
67 |               className="font-mono bg-background" data-testid="input-fecha" />
68 |           </div>
69 |
70 |           <div class="grid grid-cols-2 gap-3">
71 |             <div>
72 |               <Label class="text-[9px] font-mono text-muted-foreground tracking-[0.15em] uppercase">Litros surtidor
73 |               <Input type="number" value={form.litrosSurtidor} onChange={e => set("litrosSurtidor", e.target.value)}
74 |                 placeholder="410" class="font-mono bg-background" data-testid="input-litros-surtidor" />
75 |             </div>
76 |             <div>
77 |               <Label class="text-[9px] font-mono text-muted-foreground tracking-[0.15em] uppercase">Litros ECU Volvo
78 |               <Input type="number" value={form.litrosEcu} onChange={e => set("litrosEcu", e.target.value)}
79 |                 placeholder="280" class="font-mono bg-background" data-testid="input-litros-ecu" />
80 |             </div>
81 |             <div>
82 |               <Label class="text-[9px] font-mono text-muted-foreground tracking-[0.15em] uppercase">Km anterior</Lab
83 |               <Input type="number" value={form.kmAnterior} onChange={e => set("kmAnterior", e.target.value)}
84 |                 placeholder="301760" class="font-mono bg-background" data-testid="input-km-anterior" />
85 |             </div>
86 |             <div>
87 |               <Label class="text-[9px] font-mono text-muted-foreground tracking-[0.15em] uppercase">Km actual (odome
88 |               <Input type="number" value={form.kmActual} onChange={e => set("kmActual", e.target.value)}
89 |                 placeholder="302100" class="font-mono bg-background" data-testid="input-km-actual" />
90 |             </div>
91 |           </div>
92 |
93 |           <div>
94 |             <Label class="text-[9px] font-mono text-muted-foreground tracking-[0.15em] uppercase">Proveedor</Label>
95 |             <Select value={form.proveedor} onValueChange={v => set("proveedor", v)}>
96 |               <SelectTrigger class="font-mono bg-background" data-testid="select-proveedor">
97 |                 <SelectValue />
98 |               </SelectTrigger>
99 |               <SelectContent>
100 |                 <SelectItem value="COPEC">COPEC</SelectItem>
101 |                 <SelectItem value="SHELL">SHELL</SelectItem>
102 |               </SelectContent>
103 |             </Select>
104 |           </div>
```

```
105 |
106 |           <div className="flex gap-2 justify-end pt-2">
107 |             <Button variant="outline" onClick={onClose} data-testid="button-cancel-carga">Cancelar</Button>
108 |             <Button onClick={() => mutation.mutate()} disabled={!canSubmit || mutation.isPending}
109 |               data-testid="button-save-carga">
110 |               {mutation.isPending ? "Guardando..." : "Guardar carga"}
111 |             </Button>
112 |           </div>
113 |         </div>
114 |       </DialogContent>
115 |     </Dialog>
116 |   );
117 | }
118 |
```

client/src/components/faena-modal.tsx

84 lineas

```
1 | import { useState } from "react";
2 | import { Dialog, DialogContent, DialogHeader, DialogTitle } from "@/components/ui/dialog";
3 | import { Button } from "@/components/ui/button";
4 | import { Input } from "@/components/ui/input";
5 | import { Label } from "@/components/ui/label";
6 | import { useMutation } from "@tanstack/react-query";
7 | import { apiRequest, queryClient } from "@/lib/queryClient";
8 | import { useToast } from "@/hooks/use-toast";
9 | import type { Faena } from "@shared/schema";
10 |
11 | interface FaenaModalProps {
12 |   faena?: Faena | null;
13 |   open: boolean;
14 |   onClose: () => void;
15 | }
16 |
17 | export function FaenaModal({ faena, open, onClose }: FaenaModalProps) {
18 |   const { toast } = useToast();
19 |   const isEdit = !!faena;
20 |   const [nombre, setNombre] = useState(faena?.nombre || "");
21 |   const [color, setColor] = useState(faena?.color || "#1A8FFF");
22 |
23 |   const mutation = useMutation({
24 |     mutationFn: async () => {
25 |       if (isEdit) {
26 |         await apiRequest("PATCH", `/api/faenas/${faena!.id}`, { nombre, color });
27 |       } else {
28 |         await apiRequest("POST", "/api/faenas", { nombre, color });
29 |       }
30 |     },
31 |     onSuccess: () => {
32 |       queryClient.invalidateQueries({ queryKey: ["/api/faenas"] });
33 |       toast({ title: isEdit ? "Faena actualizada" : "Faena creada" });
34 |       onClose();
35 |     },
36 |     onError: (error: Error) => {
37 |       toast({ title: "Error", description: error.message, variant: "destructive" });
38 |     },
39 |   });
40 |
41 |   return (
42 |     <Dialog open={open} onOpenChange={() => onClose()}>
43 |       <DialogContent className="max-w-md bg-card border-border">
44 |         <DialogHeader>
45 |           <DialogTitle className="text-sm font-mono font-bold tracking-[0.15em] uppercase">
46 |             {isEdit ? "Editar Faena" : "Nueva Faena"}
47 |           </DialogTitle>
48 |         </DialogHeader>
49 |
50 |         <div className="space-y-4">
51 |           <div>
52 |             <Label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] uppercase">
53 |               Nombre de la faena / cliente
54 |             </Label>
55 |             <Input value={nombre} onChange={e => setNombre(e.target.value)}
56 |               placeholder="Anglo American — Los Bronces" className="font-mono bg-background"
57 |               data-testid="input-faena-nombre" />
58 |           </div>
59 |
60 |           <div>
61 |             <Label className="text-[9px] font-mono text-muted-foreground tracking-[0.15em] uppercase mb-1.5 block">
62 |               Color identificador
63 |             </Label>
64 |             <div className="flex items-center gap-3">
65 |               <input type="color" value={color} onChange={e => setColor(e.target.value)}
66 |                 className="w-11 h-9 border border-border bg-transparent cursor-pointer p-0.5"
67 |                 data-testid="input-faena-color" />
68 |               <div className="flex-1 h-9 opacity-30" style={{ backgroundColor: color }} />
69 |             </div>
70 |           </div>
71 |
72 |           <div className="flex gap-2 justify-end pt-2">
73 |             <Button variant="outline" onClick={onClose} data-testid="button-cancel-faena">Cancelar</Button>
74 |             <Button onClick={() => nombre.trim() && mutation.mutate()} disabled={!nombre.trim() || mutation.isPending}
75 |               data-testid="button-save-faena">
76 |               {mutation.isPending ? "Guardando..." : "Guardar"}
77 |             </Button>
78 |           </div>
79 |         </div>
80 |       </DialogContent>
81 |     </Dialog>
82 |   );
83 | }
84 |
```

client/src/components/ficha-camion-modal.tsx

296 lineas

```
1 | import { useState, useEffect } from "react";
2 | import { useQuery } from "@tanstack/react-query";
3 | import { Dialog, DialogContent, DialogHeader, DialogTitle } from "@/components/ui/dialog";
4 | import { StatusTag, getStatusVariant } from "@/components/status-tag";
5 | import { Button } from "@/components/ui/button";
6 | import { Loader2, MapPin, CheckCircle } from "lucide-react";
7 | import { apiRequest } from "@/lib/queryClient";
8 | import type { Faena } from "@shared/schema";
9 | import { type CamionStats, type Percentiles, fN, fK, fL, f$, PRECIO, rendColor, statusColor, rendColorPercentil, getNive
10 |
11 | function RutaPuntosTab({ camId, patente }: { camId: number; patente: string }) {
12 |   const [puntos, setPuntos] = useState<any[]>([]);
13 |   const [loading, setLoading] = useState(false);
14 |   const [detecting, setDetecting] = useState(false);
15 |   const [resumen, setResumen] = useState<string | null>(null);
16 |
17 |   useEffect(() => {
18 |     fetch(`/api/puntos-ruta?camionId=${camId}`)
19 |       .then(r => r.json())
20 |       .then(setPuntos)
21 |       .catch(() => {});
22 |   }, [camId]);
23 |
24 |   const detectarPuntos = async () => {
25 |     setDetecting(true);
26 |     try {
27 |       const res = await apiRequest("POST", "/api/ia/detectar-puntos", { camionId: camId });
28 |       const data = await res.json();
29 |       if (data.puntos) setPuntos(prev => [...prev, ...data.puntos]);
30 |       if (data.resumen) setResumen(data.resumen);
31 |     } catch (e: any) {
32 |       console.error("Error detecting points:", e.message);
33 |     } finally {
34 |       setDetecting(false);
35 |     }
36 |   };
37 |
38 |   const confirmarPunto = async (id: number) => {
39 |     try {
40 |       await apiRequest("POST", `/api/puntos-ruta/${id}/confirmar`, { confirmadoPor: "admin" });
41 |       setPuntos(prev => prev.map(p => p.id === id ? { ...p, confirmado: true } : p));
42 |     } catch {}
43 |   };
44 |
45 |   const tipoColors: Record<string, string> = {
46 |     COMBUSTIBLE: "text-amber-400 bg-amber-400/10 border-amber-400/30",
47 |     CARGA: "text-blue-400 bg-blue-400/10 border-blue-400/30",
48 |     ENTREGA: "text-emerald-400 bg-emerald-400/10 border-emerald-400/30",
49 |     DESCANSO: "text-purple-400 bg-purple-400/10 border-purple-400/30",
50 |   };
51 |
52 |   return (
53 |     <div className="space-y-4" data-testid="ruta-puntos-tab">
54 |       <div className="flex items-center justify-between">
55 |         <div className="text-[9px] font-mono text-primary tracking-[0.15em] uppercase">
56 |           RUTA Y PUNTOS — {patente}
57 |         </div>
58 |         <Button
59 |           variant="outline"
60 |           size="sm"
61 |           onClick={detectarPuntos}
62 |           disabled={detecting}
63 |           data-testid="btn-detectar-puntos"
64 |         >
65 |           {detecting ? <Loader2 className="w-3 h-3 mr-1 animate-spin" /> : <MapPin className="w-3 h-3 mr-1" />}
66 |           Detectar puntos automatico
67 |         </Button>
68 |       </div>
69 |
70 |       {resumen && (
71 |         <div className="p-3 border bg-primary/5 border-primary/20">
72 |           <div className="text-[8px] font-mono text-primary tracking-[0.2em] mb-1">RESUMEN RUTA</div>
73 |           <div className="text-[11px] leading-relaxed" style={{ color: "#D4E4F4" }}>{resumen}</div>
74 |         </div>
75 |       )}
76 |
77 |       {puntos.length === 0 && !detecting && (
78 |         <div className="py-8 text-center text-muted-foreground font-mono text-xs" data-testid="text-no-puntos">
79 |           Sin puntos de ruta detectados. Use "Detectar puntos automatico" para analizar.
80 |         </div>
81 |       )}
82 |
83 |       {detecting && (
84 |         <div className="flex flex-col items-center gap-3 py-8">
85 |           <Loader2 className="w-6 h-6 animate-spin text-primary" />
86 |           <span className="text-xs font-mono" style={{ color: "#5A7A9A" }}>Analizando datos de ruta con IA...</span>
87 |         </div>
88 |       )}
89 |
90 |       {puntos.length > 0 && (
91 |         <div className="bg-background border border-border">
92 |           <div className="grid grid-cols-[100px_1fr_80px_80px_80px] px-3 py-1.5 border-b border-border text-[8px] font-m
93 |             <div>TIPO</div><div>NOMBRE INFERIDO</div><div>FRECUENCIA</div><div>L. PROM</div><div>CONFIRMADO</div>
94 |           </div>
95 |           {puntos.map((p, i) => (
96 |             <div key={p.id || i} className={`grid grid-cols-[100px_1fr_80px_80px_80px] px-3 py-2 items-center ${i < punt
97 |               <span className={`px-2 py-0.5 text-[9px] font-mono font-bold border w-fit ${tipoColors[p.tipo] || "text-gr
98 |                 {p.tipo || "N/A"}
99 |               </span>
100 |               <div className="text-[11px] font-mono" style={{ color: "#D4E4F4" }}>{p.nombreInferido || "Sin nombre"}</di
101 |               <div className="text-[11px] font-mono text-muted-foreground">{p.frecuencia || "-"}</div>
102 |               <div className="text-[11px] font-mono text-muted-foreground">{p.litrosPromedio ? `${p.litrosPromedio}L` :
103 |               <div>
104 |                 {p.confirmado ? (
```

```
105 |                   <span className="text-emerald-400 text-[10px] font-mono flex items-center gap-1"><CheckCircle classNam
106 |                 ) : (
107 |                   <button onClick={() => confirmarPunto(p.id)} className="text-[10px] font-mono text-primary cursor-poin
108 |                     Confirmar
109 |                   </button>
110 |                 )}
111 |               </div>
112 |             </div>
113 |           ))}
114 |         </div>
115 |       )}
116 |     </div>
117 |   );
118 | }
119 |
120 | interface FichaCamionModalProps {
121 |   cam: CamionStats;
122 |   faenas: Faena[];
123 |   open: boolean;
124 |   onClose: () => void;
125 | }
126 |
127 | export function FichaCamionModal({ cam, faenas, open, onClose }: FichaCamionModalProps) {
128 |   const faena = faenas.find(f => f.id === cam.faenaId);
129 |   const costoRal = cam.horasRalenti ? Math.round(cam.horasRalenti * 8 * PRECIO) : 0;
130 |   const [fichaTab, setFichaTab] = useState<"detalle" | "ruta">("detalle");
131 |   const { data: percentiles } = useQuery<Percentiles>({ queryKey: ["/api/flota/percentiles"] });
132 |
133 |   const totS = cam.cargasAnalizadas.reduce((a, c) => a + c.litrosSurtidor, 0);
134 |   const totE = cam.cargasAnalizadas.reduce((a, c) => a + c.litrosEcu, 0);
135 |
136 |   return (
137 |     <Dialog open={open} onOpenChange={() => { onClose(); setFichaTab("detalle"); }}>
138 |       <DialogContent className="max-w-[860px] bg-card border-border p-0 gap-0 overflow-y-auto max-h-[92vh]">
139 |         <DialogHeader className="p-6 pb-0">
140 |           <DialogTitle className="text-sm font-mono font-bold tracking-[0.15em] uppercase text-foreground">
141 |             Ficha — {cam.patente}
142 |           </DialogTitle>
143 |         </DialogHeader>
144 |
145 |         <div className="px-6 pt-3 flex gap-1">
146 |           {(["detalle", "ruta"] as const).map(t => (
147 |             <button key={t} onClick={() => setFichaTab(t)}
148 |               data-testid={`ficha-tab-${t}`}
149 |               className={`px-4 py-2 text-[10px] font-mono font-bold tracking-[0.15em] uppercase cursor-pointer border-b-
150 |                 fichaTab === t ? "border-primary text-primary" : "border-transparent text-muted-foreground"
151 |               }`}>
152 |               {t === "detalle" ? "DETALLE" : "RUTA Y PUNTOS"}
153 |             </button>
154 |           ))}
155 |         </div>
156 |
157 |         <div className="p-6 space-y-5">
158 |           {fichaTab === "ruta" ? (
159 |             <RutaPuntosTab camId={cam.id} patente={cam.patente} />
160 |           ) : (
161 |           <>
162 |             <div
163 |               className="bg-background border border-border p-4 flex justify-between items-center"
164 |               style={{ borderLeftWidth: 4, borderLeftColor: cam.estado === "CRITICO" ? "#FF2D4A" : cam.estado === "ALERTA"
165 |             >
166 |               <div>
167 |                 <div className="flex items-center gap-2.5 mb-1">
168 |                   <span className="text-xl font-mono font-bold text-primary" data-testid="text-patente">{cam.patente}</spa
169 |                   <StatusTag label={cam.estado} variant={getStatusVariant(cam.estado)} dot />
170 |                   {faena && <StatusTag label={faena.nombre} variant="custom" color={faena.color} />}
171 |                 </div>
172 |                 <div className="text-[11px] font-mono text-muted-foreground">
173 |                   {cam.modelo} · VIN: {cam.vin || "\u2014"} · Conductor: <span className="text-foreground">{cam.conductor
174 |                 </div>
175 |               </div>
176 |               <div className="text-right">
177 |                 <div className="text-[9px] font-mono text-muted-foreground mb-0.5">SYNC VOLVO CONNECT</div>
178 |                 <div className={`text-[11px] font-mono ${cam.syncOk ? "text-emerald-400" : "text-red-400"}`}>
179 |                   {cam.syncOk ? `%Ï ${cam.syncAt}` : "'  Sin sincronizar"}
180 |                 </div>
181 |               </div>
182 |             </div>
183 |
184 |             <div className="grid grid-cols-2 gap-4">
185 |               <div>
186 |                 <div className="text-[9px] font-mono text-primary tracking-[0.15em] uppercase mb-2.5">
187 |                   VOLVO CONNECT — ECU
188 |                 </div>
189 |                 <div className="grid grid-cols-2 gap-2 mb-3">
190 |                   {[
191 |                     { l: "Odometro", v: fK(cam.odometro), c: "#1A8FFF" },
192 |                     {
193 |                       l: "Rendimiento",
194 |                       v: cam.rendProm ? `${cam.rendProm} km/L` : "\u2014",
195 |                       c: cam.rendProm && percentiles && !percentiles.calibrando
196 |                         ? (getNivelPercentil(cam.rendProm, percentiles) === "EXCELENTE" ? "#00C87A"
197 |                           : getNivelPercentil(cam.rendProm, percentiles) === "BUENO" ? "#66D9A0"
198 |                           : getNivelPercentil(cam.rendProm, percentiles) === "NORMAL" ? "#60A5FA"
199 |                           : getNivelPercentil(cam.rendProm, percentiles) === "BAJO" ? "#FFB020"
200 |                           : "#FF2D4A")
201 |                         : cam.rendProm ? (cam.rendProm >= cam.metaKmL ? "#00C87A" : cam.rendProm >= cam.metaKmL * 0.9 ? "#
202 |                       sub: percentiles && !percentiles.calibrando && cam.rendProm
203 |                         ? `${getNivelPercentil(cam.rendProm, percentiles)} · P90: ${percentiles.p90.toFixed(2)}`
204 |                         : undefined
205 |                     },
206 |                     { l: "Vel. promedio", v: cam.velPromedio ? `${cam.velPromedio} km/h` : "\u2014" },
207 |                     { l: "Horas motor", v: `${fN(cam.horasMotor)} h` },
208 |                     { l: "Horas ralenti", v: `${fN(cam.horasRalenti)} h`, c: cam.pctRal && cam.pctRal > 15 ? "#FFB020" : u
209 |                     {
210 |                       l: "Ref. Percentil",
211 |                       v: percentiles && !percentiles.calibrando ? `P90: ${percentiles.p90.toFixed(1)}` : `Meta: ${cam.meta
```

```
212 |                     sub: percentiles && !percentiles.calibrando ? `P75: ${percentiles.p75.toFixed(1)} · P50: ${percentil
213 |                   },
214 |                 ].map((k, i) => (
215 |                   <div key={i} className="bg-background border border-border p-2.5" data-testid={`ecu-${k.l.toLowerCase(
216 |                     <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-0.5 uppercase">
217 |                       {k.l}
218 |                     </div>
219 |                     <div className="text-base font-mono font-bold" style={{ color: k.c || "hsl(var(--foreground))" }}>
220 |                       {k.v}
221 |                     </div>
222 |                     {k.sub && <div className="text-[9px] text-muted-foreground mt-0.5">{k.sub}</div>}
223 |                   </div>
224 |                 ))}
225 |               </div>
226 |
227 |               {cam.pctRal && cam.pctRal > 15 && (
228 |                 <div className="bg-amber-400/5 border border-amber-400/20 p-2.5 text-[11px] text-amber-400 font-mono">
229 |                   Ralenti excesivo ({cam.pctRal}%). Consumo sin avance. Revisar habitos de conduccion.
230 |                 </div>
231 |               )}
232 |             </div>
233 |
234 |             <div>
235 |               <div className="text-[9px] font-mono text-amber-400 tracking-[0.15em] uppercase mb-2.5">
236 |                 CUADRATURA TARJETA vs ECU
237 |               </div>
238 |               <div className="grid grid-cols-2 gap-2 mb-2.5">
239 |                 {[
240 |                   { l: "Total surtidor", v: fL(totS) },
241 |                   { l: "Total ECU", v: fL(totE), c: "#00C87A" },
242 |                   { l: "Litros desviados", v: fL(cam.litDesv), c: cam.litDesv > 0 ? "#FF2D4A" : "#00C87A" },
243 |                   { l: "Costo desviado", v: f$(cam.litDesv * PRECIO), c: cam.litDesv > 0 ? "#FF2D4A" : "#00C87A" },
244 |                 ].map((k, i) => (
245 |                   <div key={i} className="bg-background border border-border p-2.5">
246 |                     <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-0.5 uppercase">{k.l}
247 |                     <div className="text-[15px] font-mono font-bold" style={{ color: k.c || "hsl(var(--foreground))" }}>
248 |                   </div>
249 |                 ))}
250 |               </div>
251 |
252 |               <div className="bg-background border border-border">
253 |                 <div className="grid grid-cols-6 px-3 py-1.5 border-b border-border text-[8px] font-mono text-muted-fore
254 |                   <div>FECHA</div><div>SURT.</div><div>ECU</div><div>DIF</div><div>REND.</div><div>ESTADO</div>
255 |                 </div>
256 |                 {cam.cargasAnalizadas.map((c, i) => (
257 |                   <div
258 |                     key={c.id}
259 |                     className={`grid grid-cols-6 px-3 py-2 items-center ${i < cam.cargasAnalizadas.length - 1 ? "border-
260 |                     data-testid={`carga-row-${c.id}`}
261 |                   >
262 |                     <div className="text-[11px] font-mono text-muted-foreground">{c.fecha.slice(5)}</div>
263 |                     <div className="text-xs font-mono text-foreground">{c.litrosSurtidor}L</div>
264 |                     <div className="text-xs font-mono text-emerald-400">{c.litrosEcu}L</div>
265 |                     <div className={`text-xs font-mono font-bold ${c.estado === "CRITICO" ? "text-red-400" : c.estado ==
266 |                       {c.dif > 0 ? `+${c.dif}` : c.dif}L
267 |                     </div>
268 |                     <div className={`text-[11px] font-mono ${rendColor(c.rend, cam.metaKmL)}`}>
269 |                       {c.rend ?? "\u2014"}
270 |                     </div>
271 |                     <StatusTag label={c.estado} variant={getStatusVariant(c.estado)} />
272 |                   </div>
273 |                 ))}
274 |               </div>
275 |             </div>
276 |           </div>
277 |
278 |           {cam.estado === "CRITICO" && (
279 |             <div className="flex gap-2 justify-end pt-3 border-t border-border">
280 |               <Button variant="outline" size="sm" data-testid="button-marcar-revisado">Marcar revisado</Button>
281 |               <Button variant="secondary" size="sm" className="bg-amber-400/20 text-amber-400 border-amber-400/30" data-
282 |                 Notificar cliente
283 |               </Button>
284 |               <Button variant="destructive" size="sm" data-testid="button-escalar">
285 |                 Escalar a gerencia
286 |               </Button>
287 |             </div>
288 |           )}
289 |           </>
290 |           )}
291 |         </div>
292 |       </DialogContent>
293 |     </Dialog>
294 |   );
295 | }
296 |
```

client/src/components/kpi-card.tsx

41 lineas

```
1 | import { cn } from "@/lib/utils";
2 |
3 | interface KpiCardProps {
4 |   label: string;
5 |   value: string | number;
6 |   unit?: string;
7 |   sub?: string;
8 |   color?: string;
9 |   icon?: React.ReactNode;
10 |   className?: string;
11 | }
12 |
13 | export function KpiCard({ label, value, unit, sub, color, icon, className }: KpiCardProps) {
14 |   return (
15 |     <div
16 |       data-testid={`kpi-${label.toLowerCase().replace(/\s/g, "-")}`}
17 |       className={cn(
18 |         "relative bg-card border border-border p-4 overflow-hidden",
19 |         className
20 |       )}
21 |       style={{ borderTopWidth: 2, borderTopColor: color }}
22 |     >
23 |       {icon && (
24 |         <div className="absolute right-3 top-2.5 text-xl opacity-10">
25 |           {icon}
26 |         </div>
27 |       )}
28 |       <div className="text-[9px] font-mono text-muted-foreground tracking-[0.2em] uppercase mb-1.5">
29 |         {label}
30 |       </div>
31 |       <div className="font-mono font-bold text-2xl leading-none" style={{ color }}>
32 |         {value}
33 |         {unit && <span className="text-[10px] text-muted-foreground ml-1">{unit}</span>}
34 |       </div>
35 |       {sub && (
36 |         <div className="text-[10px] text-muted-foreground font-mono mt-1.5">{sub}</div>
37 |       )}
38 |     </div>
39 |   );
40 | }
41 |
```

client/src/components/status-tag.tsx

59 lineas

```
1 | import { cn } from "@/lib/utils";
2 |
3 | interface StatusTagProps {
4 |   label: string;
5 |   variant?: "critico" | "alerta" | "ok" | "info" | "custom";
6 |   color?: string;
7 |   dot?: boolean;
8 |   className?: string;
9 | }
10 |
11 | const variantStyles = {
12 |   critico: "bg-red-500/10 border-red-500/25 text-red-400",
13 |   alerta: "bg-amber-400/10 border-amber-400/25 text-amber-400",
14 |   ok: "bg-emerald-400/10 border-emerald-400/25 text-emerald-400",
15 |   info: "bg-blue-500/10 border-blue-500/25 text-blue-400",
16 |   custom: "",
17 | };
18 |
19 | export function StatusTag({ label, variant = "info", color, dot, className }: StatusTagProps) {
20 |   const isCustom = variant === "custom" && color;
21 |
22 |   return (
23 |     <span
24 |       data-testid={`tag-${label.toLowerCase().replace(/\s/g, "-")}`}
25 |       className={cn(
26 |         "inline-flex items-center gap-1.5 px-2 py-0.5 text-[9px] font-mono font-bold tracking-[0.15em] uppercase border
27 |         !isCustom && variantStyles[variant],
28 |         className
29 |       )}
30 |       style={isCustom ? {
31 |         backgroundColor: `${color}1A`,
32 |         borderColor: `${color}40`,
33 |         color: color,
34 |       } : undefined}
35 |     >
36 |       {dot && (
37 |         <span
38 |           className={cn(
39 |             "w-1.5 h-1.5 rounded-full flex-shrink-0",
40 |             !isCustom && variant === "critico" && "bg-red-500",
41 |             !isCustom && variant === "alerta" && "bg-amber-400",
42 |             !isCustom && variant === "ok" && "bg-emerald-400",
43 |             !isCustom && variant === "info" && "bg-blue-500",
44 |           )}
45 |           style={isCustom ? { backgroundColor: color } : undefined}
46 |         />
47 |       )}
48 |       {label}
49 |     </span>
50 |   );
51 | }
52 |
53 | export function getStatusVariant(estado: string): StatusTagProps["variant"] {
54 |   if (estado === "CRITICO") return "critico";
55 |   if (estado === "ALERTA") return "alerta";
56 |   if (estado === "OK") return "ok";
57 |   return "info";
58 | }
59 |
```

client/src/components/volvo-truck-modal.tsx

457 lineas

```
1 | import { Dialog, DialogContent, DialogHeader, DialogTitle, DialogDescription } from "@/components/ui/dialog";
2 | import { StatusTag } from "@/components/status-tag";
3 | import { Skeleton } from "@/components/ui/skeleton";
4 | import { useQuery } from "@tanstack/react-query";
5 | import type { Camion, Faena } from "@shared/schema";
6 | import { fN, fK } from "@/lib/fuel-utils";
7 | import { MapPin, Gauge, Fuel, User, Thermometer, Zap, Navigation, ChevronDown, ChevronUp } from "lucide-react";
8 | import { useState, useMemo, useEffect, useRef } from "react";
9 | import L from "leaflet";
10 | import "leaflet/dist/leaflet.css";
11 |
12 | interface UnifiedStatus {
13 |   vin: string;
14 |   createdDateTime: string | null;
15 |   gps: {
16 |     latitude: number | null;
17 |     longitude: number | null;
18 |     altitude: number | null;
19 |     heading: number | null;
20 |     speed: number | null;
21 |     positionDateTime: string | null;
22 |   };
23 |   fuelLevel: number | null;
24 |   engineHours: number | null;
25 |   totalDistance: number | null;
26 |   totalFuelUsed: number | null;
27 |   engineSpeed: number | null;
28 |   wheelBasedSpeed: number | null;
29 |   ambientTemperature: number | null;
30 |   driverId: string | null;
31 |   driverWorkingState: string | null;
32 |   catalystFuelLevel: number | null;
33 |   grossWeight: number | null;
34 | }
35 |
36 | interface VolvoTruckModalProps {
37 |   camion: Camion;
38 |   status: UnifiedStatus | null;
39 |   faena: Faena | null;
40 |   open: boolean;
41 |   onClose: () => void;
42 | }
43 |
44 | function formatDateTime(iso?: string): string {
45 |   if (!iso) return "\u2014";
46 |   try {
47 |     const d = new Date(iso);
48 |     return d.toLocaleString("es-CL", { timeZone: "America/Santiago", day: "2-digit", month: "2-digit", year: "numeric",
49 |   } catch {
50 |     return iso;
51 |   }
52 | }
53 |
54 | function formatCoord(val?: number | null, decimals = 6): string {
55 |   if (val == null) return "\u2014";
56 |   return val.toFixed(decimals);
57 | }
58 |
59 | function DataCell({ label, value, color, sub }: { label: string; value: string; color?: string; sub?: string }) {
60 |   return (
61 |     <div className="bg-background border border-border p-2.5" data-testid={`volvo-${label.toLowerCase().replace(/\s/g, "
62 |       <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] mb-0.5 uppercase">
63 |         {label}
64 |       </div>
65 |       <div className="text-base font-mono font-bold" style={{ color: color || "hsl(var(--foreground))" }}>
66 |         {value}
67 |       </div>
68 |       {sub && <div className="text-[9px] text-muted-foreground mt-0.5">{sub}</div>}
69 |     </div>
70 |   );
71 | }
72 |
73 | function SectionHeader({ icon, label, color }: { icon: React.ReactNode; label: string; color: string }) {
74 |   return (
75 |     <div className="flex items-center gap-2 mb-2.5">
76 |       <span style={{ color }} className="opacity-70">{icon}</span>
77 |       <span className="text-[9px] font-mono tracking-[0.15em] uppercase font-bold" style={{ color }}>
78 |         {label}
79 |       </span>
80 |     </div>
81 |   );
82 | }
83 |
84 | interface TruckLocationData {
85 |   patente: string;
86 |   vin: string | null;
87 |   modelo: string;
88 |   currentGps: { latitude: number | null; longitude: number | null; speed: number | null; positionDateTime: string | null
89 |   locations: { fecha: string; lugar: string | null; litros: number; odometro: number | null; kmRecorrido: number | null;
90 |   lugarSummary: { lugar: string; count: number; totalLitros: number; lastDate: string }[];
91 |   totalCargas: number;
92 |   periodo: { from: string; to: string };
93 | }
94 |
95 | function LeafletMap({ latitude, longitude, speed, patente, locations }: {
96 |   latitude: number;
97 |   longitude: number;
98 |   speed: number | null;
99 |   patente: string;
100 |   locations?: { fecha: string; lugar: string | null; litros: number; lat?: number; lon?: number }[];
101 | }) {
102 |   const mapRef = useRef<HTMLDivElement>(null);
103 |   const mapInstanceRef = useRef<L.Map | null>(null);
104 |
```

```
105 |   useEffect(() => {
106 |     if (!mapRef.current) return;
107 |
108 |     if (mapInstanceRef.current) {
109 |       mapInstanceRef.current.remove();
110 |       mapInstanceRef.current = null;
111 |     }
112 |
113 |     const map = L.map(mapRef.current, {
114 |       zoomControl: true,
115 |       attributionControl: false,
116 |     }).setView([latitude, longitude], 13);
117 |
118 |     L.tileLayer("https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png", {
119 |       maxZoom: 19,
120 |     }).addTo(map);
121 |
122 |     const truckIcon = L.divIcon({
123 |       className: "custom-truck-marker",
124 |       html: `<div style="
125 |         width: 32px; height: 32px; background: #1A8FFF; border: 2px solid #fff;
126 |         border-radius: 50%; display: flex; align-items: center; justify-content: center;
127 |         box-shadow: 0 2px 8px rgba(0,0,0,0.4); position: relative;
128 |       ">
129 |         <svg width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="white" stroke-width="2.5" stroke-linecap="ro
130 |           <rect x="1" y="3" width="15" height="13" rx="2"/><polygon points="16 8 20 8 23 11 23 16 16 16 16 8"/><circle c
131 |         </svg>
132 |         <div style="position:absolute;top:-8px;right:-8px;width:12px;height:12px;background:${speed != null && speed > 0
133 |       </div>`,
134 |       iconSize: [32, 32],
135 |       iconAnchor: [16, 16],
136 |     });
137 |
138 |     const esc = (s: string) => s.replace(/&/g, "&amp;").replace(/</g, "&lt;").replace(/>/g, "&gt;").replace(/"/g, "&quot
139 |     L.marker([latitude, longitude], { icon: truckIcon })
140 |       .addTo(map)
141 |       .bindPopup(`<div style="font-family:monospace;font-size:11px;"><strong>${esc(patente)}</strong><br/>Lat: ${latitud
142 |
143 |     mapInstanceRef.current = map;
144 |
145 |     setTimeout(() => map.invalidateSize(), 200);
146 |
147 |     return () => {
148 |       if (mapInstanceRef.current) {
149 |         mapInstanceRef.current.remove();
150 |         mapInstanceRef.current = null;
151 |       }
152 |     };
153 |   }, [latitude, longitude, speed, patente]);
154 |
155 |   return (
156 |     <div ref={mapRef} className="w-full h-[260px] rounded-sm" data-testid="leaflet-map" />
157 |   );
158 | }
159 |
160 | function TruckLocationSection({ patente, open }: { patente: string; open: boolean }) {
161 |   const [showAll, setShowAll] = useState(false);
162 |   const from = useMemo(() => {
163 |     const d = new Date();
164 |     d.setDate(d.getDate() - 30);
165 |     return d.toISOString().slice(0, 10);
166 |   }, []);
167 |   const to = useMemo(() => new Date().toISOString().slice(0, 10), []);
168 |
169 |   const { data, isLoading, isError } = useQuery<TruckLocationData>({
170 |     queryKey: [`/api/volvo/truck-locations/${patente}?from=${from}&to=${to}`],
171 |     enabled: open,
172 |   });
173 |
174 |   if (isLoading) {
175 |     return (
176 |       <div className="space-y-2">
177 |         <Skeleton className="h-4 w-48" />
178 |         <Skeleton className="h-[200px] w-full" />
179 |         <Skeleton className="h-20 w-full" />
180 |       </div>
181 |     );
182 |   }
183 |
184 |   if (isError) {
185 |     return (
186 |       <div className="bg-red-500/10 border border-red-500/30 p-4">
187 |         <SectionHeader icon={<Navigation className="w-3.5 h-3.5" />} label="Geolocalizacion - Lugares del Mes" color="#F
188 |         <div className="text-[11px] font-mono text-red-400 text-center py-2" data-testid="text-geo-error">
189 |           Error al cargar datos de geolocalizacion. Verifique la conexion.
190 |         </div>
191 |       </div>
192 |     );
193 |   }
194 |
195 |   const gps = data?.currentGps;
196 |   const hasGps = gps && gps.latitude != null && gps.longitude != null;
197 |   const hasCargas = data && data.totalCargas > 0;
198 |   const visibleLocations = hasCargas ? (showAll ? data.locations : data.locations.slice(0, 8)) : [];
199 |
200 |   if (!hasGps && !hasCargas) {
201 |     return (
202 |       <div className="bg-background border border-border p-4">
203 |         <SectionHeader icon={<Navigation className="w-3.5 h-3.5" />} label="Geolocalizacion - Lugares del Mes" color="#1
204 |         <div className="text-[11px] font-mono text-muted-foreground text-center py-4">
205 |           Sin datos de ubicaciones ni GPS para el periodo {from} a {to}
206 |         </div>
207 |       </div>
208 |     );
209 |   }
210 |
211 |   return (
```

```
212 |     <div className="space-y-3">
213 |       <SectionHeader icon={<Navigation className="w-3.5 h-3.5" />} label="Geolocalizacion - Lugares del Mes" color="#1A8
214 |
215 |       {hasGps && (
216 |         <div className="border border-border overflow-hidden" data-testid="map-container">
217 |           <div className="bg-background px-3 py-1.5 border-b border-border flex items-center justify-between">
218 |             <div className="flex items-center gap-2">
219 |               <MapPin className="w-3 h-3 text-emerald-400" />
220 |               <span className="text-[9px] font-mono text-emerald-400 tracking-[0.1em] uppercase font-bold">Posicion actu
221 |             </div>
222 |             <span className="text-[9px] font-mono text-muted-foreground">
223 |               {gps.latitude!.toFixed(5)}, {gps.longitude!.toFixed(5)}
224 |               {gps.speed != null && gps.speed > 0 && <span className="ml-2 text-emerald-400">{Math.round(gps.speed)} km/
225 |             </span>
226 |           </div>
227 |           <LeafletMap
228 |             latitude={gps.latitude!}
229 |             longitude={gps.longitude!}
230 |             speed={gps.speed}
231 |             patente={patente}
232 |           />
233 |           {gps.positionDateTime && (
234 |             <div className="bg-background px-3 py-1 border-t border-border">
235 |               <span className="text-[8px] font-mono text-muted-foreground">Ultima posicion: {formatDateTime(gps.position
236 |             </div>
237 |           )}
238 |         </div>
239 |       )}
240 |
241 |       {hasCargas && data!.lugarSummary.length > 0 && (
242 |         <div className="bg-background border border-border" data-testid="lugar-summary">
243 |           <div className="px-3 py-2 border-b border-border">
244 |             <span className="text-[9px] font-mono text-amber-400 tracking-[0.1em] uppercase font-bold">
245 |               Resumen lugares de carga ({data!.periodo.from} a {data!.periodo.to})
246 |             </span>
247 |           </div>
248 |           <div className="divide-y divide-border/40">
249 |             {data!.lugarSummary.map((s, i) => (
250 |               <div key={i} className="grid grid-cols-[1fr_3.5rem_5rem_6rem] gap-2 px-3 py-1.5 items-center" data-testid=
251 |                 <div className="flex items-center gap-1.5 min-w-0">
252 |                   <MapPin className="w-3 h-3 text-primary flex-shrink-0" />
253 |                   <span className="text-[11px] font-mono font-bold text-foreground truncate">{s.lugar}</span>
254 |                 </div>
255 |                 <span className="text-[10px] font-mono text-muted-foreground text-right">{s.count}x</span>
256 |                 <span className="text-[10px] font-mono font-bold text-amber-400 text-right">{fN(Math.round(s.totalLitros
257 |                 <span className="text-[9px] font-mono text-muted-foreground text-right">{formatDateTime(s.lastDate).spli
258 |               </div>
259 |             ))}
260 |           </div>
261 |         </div>
262 |       )}
263 |
264 |       {hasCargas && (
265 |         <div className="bg-background border border-border" data-testid="location-detail-table">
266 |           <div className="px-3 py-2 border-b border-border flex items-center justify-between">
267 |             <span className="text-[9px] font-mono text-primary tracking-[0.1em] uppercase font-bold">
268 |               Detalle cargas ({data!.totalCargas} registros)
269 |             </span>
270 |             {data!.locations.length > 8 && (
271 |               <button
272 |                 onClick={() => setShowAll(!showAll)}
273 |                 className="flex items-center gap-1 text-[9px] font-mono text-primary hover:text-primary/80 cursor-pointe
274 |                 data-testid="btn-toggle-locations"
275 |               >
276 |                 {showAll ? "Ver menos" : `Ver todos (${data!.totalCargas})`}
277 |                 {showAll ? <ChevronUp className="w-3 h-3" /> : <ChevronDown className="w-3 h-3" />}
278 |               </button>
279 |             )}
280 |           </div>
281 |           <div className="grid grid-cols-[5.5rem_1fr_4rem_5rem_5.5rem] gap-1.5 px-3 py-1.5 border-b border-border">
282 |             <span className="text-[8px] font-mono text-muted-foreground tracking-[0.15em]">FECHA</span>
283 |             <span className="text-[8px] font-mono text-muted-foreground tracking-[0.15em]">LUGAR</span>
284 |             <span className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] text-right">LITROS</span>
285 |             <span className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] text-right">KM REC.</span>
286 |             <span className="text-[8px] font-mono text-muted-foreground tracking-[0.15em]">CONDUCTOR</span>
287 |           </div>
288 |           <div className="divide-y divide-border/30 max-h-[300px] overflow-y-auto">
289 |             {visibleLocations.map((loc, i) => (
290 |               <div key={`${loc.numGuia}-${i}`} className="grid grid-cols-[5.5rem_1fr_4rem_5rem_5.5rem] gap-1.5 px-3 py-1
291 |                 <span className="text-[10px] font-mono text-muted-foreground">{formatDateTime(loc.fecha).split(",")[0]}<
292 |                 <div className="flex items-center gap-1 min-w-0">
293 |                   <MapPin className="w-2.5 h-2.5 text-primary/60 flex-shrink-0" />
294 |                   <span className="text-[10px] font-mono text-foreground truncate">{loc.lugar || "\u2014"}</span>
295 |                 </div>
296 |                 <span className="text-[10px] font-mono font-bold text-amber-400 text-right">{fN(Math.round(loc.litros))}
297 |                 <span className="text-[10px] font-mono text-foreground text-right">{loc.kmRecorrido != null ? fN(Math.ro
298 |                 <span className="text-[9px] font-mono text-muted-foreground truncate">{loc.conductor || "\u2014"}</span>
299 |               </div>
300 |             ))}
301 |           </div>
302 |         </div>
303 |       )}
304 |     </div>
305 |   );
306 | }
307 |
308 | export function VolvoTruckModal({ camion, status: passedStatus, faena, open, onClose }: VolvoTruckModalProps) {
309 |   const { data: liveStatus, isLoading, isError } = useQuery<UnifiedStatus>({
310 |     queryKey: ["/api/volvo/vehicle-status", camion.vin],
311 |     enabled: open && !!camion.vin,
312 |   });
313 |
314 |   const status = liveStatus || passedStatus;
315 |   const gps = status?.gps;
316 |
317 |   const totalDistKm = status?.totalDistance != null ? Math.round(status.totalDistance / 1000) : null;
318 |   const totalEngineHours = status?.engineHours != null ? Math.round(status.engineHours) : null;
```

```
319 |   const totalFuelUsed = status?.totalFuelUsed != null ? Math.round(status.totalFuelUsed) : null;
320 |
321 |   return (
322 |     <Dialog open={open} onOpenChange={() => onClose()}>
323 |       <DialogContent className="max-w-[900px] bg-card border-border p-0 gap-0 overflow-y-auto max-h-[92vh]">
324 |         <DialogHeader className="p-6 pb-0">
325 |           <DialogTitle className="text-sm font-mono font-bold tracking-[0.15em] uppercase text-foreground">
326 |             Volvo Connect — {camion.patente}
327 |           </DialogTitle>
328 |           <DialogDescription className="sr-only">Datos telematicos Volvo Connect para camion {camion.patente}</DialogDes
329 |         </DialogHeader>
330 |
331 |         <div className="p-6 space-y-5">
332 |           <div
333 |             className="bg-background border border-border p-4 flex justify-between items-start gap-4 flex-wrap"
334 |             style={{ borderLeftWidth: 4, borderLeftColor: faena?.color || "#1A8FFF" }}
335 |           >
336 |             <div>
337 |               <div className="flex items-center gap-2.5 mb-1 flex-wrap">
338 |                 <span className="text-xl font-mono font-bold text-primary" data-testid="text-volvo-patente">{camion.pate
339 |                 {faena && <StatusTag label={faena.nombre} variant="custom" color={faena.color} />}
340 |                 {camion.syncOk ? (
341 |                   <StatusTag label="SYNC OK" variant="ok" dot />
342 |                 ) : (
343 |                   <StatusTag label="SIN SYNC" variant="critico" dot />
344 |                 )}
345 |               </div>
346 |               <div className="text-[11px] font-mono text-muted-foreground">
347 |                 {camion.modelo} · VIN: <span className="text-foreground" data-testid="text-volvo-vin">{camion.vin || "\u
348 |               </div>
349 |             </div>
350 |             <div className="text-right">
351 |               <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] uppercase mb-0.5">ULTIMA SYNC
352 |               <div className={`text-[11px] font-mono ${camion.syncOk ? "text-emerald-400" : "text-red-400"}`} data-testi
353 |                 {camion.syncAt || "Sin sincronizar"}
354 |               </div>
355 |               {status?.createdDateTime && (
356 |                 <>
357 |                   <div className="text-[8px] font-mono text-muted-foreground tracking-[0.15em] uppercase mb-0.5 mt-1.5">
358 |                   <div className="text-[11px] font-mono text-blue-400" data-testid="text-volvo-rfms-time">
359 |                     {formatDateTime(status.createdDateTime)}
360 |                   </div>
361 |                 </>
362 |               )}
363 |             </div>
364 |           </div>
365 |
366 |           {isLoading && (
367 |             <div className="grid grid-cols-2 gap-4">
368 |               <div className="space-y-2">
369 |                 <Skeleton className="h-4 w-40" />
370 |                 <div className="grid grid-cols-2 gap-2">
371 |                   {Array.from({ length: 4 }).map((_, i) => <Skeleton key={i} className="h-16" />)}
372 |                 </div>
373 |               </div>
374 |               <div className="space-y-2">
375 |                 <Skeleton className="h-4 w-40" />
376 |                 <div className="grid grid-cols-2 gap-2">
377 |                   {Array.from({ length: 4 }).map((_, i) => <Skeleton key={i} className="h-16" />)}
378 |                 </div>
379 |               </div>
380 |             </div>
381 |           )}
382 |
383 |           {!isLoading && (
384 |             <div className="grid grid-cols-2 gap-4">
385 |               <div>
386 |                 <SectionHeader icon={<MapPin className="w-3.5 h-3.5" />} label="Posicion GPS" color="#1A8FFF" />
387 |                 <div className="grid grid-cols-2 gap-2 mb-4">
388 |                   <DataCell label="Latitud" value={formatCoord(gps?.latitude)} color="#1A8FFF" />
389 |                   <DataCell label="Longitud" value={formatCoord(gps?.longitude)} color="#1A8FFF" />
390 |                   <DataCell label="Altitud" value={gps?.altitude != null ? `${Math.round(gps.altitude)} m` : "\u2014"} /
391 |                   <DataCell label="Rumbo" value={gps?.heading != null ? `${Math.round(gps.heading)}\u00b0` : "\u2014"} /
392 |                   <DataCell label="Velocidad GPS" value={gps?.speed != null ? `${Math.round(gps.speed)} km/h` : "\u2014"
393 |                   <DataCell label="Ultima posicion" value={formatDateTime(gps?.positionDateTime)} sub="Hora GPS" />
394 |                 </div>
395 |
396 |                 <SectionHeader icon={<User className="w-3.5 h-3.5" />} label="Conductor" color="#A78BFA" />
397 |                 <div className="grid grid-cols-2 gap-2 mb-4">
398 |                   <DataCell label="Conductor" value={camion.conductor || status?.driverId || "\u2014"} color="#A78BFA" /
399 |                   <DataCell label="Estado trabajo" value={status?.driverWorkingState || "\u2014"} color={status?.driverW
400 |                 </div>
401 |
402 |                 <SectionHeader icon={<Thermometer className="w-3.5 h-3.5" />} label="Ambiente" color="#38BDF8" />
403 |                 <div className="grid grid-cols-2 gap-2">
404 |                   <DataCell label="Temp. ambiente" value={status?.ambientTemperature != null ? `${status.ambientTemperat
405 |                   <DataCell label="AdBlue" value={status?.catalystFuelLevel != null ? `${status.catalystFuelLevel}%` : "
406 |                 </div>
407 |               </div>
408 |
409 |               <div>
410 |                 <SectionHeader icon={<Gauge className="w-3.5 h-3.5" />} label="Motor & Distancia" color="#00C87A" />
411 |                 <div class="grid grid-cols-2 gap-2 mb-4">
412 |                   <DataCell label="Distancia total" value={totalDistKm != null ? fK(totalDistKm) : (camion.odometro ? fK
413 |                   <DataCell label="Horas motor" value={totalEngineHours != null ? `${fN(totalEngineHours)} h` : (camion.
414 |                   <DataCell label="Combustible usado" value={totalFuelUsed != null ? `${fN(totalFuelUsed)} L` : "\u2014"
415 |                   <DataCell label="RPM motor" value={status?.engineSpeed != null ? `${fN(Math.round(status.engineSpeed))
416 |                 </div>
417 |
418 |                 <SectionHeader icon={<Fuel className="w-3.5 h-3.5" />} label="Combustible" color="#FFB020" />
419 |                 <div class="grid grid-cols-2 gap-2 mb-4">
420 |                   <DataCell label="Nivel tanque" value={status?.fuelLevel != null ? `${status.fuelLevel}%` : "\u2014"} c
421 |                   <DataCell label="Vel. rueda" value={status?.wheelBasedSpeed != null ? `${Math.round(status.wheelBasedS
422 |                 </div>
423 |
424 |                 <SectionHeader icon={<Zap className="w-3.5 h-3.5" />} label="Peso & Tacografo" color="#F97316" />
425 |                 <div class="grid grid-cols-2 gap-2">
```

```
426 |                   <DataCell label="Peso bruto" value={status?.grossWeight != null ? `${fN(Math.round(status.grossWeight)
427 |                   <DataCell label="Vel. tacografo" value={"\u2014"} />
428 |                 </div>
429 |               </div>
430 |             </div>
431 |           )}
432 |
433 |           <TruckLocationSection patente={camion.patente} open={open} />
434 |
435 |           {isError && (
436 |             <div className="bg-red-500/10 border border-red-500/30 p-3 text-[11px] text-red-400 font-mono" data-testid="
437 |               Error al consultar datos rFMS en tiempo real. Se muestran datos en cache si estan disponibles.
438 |             </div>
439 |           )}
440 |
441 |           {!isLoading && !isError && !status && camion.vin && (
442 |             <div className="bg-amber-400/5 border border-amber-400/20 p-3 text-[11px] text-amber-400 font-mono" data-tes
443 |               Sin datos rFMS disponibles para VIN {camion.vin}. Verifique que el vehiculo esta registrado en Volvo Conne
444 |             </div>
445 |           )}
446 |
447 |           {!camion.vin && (
448 |             <div className="bg-blue-500/5 border border-blue-500/20 p-3 text-[11px] text-blue-400 font-mono" data-testid
449 |               Este camion no tiene VIN registrado. Sincronice con Volvo Connect o ingrese el VIN manualmente para obtene
450 |             </div>
451 |           )}
452 |         </div>
453 |       </DialogContent>
454 |     </Dialog>
455 |   );
456 | }
457 |
```

client/src/hooks/use-toast.ts

192 lineas

```
1 | import * as React from "react"
2 |
3 | import type {
4 |   ToastActionElement,
5 |   ToastProps,
6 | } from "@/components/ui/toast"
7 |
8 | const TOAST_LIMIT = 1
9 | const TOAST_REMOVE_DELAY = 1000000
10 |
11 | type ToasterToast = ToastProps & {
12 |   id: string
13 |   title?: React.ReactNode
14 |   description?: React.ReactNode
15 |   action?: ToastActionElement
16 | }
17 |
18 | const actionTypes = {
19 |   ADD_TOAST: "ADD_TOAST",
20 |   UPDATE_TOAST: "UPDATE_TOAST",
21 |   DISMISS_TOAST: "DISMISS_TOAST",
22 |   REMOVE_TOAST: "REMOVE_TOAST",
23 | } as const
24 |
25 | let count = 0
26 |
27 | function genId() {
28 |   count = (count + 1) % Number.MAX_SAFE_INTEGER
29 |   return count.toString()
30 | }
31 |
32 | type ActionType = typeof actionTypes
33 |
34 | type Action =
35 |   | {
36 |       type: ActionType["ADD_TOAST"]
37 |       toast: ToasterToast
38 |     }
39 |   | {
40 |       type: ActionType["UPDATE_TOAST"]
41 |       toast: Partial<ToasterToast>
42 |     }
43 |   | {
44 |       type: ActionType["DISMISS_TOAST"]
45 |       toastId?: ToasterToast["id"]
46 |     }
47 |   | {
48 |       type: ActionType["REMOVE_TOAST"]
49 |       toastId?: ToasterToast["id"]
50 |     }
51 |
52 | interface State {
53 |   toasts: ToasterToast[]
54 | }
55 |
56 | const toastTimeouts = new Map<string, ReturnType<typeof setTimeout>>()
57 |
58 | const addToRemoveQueue = (toastId: string) => {
59 |   if (toastTimeouts.has(toastId)) {
60 |     return
61 |   }
62 |
63 |   const timeout = setTimeout(() => {
64 |     toastTimeouts.delete(toastId)
65 |     dispatch({
66 |       type: "REMOVE_TOAST",
67 |       toastId: toastId,
68 |     })
69 |   }, TOAST_REMOVE_DELAY)
70 |
71 |   toastTimeouts.set(toastId, timeout)
72 | }
73 |
74 | export const reducer = (state: State, action: Action): State => {
75 |   switch (action.type) {
76 |     case "ADD_TOAST":
77 |       return {
78 |         ...state,
79 |         toasts: [action.toast, ...state.toasts].slice(0, TOAST_LIMIT),
80 |       }
81 |
82 |     case "UPDATE_TOAST":
83 |       return {
84 |         ...state,
85 |         toasts: state.toasts.map((t) =>
86 |           t.id === action.toast.id ? { ...t, ...action.toast } : t
87 |         ),
88 |       }
89 |
90 |     case "DISMISS_TOAST": {
91 |       const { toastId } = action
92 |
93 |       // ! Side effects ! - This could be extracted into a dismissToast() action,
94 |       // but I'll keep it here for simplicity
95 |       if (toastId) {
96 |         addToRemoveQueue(toastId)
97 |       } else {
98 |         state.toasts.forEach((toast) => {
99 |           addToRemoveQueue(toast.id)
100 |         })
101 |       }
102 |
103 |       return {
104 |         ...state,
```

```
105 |         toasts: state.toasts.map((t) =>
106 |           t.id === toastId || toastId === undefined
107 |             ? {
108 |                 ...t,
109 |                 open: false,
110 |               }
111 |             : t
112 |         ),
113 |       }
114 |     }
115 |     case "REMOVE_TOAST":
116 |       if (action.toastId === undefined) {
117 |         return {
118 |           ...state,
119 |           toasts: [],
120 |         }
121 |       }
122 |       return {
123 |         ...state,
124 |         toasts: state.toasts.filter((t) => t.id !== action.toastId),
125 |       }
126 |   }
127 | }
128 |
129 | const listeners: Array<(state: State) => void> = []
130 |
131 | let memoryState: State = { toasts: [] }
132 |
133 | function dispatch(action: Action) {
134 |   memoryState = reducer(memoryState, action)
135 |   listeners.forEach((listener) => {
136 |     listener(memoryState)
137 |   })
138 | }
139 |
140 | type Toast = Omit<ToasterToast, "id">
141 |
142 | function toast({ ...props }: Toast) {
143 |   const id = genId()
144 |
145 |   const update = (props: ToasterToast) =>
146 |     dispatch({
147 |       type: "UPDATE_TOAST",
148 |       toast: { ...props, id },
149 |     })
150 |   const dismiss = () => dispatch({ type: "DISMISS_TOAST", toastId: id })
151 |
152 |   dispatch({
153 |     type: "ADD_TOAST",
154 |     toast: {
155 |       ...props,
156 |       id,
157 |       open: true,
158 |       onOpenChange: (open) => {
159 |         if (!open) dismiss()
160 |       },
161 |     },
162 |   })
163 |
164 |   return {
165 |     id: id,
166 |     dismiss,
167 |     update,
168 |   }
169 | }
170 |
171 | function useToast() {
172 |   const [state, setState] = React.useState<State>(memoryState)
173 |
174 |   React.useEffect(() => {
175 |     listeners.push(setState)
176 |     return () => {
177 |       const index = listeners.indexOf(setState)
178 |       if (index > -1) {
179 |         listeners.splice(index, 1)
180 |       }
181 |     }
182 |   }, [state])
183 |
184 |   return {
185 |     ...state,
186 |     toast,
187 |     dismiss: (toastId?: string) => dispatch({ type: "DISMISS_TOAST", toastId }),
188 |   }
189 | }
190 |
191 | export { useToast, toast }
192 |
```

client/src/hooks/use-mobile.tsx

20 lineas

```
1 | import * as React from "react"
2 |
3 | const MOBILE_BREAKPOINT = 768
4 |
5 | export function useIsMobile() {
6 |   const [isMobile, setIsMobile] = React.useState<boolean | undefined>(undefined)
7 |
8 |   React.useEffect(() => {
9 |     const mql = window.matchMedia(`(max-width: ${MOBILE_BREAKPOINT - 1}px)`)
10 |     const onChange = () => {
11 |       setIsMobile(window.innerWidth < MOBILE_BREAKPOINT)
12 |     }
13 |     mql.addEventListener("change", onChange)
14 |     setIsMobile(window.innerWidth < MOBILE_BREAKPOINT)
15 |     return () => mql.removeEventListener("change", onChange)
16 |   }, [])
17 |
18 |   return !!isMobile
19 | }
20 |
```